Mixed Holomorphic–Topological Strings

Boundary reductions, Hamiltonian actions and quantum observables

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Abstract

Bulk fields act on boundary observables through products, brackets and higher homotopies. Reconstructing the bulk from that action requires these operations together with their support, cyclic pairing and quantum state. We study the comparison through explicit polar reductions and Hamiltonian matrix probes.

On a sphere with a marked double pole, a first-jet Neumann condition leaves a two-dimensional cyclic cohomology algebra. We construct its continuous cyclic contraction and the nonlinear map from matrix connections to residual observables. The full classical reduction retains gauge stabilizers and boundary transgression terms.

Finite cyclic reductions carry higher operations. Paired smooth cutoffs and positive-scale heat tensors give explicit quantum contractions and pole contacts. Normalized finite BV elimination composes through connected cumulants. Smooth Gaussian responses converge uniformly on compact sets of matrix backgrounds. Their regularized determinants and supported normal homotopies have specified coefficient domains.

Hamiltonian probes require their own coefficient complexes. Semi-free resolutions retain the commutativity relation before passing to matrices. Coherent derivation actions lift Hamiltonian vector fields. Central rank, negative cohomology and quantum torsion distinguish the resulting recipients. Continuous, cyclic and supported comparisons must preserve these data.

At rank two, a framed cyclic chart admits an integral quantum Hamiltonian realization. Global compatibility and formal extension in the framing parameter have separate, explicit obstructions. The coordinate comparison with polyvectors precedes the Hochschild comparison: ordinary antisymmetrization applies on the zero-Poisson fiber, and a nonzero Poisson tensor requires twisted formality.

Independently prepared compact gauge and Gaussian states have exact sewing formulas. A scalar interacting preparation exists at fixed positive smoothing and is unitarily identified with a scalar Fock interaction on its full form domain. Its cutoff limit converges in trace norm. The identification of these preparations with the native quantum pushforward, the interacting continuum master equation and the full sourced closed-field comparison remain open. The required comparison must preserve observable insertions and cutting, beyond equality of partition functions or formal coordinate algebras.

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Chapter 1

Fields, boundaries and comparison

A bulk field can act on a brane without being determined by that action. Recovering the field theory requires enough boundary observables to detect its differential, products and higher operations. Traces of matrices give concrete tests of such an action. Boundary conditions and quantum contractions determine what those tests retain. The open–closed problem is to construct the action and determine the information it recovers.

The polar model imposes a first-jet boundary condition. On a marked sphere, a function is required to have zero first holomorphic derivative. The resulting Dolbeault complex has an extra cohomology class, detected by a meromorphic one-form with a double pole.

The physical matrix fields reduce classically to a surface connection and an adjoint scalar. The map extracting that scalar is nonlinear. At the quantum level, eliminating the other fields also contracts observables with one another.

Hamiltonian fields on a holomorphic symplectic surface provide a second model. Their matrix probes retain commutator equations, relations among those equations and central rank. Comparing the two models, or comparing either with a closed string field theory, requires a map between their specified fields and operations.

1.1 Mixed differential forms and local observables

Let U be an oriented real surface and Z a complex manifold. Mixed forms on $U \times Z$ are sections of

$$\mathcal{A}_{U,Z} = \Omega^\bullet(U) \mathbin{\widehat{\otimes}} \Omega^{0,\bullet}(Z).$$

The tensor product is the completed projective tensor product of the usual complex Fréchet spaces of smooth forms. On product opens it gives smooth forms in both sets of variables. Compactly supported smooth spaces carry their usual inductive-limit topologies.

On a noncompact space, integrals below use compact support. For homogeneous forms ξ, ξ' on U and η, η' on Z, the total differential and multiplication have the tensor signs

$$D(\xi \otimes \eta) = d_U \xi \otimes \eta + (-1)^{|\xi|} \xi \otimes \bar\partial \eta, \tag{1.1}$$

$$(\xi \otimes \eta)(\xi' \otimes \eta') = (-1)^{|\eta||\xi'|} (\xi \wedge \xi') \otimes (\eta \wedge \eta'). \tag{1.2}$$

Degrees are cohomological, and $V[k]^j = V^{j+k}$. Thus the shift $[1]$ lowers the degree of a field vector by one. Coordinates on a graded field space have the opposite degrees.

The adjective *mixed* refers to these two differentials. For a real vector field v on U, contraction gives $[d_U, \iota_v] = \mathcal{L}_v$. Antiholomorphic contraction gives the corresponding identity for $\bar\partial$ on a complex coordinate chart. Extending these identities to an interacting theory requires compatibility with its interaction and boundary conditions. At the quantum level it also requires compatibility with renormalization.

A classical Batalin–Vilkovisky (BV) model specifies a complex of fields, a symplectic pairing of degree -1, and an action whose Hamiltonian differential squares to zero. An infinite-dimensional pairing has a domain: a continuous functional need not have a smooth Hamiltonian vector field. The admissible observables and their coefficient topology are therefore part of the model.

Write Q for the free differential and $\{-,-\}$ for the degree-one BV bracket. Let $\hbar$ be a formal parameter of degree zero. When a regulated BV Laplacian Δ is defined, the quantum master equation for an interaction I takes the form

$$QI + \frac{1}{2}\{I, I\} + \hbar \Delta I = 0.$$

The equation is asserted only on a domain where its three terms are defined.

Local observables on an open set V form a cochain complex $\mathrm{Obs}(V)$. A unital prefactorization structure supplies chain maps

$$\mathrm{Obs}(V_1) \otimes \cdots \otimes \mathrm{Obs}(V_r) \longrightarrow \mathrm{Obs}(V)$$

for pairwise disjoint $V_i \subset V$, compatible with nested insertions and with permutations carrying the Koszul signs. Empty insertion gives the unit. Factorization also requires descent for Weiss covers: every finite subset of the open lies in some member of the cover.

Local constancy in the real directions means that the maps associated to inclusions of sufficiently small real disks are quasi-isomorphisms, with the holomorphic open fixed. It is a property of these observable maps. A formula for the differential on fields alone does not establish it.

Support fixes the variance of the maps. A smooth field on V restricts to a smaller open. A compactly supported distributional coefficient of an observable extends by zero to a larger open. These maps have opposite directions.

The continuous dual of all smooth test functions consists of compactly supported distributions. The dual of compactly supported smooth tests contains distributions of arbitrary support. Each current construction specifies which test space it uses.

1.2 The polar field theory

Take $C = \mathbb{P}^1$, with marked point $o = 0$ in the coordinate z, and fix $\kappa \in \mathbb{C}^\times$. For $f \in C^\infty(C)$, put

$$\alpha = \kappa z^{-2} dz, \qquad j(f) = \partial_z f(o), \qquad E^0 = \ker j, \quad E^1 = \Omega^{0,1}(C), \quad d_E = \bar\partial.$$

All functions and forms here are globally smooth, and $E^k = 0$ outside degrees zero and one. The condition $j = 0$ is imposed on degree-zero elements. It is preserved by multiplication, so E is a commutative differential graded algebra.

Its degree-minus-one trace is the circular principal value

$$\tau_E(\eta) = \frac{1}{2\pi i} \lim_{r \downarrow 0} \int_{C \setminus \{|z| < r\}} \alpha \wedge \eta, \qquad \tau_E(E^0) = 0.$$

The marked coordinate and the shrinking circles are part of the prescription. For every smooth function f, polar Stokes gives

$$\tau_E(\bar\partial f) = \kappa\, j(f). \tag{1.3}$$

The trace is therefore closed on E. Chapter 2 proves its continuity and computes the resulting cohomology.

For a positive integer N, matrix multiplication and the ordinary trace give the classical field space

$$\mathcal{F}_E = \big(\Omega^\bullet(U)\widehat{\otimes}E \otimes \mathrm{Mat}_N(\mathbb{C})\big)[1]. \tag{1.4}$$

Integration over U, followed by τ_E tr, has degree -3 before the shift. The induced BV pairing has degree -1.

Let A denote the full graded field, including its ghost and antifield components. For a closed surface, or compact support in its interior, the cyclic action is

$$S_E(\mathsf{A}) = \int_U \tau_E \,\mathrm{tr}\left(\frac{1}{2}\mathsf{A}D\mathsf{A} + \frac{1}{3}\mathsf{A}^3\right), \qquad D = d_U + d_E. \tag{1.5}$$

The differential graded algebra identities give its classical gauge complex. This is the partial-connection form of four-dimensional Chern–Simons theory, with the polar trace and boundary domain just specified.[1]

A physical field is $\mathsf{A} = b + a$, where b is a surface one-form with coefficients in E^0 and a is a curve $(0,1)$-form. Gauge transformations take values in invertible matrices over E^0 and act by

$$\mathsf{A}^g = g^{-1}\mathsf{A}g + g^{-1}Dg.$$

Unless stated otherwise, gauges and fields are considered formally near the zero field and the trivial curve bundle. The analytic normal form is asserted on its separately specified small Sobolev neighborhood.

The residual curve algebra is

$$A_\kappa = \mathbb{C} \oplus \mathbb{C}\varepsilon, \qquad |\varepsilon| = 1, \quad \varepsilon^2 = 0, \quad d_A = 0, \quad \tau_A(1) = 0, \quad \tau_A(\varepsilon) = \kappa.$$

It is represented in E by

$$f_0 = \frac{z}{1+|z|^2}, \qquad \eta_0 = \bar\partial f_0, \qquad 1 \longmapsto 1, \quad \varepsilon \longmapsto \eta_0.$$

Although η_0 is exact among unrestricted smooth forms, (1.3) excludes a Neumann primitive. The inclusion is a trace-preserving quasi-isomorphism.

The residual matrix of a curve connection is extracted from its normalized frame:

$$\bar\partial H = -aH, \qquad H(o) = I_N, \qquad \mathcal{J}(a) = -j(H).$$

Let G be the right inverse of $\bar\partial$ normalized by $(G\eta)(o) = 0$. Formally,

$$H = I_N - Ga + G(aGa) - \cdots, \qquad \mathcal{J}(a) = j(Ga) - jG(aGa) + \cdots. \tag{1.6}$$

The order of the matrix factors is retained. The first term is the linear cohomology projection. The other terms record how the connection changes that projection. Chapter 3 constructs the frame, its equivariance and explicit nonzero nonlinear trace invariants.

For a surface connection $b \in \Omega^1(U, \mathfrak{gl}_N)$ and $J \in C^\infty(U, \mathfrak{gl}_N)$, the residual field is $b + \eta_0 J$. The same physical density on a surface with boundary gives

$$S_E(b + \eta_0 J) = \kappa \int_U \mathrm{tr}(JF_b) - \frac{\kappa}{2}\int_{\partial U} \mathrm{tr}(bJ), \qquad F_b = d_U b + b \wedge b. \tag{1.7}$$

[1] K. Costello, E. Witten and M. Yamazaki, *Gauge Theory and Integrability, I*, arXiv:1709.09993, Section 3.1, especially equations (3.1)–(3.4). The action here uses the normalization $\frac{1}{2}\mathsf{A}D\mathsf{A} + \frac{1}{3}\mathsf{A}^3$ and the displayed principal-value trace.

The boundary term contributes to the variational problem. A boundary polarization must also be chosen.

The full shifted residual complex also contains ghosts and antifields. At a constant matrix J, its curve differential is the two-term adjoint complex $[\mathfrak{gl}_N \xrightarrow{\mathrm{ad}_J} \mathfrak{gl}_N \varepsilon]$. Retaining this complex preserves the stabilizers when eigenvalues coincide. Chapter 4 proves the full reduction and the two boundary transgressions, at the pole and at the boundary of U.

The coefficient ring matters in this reduction. Formal deformation parameters admit recursive inversion, and sufficiently small analytic parameters admit the corresponding convergent inverse. A polynomial parameter does not permit either step. The same chapter exhibits an infinite-dimensional cokernel for the polynomial family while retaining the formal and analytic equivalences. Thus the residual model comes with its parameter domain.

1.3 Hamiltonian fields and matrix probes

The symplectic model has a different holomorphic factor. Let W be a complex surface with holomorphic symplectic form. On a Darboux chart write this form as $dq \wedge dp$ and use

$$\{f, g\} = f_q g_p - f_p g_q, \qquad X_f = \{f, -\}.$$

The spacetime $U \times W$ has real dimension six, whereas $U \times C$ in (1.4) has real dimension four. A passage between them requires a reduction with its fields, integration domain and boundary conditions.

At a fixed point of W, put $\mathfrak{m} = (q, p)$ in $\mathbb{C}[[q, p]]$. Formal Hamiltonian vector fields are represented by $\mathbb{C}[[q, p]]/\mathbb{C}1$. Those fixing the point are represented by $\mathfrak{m}^2$. The latter have finite Lie quotients

$$\mathfrak{h}_d = \mathfrak{m}^2/\mathfrak{m}^{d+1}, \qquad d \geq 2,$$

because the Poisson bracket lowers polynomial degree by two. Translations do not preserve these quotients: $\{q, p^{d+1}\} = (d+1)p^d$. The pointed and moving-point problems therefore require different coefficient data.

An internal formal Hamiltonian is also distinct from a function of the physical holomorphic coordinates. If $z = (z_1, z_2)$ are physical coordinates on a chart V and $w = (w_1, w_2)$ independent formal variables, use the complex

$$O(V)[[w_1, w_2]] \otimes \Lambda(\vartheta_1, \vartheta_2), \qquad |\vartheta_i| = 1.$$

Its horizontal jet differential is

$$\nabla = \sum_{i=1}^{2} \vartheta_i (\partial_{z_i} - \partial_{w_i}).$$

Its horizontal inclusion sends $f(z)$ to the formal Taylor series $f(z+w)$. Evaluation at $w = 0$ on arbitrary coefficients is not a Poisson map: it sends $\{w_1, w_2\}_w$ to 1 and sends both inputs to zero. The horizontal complex supplies the additional constraint needed for the geometric Hamiltonian comparison.

The matrix probe of a commutative plane is the differential graded algebra

$$R_N = \mathbb{C}[X_{ij}, Y_{ij}] \otimes \Lambda(\Upsilon_{ij}), \qquad |\Upsilon_{ij}| = -1, \quad Q\Upsilon = [X, Y], \quad QX = QY = 0. \tag{1.8}$$

The Chevalley–Eilenberg cochains for simultaneous conjugation additionally record infinitesimal gauge symmetry and stabilizers. Even before that step, the derived relation matters: $\operatorname{Tr} \Upsilon$ is a closed class that

is not exact. Evaluation at $X = Y = 0$ sends it to a nonzero element of the exterior algebra with zero differential, proving the assertion. A reduction that removes this class changes the probe.

Central constants are detected independently. With $\{X_{ij}, Y_{kl}\} = \delta_{il}\delta_{jk}$, direct summation gives

$$\{\operatorname{Tr} X, \operatorname{Tr} Y\} = N. \tag{1.9}$$

The classes of q, p commute in the Lie algebra of Hamiltonians modulo constants. Their unreduced matrix traces have the nonzero bracket (1.9). A Hamiltonian-function comparison must retain this central datum or construct its reduction. The induced derivations, which annihilate constants, ask a different question.

Framed probes impose a moment equation with an additional scalar parameter. At rank two, a cyclic matrix basis gives regular coordinates even when eigenvalues collide. Chapter 12 constructs an integral quantum reduction on this chart and a regular Lie map from polynomial Hamiltonians.

Two different global obstructions remain. One fixes four low charges. The other allows every formal correction but prescribes the full trace map at zero framing. A comparison at an isolated nonzero framing level can lie outside both classes. Its nonlinear moments and its transitions between cyclic charts must still be constructed.

1.4 The bulk action and the boundary center

A boundary theory includes objects describing boundary conditions and cochain complexes of operators changing one condition into another. Their compositions are part of the data. For an associative differential graded algebra B, the corresponding algebraic center is the Hochschild cochain complex

$$C^\bullet(B, B) = \mathrm{RHom}_{B \otimes B^{\mathrm{op}}}(B, B).$$

It records coherent operations on B and its modules. A geometric boundary theory requires the analogous operations to respect the positions and support of insertions. Choosing its category of factorization bimodules and its diagonal precedes taking derived endomorphisms. That choice is not determined by the trace of a single matrix probe.

Even the polynomial coordinate comparison has two stages. Regrading the Chevalley–Eilenberg coordinates of a shifted cotangent Lie algebra gives polyvectors with their linear Poisson differential. Ordinary Hochschild–Kostant–Rosenberg antisymmetrization has zero differential on its polyvector source. It therefore supplies the Hochschild comparison only at zero Poisson structure. A nonzero Poisson tensor requires a supplied formality map, twisted by that tensor, and the corresponding deformed associative product. Chapter 15 proves the coordinate identity and gives the explicit failure of the untwisted comparison.

The bulk is specified independently by its fields, pairing, action and boundary domain. Its action on a boundary is a family of maps from bulk observables in a collar to coherent boundary operations. The maps must commute with the differential and preserve disjoint products. Moving and colliding insertions impose further homotopies.

For example, let $\mathfrak{l}$ be a Lie algebra in degree zero and $(\mathfrak{a}, d)$ a differential graded Lie algebra. Let $F_1 : \mathfrak{l} \to \mathfrak{a}^0$ take values in closed elements. A binary correction is a map $F_2 : \Lambda^2\mathfrak{l} \to \mathfrak{a}^{-1}$ required to solve

$$dF_2(f, g) = F_1([f, g]) - [F_1(f), F_1(g)].$$

The next equation tests the compatibility of these binary corrections. An action on cohomology alone leaves these equations undecided.

A dynamical coupling must also determine the source in the bulk equations. If $i : Y \hookrightarrow M$ is a defect inclusion, the transpose of restriction sends a defect density μ to a current supported on Y. For a real normal coordinate r, oriented before the coordinates of Y, this pushforward has the form $i_*\mu = \delta(r)\, dr\, \mu$. Coupling a Hamiltonian bulk field to a cyclic action on defect fields produces both their gauge action and this transposed source. A product of singular currents at several insertions requires its own extension across their collisions.

The geometric Hamiltonian model uses a six-dimensional bulk and a five-dimensional defect of this kind. Its cotangent field admits normal distributions, while the field multiplying it is smooth. This domain permits the classical action without multiplying two normal distributions. A closed theory of polyvector fields, such as Bershadsky–Cecotti–Ooguri–Vafa (BCOV) theory, starts with a different complex and its own divergence, descendants and background. Identifying its sourced observables with Hamiltonian or polar boundary operations requires a map of those complexes, preserving the source terms and the cyclic operations. The general comparison remains open.

1.5 Quantum reduction and prepared states

Classical reduction supplies a residual field and the homotopies of its gauge complex. Quantum elimination additionally supplies contractions. For a normalized Gaussian functional on one variable x, with mean m and covariance $\hbar C$,

$$\langle x^2 \rangle = m^2 + \hbar C.$$

Substitution of m alone misses the second term. For several observables the corresponding corrections are their connected cumulants. The normalized BV pushforward and its composition law in Chapter 8 retain these terms.

Infinite-dimensional elimination requires more than the finite identity. The paired native heat tensors give continuous contractions at positive scale, including pole jets. Their generator is nonlocal on the curve. Extending coincident products to zero scale therefore requires estimates and counterterms on the stated support domain. The free identities of Chapter 7 do not supply the interacting quantum master equation.

That chapter also computes the full supported normal-jet complex. Antiholomorphic derivative jets have explicit Dolbeault primitives. Holomorphic residue jets survive in cohomology. Boundary faces and their intersections require the relative differential. Admissibility of these primitives for the native cyclic field theory is a further condition.

For the Gaussian sector, the staggered inverse converges uniformly on compact sets of matrix backgrounds. Its finite background derivatives converge on bounded smooth tests. Chapter 8 uses these estimates to obtain smooth Ward identities and the regularized determinant. An inserted source detector remains nonzero even when the uninserted regularized determinant equals one.

A state is further data: a functional on observables compatible with their differential and the chosen boundary preparation. The unnormalized functional evaluated on the unit gives the partition function. Normalizing that value leaves the other observable values to be determined.

Chapter 9 constructs compact gauge and Gaussian preparations, keeps their normalization lines, and proves their sewing formulas. Its interacting scalar preparation has a fixed positive smoothing parameter. A Hermite unitary identifies it with an independently defined scalar Fock interaction, including the full form domains, semigroups, traces and normalized positive states. The finite cutoff semigroups converge in trace norm.

The remaining physical comparison must identify the state produced by the native field theory with an independently prepared residual state on common observables. It must also intertwine cutting and sewing. This requirement connects the boundary algebra to the quantum field theory: the algebra specifies which operations can be composed, and the preparation specifies their amplitudes.

Chapter 2

The polar Neumann complex

A boundary condition can leave a state even when the unconstrained Dolbeault complex has no first cohomology. On the sphere, every smooth $(0,1)$-form has a primitive. If the primitive must have zero first holomorphic derivative at a marked point, one scalar obstruction remains. The meromorphic form with a double pole at that point measures precisely this obstruction. We compute the resulting algebra and retain the trace needed for its classical action.

2.1 A first jet and its boundary integral

Let $C = \mathbb{P}^1$ have its complex orientation, and let $o = 0$ in the affine coordinate z. Fix $\kappa \in \mathbb{C}^\times$ and put

$$\alpha = \kappa z^{-2} dz, \qquad j(f) = \partial_z f(o) \quad \text{for } f \in C^\infty(C).$$

The form α extends across infinity and has a single double pole, with zero residue. All smooth function and form spaces below carry their usual Fréchet topologies.

Definition 2.1. The Neumann Dolbeault algebra is the cohomologically graded algebra

$$E^0 = \ker j, \qquad E^1 = \Omega^{0,1}(C), \qquad E^k = 0 \ (k \neq 0,1), \qquad d_E = \bar\partial,$$

with pointwise multiplication and wedge product.

The condition is imposed on functions. It is preserved by multiplication, since $j(fg) = f(o)\,j(g) + j(f)\,g(o)$. A product of two degree-one elements vanishes. Thus E is a unital commutative differential graded algebra. Its holomorphic germs at o form $\mathbb{C} \oplus z^2\mathbb{C}\{z\}$, whose completion is $\mathbb{C} \oplus z^2\mathbb{C}[[z]]$.

The polar form defines a trace by deleting round disks in the fixed coordinate:

$$\tau_E(\eta) = \frac{1}{2\pi i} \lim_{r \downarrow 0} \int_{C \setminus \{|z| < r\}} \alpha \wedge \eta \quad (\eta \in E^1), \qquad \tau_E(E^0) = 0. \tag{2.1}$$

The coordinate and the circular prescription are part of this definition.

Lemma 2.2 (Polar Stokes identity)**.** *The limit in* (2.1) *exists and defines a continuous functional of degree* -1*. For every* $f \in C^\infty(C)$*,*

$$\tau_E(\bar\partial f) = \kappa\, j(f). \tag{2.2}$$

In particular, $\tau_E d_E = 0$ *on* E*, and* $\tau_E(xy) = (-1)^{|x||y|}\tau_E(yx)$ *for homogeneous* $x, y \in E$*.*

Proof. The vanishing of $H^{0,1}(\mathbb{P}^1)$ gives a continuous solution operator

$$G : \Omega^{0,1}(C) \longrightarrow C^\infty(C), \qquad \bar\partial G\eta = \eta, \qquad (G\eta)(o) = 0. \tag{2.3}$$

One obtains it from a Dolbeault Green operator by subtracting the value at o. The normalization makes it unique, because the global holomorphic functions on C are constant.

Away from o, one has $d(\alpha f) = -\alpha \wedge \bar\partial f$. The inner boundary of the deleted disk is oriented clockwise, so Stokes' theorem gives

$$\frac{1}{2\pi i}\int_{C\setminus\{|z|<r\}} \alpha \wedge \bar\partial f = \frac{1}{2\pi i}\oint_{|z|=r}^{\text{counterclockwise}} \kappa z^{-2} f\, dz.$$

In the expansion $f = f(o) + z\,j(f) + \bar z\,\partial_{\bar z} f(o) + O(|z|^2)$, only the holomorphic linear term contributes in the limit. This proves (2.2).

Every η is $\bar\partial G\eta$, so the same identity proves existence and continuity of the trace on all of E^1. It vanishes on $d_E E^0$ by the defining condition $j = 0$. Graded commutativity gives cyclicity. □

The trace is therefore the obstruction to finding a Neumann primitive. To exhibit its class, choose a smooth function f_0 with $f_0(o) = 0$ and $j(f_0) = 1$, and set $\eta_0 = \bar\partial f_0$. A useful choice, used in the explicit calculations below, is

$$f_0 = \frac{z}{1 + z\bar z}, \qquad \eta_0 = -\frac{z^2}{(1 + z\bar z)^2}\, d\bar z. \tag{2.4}$$

In the coordinate $w = 1/z$, the function is $\bar w/(1 + w\bar w)$, so it is smooth at infinity. Although η_0 is exact in the unconstrained Dolbeault complex, it is not exact in E, since $\tau_E(\eta_0) = \kappa$.

2.2 The two-dimensional state algebra

Let

$$A_\kappa = \mathbb{C} \oplus \mathbb{C}\varepsilon, \qquad |\varepsilon| = 1, \quad \varepsilon^2 = 0, \quad d_A = 0, \qquad \tau_A(1) = 0, \quad \tau_A(\varepsilon) = \kappa. \tag{2.5}$$

Its inclusion into the smooth complex sends the odd generator to the class just constructed:

$$i : A_\kappa \longrightarrow E, \qquad i(1) = 1, \quad i(\varepsilon) = \eta_0.$$

Proposition 2.3. *The map i is a continuous unital differential graded algebra quasi-isomorphism, and $\tau_E i = \tau_A$.*

Proof. It preserves the differential and product because η_0 is closed and $\eta_0^2 = 0$. Define

$$q(f) = f(o), \qquad q(\eta) = j(G\eta)\varepsilon, \qquad h_0(\eta) = G\eta - j(G\eta) f_0, \qquad h_0(E^0) = 0.$$

The first jet of $h_0\eta$ is zero.

For $f \in E^0$, the normalization of G gives $G d_E f = f - f(o)$, hence $h_0 d_E f = f - iq(f)$ and $q d_E f = 0$. On degree one, $d_E h_0\eta = \eta - iq(\eta)$. Also $G\eta_0 = f_0$, so $qi = 1$, $h_0 i = q h_0 = h_0^2 = 0$. These formulas give a continuous contraction of complexes and prove the quasi-isomorphism. The polar Stokes identity gives $\tau_E i = \tau_A$. □

Point evaluation gives the simplest contraction, but an action also uses the bilinear pairing $\langle x, y\rangle_E = \tau_E(xy)$. For transfer of that pairing, the projection must be adjoint to i. This requirement changes its degree-zero component.

Theorem 2.4 (Cyclic contraction)**.** *Define $p(f) = p_0(f)1$ and $p(\eta) = p_1(\eta)\varepsilon$ by*

$$p_0(f) = \kappa^{-1}\tau_E(f\eta_0), \qquad p_1(\eta) = \kappa^{-1}\tau_E(\eta), \tag{2.6}$$

and define a degree--1 map by

$$h\eta = G\eta - p_1(\eta) f_0 + \kappa^{-1}\tau_E(f_0\eta)1, \qquad h(E^0) = 0. \tag{2.7}$$

These continuous maps satisfy

$$pi = 1, \qquad d_E h + h d_E = 1 - ip, \qquad h^2 = hi = ph = 0. \tag{2.8}$$

The maps p and i are adjoint for $\tau_E(x\,y)$ and $\tau_A(x\,y)$. The homotopy satisfies

$$\tau_E((h\eta)\xi) + \tau_E(\eta h\xi) = 0 \qquad (\eta, \xi \in E^1). \tag{2.9}$$

Proof. By (2.2), $j(G\eta) = p_1(\eta)$, so $h\eta \in E^0$. For $f \in E^0$, the same identity applied to $f f_0$ yields

$$\tau_E(f\eta_0) + \tau_E(f_0 d_E f) = \kappa f(o).$$

Together with $G d_E f = f - f(o)$, this proves $h d_E f = f - p_0(f)$. Differentiating (2.7) gives $d_E h\eta = \eta - p_1(\eta)\eta_0$.

One has $p_0(1) = p_1(\eta_0) = 1$ and $\tau_E(f_0\eta_0) = \frac{1}{2}\tau_E d_E(f_0^2) = 0$. Consequently $pi = 1$ and $hi = 0$. Applying the polar Stokes identity to $(G\eta)f_0$, whose value and first derivative at o vanish, gives

$$\tau_E((G\eta)\eta_0) = -\tau_E(f_0\eta).$$

This proves $ph = 0$. Degree considerations give $h^2 = 0$. The two formulas in (2.6) are exactly the adjointness identities.

For cyclicity, set $F = G\eta$, $K = G\xi$, $r = p_1(\eta)$, $s = p_1(\xi)$, $u = \kappa^{-1}\tau_E(f_0\eta)$ and $v = \kappa^{-1}\tau_E(f_0\xi)$. The left side of (2.9) becomes

$$\tau_E(F\xi + \eta K) - \kappa rv - \kappa su + \kappa us + \kappa vr.$$

The last four terms cancel. The first is $\tau_E d_E(FK) = \kappa\, j(FK) = 0$, since $F(o) = K(o) = 0$. Continuity follows from that of G, τ_E, and multiplication by the fixed smooth function f_0. □

Example 2.5. For the choice (2.4), put $v = z\bar{z}/(1 + z\bar{z})$. Then $v \in E^0$ and $v(o) = 0$, whereas

$$G(v\eta_0) = \frac{z}{1 + z\bar{z}} - \frac{z}{2(1 + z\bar{z})^2}, \qquad p_0(v) = \frac{1}{2}.$$

The displayed primitive is smooth in both charts, vanishes at o, and differentiates to $v\eta_0$. Its first jet computes $p_0(v)$. Thus point evaluation and the cyclic projection impose different normalizations on the eliminated functions.

The contraction has a particularly simple effect on the internal algebra. In the binary-tree formulas for transfer, leaves are labeled by i, vertices by multiplication, internal edges by h, and the root by p. Every tree with at least three leaves has a non-root vertex joining two leaves. Their product lies in $i(A_\kappa)$ and is killed by the following h. Hence all transferred products of arity at least three vanish. The binary product is the one in (2.5).

This statement concerns the internal coefficient algebra. It supplies neither an inverse pairing on arbitrary distributional covectors nor operations at coincident field insertions.

2.3 Choice of representative and surface fields

Changing f_0 changes the representative of the odd state, but leaves its trace fixed. The comparison can be made over parameter simplices, so the choices admit compatible homotopies.

Proposition 2.6. *An affine family f of smooth functions satisfying $f(o) = 0$ and $j(f) = 1$, parametrized by a simplex P, defines a trace-preserving unital differential graded algebra map*

$$A_\kappa \longrightarrow \Omega^\bullet_{\mathrm{poly}}(P) \otimes E, \qquad \varepsilon \longmapsto (d_P + \bar{\partial})f.$$

The maps commute with face and degeneracy restrictions.

Proof. Use the indicated tensor order and its Koszul signs. The coefficients of $d_P f$ have zero first jet, so they lie in E^0. The image of ε is closed and odd, so its square is zero. The trace annihilates its E^0 component and sends $\bar\partial f$ to κ. Restriction of the affine family commutes with the formula.

On an interval, writing $u = f_1 - f_0$, the image is $\eta_0 + t\,\bar\partial u + dt\,u$. The two differential terms $dt\,\bar\partial u$ cancel. □

Let U be an oriented smooth surface and $N \geq 1$ an integer. Tensor the comparison with $\Omega^\bullet(U)$ and $\mathrm{Mat}_N(\mathbb{C})$, using the completed projective tensor product of the smooth spaces and the graded tensor product of algebras. The homotopy on a simple tensor is

$$h_U(\omega \otimes \eta) = (-1)^{|\omega|}\omega \otimes h\eta.$$

The identities (2.8) then hold for the total differential. These maps commute with restriction in U and preserve support there.

For compact support, take the inductive limit over fixed compact subsets of U. The Green operator acts on the whole compact curve, so it is not local in the curve variable.

The trace also determines the classical action. Let

$$L_U = \Omega^\bullet(U)\widehat{\otimes}E \otimes \mathrm{Mat}_N(\mathbb{C})$$

with total form degree and the graded commutator bracket.

A physical connection is an element $\mathcal{A} \in L_U^1$. Its surface component has Neumann function coefficients, while its curve component is an arbitrary smooth $(0,1)$-form. Define

$$S_E(\mathcal{A}) = \int_U \tau_E \,\mathrm{tr}\left(\frac{1}{2}\mathcal{A}(d_U + d_E)\mathcal{A} + \frac{1}{3}\mathcal{A}^3\right). \tag{2.10}$$

Only surface degree two and internal degree one contribute. On a noncompact surface take compactly supported fields.

For the residual fields, write $a \in \Omega^1(U) \otimes \mathrm{Mat}_N(\mathbb{C})$ and $J \in C^\infty(U) \otimes \mathrm{Mat}_N(\mathbb{C})$. Their image is $\mathcal{A} = a + J\eta_0$.

Proposition 2.7. *With $F_a = d_U a + a \wedge a$, the density of (2.10) is*

$$\kappa\,\mathrm{tr}(J F_a) - \frac{\kappa}{2} d_U\,\mathrm{tr}(aJ).$$

On a closed surface this is the BF action $\kappa \int_U \mathrm{tr}(J F_a)$. If U has boundary, the additional term is $-\frac{\kappa}{2}\int_{\partial U}\mathrm{tr}(aJ)$.

Proof. The kinetic term has trace

$$\frac{\kappa}{2}\,\mathrm{tr}(a \wedge d_U J + J d_U a) = \kappa\,\mathrm{tr}(J d_U a) - \frac{\kappa}{2} d_U\,\mathrm{tr}(aJ).$$

The three terms of $\mathcal{A}^3$ containing one internal form are $a^2 J\eta_0$, $-aJa\eta_0$, and $Ja^2\eta_0$. Graded cyclicity makes their traces equal, so the cubic term contributes $\kappa\,\mathrm{tr}(Ja^2)$. All other terms vanish by degree. Stokes' theorem gives the boundary integral. □

The inclusion now identifies states, products, trace, and the action on the chosen residual fields. To evaluate an invariant on an arbitrary smooth connection, we still need its residual matrix. The linear projection need not compute it: the gauge transformation that eliminates the other modes is nonlinear.

Chapter 3

Nonlinear residual observables

The residual algebra has no higher products, but recovering a residual field is a different operation. A matrix-valued connection must first be put into Neumann gauge. Its normalized frame solves a linear equation whose coefficient is the connection. The solution depends nonlinearly on that coefficient. The first jet of this frame gives the residual matrix and all its invariant polynomials. An explicit sphere family will have zero linear projection and a nonzero quadratic invariant.

3.1 The normalized frame

Retain C, o, j, E, G, f_0, η_0 and κ from Chapter 2. Fix $N \geq 1$ and let t be a formal parameter of degree zero. A formal curve connection is

$$a \in t\big(E^1 \otimes \mathrm{Mat}_N(\mathbb{C})\big)[[t]].$$

All series in this section are t-adically complete. The operator G acts entrywise on matrix-valued one-forms.

Definition 3.1. Let $L_a(F) = aF$. The normalized frame and the residual matrix of a are

$$H(a) = (1 + GL_a)^{-1} I_N = I_N - Ga + G(aGa) - G(aG(aGa)) + \cdots, \tag{3.1}$$

$$\mathcal{J}(a) = -jH(a). \tag{3.2}$$

Each coefficient uses finitely many applications of G and multiplications of a one-form by a function. The recursion and the normalization of G show that $H(a)$ is the unique formal solution of

$$\bar{\partial} H = -aH, \qquad H(o) = I_N. \tag{3.3}$$

Its constant term is I_N, so it is formally invertible. On a complex curve there is no further $(0,2)$ integrability equation.

A formal Neumann gauge is a matrix $k \in \mathrm{GL}_N(C^\infty(C)[[t]])$ whose reduction modulo t is a constant invertible matrix and whose first jet satisfies $j(k) = 0$. It acts by

$$a^k = k^{-1} a k + k^{-1} \bar{\partial} k.$$

The gauge is based if $k(o) = I_N$. The condition on its constant term ensures that a^k again vanishes modulo t.

Theorem 3.2. *For every formal curve connection, the matrix*

$$g(a) = H(a) \exp(f_0 \mathcal{J}(a)) \tag{3.4}$$

is a based Neumann gauge and satisfies $a^{g(a)} = \eta_0 \mathcal{J}(a)$. *Under a Neumann gauge,*

$$H(a^k) = k^{-1} H(a) k(o), \qquad \mathcal{J}(a^k) = k(o)^{-1} \mathcal{J}(a) k(o). \tag{3.5}$$

More precisely, gauges with $a^k = b$ *correspond bijectively to* $K \in \mathrm{GL}_N(\mathbb{C}[[t]])$ *satisfying* $\mathcal{J}(a)K = K\mathcal{J}(b)$, *by*

$$k = H(a) K H(b)^{-1}, \qquad K = k(o). \tag{3.6}$$

Proof. The frame equation gives $a^{H(a)} = 0$, and $0^{\exp(f_0 \mathcal{J})} = \eta_0 \mathcal{J}$. The factors in (3.4) have value I_N at o and first jets $-\mathcal{J}$ and $\mathcal{J}$. Thus their product is a based Neumann gauge.

The matrix $k^{-1} H(a) k(o)$ solves the normalized frame equation for a^k. Uniqueness gives the first identity in (3.5). Taking its first jet, using $j(k^{-1}) = 0$, gives the second.

If $a^k = b$, both $kH(b)$ and $H(a)$ are frames for a. Their ratio is a global holomorphic matrix and hence is constant. This yields (3.6). Conversely, that formula satisfies $\bar\partial k = -ak + kb$, and

$$j(k) = -\mathcal{J}(a)K + K\mathcal{J}(b).$$

It is Neumann precisely when the indicated intertwining relation holds. □

Every $J \in t\,\mathrm{Mat}_N(\mathbb{C})[[t]]$ occurs, since $H(\eta_0 J) = \exp(-f_0 J)$ and $\mathcal{J}(\eta_0 J) = J$. Thus the quotient by based gauges is exactly this formal matrix space. Constant gauges then act by conjugation. The construction of H and $\mathcal{J}$ does not use f_0. That choice enters only the representative $\eta_0 \mathcal{J}$.

The theorem describes a neighborhood of the trivial holomorphic bundle. It does not assert that every holomorphic bundle on the sphere is trivial. In a small Sobolev neighborhood, the same construction is convergent.

Proposition 3.3. *Fix an integer* $s \geq 3$ *and smooth metrics for the Sobolev norms on* C. *For* a *in a sufficiently small neighborhood of zero in* $H^s(\Omega^{0,1}_C \otimes \mathrm{Mat}_N(\mathbb{C}))$, *the series* (3.1) *converges in* H^{s+1} *to an invertible frame. The maps* $a \mapsto H(a)$ *and* $a \mapsto \mathcal{J}(a)$ *are analytic. Smooth inputs give smooth frames, also for smooth families parametrized by compact subsets of a surface.*

Proof. The Dolbeault solution operator gains one derivative. Subtracting its value at o preserves this bound, since evaluation is continuous on H^{s+1}. Sobolev multiplication on the compact real surface gives

$$\|G(aF)\|_{H^{s+1}} \leq C_s \|a\|_{H^s} \|F\|_{H^{s+1}}.$$

The Neumann series converges when $C_s \|a\|_{H^s} < 1$.

After shrinking the neighborhood, Sobolev embedding makes $H - I_N$ uniformly small, so H is invertible. First-jet evaluation is continuous on H^{s+1}, proving analyticity of $\mathcal{J}$. The frame equation bootstraps smooth inputs to smooth solutions. Differentiation of the inverse operator with respect to surface parameters gives the same estimates on each compact parameter set. □

Gauge equivariance continues to hold whenever both a and a^k belong to this neighborhood. No smallness assumption on an unrelated gauge orbit is implicit.

3.2 The linear trace and the cyclic representative

The first term of (3.2) is $p_1 a = \kappa^{-1} \tau_E(a)$. The trace removes all its higher corrections, although the matrix itself usually retains them.

Proposition 3.4. *Formally, and in the analytic neighborhood above,*

$$\operatorname{tr} \mathcal{J}(a) = \kappa^{-1} \tau_E(\operatorname{tr} a).$$

If the coefficient matrices of a lie in one commutative subalgebra of $\mathrm{Mat}_N(\mathbb{C})$*, then* $\mathcal{J}(a) = \kappa^{-1}\tau_E(a)$.

Proof. The formal or near-identity logarithm satisfies

$$\bar\partial \log\det H = \operatorname{tr}(H^{-1}\bar\partial H) = -\operatorname{tr} a.$$

Its first jet is $-\operatorname{tr}\mathcal{J}(a)$. The polar Stokes identity proves the first assertion. In a commutative coefficient algebra, $\exp(-Ga)$ solves the normalized frame equation. Taking its first jet proves the second. □

The cyclic homotopy of Theorem 2.4 gives a different gauge normalization. Seek $\xi \in t(E^0 \otimes \mathrm{Mat}_N(\mathbb{C}))[[t]]$ with $p_0(\xi) = 0$, and expand

$$a^{e^{\xi}} = e^{-\operatorname{ad}\xi} a + \sum_{k\ge 0} \frac{(-1)^k}{(k+1)!} (\operatorname{ad}\xi)^k d_E \xi, \qquad \operatorname{ad}\xi(x) = [\xi, x].$$

At order n, let R_n denote the coefficient with the unknown ξ_n omitted. Set

$$\xi_n = -h R_n, \qquad J_n^{\mathrm{cyc}} = p_1 R_n. \tag{3.7}$$

The contraction identities give $R_n + d_E\xi_n = \eta_0 p_1 R_n$ and $p_0(\xi_n) = 0$, so this recursion puts the connection in the form $\eta_0 J^{\mathrm{cyc}}$. It is unique: a first difference ζ of two normalized choices satisfies $p_0\zeta = 0$ and $h d_E \zeta = 0$, hence $\zeta = 0$.

For an input of field degree one, the first two homogeneous terms are

$$J_1^{\mathrm{cyc}} = p_1 a, \qquad J_2^{\mathrm{cyc}} = -\frac{1}{2} p_1 [a, ha]. \tag{3.8}$$

Indeed, substituting $\xi_1 = -ha$ in the quadratic coefficient leaves $-\frac{1}{2}[a, ha] - \frac{1}{2}[\eta_0 p_1 a, ha]$ before projection. The projection of the second commutator is zero, since $p_1(\eta_0 h\eta) = p_0(h\eta) = 0$ entrywise. Theorem 3.2 makes J^{cyc} conjugate to $\mathcal{J}(a)$. Their invariant polynomials therefore agree.

3.3 An exact noncommuting sphere family

Take $N = 2$, write $s_0 = z\bar z$, and put

$$f = \frac{z^2}{(1+s_0)^2}, \qquad g = \frac{\bar z}{1+s_0}, \qquad A = E_{12}, \quad B = E_{21}, \quad M_0 = [A, B] = \operatorname{diag}(1, -1).$$

Here E_{ij} are the elementary matrices. Both functions are globally smooth: in the coordinate $w = 1/z$ they become $\bar w^2/(1+w\bar w)^2$ and $w/(1+w\bar w)$. Their values and first holomorphic jets at o vanish. Consequently

$$a = t(A\bar\partial f + B\bar\partial g)$$

is exact in the linear Neumann complex, and $p_1 a = 0$.

Theorem 3.5. *For this family, as a formal identity and for sufficiently small complex t,*

$$\mathcal{J}(a) = -\frac{t^2}{3}M_0 - \frac{13t^5}{315}B. \tag{3.9}$$

It satisfies $\mathcal{J}(a)^2 = (t^4/9)I_2$. For every $m \geq 1$,

$$\operatorname{tr}\mathcal{J}(a)^{2m+1} = 0, \qquad \frac{1}{2m}\operatorname{tr}\mathcal{J}(a)^{2m} = \frac{t^{4m}}{m\,3^{2m}}. \tag{3.10}$$

The constant Neumann gauge $K = I_2 + (13/210)t^3B$ obeys $K^{-1}\mathcal{J}(a)K = -(t^2/3)M_0$.

Proof. The normalized primitive

$$G(f\,\bar\partial g) = -\frac{z}{3(1+s_0)^3}, \qquad f\,\bar\partial g = \frac{z^2}{(1+s_0)^4}\,d\bar z$$

has first jet $-1/3$. These identities follow by differentiation, smoothness in both charts, and the zero value at o. The polar Stokes identity applied to $f\,g$ gives $\tau_E(g\bar\partial f) = -\tau_E(f\,\bar\partial g)$. The quadratic term $-\kappa^{-1}\tau_E(aGa)$ of $\mathcal{J}(a)$ is therefore $-t^2M_0/3$.

Write $H = \sum_{n\geq 0} t^n H_n$, with $H_0 = I_2$. The recursion $\bar\partial H_n = -(a/t)H_{n-1}$ gives

$$\begin{aligned}
(H_2)_{11} &= \frac{z(3s_0+1)}{3(1+s_0)^3},\\
(H_3)_{21} &= -\frac{s_0^4+4s_0^3+6s_0^2+2s_0}{6(1+s_0)^4},\\
(H_4)_{11} &= \frac{z^2(15s_0^4+60s_0^3+90s_0^2+48s_0+8)}{90(1+s_0)^6},\\
(H_5)_{21} &= \frac{z(35s_0^4+140s_0^3+210s_0^2+126s_0+26)}{630(1+s_0)^7}.
\end{aligned}$$

Each right side is smooth in both charts and vanishes at o. Differentiating, with $\bar\partial f = -2z^3(1+s_0)^{-3}d\bar z$ and $\bar\partial g = (1+s_0)^{-2}d\bar z$, verifies the recursion and fixes each primitive uniquely. The last first jet is $13/315$.

To exclude all other orders, use rotations $z \mapsto e^{i\theta}z$. The normalized solution operator commutes with their pullback, and f, g have weights 2, -1. Since $A^2 = B^2 = 0$, only alternating matrix words occur. A word of length $2k$ has weight k, while words of length $2k+1$ beginning with A and B have weights $k+2$ and $k-1$.

A smooth function can have nonzero first holomorphic jet at o only in weight one. The only surviving orders are therefore $n = 2$ and the B word at $n = 5$. This proves (3.9) in every formal order.

Proposition 3.3 gives the analytic identity.

Now $M_0^2 = I_2$, $M_0B + BM_0 = 0$, and $B^2 = 0$ give the square and all trace identities. Finally $K^{-1} = I_2 - (13/210)t^3B$ and $[M_0, B] = -2B$ give the asserted conjugation. □

The first nonzero quadratic measurement is thus $\frac{1}{2}\operatorname{tr}\mathcal{J}(a)^2 = t^4/9$. The linear exactness of a does not contradict this invariant: a straight line of infinitesimal gauge variations need not be a finite gauge orbit. A pure-gauge path has higher terms that enforce zero residual.

The kinetic differential matters even for this small matrix example. For comparison, take the Laurent complex

$$E_{\mathrm{L}}^0 = \mathbb{C}[z, z^{-1}], \quad E_{\mathrm{L}}^1 = E_{\mathrm{L}}^0\eta, \quad d_{\mathrm{L}}u = (z\partial_z u)\eta, \quad \tau_{\mathrm{L}}(u\eta) = [z^0]u.$$

Here η has degree one and square zero, and $[z^0]$ extracts the constant coefficient. Its contraction projects to constant modes and divides a nonzero mode by its integer exponent.

Set

$$a_{\mathrm{L}} = t\begin{pmatrix} 0 & z \\ z^{-1} & 0 \end{pmatrix}\eta, \quad b(t) = \frac{\sqrt{1+4t^2}-1}{2}, \quad r(t) = b(t)/t, \quad G_0 = \begin{pmatrix} 1 & -rz \\ rz^{-1} & 1 \end{pmatrix},$$

using the formal square root with constant term one. Since $b^2 + b = t^2$,

$$z\partial_z G_0 + (a_{\mathrm{L}}/\eta)G_0 = G_0\,\mathrm{diag}(b,-b).$$

Multiplying G_0 by $(1+r^2)^{-1/2}$ leaves the transformed connection unchanged. Its logarithm is $\arctan(r)\left(\begin{smallmatrix} 0 & -z \\ z^{-1} & 0 \end{smallmatrix}\right)$, which has zero constant Laurent coefficient. The resulting quadratic invariant is

$$b(t)^2 = t^4 - 2t^6 + 5t^8 - 14t^{10} + O(t^{12}).$$

It belongs to this Laurent complex. The sphere invariant is (3.10).

3.4 Invariant functions and their smooth derivatives

For each integer $r \geq 1$, define

$$O_r(a) = \frac{1}{r}\,\mathrm{tr}\,\mathcal{J}(a)^r. \tag{3.11}$$

These functions are gauge invariant and satisfy $O_r(\eta_0 J) = \mathrm{tr}(J^r)/r$. Their Taylor coefficients are iterated Green expressions. For example, superscripts denoting homogeneous field degree,

$$\begin{aligned} O_2^{[2]} &= \frac{1}{2\kappa^2}\,\mathrm{tr}\,\tau_E(a)^2, \\ O_2^{[3]} &= -\frac{1}{\kappa^2}\,\mathrm{tr}\big(\tau_E(a)\tau_E(aGa)\big), \\ O_2^{[4]} &= \frac{1}{\kappa^2}\,\mathrm{tr}\big(\tau_E(a)\tau_E(aG(aGa))\big) + \frac{1}{2\kappa^2}\,\mathrm{tr}\,\tau_E(aGa)^2. \end{aligned}$$

The polar trace acts entrywise. The final trace is a matrix trace.

Proposition 3.6. *The variation of O_r is represented by the smooth matrix function*

$$X_r(a) = \kappa^{-1}H(a)\mathcal{J}(a)^{r-1}H(a)^{-1}, \qquad \delta O_r = \tau_E\,\mathrm{tr}(X_r\,\delta a). \tag{3.12}$$

It lies in $E^0 \otimes \mathrm{Mat}_N(\mathbb{C})$ and obeys $\bar\partial X_r + [a, X_r] = 0$.

Proof. Write $\delta H = HY$. Variation of the frame equation gives

$$\bar\partial Y = -H^{-1}\delta a H, \qquad Y(o) = 0, \qquad Y = -G(H^{-1}\delta a H).$$

Taking the first jet, using $H(o) = I_N$, yields $\delta\mathcal{J} = \kappa^{-1}\tau_E(H^{-1}\delta a H)$. Differentiate the invariant polynomial and use cyclicity of matrix trace to obtain (3.12). The first jet of X_r is $\kappa^{-1}[-\mathcal{J}, \mathcal{J}^{r-1}] = 0$. Differentiating $H\mathcal{J}^{r-1}H^{-1}$ by $\bar\partial$ proves covariant closedness. □

Let U be a closed oriented surface with a smooth two-form ρ. Let a_C be a smoothly U-dependent curve connection in the formal or analytic domain above. The functionals

$$I_r = \int_U \rho\, O_r(a_C)$$

have smooth representatives ρX_r in the conjugate antifield component. They are local in U and generally nonlocal in C. The same construction applies with compact support on an open surface, or with boundary conditions that annihilate the boundary variations.

For the Batalin–Vilkovisky (BV) extension, use the shifted space $L_U[1]$. Its components of total form degrees 0, 1, 2, 3 are respectively the ghost c, physical connection $\mathcal{A}$, connection antifield, and ghost antifield. Their coordinate ghost numbers are 1, 0, −1, −2. The pairing $\int_U \tau_E$ tr pairs complementary total degrees summing to three. After the shift, it has degree −1. We use its degree-one antibracket on functionals whose differentials admit smooth Hamiltonian representatives.

Proposition 3.7. *Let S_{BV} be the minimal classical BV extension of (2.10), with field–antifield pairing induced by $\int_U \tau_E$ tr. Assume a boundary condition for which its classical master equation holds. Then*

$$\{S_{\mathrm{BV}}, I_r\} = 0, \qquad \{I_r, I_s\} = 0.$$

For any fixed finite set of degree-zero parameters σ_r, the action $S_{\mathrm{BV}} + \sum_r \sigma_r I_r$ satisfies the classical master equation. This remains true after completion in the ideal generated by these sources.

Proof. On the physical curve connection, the minimal BV differential is its infinitesimal gauge transformation with the ghost as parameter. Gauge invariance of O_r gives the first identity, and Proposition 3.6 supplies its admissible smooth Hamiltonian representative. The I_r depend only on fields, so their mutual field–antifield contractions vanish. Expansion of the master equation gives the assertion. □

One may also adjoin source antifields σ_r^+ of degree −1. With antibracket convention $\{\sigma_r, \sigma_r^+\} = 1$ and $Q = \{S_{\mathrm{BV}} + \sum_r \sigma_r I_r, -\}$, the extended differential satisfies $Q\sigma_r = 0$ and $Q\sigma_r^+ = I_r$. This is a classical source deformation. It specifies no quantum measure or coincident-point contraction.

3.5 Direct sums and the closed-field comparison

The frame equation respects block sums. Consequently the same normalization of (3.11) at every rank gives

$$O_r(a \oplus b) = O_r(a) + O_r(b).$$

This property distinguishes single traces among homogeneous polynomial families.

Proposition 3.8. *Let F_N be a conjugation-invariant polynomial on $\mathrm{Mat}_N(\mathbb{C})$ for every $N \geq 1$, homogeneous of one fixed degree $d > 0$. If*

$$F_{N+M}(X \oplus Y) = F_N(X) + F_M(Y),$$

then $F_N(X) = c_d \operatorname{tr}(X^d)$ with a scalar c_d independent of N. In degree zero, the additive constant families are $F_N = c_0 N$.

Proof. Repeated direct-sum additivity gives $F_N(\operatorname{diag}(x_1, \dots, x_N)) = \sum_i F_1(x_i)$. Homogeneity gives $F_1(x) = c_d x^d$. Conjugation invariance then determines F_N on matrices with distinct eigenvalues, a Zariski dense set. Equality holds as polynomials. In degree zero, additivity determines the constant at rank N as N times the constant at rank one. □

For example, the direct-sum defect of $u \operatorname{tr}(J^2) + v(\operatorname{tr} J)^2$ is $2v \operatorname{tr} X \operatorname{tr} Y$. Additivity forces $v = 0$, while leaving u undetermined. It tests block-diagonal inputs and removes no off-diagonal fields at the larger rank.

The invariant functions and their classical Hamiltonians are now constructed. Identifying them as the image of geometric closed fields requires another map, whose normalization and locality are independent conditions.

Problem 3.9. Construct an action-derived map from an independently specified geometric closed-field complex to the smooth Neumann insertions. Determine its scalar normalization and whether its disk operation is additive under block sums at fixed sources. Compare its full nonlinear image with (3.11), including the family of Theorem 3.5. The comparison must preserve the chosen normal marking, boundary terms, two-source collisions, and sewing. If it is required to be local in the curve variable, construct that local comparison before using the global operator G.

Chapter 4

Reduction and boundary fields

Putting the curve connection in normal form also changes the surface connection. Before imposing the field equations, the transformed surface field has an error determined by the mixed curvature. On flat fields this error vanishes, giving the full formal BF reduction. At a fixed residual matrix, the linearized reduction retains its entire adjoint complex, including the stabilizer directions that appear when eigenvalues coincide.

The action retains two kinds of boundary data. At the pole, its Stokes term pairs the value of a function with its first holomorphic jet. Along the boundary of the surface, a varying gauge transformation contributes a cubic term in its Maurer–Cartan form. We derive both terms after computing the exact quadratic action of the eliminated surface fields.

4.1 Curvature and the surface connection

Keep the sphere, trace, normalized Green operator, and representative f_0, η_0 of Chapter 2. Write $\mathfrak{g} = \mathfrak{gl}_N(\mathbb{C})$ and let U be an oriented smooth surface. Surface forms precede curve forms in the graded tensor product. Decompose a physical mixed connection as

$$\mathsf{A} = b + a, \qquad b \in \Omega^1(U)\widehat{\otimes}E^0 \otimes \mathfrak{g}, \qquad a \in C^\infty(U)\widehat{\otimes}E^1 \otimes \mathfrak{g}.$$

Work formally in the field amplitude, or in the small Sobolev neighborhood of Proposition 3.3, smoothly parametrized over U. Let $H = H(a)$ and $J = \mathcal{J}(a)$, so $\bar\partial H = -aH$, $H(o) = I_N$, and $j(H) = -J$. Put $b_o = b(o)$.

Transformation by H kills the curve connection and leaves the surface one-form

$$B = H^{-1}bH + H^{-1}d_U H. \tag{4.1}$$

To recover it, define the curvature components and their residual value:

$$\begin{aligned} F_{UU} &= d_U b + b^2, & F_{UC} &= d_U a - \bar\partial b + [b, a], \\ C_a &= H^{-1}F_{UC}H, & r &= d_U J + [b_o, J]. \end{aligned} \tag{4.2}$$

Here $\bar\partial b$ differentiates the curve coefficients. Its minus sign is the tensor sign in the total differential.

Proposition 4.1. *Without imposing flatness,*

$$B = b_o - GC_a, \qquad r = j(GC_a) = \kappa^{-1}\tau_E(C_a), \qquad (H^{-1}F_{UU}H)(o) = d_U b_o + b_o^2. \tag{4.3}$$

All curve operators act coefficientwise on surface forms.

Proof. The transformed curve connection is zero, so covariance of mixed curvature gives $-\bar\partial B = C_a$. Since $B(o) = b_o$, normalization of G gives the first identity. Differentiate (4.1) at o, using $j(H^{-1}) = J$, $j(b) = 0$, and $j(d_U H) = -d_U J$. The result is $j(B) = -r$, which gives the second identity by polar Stokes. Surface curvature transforms by conjugation, and evaluation at o gives the last identity. □

The frame H need not be Neumann. The admissible transformation is $g = H\exp(f_0 J)$ from Theorem 3.2. For $L = \mathrm{ad}_J$, introduce the entire operator

$$T_J = f_0 \int_0^1 e^{-u f_0 L}\, du = \sum_{n\geq 0} \frac{(-1)^n f_0^{n+1} L^n}{(n+1)!}. \tag{4.4}$$

This formula is defined even when L has a kernel.

Corollary 4.2. *For the convention* $\mathsf{A}^g = g^{-1}\mathsf{A}g + g^{-1}(d_U + \bar\partial)g$,

$$\mathsf{A}^g = b_o + \eta_0 J + R_a, \qquad R_a = -e^{-f_0 L} G C_a + T_J r. \tag{4.5}$$

The surface one-form R_a has value and first holomorphic jet zero at o.

Proof. After transformation by H, transform B by $e^{f_0 J}$. Differentiation of the matrix exponential gives

$$e^{-f_0 J} B e^{f_0 J} + e^{-f_0 J} d_U e^{f_0 J} - b_o = e^{-f_0 L}(B - b_o) + T_J(d_U J + [b_o, J]).$$

Substitute (4.3). The curve component becomes $\eta_0 J$. Both GC_a and T_J vanish at o, and their first jets are r and the identity operator. Their contributions to $j(R_a)$ cancel. □

Thus mixed curvature measures exactly the failure of a general field to be represented by residual BF fields. On flat fields, $C_a = r = 0$, and the normal form is $b_o + \eta_0 J$.

4.2 Formal fields and gauge homotopies

To retain ghosts and higher gauge homotopies, let R be a local Artin commutative differential graded algebra over $\mathbb{C}$, with nilpotent maximal ideal $\mathfrak{m}$. A formal field is a total-degree-one element of

$$\Omega^\bullet(U)\widehat{\otimes} E \otimes \mathfrak{g} \otimes \mathfrak{m}.$$

It is a Maurer–Cartan field if $D\mathsf{A} + \mathsf{A}^2 = 0$, where D is the total differential. Tensoring additionally with $\Omega^\bullet_{\mathrm{poly}}(\Delta^n)$ defines its simplicial space of fields. The residual complex replaces E by $A_\kappa = \mathbb{C} \oplus \mathbb{C}\varepsilon$.

Theorem 4.3. *The maps*

$$\mathsf{A} = \mathbf{b} + a \longmapsto (\mathbf{b}(o), \mathcal{J}(a)), \qquad (\mathbf{b}_o, \mathbf{J}) \longmapsto \mathbf{b}_o + \mathbf{J}\eta_0$$

give mutually inverse equivalences of these formal deformation problems, up to a natural based Neumann gauge homotopy. The residual equations are

$$d\mathbf{b}_o + \mathbf{b}_o^2 = 0, \qquad d\mathbf{J} + [\mathbf{b}_o, \mathbf{J}] = 0. \tag{4.6}$$

Here d includes the surface, Artin, and simplex differentials.

Proof. The coefficient of a curve one-form has total coefficient degree zero. Its normalized frame therefore has degree zero, and the Green series and exponential are defined over R. Nilpotence makes the series finite. The curvature calculation uses only the Leibniz rule and graded matrix multiplication, so it applies with the full coefficient differential. On a Maurer–Cartan field, both curvature components vanish. Equation (4.5) then gives the residual field and (4.6).

For a residual field, $H = \exp(-f_0\mathbf{J})$, so the residual composite is the identity and the correcting gauge is $g = I_N$. For a general field, set $g_u = \exp(u \log g)$. The connection

$$\mathsf{A}^{g_u} + g_u^{-1}\partial_u g_u \, du$$

is Maurer–Cartan on the parameter interval. It joins A to its residual normal form. Its residual image is constant, since $g_u(o) = I_N$, $j(g_u) = 0$, and its additional du coefficient vanishes at o.

For a Neumann gauge k, uniqueness of the frame gives $H(a^k) = k^{-1}H(a)k(o)$. Hence $\mathcal{J}$ transforms by conjugation with $k(o)$, and $\mathbf{b}(o)$ transforms as a connection. The formulas commute with coefficient morphisms and simplex restrictions. They consequently give the stated equivalence with its higher gauge homotopies. □

The strict inclusion preserves the cyclic pairing. On a closed surface, or with compact support, substitution in the cyclic action gives

$$S_{\mathrm{BF}} = \kappa \int_U \operatorname{tr}\big(\mathbf{J}(d_U\mathbf{b} + \mathbf{b}^2)\big), \tag{4.7}$$

$$\omega_{\mathrm{BF}} = \kappa \int_U \operatorname{tr}(\delta\mathbf{J}\,\delta\mathbf{b}). \tag{4.8}$$

The pairing has degree -1. In surface degrees $0, 1, 2$, write $\mathbf{b} = (c, b, J^+)$ and $\mathbf{J} = (J, b^+, c^+)$. Their coordinate ghost numbers are respectively $(1, 0, -1)$ and $(0, -1, -2)$. These are all components of the residual shifted complex.

The projection removes contractible directions, so it is not a symplectomorphism of the full spaces of fields before the equations of motion are imposed. The equivalence concerns the trivial curve-bundle neighborhood. Its maps preserve restriction and support in U, while the Green operator still uses the whole curve.

4.3 The adjoint complex at a residual matrix

The residual field J may have repeated eigenvalues or be nonsemisimple. The correct linearized complex keeps both copies of $\mathfrak{g}$:

$$K_J = \big[\mathfrak{g} \xrightarrow{\mathrm{ad}_J} \mathfrak{g}\varepsilon\big]. \tag{4.9}$$

They lie in degrees zero and one. On the smooth curve complex, the differential is

$$d_J f = \bar\partial f + \eta_0[J, f], \qquad d_J(E^1) = 0.$$

The inclusion $I : K_J \to E \otimes \mathfrak{g}$ is $I_0 x = x$, $I_1 y = \eta_0 y$. Extend the cyclic projections entrywise:

$$P_0 f = \kappa^{-1}\tau_E(f\,\eta_0), \qquad P_1\eta = \kappa^{-1}\tau_E(\eta).$$

The factor ε in the target of P_1 is understood.

Conjugation by $e^{f_0 \,\mathrm{ad}_J}$ removes the twist from d_J. It changes the domain condition, so a correction is still needed. With $L = \mathrm{ad}_J$ and T_J from (4.4), define

$$\begin{aligned} p^{\mathrm{fr}}_{J,0} f &= f(o), & p^{\mathrm{fr}}_{J,1}\eta &= jG(e^{f_0 L}\eta), \\ h^{\mathrm{fr}}_J\eta &= e^{-f_0 L}G(e^{f_0 L}\eta) - T_J\, p^{\mathrm{fr}}_{J,1}\eta, & h^{\mathrm{fr}}_J(E^0) &= 0. \end{aligned} \tag{4.10}$$

Theorem 4.4. *For every finite matrix J, the map*

$$h_J = (1 - I_0 P_0) h_J^{\mathrm{fr}} \tag{4.11}$$

is a continuous cyclic contraction onto K_J:

$$PI = 1, \qquad d_J h_J + h_J d_J = 1 - IP, \qquad h_J^2 = h_J I = P h_J = 0.$$

For matrix-valued curve one-forms η, ξ,

$$\tau_E \operatorname{tr}((h_J \eta)\xi) + \tau_E \operatorname{tr}(\eta h_J \xi) = 0. \tag{4.12}$$

The maps depend analytically on J, including at nonsemisimple matrices, and define coefficientwise maps for formal matrix parameters. The cohomology is $\ker \operatorname{ad}_J$ *in degree zero and* $\operatorname{coker} \operatorname{ad}_J$ *in degree one.*

Proof. For $F = e^{f_0 L} f$, the Neumann condition $j(f) = 0$ becomes $j(F) = LF(o)$. Direct differentiation on unconstrained smooth functions gives

$$d_J(T_J y) = \eta_0 y, \qquad T_J L = 1 - e^{-f_0 L}.$$

The two first jets in (4.10) cancel. Thus h_J^{fr} takes values in E^0 and vanishes at o. The same identities give

$$d_J h_J^{\mathrm{fr}} \eta = \eta - \eta_0 p_{J,1}^{\mathrm{fr}} \eta, \qquad h_J^{\mathrm{fr}} d_J f = f - f(o).$$

For the latter equality use $G(e^{f_0 L} d_J f) = e^{f_0 L} f - f(o)$.

The normalized primitive of $e^{f_0 L} \eta_0 y$ is $\int_0^{f_0} e^{uL} y \, du$. It proves $p_J^{\mathrm{fr}} I = 1$ and $h_J^{\mathrm{fr}} I = 0$. The adjoint projection is a chain map, since $P_1 d_J f = [J, P_0 f]$. Hence $\Pi = 1 - IP$ commutes with d_J, and $h_J = \Pi h_J^{\mathrm{fr}} \Pi$. Therefore

$$d_J h_J + h_J d_J = \Pi(1 - I p_J^{\mathrm{fr}})\Pi = \Pi.$$

The side conditions follow from $\Pi I = P\Pi = 0$, $h_J^{\mathrm{fr}} I = 0$, and degree.

For cyclicity, put $v = h_J \eta$ and $w = h_J \xi$. Both have zero P_0. Adjointness of P and I, followed by polar integration by parts, gives

$$\tau_E \operatorname{tr}(v\xi) = \tau_E \operatorname{tr}(v \, d_J w) = -\tau_E \operatorname{tr}((d_J v) w) = -\tau_E \operatorname{tr}(\eta w).$$

The commutator contribution has zero matrix trace, and the degree-zero products remain in the Neumann domain.

Finally, multiplication by the smooth matrix exponentials, G, and the finite-rank projections is continuous. Parameter derivatives are bounded on compact sets of matrices in every smooth seminorm. This proves the analytic assertion and the coefficientwise formal construction. The contraction identifies cohomology with that of (4.9). □

In particular, put

$$V = \ker P_0 \subset E^0 \otimes \mathfrak{g}, \qquad W = \ker P_1 \subset E^1 \otimes \mathfrak{g}.$$

Then $d_J : V \to W$ is an isomorphism with inverse h_J. Write $h = h_{J=0}$ for the cyclic homotopy of Theorem 2.4. As a formal series at $J = 0$,

$$h_J = h(1 + \eta_0 \operatorname{ad}_J h)^{-1}.$$

The exact formula (4.11) does not require convergence of this particular series.

4.3.1 *An eigenvalue collision*

The family $J_x = \operatorname{diag}(x, -x)$ shows what would be lost by dividing by nonzero eigenvalue differences. Over either $R = \mathbb{C}[x]$ or $R = \mathbb{C}[[x]]$, its residual complex has

$$H^0(K_{J_x}) = R^2, \qquad H^1(K_{J_x}) = R^2 \oplus (R/(x))^2. \tag{4.13}$$

Indeed, the two diagonal directions have zero differential, and the two off-diagonal arrows are multiplication by $2x$ and $-2x$. They are injective over R with cokernel $R/(x)$.

At $x = 0$, the specialized complex has four-dimensional kernel and cokernel. Each two-term free resolution of $R/(x)$ contributes a $\operatorname{Tor}_1^R(R/(x), \mathbb{C})$ class in degree zero. This accounts for the two additional stabilizer directions. Theorem 4.4 transports the calculation to the native formal family $(E \otimes \mathfrak{g})[[x]]$.

The completion in this last assertion is necessary. For $f_0 = z/(1 + z\bar{z})$, the differential in the E_{12} direction is $D_x = \bar{\partial} + 2x\eta_0$ on $E^0[x]$. The cokernel of the residual inclusion on first cohomology is

$$Q = \frac{E^1[x]}{D_x E^0[x] + \mathbb{C}[x]\eta_0}.$$

Proposition 4.5. *Inclusion as constant polynomials induces an injective complex linear map $E^1/\mathbb{C}\eta_0 \longrightarrow Q$. For $v = z\bar{z}/(1 + z\bar{z})$, the classes $[\bar{\partial}(v^m)]$, $m \geq 1$, are linearly independent in Q. Thus the residual inclusion over $\mathbb{C}[x]$ has an infinite-dimensional cokernel on first cohomology of $(E \otimes \mathfrak{gl}_2)[x]$ with differential d_{J_x}.*

Proof. Suppose that $\eta \in E^1$, viewed as constant in x, has zero class in Q. There are polynomials $c(x) \in \mathbb{C}[x]$ and $u(x) \in E^0[x]$ with

$$\eta = c(x)\eta_0 + D_x u(x).$$

Multiplication by η_0 on smooth functions is injective because the form is nonzero on a dense open set.

If u has degree n, comparison above degree $n + 1$ first gives $\deg c \leq n + 1$. The coefficient of x^{n+1} says that the top coefficient of u is a scalar constant. Its Dolbeault derivative vanishes. Descending through powers of x then makes every coefficient of u constant. The constant term gives $\eta = c(0)\eta_0$, which also follows directly when $u = 0$. This proves the injection.

A finite relation among the displayed classes would therefore give

$$\bar{\partial}\left(\sum_{m=1}^{M} a_m v^m - c f_0\right) = 0.$$

The function in parentheses is smooth on the sphere, so it is holomorphic and constant. Its first holomorphic jet at o is $-c$, so $c = 0$, and its value at o makes the constant zero. Since v takes every value in $[0, 1)$ on the affine chart, the polynomial $\sum_{m=1}^{M} a_m t^m$ vanishes identically. Every coefficient a_m is zero. □

Thus a formal or analytic parameter family retains the collision through the full adjoint complex. The exponential contraction is not a map on the uncompleted polynomial family.

4.4 The exact action of the eliminated fields

On the curve slice $a = \eta_0 J$, split the surface field by the cyclic projection:

$$b = b_c + u, \qquad b_c = P_0 b, \qquad P_0 u = 0.$$

Here b_c is a surface connection with constant curve coefficients. Before imposing flatness, it can differ from $b(o)$.

Proposition 4.6. *The physical action is exactly*

$$S_E(b+\eta_0 J) = \kappa \int_U \operatorname{tr}(J F_{b_c}) - \frac{1}{2}\int_U \tau_E \operatorname{tr}(u\, d_J u) - \frac{\kappa}{2}\int_{\partial U} \operatorname{tr}(b_c J). \tag{4.14}$$

The boundary term is absent on a closed surface or with compact support in its interior. The differential d_J acts on curve coefficients. In oriented surface coordinates (s,t), the middle term is $-\int_U \tau_E \operatorname{tr}(u_s d_J u_t)\, ds\, dt$.

The infinitesimal gauge derivative of the condition $ha = 0$, on gauge parameters $\xi \in V$, is

$$M_J = h d_J|_V = 1 + h\eta_0 \operatorname{ad}_J|_V, \qquad M_J^{-1} = h_J \bar\partial|_V. \tag{4.15}$$

Proof. Expand the kinetic and cubic terms in $b_c + u + \eta_0 J$. Every term linear in u vanishes because $\tau_E(u\eta_0) = \kappa P_0 u = 0$, $\tau_E(\bar\partial u) = 0$, and P_0 commutes with d_U. The terms independent of u are the BF density and its boundary term from Proposition 2.7.

The remaining cubic coefficient is $\tau_E \operatorname{tr}(\eta_0 J u^2)$. Cyclicity identifies it with the $\eta_0 \operatorname{ad}_J$ part of the displayed quadratic form. The remaining kinetic coefficient is its $\bar\partial$ part. A product of three surface fields has no curve one-form and contributes zero. This proves the exact identity for surface-dependent J as well.

At the slice, the curve component of a gauge variation is $d_J\xi$. Applying h gives M_J. Since $d_J V \subset W$ and $\bar\partial h = 1$ on W, one has $\bar\partial M_J = d_J$ on V. Composition of the two isomorphisms $d_J, \bar\partial : V \to W$ gives the inverse in (4.15). □

The eliminated surface fields therefore have a quadratic action whose kernel depends on J. A quantum elimination must use this same background dependence both for the gauge determinant and for observable contractions. The classical splitting alone assigns neither quantity.

4.5 The polar first-jet boundary

The condition $j(f) = 0$ makes the polar trace closed. To allow the first jet to vary, retain it as a boundary variable. Write

$$\mathcal{A} = \Omega^{0,\bullet}(C), \qquad B = \mathbb{C}[\xi]/(\xi^2), \qquad |\xi| = 0, \qquad d_B = 0.$$

Here ξ is even, unlike the odd residual class $\varepsilon \in A_\kappa$. Define differential graded algebra maps

$$\begin{aligned} r &: \mathcal{A} \longrightarrow B, \qquad r(f) = f(o) + \xi\, j(f), \quad r(\eta) = 0, \quad |\eta| = 1, \\ \ell &: \mathbb{C} \longrightarrow B, \qquad \ell(c) = c. \end{aligned} \tag{4.16}$$

The product rule for j proves multiplicativity of r. The map annihilates positive degree, so it commutes with the differentials. The Neumann algebra is the strict pullback

$$E = \mathcal{A} \times_B \mathbb{C}. \tag{4.17}$$

Indeed, the equation $r(f) = \ell(c)$ says $j(f) = 0$ and $c = f(o)$. There is no restriction in degree one.

Let $\tau_{\mathcal{A}}$ be the polar functional on $\mathcal{A}^1$, extended by zero on $\mathcal{A}^0$, and define

$$\tau_B(c_0 + \xi c_1) = \kappa c_1.$$

The polar Stokes identity becomes the relative trace identity

$$\tau_{\mathcal{A}}\bar\partial = \tau_B r, \qquad \tau_B \ell = 0. \tag{4.18}$$

The two traces have degrees -1 and 0, respectively.

Proposition 4.7. *The pairing $(x, y) \mapsto \tau_B(x\,y)$ on B is nondegenerate, and $\ell(\mathbb{C})$ is Lagrangian. The same holds after tensoring with $\mathfrak{gl}_N$ and the matrix trace: the constant matrices form a Lagrangian Lie subalgebra of $\mathfrak{gl}_N(B)$.*

Proof. In the basis $(1, \xi)$ the pairing has matrix

$$\kappa\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The constant line is isotropic and equals its orthogonal complement. For matrices the pairing is

$$\kappa\,\mathrm{tr}(X_0 Y_1 + X_1 Y_0).$$

Nondegeneracy of the ordinary matrix trace gives the assertion, and cyclicity makes the pairing invariant under the Lie bracket. □

Thus the Neumann condition selects the constant Lagrangian subalgebra of the first-jet boundary. This boundary records the integration term at the double pole without choosing a real boundary circle.

4.5.1 The boundary action

For this subsection assume that U is closed and oriented. Tensor the relative diagram (4.16) with $\Omega^\bullet(U) \otimes \mathfrak{g}$, retaining the full graded fields. Let $\mathbf{A}$ be the total-degree-one bulk superfield, $D = d_U + \bar\partial$, and $a = r(\mathbf{A})$. Put

$$T_{\mathcal{A}} = \int_U \tau_{\mathcal{A}}\,\mathrm{tr}, \qquad T_B = \int_U \tau_B\,\mathrm{tr}, \qquad S_{\mathcal{A}}(\mathbf{A}) = T_{\mathcal{A}}\Big(\frac{1}{2}\mathbf{A}D\mathbf{A} + \frac{1}{3}\mathbf{A}^3\Big).$$

The traces $T_{\mathcal{A}}$ and T_B have degrees -3 and -2. After shifting the field complexes by one, their induced pairings have degrees -1 and 0. These are the bulk Batalin–Vilkovisky pairing and the boundary Batalin–Fradkin–Vilkovisky pairing, respectively.

Proposition 4.8. *In the cyclic suspension convention for the full fields,*

$$\delta S_{\mathcal{A}} = T_{\mathcal{A}}\big(\delta\mathbf{A}(D\mathbf{A} + \mathbf{A}^2)\big) - \frac{1}{2}T_B(a\,\delta a). \tag{4.19}$$

The boundary one-form and Hamiltonian are

$$\Theta_B = \frac{1}{2}T_B(a\,\delta a), \qquad \omega_B = \delta\Theta_B, \tag{4.20}$$

$$S_B(a) = T_B\Big(\frac{1}{2}a\,d_U a + \frac{1}{3}a^3\Big). \tag{4.21}$$

Writing $a = a_0 + \xi a_1$, one obtains

$$S_B = \kappa\int_U \mathrm{tr}\big(a_1(d_U a_0 + a_0^2)\big). \tag{4.22}$$

The boundary condition $a_1 = 0$ is Lagrangian and preserved by the cohomological vector field. Its restriction recovers the Neumann action.

Proof. The kinetic variation is $\frac{1}{2}T_{\mathcal{A}}(\partial\mathbf{A}D\mathbf{A} + \mathbf{A}D\partial\mathbf{A})$. Use the identity

$$D(\mathbf{A}\partial\mathbf{A}) = (D\mathbf{A})\partial\mathbf{A} - \mathbf{A}D\partial\mathbf{A}$$

and (4.18). The resulting endpoint term is $-\frac{1}{2}T_B(a\partial a)$, and the interior kinetic terms combine. Graded cyclicity makes the cubic variation $T_{\mathcal{A}}(\partial\mathbf{A}\,\mathbf{A}^2)$. This proves (4.19) and gives the relative Hamiltonian identity $\iota_Q\omega_{\mathcal{A}} = \partial S_{\mathcal{A}} + r^*\Theta_B$.

On the boundary, the vector field is $Qa = d_U a + a^2$. Associativity and the derivation identity give $Q^2 = 0$ in this suspension convention. To expand its Hamiltonian, retain the coefficient of ξ:

$$\frac{\kappa}{2}\int_U \operatorname{tr}(a_0 d_U a_1 + a_1 d_U a_0) + \frac{\kappa}{3}\int_U \operatorname{tr}(a_0^2 a_1 + a_0 a_1 a_0 + a_1 a_0^2).$$

Surface integration by parts identifies the two kinetic terms, and graded cyclicity identifies the three cubic terms. This yields (4.22). Moreover,

$$Qa_0 = d_U a_0 + a_0^2, \qquad Qa_1 = d_U a_1 + a_0 a_1 + a_1 a_0.$$

Hence $a_1 = 0$ is invariant. Its Lagrangian property follows from Proposition 4.7. Both S_B and the boundary one-form restrict to zero there. □

The degrees in this calculation matter. The variables a_0 and a_1 include ghosts and antifields as well as ordinary surface forms. Restricting them both to one-forms would erase the boundary Hamiltonian by surface degree. If U has a boundary, its Stokes term must also be retained. The polar calculation alone does not give the corner data.

4.6 A closed path model for the relative trace

The pullback (4.17) fixes the boundary jet to be constant. A path from the unrestricted jet to a constant instead retains the boundary comparison itself. Let $B[t, dt]$ be the polynomial de Rham algebra of the interval, with $|t| = 0$, $|dt| = 1$ and $d_t t = dt$. Define

$$\widetilde{E} = \big\{(a, b, c) \in \mathcal{A} \oplus B[t, dt] \oplus \mathbb{C} : b(0) = r(a),\ b(1) = \ell(c)\big\}. \tag{4.23}$$

Endpoint evaluation sets $dt = 0$. The product is componentwise and the differential is $D = (\bar{\partial}, d_t, 0)$.

Write a degree-one element as $y = (\eta, dt(\beta_0(t) + \xi\beta_1(t)), 0)$, and set

$$\widetilde{\tau}(y) = \tau_{\mathcal{A}}(\eta) + \kappa\int_0^1 \beta_1(t)\,dt. \tag{4.24}$$

Extend this functional by zero in degree zero. It is a graded cyclic trace of degree -1. For $x = (a, b, c)$ in degree zero,

$$\widetilde{\tau}(Dx) = \tau_B r(a) + \tau_B\big(b(1) - b(0)\big) = \tau_B\ell(c) = 0.$$

Thus the path integral cancels the failure of the bulk trace to be closed.

Proposition 4.9. *The multiplicative inclusion $\iota : E \to \widetilde{E}$, given by*

$$\iota(f) = (f, f(o), f(o)), \qquad \iota(\eta) = (\eta, 0, 0),$$

is a trace-preserving quasi-isomorphism. It has the following cochain retraction. For $x = (a, b_0(t) + \xi b_1(t), c)$ and y as above, put

$$p_0 x = a - f_0 j(a), \qquad p_1 y = \eta + \eta_0 I_1, \qquad I_i = \int_0^1 \beta_i(t)\,dt. \tag{4.25}$$

Let K vanish in degree zero and define

$$K\,y = \left(-f_0 I_1, \int_0^t \beta_0(u)\,du + \xi\left[-I_1 + \int_0^t \beta_1(u)\,du\right], I_0\right). \tag{4.26}$$

Then

$$DK + KD = 1 - \iota p, \qquad p\iota = 1, \qquad K^2 = K\iota = pK = 0, \qquad \widetilde{\tau} = \tau_E p.$$

Proof. The zero endpoint of the middle component of $K\,y$ is $-\xi I_1$, which is $r(-f_0 I_1)$. The endpoint at one is I_0. Thus $K\,y$ belongs to (4.23).

On degree one, $DK\,y$ has first component $-\eta_0 I_1$ and path component $dt(\beta_0 + \xi\beta_1)$. These are the components of $y - \iota p_1 y$. On degree zero, the endpoint conditions give

$$\int_0^1 b_1'(t)\,dt = -j(a), \qquad \int_0^1 b_0'(t)\,dt = c - a(o).$$

Substitution in (4.26) gives $KDx = x - \iota p_0 x$. The chain-map identity for p also follows directly:

$$p_1 Dx = \bar{\partial} a - \eta_0 j(a) = \bar{\partial} p_0 x.$$

The identities $p\iota = 1$ and $K\iota = 0$ follow from the constant path in the inclusion. Degree gives $K^2 = 0$, while $p_0(-f_0 I_1) = 0$ gives $pK = 0$. Finally, $\tau_E(\eta_0) = \kappa$ proves the trace identity. □

Example 4.10. This retraction does not preserve the cyclic pairings. For

$$v = \frac{|z|^2}{1 + |z|^2} \in E^0, \qquad y = (0, \xi\,dt, 0),$$

one has

$$\widetilde{\tau}(\iota(v)\,y) = \kappa v(o) = 0, \qquad \tau_E(v\,p_1 y) = \tau_E(v\eta_0) = \frac{\kappa}{2}.$$

The last value is the integral $\kappa \int_0^\infty r(1 + r)^{-3}\,dr$.

In fact no projection into unchanged smooth E can be the cyclic adjoint of ι on this y. Such a projection would give a smooth form η satisfying $\tau_E(v\eta) = \kappa v(o)$ for every $v \in E^0$. Functions supported away from o belong to E^0 and would force $\eta = 0$ away from o, hence everywhere by smoothness. The test $v = 1$ then contradicts the required value κ.

The closed trace on $\widetilde{E}$ nevertheless gives a classical cyclic action after tensoring with the surface and matrix algebras. Its restriction through ι is the Neumann action. The cochain retraction proves equivalence of the complexes, but supplies no smooth cyclic inverse pairing or elliptic propagator on the path algebra.

4.7 The residual class is global on the curve

The global normal form depends on solving a problem over the whole sphere. A local gauge calculation explains why pole jets alone cannot replace that step. In this calculation, w is a formal antiholomorphic variable, independent of z. Write a matrix connection germ as

$$a = A(z, w)\,dw, \qquad A(z, w) = \sum_{n \geq 0} A_n(z) w^n, \qquad A_n(z) \in \mathrm{Mat}_N(\mathbb{C}[[z]]).$$

Proposition 4.11. *There is a unique formal frame g with $g(z, 0) = I_N$ and $\partial_w g = -Ag$. It is an admissible Neumann gauge and sends a to zero. Consequently, every gauge-invariant formal function of this connection germ alone is constant.*

Proof. For $g = \sum_{n\geq 0} g_n(z)w^n$, the equation is equivalent to

$$g_0 = I_N, \qquad g_{n+1} = -\frac{1}{n+1}\sum_{j=0}^{n} A_j g_{n-j}. \tag{4.27}$$

The recursion constructs each coefficient uniquely without commuting matrix factors. The constant term makes g invertible, and $g(z,0) = I_N$ gives $j(g) = 0$. Its gauge action sends a to zero by the defining differential equation. An invariant function therefore takes on every germ its value at zero. □

The same distinction appears linearly. The exact sequence of global complexes

$$0 \longrightarrow E \longrightarrow \mathcal{A} \xrightarrow{j} \mathbb{C}[0] \longrightarrow 0$$

gives $H^1(E) = \mathbb{C}[\eta_0]$, because a global holomorphic function has zero derivative and $H^1(\mathcal{A}) = 0$. On a disk, the holomorphic function z has derivative one, so the analogous local first cohomology vanishes.

There is an explicit representative invisible to every jet at the pole. Choose a smooth cutoff $\chi(r)$, equal to one near $r = 0$ and zero outside a small coordinate disk, and define

$$f_* = z\chi(|z|^2), \qquad \eta_* = \bar\partial f_* = z^2\chi'(|z|^2)\, d\bar z. \tag{4.28}$$

Extend f_* by zero to the sphere. The form η_* is supported in an annulus, so all of its jets at o vanish. Nevertheless,

$$\tau_E(\eta_*) = -\kappa \int_0^\infty \chi'(r)\, dr = \kappa.$$

For a constant matrix J, the connection $a = \eta_* J$ has normalized global frame $H = e^{-f_* J}$ and residual matrix $\mathcal{J}(a) = J$. This formula holds formally and, for this family, at every finite J. Thus even all local jets of a fail to determine the global residual. The assertion concerns the curve connection alone. Surface curvature and additional fields may carry their own local invariants.

4.8 Surface-dependent gauge transformations

Let $D = d_U + \bar\partial$ and let g be a smooth formal Neumann gauge near the identity, defined on $U \times C$. Set

$$\theta = g^{-1}Dg, \qquad \theta_R = (Dg)g^{-1}, \qquad \mathrm{CS}(A) = \operatorname{tr}\left(\frac{1}{2}A\,DA + \frac{1}{3}A^3\right).$$

The one-forms θ and θ_R are conjugate, but their placement in the finite gauge identity is fixed by the convention $A^g = g^{-1}Ag + \theta$.

Lemma 4.12. *As a cyclic three-form density,*

$$\mathrm{CS}(A^g) - \mathrm{CS}(A) = -\frac{1}{2}D\operatorname{tr}(\theta_R A) - \frac{1}{6}\operatorname{tr}(\theta^3). \tag{4.29}$$

Proof. It is convenient to set $k = g^{-1}$ and $\vartheta = k^{-1}Dk = -\theta_R$. The transformed connection is $kAk^{-1} - (Dk)k^{-1}$. Conjugating it and its derivative by k^{-1} gives $A - \vartheta$ and $DA + \vartheta A + A\vartheta - \vartheta^2$. Here $D\vartheta = -\vartheta^2$.

Substitution in the kinetic and cubic terms, with graded cyclicity, reduces their difference to

$$-\frac{1}{2}\operatorname{tr}(\vartheta^2 A + \vartheta DA) + \frac{1}{6}\operatorname{tr}(\vartheta^3) = \frac{1}{2}D\operatorname{tr}(\vartheta A) + \frac{1}{6}\operatorname{tr}(\vartheta^3).$$

Substitute $\vartheta = -\theta_R$ and use conjugacy of θ_R and θ. □

Choose a formal Neumann path g_u, $0 \le u \le 1$, from 1 to g. The parameter u is independent of the surface coordinates. Put

$$v_u = g_u^{-1}\partial_u g_u, \qquad \theta_u = g_u^{-1} D g_u.$$

Path integrals are interpreted coefficientwise in the formal amplitude.

Proposition 4.13. *For $S_E(A) = \int_U \tau_E \,\mathrm{CS}(A)$,*

$$S_E(A^g) - S_E(A) = \mathcal{B}(A, g), \tag{4.30}$$

$$\mathcal{B}(A, g) = -\frac{1}{2}\int_{\partial U} \tau_E \,\mathrm{tr}(\theta_R A) - \frac{1}{2}\int_{\partial U}\int_0^1 \tau_E \,\mathrm{tr}(v_u \theta_u^2)\, du. \tag{4.31}$$

This integrated functional is independent of the chosen formal path and satisfies

$$\mathcal{B}(A, gh) = \mathcal{B}(A, g) + \mathcal{B}(A^g, h). \tag{4.32}$$

These assertions concern small formal gauges defined on $U \times C$.

Proof. The Maurer–Cartan identity and its parameter derivative are

$$D\theta_u = -\theta_u^2, \qquad \partial_u \theta_u = D v_u + [\theta_u, v_u].$$

Graded cyclicity removes the commutator in the trace, while $D(\theta_u^2) = 0$. Hence

$$\partial_u \,\mathrm{tr}(\theta_u^3) = 3D \,\mathrm{tr}(v_u \theta_u^2).$$

Integrate along the path and substitute in (4.29).

The surface coefficients of θ_u, θ_R, A, and the function v_u, have zero first holomorphic jet. Their degree-zero curve products are therefore in E^0. Polar Stokes contributes zero. Surface Stokes leaves exactly the oriented boundary expression (4.31).

The action difference depends only on the endpoints, proving independence of the integrated functional from the path. Finally, $(A^g)^h = A^{gh}$ makes the action differences telescope and gives (4.32). □

This construction does not assign periods to large gauge transformations. On its formal domain, it gives a classical boundary compensator:

$$\widetilde{S}(A, g) = S_E(A) + \mathcal{B}(A, g) = S_E(A^g).$$

Under $(A, g) \mapsto (A^k, k^{-1}g)$, the connection A^g stays fixed. Thus the group-valued edge field g makes $\widetilde{S}$ invariant. Its identification with a geometric closed-string field would require a separate map.

4.8.1 A nonzero cubic boundary response

Take $U = [0, 1]^2$ with coordinates (s, t) and orientation $ds \wedge dt$. On the sphere set

$$f = \frac{z^2}{(1 + |z|^2)^2}, \qquad g_1 = \frac{\bar{z}}{1 + |z|^2}, \qquad v = \frac{|z|^2}{1 + |z|^2}.$$

These are smooth Neumann functions. Let $E = E_{12}$, $F = E_{21}$, $Z = [E, F]$, and define

$$X = sf E + t g_1 F + st\, v Z, \qquad g_\lambda = \exp(\lambda X),$$

where λ is a formal amplitude. Since $j(X) = 0$, g_λ is an admissible gauge.

Matrix multiplication, with surface forms before $d\bar z$, gives

$$\begin{aligned}\tau_E \operatorname{tr}((DX)^3) &= 6st\ \tau_E\big(f\,g_1\bar\partial v - fv\bar\partial g_1 - g_1 v\bar\partial f\big)\,ds\,dt\\ &= 12st\ \tau_E(f\,g_1\bar\partial v)\,ds\,dt.\end{aligned}$$

The second equality uses polar Stokes for $f\,g_1 v$, whose first jet is zero. The remaining integral is explicit:

$$f\,g_1\bar\partial v = \frac{z^2|z|^2}{(1+|z|^2)^5}\,d\bar z, \qquad \tau_E(f\,g_1\bar\partial v) = -\kappa\int_0^\infty \frac{r}{(1+r)^5}\,dr = -\frac{\kappa}{12}.$$

Thus $\tau_E \operatorname{tr}((DX)^3) = -\kappa st\,ds\,dt$, and

$$S_E(0^{g_\lambda}) = -\frac{\lambda^3}{6}\int_U \tau_E \operatorname{tr}((DX)^3) + O(\lambda^4) = \frac{\kappa\lambda^3}{24} + O(\lambda^4). \tag{4.33}$$

For the path $\exp(u\lambda X)$, the leading boundary primitive is $-\lambda^3\tau_E \operatorname{tr}(X(DX)^2)/6$. Its derivative is the leading bulk density, so the four oriented boundary edges give the same coefficient.

A frame depending only on the curve has $\theta^2 = 0$ by curve degree. Equation (4.33) shows why that simplification fails for a surface-dependent frame. The residual comparison must retain the boundary functional and its composition law. A quantum comparison must in addition account for the background-dependent quadratic complement in (4.14).

Chapter 5

Finite cyclic reductions

A finite collection of curve modes can retain the kinetic pairing without retaining the product. The projected product then has an associator, and the eliminated modes determine the higher operations that cancel it. We construct this cancellation on the smooth Neumann complex. A six-dimensional example already has nonzero ternary and quartic operations. We then vary the residual matrix and transport the finite subspace so that its pairing stays nondegenerate through every eigenvalue collision.

5.1 Finite subspaces and the polar trace

Retain E, d_E, τ_E, p_0, p_1, and h from Chapter 2. Throughout this chapter choose

$$f_0 = \frac{z}{1 + |z|^2}, \qquad \eta_0 = d_E f_0,$$

and keep the circular prescription for the double-pole trace. There are two elementary restrictions on a finite approximation.

Proposition 5.1. *On a connected smooth manifold, every finite-dimensional unital subalgebra of complex-valued smooth functions consists of constants. On $C = \mathbb{P}^1$, no operator $R : \Omega^{0,1}(C) \to \Omega^{0,1}(C)$ with a globally smooth Schwartz kernel preserves τ_E.*

Proof. The powers of a function in a finite-dimensional algebra satisfy a nonzero polynomial relation. The function therefore takes values in a finite set. Connectedness makes it constant.

The polar trace is a distribution of finite order. Applying it in the output variable of a smooth kernel gives a smooth covector in the input variable. If $\tau_E R = \tau_E$, that covector would equal $\kappa z^{-2} dz/(2\pi i)$ away from o. This coefficient has no smooth extension across o. □

Thus a finite space containing a nonconstant scalar mode cannot retain the pointwise product by restriction. A projection preserving the trace must also retain its distributional input covector. Neither restriction prevents a cyclic contraction.

The acyclic part of the complex is

$$V = \ker p_0 \subset E^0, \qquad W = \ker p_1 \subset E^1, \qquad d_E : V \xrightarrow{\sim} W,$$

with inverse $h|_W$. On V define

$$\omega_0(v, w) = \tau_E(v d_E w). \tag{5.1}$$

Lemma 5.2. *The form ω_0 is continuous, skew and weakly nondegenerate. Let $\mathcal{R}$ be the restrictions to the round sphere of complex polynomials in its three Cartesian coordinates. Then*

$$E^0_{\mathrm{alg}} = \mathcal{R} \cap \ker j, \qquad E^1_{\mathrm{alg}} = d_E \mathcal{R}$$

form a dense subalgebra and subcomplex of E, preserved by h and the residual projections. The restriction of ω_0 to $V_{\mathrm{alg}} = V \cap \mathcal{R}$ is nondegenerate.

Proof. For $v, w \in E^0$, polar Stokes applied to vw gives $\tau_E(v d_E w) = -\tau_E((d_E v)w)$. If $v \in V$ annihilates V under this form, it pairs to zero with $W = d_E V$. The equality $p_0(v) = 0$ also gives $\tau_E(v\eta_0) = 0$. Hence $\tau_E(v\eta) = 0$ for every smooth one-form η. Tests supported away from o imply $v = 0$ there, and smoothness gives $v = 0$ on C.

The space $\mathcal{R}$ consists of finite sums of spherical harmonics. Smooth functions and forms have convergent expansions in rotation types in every smooth seminorm. The Dolbeault differential and the round Dolbeault Laplacian preserve these types. Since $H^{0,1}(C) = 0$, the Dolbeault inverse sends a finite sum of form types to a finite sum of function types. Subtracting its value at o gives G and preserves this property.

Products of finite sums of types are again finite sums, also for a function times a form. This proves closure under products and G, hence under h and p.

For a sequence $f_n \in \mathcal{R}$ converging to a Neumann function, replace f_n by $f_n - j(f_n)f_0$. Subtracting its p_0 projection also imposes the condition defining V. These continuous corrections prove the required density. A vector of V_{alg} that annihilates V_{alg} annihilates its closure V, so the first paragraph proves nondegeneracy on V_{alg}. □

Choose a basis of the countable-dimensional space V_{alg}. Starting with a finite nondegenerate subspace, take the first basis vector outside it and subtract its symplectic projection. The nonzero remainder has a partner by Lemma 5.2. Subtract the partner's projection as well and adjoin the resulting two-dimensional symplectic space. Iteration gives

$$V_1 \subset V_2 \subset \cdots, \qquad \dim V_K = 2K, \qquad \bigcup_K V_K = V_{\mathrm{alg}}.$$

For a basis $(v_a)_{a=1}^{2K}$ of V_K, its symplectic projection is

$$\Omega_{ab} = \omega_0(v_a, v_b), \qquad \Pi_K v = \sum_{a,b} v_a (\Omega^{-1})_{ab} \omega_0(v_b, v). \tag{5.2}$$

Only a finite matrix is inverted.

Definition 5.3. The finite Neumann complex and its projection are

$$\begin{aligned} E_K^0 &= \mathbb{C}1 \oplus V_K, & E_K^1 &= \mathbb{C}\eta_0 \oplus d_E V_K, \\ \mathsf{p}_K^0 f &= p_0(f)1 + \Pi_K(f - p_0(f)1), & \mathsf{p}_K^1 \eta &= p_1(\eta)\eta_0 + d_E \Pi_K h \eta. \end{aligned} \tag{5.3}$$

Write $i_K : E_K \hookrightarrow E$ for inclusion and set

$$H_K \eta = (1 - \Pi_K) h \eta, \qquad H_K(E^0) = 0.$$

Theorem 5.4. *The pairing $\tau_E(xy)$ is nondegenerate on E_K. The maps above give a continuous cyclic strong deformation retract:*

$$\mathsf{p}_K i_K = 1, \qquad d_E H_K + H_K d_E = 1 - i_K \mathsf{p}_K, \qquad H_K^2 = H_K i_K = \mathsf{p}_K H_K = 0.$$

Projection is adjoint to inclusion, and both preserve the unit and trace. For $K \le L$, the induced contraction from E_L to E_K has homotopy

$$H_{LK} = (\Pi_L - \Pi_K) h|_{E_L}, \qquad H_K = H_L + i_L H_{LK} \mathsf{p}_L. \tag{5.4}$$

Proof. The decomposition

$$E = (\mathbb{C}1 \oplus \mathbb{C}\eta_0) \oplus (V \xrightarrow{d_E} W)$$

is orthogonal for the cyclic pairing. The first summand has $\langle 1, \eta_0 \rangle = \kappa$. Identifying W with V by h identifies the pairing between V_K and $d_E V_K$ with Ω. This proves nondegeneracy on E_K.

The symplectic projection Π_K is self-adjoint for ω_0. Its transport to W is $d_E \Pi_K h$, and the inverse differential on the orthogonal complement is $(1 - \Pi_K)h$. This proves the homotopy identity and its side conditions. The same orthogonality and polar Stokes prove cyclicity of H_K. The unit and its dual η_0 are retained, giving preservation of the trace. All the coefficient functionals in the projection are continuous.

Nested symplectic projections satisfy $\Pi_K \Pi_L = \Pi_L \Pi_K = \Pi_K$. Decompose the acyclic summand into V_K, its complement in V_L, and the complement of V_L. The inverse differential on these summands gives (5.4). □

5.2 A six-dimensional example

The first finite complex can be made explicit. Put $r = |z|^2$ and

$$v = \frac{r}{1+r} - \frac{1}{2}, \qquad w = \frac{zr}{(1+r)^2} + \frac{\bar{z}}{1+r}, \qquad V_1 = \operatorname{span}\{v, w\}. \tag{5.5}$$

Both functions are smooth in the two sphere charts, satisfy $j = 0$, and have zero p_0 projection. Their pairings are

$$\Omega = \kappa \begin{pmatrix} 0 & 1/6 \\ -1/6 & 0 \end{pmatrix}, \qquad S = \big(\tau_E(\eta_0 v_a v_b)\big) = \kappa \begin{pmatrix} 1/12 & 0 \\ 0 & 1/6 \end{pmatrix}. \tag{5.6}$$

These values follow from one radial integral. For integers $0 \le a \le k$ and $0 \le b \le k-2$, the indicated form is smooth at infinity, and angular integration gives

$$\kappa^{-1} \tau_E \left(\frac{z^a \bar{z}^b}{(1+|z|^2)^k} d\bar{z} \right) = \begin{cases} -\dfrac{b!(k-b-2)!}{(k-1)!}, & a - b = 2, \\ 0, & a - b \neq 2. \end{cases} \tag{5.7}$$

In the nonzero case the remaining integral is $-\int_0^\infty r^b (1+r)^{-k}\, dr$.

Write π_1 for the projection to the complex E_1 in both degrees, and put $m_2(x, y) = \pi_1(xy)$. Substituting (5.6) into (5.3) gives

$$m_2(v, v) = \frac{1}{12}, \qquad m_2(w, w) = \frac{1}{6}, \qquad m_2(v, w) = 0. \tag{5.8}$$

Consequently

$$m_2(m_2(v, v), w) - m_2(v, m_2(v, w)) = \frac{w}{12}. \tag{5.9}$$

For the projected matrix commutator, the cyclic Jacobi sum $[[x, y], z] + [[y, z], x] + [[z, x], y]$ on $x = vE_{12}$, $y = vE_{21}$, $z = wE_{12}$ is $wE_{12}/6$. Thus keeping the projected binary operation alone loses the gauge identity already on these smooth modes.

5.3 Higher operations from the eliminated modes

Let $\sigma : E \to E[1]$ have degree -1. On the shifted complex use

$$b_1 = -\sigma d_E \sigma^{-1}, \qquad b_2(\sigma x, \sigma y) = (-1)^{|x|} \sigma(xy), \qquad \mathbf{H}_K = -\sigma H_K \sigma^{-1}.$$

Shift i_K and p_K as well. Define maps on shifted tensor powers by the recursion

$$\begin{aligned} I_1 &= i_K, & \beta_1 &= b_1|_{E_K[1]}, \\ \lambda_n &= \sum_{r=1}^{n-1} b_2(I_r \otimes I_{n-r}), & I_n &= -\mathbf{H}_K \lambda_n, & \beta_n &= \mathsf{p}_K \lambda_n \quad (n \ge 2). \end{aligned} \tag{5.10}$$

Tensor products of maps have their Koszul signs. Each I_n has degree zero on the shifted spaces, and each β_n has degree one. At a fixed arity the recursion is a finite sum over planar binary trees.

Theorem 5.5. *The maps β_n extend to a square-zero coderivation on the conilpotent tensor coalgebra $T^c(E_K[1])$. They define a strictly unital cyclic A_∞ algebra, and the maps I_n give an A_∞ quasi-isomorphism to E. Tensoring with matrices and antisymmetrizing gives cyclic L_∞ operations with the full matrix trace.*

For the nested contractions of (5.4), successive transfer through E_L gives the same operations on E_K as direct transfer. The tree comparisons are compatible with every further nested elimination.

Proof. The differential, Leibniz rule and associativity are the coderivation identities

$$b_1^2 = 0, \qquad b_1 b_2 + b_2 b_1 = 0, \qquad b_2^2 = 0.$$

The shifted homotopy satisfies $b_1 \mathbf{H}_K + \mathbf{H}_K b_1 = 1 - i_K \mathsf{p}_K$.

Move b_1 through a tree in (5.10). Its terms at vertices cancel by the Leibniz rule. At an internal edge, replace the commutator with $\mathbf{H}_K$ by $1 - i_K \mathsf{p}_K$. The identity contracts the edge. The two resolutions of the resulting ternary vertex cancel by associativity. The $i_K \mathsf{p}_K$ term cuts the tree into two transferred trees.

Summing all cuts gives $\beta^2 = 0$ at the root and $(b_1 + b_2)I = I\beta$ at the inclusion. The linear component is a quasi-isomorphism by Theorem 5.4.

Pair the root with one additional input. Adjointness of p_K and cyclicity of H_K allow the root to move across every edge, with the graded sign of the pairing. The resulting unrooted tensor is cyclic.

This is the cyclic tree transfer associated with an abstract Hodge decomposition.[1]

Suppose a unit enters a tree of arity at least three. Its neighboring vertex sends either a retained input into $H_K i_K = 0$ or a homotopy output into $H_K^2 = 0$. If that vertex is the root, the same term vanishes by $\mathsf{p}_K H_K = 0$. The binary operation preserves the unit. Antisymmetrization of the associative identities gives the Lie identities and preserves the cyclic tensor.

For successive transfer, expand each internal homotopy using (5.4). Mark the edge by its chosen summand. The maximal subtrees whose edges use H_L are exactly the transferred vertices on E_L. The remaining edges use H_{LK}. This is a bijection of signed planar trees with those of successive transfer. With three or more cutoffs, grouping the connected subtrees in either order gives the same marked tree. □

The missing operations in the six-dimensional example can now be calculated. In the next formulas, $\beta_n(x_1, \dots, x_n)$ denotes $\sigma^{-1}\beta_n(\sigma x_1, \dots, \sigma x_n)$, with this convention fixed rather than an additional desuspension sign.

For inputs of degrees a, b, c, the cubic recursion reads

$$\beta_3(x, y, z) = (-1)^{a+b}\pi_1\big(xH_1(yz)\big) - (-1)^b \pi_1\big(H_1(xy)z\big).$$

It gives

$$\begin{aligned} \beta_3(v, v, d_E v) &= v/30, & \beta_3(v, v, d_E w) &= w/30, \\ \beta_3(d_E v, v, w) &= w/60, & \beta_3(v, d_E v, w) &= -w/30. \end{aligned} \tag{5.11}$$

For example, direct Green integration and (5.7) give

$$H_1(v d_E v) = \frac{r^2 - 4r + 1}{12(1 + r)^2}, \qquad \pi_1\big(vH_1(v d_E v)\big) = v/30.$$

On (v, v, w) the complete cubic identity reduces to

$$\frac{1}{12} - \frac{1}{60} - \frac{1}{30} - \frac{1}{30} = 0.$$

[1] J. Chuang and A. Lazarev, *Abstract Hodge decomposition and minimal models for cyclic algebras*, arXiv:0810.2393. See Definition 2.1 and Theorem 4.2. Here the retained differential need not vanish.

The unary differential is essential here: v and w are not closed cohomology classes.

The fourth operation also survives:

$$\begin{aligned}\beta_4(d_E v, v, w, d_E w) &= \frac{1}{720} - \frac{v}{280},\\ \beta_4(v, d_E v, w, d_E w) &= \frac{1}{720} + \frac{11v}{630}.\end{aligned} \tag{5.12}$$

There are five planar binary trees. Grouping them by the three root splittings, their projected contributions for the first input are 0, $1/720 - v/280$, 0. For the second input they are $1/360 + v/72$, $-1/720 + v/280$, 0. Each is evaluated by the same radial moment, and their sums give (5.12).

5.4 Algebraic recovery and analytic convergence

The choice of symplectic exhaustion gives exact recovery on finite algebraic expressions. It need not control the norms of its projections.

Proposition 5.6. *Fix finitely many inputs in E_{alg} and finitely many expressions formed from their products and h. Their projections to E_K equal the original expressions for every sufficiently large K.*

For fixed K and integer $s \geq 2$, give functions the H^{s+1} norm and one-forms the H^s norm. The tree series in (5.10) *converges on a sufficiently small field in this graded Sobolev space. Its radius can depend on K.*

Proof. Lemma 5.2 keeps every intermediate coefficient in E_{alg}. A sufficiently large V_K contains the acyclic components of this finite collection. Projection fixes them, and the complementary homotopy vanishes on them. This proves eventual recovery, including vanishing of higher corrections on the fixed collection when every required intermediate coefficient has entered the finite complex.

The Green operator gains one derivative. The identity $\tau_E(\eta) = \kappa\, j(G\eta)$ makes the trace continuous in these norms, since first-jet evaluation is continuous on H^{s+1} for $s > 1$ in real dimension two. Multiplication is continuous, and the finite-rank projections and H_K are bounded.

Let C bound multiplication and let C_{n-1} be the number of planar binary trees with n inputs. Then

$$\|I_n\| \leq C_{n-1} C^{n-1} \|H_K\|^{n-1} \|i_K\|^n, \qquad C_{n-1} \leq 4^{n-1}.$$

For β_n, replace the last homotopy by p_K. Both sums are bounded by a geometric series for a sufficiently small input. □

No bound uniform in K has been used. The proposition therefore separates eventual algebraic recovery from analytic removal of the cutoff. A family with quantitative projection estimates is needed for the latter.

5.5 Backgrounds and artificial singularities

Tensor V and W with $\mathfrak{g} = \mathfrak{gl}_N(\mathbb{C})$. Extend p_0, p_1, h entrywise, writing P_0, P_1 for the matrix projections. For a residual matrix J, set

$$d_J = d_E + \eta_0 \operatorname{ad}_J, \qquad M_J = h d_J|_V = 1 + h\eta_0 \operatorname{ad}_J|_V.$$

The full contraction of Theorem 4.4 gives $M_J^{-1} = h_J d_E|_V$ at every finite J. A fixed compression of this operator need not be invertible.

Example 5.7. In a root direction with $\mathrm{ad}_J = \lambda$, compression to the modes (5.5) gives

$$M_{J,1} = 1 + \lambda\Omega^{-1}S = \begin{pmatrix} 1 & -\lambda \\ \lambda/2 & 1 \end{pmatrix}, \qquad \det M_{J,1} = 1 + \lambda^2/2. \tag{5.13}$$

The compression is singular at $\lambda = i\sqrt{2}$. For $J = j\,\mathrm{diag}(1, -1)$, the two opposite root spaces give determinant $(1 + 2j^2)^2$, and the two Cartan directions contribute determinant one. The matrix trace pairs opposite roots. One scalar root block is not itself a skew kinetic form.

The mode w in this example mixes rotation weights 1 and -1. The singularity depends on the chosen subspace. To avoid it for an arbitrary finite subspace, first change the full kinetic form to a constant one and then transport that subspace.

5.6 An entire family of cyclic frames

Fix an integer $s \geq 2$ and a smooth metric on C. On the scalar Sobolev space

$$V_s = \{f \in H^{s+1}(C) : j(f) = 0,\ p_0(f) = 0\}$$

define $Xf = h(\eta_0 f)$.

Lemma 5.8. *The operator $X : V_s \to V_s$ is compact and has spectrum $\{0\}$. For every finite matrix J and $\alpha \in \{-1, -\frac{1}{2}, \frac{1}{2}\}$, the series*

$$(1 + X \otimes \mathrm{ad}_J)^\alpha = \sum_{n \geq 0} \binom{\alpha}{n} X^n \otimes (\mathrm{ad}_J)^n \tag{5.14}$$

converges in operator norm, locally uniformly in the entries of J. It is an entire operator-valued function of those entries.

Proof. Multiplication by η_0 is smooth and h gains one derivative. Its finite-rank corrections obey the trace estimate in Proposition 5.6. Hence X maps V_s continuously into H^{s+2}, and Rellich compactness proves that it is compact on V_s.

Suppose $Xv = \mu v$ with $\mu \neq 0$. The gain of regularity in this equation makes v smooth. Put $\lambda = -\mu^{-1}$. Since $hd_E v = v$, one has $h(d_E + \lambda\eta_0)v = 0$. Moreover

$$p_1((d_E + \lambda\eta_0)v) = \lambda p_0(v) = 0.$$

The form lies in W, where h is injective, so $(d_E + \lambda\eta_0)v = 0$. Multiplication by $e^{\lambda f_0}$ gives a holomorphic function on $\mathbb{P}^1$, hence $v = ce^{-\lambda f_0}$. The first jet is $j(v) = -\lambda c$. The Neumann condition forces $c = 0$, excluding every nonzero eigenvalue. Compactness and infinite dimension imply $\mathrm{spec}(X) = \{0\}$.

The spectral-radius formula gives $\lim_n \|X^n\|^{1/n} = 0$. On each compact set of matrices, $\|\mathrm{ad}_J\|$ is bounded. The binomial coefficients have at most polynomial growth, so the root test gives normal convergence of (5.14). Each summand is polynomial in J, proving the entire dependence. □

On the matrix-valued acyclic space put

$$\omega_J(v, w) = \tau_E\,\mathrm{tr}(vd_J w), \qquad \omega_0(v, w) = \tau_E\,\mathrm{tr}(vd_E w).$$

Both forms are skew by cyclicity, and

$$\omega_J(v, w) = \omega_0(v, M_J w), \qquad \omega_0(M_J v, w) = \omega_0(v, M_J w). \tag{5.15}$$

Define the full frame

$$\mathcal{T}_J = M_J^{-1/2} = 1 - \frac{1}{2}A_J + \frac{3}{8}A_J^2 - \frac{5}{16}A_J^3 + \cdots, \qquad A_J = X \otimes \mathrm{ad}_J. \tag{5.16}$$

The scalar projection Π_K is now tensored with the identity on $\mathfrak{g}$. Set

$$\begin{aligned} V_K(J) &= \mathcal{T}_J(V_K \otimes \mathfrak{g}), & \Pi_K^J &= \mathcal{T}_J \Pi_K \mathcal{T}_J^{-1}, \\ E_K^0(J) &= \mathfrak{g} \oplus V_K(J), & E_K^1(J) &= \eta_0 \mathfrak{g} \oplus d_J V_K(J). \end{aligned} \tag{5.17}$$

The residual summand is the full complex $[\mathfrak{g} \xrightarrow{\mathrm{ad}_J} \mathfrak{g}\eta_0]$.

Theorem 5.9. *For every finite J,*

$$\omega_J(\mathcal{T}_J v, \mathcal{T}_J w) = \omega_0(v, w). \tag{5.18}$$

The complex (5.17) *has a nondegenerate cyclic pairing and the cyclic strong contraction*

$$\begin{aligned} \mathsf{p}_{K,J}^0 f &= P_0 f + \Pi_K^J(f - P_0 f), \\ \mathsf{p}_{K,J}^1 \eta &= \eta_0 P_1 \eta + d_J \Pi_K^J h_J \eta, & H_{K,J}\eta &= (1 - \Pi_K^J) h_J \eta. \end{aligned} \tag{5.19}$$

The homotopy is zero on degree zero. These maps depend analytically on all entries of J and preserve smooth vectors. Their formal coefficients involve finitely many Green operations and commutators, with no inverse eigenvalue difference.

Proof. The binomial series and (5.15) give $\mathcal{T}_J^\dagger M_J \mathcal{T}_J = 1$, first near zero by series multiplication and then everywhere by analyticity. Here $\dagger$ is the adjoint for ω_0. This proves (5.18). Thus Π_K^J is the ω_J-orthogonal projection onto $V_K(J)$.

The residual summand remains orthogonal to the acyclic summand. On W, conjugate the degree-zero projection by the isomorphism $d_J : V \to W$, whose inverse is h_J. This gives the degree-one projection in (5.19). The inverse differential on the remaining orthogonal summand is $H_{K,J}$. The decomposition proves the contraction identity, adjointness, cyclicity and all side conditions. The residual differential is retained because $P_1 d_J f = [J, P_0 f]$.

The series converge in every V_s, and their limits agree under Sobolev inclusion. Hence they preserve smooth vectors. Near zero, $h_J = M_J^{-1} h$, so every formal coefficient uses only finitely many copies of $h\eta_0 \,\mathrm{ad}_J$. This also describes the coefficients of the transported projection and homotopy. □

Taking a square root of the compressed matrix (5.13) would retain its singularities. The full frame supplies a nondegenerate moving subspace at every finite background, including nonsemisimple matrices.

5.7 The parameter connection and its curvature

Let d_{par} differentiate the entries of J, and put $\mathcal{R}_J = \mathcal{T}_J^{-1} = M_J^{1/2}$. The frame connection is

$$\nabla = d_{\mathrm{par}} + \mathcal{A}, \qquad \mathcal{A} = \mathcal{R}_J^{-1} d_{\mathrm{par}} \mathcal{R}_J. \tag{5.20}$$

This differential acts on parameters, separately from d_J on curve coefficients.

Proposition 5.10. *The connection* (5.20) *is flat and compatible with the varying form:*

$$d_{\mathrm{par}}\mathcal{A} + \mathcal{A} \wedge \mathcal{A} = 0, \qquad d_{\mathrm{par}}\omega_J = \mathcal{A}^{\mathsf{T}} \omega_J + \omega_J \mathcal{A}.$$

In contrast, for $\theta = M_J^{-1} d_{\mathrm{par}} M_J$,

$$F_{\theta/2} = -\frac{1}{4}\theta \wedge \theta, \qquad \mathcal{A} = \frac{1}{2}\theta + \frac{1}{8}[A_J, d_{\mathrm{par}} A_J] + O(A_J^2 d_{\mathrm{par}} A_J). \tag{5.21}$$

The last remainder has at least two undifferentiated factors A_J.

Proof. The Maurer–Cartan identity for $\mathcal{R}_J^{-1} d_{\mathrm{par}} \mathcal{R}_J$ gives flatness. Differentiating $\omega_J = \mathcal{R}_J^{\mathsf{T}} \omega_0 \mathcal{R}_J$ proves compatibility. The same identity for θ gives the curvature of $\theta/2$. Expanding $\mathcal{R}_J = 1 + A_J/2 - A_J^2/8 + \cdots$ and its inverse gives, through the displayed order,

$$\mathcal{A} = \frac{1}{2} d_{\mathrm{par}} A_J - \frac{3}{8} A_J d_{\mathrm{par}} A_J - \frac{1}{8} (d_{\mathrm{par}} A_J) A_J + \cdots .$$

Subtract $\theta/2 = (d_{\mathrm{par}} A_J - A_J d_{\mathrm{par}} A_J)/2 + \cdots$ to obtain the commutator term. □

The curvature is nonzero for the curve operator itself. On the mode v of (5.5), Green integration gives

$$X^2 v = \frac{z^2(1 + 3z\bar{z})}{12(1 + z\bar{z})^3}, \qquad F_{\theta/2}|_{J=0}(U, V) = -\frac{1}{4} X^2 \otimes \mathrm{ad}_{[U,V]} . \tag{5.22}$$

For $U = E_{12}$ and $V = E_{21}$, the commutator is $\mathrm{diag}(1, -1)$ and its adjoint action is nonzero. Thus the half-logarithmic formula already fails to give a flat connection in the native matrix family.

5.8 Source antifields and change of density

A parameter-dependent change of finite field coordinates also changes the antifields of those parameters. Let q be graded coordinates, let J^i be even parameters, and give their odd cotangent coordinates the degrees $|q^+| = -1 - |q|$ and $|J_i^+| = -1$.

Let $T(J)$ be an invertible degree-preserving formal matrix, with $T(0) = 1$, and set $q = T(J)Q$. Its cotangent lift is determined by the one-form identity

$$q^+ \delta q + J_i^+ \delta J^i = Q^+ \delta Q + \widetilde{J}_i^+ \delta J^i. \tag{5.23}$$

Dual coordinates are placed on the left. Expanding the left side gives

$$Q^+ = q^+ T, \qquad \widetilde{J}_i^+ = J_i^+ + q^+ (\partial_i T) Q. \tag{5.24}$$

For the moving cutoff, T is a finite coordinate comparison between chosen transported bases. No infinite-dimensional determinant of $\mathcal{T}_J$ is required.

Proposition 5.11. *Write* $b = \mathrm{Ber}\, T$, *with formal branch* $b(0) = 1$. *The odd cotangent lift has Berezinian* b^2 *and sends a coordinate semidensity to* b *times that semidensity. If a master weight is* $e^{I/\hbar}$ *times the old coordinate semidensity, its new action is*

$$I' = I \circ F + \hbar \log \mathrm{Ber}\, T, \tag{5.25}$$

where F *is the full coordinate change including* (5.24).

Proof. The base Jacobian in (Q, J) is triangular and has Berezinian b. The inverse transpose on the dual variables would have reciprocal Berezinian, but parity reversal for the odd cotangent space inverts it again. The full Berezinian is therefore b^2. Its normalized square root is b, and absorbing that factor into the exponential gives (5.25). □

For one even coordinate, $q = e^J Q$ gives $q^+ = e^{-J} Q^+$ and $\widetilde{J}^+ = J^+ + Q^+ Q$. The coordinate Berezinian is e^{2J}, and the action acquires $\hbar J$. If equal even and odd copies carry the same matrix T, their base Berezinians cancel. Their source-antifield shifts still remain. A varying cutoff therefore requires its cotangent lift even when its density contribution is one.

Chapter 6

Angular cutoffs and smooth limits

The finite reductions of Chapter 5 preserve the cyclic pairing. To remove the cutoff, their projections must also approximate smooth fields with controlled norms. On the round sphere the polar pairing determines such a family. Its rotation weights pair to one, so a symmetric harmonic cutoff omits the partners of its highest modes. Retaining one additional band restores the pairing and gives uniform estimates for each fixed higher operation.

6.1 Rotation weights and the Green polarization

Keep the round-sphere representative

$$f_0 = \frac{z}{1+|z|^2}, \qquad \eta_0 = d_E f_0 = -\frac{z^2}{(1+|z|^2)^2}\, d\bar{z}$$

and the circular polar trace of Chapter 2. The scalar cyclic projection is the spherical mean:

$$p_0(f) = \int_{S^2} f \, d\mu, \qquad d\mu = \frac{dx\, dy}{\pi(1+|z|^2)^2}, \qquad \mu(S^2) = 1. \tag{6.1}$$

Indeed, substitute η_0 into $p_0(f) = \kappa^{-1}\tau_E(f\eta_0)$ and use $dz \wedge d\bar{z} = -2i\, dx \wedge dy$.

As before, $V = \ker p_0 \subset E^0$, $W = \ker p_1 \subset E^1$, and $d_E : V \to W$ has inverse h. Write

$$\omega(v, w) = \tau_E(v d_E w).$$

A function has rotation weight $m \in \mathbb{Z}$ if $f(e^{i\theta}z) = e^{im\theta} f(z)$. Let $\mathcal{P}_m$ be the corresponding Fourier projector and put $V_m = \mathcal{P}_m V$. The polar integral gives the selection rule

$$\omega(V_m, V_n) = 0 \qquad \text{unless } m + n = 1. \tag{6.2}$$

To see the shift, differentiation by $\bar{z}$ raises the coefficient weight by one. The coefficient of $v d_E w$ therefore has weight $m + n + 1$, whereas the factor z^{-2} requires weight two.

Let V_- and V_+ be the closed subspaces with nonpositive and positive weights, respectively. The mean condition affects weight zero, and the first-jet condition affects weight one. Hence

$$V = V_- \oplus V_+.$$

Both summands are isotropic. The Green operator identifies their pairing with the ordinary Hermitian pairing on one summand.

Proposition 6.1. *The antilinear maps*

$$Cv = h(\eta_0 \bar{v}) \quad (v \in V_-), \qquad \mathcal{R}f = \frac{\overline{d_E f}}{\eta_0} \quad (f \in V_+) \tag{6.3}$$

are inverse continuous isomorphisms. For $u, v \in V_-$,

$$\omega(u, Cv) = \kappa \int_{S^2} u\bar{v}\, d\mu. \tag{6.4}$$

For every integer $s \geq 2$, the round Sobolev norms satisfy

$$\|\mathcal{R}f\|_{H^s} \leq C_s \|f\|_{H^{s+6}}, \qquad \|Cv\|_{H^{s+1}} \leq C_s \|v\|_{H^s}.$$

The loss of six derivatives is not asserted to be optimal.

Proof. The mean of $\bar{v}$ is zero. In the formula for $h(\eta_0 \bar{v})$, the other scalar correction is the mean of $f_0 \bar{v}$. Its weights are strictly positive, so it also vanishes. Thus

$$Cv = G(\eta_0 \bar{v}), \qquad d_E Cv = \eta_0 \bar{v}.$$

This function has positive weights. Its first jet is zero by polar Stokes and the zero mean of v. Equation (6.1) now proves (6.4).

The quotient in (6.3) requires care only at o. If $f \in V_+$, the coefficient of $d_E f$ has weights at least two. Every monomial $z^a \bar{z}^b$ in its Taylor polynomial satisfies $a - b \geq 2$, and hence is divisible by z^2.

Fix a nonnegative integer r. Expand that coefficient through degree $r + 2$. The remainder and each derivative of order $j \leq r$ are bounded by a constant times $|z|^{r+3-j}$. After division by z^2, the derivatives of order at most r tend to zero at o. The divided polynomial supplies their limits. This proves smooth division and bounds the quotient in C^r by the original coefficient in C^{r+3}.

In the chart $w = 1/z$, one has $\eta_0 = (1 + |w|^2)^{-2} d\bar{w}$, so division there is ordinary smooth division. Apply the local estimate with $r = s$, differentiate f once, and use the Sobolev embedding $H^{s+6} \hookrightarrow C^{s+4}$ in real dimension two. This gives the estimate for $\mathcal{R}$. The estimate for C follows from the elliptic gain of one derivative for G.

The quotient defining $\mathcal{R}f$ has nonpositive weights after conjugation. Its mean is zero because $\tau_E(d_E f) = \kappa J(f) = 0$. Thus $\mathcal{R}f \in V_-$. The differential formula proves $\mathcal{R}C = 1$. Conversely, $G d_E f = f - f(o) = f$ on positive weights, proving $C\mathcal{R} = 1$. □

The normalized form $\kappa^{-1}\omega(u, Cv)$ is positive Hermitian on V_-. This statement uses the two isotropic sectors. The original bilinear form ω remains skew.

6.2 The additional harmonic band

Use the sphere of radius one, whose scalar Laplacian has eigenvalues $\ell(\ell + 1)$. For an integer $L \geq 0$, let $\mathscr{H}_{\leq L}$ be its harmonics of degrees at most L, and set

$$W_L = V_- \cap \mathscr{H}_{\leq L}, \qquad d_L = \dim W_L = \frac{(L+1)(L+2)}{2} - 1.$$

There are $\ell + 1$ nonpositive weights at degree ℓ, and the constant has been removed. In particular, $W_0 = 0$.

Proposition 6.2. *For $L \geq 1$,*

$$V \cap \mathscr{H}_{\leq L} = W_L \oplus CW_{L-1}. \tag{6.5}$$

The restricted form has radical $W_L \cap W_{L-1}^{\perp_{L^2}}$, of dimension $L + 1$, and rank $L(L + 1) - 2$. The subspace

$$\widehat{V}_L = W_L \oplus CW_L, \qquad \dim \widehat{V}_L = L(L + 3), \tag{6.6}$$

is nondegenerate, and the spaces $\widehat{V}_L$ are nested.

Proof. The form $\eta_0 = d_E f_0$ belongs to the degree-one angular representation. Multiplication by it takes degrees at most $L-1$ to degrees at most L. The round Dolbeault inverse preserves each nonconstant angular representation. Its normalization adds no constant on positive weights. Therefore

$$CW_{L-1} \subset V_+ \cap \mathscr{H}_{\le L}.$$

The space on the right has dimension $L(L+1)/2 - 1$: the first-jet condition removes one dimension from the positive-weight harmonics. This equals d_{L-1}, so the inclusion is an equality.

Equation (6.4) identifies the cross pairing with the Hermitian pairing between W_L and W_{L-1}. Its radical is the orthogonal complement of W_{L-1} in W_L. The dimension difference is $d_L - d_{L-1} = L+1$, and the rank is $2d_{L-1} = L(L+1) - 2$. Retaining CW_L instead makes the cross pairing nondegenerate. Nesting follows from $W_L \subset W_{L+1}$. □

At the first cutoff, take

$$v = \frac{|z|^2}{1+|z|^2} - \frac{1}{2}, \qquad u = \frac{\bar z}{1+|z|^2}. \tag{6.7}$$

Their partners are

$$Cv = \frac{z^2\bar z}{2(1+|z|^2)^2}, \qquad Cu = \frac{z^2}{2(1+|z|^2)^2}.$$

Differentiation proves the defining equations, and each partner vanishes at o. All four functions are smooth on the sphere and satisfy the Neumann condition. In the basis (v, u, Cv, Cu) of $\widehat{V}_1$, the pairing is

$$\kappa \begin{pmatrix} 0 & 0 & 1/12 & 0 \\ 0 & 0 & 0 & 1/6 \\ -1/12 & 0 & 0 & 0 \\ 0 & -1/6 & 0 & 0 \end{pmatrix}. \tag{6.8}$$

The two nonzero entries are the spherical norms of v and u. The mode w of (5.5) is $2Cv + u$. Thus the earlier six-dimensional complex retains a mixed-weight subspace of this first ten-dimensional coefficient complex.

6.3 Uniform estimates for the projections

Let S_L^- be the orthogonal spectral projector onto W_L. For $f = f_- + f_+ \in V$, define

$$\widehat{\Pi}_L f = S_L^- f_- + C S_L^- \mathcal{R} f_+. \tag{6.9}$$

The two antilinear maps make the second term complex linear. The cross pairing (6.4) shows that this is the ω-orthogonal projection onto $\widehat{V}_L$. Consequently

$$\widehat{\Pi}_L \widehat{\Pi}_M = \widehat{\Pi}_{\min(L,M)}, \qquad \omega(\widehat{\Pi}_L f, g) = \omega(f, \widehat{\Pi}_L g).$$

Theorem 6.3. *For integers $s \ge 2$ and $r \ge 0$, there are constants independent of L such that*

$$\begin{aligned} \|\widehat{\Pi}_L f\|_{H^s} &\le C_s \|f\|_{H^{s+6}}, \\ \|(1-\widehat{\Pi}_L) f\|_{H^s} &\le C_{s,r} (1+L)^{-r} \|f\|_{H^{s+r+6}}. \end{aligned} \tag{6.10}$$

The projections converge to the identity uniformly on bounded smooth sets in every smooth seminorm.

Proof. The projections to positive and nonpositive rotation weights commute with the round Laplacian and are bounded on every Sobolev space. Spectral expansion on V_- gives

$$\|(1 - S_L^-)v\|_{H^s} \le C_r(1 + L)^{-r}\|v\|_{H^{s+r}}.$$

Indeed, every omitted harmonic has degree greater than L. Apply this inequality to f_- and $\mathcal{R}f_+$, and use both estimates in Proposition 6.1. The same argument without the tail factor bounds the projection. Every bounded smooth set has the higher Sobolev bounds on the right. □

The norms

$$\|f\|_{\mathfrak{B}_s}^2 = \|f_-\|_{H^s}^2 + \|\mathcal{R}f_+\|_{H^s}^2 \tag{6.11}$$

make $\widehat{\Pi}_L$ a contraction. The two finite-loss estimates show that their projective-limit topology is the original smooth topology. The inverse polarization is not bounded in the unchanged L^2 norms.

For an explicit test, let $L \ge 1$ and put

$$v_L = \frac{\bar{z}^L}{(1 + |z|^2)^L}, \qquad Cv_L = \frac{z^{L+1}}{(L + 1)(1 + |z|^2)^{L+1}}.$$

The second equality follows by differentiation and normalization at o. With $r = |z|^2$, radial integration gives

$$\|v_L\|_{L^2}^2 = \int_0^\infty \frac{r^L}{(1 + r)^{2L+2}}\, dr = \frac{(L!)^2}{(2L + 1)!}, \qquad \|Cv_L\|_{L^2}^2 = \frac{(L!)^2}{(2L + 3)!}.$$

Hence

$$\frac{\|Cv_L\|_{L^2}^2}{\|v_L\|_{L^2}^2} = \frac{1}{(2L + 2)(2L + 3)}. \tag{6.12}$$

Since $\mathcal{R}Cv_L = v_L$, these ratios exclude a uniform L^2 bound for $\mathcal{R}$.

6.4 Removal of the cutoff at each fixed arity

Use the finite complexes

$$\widehat{E}_L^0 = \mathbb{C}1 \oplus \widehat{V}_L, \qquad \widehat{E}_L^1 = \mathbb{C}\eta_0 \oplus d_E\widehat{V}_L.$$

Their inclusion $\widehat{i}_L$, projection $\widehat{p}_L$ and homotopy $\widehat{H}_L$ are

$$\begin{aligned} \widehat{p}_L^0 f &= p_0(f)1 + \widehat{\Pi}_L(f - p_0(f)1), \\ \widehat{p}_L^1\eta &= \eta_0 p_1(\eta) + d_E\widehat{\Pi}_L h\eta, \qquad\qquad \widehat{H}_L\eta = (1 - \widehat{\Pi}_L)h\eta. \end{aligned} \tag{6.13}$$

The homotopy is zero on degree zero. By Theorem 5.4, these maps give a cyclic strong deformation retract preserving the unit and trace. Their nested homotopies satisfy

$$\widehat{H}_L = \widehat{H}_M + \widehat{i}_M\widehat{H}_{ML}\widehat{p}_M, \qquad \widehat{H}_{ML} = (\widehat{\Pi}_M - \widehat{\Pi}_L)h|_{\widehat{E}_M} \quad (L \le M).$$

Give the coefficient space the graded Sobolev norms

$$\mathcal{E}_s = (E^0 \cap H^{s+1}) \oplus (E^1 \cap H^s), \qquad s \ge 2.$$

Theorem 6.3 gives uniform loss six for $\widehat{p}_L$ and $\widehat{H}_L$. Both $1 - \widehat{i}_L\widehat{p}_L$ and $\widehat{H}_L$ have the rapid tail estimate with this same loss. The additional derivative on functions matches the gain of h and the derivative in d_E.

Let $\beta_{n,L}$ be the shifted operations of (5.10) for this contraction. For a common domain, write $P_L = \widehat{i}_L \widehat{p}_L$ and set

$$B_{n,L} = \widehat{i}_L \beta_{n,L} \widehat{p}_L^{\otimes n} : (E[1])^{\otimes n} \longrightarrow E[1].$$

Suspension is understood on the inclusion and projection. Let $B_1 = b_1$, $B_2 = b_2$, and $B_n = 0$ for $n \geq 3$, using the signs of Chapter 5.

Theorem 6.4. *For every fixed $n \geq 1$, $s \geq 2$ and integer $r \geq 0$,*

$$\|(B_{n,L} - B_n)(a_1, \dots, a_n)\|_{\mathcal{E}_s} \leq C_{n,s,r}(1+L)^{-r} \prod_{j=1}^{n} \|a_j\|_{\mathcal{E}_{s+r+12n+12}}. \tag{6.14}$$

In particular, the finite operations converge uniformly on bounded smooth sets at each arity. Their binary limit is the native product, and their higher limits vanish.

Proof. Multiplication is continuous on $\mathcal{E}_s$. At fixed arity there are finitely many planar binary trees. When $n \geq 3$, each tree contributing to $\beta_{n,L}$ contains at least one internal $\widehat{H}_L$ edge. Use its rapid tail estimate on one edge and the uniform finite-loss estimates on every other homotopy and projection. Each root-to-input path meets at most $n-2$ internal homotopies, one input projection and one output projection. The displayed derivative budget exceeds their combined losses.

For $n = 2$, the extended product is $P_L b_2(P_L \otimes P_L)$. Subtract b_2 and telescope over the output and the two input projections. Each difference contains a rapid tail $P_L - 1$. For $n = 1$, commutation with b_1 gives $B_{1,L} - b_1 = (P_L - 1)b_1$, to which the same estimate applies. This proves (6.14).

The cyclic and higher-associativity identities involve finitely many compositions at each arity. The same estimates allow those finite compositions to pass to the limit. □

The loss grows with arity. Thus all arities together define a formal series in the field amplitude. The theorem gives no uniform analytic sum of that series. A fixed finite-order current can be applied to the smooth output. Products of currents on a common support and traces around loops require further extension estimates.

6.5 Polynomial dependence on the matrix background

The additional band also removes the artificial kinetic poles of a fixed compression. Let

$$X_L = \widehat{\Pi}_L h \eta_0|_{\widehat{V}_L}.$$

For $J \in \mathfrak{g} = \mathfrak{gl}_N(\mathbb{C})$, define the compressed kinetic form by

$$K_{J,L}(a,b) = \tau_E \operatorname{tr}\big(a(d_E b + \eta_0[J,b])\big), \qquad a, b \in \widehat{V}_L \otimes \mathfrak{g}.$$

Write $K_{0,L}$ for the form at $J = 0$.

Proposition 6.5. *The operator X_L raises rotation weight by one and satisfies $X_L^{2L+2} = 0$. Put $N_{J,L} = X_L \otimes \mathrm{ad}_J$. For every finite J,*

$$K_{J,L}(a,b) = K_{0,L}(a, (1 + N_{J,L})b),$$
$$(1+N_{J,L})^{-1} = \sum_{q=0}^{2L+1} (-N_{J,L})^q, \qquad \det(1+N_{J,L}) = 1. \tag{6.15}$$

Thus the compressed kinetic form is nondegenerate, with inverse polynomial in the entries of J.

Proof. Multiplication by η_0 raises total rotation weight by one. Here a one-form includes the weight of $d\bar{z}$. The normalized Green operator commutes with rotations fixing o, and the finite corrections defining h occur in the matching weight. The projection $\widehat{\Pi}_L$ preserves weights. The weights of $\widehat{V}_L$ lie between $-L$ and $L+1$, so $2L+2$ successive applications of X_L give zero.

For $b \in V$, one has $d_E h(\eta_0 b) = \eta_0 b$ because $p_0(b) = 0$. Adjointness of $\widehat{\Pi}_L$ for ω gives the kinetic-form identity. The operator $N_{J,L}$ has the same nilpotence bound. Its finite geometric series is the inverse, and all eigenvalues of $1 + N_{J,L}$ equal one. Nondegeneracy of $K_{0,L}$ follows from the cyclic pairing and the full matrix trace. □

For the first four modes, the compressed scalar operator is

$$X_1 u = -\frac{1}{2} v, \qquad X_1 v = Cv, \qquad X_1 Cv = \frac{1}{4} Cu, \qquad X_1 Cu = 0. \tag{6.16}$$

Indeed, its matrix is $\Omega^{-1} S$, where Ω is (6.8) and $S_{ab} = \kappa \int v_a v_b \, d\mu$. Besides $\int v^2 d\mu = 1/12$, the only nonzero entry needed is $\int uCv \, d\mu = 1/24$. These identities give (6.16) directly.

The inverse still depends on J, despite the unit determinant. The nilpotence bound alone gives no estimate for that inverse uniform in L. This finite compression therefore leaves the quantum contraction and cutoff limits as separate questions.

Nor does the rotation grading restore strict associativity. For $m_2(a,b) = \widehat{p}_1^{\,0}(ab)$ on $\widehat{E}_1^{\,0}$, one has

$$m_2(v,v) = \frac{1}{12}, \qquad m_2(v,u) = 0, \qquad m_2(m_2(v,v),u) - m_2(v,m_2(v,u)) = \frac{u}{12}.$$

The retained weight-zero functions are 1, v, and the retained weight-minus-one functions are multiples of u. The moments $\int v^3 d\mu = 0$ and $\int v|u|^2 d\mu = 0$ give the two projected products. The higher operations remain necessary even for a cutoff whose kinetic form is regular at every background.

Chapter 7

Heat tensors and contact terms

Classical transfer applies a homotopy to smooth fields. Quantum contraction instead evaluates two input covectors on a tensor of fields, so it requires control of both factors at once. The paired modes of Chapter 6 provide this control at positive heat time. They give a free BV complex and explicit contractions of jets at the pole. Their generator also identifies the limitation of this construction: it contains an angular projection and is therefore nonlocal on the curve.

The tensor contracts all surface modes and evaluates jets at the pole. Products of coincident kernels and collisions of poles require further choices. A subtraction prescription determines finite contacts, and convergence of a polar trace need not preserve the boundary product. The differential on supported normal jets determines which finite contacts survive as classes. At a matrix background, a separate Hilbert-scale estimate controls the inverse after the angular cutoff is removed.

7.1 The paired heat tensor

Retain $E, d_E, \tau_E, h, V_\pm, C, \mathcal{R}$ and the round probability measure $d\mu$ from the preceding chapter. Choose an orthonormal spherical basis $e_{\ell m}$ of V_-, where $\ell \geq 1$, $0 \leq m \leq \ell$, and $e_{\ell m}$ has rotation weight $-m$. Put

$$\lambda_\ell = \ell(\ell+1), \qquad f_{\ell m} = Ce_{\ell m}.$$

The polarization identity gives

$$\omega(e_{\ell m}, f_{\ell' m'}) = \kappa \delta_{\ell\ell'}\delta_{mm'}, \qquad \omega(e, e') = \omega(f, f') = 0.$$

Cyclic self-adjointness therefore requires equal heat weights on the two members of each pair.

Define R_t to fix 1, η_0 and to multiply $e_{\ell m}, f_{\ell m}, d_E e_{\ell m}, d_E f_{\ell m}$ by $e^{-t\lambda_\ell}$. If Δ_- is the round scalar Laplacian on V_-, its restriction to $V = \ker p_0$ is

$$R_t|_V = e^{-t\Delta_-}\mathcal{P}_- + Ce^{-t\Delta_-}\mathcal{R}\mathcal{P}_+. \tag{7.1}$$

On functions include the residual term $p_0(f)1$, and on one-forms set $R_t\eta = \eta_0 p_1(\eta) + d_E R_t h\eta$.

The signs of the tensor can be fixed on one pair. Write $\langle x, y\rangle = \tau_E(xy)$. For even vectors a, b and odd vectors $A = d_E a, B = d_E b$, suppose $\langle a, B\rangle = \kappa$ and $\langle b, A\rangle = -\kappa$. Define

$$\begin{aligned} P^{\mathrm{pair}} &= \kappa^{-1}(a\otimes b - b\otimes a), \\ K^{\mathrm{pair}} = d_E P^{\mathrm{pair}} &= \kappa^{-1}(a\otimes B - B\otimes a - b\otimes A + A\otimes b). \end{aligned} \tag{7.2}$$

The differential on a tensor has its Koszul sign.

In the convention

$$R_K x = (-1)^{|x|+1}\langle x, K_{(1)}\rangle K_{(2)},$$

K^{pair} contracts to the identity on these four vectors. The residual tensor is $K_{\mathrm{res}} = \kappa^{-1}(1\otimes\eta_0 - \eta_0\otimes 1)$.

For $t > 0$ define

$$\mathcal{P}_t = \frac{1}{\kappa} \sum_{\ell \geq 1} \sum_{m=0}^{\ell} e^{-t\lambda_\ell} (e_{\ell m} \otimes f_{\ell m} - f_{\ell m} \otimes e_{\ell m}),$$
$$\mathcal{K}_t = K_{\mathrm{res}} + d_E \mathcal{P}_t. \tag{7.3}$$

For $0 < \epsilon < T$, put $P_{\epsilon,T} = \mathcal{P}_\epsilon - \mathcal{P}_T$.

Theorem 7.1. *The series in* (7.3) *converge with all derivatives on the product sphere. Their limits lie in the completed projective tensor product of smooth Neumann fields and satisfy*

$$d_E P_{\epsilon,T} = \mathcal{K}_\epsilon - \mathcal{K}_T, \qquad P_{\epsilon,T} + P_{T,S} = P_{\epsilon,S}. \tag{7.4}$$

Contraction with $\mathcal{K}_t$ is R_t. These operators form a chain semigroup, preserve the unit, trace and Neumann condition, and satisfy

$$\langle R_t x, y \rangle = \langle x, R_t y \rangle.$$

They tend to the identity uniformly on bounded subsets of the smooth field space.

For total derivative order $r \geq 0$ and $0 < t \leq 1$,

$$\|\mathcal{K}_t\|_{C^r} \leq C_r t^{-(r+7)/2}. \tag{7.5}$$

The part with $\ell > L$ satisfies the same bound times $e^{-t(L+1)^2/2}$, with a different constant.

Proof. An L^2-normalized spherical harmonic has H^a norm bounded by $C_a(1+\ell)^a$. Sobolev embedding therefore bounds its C^r norm by $C_r(1+\ell)^{r+2}$. The estimate for C in Proposition 6.1 gives this same safe bound for $f_{\ell m}$. One application of d_E costs at most one further power.

At total derivative order r, a summand of $\mathcal{K}_t$ is bounded by $C_r(1+\ell)^{r+5} e^{-t\lambda_\ell}$. There are $\ell + 1$ summands at degree ℓ. A Gaussian integral bounds their sum by (7.5). Splitting the exponential into two equal factors gives the tail estimate. The same argument applies to every derivative of $\mathcal{P}_t$.

Smooth functions and forms on a compact product identify with the completed projective tensor products of their smooth spaces. The Neumann jet constraints define closed complemented subspaces, so they persist in these limits. The four-vector calculation (7.2) and the residual tensor now give the asserted contraction operator.

All chain identities hold on each pair. Equal weights on paired vectors give cyclic self-adjointness. Since $R_t 1 = 1$, that identity also gives $\tau_E R_t = \tau_E$. Finally, spectral convergence on V_-, composed with the continuous maps $C, \mathcal{R}$, proves convergence in every smooth seminorm. The spectral tail estimates are uniform when the input ranges over a bounded smooth set. □

The tensor is smooth, but its contraction with an input still uses τ_E. Relative to an ordinary smooth density, the input covectors retain the polar singularity. Already the residual tensor distinguishes these two assertions about a kernel.

7.2 The generator and curve locality

Let $\Lambda_E = -\partial_t R_t|_{t=0}$. It acts by λ_ℓ on each acyclic pair and by zero on $1, \eta_0$. On scalars, write $r = |z|^2$ and

$$\Delta = -(1+r)^2 \partial_z \partial_{\bar{z}}.$$

This is the positive round Laplacian in our normalization. Formula (7.1) determines $\Lambda_E = \Delta \mathcal{P}_- + C \Delta \mathcal{R} \mathcal{P}_+$ on V.

Proposition 7.2. *On the smooth scalar Neumann domain,*

$$\Lambda_E f = \Delta f + \frac{2(1+|z|^2)}{z}\partial_{\bar z}\mathcal{P}_+ f. \tag{7.6}$$

The second term extends smoothly across the marked point.

Proof. For a positive weight m, write $f = z^m F(r)$ and $V(r) = -(1+r)^2 F'(r)$. Then $\mathcal{R}f = \bar z^{m-1}\overline{V(r)}$. The equation $d_E C v = \eta_0 \bar v$ gives $C\Delta\mathcal{R}f = z^m G(r)$, with

$$G' = rV'' + mV', \qquad G = rV' + (m-1)V$$

up to a holomorphic term. Substitution gives $z^m G = \Delta(z^m F) + 2(1+r)z^m F'$.

For $m = 1$, the Neumann condition is $F(0) = 0$, and the two first-jet contributions in this expression cancel. For $m > 1$ the first jet vanishes by weight. In the chart $w = 1/z$, the extra differential expression is $-2(1+|w|^2)\bar w \partial_{\bar w}$, so it is smooth there. The possible holomorphic difference is a global positive-weight holomorphic function, and hence zero. Nonpositive weights give Δ, and constants give zero. Smooth Fourier convergence and the estimates for $C, \mathcal{R}$ extend the identity to every smooth Neumann input. □

The projection $\mathcal{P}_+$ prevents locality. To see its effect between disjoint supports, put $z = \rho e^{i\theta}$. Choose an even nonnegative smooth function b, supported in a small arc about zero, with mean $\bar b = (2\pi)^{-1}\int b\, d\theta > 0$. Choose an annular function a with $a(\rho_0) = 0$ and $a'(\rho_0) = 1$, and put $f(\rho,\theta) = a(\rho)b(\theta)$. This function vanishes near the opposite arc and near both poles.

Proposition 7.3. *For this function,*

$$(\Lambda_E f)(\rho_0,\pi) = -\frac{1+\rho_0^2}{2\rho_0}\bar b, \qquad (R_t f)(\rho_0,\pi) = t\frac{1+\rho_0^2}{2\rho_0}\bar b + O(t^2). \tag{7.7}$$

No finite-rank linear correction to Λ_E makes it support-preserving.

Proof. Off the angular diagonal, the kernel of the positive Fourier projection is

$$k_+(\theta-\varphi) = \frac{1}{2\pi}\frac{e^{i(\theta-\varphi)}}{1-e^{i(\theta-\varphi)}}.$$

Its real part is $-1/(4\pi)$. At $\theta = \pi$ its imaginary part is odd in φ, so $(\mathcal{P}_+ b)(\pi) = -\bar b/2$. The local Laplacian vanishes on the opposite arc.

The extra term in (7.6) is

$$\frac{1+\rho^2}{\rho}\left(\partial_\rho + \frac{i}{\rho}\partial_\theta\right)\mathcal{P}_+ f.$$

At ρ_0, only the radial derivative remains. This proves the first formula. On smooth inputs the spectral series is twice differentiable at zero, giving the second.

Repeat the construction in arbitrarily many disjoint radial annuli. On the opposite arc the resulting outputs are nonzero and have disjoint radial supports, so they are linearly independent. A correction restoring support preservation would have to cancel all of them there. Its rank cannot be finite. □

A local cyclic operator exists if uniform ellipticity at the pole is relinquished. Consider a first-order operator on one-forms,

$$q(\eta\, d\bar z) = A\partial_z\eta + B\eta,$$

with smooth coefficients, and set $q(E^0) = 0$. Cyclicity means $\tau_E((q\eta)\zeta) + \tau_E(\eta q\zeta) = 0$.

Proposition 7.4. *Cyclicity forces*

$$2B = \partial_z A - \frac{2A}{z}. \tag{7.8}$$

In particular $A(0) = 0$. *A real coefficient* $A = -a$, *with* $a > 0$ *off the pole, therefore cannot give a uniformly elliptic Dolbeault–Laplace principal part there. The operator*

$$q_{\mathrm{loc}}(\eta\, d\bar z) = -r(1+r)\partial_z\eta + \tfrac{1}{2}\bar z\eta \tag{7.9}$$

is globally smooth, maps E^1 *to* E^0, *is cyclic, and has square zero. Its commutator* $\Lambda_{\mathrm{loc}} = [d_E, q_{\mathrm{loc}}]$ *commutes with* d_E.

Proof. Integration by parts on tests away from the pole gives $2z^{-2}B - \partial_z(z^{-2}A) = 0$. Thus A/z extends smoothly, proving (7.8) and $A(0) = 0$. The coefficients in (7.9) satisfy this identity. Its output and first holomorphic jet vanish at zero.

In the coordinate $w = 1/z$, its action on $\eta_w d\bar w$ is

$$-(1+|w|^2)\partial_w\eta_w - \tfrac{1}{2}\bar w\eta_w,$$

which proves global smoothness.

Since $r(1+r)/z^2$ is bounded near zero, the boundary term in polar integration by parts tends to zero with the circle length. This proves cyclicity on all smooth inputs. The square vanishes by degree, and the commutation assertion follows from $d_E^2 = 0$. □

The degeneracy opens an infinite radial end for a natural weighted realization. On radial scalars $f = F(r)$, put $s = -\log r$ and $g(s) = F(e^{-s})$. Direct differentiation gives

$$\Lambda_{\mathrm{loc}} g = -(1+e^{-s})g'' - \tfrac{1}{2}g'. \tag{7.10}$$

Fix $s_0 \geq 0$ and set $a(s) = 1 + e^{-s}$, $w(s) = e^{s/2}/\sqrt{1+e^{-s}}$. Since $(aw)' = w/2$, the expression is $-w^{-1}\partial_s(aw\partial_s)$. Let L be the Friedrichs operator on $L^2([s_0,\infty), w\,ds)$ obtained by closing $\int aw|g'|^2 ds$ on $C_c^\infty((s_0,\infty))$. This specifies a Dirichlet condition at s_0.

Proposition 7.5. *For this realization,* $\sigma_{\mathrm{ess}}(L) = [1/16, \infty)$. *Its resolvent is not compact, and* e^{-tL} *is not trace class for any* $t > 0$.

Proof. The unitary substitution $u = w^{1/2}g$ gives

$$\widehat L = -\partial_s(a\partial_s) + V(s), \qquad V(s) = \frac{1 - 4e^{-s} - 4e^{-2s}}{16(1+e^{-s})}.$$

The coefficients approach those of $-\partial_s^2 + 1/16$ exponentially, with all derivatives. For $\xi \in \mathbb{R}$ and $\chi \in C_c^\infty(0,1)$ of norm one, the unit vectors

$$u_R(s) = R^{-1/2}\chi((s-R^2)/R)e^{i\xi s}$$

escape every compact interval and satisfy $\|(\widehat L - \xi^2 - 1/16)u_R\| = O(R^{-1}) + O(e^{-R^2})$. Weyl's criterion includes $[1/16,\infty)$ in the essential spectrum.

For any $\varepsilon > 0$, the quadratic form on a sufficiently distant tail is bounded below by $(1/16 - \varepsilon)\|u\|^2$. Dirichlet–Neumann bracketing separates that tail from a compact interval, whose resolvent is compact. Hence no essential spectrum lies below $1/16 - \varepsilon$. Letting ε decrease to zero proves equality. This nonempty essential spectrum excludes a compact resolvent and a compact, hence trace-class, heat operator. □

After translating the endpoint to zero, the limiting operator $-\partial_s^2 + 1/16$ has Dirichlet heat kernel

$$\frac{e^{-t/16}}{\sqrt{4\pi t}}\Big(e^{-(s-s')^2/(4t)} - e^{-(s+s')^2/(4t)}\Big).$$

Odd reflection of the line Gaussian proves this formula. It describes the limiting end, while (7.10) retains variable coefficients. The weighted Hilbert domain is additional analytic data. Its spectrum makes no assertion about the cohomology of the smooth Neumann complex.

7.3 All surface modes and the free BV complex

Return to the paired spectral regulator. Define $Q_E^{\mathrm{GF}} = \Lambda_E h$. The contraction identities give

$$[d_E, Q_E^{\mathrm{GF}}] = \Lambda_E, \qquad (Q_E^{\mathrm{GF}})^2 = 0.$$

Both operators are continuous on smooth fields.

Let U be a closed compact oriented Riemannian surface and use the full space

$$\mathcal{F} = \big(\Omega^\bullet(U)\widehat{\otimes}E \otimes \mathfrak{gl}_N\big)[1]. \tag{7.11}$$

The tensor product is completed projectively. Surface forms precede coefficient forms, and tensor operators use the Koszul rule. Integration over U, the trace τ_E, and $\mathrm{tr}(XY)$ give a pairing of degree -3 before the shift, and the odd BV pairing after it.

Set

$$Q = d_U + d_E, \qquad Q^{\mathrm{GF}} = d_U^* + Q_E^{\mathrm{GF}}, \qquad [Q, Q^{\mathrm{GF}}] = \Delta_U + \Lambda_E.$$

The cross commutators vanish by the tensor signs.

Let $\mathbf{K}_t$ be the de Rham heat tensor on U, tensored with $\mathcal{K}_t$ and the matrix coevaluation, with the suspension signs induced by the pairing. Thus every smooth surface mode is retained. Put

$$\mathbf{P}_{\epsilon,T} = \int_\epsilon^T (Q^{\mathrm{GF}} \otimes 1)\mathbf{K}_t \, dt. \tag{7.12}$$

For $0 < \epsilon < T$ this is smooth, and

$$Q\mathbf{P}_{\epsilon,T} = \mathbf{K}_\epsilon - \mathbf{K}_T. \tag{7.13}$$

Indeed $Q\mathbf{K}_t = 0$, and the commutator in the integral acts as $-\partial_t$ on the heat semigroup.

Write $\mathcal{O}_n(\mathcal{F})$ for continuous homogeneous polynomials of degree n, defined by continuous graded symmetric multilinear forms, and set $\mathcal{O}_{\mathrm{poly}} = \oplus_{n\geq 0}\mathcal{O}_n$. Give $\mathcal{O}_n$ the strong dual topology of the completed projective graded symmetric tensor power $\widehat{\mathrm{Sym}}_\pi^n\mathcal{F}$: its seminorms are uniform bounds on bounded sets of that space. Use the product topology in field degree and formal completion in $\hbar$; continuity in $\hbar$ is understood coefficientwise. Contraction of two inputs with $\mathbf{K}_t$ defines an odd constant second-order operator Δ_t. Contraction with $\mathbf{P}_{\epsilon,T}$ defines the even operator $\partial_{\mathbf{P}_{\epsilon,T}}$, with convention

$$[Q, \partial_{\mathbf{P}_{\epsilon,T}}] = \Delta_\epsilon - \Delta_T.$$

The derived bracket is

$$\{F, G\}_t = (-1)^{|F|}\big(\Delta_t(FG) - (\Delta_t F)G - (-1)^{|F|}F\Delta_t G\big).$$

Proposition 7.6. *On* $\mathcal{O}_{\mathrm{poly}}(\mathcal{F})[[\hbar]]$,

$$D_t = Q + \hbar\Delta_t, \qquad D_t^2 = 0, \qquad D_T e^{\hbar\partial_{\mathbf{P}_{\epsilon,T}}} = e^{\hbar\partial_{\mathbf{P}_{\epsilon,T}}} D_\epsilon. \tag{7.14}$$

These formulas also define continuous coefficientwise operations on $\big(\prod_{n\geq 0}\mathcal{O}_n(\mathcal{F})\big)[[\hbar]]$.

Proof. The smooth projective tensor can be evaluated on every continuous multilinear coefficient, including marked jets. The identity $Q\mathbf{K}_t = 0$ gives $[Q,\Delta_t] = 0$. An odd constant contraction has square zero, and constant contraction operators graded commute. Hence $D_t^2 = 0$. Conjugating D_T by the even exponential adds precisely $\hbar(\Delta_\epsilon - \Delta_T)$, proving the intertwiner.

For output degree n and coefficient $\hbar^r$, its exponential uses only input degrees $n + 2j$ with $0 \leq j \leq r$. Insertion of a fixed smooth tensor is a continuous linear map between the completed symmetric tensor powers and preserves bounded sets. Its transpose is therefore continuous for their strong dual topologies. Each output coefficient is a finite sum of these continuous contractions. This proves the assertion about the product completion without summing infinitely many contractions at one power of $\hbar$. □

At positive lower scale every fixed finite graph with continuous polar interaction vertices is defined by these tensors. A local counterterm theorem at zero scale requires a further argument: Proposition 7.3 excludes an appeal to curve-local elliptic heat asymptotics for this generator. If U has boundary, the surface Stokes term also requires a specified compatible BV–BFV domain.

7.4 Pole jets and their subtractions

The positive-scale tensor may be evaluated on the continuous functionals

$$L_m(f) = \frac{\partial_{\bar z}^m f(0)}{m!}, \qquad R_{m+1}(f) = \frac{\partial_z^{m+1} f(0)}{(m+1)!}, \qquad m \geq 1.$$

They act on scalar acyclic fields and avoid the excluded first holomorphic jet. Their oriented coefficient contraction is $C_m(t) = (L_m \otimes R_{m+1})\mathcal{P}_t$. After choosing a surface Gaussian polarization, the corresponding component covariance is $\hbar C_m(t)$.

For an explicit calculation set $x = (1-|z|^2)/(1+|z|^2)$ and let P_ℓ denote the Legendre polynomial normalized by $P_\ell(1) = 1$. Put

$$N_{\ell m} = \sqrt{\frac{(2\ell+1)(\ell-m)!}{(\ell+m)!}}, \qquad e_{\ell m} = 2^m N_{\ell m}\left(\frac{\bar z}{1+|z|^2}\right)^m P_\ell^{(m)}(x).$$

Here $P_\ell^{(m)}$ means the ordinary mth derivative. Legendre orthogonality gives norm one for $d\mu$.

The radial Dolbeault equation has solution

$$Ce_{\ell m} = \frac{N_{\ell m} z^{m+1}}{2} \int_{-1}^{x} (1+y)^m P_\ell^{(m)}(y)\, dy. \tag{7.15}$$

Differentiating verifies $d_E C e = \eta_0 \bar e$. The integral vanishes to order at least $m+1$ at $x = -1$, which gives smoothness at infinity. For $m = 0$, its value at $x = 1$ is zero when $\ell \geq 1$, giving the Neumann first jet.

Lemma 7.7. *For* $\ell, m \geq 1$,

$$\int_{-1}^{1} (1+x)^m P_\ell^{(m)}(x)\, dx = \frac{2}{(m-1)!} \prod_{j=1}^{m-1} \big(\ell(\ell+1) - j(j+1)\big). \tag{7.16}$$

Proof. Integrate by parts m times. All boundary terms at -1 vanish, and the remaining integral of P_ℓ is zero. The derivatives at 1 make the answer a polynomial in $\lambda = \ell(\ell+1)$ of degree $m-1$, with leading coefficient $2/(m-1)!$. For $\ell = 1, \dots, m-1$ the original derivative vanishes. These roots determine the polynomial. The empty product gives the case $m = 1$. □

Let $\Theta(t) = \sum_{\ell \geq 0} (2\ell+1) e^{-t\ell(\ell+1)}$.

Proposition 7.8. *For $t > 0$ and $m \geq 1$,*

$$C_m(t) = \frac{1}{\kappa m!(m-1)!} \prod_{j=1}^{m-1} (-\partial_t - j(j+1))[\Theta(t) - 1]. \tag{7.17}$$

In particular,

$$\begin{aligned} \kappa C_1(t) &= t^{-1} - \tfrac{2}{3} + O(t), \\ \kappa C_2(t) &= \tfrac{1}{2t^2} - \tfrac{1}{t} + \tfrac{19}{30} + O(t), \\ \kappa C_3(t) &= \tfrac{1}{6t^3} - \tfrac{2}{3t^2} + \tfrac{1}{t} - \tfrac{586}{945} + O(t). \end{aligned} \tag{7.18}$$

For general m, the leading term is $1/(\kappa m! t^m)$.

Proof. The leading coefficients of $\bar{z}^m$ in $e_{\ell m}$ and of z^{m+1} in $C e_{\ell m}$ are

$$a_{\ell m} = N_{\ell m} \frac{(\ell+m)!}{m!(\ell-m)!}, \qquad b_{\ell m} = \frac{N_{\ell m}}{2} \int_{-1}^{1} (1+x)^m P_\ell^{(m)}(x)\, dx.$$

Their product is $(2\ell+1) \prod_{j=1}^{m-1} (\lambda_\ell - j(j+1))/(m!(m-1)!)$. The second orientation in $\mathcal{P}_t$ contributes zero by weights. Sum the first against $\kappa^{-1} e^{-t\lambda_\ell}$. Modes $1 \leq \ell < m$ may be added because their coefficients vanish, proving (7.17).

Euler–Maclaurin summation gives

$$\Theta(t) \sim t^{-1} + \tfrac{1}{3} + \tfrac{t}{15} + \tfrac{4t^2}{315} + \tfrac{t^3}{315} + \tfrac{4t^4}{3465} + \cdots.$$

The integral of $(2x+1)e^{-tx(x+1)}$ on $[0, \infty)$ is $1/t$. The endpoint half-value and the Bernoulli derivative terms give the remaining coefficients. Its remainder estimate also holds after any fixed number of t derivatives. Applying the differential polynomial gives (7.18). Its highest derivative on t^{-1} gives the general leading term. □

For a finite set of pairs choose commuting formal sources J_m, K_m in the Gaussian polarization. Rotation weights kill the pairings with distinct indices, so their generating series is

$$Z_t(J, K) = \exp\left(\hbar \sum_{m=1}^{M} C_m(t) J_m K_m \right).$$

Let PP and FP denote the negative-power part and constant part of the small-t expansion. Then, coefficientwise in sources and $\hbar$,

$$e^{-\hbar \sum_m \mathrm{PP}\, C_m(t) J_m K_m} Z_t \longrightarrow e^{\hbar \sum_m \mathrm{FP}\, C_m(t) J_m K_m}.$$

This defines the Gaussian multipoint coefficients in the selected component sector. Gauge invariance and comparison with sourced closed observables require their own maps.

Proposition 7.9. *On a finite jet system closed under the dual differential, the divergent parts of the paired tensors obey* $\mathrm{CT}_K = d_E \mathrm{CT}_P$. *Their subtraction preserves the free Ward identity. That identity does not fix the finite parts.*

Proof. Choose $N \geq M + 1$, take scalar jets through order N and one-form jets through order $N - 1$, and impose the Neumann relation on scalar jets. The dual of d_E sends a one-form jet to a scalar jet of at most order N. These tests therefore form a finite differential system containing the selected pairs. Evaluate $d_E \mathcal{P}_t = \mathcal{K}_t - K_{\mathrm{res}}$ on it. Each asymptotic coefficient commutes with the finite differential, proving the subtraction identity.

Replacing the heat parameter by $t + ct^2$ retains the paired identity but changes the first covariance to $\kappa^{-1}(t^{-1} - 2/3 - c + O(t))$. Its finite part varies by $-c/\kappa$. This is a reparametrized prescription. It need not be a semigroup in the new parameter. □

7.5 Coincident kernels

Linear approximation on smooth inputs does not by itself control products of kernels. Consider the ordinary scalar sharp projector on the sphere, independently of the paired projection of Chapter 6. At the north pole its kernel relative to $d\mu$ is

$$D_L(x) = \sum_{\ell=0}^{L} (2\ell + 1) P_\ell(x), \qquad x = \cos\theta.$$

Proposition 7.10. *No sequence of counterterms supported at the north pole makes* D_L^2 *converge as a distribution. If* a *is a nonzero nonnegative smooth radial test supported in a compact annulus in* $0 < \theta < \pi$, *then*

$$\lim_{L\to\infty} \frac{1}{L} \int a(\theta) D_L(\cos\theta)^2 d\mu = \frac{1}{2\pi} \int \frac{a(\theta)}{\sin^2(\theta/2)} d\theta > 0. \tag{7.19}$$

Proof. Christoffel–Darboux gives $(1 - x) D_L(x) = (L + 1)(P_L(x) - P_{L+1}(x))$. Uniformly on a compact interior interval, Darboux's formula is

$$P_L(\cos\theta) = \sqrt{\frac{2}{\pi L \sin\theta}} \cos((L + \tfrac{1}{2})\theta - \tfrac{\pi}{4}) + O(L^{-3/2}).$$

This is the zero-parameter Jacobi case of DLMF 18.15.4_5. Consequently the leading term of D_L is

$$\sqrt{\frac{2L}{\pi \sin\theta}} \frac{\sin((L + 1)\theta - \pi/4)}{\sin(\theta/2)}.$$

Squaring and integrating against the radial measure $\frac{1}{2} \sin\theta \, d\theta$ gives (7.19). The oscillatory cosine has vanishing integral in the limit. Every pole-supported counterterm vanishes on a, while this positive integral grows linearly. □

The same kernel has exact moments

$$\begin{aligned}
\tfrac{1}{2} \int_{-1}^{1} D_L^2 \, dx &= (L + 1)^2, \\
\tfrac{1}{2} \int_{-1}^{1} (x - 1) D_L^2 \, dx &= -(L + 1), \\
\tfrac{1}{2} \int_{-1}^{1} (1 - x)^2 D_L^2 \, dx &= (L + 1)^2 \left(\frac{1}{2L + 1} + \frac{1}{2L + 3} \right).
\end{aligned}$$

Orthogonality proves the first. The three-term recurrence for xP_ℓ proves the second, and the squared Christoffel–Darboux identity proves the third. These moments belong to the ordinary sharp kernel. A comparison with the paired regulator would require its individual components.

There is also a finite comparison between different heat prescriptions. In a separate complex normal plane, with real coordinates (x, y), put

$$\Delta_\perp = \partial_x^2 + \partial_y^2, \qquad \delta_t(x, y) = \frac{1}{4\pi t} e^{-(x^2+y^2)/(4t)}.$$

These are scalar coefficient densities after contraction.

Proposition 7.11. *For $n \geq 2$, the distributional finite part is*

$$\mathrm{FP}_{t=0}\, \delta_t^n = \frac{\Delta_\perp^{n-1}\delta_0}{(4\pi)^{n-1} n^n (n-1)!}. \tag{7.20}$$

The joint cubic prescription minus the prescription which first takes the binary finite part and then heat-regularizes both remaining factors is

$$\frac{37}{55296\pi^2}\Delta_\perp^2 \delta_0. \tag{7.21}$$

Proof. The identities

$$\delta_t^n = \frac{\delta_{t/n}}{n(4\pi t)^{n-1}}, \qquad \delta_{t/n} = e^{(t/n)\Delta_\perp}\delta_0$$

give (7.20) by taking the constant coefficient. The binary and joint cubic finite parts are respectively $\Delta_\perp \delta_0/(16\pi)$ and $\Delta_\perp^2 \delta_0/(864\pi^2)$.

For the successive prescription, $\delta_t \Delta_\perp \delta_t = \frac{1}{2}\partial_t(\delta_t^2)$, whose finite part is $\Delta_\perp^2 \delta_0/(128\pi)$. Multiplication by the preceding binary coefficient $1/(16\pi)$ gives $\Delta_\perp^2 \delta_0/(2048\pi^2)$. Subtracting yields (7.21). □

The comparison is between two declared prescriptions on scalar densities. It supplies a local contact between them. A product of arbitrary singular Dolbeault currents, or coherent collision operations in all arities, requires additional extension data.

7.6 Two poles approaching one

The spectral coordinate has its own contact, separate from the normal plane. Work in a fixed chart and use compactly supported smooth coefficients η with measure $d^2z = dx\, dy$. For $a \neq 0$ define the locally integrable distribution $K_a(z) = (z^2 - a^2)^{-1}$ and functionals

$$\tau_a(\eta) = -\frac{\kappa}{\pi}\langle K_a, \eta\rangle, \qquad \tau_0(\eta) = -\frac{\kappa}{\pi}\langle \mathrm{PV}(z^{-2}), \eta\rangle.$$

Here $\mathrm{PV}(z^{-2}) = -\partial_z(1/z)$, and the orientation fixes $\partial_{\bar z}(1/z) = \pi\delta_0$. Thus $\tau_0(\eta)$ is the circular double-pole trace of $\eta\, d\bar z$.

Theorem 7.12. *There is an exact distributional identity*

$$K_a = \frac{1}{2}\int_{-1}^{1} \mathrm{PV}\,\frac{1}{(z - ta)^2}\, dt - \frac{\pi \bar a}{2a}\int_{-1}^{1} \delta_{ta}\, dt. \tag{7.22}$$

Along $a = \rho e^{i\theta}$ with $\rho \downarrow 0$,

$$\tau_a = \tau_0 + \kappa e^{-2i\theta}\,\mathrm{ev}_0 + O(\rho^2).$$

For tests in a fixed compact chart the error is bounded by $C|a|^2\|\eta\|_{C^3}$, uniformly in θ. The corrected functional

$$\tau_a^{\mathrm{seg}}(\eta) = \tau_a(\eta) - \frac{\kappa\bar a}{2a}\int_{-1}^{1}\eta(ta)\,dt \tag{7.23}$$

converges to τ_0 with the same bound and satisfies

$$\tau_a^{\mathrm{seg}}(\partial_{\bar z}f) = \frac{\kappa}{2}\int_{-1}^{1}\partial_z f(ta)\,dt \tag{7.24}$$

for compactly supported smooth f.

Proof. Differentiate the translated locally integrable distribution:

$$\frac{d}{dt}\frac{1}{z-ta} = a\,\mathrm{PV}\,\frac{1}{(z-ta)^2} - \pi\bar a\,\delta_{ta}.$$

Integration from -1 to 1, followed by division by $2a$, gives (7.22). Translation is continuous on distributions, so the first average tends to $\mathrm{PV}(z^{-2})$ and the second gives the stated directional contact.

On a fixed compact neighborhood, $1/z$ is integrable and its derivative has order at most one. Symmetric Taylor expansion of the translated test cancels the linear term. The remainder in C^1 is bounded by $C|a|^2\|\eta\|_{C^3}$. The symmetric delta average differs from evaluation at zero by $C|a|^2\|\eta\|_{C^2}$. These two estimates prove both convergence bounds.

The simple-pole distributional identity gives $\tau_a(\partial_{\bar z}f) = \kappa(f(a)-f(-a))/(2a)$. The chain rule along ta separates this difference into holomorphic and antiholomorphic derivatives. Subtracting the segment term leaves (7.24). □

For $f = \bar z$ near the segment, cut off outside a larger disc, the raw expression is $\kappa\bar a/a$, while the double-pole expression is zero. The contact is therefore detected on smooth tests even though holomorphic residue formulas alone do not see it.

A correction to the trace does not automatically specify an algebra of boundary functions. Put $\ell_a(f) = \frac{1}{2}\int_{-1}^{1}\partial_z f(ta)dt$. With the same cutoff convention,

$$\ell_a(\bar z) = \ell_a(z^2) = 0, \qquad \ell_a(\bar z z^2) = \tfrac{2}{3}|a|^2. \tag{7.25}$$

Hence $\ker\ell_a$ is not a subalgebra when $a \neq 0$.

In contrast, the raw condition $f(a) = f(-a)$ is multiplicative. Its normalized difference tends, along the ray, to $\partial_z f(0) + e^{-2i\theta}\partial_{\bar z}f(0)$. This is a point derivation, so its kernel is an algebra, but it imposes a directional boundary condition. Comparing it with the holomorphic Neumann condition requires a map preserving the boundary product as well as the corrected trace.

7.7 Relative normal insertions and supported contacts

The normal contact in (7.21) is a nonzero scalar distribution. To decide whether it represents an obstruction, one must place it in a complex with its allowed primitives. We first normalize its insertion on a disk, retaining the outer boundary, and then compute the supported Dolbeault complex.

Write $p = x + iy$ for this normal coordinate, independently of the spectral coordinate z. Orient $D_R = \{|p| \le R\}$ by $dx\wedge dy$ and its boundary S_R by the induced orientation. Normalize the area delta by $\langle\delta_0,\phi\rangle = \phi(0)$. Then

$$\bar\partial(1/p) = \pi\delta_0\,d\bar p, \qquad d(dp/p) = 2\pi i\,\delta_0\,dx\wedge dy.$$

Let $\mathcal{K}^\bullet(D_R)$ be the de Rham currents that are smooth on a fixed outer collar. Give them the strong distribution topology in the interior and the smooth topology on that collar. Restriction i^* to S_R is therefore continuous. The relative complex and its charge are

$$\begin{aligned} \mathcal{N}_R^n &= \mathcal{K}^n(D_R) \oplus \Omega^{n-1}(S_R), \\ d_{\mathrm{rel}}(\omega, \gamma) &= (d\omega, i^*\omega - d\gamma), \\ \mathrm{Ch}(\omega, \gamma) &= \frac{1}{2\pi i}\left(\int_{D_R} \omega - \int_{S_R} \gamma\right) \quad (n = 2). \end{aligned} \tag{7.26}$$

The charge is zero in other degrees. Stokes' theorem makes it a cochain map to $\mathbb{C}[-2]$, with its generator in degree two.

For normal heat time $t > 0$, independently of the spectral heat time, put

$$Q_t = 1 - e^{-R^2/(4t)}, \qquad \beta_t^H = (1 - e^{-|p|^2/(4t)})\frac{dp}{p}, \qquad \widetilde{\beta}_t^H = Q_t^{-1}\beta_t^H,$$

$$q_\epsilon = \frac{R^2}{R^2 + \epsilon^2}, \qquad \beta_\epsilon^R = \frac{\bar{p}\, dp}{|p|^2 + \epsilon^2}, \qquad \widetilde{\beta}_\epsilon^R = q_\epsilon^{-1}\beta_\epsilon^R \quad (\epsilon > 0).$$

The superscripts distinguish heat and rational representatives. Define $U_t^H = (d\widetilde{\beta}_t^H, 0)$ and $U_\epsilon^R = (d\widetilde{\beta}_\epsilon^R, 0)$ in $\mathcal{N}_R^2$.

Proposition 7.13. *Both insertions have charge one and boundary primitive $i\, d\theta$. Their difference is the relative boundary*

$$U_t^H - U_\epsilon^R = d_{\mathrm{rel}}(\widetilde{\beta}_t^H - \widetilde{\beta}_\epsilon^R, 0). \tag{7.27}$$

Differences of normalized primitives give additive homotopies between any two positive scales. Moreover, $\widetilde{\beta}_t^H \to dp/p$ in the declared current and collar topologies as $t \downarrow 0$.

Proof. The function $(1 - e^{-r^2/(4t)})/r^2$ is smooth at zero, so β_t^H is smooth there. Polar differentiation gives

$$d\beta_t^H = \frac{ir}{2t} e^{-r^2/(4t)}\, dr \wedge d\theta = 2\pi i \delta_t\, dx \wedge dy.$$

Both normalized primitives restrict to $i\, d\theta$ on S_R. Stokes gives charge one, and equality of the boundary restrictions proves (7.27). Differences telescope, which proves additivity.

Away from zero the heat primitive converges in every smooth seminorm. Its local L^1 difference from dp/p is bounded by a constant times $\int_0^R e^{-r^2/(4t)}\, dr = O(\sqrt{t})$. This gives convergence uniformly on bounded test sets. Since $Q_t \to 1$, normalization preserves that convergence and the smooth collar limit. □

This is a comparison of independently specified insertion maps. It preserves relative charge and the stated support and boundary conditions. It does not define a coincident product of singular currents.

7.7.1 The differential on supported jets

Set $\rho_t = Q_t^{-1}\delta_t$ on D_R and

$$c_n = \frac{1}{(4\pi)^{n-1} n^n (n-1)!}, \qquad n \geq 2.$$

The factor $Q_t^{-n} - 1$ is exponentially small with all fixed t derivatives. The finite-part calculation of Proposition 7.11 therefore gives $\mathrm{FP}_{t=0}\, \rho_t^n = c_n \Delta_\perp^{n-1} \delta_0$. Its relative de Rham primitive is supported at the origin:

$$\begin{aligned} V_n &= c_n \big(\partial_x \Delta_\perp^{n-2} \delta_0 \, dy - \partial_y \Delta_\perp^{n-2} \delta_0 \, dx \big), \\ d_{\mathrm{rel}}(V_n, 0) &= (c_n \Delta_\perp^{n-1} \delta_0 \, dx \wedge dy, 0). \end{aligned} \tag{7.28}$$

Direct differentiation proves the identity, and support makes the boundary restriction zero.

The Dolbeault quotient retains more information than this real charge. Define the finite supported normal-jet complex by

$$\begin{aligned} \mathcal{J}_p^0 &= \mathbb{C}[\xi, \eta] \delta_0, & \mathcal{J}_p^1 &= \mathbb{C}[\xi, \eta] \delta_0 \, d\bar{p}, \\ \xi &= \partial_p, & \eta &= \partial_{\bar{p}}, \\ \bar\partial(\xi^a \eta^b \delta_0) &= \xi^a \eta^{b+1} \delta_0 \, d\bar{p}. \end{aligned} \tag{7.29}$$

The symbols denote distribution derivatives without factorial normalization. Degree is Dolbeault degree. Use the inductive topology of finite normal-order spaces, with smooth tangential coefficients at fixed compact supports when these are present. Formal completions are coefficientwise, with finite normal order at each coefficient.

Theorem 7.14. *The complex $\mathcal{J}_p$ contracts onto $\mathbb{C}[\xi] \delta_0 \, d\bar{p}$ in degree one. Let π retain only degree-one terms with antiholomorphic derivative order zero, and let ι be their inclusion. The homotopy is zero in degree zero and is*

$$h_p(\xi^a \eta^b \delta_0 \, d\bar{p}) = \begin{cases} \xi^a \eta^{b-1} \delta_0, & b \geq 1, \\ 0, & b = 0. \end{cases}$$

It satisfies

$$\bar\partial h_p + h_p \bar\partial = 1 - \iota\pi, \qquad h_p^2 = \pi h_p = h_p \iota = 0.$$

Thus $H^0(\mathcal{J}_p) = 0$ and $H^1(\mathcal{J}_p) = \mathbb{C}[\partial_p] \delta_0 \, d\bar{p}$. The maps are continuous, commute with holomorphic normal differentiation, and extend after tensoring with an independent cochain coefficient space in the stated finite-order topology.

Proof. On a degree-zero monomial $h_p \bar\partial$ is the identity. On a degree-one monomial with $b \geq 1$, $\bar\partial h_p$ is the identity. On the remaining monomials both sides of $1 - \iota\pi$ vanish. These cases also prove the side conditions.

The homotopy lowers normal order, so it is continuous at every finite stage. The graded tensor rule cancels the two terms involving an independent coefficient differential. This proves the tensor assertion without passing to an unrestricted completion. □

In particular, $\delta_0 \, d\bar{p}$ has no supported primitive. Such a primitive would require negative antiholomorphic derivative order. The higher heat contacts behave differently, since $\Delta_\perp = 4 \partial_p \partial_{\bar{p}}$ gives

$$\Delta_\perp^{n-1} \delta_0 \, d\bar{p} = \bar\partial \big(4^{n-1} \partial_p^{n-1} \partial_{\bar{p}}^{n-2} \delta_0 \big). \tag{7.30}$$

The cubic difference in (7.21) therefore satisfies

$$\frac{37}{55296\pi^2} \Delta_\perp^2 \delta_0 \, d\bar{p} = \bar\partial \left(\frac{37}{3456\pi^2} \partial_p^2 \partial_{\bar{p}} \delta_0 \right). \tag{7.31}$$

The primitive requires a mixed normal jet in degree zero. A pure holomorphic residue module does not contain the heat contact. Adjoining the contact while excluding its displayed primitive changes the complex and hence the obstruction question.

7.7.2 Two normal directions and a surface boundary

Tensor products below use independent normal variables, at finite normal order and fixed compact test support. They do not multiply arbitrary distributions in a single variable. Suppose U_i, U_i' are closed cochains of degree a_i, and $dH_i = U_i - U_i'$ with $|H_i| = a_i - 1$. The two ordered homotopies are

$$H_{12} = H_1 \otimes U_2 + (-1)^{a_1} U_1' \otimes H_2,$$
$$H_{21} = (-1)^{a_1} U_1 \otimes H_2 + H_1 \otimes U_2'.$$

Both differentiate to $U_1 \otimes U_2 - U_1' \otimes U_2'$. Their difference is

$$H_{12} - H_{21} = (-1)^{a_1+1} d(H_1 \otimes H_2). \tag{7.32}$$

Indeed, the first difference expands as $H_1 \otimes dH_2 - (-1)^{a_1} dH_1 \otimes H_2$, which is the displayed differential. The normal charges have $a_i = 2$. Adjoining an independent closed odd label to each factor, placed last, changes a_i to three and reverses the corner sign.

The tensor contraction for $\mathcal{J}_u \otimes \mathcal{J}_v$ is

$$h_u \otimes 1 + (\iota_u \pi_u) \otimes h_v.$$

It is interpreted as a graded tensor of maps, so the second term has a minus sign after a degree-one residue in the first factor. The retained cohomology is

$$\mathbb{C}[\partial_u, \partial_v]\, \delta_u \delta_v \, d\bar{u}\, d\bar{v}$$

in degree two. Expanding the tensor differential and using the two one-variable contraction identities proves this assertion.

To retain contacts along the union of the two divisors, one must also retain its faces. In one normal variable let $\mathcal{L}$ be the Dolbeault coefficient complex localized at the divisor, with finite pole order. Its principal-value section uses

$$E_m(p) = \frac{(-1)^{m-1}}{(m-1)!} \partial_p^{m-1}(1/p).$$

Let $\mathcal{E} = \sigma(\mathcal{L}) + \mathcal{J}$ inside currents, where σ is this graded section and $\mathcal{J}$ is the full supported jet complex. In the graded splitting,

$$\mathcal{E} = \mathcal{L} \oplus \mathcal{J}, \qquad d(a, s) = (d_{\mathcal{L}} a, d_{\mathcal{J}} s + \mathsf{R} a), \qquad \mathsf{R} = d\sigma - \sigma d_{\mathcal{L}}.$$

The identities $pE_1 = 1$ and $pE_m = E_{m-1}$ make the section well-defined on localized coefficients. Restriction off the divisor identifies the quotient with $\mathcal{L}$. The equation $d_{\mathcal{J}} \mathsf{R} + \mathsf{R} d_{\mathcal{L}} = 0$ follows from $d^2 = 0$.

For two variables the supported kernel of $\mathcal{E}_u \boxtimes \mathcal{E}_v \to \mathcal{L}_u \boxtimes \mathcal{L}_v$ has graded space

$$\mathcal{J}_{uv} = (\mathcal{J}_u \boxtimes \mathcal{L}_v) \oplus (\mathcal{L}_u \boxtimes \mathcal{J}_v) \oplus (\mathcal{J}_u \boxtimes \mathcal{J}_v). \tag{7.33}$$

Its differential retains both face-to-corner maps. With $r_1 = \bar{\partial}(1/p)$, the signs are

$$\bar{\partial}(E_1(u) E_1(v)) = r_1(u) E_1(v) + E_1(u) r_1(v),$$
$$\bar{\partial}(r_1(u) E_1(v)) = -r_1(u) r_1(v),$$
$$\bar{\partial}(E_1(u) r_1(v)) = +r_1(u) r_1(v).$$

Deleting either face destroys the cancellation in the second differential.

Now let $\mathcal{B}$ be an independently given cochain complex with cochain map $\mathsf{W} : \mathcal{B} \to \mathcal{L}$. The pullback of the extension is

$$\mathcal{E}_{\mathsf{W}} = \mathcal{B} \oplus \mathcal{J}, \qquad d(b, s) = (d_{\mathcal{B}} b, d_{\mathcal{J}} s + \mathsf{R}\mathsf{W} b).$$

It maps to $\mathcal{E}$ by $(b, s) \mapsto (\mathsf{W}b, s)$. Both the square-zero and chain-map identities follow from $d_{\mathcal{L}}\mathsf{W} = \mathsf{W}d_{\mathcal{B}}$ and the identity for R. This construction supplies a coefficient extension, without a product of meromorphic functions with arbitrary supported currents.

Proposition 7.15. *Suppose these complexes and maps are given on a surface U and its boundary Γ. Let the degree-zero cochain restrictions ρ commute with W and satisfy $\rho_{\mathcal{J}}\mathsf{R}_U = \mathsf{R}_\Gamma \rho_{\mathcal{L}}$. The relative complex has differential*

$$D(b, s; c, v) = \begin{pmatrix} d_{\mathcal{B}} b, & d_{\mathcal{J}} s + \mathsf{R}_U \mathsf{W}_U b; \\ \rho_{\mathcal{B}} b - d_{\mathcal{B}} c, & \rho_{\mathcal{J}} s - d_{\mathcal{J}} v - \mathsf{R}_\Gamma \mathsf{W}_\Gamma c \end{pmatrix}. \tag{7.34}$$

Here (b, s) has degree n over U, and (c, v) has degree $n - 1$ over Γ. One has $D^2 = 0$. For two normal directions, $\mathcal{J}$ includes the full kernel (7.33).

Proof. The restriction hypotheses make $\mathcal{E}_{\mathsf{W},U} \to \mathcal{E}_{\mathsf{W},\Gamma}$ a cochain map. For any cochain map $f : A \to B$, applying $(a, b) \mapsto (d_A a, f a - d_B b)$ twice gives zero, since $f d_A = d_B f$. Substitution of the two extension differentials gives (7.34). □

Thus a closed degree-one element satisfies all four equations

$$\begin{aligned} d_{\mathcal{B}} b = 0, \qquad & d_{\mathcal{J}} s + \mathsf{R}_U \mathsf{W}_U b = 0, \\ \rho_{\mathcal{B}} b - d_{\mathcal{B}} c = 0, \quad & \rho_{\mathcal{J}} s - d_{\mathcal{J}} v - \mathsf{R}_\Gamma \mathsf{W}_\Gamma c = 0. \end{aligned}$$

If a sourced BV–BFV action realizes this diagram in its admissible local cyclic counterterm complex, a degree-zero counterterm changes its first quantum residual by DX. For $X = (x, u; y, w)$ the boundary contact component is

$$\rho_{\mathcal{J}} u - d_{\mathcal{J}} w - \mathsf{R}_\Gamma \mathsf{W}_\Gamma y.$$

An admissible regulator transport giving such a boundary preserves the residual class.

The primitive (7.31), tensored with a closed independent coefficient, kills that supported coefficient class. It does not kill additional field or boundary components. Constructing the native sourced action, its local cyclic domain, compatible restrictions and quantum measure is still required. A compact-group disk preparation requires its own real integration cycle, measure and sewing comparison. Its state is not determined by the charge-one normal insertion.

7.8 Uniform inverses at a matrix background

The finite inverse in (6.15) is polynomial in J, but its degree grows with the cutoff. To take a limit one needs a common bound. The adapted norm (6.11) reveals an infinite square-zero block whose removal would destroy the operator-norm approximation.

For $s \geq 3$, let H_s^- be the closure of V_- in ordinary round H^s, and complete $V_- \oplus V_+$ in the adapted norm to obtain $\mathfrak{B}_s$. Its complex linear Hilbert coordinates are

$$\mathfrak{B}_s \longrightarrow H_s^- \oplus \overline{H_s^-}, \qquad f \longmapsto (f_-, \overline{\mathcal{R} f_+}).$$

The second bar denotes the conjugate vector space, compensating for the antilinearity of $\mathcal{R}$. The one-derivative gain of C embeds $\mathfrak{B}_s$ continuously in ordinary H^s. Its vectors satisfy the mean and Neumann constraints, and the intersection of the scale is the original smooth space V.

Let $\mathcal{P}_0^{\mathrm{rot}}$ project to rotation weight zero, distinct from the scalar residual mean p_0. Put

$$X = h\eta_0, \qquad \mathsf{N}f = h(\eta_0 \mathcal{P}_0^{\mathrm{rot}} f_-), \qquad K = X - \mathsf{N}.$$

Proposition 7.16. *The operator* N *is bounded on* $\mathfrak{B}_s$ *and satisfies* $\mathsf{N}^2 = 0$. *The remainder* K *maps* $\mathfrak{B}_s$ *boundedly to* $\mathfrak{B}_{s+1}$. *The actual staggered projections* $\widehat{\Pi}_L$ *are orthogonal at every index, converge strongly to the identity and commute with* N.

Proof. Write $f_- = r + w$, with r of weight zero and w of strictly negative weights. Then

$$\mathsf{N}r = C\overline{r}, \qquad \mathcal{R}\mathsf{N}r = \overline{r}.$$

These bars are pointwise conjugation. Thus N sends the radial summand isometrically to its positive partner in the adapted norm. It kills positive weights, proving boundedness and square zero.

The vector Xw has nonpositive weights and gains one ordinary Sobolev derivative. The finite-rank corrections in h obey the same bound: $\tau_E(\eta) = \kappa\, j(G\eta)$, the Green operator gains one derivative, and first-jet evaluation is continuous at these indices. For $f_+ = Cv$, one has

$$d_E X f_+ = \eta_0 f_+, \qquad \mathcal{R}X f_+ = \overline{f_+}, \qquad \|\mathcal{R}X f_+\|_{H^{s+1}} \le C_s \|v\|_{H^s}.$$

The first identity uses $p_0(f_+) = 0$. These bounds prove the one-derivative gain of K on both summands.

In the Hilbert coordinates, $\widehat{\Pi}_L$ is the ordinary spherical spectral projection on each summand, with conjugation on the second. It is orthogonal and converges strongly. Radial spherical projection commutes with pointwise conjugation, so the formulas for N also give $\mathsf{N}\widehat{\Pi}_L = \widehat{\Pi}_L \mathsf{N}$. □

The inclusion $\mathfrak{B}_{s+1} \hookrightarrow \mathfrak{B}_s$ is compact, as the sum of two ordinary Sobolev inclusions. Hence K and $X^2 = \mathsf{N}K + K\mathsf{N} + K^2$ are compact. The operator X is not compact on this scale, because N is an isometry on an infinite radial summand.

Lemma 7.17. *The spectrum of* X *on* $\mathfrak{B}_s$ *is* $\{0\}$.

Proof. Rotation pullback U_θ is unitary. On the positive summand, $\mathcal{R}U_\theta f_+ = e^{-i\theta} U_\theta \mathcal{R} f_+$, whose extra phase preserves the norm. Since X raises weight by one,

$$U_\theta X U_\theta^{-1} = e^{i\theta} X.$$

A nonzero spectral value would bring its entire rotation circle into the spectrum. Polynomial spectral mapping would then put a nonzero circle in the spectrum of the compact operator X^2. The nonzero spectrum of a compact operator consists of isolated finite-multiplicity eigenvalues, so this is impossible. Nonemptiness of the spectrum of a bounded operator gives $\{0\}$. □

Fix a positive Hilbert norm on $\mathfrak{g} = \mathfrak{gl}_N(\mathbb{C})$ and retain ordinary matrix trace in the cyclic pairing. Define

$$M_J = 1 + X \otimes \mathrm{ad}_J, \qquad B_J = M_J^{-1}, \qquad B_{J,L} = (1 + X_L \otimes \mathrm{ad}_J)^{-1}.$$

The finite inverse acts on $\widehat{V}_L \otimes \mathfrak{g}$. The tensor with $\mathfrak{g}$ is understood in the norms below.

Theorem 7.18. *For $s \geq 3$ and every compact matrix set $\mathcal{K}$,*

$$\sup_{L\geq 0,\ J\in\mathcal{K}} \|B_{J,L}\|_{\mathfrak{B}_s\otimes\mathfrak{g}} < \infty.$$

The full inverse B_J is entire in the entries of J. For every integer $r \geq 0$ and finite parameter derivative ∂_J^α,

$$\sup_{J\in\mathcal{K}} \|\partial_J^\alpha (B_{J,L}\widehat{\Pi}_L - B_J) f\|_{\mathfrak{B}_s} \leq C_{s,r,\alpha,\mathcal{K}} (1+L)^{-r} \|f\|_{\mathfrak{B}_{s+r}}. \tag{7.35}$$

Thus the inverses and all finite background responses converge uniformly on bounded smooth sets, locally uniformly in J.

Proof. The spectral-radius formula gives $\|X^n\|^{1/n} \to 0$. Consequently

$$B_J = \sum_{n\geq 0} (-X\otimes \mathrm{ad}_J)^n$$

converges in operator norm uniformly on compact matrix sets and defines an entire inverse.

Retain the nonsmoothing block before compressing:

$$\widehat{X}_L = \mathsf{N} + \widehat{\Pi}_L K \widehat{\Pi}_L.$$

Compactness of K and strong convergence of the orthogonal projections imply $\widehat{X}_L \to X$ in operator norm. Indeed, $(1-\widehat{\Pi}_L)K$ converges uniformly on the unit ball because its image under K is relatively compact. Apply the same argument to the adjoint compact operator to control $K(1-\widehat{\Pi}_L)$.

The operator $\widehat{X}_L$ commutes with $\widehat{\Pi}_L$. On its range it is X_L, and on its complement it is N. Uniform boundedness of B_J on $\mathcal{K}$ and the resolvent Neumann series therefore give a uniform inverse bound for $1+\widehat{X}_L\otimes \mathrm{ad}_J$ at large L. Restriction gives the required bound for $B_{J,L}$. The finitely many remaining polynomial inverses are bounded on $\mathcal{K}$, so one constant works for every L.

For the rate set $u = B_J f$ and $u_L = B_{J,L}\widehat{\Pi}_L f$. Projecting $M_J u = f$ and subtracting the finite equation gives

$$u_L - u = -(1-\widehat{\Pi}_L)u + B_{J,L}\widehat{\Pi}_L (X\otimes \mathrm{ad}_J)(1-\widehat{\Pi}_L)u. \tag{7.36}$$

The adapted spectral tail obeys

$$\|(1-\widehat{\Pi}_L)v\|_{\mathfrak{B}_s} \leq C_r (1+L)^{-r} \|v\|_{\mathfrak{B}_{s+r}}.$$

Apply it to u, using the finite inverse bound at index s and the full inverse bound at index $s+r$. This proves the rate for $\alpha = 0$.

Enlarge $\mathcal{K}$ by a fixed small compact complex neighborhood. The same rate holds uniformly there. Cauchy estimates in the finite-dimensional matrix entries give every fixed parameter derivative in (7.35). The inverses at different indices agree on smooth vectors, since the same norm-convergent Neumann series defines them. □

These estimates concern the actual linear inverses on the declared Hilbert scale. They will control smoothly smeared Gaussian contractions and their source derivatives. A local interacting quantum comparison still requires its cyclic counterterm domain, normal contacts, boundary data and prepared state.

Chapter 8

Quantum observables under elimination

The eliminated curve modes have a quadratic action on the Neumann slice. Integrating them therefore determines both a determinant and an inverse quadratic form. The determinant can cancel against the ghost determinant while the inverse form still contracts observables. This is the first quantum information that a classical residual projection loses.

We begin with that Gaussian calculation, then retain its contractions in the full observable map. Connected insertions give its higher components. Background variation gives a connection on them. The finite torus model and the first-jet boundary completion provide quantum master equations on explicit finite spaces. Their proofs also identify which additional constructions the smooth interacting theory requires.

We then remove the curve and surface cutoffs on smooth Gaussian observables, retaining their source-antifield corrections. The continuum determinant uses an ordinary Sobolev space.

8.1 The determinant and the covariance

Use the cyclic Neumann algebra E, trace τ_E, and residual projections p_0, p_1 of Chapter 2. Tensor with $\mathfrak{g} = \mathfrak{gl}_N(\mathbb{C})$ and write

$$V = \ker p_0, \qquad W = \ker p_1, \qquad d_J = \bar{\partial} + \eta_0 \operatorname{ad}_J .$$

Here V consists of matrix-valued functions and W of matrix-valued curve one-forms.

Theorem 4.4 supplies the inverse $h_J : W \to V$. Put $h = h_0$ and

$$K_J(v, w) = \tau_E \operatorname{tr}(v d_J w), \qquad M_J = h d_J|_V .$$

The forms K_J are skew. On the physical slice $a = \eta_0 J$, Proposition 4.6 identifies the eliminated quadratic action with K_J and the gauge derivative with M_J.

Choose a finite-dimensional subspace $V_K \subset V$ on which K_0 is nondegenerate. Let π_K be its K_0-orthogonal projection and let $K_{J,K}$ be the matrix of $K_J|_{V_K}$ in one fixed basis. Initially all expressions are formal in the entries of J at zero. The loop parameter $\hbar$ is independent of this completion.

Proposition 8.1. *The compressed gauge derivative is*

$$M_{J,K} = \pi_K M_J|_{V_K} = K_{0,K}^{-1} K_{J,K}. \tag{8.1}$$

Normalize the bosonic action $y^{\mathsf{T}} K_{J,K} x$ relative to its Gaussian at $J = 0$. Normalize the ghost Gaussian with the same matrix $M_{J,K}$. With the reference orientations fixed, their vacuum factors are

$$Z_{\mathrm{bos}} = \det(M_{J,K})^{-1}, \qquad Z_{\mathrm{gh}} = \det(M_{J,K}), \qquad Z_{\mathrm{bos}} Z_{\mathrm{gh}} = 1.$$

In the Wick convention with weight $e^{-y^{\mathsf T}K_{J,K}x/\hbar}$, the normalized bosonic expectation satisfies

$$\langle x_i y_j\rangle = \hbar (K_{J,K}^{-1})_{ij},$$
$$\langle x_i y_j x_k y_l\rangle - \langle x_i y_j\rangle\langle x_k y_l\rangle = \hbar^2 (K_{J,K}^{-1})_{il}(K_{J,K}^{-1})_{kj}. \tag{8.2}$$

Proof. Since $\bar\partial M_J = d_J$ on V, one has $K_0(v, M_J w) = K_J(v,w)$. Pairing with every $v \in V_K$ and using the definition of π_K proves (8.1). Its constant term is the identity. The block Hessian of the bosonic action has determinant equal to a fixed sign times $\det(K_{J,K})^2$. The square-root branch continued from the reference Gaussian gives the inverse determinant ratio. Berezin integration gives the ghost determinant ratio.

For commuting source columns F, G, completing the square gives

$$\left\langle e^{F^{\mathsf T}x + G^{\mathsf T}y}\right\rangle = \exp\bigl(\hbar F^{\mathsf T} K_{J,K}^{-1} G\bigr).$$

Two derivatives give the covariance. Four derivatives give two pairings. Subtraction of the product of expectations leaves the crossed pairing in (8.2). □

These formulas define a formal complex Gaussian, without a choice of positive real contour. Numerical evaluation requires $K_{J,K}$ to be invertible. The artificial singularity of Example 5.7 shows why invertibility of the full smooth operator does not settle that requirement for an arbitrary finite compression.

Example 8.2. On the round sphere take

$$f = \frac{z^2}{(1+|z|^2)^2}, \qquad g = \frac{\bar z}{1+|z|^2}.$$

Both have zero p_0 projection. With $r = |z|^2$, polar integration gives

$$\tau_E(f\,\bar\partial g) = -\kappa \int_0^\infty \frac{dr}{(1+r)^4} = -\frac{\kappa}{3}.$$

Thus on their span

$$K_{0,K} = \kappa \begin{pmatrix} 0 & -1/3 \\ 1/3 & 0 \end{pmatrix}, \qquad \langle x_1 y_2\rangle = \frac{3\hbar}{\kappa}.$$

The combined vacuum is one and this covariance is nonzero.

For a background-dependent example put

$$v = \frac{r}{1+r} - \frac{1}{2}, \qquad u = \frac{zr}{(1+r)^2}.$$

The two polar matrices on (v,u) are

$$\kappa^{-1}\bigl(\tau_E(v_i\,\bar\partial v_j)\bigr) = \begin{pmatrix} 0 & 1/6 \\ -1/6 & 0 \end{pmatrix}, \qquad \kappa^{-1}\bigl(\tau_E(\eta_0 v_i v_j)\bigr) = \begin{pmatrix} 1/12 & 0 \\ 0 & 0 \end{pmatrix}.$$

The nonzero entry of the second matrix is the integral $\int_0^\infty (r/(1+r) - 1/2)^2 (1+r)^{-2}\,dr = 1/12$. The other entries vanish by angular weight.

Tensor with $\mathfrak{gl}_2$ and take $J = j\,\mathrm{diag}(1,-1)$. In the order $E_{11}, E_{12}, E_{21}, E_{22}$, let G be the trace matrix and let $L = \mathrm{ad}_J$. Direct multiplication gives

$$M_{J,K} = \begin{pmatrix} 1 & 0 \\ L/2 & 1 \end{pmatrix}, \qquad K_{0,K} = \kappa \begin{pmatrix} 0 & G/6 \\ -G/6 & 0 \end{pmatrix}, \qquad K_{J,K}^{-1} = \kappa^{-1} \begin{pmatrix} 0 & -6G^{-1} \\ 6G^{-1} & 3LG^{-1} \end{pmatrix}. \tag{8.3}$$

Although $\det M_{J,K} = 1$, the last block has entries $6j/\kappa$ and $-6j/\kappa$. The covariance remembers the background.

The inverse quadratic form also has a smooth domain independent of finite compression. Let $U \subset \mathbb{R}^2$ be an oriented surface patch with coordinates (s,t) and let $J : U \to \mathfrak{g}$ be smooth. For smooth compactly supported maps $\phi, \gamma : U \to W$, define

$$F_\phi(x) = \int_U \tau_E \,\mathrm{tr}(x\phi)\, ds\, dt, \qquad G_\gamma(y) = \int_U \tau_E \,\mathrm{tr}(y\gamma)\, ds\, dt$$

and

$$C_J(\phi,\gamma) = \int_U \tau_E \,\mathrm{tr}\big((h_{J(s,t)}\gamma)\phi\big)\, ds\, dt.$$

These covectors are represented by the cyclic pairing.

Proposition 8.3. *The generating rule*

$$\mathcal{P}_J e^{F_\phi + G_\gamma} = e^{\hbar C_J(\phi,\gamma)}$$

defines a normalized Gaussian functional on polynomials in the displayed linear observables. Each Wick coefficient is continuous on the smooth test spaces with a common compact surface support.

Proof. The equation $d_J h_J \gamma = \gamma$ identifies $h_J\gamma$ with the inverse kinetic form on these covectors. The background Green map is continuous, with seminorm bounds uniform on compact matrix sets. The image of J on a common compact surface support is such a set. Continuity of multiplication and the polar trace therefore proves continuity of C_J. Every polynomial coefficient is a finite sum of products of this bilinear form. □

The surface delta distribution has been paired with tests before evaluation. The curve inverse still uses the global Green map. Products of coincident local insertions require a further extension of this domain.

8.2 Finite BV integration

Let $\mathcal{F} = \mathcal{F}' \oplus \mathcal{F}''$ be a finite-dimensional graded symplectic complex with pairing of degree -1, split into symplectic subcomplexes. Assume $\mathcal{F}''$ is acyclic. A cyclic strong deformation retract onto $\mathcal{F}'$ fixes a gauge-fixing Lagrangian in $\mathcal{F}''$ on which the free quadratic action is nondegenerate. Fix compatible constant reference semidensities. Integration on this Lagrangian, denoted $\mathcal{I}$, is the coefficientwise Gaussian integral, including Berezin integration in odd coordinates.

Write Δ, Δ' for the degree-one BV Laplacians on the two spaces. The odd bracket is normalized by

$$\Delta(fg) = (\Delta f)g + (-1)^{|f|} f\Delta g + (-1)^{|f|}\{f,g\}. \tag{8.4}$$

The constant semidensity identifies functions with semidensities. An even *master weight* is an element U of the integration domain satisfying $\Delta U = 0$. For $U = e^{I/\hbar}$ this is

$$\frac{1}{2}\{I, I\} + \hbar\Delta I = 0.$$

The weight $e^{-S/\hbar}$ corresponds to $I = -S$ in this convention.

We use polynomial observables completed in $\hbar$ and in a positive interaction parameter, with polynomial coefficients at each interaction order. Alternatively, interactions of field degree at least three use the completed field algebra $\widehat{\mathrm{Sym}}(\mathcal{F}^\vee)[[\hbar]]$. In the latter case the connected expansion is finite at each fixed field degree and loop order. These are perturbative completions. An analytic integration cycle requires its own convergence conditions.

Proposition 8.4. *Finite BV integration satisfies $\Delta' \mathcal{I} = \mathcal{I}\Delta$. Let $I = I_0 + I_{\mathrm{int}}$ be a degree-zero action with free quadratic part I_0 and no negative powers of $\hbar$. Require I_{int} to have positive interaction order in the polynomial completion, or field degree at least three in the field completion. For the master weight $U = e^{I/\hbar}$, assume $U' = \mathcal{I}(U)$ has invertible leading coefficient after the chosen Gaussian normalization, and put*

$$D = \hbar U^{-1}\Delta U, \qquad D' = \hbar (U')^{-1}\Delta' U', \qquad p(f) = (U')^{-1}\mathcal{I}(Uf). \tag{8.5}$$

Then $D^2 = (D')^2 = 0$, $D1 = D'1 = 0$, $p1 = 1$, and $D'p = pD$. Successive normalized integrations compose when each uses the effective weight produced by the preceding integration.

Proof. The Laplacian splits into retained and eliminated terms. Retained derivatives commute with the integral. Each eliminated term is an integrated total derivative. It vanishes by Gaussian integration by parts in even coordinates and by the Berezin derivative identity in odd coordinates.

This proves the Stokes relation and implies $\Delta' U' = 0$. The exponential weights give $D = \{I, -\} + \hbar\Delta$ and $D' = \{I', -\}' + \hbar\Delta'$, where $U' = e^{I'/\hbar}$ is interpreted in the same perturbative Gaussian domain. These operators preserve the stated observable completions. Conjugating the square-zero Laplacians proves the differential identities, and

$$D' p(f) = \hbar (U')^{-1}\Delta' \mathcal{I}(Uf) = \hbar (U')^{-1}\mathcal{I}\Delta(Uf) = pD(f).$$

For two integrations put $U_1 = \mathcal{I}_1 U$ and $U_2 = \mathcal{I}_2 U_1$, so the two intermediate factors cancel in

$$U_2^{-1}\mathcal{I}_2\big(U_1 U_1^{-1}\mathcal{I}_1(Uf)\big) = U_2^{-1}\mathcal{I}_2\mathcal{I}_1(Uf).$$

All identities are coefficientwise in the declared perturbative domain. □

In (8.5), $U^{-1}\Delta U$ denotes conjugation of operators. It is not multiplication by $U^{-1}\Delta(U)$. For this finite symplectic splitting, the perturbative pushforward is a quasi-isomorphism. Homological perturbation gives its inverse and homotopy.[1] The eliminated acyclic complex and the coefficient completion are part of this theorem's hypotheses.

For a free propagator P, include the factor $1/2$ in the contraction operator ∂_P. Normalize its sign by the free Stokes identity

$$(Q' + \hbar\Delta')\,\pi e^{\hbar\partial_P} = \pi e^{\hbar\partial_P}(Q + \hbar\Delta),$$

where π restricts to retained fields and Q, Q' are the free differentials. It follows by commuting Q through the exponential and using the contraction identity.

[1] Cattaneo–Mnev, *BV pushforward as a quasi-isomorphism*, Theorem 5.1 and Corollary 5.2, arXiv:2605.30558v2.

If an interaction I obeys $(Q+\hbar\Delta)e^{I/\hbar}=0$, its effective interaction is

$$I' = \hbar\log\bigl(\pi e^{\hbar\partial_P} e^{I/\hbar}\bigr). \tag{8.6}$$

The same intertwining proves the effective master equation. The logarithm selects connected graphs. A graph with v interaction vertices and e contractions has power $\hbar^{e-v+1}$, its loop number, and coefficient $1/|\operatorname{Aut}\Gamma|$ times its contracted vertex tensor. Thus (8.6) constructs interaction vertices as well as the unary differential.

Example 8.5. Let $a\neq 0$ and set

$$S(x,y)=\frac{a+\lambda y}{2}x^2+gxy^2+\frac{b}{2}y^2.$$

Treat g,λ formally. Completing the square and dividing the x integral by its reference Gaussian gives

$$S_{\mathrm{eff}}(y)=\frac{b}{2}y^2-\frac{g^2y^4}{2(a+\lambda y)}+\frac{\hbar}{2}\log\frac{a+\lambda y}{a}. \tag{8.7}$$

The conditional mean and variance are

$$p_y(x)=-\frac{gy^2}{a+\lambda y},\qquad p_y(x^2)-p_y(x)^2=\frac{\hbar}{a+\lambda y}.$$

The rational correction in (8.7) is its tree series. The logarithm is its one-loop series. Evaluation at the conditional mean recovers the former and loses the latter and the variance.

8.3 Connected insertions and their compositions

Let $p:A\to B$ be the normalized map of Proposition 8.4. The observable algebras A, B are graded commutative and $\hbar$-adically complete. Their differentials satisfy

$$D(fg)=(Df)g+(-1)^{|f|}fDg+(-1)^{|f|}\hbar\{f,g\}, \tag{8.8}$$

and the corresponding identity with primes on B.

For homogeneous inputs $f_1,\dots,f_n$, let $\mathrm{Part}(n)$ denote the partitions of $\{1,\dots,n\}$. Order each block by its input order and order blocks by their least element. The sign $\epsilon(\pi;f)$ is the Koszul sign of this reordering. Define the cumulant

$$\kappa_n^p(f_1,\dots,f_n)=\sum_{\pi\in\mathrm{Part}(n)}(-1)^{|\pi|-1}(|\pi|-1)!\,\epsilon(\pi;f)\prod_{C\in\pi}p\Bigl(\prod_{i\in C}f_i\Bigr). \tag{8.9}$$

Theorem 8.6. *In the finite perturbative Gaussian domain, with observables having no negative powers of $\hbar$, one has $\kappa_n^p\in\hbar^{n-1}B$. Hence the maps*

$$\Phi_n^p=\hbar^{1-n}\kappa_n^p$$

are defined over the original quantum power-series ring. Their first component is p, and every higher component vanishes when an input is the unit.

These maps give a coalgebra morphism for the degree-one coderivations whose unary parts are D, D' and whose binary parts are $b_2(f,g)=(-1)^{|f|}\{f,g\}$. In particular,

$$D'\Phi_2^p(f,g)-\Phi_2^p(Df,g)-(-1)^{|f|}\Phi_2^p(f,Dg)=(-1)^{|f|}\bigl(p\{f,g\}-\{pf,pg\}'\bigr). \tag{8.10}$$

For successive normalized integrations p, q, their full Taylor series compose: $\Phi^{qp}=\Phi^q\circ\Phi^p$.

Proof. Introduce formal graded parameters t_i with $|t_i| = -|f_i|$, so $F = \sum_i t_i f_i$ has degree zero. The partition formula is equivalent to

$$\Phi^p(F) = \hbar \log p(e^{F/\hbar}) = \sum_{n\geq 1} \frac{1}{n!} \Phi^p_n(F, \dots, F). \tag{8.11}$$

The logarithm retains connected components containing all marked observables. A connected graph with n marks, v interaction vertices and e contractions has $e \geq n + v - 1$. Its quantum power in the unscaled cumulant is $e - v$, proving divisibility by $\hbar^{n-1}$. Normalization removes components containing only interaction vertices.

At fixed interaction order the inputs are polynomials, so the contributing contractions are finite. The field-completed alternative is finite at each specified field degree and loop order.

For completeness, let $S^c(A) = \bigoplus_{n\geq 1} S^n(A)$ be the conilpotent symmetric coalgebra. The coalgebra automorphism τ_A whose Taylor coefficients are the products in A sums over partitions. Its inverse is finite on each tensor length.

Since D has order at most two, its conjugated coderivation has only unary part D and binary part $\hbar b_2$. The conjugate of $S^c(p)$ has Taylor coefficients (8.9). The relation $D'p = pD$ gives its coalgebra identity. Dividing the arity-n identity by $\hbar^{n-1}$ gives the displayed normalized coderivations and their morphism. Direct subtraction of (8.8) before and after p also proves (8.10).

For a scalar mark t_1, the generating function factors by $e^{t/\hbar}$. Its logarithm has no mixed coefficient with this mark, proving the unit assertion. Finally

$$e^{\Phi^p(F)/\hbar} = p(e^{F/\hbar})$$

and substitution into the next logarithm proves composition, including every partition coefficient and Koszul sign. □

The quantum division is performed before specializing to a quotient by $\hbar^r$. Knowing an unscaled cumulant only modulo $\hbar^r$ does not determine its divided component to that same precision.

Example 8.7. For one even Gaussian coordinate x with invertible stiffness A,

$$p_A(x^{2m}) = (2m-1)!!(\hbar/A)^m, \qquad p_A(x^{2m+1}) = 0.$$

Its BV complex adjoins η of degree -1, with $D(\eta f) = Axf - \hbar f'$ and $D(f) = 0$. Gaussian integration by parts proves its Stokes identity. The first nonzero connected maps are

$$\Phi_2(x,x) = A^{-1}, \qquad \Phi_3(x^2,x,x) = 2A^{-2}, \qquad \Phi_n(x^2,\dots,x^2) = 2^{n-1}(n-1)!\hbar A^{-n}. \tag{8.12}$$

The second formula is $\hbar^{-2}(p_A(x^4) - p_A(x^2)^2)$. For the last, expand $\log p_A(e^{tx^2}) = -\frac{1}{2}\log(1 - 2\hbar t/A)$. Thus the higher comparison remains nontrivial even when $p_A(x) = 0$.

8.4 Changing the covariance and lifting sources

Let $C^{ab}(J)$ be a symmetric matrix of formal functions of even parameters J. On polynomials in even coordinates x^a, define

$$p_J = \mathrm{ev}_{x=0} \exp\left(\frac{\hbar}{2} C^{ab}(J)\,\partial_a \partial_b\right).$$

Repeated indices are summed. On each polynomial the exponential contains finitely many contractions. Write d_{par} for exterior differentiation in the parameters.

Proposition 8.8. *The connection*

$$\nabla^{\mathrm{W}} = d_{\mathrm{par}} + \frac{\hbar}{2}(d_{\mathrm{par}} C^{ab})\,\partial_a \partial_b \tag{8.13}$$

is flat and obeys $d_{\mathrm{par}}(p_J F) = p_J \nabla^{\mathrm{W}} F$. *Its transport from* J_0 *to* J_1 *is*

$$F_{J_1} = \exp\left[-\frac{\hbar}{2}\big(C(J_1) - C(J_0)\big)^{ab}\,\partial_a \partial_b\right] F_{J_0}, \qquad p_{J_1} F_{J_1} = p_{J_0} F_{J_0}. \tag{8.14}$$

Proof. All constant-coefficient contraction operators commute. Differentiate their exponential to obtain (8.13). The exterior derivative of its one-form is zero and its commutator square is zero, proving flatness. Multiplication of the two commuting exponentials proves (8.14). □

For $C(J) = (1 + J)^{-1}$, the quadratic expectation has derivative $\partial_J p_J(x^2) = -\hbar/(1 + J)^2$. The connection accounts for this term. A varying field frame also has the cotangent source shift and the semidensity factor of (5.24) and (5.25). These data enter the same observable comparison, with different roles.

A finite gauge quartet displays the source shift directly. Let J be even, with conjugate J^+ of degree -1. Take even coordinates x, b and odd coordinates c, $\bar{c}$ of degrees 1, -1. For vector-valued coordinates all following expressions sum over components. Set

$$Q_0 x = c, \qquad Q_0 \bar{c} = b, \qquad Q_0 b = Q_0 c = Q_0 J = 0.$$

Adjoin cotangent antifields q^+ of degree $-1 - |q|$. Fix $\{q, q^+\} = 1$, use left odd derivatives, and set

$$\Delta = \sum_q (-1)^{|q|}\,\partial_q \partial_{q^+}, \qquad S_0 = \sum_i (x_i^+ c_i - \bar{c}_i^+ b_i).$$

This convention gives $\{S_0, q\} = Q_0 q$. For an invertible formal matrix $M(J)$ with $M(0) = 1$, let

$$\Psi = \bar{c} M(J) x, \qquad Q_0 \Psi = b M(J) x - \bar{c} M(J) c.$$

Proposition 8.9. *The canonical transformation* $T = \exp(-\{\Psi, -\})$ *has unit Berezinian, commutes with* Δ, *and sends* S_0 *to* $S_\Psi = S_0 + Q_0 \Psi$. *Both actions satisfy the quantum master equation. For* $D_\Psi = \{S_\Psi, -\} + \hbar\Delta$, *the transformed source*

$$T(J^+) = J^+ - \bar{c} M'(J) x \tag{8.15}$$

is closed. For several parameters the corresponding term is $\bar{c}(\partial_{J_a} M) x$, *and* T *transports every polynomial in their conjugates.*

Proof. The cotangent lift of Q_0 squares to zero and has zero divergence, so $\{S_0, S_0\} = \Delta S_0 = 0$. The fermion has no antifields. Thus $\Delta\Psi = 0$ and its Hamiltonian flow commutes with Δ. The flow fixes fields and translates antifields, giving a triangular Jacobian with identity diagonal. The transformation of S_0 terminates at its first commutator because $Q_0 \Psi$ has no antifields. This proves the action and master equations.

The derivative of $Q_0 \Psi$ with respect to J is $b M' x - \bar{c} M' c$. It equals $Q_0(\bar{c} M' x)$, whereas $\{S_\Psi, J^+\} = \partial_J (Q_0 \Psi)$. Thus the Hamiltonian differential kills (8.15). Its Laplacian is zero as well. More generally, T intertwines the two quantum differentials and is an algebra map. The original polynomials in the J_a^+ are closed, proving the last assertion. □

8.5 A homotopy for a changing master weight

The source correction is a special case of a normalized variation. Keep a finite BV integration map $\mathcal{I}$ and the Laplacians fixed. Let t be a parameter and let R_t have degree -1. Assume a family of master weights satisfies

$$\partial_t U = \Delta(R_t U), \qquad \Delta U = 0. \tag{8.16}$$

Use the normalized maps of (8.5) and put $r_t = p_t(R_t)$. Stokes gives $\partial_t U' = \Delta'(r_t U')$.

For multiplication m_R, set $L_R = [\Delta, m_R]$, the graded commutator. Define degree-zero operators

$$v_A = U^{-1} L_R U - m_{U^{-1}\partial_t U}, \qquad v_B = (U')^{-1}[\Delta', m_r]U' - m_{(U')^{-1}\partial_t U'}.$$

An order-two operator commuted with multiplication has order at most one. Subtracting its value at 1 therefore makes each v a derivation. Moreover $\partial_t D = [v_A, D]$ and $\partial_t D' = [v_B, D']$, by (8.16) and $\Delta^2 = (\Delta')^2 = 0$.

Proposition 8.10. *The degree-minus-one map*

$$h_t(f) = \hbar^{-1}\big(p_t(R_t f) - p_t(R_t)\,p_t(f)\big)$$

satisfies

$$\partial_t p_t + p_t v_A - v_B p_t = D' h_t + h_t D, \qquad h_t 1 = 0. \tag{8.17}$$

If R_t has no negative quantum powers in the perturbative Gaussian domain, the division by $\hbar$ is integral.

Proof. On unnormalized weights put $K = \mathcal{I} m_R - m_r \mathcal{I}$. The Stokes relation gives

$$\mathcal{I} L_R - L_r \mathcal{I} = \Delta' K + K \Delta.$$

Differentiate $p_t = (U')^{-1}\mathcal{I} U$. In $\partial_t p_t + p_t v_A - v_B p_t$ the two logarithmic derivatives cancel, leaving the conjugate of the left side. The right side becomes $D' h_t + h_t D$ because $h_t = \hbar^{-1}(U')^{-1} K U$. Normalization gives $K(U) = 0$. The numerator of h_t is a two-input cumulant, so the integrality follows from Theorem 8.6. □

The homotopy retains higher insertions as well. Write F_p for the undivided cumulant coalgebra map, and Q_A, Q_B for the coderivations with unary parts D, D' and binary parts $\hbar b_2$.

Let $S^c(p_t; h_t)$ apply h_t to one factor and p_t to every other factor, summed with the graded tensor signs. Define

$$\mathcal{H}_t = \tau_B^{-1} S^c(p_t; h_t)\tau_A.$$

Derivations commute with the multiplication coalgebra maps. Conjugating (8.17) gives

$$\partial_t F_p + F_p \widetilde{v}_A - \widetilde{v}_B F_p = Q_B \mathcal{H}_t + \mathcal{H}_t Q_A,$$

where $\widetilde{v}$ is the unary coderivation induced by v. The arity-n coefficient of $\mathcal{H}_t$ is

$$(\mathcal{H}_t)_n(f_1, \dots, f_n) = \hbar^{-1}\kappa_{n+1}^{p_t}(R_t, f_1, \dots, f_n). \tag{8.18}$$

Indeed, differentiating a partition term replaces one block expectation by its connected expectation with R_t. The first summands enumerate partitions in which the block of R_t is not a singleton. For a partition into k other blocks, the k subtracted summands contribute $(-1)^k k!$, the coefficient of the additional singleton block. This proves the formula, with R_t placed first to fix the odd signs.

Connected graphs now have $n+1$ marks, so $(\mathcal{H}_t)_n \in \hbar^{n-1} B$. Its divided coefficients give the homotopy of the normalized higher maps. For two successive eliminations, the undivided coalgebra formula is

$$\mathcal{H}^{qp} = F_q \mathcal{H}^p + \mathcal{H}^q F_p,$$

where F_p, F_q are the cumulant coalgebra maps. This follows by differentiating $F_{qp} = F_q F_p$, and is associative under three eliminations.

A higher parameter simplex requires a total master weight satisfying its parameter-dependent master equation. Separate paths alone do not provide that datum.

8.6 Finite surface actions and the torus

Let ℓ_n be the cyclic L_∞ operations on $E_K \otimes \mathfrak{gl}_N$ obtained in Theorem 5.5. Their degrees are $2 - n$. For an oriented surface U, tensor with $\Omega^\bullet(U)$, placing surface forms first, and include d_U in the unary differential. Write $\langle x, y\rangle = \tau_E \operatorname{tr}(xy)$ for the internal pairing, extended with the tensor signs.

For a total-degree-one superfield A, define

$$S_{K,U} = \frac{1}{2}\int_U \langle \mathsf{A}, (d_U + d_E)\mathsf{A}\rangle + \sum_{n\geq 2} \frac{1}{(n+1)!}\int_U \langle \mathsf{A}, \ell_n(\mathsf{A}^{\otimes n})\rangle. \tag{8.19}$$

The series is completed in field amplitude. Its odd symplectic form and cohomological vector field are

$$\omega_{K,U} = \frac{1}{2}\int_U \langle \delta\mathsf{A}, \delta\mathsf{A}\rangle, \qquad Q\mathsf{A} = (d_U + d_E)\mathsf{A} + \sum_{n\geq 2}\frac{1}{n!}\ell_n(\mathsf{A}^{\otimes n}).$$

Proposition 8.11. *For closed U, or compact support in its interior, (8.19) satisfies the classical master equation. If U has boundary, its cohomological vector field and field-space symplectic form satisfy*

$$\iota_Q\omega_{K,U} = \delta S_{K,U} + \pi^*\alpha_{K,\partial U}, \qquad \alpha_{K,\partial U} = \frac{1}{2}\int_{\partial U}\langle \mathsf{A}, \delta\mathsf{A}\rangle,$$

where π is boundary restriction. Internal transfer preserves surface support and commutes with restriction. The boundary terms cancel on an interface with opposite induced orientations.

Proof. Cyclicity makes the variation of each $(n+1)$-field term equal to its Hamiltonian contraction with $\ell_n/n!$. The transferred identities give $Q^2 = 0$. Integration by parts in d_U leaves the displayed boundary primitive. The other operations use only internal coefficients and commute with d_U and restriction, with their Koszul signs. Stokes' theorem supplies the opposite interface signs. □

To obtain a finite quantum model, replace the surface complex by its specified Frobenius algebra

$$B = H^*(T^2;\mathbb{C}) = \Lambda(a,b), \qquad |a| = |b| = 1, \qquad \tau_B(ab) = 1.$$

Its differential and its other basis traces are zero. The shifted field space

$$\mathcal{F}_K = (B \otimes E_K \otimes \mathfrak{gl}_N)[1]$$

has dimension $(8 + 16K)N^2$. The pairing $\tau_B\tau_E$ tr induces an odd symplectic form. Fix a constant Darboux semidensity, its Laplacian Δ_K, and the action S_K obtained from (8.19) by replacing integration over U with τ_B.

Theorem 8.12. *The action obeys*

$$\frac{1}{2}\{S_K, S_K\} + \hbar\Delta_K S_K = 0.$$

For $K \leq L$, normalized finite BV integration of the acyclic complement of $\mathcal{F}_K$ in $\mathcal{F}_L$ gives S_K with no loop correction to its action. These integrations compose.

Proof. For a homogeneous Frobenius basis (e_i), use right trace duals $\tau_B(e_j e^i) = \delta^i_j$. Closing two inputs of a multiplication tensor inserts the Euler element

$$\mathfrak{e}_B = \sum_i (-1)^{|e_i|} e_i e^i.$$

The basis $(1, a, b, ab)$ has dual $(ab, b, -a, 1)$, so

$$\mathfrak{e}_B = ab - ab - ab + ab = 0. \tag{8.20}$$

Each vertex of S_K factors into a multiplication tensor of B and an internal cyclic tensor. Applying Δ_K closes two inputs and therefore gives zero. The classical master equation follows from the transferred identities, proving the quantum one.

The eliminated propagator is the identity on B tensored with the internal complement homotopy. In a connected interaction graph, contract a spanning tree on the B factors. The Frobenius dual-basis identity joins the multiplication tensors into one. Every remaining edge inserts $\mathfrak{e}_B$. Thus every graph with a loop vanishes.

The surviving trees are exactly the nested transfer of (5.4). Normalizing the free Gaussian to one and using coefficientwise integration proves composition. □

Open contractions survive, since $\tau_B(ab) = 1$. They enter the cumulant maps of Theorem 8.6. For a closed genus-g surface, the Euler element of its cohomology algebra is $(2 - 2g)$ times the normalized top class. Thus the loop cancellation uses the chosen torus coefficient algebra.

The full surface heat propagator imposes a second restriction. On a flat patch, let L_a, L_b denote wedge multiplication by the parallel one-forms and let ι_a, ι_b be the dual exterior contractions. Pure wedge multiplication on $\Lambda(a, b)$ has supertrace zero: positive-degree maps are strictly upper triangular, and the identity has supertrace $1 - 2 + 1 = 0$.

Exterior contractions change this trace:

$$\operatorname{Str}(L_a L_b \iota_b \iota_a) = 1. \tag{8.21}$$

Indeed, in the order $1, a, b, ab$ this operator is the projection onto ab. The adjoint surface differential in the full heat propagator contains such exterior contractions. The Euler argument therefore does not cancel every coefficient tensor of its loops. The spacetime kernels and the full graph sum are still needed to compute an anomaly.

8.7 The compensating first-jet field

The first-jet boundary in (4.17) permits an explicit enlargement of the gauge group. Work in the formal normalized-frame neighborhood of the polar connection a. Its frame $H(a)$ obeys $\bar\partial H = -aH$, $H(o) = I_N$, and its residual is $\mathcal{J}(a) = -j(H(a))$. For a formal smooth gauge k whose amplitude-zero coefficient is an invertible constant matrix, put $k_0 = k(o)$ and $k_1 = j(k)$, without imposing $k_1 = 0$. Frame uniqueness gives $H(a^k) = k^{-1} H(a) k_0$, and hence

$$\mathcal{J}(a^k) = k_0^{-1} \mathcal{J}(a) k_0 + k_0^{-1} k_1. \tag{8.22}$$

Here the derivative of k^{-1} is $-k_0^{-1} k_1 k_0^{-1}$, fixing the affine sign.

Introduce a degree-zero edge matrix ϕ with the same affine law. Then

$$X = \mathcal{J}(a) - \phi, \qquad X^k = k_0^{-1} X k_0, \qquad O_r^{\mathrm{rel}} = \frac{1}{r} \operatorname{tr}(X^r) \quad (r \geq 1).$$

For formal ϕ of positive amplitude, the smooth global gauge $k = e^{-f_0\phi}$ has $k_0 = I_N$, $k_1 = -\phi$ and sends ϕ to zero. The subgroup preserving $\phi = 0$ is exactly the Neumann group. The edge field compensates the translation part of $\mathrm{GL}_N(\mathbb{C}[\xi]/\xi^2)$. It is different from the coefficient a_1 of the boundary superconnection in (4.22).

The finite residual complex has even matrices J, ϕ and odd ghost matrices c_0, c_1, of degrees 1, 1. Its algebra is polynomial in even entries and exterior in odd entries. Define the degree-one derivation

$$\begin{aligned} QJ &= c_1 + [J, c_0], & Q\phi &= c_1 + [\phi, c_0], \\ Qc_0 &= -c_0^2, & Qc_1 &= -c_0c_1 - c_1c_0. \end{aligned} \tag{8.23}$$

Matrix multiplication includes the exterior signs of ghost entries. The comparison with smooth fields sets $J = \mathcal{J}(a)$.

Theorem 8.13. *The derivation Q squares to zero. In the polynomial coordinates*

$$X = J - \phi, \qquad \gamma = c_1 + [\phi, c_0]$$

it is

$$QX = [X, c_0], \quad Qc_0 = -c_0^2, \quad Q\phi = \gamma, \quad Q\gamma = 0.$$

Let p kill ϕ, γ, let i include the retained Chevalley–Eilenberg complex

$$C_N = \mathbb{C}[X_{ij}] \otimes \Lambda((c_0)_{ij}),$$

and let N_{pair} count the added variables. On positive pair degree set

$$h = N_{\mathrm{pair}}^{-1} \sum_{ij} \phi_{ij}\, \partial_{\gamma_{ij}},$$

with $h = 0$ in degree zero. Then $Qh + hQ = 1 - ip$.

Proof. The graded Leibniz rule verifies $Q^2 = 0$ on the four matrix generators. It gives the displayed equations after the coordinate change, whose inverse is $J = X + \phi, c_1 = \gamma - [\phi, c_0]$. With left odd derivatives, the anticommutator of $\sum \gamma_{ij}\partial_{\phi_{ij}}$ and $\sum \phi_{ij}\partial_{\gamma_{ij}}$ is N_{pair}. The retained differential involves only X, c_0 and graded-commutes with the added homotopy. Division on each positive eigenspace proves the identity. □

The retained complex contains its full ghost cohomology. Replacing it by invariant functions of X would discard part of this retraction. The first invariant has the particularly simple form

$$O_1^{\mathrm{rel}} = \kappa^{-1}\tau_E(\operatorname{tr} a) - \operatorname{tr}\phi. \tag{8.24}$$

To see this, apply polar Stokes to $\bar\partial \log\det H = -\operatorname{tr} a$ and use $j(\log\det H) = -\operatorname{tr}\mathcal{J}(a)$. Both terms on the right of (8.24) have variation $\operatorname{tr} c_1$ under the enlarged gauge group, so their difference is closed.

8.8 The finite edge master equation and local primitives

Adjoin antifields q^+ of degree $-1 - |q|$ to every coordinate of (8.23). Use the cotangent bracket and Laplacian of Proposition 8.9. The cotangent Hamiltonian is

$$S_Q = \sum_q (-1)^{|q|} q^+ Qq, \qquad \{S_Q, q\} = Qq.$$

Theorem 8.14. *The coordinate change of Theorem 8.13 and its cotangent lift have Berezinian one. They split S_Q into the retained cotangent Hamiltonian and*

$$S_{\mathrm{pair}} = \sum_{ij} \phi^+_{ij}\gamma_{ij}.$$

Both $\{S_Q, S_Q\}$ and ΔS_Q vanish. For every conjugation-invariant polynomial or formal field-only potential $V(X)$,

$$\frac{1}{2}\{S_Q + V, S_Q + V\} + \hbar\Delta(S_Q + V) = 0. \tag{8.25}$$

Normalized BV integration of the separated added pair leaves the retained potential unchanged.

Proof. The even change $J = X + \phi$ is a determinant-one shear. The odd change $c_1 = \gamma - [\phi, c_0]$ is triangular with identity diagonal. Their base Berezinian, and hence that of the cotangent lift, is one. The split differential determines S_{pair} and also gives $Q\gamma^+ = \phi^+$, $Q\phi^+ = 0$.

The cotangent master bracket is the lift of $2Q^2$, so vanishes. For the Laplacian, the adjoint action on $\mathfrak{gl}_N$ has trace zero. The first-jet Lie algebra has adjoint matrix

$$\begin{pmatrix} \mathrm{ad}_x & 0 \\ \mathrm{ad}_y & \mathrm{ad}_x \end{pmatrix}$$

at $x + \xi\, y$, and therefore also has trace zero. These identities give zero divergence for the even representation and ghost parts of (8.23). The affine terms have zero divergence. Thus $\Delta S_Q = 0$.

The potential has no antifields, so $\Delta V = \{V, V\} = 0$. Invariance gives $\{S_Q, V\} = 0$, completing the master equation.

On the added Lagrangian $\phi = \gamma^+ = 0$, the remaining coordinates ϕ^+, γ are odd and have the nondegenerate quadratic pairing $\phi^+\gamma$. Orient each Berezin integral by $\int \phi^+\gamma = 1$ and normalize it by $\hbar$. Then $\hbar \int e^{\phi^+\gamma/\hbar} = 1$. The product over entries is one and the retained weight factors out. □

The contraction also gives supported primitives in a precisely defined local algebra. On a surface U, take polynomial densities in finitely many derivatives of the retained fields, the added fields and all their antifields. For a derivative multi-index I, write $\phi_I, \gamma_I, \phi^+_I, \gamma^+_I$ for the added jets. Prolongation of the differential gives

$$Q\phi_I = \gamma_I, \quad Q\gamma_I = 0, \quad Q\gamma^+_I = \phi^+_I, \quad Q\phi^+_I = 0.$$

Let N_{add} count these four sorts of variables, and let p kill them. Define

$$h_0 = \sum_{I,i,j} \left(\phi_{I,ij}\, \partial_{\gamma_{I,ij}} + \gamma^+_{I,ij}\, \partial_{\phi^+_{I,ij}} \right), \qquad h = N_{\mathrm{add}}^{-1} h_0$$

on positive added degree, and set $h = 0$ on degree zero.

Theorem 8.15. *On these densities and their quotient by total surface derivatives, one has $Qh + hQ = 1 - i\,p$. The maps preserve surface support and maximal derivative order. They act coefficientwise on formal series of finite jet order in each coefficient. They also extend to a fixed normal-current coefficient complex with defined products and total Koszul signs. Consequently a defined local residual $\mathcal{A}$ satisfying $Q\mathcal{A} = 0$ and $p\mathcal{A} = 0$ has primitive*

$$\mathcal{A} = Q(h\mathcal{A}). \tag{8.26}$$

Proof. Each field pair and each antifield pair contributes its Euler operator to $Qh_0 + h_0Q$. The retained differential uses only retained jets, so graded-commutes with h_0. Division by the positive count proves the identity.

A total derivative shifts each pair to the next jet pair and commutes with h_0 and the count. The identity therefore descends modulo total derivatives. No inverse differential operator or new derivative is introduced. Coefficient functions remain at the same surface point. This proves the support and derivative-order assertions.

Tensoring with the specified coefficient complex preserves the identity. Apply it to the closed residual with zero retained projection to obtain (8.26). □

For a compactly supported smooth density ρ,

$$\int_U \rho \ \operatorname{tr} c_1 = Q\Big(\int_U \rho \ \operatorname{tr} \phi\Big).$$

The same homotopy leaves a retained component $p\mathcal{A}$ untouched. Its coefficient and cohomology class must be computed in the retained theory.

8.9 A retained current anomaly

The first-jet boundary has an algebraic deformation that makes this last restriction concrete. Let

$$R = \mathbb{C}[\sigma, \nu], \qquad B_\sigma = R[\zeta]/(\zeta^2 - \sigma), \qquad \tau(1) = 0, \quad \tau(\zeta) = \kappa \in \mathbb{C}^\times.$$

Here σ, ν have degree zero and ζ is a coefficient coordinate, distinct from a surface or current coordinate. The fiber at $\sigma = 0$ is the first-jet Frobenius algebra.

Let $\mathfrak{g}$ be a finite-dimensional complex Lie algebra with nondegenerate invariant symmetric form κ_0, and set $\mathfrak{d}_\sigma = \mathfrak{g} \otimes B_\sigma$. Writing $(x, v) = x + \zeta v$, its bracket and form are

$$\begin{aligned} [(x, v), (y, w)] &= ([x, y] + \sigma[v, w],\ [x, w] + [v, y]), \\ B_0((x, v), (y, w)) &= \kappa\big(\kappa_0(x, w) + \kappa_0(v, y)\big). \end{aligned} \tag{8.27}$$

The subalgebra $\mathfrak{l} = R \otimes_{\mathbb{C}} \mathfrak{g}$, embedded through the coefficient 1, is Lagrangian. Denote the Killing form by $K_{\mathfrak{g}}(x, y) = \operatorname{Tr}_{\mathfrak{g}}(\operatorname{ad}_x \operatorname{ad}_y)$.

For an invariant level k, the affine current algebra has modes

$$[J_m(X), J_n(Y)] = J_{m+n}([X, Y]) + m\delta_{m+n,0}k(X, Y)\mathbf{1}.$$

Induce its vacuum module from the one-dimensional module on which all nonnegative modes vanish and the central element acts as one. Ordered negative modes give its PBW basis. Choose a basis t_a of $\mathfrak{l}$ with structure constants $f_{ab}{}^c$, and odd ghosts c^a, b_a of degrees 1, −1 with $b_a(w)c^b(w') \sim \delta_a^b/(w - w')$. The current coordinate is w and the BRST residue is

$$Q_{\mathrm{cur}} = \operatorname{Res}_w\Big(:c^a J_a: - \frac{1}{2} f_{ab}{}^c :c^a c^b b_c:\Big). \tag{8.28}$$

Proposition 8.16. *The invariant level*

$$k_\sigma = \nu B_0 - \frac{1}{2} K_{\mathfrak{d}_\sigma} = \begin{pmatrix} -K_{\mathfrak{g}} & \nu\kappa\kappa_0 \\ \nu\kappa\kappa_0 & -\sigma K_{\mathfrak{g}} \end{pmatrix} \tag{8.29}$$

defines a universal affine vacuum algebra free over R. On its tensor product with the ghosts, $Q_{\mathrm{cur}}^2 = 0$ over R and in every fiber. After $\sigma = a^2$ and inversion of a, the two separated levels are

$$k_\pm = \pm\frac{\nu\kappa}{2a}\kappa_0 - \frac{1}{2}K_{\mathfrak{g}}.$$

Proof. The adjoint matrix of (x, v) is

$$\begin{pmatrix} \operatorname{ad}_x & \sigma \operatorname{ad}_v \\ \operatorname{ad}_v & \operatorname{ad}_x \end{pmatrix}.$$

The trace of a product gives $K_{\mathfrak{d}_\sigma}((x,v),(y,w)) = 2K_{\mathfrak{g}}(x,y) + 2\sigma K_{\mathfrak{g}}(v,w)$. This proves (8.29) and its restriction $k_\sigma|_{\mathfrak{l}} = -K_{\mathfrak{g}}$. Invariance proves the Jacobi identity of the mode algebra. PBW and induction give the free vacuum module without inverting either parameter.

Compute the two residues of the product of (8.28) with itself. The current double pole contributes $\frac{1}{2}k|_{\mathfrak{l},ab} \operatorname{Res} :\partial_w c^a c^b:$. The terms containing a current and two ghosts cancel against the mixed contractions. The remaining single contractions cancel by the Lie Jacobi identity.

The two ghost contractions give the adjoint trace, with contribution $\frac{1}{2}K_{\mathfrak{g},ab} \operatorname{Res} :\partial_w c^a c^b:$. Thus, in these conventions,

$$Q^2_{\mathrm{cur}} = \frac{1}{2}(k|_{\mathfrak{l}} + K_{\mathfrak{g}})_{ab} \operatorname{Res} :\partial_w c^a c^b:. \tag{8.30}$$

The chosen level makes this zero. Finally the idempotents $(1 \pm \zeta/a)/2$ have traces $\pm\kappa/(2a)$ and separate the two adjoint representations. They give the stated levels on the split fiber. □

At the bare level νB_0, the restriction to $\mathfrak{l}$ vanishes and leaves the ghost term. For $\mathfrak{g} = \mathfrak{sl}_2$ with $[h,e] = 2e$, $[h,f] = -2f$, one has $K_{\mathfrak{g}}(h,h) = 8$. Set $c^h = \eta_+ w + \eta_- w^{-1}$ with independent odd coefficients. Then

$$\operatorname{Res}(\partial_w c^h) c^h = 2\eta_+\eta_-, \qquad Q^2_{\mathrm{cur}} = 8\eta_+\eta_- \neq 0.$$

This polynomial in retained ghosts survives the projection that kills the added edge pairs. The zero-projection hypothesis of Theorem 8.15 therefore fails. The half-Killing level supplies the algebraic correction.

Its polynomial dependence on ν does not identify ν with an inverse physical coupling at zero coupling. That normalization still requires the native field-theory comparison.

8.10 Smooth Gaussian Ward identities

The smooth functional of Proposition 8.3 is defined without a cutoff. We now obtain it from finite cyclic spaces, with control of background derivatives. This comparison concerns the quadratic complement in a prescribed smooth residual field.

Use the staggered spaces $\widehat{V}_L$, projections $\widehat{\Pi}_L$, and operators X_L of Chapter 6. Put

$$M_{J,L} = 1 + X_L \otimes \operatorname{ad}_J, \qquad h_{J,L} = M_{J,L}^{-1} \widehat{\Pi}_L h.$$

The inverse exists for every finite matrix J. If $a \in \widehat{V}_L \otimes \mathfrak{g}$ and $\gamma \in W$, then

$$K_{J,L}(a, h_{J,L}\gamma) = \tau_E \operatorname{tr}(a\gamma). \tag{8.31}$$

Indeed, $K_{J,L}(a,v) = K_0(a, M_J v)$ and $\widehat{\Pi}_L M_J h_{J,L}\gamma = \widehat{\Pi}_L h\gamma$. Adjointness of the projection for K_0 proves the identity. Thus $h_{J,L}$ inverts the restricted kinetic form on the smooth covectors represented by this pairing.

For the compactly supported tests ϕ, γ used above, define

$$C_{J,L}(\phi,\gamma) = \int_U \tau_E \operatorname{tr}\big((h_{J(s,t),L}\gamma)\phi\big)\, ds\, dt.$$

All tests have common compact surface support. Backgrounds and variations range over bounded smooth sets, with the background values in a fixed compact subset of $\mathfrak{g}$.

Theorem 8.17. *At every fixed polynomial degree, the normalized Wick functionals*

$$\mathcal{P}_{J,L} e^{F_\phi + G_\gamma} = e^{\hbar C_{J,L}(\phi,\gamma)}$$

converge to $\mathcal{P}_J$, rapidly in L on the stated bounded sets. The convergence holds after every fixed finite number of background variations and surface derivatives. For fixed smooth matrix variations A, B,

$$\begin{aligned} \partial_A h_J &= -h_J \eta_0 \operatorname{ad}_A h_J, \\ \partial_B \partial_A h_J &= h_J \eta_0 \operatorname{ad}_B h_J \eta_0 \operatorname{ad}_A h_J + h_J \eta_0 \operatorname{ad}_A h_J \eta_0 \operatorname{ad}_B h_J. \end{aligned} \tag{8.32}$$

The same identities hold for $h_{J,L}$ and converge to these formulas. The matrices A, B, J need not commute.

Proof. Apply Theorem 7.18 to $h\gamma$. Its adapted Hilbert norms control ordinary Sobolev norms, and the polar pairing is continuous on the smooth core. For $s \geq 3$, Sobolev multiplication and $\tau_E(\eta) = \kappa j(G\eta)$ give

$$|\tau_E \operatorname{tr}(v\phi)| \leq C_s \|v\|_{H^s} \|\phi\|_{H^s}.$$

Integration over the fixed compact support preserves the estimate. Surface derivatives act on the tests and on the finite matrix parameters of the inverse. The theorem already controls every fixed number of those derivatives. Each Wick coefficient is a finite sum of products of covariances, so it has the same rapid convergence, with a larger seminorm bound.

Differentiating $M_J M_J^{-1} = 1$ gives

$$\partial_A M_J^{-1} = -M_J^{-1}(X \otimes \operatorname{ad}_A) M_J^{-1}.$$

Use $X = h\eta_0$ and $h_J = M_J^{-1} h$ to obtain the first response. A second differentiation gives both ordered terms in (8.32). The finite projection is independent of J, so the same calculation applies at finite L. The parameter-derivative estimates identify their limits. □

For a finite list of smooth smearings, write x_a, y_b for the resulting even linear observables and $C_{ab}(J)$ for their covariance without $\hbar$. On their polynomial algebra,

$$\mathcal{P}_J = \operatorname{ev}_0 \exp\left(\hbar \sum_{a,b} C_{ab}(J) \partial_{x_a} \partial_{y_b} \right).$$

The corresponding Ward connection is

$$\nabla^{\mathrm{W}}_A = \partial_A + \hbar \sum_{a,b} (\partial_A C_{ab}) \partial_{x_a} \partial_{y_b}, \qquad \partial_A(\mathcal{P}_J F) = \mathcal{P}_J \nabla^{\mathrm{W}}_A F. \tag{8.33}$$

Mixed parameter derivatives and the constant-coefficient contraction operators commute. The connection is therefore flat for fixed parameter directions, as in Proposition 8.8. It is compatible with changing or enlarging the list of smearings, so defines a linear connection on the smooth cylindrical polynomial core. Its second-order term retains the contractions in the observable map. The connection is not a derivation of the ordinary product.

8.10.1 Surface cells and source antifields

The quadratic complement has no surface derivatives. This permits an independent finite surface regulator. Choose a bounded oriented coordinate rectangle $R \Subset U$ containing the support of the tests, and partition it into cells Q_ν of diameter at most a. Write $w_\nu = \int_{Q_\nu} d\sigma > 0$ and choose $\sigma_\nu \in Q_\nu$.

At each cell take the quartet of Proposition 8.9. Its even fields are x_ν, b_ν, and its odd fields c_ν, $\bar{c}_\nu$ have degrees 1, −1. Their components lie in the finite curve space and its dual. The reference kinetic pairing identifies b_ν with the second surface component. The quadratic action and gauge fermion are

$$S_{J,L,a} = \sum_\nu w_\nu \big(b_\nu M_{J(\sigma_\nu),L} x_\nu - \bar{c}_\nu M_{J(\sigma_\nu),L} c_\nu\big),$$
$$\Psi_{J,L,a} = \sum_\nu w_\nu \bar{c}_\nu M_{J(\sigma_\nu),L} x_\nu. \tag{8.34}$$

Use ordinary Darboux brackets in each cell and the reference Berezin orientation for which, with $M = M_{J(\sigma_\nu),L}$,

$$\mathcal{P}(x_{\nu,i} b_{\nu,j}) = \hbar w_\nu^{-1} (M^{-1})_{ij}, \qquad \mathcal{P}(\bar{c}_\nu B c_\nu) = \hbar w_\nu^{-1} \operatorname{tr}(BM^{-1}).$$

Different cells have no contractions. Smooth smearings use the weights w_ν, so their covariance is the Riemann sum for $C_{J,L}$.

Before evaluating the residual matrices at $J(\sigma_\nu)$, give each J_ν its ordinary cotangent antifield J_ν^+. For a prescribed smooth direction A, put

$$J_A^+ = \sum_\nu \langle A(\sigma_\nu), J_\nu^+ \rangle, \qquad A_{\nu,L} = X_L \otimes \mathrm{ad}_{A(\sigma_\nu)}\,.$$

The gauge fermion has the closed source lift

$$\mathcal{L}_{A,L,a} = J_A^+ - \sum_\nu w_\nu \bar{c}_\nu A_{\nu,L} x_\nu. \tag{8.35}$$

The same cell weights occur in the action, fermion and source correction. Closure follows by the canonical transformation of Proposition 8.9 before any limit.

For a fixed finite parameter family $J + \sum_i z_i A_i$, the same formula uses the retained antifield z_i^+ in place of $J_{A_i}^+$. We keep this residual coefficient algebra fixed as the cells are refined.

The quartet cylindrical core consists of polynomials in smoothly smeared x, b, c, $\bar{c}$ and unintegrated residual parameters and their antifields. Eliminated cotangent antifields and unsmeared coincident surface composites are excluded. Let $\mathcal{P}_J^{\mathrm{q}}$ be the Wick functional on this core with continuum covariance C_J, the displayed ghost signs, and the identity on residual variables. It extends $\mathcal{P}_J$. All formal moments are interpreted coefficientwise in $\hbar$.

Proposition 8.18. *On every fixed cylindrical polynomial, the finite functionals $\mathcal{P}_{J,L,a}$ converge to $\mathcal{P}_J^{\mathrm{q}}$ as $L \to \infty$ and $a \to 0$. For fixed polynomial degree, every integer $r \geq 0$ and bounded smooth sets, the error is at most*

$$C_r\big((1+L)^{-r} + a\big).$$

The estimate holds after any fixed finite number of background variations. Moments containing finitely many source primitives (8.35) *also have limits, with the same estimate. The finite Ward insertions define convergent operations on this core.*

Proof. For a smooth function f on R, the mean value estimate gives

$$\left| \sum_\nu w_\nu f(\sigma_\nu) - \int_R f \, d\sigma \right| \leq a \ \mathrm{Area}(R) \sup_R |\nabla f|.$$

Theorem 8.17 bounds the required derivatives of the finite covariance integrands uniformly in L. Its rapid curve estimate and this quadrature bound prove the claim for covariances. Fixed Wick polynomials have finitely many contractions, proving the first assertion and its differentiated versions.

Each correction $\bar{c}A_{\nu,L}x$ has no internal contraction, and two such corrections cannot contract with each other. In a moment against a fixed cylindrical polynomial, both ends must therefore attach to external fields. The resulting open chains involve the bounded direction operator $\widehat{\Pi}_L(X \otimes \mathrm{ad}_A)\widehat{\Pi}_L$ and the convergent inverse. Each source primitive contributes one cell weight. The contractions leave the weights of an ordinary Riemann sum. This proves the assertion for their products.

For Ward insertions, differentiate the actual finite Gaussian. Its normalized determinant is one in every cell. For a test F independent of J, let $W_{A,L,a} = \partial_A S_{J,L,a}$. Then

$$\begin{aligned} \mathcal{P}_{J,L,a}(W_{A,L,a}) &= 0, \\ \partial_A \mathcal{P}_{J,L,a}(F) &= -\hbar^{-1}\mathcal{P}_{J,L,a}(W_{A,L,a}F), \\ \partial_B\partial_A \mathcal{P}_{J,L,a}(F) &= \hbar^{-2}\mathcal{P}_{J,L,a}(W_{B,L,a}W_{A,L,a}F). \end{aligned} \tag{8.36}$$

The last identity uses the linear dependence of $M_{J,L}$ on J. The differentiated left sides converge with the stated estimate, so the finite insertions on the right have those limits. This constructs the inserted operations without multiplying surface delta distributions after the limit. □

This regulator is local on the surface for the displayed quadratic action, and global on the curve. Extending it to nonlinear surface vertices requires their differential and local products as further data.

8.10.2 A source response with unit vacuum

The first four-mode space gives a nonzero inserted response. Use (u, v, Cv, Cu) from (6.16), tensor with $(E_{11}, E_{12}, E_{21}, E_{22})$, and put

$$H = \mathrm{diag}(1, -1), \qquad A = X_1 \otimes \mathrm{ad}_H, \qquad M = 1 + \lambda A, \qquad C = M^{-1} = 1 - \lambda A + \lambda^2 A^2 - \lambda^3 A^3.$$

Here $A^4 = 0$ and λ is an even residual parameter. For the quartet action $S = bMx - \bar{c}Mc$, normalize the Wick functional $\mathcal{P}$ with weight $e^{-S/\hbar}$ and the Berezin orientation used above. Adjoin cotangent antifields and write

$$S_{\mathrm{cot}} = \sum_i (x_i^+ c_i - \bar{c}_i^+ b_i), \qquad D^- = \{S_{\mathrm{cot}} + S, -\} - \hbar\Delta.$$

The source lift is $\mathcal{L} = \lambda^+ - \bar{c}Ax$ and satisfies $D^-\mathcal{L} = 0$.

Let B take $v \otimes E_{12}$ to $u \otimes E_{12}$ and vanish on the other basis vectors. Set $F = \bar{c}Bc$ and $W = bAx - \bar{c}Ac$.

Proposition 8.19. *The normalized moments satisfy*

$$\mathcal{P}(F) = \hbar\lambda, \qquad \mathcal{P}(W) = 0, \qquad \mathcal{P}(WF) = -\hbar^2, \qquad \partial_\lambda \mathcal{P}(F) = -\hbar^{-1}\mathcal{P}(WF) = \hbar. \tag{8.37}$$

The bare residual antifield obeys $D^-\lambda^+ = W$, detected by $\mathcal{P}((D^-\lambda^+)F) = -\hbar^2$. The corrected lift is closed before integration.

Proof. The four-mode chain gives

$$\mathrm{tr}(AC) = 0, \qquad \mathrm{tr}(BC) = \lambda, \qquad \mathrm{tr}(ACBC) = -1.$$

Berezin contraction, with the order of odd factors fixed, gives

$$\mathcal{P}((\bar{c}Ac)(\bar{c}Bc)) = \hbar^2\big(\operatorname{tr}(AC)\operatorname{tr}(BC) - \operatorname{tr}(ACBC)\big).$$

The independent bosonic factor gives $\mathcal{P}((bAx)F) = \hbar^2 \operatorname{tr}(AC)\operatorname{tr}(BC)$. Subtraction proves the moment formulas. Finally $D^-\lambda^+ = W = D^-(\bar{c}Ax)$ by differentiation of the gauge fermion, proving both closure statements. □

This is a test of the inserted cochain map. The ghosts remain contractible. It persists on the smooth cylindrical core for a constant background $J = \lambda H$: smear the two ghost factors by smooth surface tests whose pairing is one. The weight change from u to v permits exactly one X step, with the same coefficient in every larger staggered space. Thus the nonzero response survives Proposition 8.18, although the vacuum determinant is one.

8.11 The regularized continuum determinant

The adapted Hilbert space controls the actual staggered inverse. For a determinant, use instead the ordinary constrained Sobolev space

$$V_s^{\mathrm{ord}} = \{f \in H^{s+1}(C) : j(f) = p_0(f) = 0\}, \qquad s \geq 2.$$

The operators agree on their common smooth vectors. On this space $X = h\eta_0$ gains one ordinary derivative. Fix a positive Hilbert norm on $\mathfrak{g}$ and put $A_J = X \otimes \mathrm{ad}_J$, so $M_J = 1 + A_J$.

The Schatten ideal $\mathcal{S}_p$ consists of compact operators whose singular values have summable pth powers. The ideal $\mathcal{S}_1$ is the trace class. For an integer $p \geq 1$ and $A \in \mathcal{S}_p$, the regularized Fredholm determinant is

$$\det{}_p(1 + A) = \det\left((1 + A)\exp\left(\sum_{r=1}^{p-1} \frac{(-1)^r}{r} A^r\right)\right). \tag{8.38}$$

The operator inside the determinant is identity plus trace class. The determinant on the right is the ordinary Fredholm determinant.

Theorem 8.20. *On V_s^{ord}, the operator X belongs to $\mathcal{S}_p$ for every real $p > 2$. For every integer $n \geq 3$, A_J^n is trace class and $\operatorname{Tr}(A_J^n) = 0$. For every integer $p \geq 3$ and every finite matrix J,*

$$\det{}_p M_J = 1.$$

The trace-log series and all its finite matrix-parameter derivatives converge locally uniformly in J, in trace norm before taking the trace.

Proof. The spherical Sobolev inclusion $H^{s+2}(C) \hookrightarrow H^{s+1}(C)$ has singular values $(1 + \ell(\ell+1))^{-1/2}$ with multiplicity $2\ell + 1$. Their pth powers are summable for $p > 2$, since the tail is bounded by a multiple of $\sum_\ell \ell^{1-p}$.

Factor X through this inclusion. The Green operator gains one derivative, and its finite-rank corrections obey the same bound because first-jet evaluation is continuous at these indices. Fixed smooth corrections impose $j = p_0 = 0$. The ideal property therefore gives $X \in \mathcal{S}_p$ for $p > 2$.

The Schatten product inequality makes X^3 trace class, and bounded composition gives the same for every X^n, $n \geq 3$. [2]

[2] B. Simon, *Notes on infinite determinants of Hilbert space operators*. Advances in Mathematics **24** (1977), 244–273, §§1, 3, 6.

Rotation is unitary on the ordinary constrained space and $U_\theta X U_\theta^{-1} = e^{i\theta} X$. Trace invariance gives

$$\operatorname{Tr}(X^n) = e^{in\theta} \operatorname{Tr}(X^n) = 0 \qquad (n \geq 3),$$

where the last equality follows by choosing $e^{in\theta} \neq 1$. The finite matrix factor can now be traced:

$$\operatorname{Tr}(A_J^n) = \operatorname{Tr}(X^n) \operatorname{tr}_{\mathfrak{g}}((\operatorname{ad}_J)^n) = 0.$$

On this ordinary carrier X is compact. A nonzero spectral value would bring an entire nonzero rotation circle into its spectrum, contradicting the compact-operator spectrum theorem. Thus $\|X^n\|^{1/n} \to 0$. For $n \geq p \geq 3$,

$$\|X^n\|_1 \leq \|X^p\|_1 \|X^{n-p}\|.$$

The root test gives absolute trace-norm convergence, locally uniformly in J, of

$$\sum_{n \geq p} \frac{(-1)^{n+1}}{n} A_J^n. \tag{8.39}$$

A fixed number k of matrix derivatives introduces at most $n(n-1)\cdots(n-k+1)$ ordered placements of directions. This polynomial factor leaves the root test unchanged.

The scalar series defining the operator in (8.38) differs from one first at order p. Factoring out A^p proves its trace-class property. Near $J = 0$, its logarithmic determinant is the trace of (8.39), and hence zero. Analyticity of the regularized Fredholm determinant gives the identity for every J. □

For a matrix direction U, set $A_U = X \otimes \operatorname{ad}_U$. Differentiating the trace-log, using cyclicity only for trace-class products, gives

$$d \log \det_p M_J(U) = (-1)^{p+1} \operatorname{Tr}(M_J^{-1} A_J^{p-1} A_U) = 0. \tag{8.40}$$

Indeed, the derivative series is $\sum_{n \geq p} (-1)^{n+1} \operatorname{Tr}(A_J^{n-1} A_U)$. Summing its geometric factor proves the formula. Each summand has positive rotation weight n and zero trace. The same trace-norm estimates justify every further finite source derivative, retaining all ordered placements of its matrix directions.

This determinant is a scalar entire function on the declared Hilbert family. It assigns no ordinary trace to X or X^2, and does not define the full native quantum measure. The nonzero response in (8.37) remains part of the observable comparison.

8.11.1 The two subtracted orders

A finite rotation-equivariant projector Π gives $\operatorname{Tr}((\Pi A_J \Pi)^n) = 0$ for every $n \geq 1$ on its finite range. A trace-class regulator R_ϵ commuting with rotations likewise gives $\operatorname{Tr}(R_\epsilon A_J^n) = 0$ for $n \geq 1$ on $V_s^{\mathrm{ord}} \otimes \mathfrak{g}$. These prescriptions assign zero to the first two traces. Relaxing that normalization leaves the following precise freedom for constant backgrounds.

Proposition 8.21. *Let $N \geq 2$ and let $F(J)$ be a holomorphic germ vanishing at zero. Suppose its Taylor coefficients vanish in degrees at least three, and it is invariant under conjugation and central translation $J \mapsto J + cI_N$. Then, for a unique scalar β,*

$$F(J) = \beta \operatorname{tr}_{\mathfrak{g}}((\operatorname{ad}_J)^2) = \beta\big(2N \operatorname{tr}(J^2) - 2(\operatorname{tr} J)^2\big). \tag{8.41}$$

If $F(e^{i\theta} J) = F(J)$ also holds, then $\beta = 0$. For $N = 1$, central-translation invariance already gives $F = 0$.

Proof. The Taylor hypothesis makes F a polynomial of degree at most two. On diagonal matrices it is symmetric. Matrices with distinct eigenvalues are dense, so conjugation invariance gives

$$F = a \operatorname{tr} J + b \operatorname{tr}(J^2) + c(\operatorname{tr} J)^2.$$

Central translation forces $a = 0$ and $c = -b/N$. On $\operatorname{End}(\mathbb{C}^N)$ one has $\operatorname{ad}_J = J \otimes 1 - 1 \otimes J^{\mathsf{T}}$. Taking the trace of its square proves the displayed formula, with $\beta = b/(2N)$. It is nonzero for some traceless J when $N \geq 2$, proving uniqueness. Rotation invariance kills this homogeneous term of degree two. In rank one every matrix is a central translate of zero. □

The hypotheses compare constant-background logarithms with $\log \det_3 M_J = 0$. They do not classify local cyclic counterterms depending on surface derivatives, normal sources or boundary fields.

8.12 The interacting comparison

The preceding constructions supply finite master weights, their connected observables, supported contractions of an added edge sector, and smooth Gaussian source limits. The regularized scalar determinant and the inserted Ward responses have separate roles. On the smooth theory the next equation involves the actual regulated interacting fields. For a positive-scale kernel, an effective interaction $I_t = \sum_{g \geq 0} \hbar^g I_{g,t}$ must satisfy

$$Q I_t + \frac{1}{2}\{I_t, I_t\}_t + \hbar \Delta_t I_t = 0.$$

After lower orders have been solved, the next residual is closed for their linearized differential. Its primitive must lie in the prescribed local cyclic counterterm complex, with the same normal support and boundary data.

Problem 8.22 (The native quantum comparison)**.** Extend the smooth Gaussian regulator and source-antifield map to one regulator for the interacting Neumann theory and its nonminimal fields. Require compatibility with cyclic transfer and the polar trace, estimates uniform on compact residual-matrix sets, and Ward identities on the enlarged observable domain. Construct local cyclic extensions at every simultaneous source collision. Compute the retained interacting master residual and determine whether it has an allowed supported primitive.

Construct the map from independently specified dynamical closed fields and descendants to these observables, preserving the normal marking, relative boundary operations and connected insertions. Compare its determinant, residual measure, boundary polarization, disk normalization and heat operator with an independently prepared state, including sewing. In the affine collision family, derive the level and its parameter normalization from the native regulator.

The free kernels of Chapter 7 determine the positive-scale contractions. The finite torus theorem determines the action after internal elimination on its specified coefficient algebra. The remaining state comparison requires a preparation on both sides of the elimination map.

Chapter 9

States and sewing

The residual action determines equations of motion and their gauge symmetries. To obtain an amplitude, one must also choose a state. For a boundary circle this requires a real form, a polarization, a measure and a Hamiltonian. These choices determine the disk amplitude and the pairing used when two boundaries are sewn.

We first prepare the compact residual gauge theory and compute its character amplitudes. Holonomy density and invariant ordering then exhibit two quantities that a normalized Gaussian vacuum does not determine. Finite Gaussian integration, a free gauge quartet and an infinite family of scalar oscillators provide explicit sewing maps. A nonnegative interaction preserves scalar sewing at fixed positive smoothing. Its divergent coupling series has a Borel reconstruction, and a Hermite unitary identifies the interacting Fock Hamiltonian. The final comparison asks which of these data the native Neumann quantum theory supplies.

9.1 The residual boundary and its real quantization

Write $b \in \Omega^1(U, \mathfrak{gl}_N)$ and $J \in \Omega^0(U, \mathfrak{gl}_N)$ for the physical residual fields. The action of Proposition 4.6 is

$$S_{\mathrm{res}} = \kappa \int_U \operatorname{tr}(J F_b) - \frac{\kappa}{2} \int_{\partial U} \operatorname{tr}(bJ), \qquad F_b = d_U b + b \wedge b.$$

Its boundary one-form is

$$\Theta_{\partial U} = \frac{\kappa}{2} \int_{\partial U} \operatorname{tr}(J \partial b - b \partial J).$$

Adding $\kappa \int_{\partial U} \operatorname{tr}(bJ)/2$ changes this primitive to $\kappa \int_{\partial U} \operatorname{tr}(J \partial b)$. The boundary two-form is unchanged. We use the resulting connection polarization, in which b is the boundary configuration variable.

Choose a connected compact Lie group G with Lie algebra $\mathfrak{g}$, a positive invariant inner product $\langle -, - \rangle$, and numerical parameters $\kappa \in \mathbb{R} \setminus \{0\}$, $v > 0$, $\hbar > 0$. Let U be an oriented surface with a positive area form. We declare the real Euclidean preparation by the first-order action

$$S_E = \int_U \Big(i\kappa \langle J, F_b \rangle + \frac{v}{2} \langle J, J \rangle \, d\mathrm{vol} \Big). \tag{9.1}$$

Here J is a real adjoint scalar and b is a G-connection. The Gaussian measure in J is normalized as part of this preparation. Completing its square gives the density $\kappa^2 |F_b|^2/(2v)$ for the second-order action.

The corresponding canonical momentum is $P = \kappa J$. Its kinetic energy per unit spatial length is $v\langle P, P \rangle/(2\kappa^2)$. Choose the positive Laplace–Beltrami quantization and assign energy zero to the trivial representation. For the positive Laplacian Δ_G of the chosen metric, this gives

$$\mathsf{H} = \frac{v\hbar^2}{2\kappa^2} \Delta_G, \qquad t = \frac{v\hbar}{\kappa^2} \operatorname{Area}(U), \qquad e^{-\operatorname{Area}(U)\mathsf{H}/\hbar} = e^{-t\Delta_G/2}. \tag{9.2}$$

An additive energy constant would multiply each amplitude by an area factor. The zero-energy convention fixes this freedom. We also fix normalized Haar measure and the disk normalization below. These are the representation-theoretic choices in two-dimensional Yang–Mills theory.[1]

9.1.1 The conjugation constraint

Let $\mathcal{R}(G)$ be the complex representative functions on G. Choose a basis of $\mathfrak{g}$ with structure constants $f^a{}_{bc}$. Let G_a denote the infinitesimal conjugation operators, with $[G_a, G_b] = f^c{}_{ab} G_c$. On the graded algebra $\mathcal{R}(G) \otimes \Lambda(\mathfrak{g}^*)$, put $|c^a| = 1$ and use left odd derivatives. Its BRST differential is

$$Q = c^a G_a - \frac{1}{2} f^a{}_{bc} c^b c^c \partial_{c^a}. \tag{9.3}$$

Proposition 9.1. *The differential Q squares to zero and commutes with $e^{-t\Delta_G/2}$. Its degree-zero cohomology is $\mathcal{R}(G)^G$. For normalized Haar measure dg, the Hilbert completion of this cohomology is*

$$\mathcal{H}_G = L^2(G, dg)^G.$$

The orthogonal projection onto $\mathcal{H}_G$ is $P_{\mathrm{inv}} f(U) = \int_G f(hUh^{-1})\, dh$.

Proof. The coefficient of two ghosts in Q^2 is $c^a c^b([G_a, G_b] - f^c{}_{ab} G_c)/2$. The coefficient of three ghosts is the Jacobi identity. Both vanish on the whole exterior algebra. The bi-invariant Laplacian commutes with each G_a and acts trivially on the ghosts, proving the heat commutation relation.

In degree zero, $Qf = 0$ is the system $G_a f = 0$ for all a. Connectedness of G identifies its solutions with conjugation invariants. There are no boundaries from negative degrees. Haar invariance proves that P_{inv} is idempotent and self-adjoint. Jensen's inequality gives its L^2 bound. Peter–Weyl approximation, followed by this projection, shows that invariant representative functions are dense in its range. □

This identifies ghost degree zero. Higher Lie algebra cohomology is retained by the BRST complex and need not vanish.

9.1.2 Disk, cylinder and surface amplitudes

For $\lambda \in \widehat{G}$, let d_λ, χ_λ and $C_2(\lambda)$ denote its dimension, character and positive Laplacian eigenvalue. The characters are an orthonormal basis of $\mathcal{H}_G$. Equip this space on its smooth core with Haar convolution,

$$(f * g)(U) = \int_G f(UW^{-1}) g(W)\, dW.$$

The preparation uses the complete sum over $\widehat{G}$. It has no additional discrete theta weight or fixed-bundle projection. For nonsimply connected G, prescribing such a sector would change this global choice.

Theorem 9.2. *At positive heat time, the cylinder kernel and disk state are*

$$K_t(U, V) = \sum_{\lambda \in \widehat{G}} e^{-tC_2(\lambda)/2} \chi_\lambda(U) \overline{\chi_\lambda(V)}, \tag{9.4}$$

$$D_t(U) = \sum_{\lambda \in \widehat{G}} d_\lambda e^{-tC_2(\lambda)/2} \chi_\lambda(U). \tag{9.5}$$

[1] E. Witten, *Two Dimensional Gauge Theories Revisited*, §4, including §§4.1–4.2, https://arxiv.org/abs/hep-th/9204083. The Haar and vacuum normalizations used here are specified explicitly.

They are smooth, and for $s, t > 0$ satisfy

$$\int_G K_s(U,W)K_t(W,V)\,dW = K_{s+t}(U,V). \tag{9.6}$$

The disk tends to the identity point source distributionally as $t \downarrow 0$. With this normalization, the amplitude of a genus-g surface with b outgoing boundaries is

$$Z_{g,b}(t;U_1,\dots,U_b) = \sum_\lambda d_\lambda^{2-2g-b} e^{-tC_2(\lambda)/2} \prod_{i=1}^{b} \chi_\lambda(U_i). \tag{9.7}$$

An incoming boundary carries the conjugate character. These amplitudes satisfy boundary and handle sewing.

Proof. The heat operator is diagonal in the character basis, giving (9.4). The identity point source has Fourier coefficients $\chi_\lambda(1) = d_\lambda$. Heat evolution therefore gives (9.5). Character orthogonality proves (9.6) for finite sums.

Highest weights have lattice growth. Representation dimensions and the coefficients of differentiated matrix elements grow polynomially in the weight. The Casimir grows quadratically for a positive invariant metric. Consequently the sums and all their derivatives converge locally uniformly for $t > 0$. These bounds justify Haar integration and prove smoothness. On a smooth test function the Fourier coefficients also give the asserted distributional limit of the disk.

Schur orthogonality gives

$$\chi_\lambda * \chi_\mu = \delta_{\lambda\mu} \frac{\chi_\lambda}{d_\lambda}.$$

Thus a three-holed sphere contributes d_λ^{-1}. Its product with the disk coefficient d_λ is the unit cylinder coefficient. Decomposition into these pieces gives (9.7).

Sewing opposed boundary characters identifies their labels. For two surfaces, their dimension exponents add to the Euler characteristic of the glued surface. Gluing two boundaries of one surface leaves that exponent unchanged: g increases by one and b decreases by two. These observations prove both sewing laws. The positive-time bounds justify each passage from finite sums to the displayed amplitudes. □

In particular, the sphere and torus amplitudes are

$$Z_{0,0}(t) = \sum_\lambda d_\lambda^2 e^{-tC_2(\lambda)/2}, \qquad Z_{1,0}(t) = \operatorname{Tr}_{\mathcal{H}_G} e^{-t\Delta_G/2} = \sum_\lambda e^{-tC_2(\lambda)/2}. \tag{9.8}$$

For a nontrivial connected compact group, the torus trace is strictly greater than one. If $\dim G > 0$, the zero-area disk is not an L^2 state. Indeed, invariant smooth functions equal to one at the identity can have arbitrarily small L^2 norm, by shrinking an invariant bump function to the identity. Point evaluation is therefore unbounded on the invariant core. For the trivial group the point source is the normalized constant.

On a closed surface the positive summands define a probability distribution after division by $Z_{g,0}(t)$. Differentiation gives

$$-2\partial_t \log Z_{g,0} = \langle C_2 \rangle, \qquad 4\partial_t^2 \log Z_{g,0} = \langle C_2^2 \rangle - \langle C_2 \rangle^2. \tag{9.9}$$

The same positive-time majorants justify both derivatives. These are moments of the chosen quadratic quantum operator. Higher invariant operators require an ordering convention.

9.1.3 *A normalization and tail bound for $SU(2)$*

Give $SU(2)$ its round-S^3 metric. Its irreducibles have labels $n \geq 0$ and data

$$d_n = n + 1, \qquad C_2(n) = n(n+2), \qquad \chi_n(\theta) = \frac{\sin((n+1)\theta)}{\sin\theta}.$$

The endpoint values of the quotient are its continuous limits. On class functions, normalized Haar measure is $(2/\pi)\sin^2\theta\, d\theta$ for $0 \leq \theta \leq \pi$. The character inner product is therefore the elementary sine inner product.

At $t = 1$, the convergent sums give

$$Z_{1,0}(1) \simeq 1.242005052767 43, \qquad Z_{0,0}(1) \simeq 2.066365251634 37.$$

The corresponding torus mean and variance of C_2 are approximately 0.66373337 and 2.22320420.

To control truncation, set $\alpha = t/2$ and retain $n \leq K$, where $K \geq 0$ is integral. Write erfc for the complementary error function. Since $n(n+2) \geq n^2$,

$$0 \leq \sum_{n>K} e^{-tn(n+2)/2} \leq \int_K^\infty e^{-\alpha x^2}\, dx = \frac{\sqrt{\pi}}{2\sqrt{\alpha}} \operatorname{erfc}(\sqrt{\alpha}K). \tag{9.10}$$

For $K + 1 \geq \alpha^{-1/2}$, the function $x^2 e^{-\alpha x^2}$ decreases on $[K+1, \infty)$. Putting $m = n + 1$ and integrating gives

$$\begin{aligned}\sum_{n>K} (n+1)^2 e^{-tn(n+2)/2} &\leq e^{\alpha} \int_{K+1}^\infty x^2 e^{-\alpha x^2}\, dx \\ &= e^{\alpha}\left[\frac{(K+1)e^{-\alpha(K+1)^2}}{2\alpha} + \frac{\sqrt{\pi}\operatorname{erfc}(\sqrt{\alpha}(K+1))}{4\alpha^{3/2}}\right].\end{aligned} \tag{9.11}$$

Because $|\chi_n| \leq n + 1$, the same bound controls the absolute disk-character tail. Both estimates require $t > 0$.

9.2 Holonomy density and invariant ordering

The heat state uses a group holonomy and a quantum Hamiltonian. Exponential coordinates contribute a Jacobian, while ordering changes products of invariant insertions. We compute both with their normalizations before asking for a map from the polar fields.

9.2.1 *The exponential half-density*

Let $\mathfrak{g}$ now be a finite-dimensional unimodular complex Lie algebra. For a dimensionless exponential coordinate X near zero, write $L = \operatorname{ad}_X$. Left trivialization gives

$$\operatorname{dexp}_X = \int_0^1 e^{-tL}\, dt = e^{-L/2}\frac{\sinh(L/2)}{L/2}.$$

The quotient denotes its entire power series. Normalize the tangent and invariant volume forms to agree at the identity. Choose every square-root branch to have value one there.

Proposition 9.3. *The exponential Jacobian and its half-density are*

$$j(X) = \det_{\mathfrak{g}} \frac{\sinh(L/2)}{L/2}, \qquad \mathfrak{j}(X) = \exp\left(\frac{1}{2}\operatorname{Tr}_{\mathfrak{g}} \log \frac{\sinh(L/2)}{L/2}\right). \tag{9.12}$$

On a sufficiently small neighborhood of zero,

$$\mathfrak{j}(X) = \lim_{K\to\infty} \prod_{n=1}^{K} \det_{\mathfrak{g}}\left(1 + \frac{L^2}{4\pi^2 n^2}\right)^{1/2} \tag{9.13}$$

locally uniformly. Let $\mathfrak{j}_K$ denote this product through $n = K$. If L is real diagonalizable with real eigenvalues, then, for $K \geq 1$,

$$0 \leq \log \mathfrak{j} - \log \mathfrak{j}_K \leq \frac{\operatorname{Tr}_{\mathfrak{g}} L^2}{8\pi^2 K}. \tag{9.14}$$

Proof. Unimodularity gives $\operatorname{Tr} L = 0$, so $\det(e^{-L/2}) = 1$. This proves the Jacobian formula. The scalar product for $\sinh(z/2)/(z/2)$ gives (9.13) on eigenvalues, including their algebraic multiplicities. Each logarithmic factor is locally bounded by a constant times n^{-2}. Hence the product converges locally uniformly on its normalized branch.

For real eigenvalues λ_a, apply $0 \leq \log(1+u) \leq u$ to $u = \lambda_a^2/(4\pi^2 n^2)$. The logarithmic tail is bounded by $\sum_a \lambda_a^2 \sum_{n>K} n^{-2}/(8\pi^2)$. The estimate $\sum_{n>K} n^{-2} \leq K^{-1}$ proves the result. □

On a compact real form, the eigenvalues of L are imaginary and the factors become sine factors. The branch is local, before their first zeros. For a quadratic Lie algebra, the same half-determinant occurs in the effective action of one-dimensional Chern–Simons theory.[2] That formula concerns the circle field complex. A comparison with the Neumann sphere must identify the corresponding fields.

Expansion at $X = 0$ gives

$$\log \mathfrak{j}(X) = \frac{1}{48}\operatorname{Tr} L^2 - \frac{1}{5760}\operatorname{Tr} L^4 + \frac{1}{362880}\operatorname{Tr} L^6 + O(X^8). \tag{9.15}$$

For $\mathfrak{g} = \mathfrak{gl}_N$, put $p_r = \operatorname{tr}(X^r)$, with $p_0 = N$.

Proposition 9.4. *For every integer $m \geq 0$,*

$$\operatorname{Tr}_{\mathfrak{gl}_N} (\operatorname{ad}_X)^m = \sum_{r=0}^{m} (-1)^r \binom{m}{r} p_{m-r}\, p_r. \tag{9.16}$$

In particular, $\operatorname{Tr} L^2 = 2N p_2 - 2p_1^2$ *and* $\operatorname{Tr} L^4 = 2N p_4 - 8 p_1 p_3 + 6 p_2^2$. *For matrices X, Y of sizes N, M, the quadratic part of* $\log \mathfrak{j}(X \oplus Y) - \log \mathfrak{j}(X) - \log \mathfrak{j}(Y)$ *is*

$$\frac{1}{24}(M \operatorname{tr} X^2 + N \operatorname{tr} Y^2 - 2 \operatorname{tr} X \operatorname{tr} Y). \tag{9.17}$$

Proof. Left and right multiplication on $\operatorname{End}(\mathbb{C}^N)$ commute. Their difference is ad_X, and $\operatorname{Tr}(L_X^a R_X^b) = \operatorname{tr}(X^a)\operatorname{tr}(X^b)$. The binomial theorem proves (9.16) without diagonalizing X. Substituting $m = 2, 4$ gives the displayed traces. The quadratic half-density logarithm is $(N p_2 - p_1^2)/24$. Its block-sum difference is (9.17). □

The two off-diagonal Hom spaces between the blocks contribute to this difference. Thus this quantum density is not an additive family of classical invariant polynomials. Its holonomy coordinate X has not been identified with the residual matrix J.

[2] A. Alekseev and P. Mnev, *One-dimensional Chern–Simons theory*, Proposition 1, https://arxiv.org/abs/1005.2111.

9.2.2 *The universal quadratic and quartic contacts*

Use Hermitian generators in the complex Rees enveloping algebra, completed in $\hbar$, with relations

$$[\widehat{J}_i, \widehat{J}_j] = i\hbar\epsilon_{ijk}\widehat{J}_k.$$

Put $p = x_1^2 + x_2^2 + x_3^2$, $C = \sum_i \widehat{J}_i^2$, and $\Delta = \sum_i \partial_{x_i}^2$. PBW symmetrization sym averages all orderings of a commutative monomial in the generators $\widehat{J}_i$.

In this convention define

$$\mathrm{Duf}_\hbar = \mathrm{sym} \circ \left(1 + \frac{\hbar^2}{24}\Delta + \frac{\hbar^4}{1920}\Delta^2 + \cdots\right). \tag{9.18}$$

The series in parentheses is $\sinh(\hbar\sqrt{\Delta}/2)/(\hbar\sqrt{\Delta}/2)$, interpreted as a power series in Δ. The adjoint eigenvalues in the unscaled Hermitian basis are $0, \sqrt{p}, -\sqrt{p}$, so this is the half-Jacobian factor. The Duflo theorem identifies its invariant restriction as an algebra isomorphism.[3]

Proposition 9.5. *The universal Rees algebra satisfies*

$$\mathrm{Duf}_\hbar(p) = C + \hbar^2/4, \tag{9.19}$$

$$\mathrm{sym}(p^2) = C^2 - \hbar^2 C/3, \tag{9.20}$$

$$\mathrm{Duf}_\hbar(p^2) = (C + \hbar^2/4)^2. \tag{9.21}$$

For the multiplicative ordering $p \mapsto C$, the collapsed double-insertion contact relative to raw PBW symmetrization is $\hbar^2 C/3$.

Proof. The relations make C central and give

$$\sum_j [\widehat{J}_j, [\widehat{J}_j, \widehat{J}_i]] = 2\hbar^2\widehat{J}_i, \qquad \sum_j \widehat{J}_j\widehat{J}_i\widehat{J}_j = (C - \hbar^2)\widehat{J}_i.$$

Grouping the six orders of two repeated indices yields

$$\mathrm{sym}(p^2) = \frac{1}{3}\sum_{i,j}\left(\widehat{J}_i^2\widehat{J}_j^2 + \widehat{J}_i\widehat{J}_j\widehat{J}_i\widehat{J}_j + \widehat{J}_i\widehat{J}_j^2\widehat{J}_i\right).$$

The three sums equal C^2, $C^2 - \hbar^2 C$ and C^2. This proves (9.20) before any representation is chosen. Finally, $\Delta p = 6$, $\Delta p^2 = 20p$, and $\Delta^2 p^2 = 120$. Substitution into (9.18) gives the other two identities. □

The invariant ordering with zero vacuum energy is

$$Q(F(p)) = \mathrm{Duf}_\hbar(F(p - \hbar^2/4)), \qquad Q(p) = C.$$

Translation in p and the Duflo theorem prove multiplicativity. In spin j, the quadratic eigenvalues before and after the Duflo shift are $\hbar^2 j(j+1)$ and $\hbar^2(j+1/2)^2$. At numerical $\hbar = 1$, the two half-Casimir evolutions differ by $e^{-t/8}$. This scalar factor respects addition of t, so sewing does not select between the two vacuum energies. The round-S^3 Casimir above is four times this spin Casimir.

[3] A. Alekseev and E. Meinrenken, *The non-commutative Weil algebra*, §6.1, https://arxiv.org/abs/math/9903052.

9.3 Gaussian states with their normalization lines

The distinction between a state and its normalized correlations already occurs in one variable. Gaussian integration gives a nonzero quadratic moment, while its value on the unit carries a square root of a determinant. Under elimination, the moment and determinant compose by different parts of the same block matrix identity.

9.3.1 The polynomial Gaussian complex

Let $R = \mathbb{C}[\hbar]$ and let Q be an invertible symmetric complex $n \times n$ matrix. Set

$$B_Q = R[x_1, \dots, x_n] \otimes \Lambda(\xi_1, \dots, \xi_n), \qquad |x_i| = 0, \quad |\xi_i| = -1.$$

Using left odd derivatives, define

$$D_Q = \sum_i \big(\hbar \partial_{x_i} - (Qx)_i\big)\, \partial_{\xi_i}. \tag{9.22}$$

The even factors commute because Q is symmetric, whereas the odd derivatives anticommute. Thus $D_Q^2 = 0$. These formulas define B_Q, D_Q for every symmetric matrix. Invertibility is required for the integration map below.

Over the space of nondegenerate symmetric matrices, let $\mathscr{G}_Q$ be the degree-zero Gaussian square-root line. Its local sections $\gamma_Q = (\det Q)^{-1/2}$ have the sign transitions of the square root. On real positive matrices use the positive section, corresponding to the density $dx/(2\pi\hbar)^{n/2}$ for numerical $\hbar > 0$. Continuation retains this line and its chosen branch.

Proposition 9.6. *The R-linear map*

$$\Pi_Q(F) = \gamma_Q \exp\!\left(\frac{\hbar}{2} \sum_{i,j} (Q^{-1})_{ij}\, \partial_{x_i} \partial_{x_j}\right) F(x, 0)\Big|_{x=0} \tag{9.23}$$

is a cochain map from (B_Q, D_Q) to $\mathscr{G}_Q \otimes R$ with zero differential. For commuting formal sources t_i,

$$Z_Q(t) = \gamma_Q e^{\hbar t^{\mathsf{T}} Q^{-1} t/2}, \qquad \sum_j Q_{ij}\, \partial_{t_j} Z_Q = \hbar t_i Z_Q. \tag{9.24}$$

These identities hold coefficientwise in $\mathscr{G}_Q \otimes R[[t_1, \dots, t_n]]$.

Proof. Write $\mathbb{E}_Q = \gamma_Q^{-1} \Pi_Q$ in a local branch. The exponential terminates on each polynomial. Commuting multiplication by x_j through it gives

$$\mathbb{E}_Q(x_j f) = \hbar \sum_l (Q^{-1})_{jl}\, \mathbb{E}_Q(\partial_{x_l} f).$$

Consequently $\Pi_Q D_Q(\xi_i f) = 0$. An input with at least two odd factors retains an odd factor after D_Q and maps to zero. An input with no odd factors has zero differential. This proves the cochain assertion.

Applying the heat exponential to the formal source gives the first identity in (9.24). Differentiation proves the second. □

The Ward equations allow every nonzero source-independent multiple of Z_Q. Integration data fix $Z_Q(0) = \gamma_Q$. Even after normalization, evaluation at zero is insufficient: for $Q = 1$ in one variable,

$$D_Q(x\xi) = \hbar - x^2, \qquad \mathbb{E}_Q(x^2) = \hbar.$$

For a real symmetric invertible Hessian H, use the damped matrix $Q_\epsilon = \epsilon I - iH$, $\epsilon > 0$. Continuation through $\operatorname{Re} Q_\epsilon > 0$ gives

$$\gamma_{-iH} = |\det H|^{-1/2} e^{i\pi\,\operatorname{sign}(H)/4}, \qquad \mathbb{E}_{-iH}(x_i x_j) = i\hbar (H^{-1})_{ij}. \tag{9.25}$$

Here $\operatorname{sign}(H)$ is the number of positive eigenvalues minus the number of negative eigenvalues. Diagonalization reduces both formulas to the one-dimensional Gaussian. No infinite determinant occurs in these identities.

9.3.2 The full residual complex under elimination

Split the even coordinates into (x, y) and their odd partners into (ξ, η). For a symmetric matrix write

$$\mathsf{Q} = \begin{pmatrix} A & B \\ B^{\mathsf{T}} & C \end{pmatrix}, \qquad C_{\text{eff}} = C - B^{\mathsf{T}} A^{-1} B,$$

assuming that A is invertible. The even shear and its cotangent lift are

$$x' = x + A^{-1} B y, \quad y' = y, \quad \xi' = \xi, \quad \eta' = \eta - B^{\mathsf{T}} A^{-1} \xi. \tag{9.26}$$

They preserve the quantum contraction $\sum_i \partial_{x_i} \partial_{\xi_i} + \sum_j \partial_{y_j} \partial_{\eta_j}$ and have Berezinian one. The quadratic form becomes $x'^{\mathsf{T}} A x' + y^{\mathsf{T}} C_{\text{eff}}\, y$.

Proposition 9.7. *Apply Π_A in the primed eliminated variables and retain the full (y, η') complex. The result is a cochain map*

$$(B_{\mathsf{Q}}, D_{\mathsf{Q}}) \longrightarrow \mathscr{G}_A \otimes (B_{C_{\text{eff}}}, D_{C_{\text{eff}}}).$$

Here the polynomial differential defines the residual complex even if C_{eff} is degenerate. On formal sources, the map is

$$e^{t_x^{\mathsf{T}} x + t_y^{\mathsf{T}} y} \longmapsto \gamma_A e^{\hbar t_x^{\mathsf{T}} A^{-1} t_x/2} e^{(t_y - B^{\mathsf{T}} A^{-1} t_x)^{\mathsf{T}} y}. \tag{9.27}$$

When C_{eff} is invertible, successive integration agrees with Π_{Q} under the branch identification

$$\mathscr{G}_{\mathsf{Q}} \simeq \mathscr{G}_A \otimes \mathscr{G}_{C_{\text{eff}}}.$$

Iterated integration is associative on that branch.

Proof. The inverse-transpose transformation of the odd variables splits D_{Q} into the A and C_{eff} differentials. Proposition 9.6 therefore proves the cochain assertion with all residual odd variables retained. Substituting $x = x' - A^{-1} B y$ proves (9.27).

The block inverse formula identifies the source exponent after the second integration. The identity $\det \mathsf{Q} = \det A \det C_{\text{eff}}$ identifies the normalization lines. Fubini on the positive real domain fixes their common branch and proves associativity. Analytic continuation preserves these identities. For real oscillatory Hessians, the same congruence adds signatures, so the Fresnel phases compose as well. □

For $\mathsf{Q} = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 3 \end{smallmatrix}\right)$, one has $C_{\text{eff}} = 5/2$ and

$$\frac{\hbar}{10}(3t_x^2 - 2t_x t_y + 2t_y^2) = \frac{\hbar}{4} t_x^2 + \frac{\hbar}{5}(t_y - t_x/2)^2.$$

The vacuum factor is $5^{-1/2} = 2^{-1/2}(5/2)^{-1/2}$. A kernel of C_{eff} instead remains in the residual complex. The partial integration never inverts that kernel.

9.3.3 Continuous Wick preparation on regular kernels

For infinitely many fields, specify a test space before applying the same contraction. Let (W, d) be a nuclear graded test complex with a continuous graded-symmetric pairing b of degree one. Assume

$$b(du, v) + (-1)^{|u|} b(u, dv) = 0.$$

Let (H, d_H) be finite-dimensional. Choose continuous maps i, p, h of degrees $0, 0, -1$, with i, p cochain maps, such that

$$pi = 1, \qquad dh + hd = 1 - ip, \qquad h^2 = hi = ph = 0.$$

Assume $P = ip$ is self-adjoint for b and $C(u, v) = -b(hu, v)$ is graded symmetric of degree zero.

Use the completed nuclear tensor spaces of the tests. For a smooth test complex these are smooth compact kernels, and covariance distributions act continuously on them. Let $\mathrm{Sym}^n_{\mathrm{reg}} W$ be the symmetric part of the chosen n-fold tensor space. Denote unordered pair contraction with C by ∂_C, and pair contraction with b by Δ_b. Put $b_H(a, a') = b(ia, ia')$.

Proposition 9.8. *The normalized Wick map*

$$\Gamma_h = \mathrm{Sym}(p) e^{\hbar \partial_C}$$

is a continuous cochain map

$$\left(\bigoplus_{n \geq 0} \mathrm{Sym}^n_{\mathrm{reg}} W[\hbar], d + \hbar \Delta_b \right) \longrightarrow \left(\mathrm{Sym}\, H[\hbar], d_H + \hbar \Delta_{b_H} \right).$$

It sends 1 *to* 1 *and extends to coefficientwise* $\hbar$*-series in these spaces.*

Proof. On homogeneous generators,

$$\begin{aligned} [d, \partial_C](uv) &= -C(du, v) - (-1)^{|u|} C(u, dv) \\ &= b((dh + hd)u, v) = b(u, v) - b(Pu, Pv). \end{aligned}$$

The last equality uses self-adjointness and $P^2 = P$. Summing over pairs gives $[d, \partial_C] = \Delta_b - \Delta_{b(P-, P-)}$. Constant contractions supercommute, and ∂_C is even. Thus

$$e^{\hbar \partial_C} (d + \hbar \Delta_b) e^{-\hbar \partial_C} = d + \hbar \Delta_{b(P-, P-)}.$$

Applying $\mathrm{Sym}(p)$ proves the cochain equation.

Each term contracts disjoint pairs of variables in one smooth compact kernel. Partial distributional evaluation leaves a smooth kernel with support in the projected compact set. The nuclear tensor topology makes these operations continuous. There is no multiplication of singular distributions on a common diagonal. At fixed field number the exponential terminates. At fixed $\hbar$ order only finitely many input coefficients contribute, proving the formal extension. □

Strong convergence of the covariance distributions on each fixed compact test space gives coefficientwise convergence of these maps. Their value on the unit remains one. An unnormalized state still needs its determinant section and a boundary-compatible family of contractions.

9.4 A free gauge state with nonzero contractions

The preceding Gaussian cochain map is compatible with a quantum second-order differential. A free BRST differential instead acts as a derivation. A quartet gives an explicit preparation for this case and separates cancellation of the graded trace from the values of its observables.

Over $\mathbb{C}[[\hbar]]$, set

$$\mathcal{A} = \mathbb{C}[x, b] \otimes \Lambda(\bar{c}, c)[[\hbar]], \qquad |x| = |b| = 0, \quad |\bar{c}| = -1, \quad |c| = 1.$$

Each coefficient of $\hbar$ is polynomial in the displayed variables. Use the odd order $\bar{c}, c$ and left derivatives. Define

$$Q = c\,\partial_x + b\,\partial_{\bar{c}}, \qquad s = x\,\partial_c + \bar{c}\,\partial_b.$$

Let N count total polynomial and exterior degree. The identities $Q^2 = s^2 = 0$ and $Qs + sQ = N$ hold on generators, and hence on the algebra by the derivation rule. Put $h_0 = s/N$ on positive degree and zero on constants. If i includes constants, then $Qh_0 + h_0 Q = 1 - i\,\mathrm{ev}_0$.

Introduce the even contraction

$$C = \partial_x \partial_b + \partial_c \partial_{\bar{c}}, \qquad W = e^{\hbar C}, \qquad P_G = \mathrm{ev}_0\, W. \tag{9.28}$$

The exponential terminates on each polynomial and is a continuous linear automorphism of the underlying formal module.

Theorem 9.9. *The functional P_G is Q-closed and has moments*

$$\begin{aligned} P_G(x^m b^n) &= \delta_{mn} m! \hbar^m, \\ P_G(\bar{c}c\, x^m b^n) &= \hbar \delta_{mn} m! \hbar^m. \end{aligned} \tag{9.29}$$

Terms with exactly one ghost have zero expectation. For $H_G = W^{-1} h_0 W$,

$$QH_G + H_G Q = 1 - i P_G, \qquad H_G^2 = H_G i = P_G H_G = 0. \tag{9.30}$$

Proof. The commutators of Q with the two terms of C are $-\partial_x \partial_{\bar{c}}$ and $\partial_x \partial_{\bar{c}}$. Hence $[Q, C] = 0$. Evaluation at zero kills Q, so $P_G Q = 0$.

The bosonic contraction survives evaluation only when it differentiates $x^m b^n$ equally often in both variables. For $m = n$ its coefficient is $(m!)^2 \hbar^m / m!$. Also $\partial_c \partial_{\bar{c}}(\bar{c}c) = 1$. These commuting contractions prove the moment formulas. Conjugation of the h_0 identities by W gives (9.30), since $Wi = i$. □

For example,

$$Q(\bar{c}x) = bx - \bar{c}c, \qquad P_G(bx) = P_G(\bar{c}c) = \hbar.$$

The Ward identity equates two nonzero contractions. The preparation is a graded algebraic functional, with no claim of a positive measure on odd variables.

9.4.1 Transfer and its Hilbert completion

For $0 < r < 1$, define $T_r = W^{-1} r^N W$ on polynomials. Because $[Q, N] = [Q, W] = 0$,

$$[Q, T_r] = 0, \qquad T_{r_1} T_{r_2} = T_{r_1 r_2}, \qquad P_G T_r = P_G, \qquad T_r 1 = 1. \tag{9.31}$$

For an analytic trace, fix a numerical value of $\hbar$. Declare the vectors

$$W^{-1}(x^a b^d \bar{c}^{\nu_1} c^{\nu_2}), \qquad a, d \geq 0, \quad \nu_1, \nu_2 \in \{0, 1\},$$

orthonormal, with parity $\nu_1 + \nu_2$. This defines the Hilbert completion. The operator T_r is diagonal there, with eigenvalue $r^{a+d+\nu_1+\nu_2}$, so

$$\operatorname{Tr}|T_r| = \frac{(1+r)^2}{(1-r)^2}, \qquad \operatorname{Str} T_r = 1. \tag{9.32}$$

The differential and homotopy identities hold on the invariant polynomial core. No boundedness of W on a different Hilbert space is assumed.

This free differential also has a finite BFV charge. Give the cotangent momenta of $(x, b, \bar{c}, c)$ complementary degrees so that the canonical Poisson bracket has degree zero. Let Θ be the degree-one fiber-linear Hamiltonian of Q. The correspondence from vector fields to these Hamiltonians preserves brackets, so $\{\Theta, \Theta\} = 0$. Coordinate quantization gives the square-zero operator $\hbar Q$.

9.4.2 Nonzero Fourier modes and the residual sector

Take a flat product

$$Y = S^1_s \times (\mathbb{C}/\Lambda)^2 \cong \mathbb{T}^5, \qquad D = d_s + \bar{\partial}.$$

For a real dual-lattice vector $k = (k_s, k_{x_1}, k_{y_1}, k_{x_2}, k_{y_2})$, the mode complex is $\Lambda^\bullet\mathbb{C}^3$ with differential $D_k = \eta(k)\wedge$, where

$$\eta(k) = ik_s\, ds + \frac{ik_{x_1} - k_{y_1}}{2}\, d\bar{z}_1 + \frac{ik_{x_2} - k_{y_2}}{2}\, d\bar{z}_2.$$

For $k \neq 0$, this covector is nonzero. The Hermitian dual vector $\eta(k)^\sharp$ satisfies $\eta(k)(\eta(k)^\sharp) = \|\eta(k)\|^2$, and

$$h_k = \frac{\iota_{\eta(k)^\sharp}}{\|\eta(k)\|^2}, \qquad D_k h_k + h_k D_k = 1. \tag{9.33}$$

To see the exact pairs, choose $u_1 = \eta(k)/\|\eta(k)\|$ and an orthonormal complement u_2, u_3. For each $I \subset \{2, 3\}$,

$$D_k u_I = \|\eta(k)\| u_1 \wedge u_I, \qquad D_k(u_1 \wedge u_I) = 0.$$

There are four pairs.

Choose grading shifts and cotangent partners that turn each pair into the quartet just constructed. This specifies the free paired sector. Give its quartets positive frequencies ω_k with $c|k| \le \omega_k \le C|k|$ outside a finite set, and define, for $\ell > 0$,

$$W_k = \exp\!\left(\frac{\hbar C_k}{2\omega_k}\right), \qquad r_k = e^{-\ell\omega_k}.$$

Here C_k is the quartet contraction of (9.28). The paired covariance is $\hbar/(2\omega_k)$ and commutes with the mode differential. Tensor products define preparation and transfer on polynomials in finitely many modes. Smoothly smeared contractions converge by rapid decay of their Fourier coefficients.

Complete the occupation basis with the vacuum as reference vector. The logarithm of the absolute transfer trace is a fixed finite multiple of

$$\sum_{k\neq 0} 2\log\frac{1+r_k}{1-r_k} < \infty.$$

Indeed, $\sum_{k\neq 0} e^{-\ell\omega_k} < \infty$ by lattice shell counting. Absolute convergence permits the graded product. Each quartet contributes one, so the nonzero-mode supertrace is one.

At $k = 0$ the differential vanishes. Before field shifts and cotangent partners, the residual complex is

$$\Lambda^\bullet\langle ds, d\bar{z}_1, d\bar{z}_2\rangle,$$

with eight components per coefficient copy. Its polarization and state remain separate data. This free mixed Fourier preparation has not identified those data with the compact residual state of (9.4).

9.5 A scalar collar with infinitely many modes

We next construct a positive scalar preparation on the compact five-dimensional space. Its reference measure and vacuum energy are fixed before the mode cutoff is removed. This makes it possible to prove sewing and then add a nonnegative interaction. The scalar sector is independently defined from the gauge quartets and their residual cohomology.

Let $\mathbb{T}^5 = (\mathbb{R}/2\pi\mathbb{Z})^5$ with its flat volume measure. Fix $m > 0$ and numerical $\hbar > 0$. Put

$$\Omega_0 = \sqrt{-\Delta_{\mathbb{T}^5} + m^2}, \qquad \Omega_0 e_j = \omega_j e_j,$$

where e_j is a real orthonormal Fourier basis. Write k_j for a frequency of e_j. The eigenvalue multiset is $\omega_k = \sqrt{|k|^2 + m^2}$, $k \in \mathbb{Z}^5$. The sine and cosine multiplicities give this same count.

Take independent centered Gaussian coordinates x_j with variances $c_j = \hbar/(2\omega_j)$.

For $s > 2$, the random series $\phi = \sum_j x_j e_j$ defines a Gaussian probability measure μ on $H^{-s}(\mathbb{T}^5)$. Indeed,

$$\mathbb{E}\|\phi\|^2_{H^{-s}} = \frac{\hbar}{2} \sum_{k\in\mathbb{Z}^5} \frac{(1 + |k|^2)^{-s}}{\sqrt{|k|^2 + m^2}} < \infty. \tag{9.34}$$

Shell counting gives summands of order n^{3-2s} after grouping lattice points of radius n. The finite sums converge in mean square in this Hilbert space. For real smooth f,

$$\int e^{i\langle \phi, f\rangle}\, d\mu(\phi) = \exp\!\left(-\frac{\hbar}{4}\langle f, \Omega_0^{-1} f\rangle\right). \tag{9.35}$$

Proposition 9.10 (The exact Sobolev threshold)**.** *The measure μ is concentrated on $H^{-s}(\mathbb{T}^5)$ if and only if $s > 2$.*

Proof. The finite expectation in (9.34) proves sufficiency. For necessity put $a_j = (1 + |k_j|^2)^{-s} c_j$ and $\xi_j = x_j/\sqrt{c_j}$. Independence gives

$$\mathbb{E}\exp\!\left(-\sum_{j\le K} a_j \xi_j^2\right) = \prod_{j\le K} (1 + 2a_j)^{-1/2}.$$

For $s \le 2$, shell counting gives $\sum_j a_j = \infty$. Hence $\sum_j \log(1 + 2a_j) = \infty$. Indeed, either infinitely many a_j are bounded below, or the logarithms are comparable to a_j on the tail. The product tends to zero.

Bounded convergence shows that the exponential of the negative full square sum is zero almost surely. Thus that square sum is infinite almost surely, proving necessity. □

9.5.1 *The measure in the Mehler kernel*

For a collar of length $\ell > 0$, let $r_j = e^{-\ell\omega_j}$. Relative to the fixed measure μ, the first K modes have kernel

$$p_\ell^{(K)}(x, y) = \prod_{j=1}^{K} (1 - r_j^2)^{-1/2} \exp\!\left[-\sum_{j=1}^{K} \frac{r_j^2(x_j^2 + y_j^2) - 2r_j x_j y_j}{2c_j(1 - r_j^2)}\right]. \tag{9.36}$$

Unused modes have kernel one on the full probability space. In one coordinate, $p_\ell(x, y)\,d\mu(y)$ is the Gaussian law with mean rx and variance $c(1 - r^2)$.

The ordinary oscillator amplitude relative to Lebesgue measure is

$$Z_\ell(a, b) = \left(\frac{\omega}{2\pi\hbar \sinh(\omega\ell)}\right)^{1/2} \exp\left[-\frac{\omega((a^2 + b^2)\cosh(\omega\ell) - 2ab)}{2\hbar \sinh(\omega\ell)}\right]. \tag{9.37}$$

If ψ_0^2 is the Gaussian density of variance $\hbar/(2\omega)$, direct substitution gives

$$p_\ell(a, b) = e^{\ell\omega/2}\frac{Z_\ell(a, b)}{\psi_0(a)\psi_0(b)}. \tag{9.38}$$

Thus (9.36) includes a change of reference measure and a ground-energy subtraction.

Theorem 9.11. *The kernels $p_\ell^{(K)}$ converge in $L^2(\mu \otimes \mu)$ to a kernel p_ℓ that is positive almost everywhere. For each pair of positive lengths, almost everywhere,*

$$\int p_\ell(x, y)\, d\mu(y) = 1,$$
$$\int p_{\ell_1}(x, z)\, p_{\ell_2}(z, y)\, d\mu(z) = p_{\ell_1+\ell_2}(x, y). \tag{9.39}$$

The associated operators P_ℓ form a strongly continuous, self-adjoint Markov semigroup on $L^2(\mu)$. At every positive length they are trace class, with

$$\operatorname{Tr} P_\ell = \det(1 - e^{-\ell\Omega_0})^{-1},$$
$$\|p_\ell\|^2_{L^2(\mu\otimes\mu)} = \det(1 - e^{-2\ell\Omega_0})^{-1}. \tag{9.40}$$

These Fredholm determinants converge on the one-particle space $L^2(\mathbb{T}^5)$.

Proof. Composition of one-mode transitions gives mean $r_1 r_2 x$ and variance $c(1 - r_1^2 r_2^2)$. This proves finite-mode sewing with its normalization. Direct Gaussian integration gives

$$\int p_r(x, y)^2\, d\mu(x)d\mu(y) = (1 - r^2)^{-1}.$$

Each independent coordinate factor has expectation one. Consequently $p_\ell^{(K)}$ is a nonnegative martingale in the first K pairs of coordinates, with squared norm $\prod_{j\le K}(1 - r_j^2)^{-1}$.

For every $a > 0$, shell counting gives $\sum_j e^{-a\omega_j} < \infty$. Also $r_j \le e^{-\ell m} < 1$. The squared norms are bounded, so the martingale converges in L^2 to a nonnegative kernel.

To prove positivity, take the logarithm of the product. The diagonal quadratic terms have summable absolute expectations. The independent centered cross terms have summable variances, and the constant logarithmic terms converge. The logarithm is therefore finite almost everywhere.

Integration in one variable is an L^2 contraction, so normalization passes to the limit. Kernel convergence gives Hilbert–Schmidt convergence of the operators. The finite operators are contractions. Their composition identity therefore passes to the limit, proving (9.39). Symmetry follows from the kernel formula.

Normalized Hermite polynomials in one coordinate have eigenvalues r^n, as follows from their generating function. Finite products give an orthonormal basis of $L^2(\mu)$: conditional expectations onto finitely many coordinates converge in L^2, and the finite-dimensional Hermite bases are complete. The eigenvalues are $\prod_j r_j^{n_j}$ for finite-support occupation sequences. They tend to one as $\ell \downarrow 0$ on each basis vector. Density and contraction prove strong continuity.

Monotone summation of these positive eigenvalues gives

$$\operatorname{Tr} P_\ell = \prod_j \sum_{n\ge 0} r_j^n = \prod_j (1 - r_j)^{-1} < \infty.$$

The one-particle operator $e^{-\ell\Omega_0}$ is trace class, so this is its stated Fredholm determinant. The identity $P_{2\ell} = P_\ell^* P_\ell$ gives the second one. □

For a Fourier cutoff $\|k\|_\infty \le K$, let $Z_\ell^{(K)}$ be the truncated trace and $Z_\ell = \operatorname{Tr} P_\ell$. The shell cardinality is

$$(2n+1)^5 - (2n-1)^5 = 160n^4 + 80n^2 + 2.$$

Since $\omega_k \ge \|k\|_\infty$ and $-\log(1-x) \le x/(1-x)$,

$$0 \le \log Z_\ell - \log Z_\ell^{(K)} \le \frac{\sum_{n>K} (160n^4 + 80n^2 + 2) e^{-\ell n}}{1 - e^{-\ell(K+1)}}. \tag{9.41}$$

The massive zero mode is included. At $m = 0$, its variance and trace factor are undefined in this construction. It must then be prepared as a residual variable.

9.6 An interaction at fixed positive smoothing

Fix $\sigma > 0$. Define a smoothing operator and potential by

$$R_\sigma = e^{-\sigma\Omega_0}, \qquad V_\sigma(\phi) = \frac{1}{24} \int_{\mathbb{T}^5} (R_\sigma \phi)(z)^4 \, dz. \tag{9.42}$$

The map R_σ sends H^{-s} continuously into every fixed smooth Sobolev space. Thus V_σ is continuous, finite and nonnegative on H^{-s}. Smoothing bounds it by a constant times $\|\phi\|_{H^{-s}}^4$. All its Gaussian moments are finite.

For paths fix $s > 2$. Let Φ_t be the stationary Ornstein–Uhlenbeck process with coordinate covariance

$$\mathbb{E}(\Phi_j(t)\Phi_j(t')) = c_j e^{-\omega_j |t-t'|}.$$

Proposition 9.12. *The process has a continuous $H^{-s}(\mathbb{T}^5)$ modification, locally Hölder of every order $\alpha < \min\{1/2, s-2\}$. For $0 < h \le 1/2$ its increment variance satisfies*

$$\mathbb{E}\|\Phi_{t+h} - \Phi_t\|_{H^{-s}}^2 \le C_s \begin{cases} h^{2s-4}, & 2 < s < 5/2, \\ h \log(e/h), & s = 5/2, \\ h, & s > 5/2. \end{cases} \tag{9.43}$$

Proof. The increment variance is

$$2 \sum_j (1 + |k_j|^2)^{-s} c_j (1 - e^{-\omega_j h}).$$

The bound $1 - e^{-x} \le \min\{1, x\}$ and shell counting reduce it to a constant times $\sum_{n\ge 1} n^{3-2s} \min\{1, nh\}$, plus a bounded multiple of h from the massive zero mode. Splitting this sum at $n = h^{-1}$ gives (9.43). For $s > 5/2$, the simpler bound $\hbar h \sum_j (1 + |k_j|^2)^{-s}$ also converges.

For a centered Hilbert-valued Gaussian G, diagonalize its covariance and apply the L^q triangle inequality to its squared norm. This gives $\mathbb{E}\|G\|^{2q} \le C_q (\mathbb{E}\|G\|^2)^q$. Apply this to the increments and use the moment continuity criterion with arbitrarily large integers q. At $s = 5/2$, the logarithm is bounded by every negative power of h, giving each strict exponent $\alpha < 1/2$. □

The transition kernel of this continuous modification is p_t.

The continuous path space is separable, so regular conditional Gaussian bridge laws exist given $\Phi_0 = x$, $\Phi_\ell = y$. Their endpoint law is $p_\ell(x, y)\,d\mu(x)\,d\mu(y)$. Positivity of p_ℓ makes this measure equivalent to $\mu \otimes \mu$. For $g \geq 0$ define

$$p_\ell^{g,\sigma}(x, y) = p_\ell(x, y)\,\mathbb{E}_{x,y}^{\text{bridge}} \exp\left(-g \int_0^\ell V_\sigma(\Phi_t)\,dt\right). \tag{9.44}$$

Here g is the coefficient in the Euclidean weight. An action coupling g_{phys} corresponds to $g = g_{\text{phys}}/\hbar$.

Theorem 9.13. *For $g \geq 0$ and $\sigma > 0$, the kernel (9.44) is symmetric, belongs to $L^2(\mu \otimes \mu)$, and satisfies $0 \leq p_\ell^{g,\sigma} \leq p_\ell$. For each pair of positive lengths, almost everywhere,*

$$\int p_{\ell_1}^{g,\sigma}(x, z)\,p_{\ell_2}^{g,\sigma}(z, y)\,d\mu(z) = p_{\ell_1+\ell_2}^{g,\sigma}(x, y). \tag{9.45}$$

The operators form a strongly continuous, self-adjoint, positivity-preserving contraction semigroup. At positive length they are trace class and satisfy

$$\operatorname{Tr} P_\ell^{g,\sigma} = \|P_{\ell/2}^{g,\sigma}\|_{\text{HS}}^2 \leq \det(1 - e^{-\ell\Omega_0})^{-1}. \tag{9.46}$$

Proof. First use a bounded nonnegative potential. Its path weight lies between zero and one. Conditioning at an intermediate time and using the Markov property gives the sewing law. Reversal of the stationary path gives symmetry.

Apply this construction to $V_\sigma \wedge M$ and let $M \to \infty$. On every continuous path, the potential integral is finite by continuity and compactness of the time interval. The weights decrease to the weight in (9.44). Domination by p_ℓ gives convergence in L^2. In the sewing integral the product is dominated by the free product, whose integral is $p_{\ell_1+\ell_2}$. Dominated convergence proves the claimed sewing law.

The same kernel is obtained by spatial mode cutoffs. Fourier projections converge uniformly on the compact image of a continuous H^{-s} path. After smoothing, the potentials and their integrated weights therefore converge. The finite-coordinate marginal of the full bridge is precisely the corresponding finite-mode bridge. Its conditional weights converge almost everywhere and are bounded by one. Together with $p_\ell^{(K)} \to p_\ell$ in L^2, this proves the cutoff assertion.

Kernel domination gives the Hilbert–Schmidt bound. Symmetry and sewing imply

$$P_\ell^{g,\sigma} = (P_{\ell/2}^{g,\sigma})^* P_{\ell/2}^{g,\sigma}.$$

This proves trace class, operator positivity and (9.46). The inequality $P_\ell^{g,\sigma} 1 \leq 1$ and symmetry give contraction on $L^2(\mu)$. On bounded continuous cylinder functions, path continuity and bounded convergence prove strong continuity at zero. Density and contraction extend it to all of $L^2(\mu)$. □

Corollary 9.14. *At fixed $g \geq 0$, $\sigma > 0$ and $\ell > 0$, the interacting mode-cutoff operators converge to $P_\ell^{g,\sigma}$ in trace norm. The finite kernels are extended by the constant kernel one in the unused modes.*

Proof. The proof of Theorem 9.13 gives Hilbert–Schmidt convergence at length $\ell/2$. Write A_K and A for the half-length operators. The finite sewing identity includes the Gaussian conditional-expectation projection on unused modes. Consequently its length-ℓ operator is A_K^2, and the limiting sewing identity gives A^2.

Decompose the difference as $(A_K - A)A_K + A(A_K - A)$. The trace-ideal product estimate gives

$$\|A_K^2 - A^2\|_1 \leq \|A_K - A\|_2(\|A_K\|_2 + \|A\|_2),$$

which tends to zero. □

At positive coupling these operators are generally sub-Markov: their mass can be less than one. Dividing each interval kernel by its own mass would generally change the sewing law. The path weights in (9.44) remain part of the amplitude.

9.6.1 *The first insertion and the ordering of composites*

Conditioning the stationary covariance on both endpoints gives

$$m_j(t) = \frac{\sinh(\omega_j(\ell - t))}{\sinh(\omega_j \ell)} x_j + \frac{\sinh(\omega_j t)}{\sinh(\omega_j \ell)} y_j, \tag{9.47}$$

$$v_j(t) = \frac{\hbar \sinh(\omega_j t)\sinh(\omega_j(\ell - t))}{\omega_j \sinh(\omega_j \ell)}. \tag{9.48}$$

For a direct derivation, put $r = e^{-\omega_j t}$ and $s = e^{-\omega_j(\ell - t)}$. The endpoint covariance is $c_j \left(\begin{smallmatrix} 1 & rs \\ rs & 1 \end{smallmatrix}\right)$ and the cross covariance is $c_j(r, s)$. Multiplying by the inverse endpoint matrix gives the mean. The Schur complement is $c_j(1 - r^2)(1 - s^2)/(1 - r^2 s^2)$, giving the variance.

Define the smoothed mean and pointwise variance by

$$m_\sigma(t, z) = \sum_j e^{-\sigma\omega_j} m_j(t) e_j(z), \qquad v_\sigma(t, z) = \sum_j e^{-2\sigma\omega_j} v_j(t) e_j(z)^2.$$

These series converge with every fixed spatial derivative. The conditional variance is bounded by c_j, and the smoothed mean is bounded by the endpoint H^{-s} norms. Consequently all moments of the integrated potential are finite for almost every endpoint pair.

Proposition 9.15. *The first right response at zero coupling is*

$$-\partial_g \log \frac{p_\ell^{g,\sigma}(x, y)}{p_\ell(x, y)}\bigg|_{g=0+} = \frac{1}{24} \int_0^\ell dt \int_{\mathbb{T}^3} dz \left(m_\sigma^4 + 6 m_\sigma^2 v_\sigma + 3 v_\sigma^2\right). \tag{9.49}$$

All higher right derivatives at zero exist as finite bridge moments and cumulants.

Proof. Put $\mathcal{V} = \int_0^\ell V_\sigma(\Phi_t)\, dt$. The quotient $(1 - e^{-g\mathcal{V}})/g$ is bounded by $\mathcal{V}$ and tends to it as $g \downarrow 0$. Dominated convergence gives the derivative of its conditional expectation. At zero coupling that expectation is one, so the negative logarithmic derivative is $\mathbb{E}^{\mathrm{bridge}}_{x,y} \mathcal{V}$.

For a Gaussian of mean m and variance v, $\mathbb{E}X^4 = m^4 + 6m^2 v + 3v^2$. Integrating this identity proves the formula. Higher finite moments justify repeated right differentiation. Differentiating the logarithm gives their cumulants. □

The last two terms retain fluctuations about the conditional mean. A change of covariance has its own polynomial ordering rule. For a scalar variance C, define $:F:_C = e^{-C\partial_x^2/2} F$. Then

$$:F:_{C_2} = e^{-(C_2 - C_1)\partial_x^2/2} :F:_{C_1}, \qquad :x^4:_C = x^4 - 6Cx^2 + 3C^2. \tag{9.50}$$

The exponentials terminate on polynomials.

On smeared kernels the same identity uses the corresponding covariance contraction. It does not define unsmoothed coincident products.

9.7 Divergence and Borel reconstruction

The finite bridge moments determine an asymptotic expansion at zero coupling. Their factorial growth also determines a Borel reconstruction of the positive-coupling kernel. Write $X = \mathbb{T}^3$, $\mathcal{V} = \int_0^\ell V_\sigma(\Phi_t)\,dt$ and $F_{\ell,x,y}(g) = \mathbb{E}^{\mathrm{bridge}}_{x,y} e^{-g\mathcal{V}}$. This expectation is defined for $\operatorname{Re} g \geq 0$. Let $J_0(z) = \sum_{n\geq 0}(-1)^n (z^2/4)^n/(n!)^2$ be the Bessel function of order zero.

Lemma 9.16. *Let $\mathcal{V} \geq 0$ be a random variable with $\mathbb{E}e^{\varepsilon\sqrt{\mathcal{V}}} < \infty$ for some $\varepsilon > 0$. Put $M_n = \mathbb{E}\mathcal{V}^n$ and $F(g) = \mathbb{E}e^{-g\mathcal{V}}$ for $g \geq 0$. The Borel germ of its formal expansion is*

$$\widehat{F}(\zeta) = \sum_{n\geq 0} \frac{(-1)^n M_n}{(n!)^2}\zeta^n.$$

It continues along the nonnegative ray by

$$\widehat{F}(\zeta) = \mathbb{E}J_0(2\sqrt{\zeta\mathcal{V}}), \qquad |\widehat{F}(\zeta)| \leq 1 \quad (\zeta \geq 0), \tag{9.51}$$

and for every $g > 0$,

$$F(g) = \int_0^\infty e^{-t}\widehat{F}(gt)\,dt. \tag{9.52}$$

The continuation is analytic wherever $2|\operatorname{Im}\sqrt{\zeta}| < \varepsilon$, together with its germ at zero.

Proof. The function $J_0(2\sqrt{\zeta\mathcal{V}})$ is entire in ζ, by its power series. The integral representation

$$J_0(z) = \frac{1}{2\pi}\int_0^{2\pi} e^{iz\cos\theta}\,d\theta$$

gives $|J_0(z)| \leq e^{|\operatorname{Im} z|}$. For small $|\zeta|$, absolute summation is dominated by $e^{2\sqrt{|\zeta|}\sqrt{\mathcal{V}}}$. This proves the germ identity. The same bound on compact subsets proves the stated analytic continuation, while for nonnegative ζ the bound is one.

For fixed $w \geq 0$, termwise integration is justified by the convergent sum $\sum_n (gw)^n/n! = e^{gw}$ and gives

$$\int_0^\infty e^{-t}J_0(2\sqrt{gtw})dt = e^{-gw}.$$

To average this identity, use $|J_0(2\sqrt{gt\mathcal{V}})| \leq 1$. Fubini now requires only the integrable bound e^{-t} and proves (9.52). No exponential moment $\mathbb{E}e^{g\mathcal{V}}$ is required. □

Theorem 9.17. *Fix $m, \hbar, \ell, \sigma > 0$ and $s > 2$ in (9.44). For almost every endpoint pair (x, y) there are constants $0 < c_- \leq c_+ < \infty$ such that, for $n \geq 1$,*

$$c_-^n (n!)^2 \leq \mathbb{E}^{\mathrm{bridge}}_{x,y}\mathcal{V}^n \leq c_+^n (n!)^2. \tag{9.53}$$

The coupling series of $F_{\ell,x,y}$ has radius zero and exact Gevrey order one. Its coefficients are bounded by an exponential times $n!$. No exponential multiple of $(n!)^a$, with $a < 1$, bounds them. It is an asymptotic expansion on $\operatorname{Re} g \geq 0$, with a remainder bounded by $c_+^N N!|g|^N$ after N terms.

Its endpointwise Borel continuation and Laplace integral are given by Lemma 9.16. Multiplication by p_ℓ gives $L^2(\mu\otimes\mu)$ kernels along the positive Borel ray, whose Bochner Laplace integral is exactly $p_\ell^{g,\sigma}$. Thus this reconstruction preserves the sewing of Theorem 9.13.

Proof. Write $D = \ell \operatorname{vol}(X)$ and $u(t,z) = (R_\sigma \Phi_t)(z)$. Its conditional mean has a finite uniform bound M, and its conditional pointwise variance has a finite uniform bound v_*, because $\sigma > 0$ and $x, y \in H^{-s}$. For a standard real Gaussian Z, let $b_n = \mathbb{E} Z^{4n} = (4n-1)!!$. The exact ratio

$$\frac{b_{n+1}}{b_n} = (4n+1)(4n+3)$$

lies between $3(n+1)^2$ and $16(n+1)^2$. Hence $3^n (n!)^2 \le b_n \le 16^n (n!)^2$. Using $|a+b|^{4n} \le 2^{4n-1}(|a|^{4n} + |b|^{4n})$ and Jensen's inequality on the finite measure space $[0,\ell] \times X$ yields

$$\mathbb{E}\mathcal{V}^n \le \left[\frac{D}{24}\, 16(M^4 + 16 v_*^2)\right]^n (n!)^2.$$

For a lower bound use the space-time average $Y = D^{-1} \int_0^\ell \int_X u(t,z)\, dz\, dt$. Hölder's inequality gives $\mathcal{V} \ge (D/24) Y^4$. The constant spatial mode is $e_0 = \operatorname{vol}(X)^{-1/2}$, so

$$Y = \frac{e^{-\sigma m}}{\ell \sqrt{\operatorname{vol}(X)}} \int_0^\ell \Phi_0(t)\, dt.$$

Its conditional variance is

$$v_Y = \frac{\hbar e^{-2\sigma m}}{\ell^2 \operatorname{vol}(X) m^2} \left(\ell - \frac{2}{m} \tanh \frac{m\ell}{2}\right) > 0. \tag{9.54}$$

To obtain this formula, the covariance of the pinned constant mode is $\hbar$ times the Dirichlet Green kernel of $-\partial_t^2 + m^2$. Its integral solves $(-\partial_t^2 + m^2) f = 1$, $f(0) = f(\ell) = 0$, and is

$$f(t) = \frac{1}{m^2}\left(1 - \frac{\cosh(m(t - \ell/2))}{\cosh(m\ell/2)}\right).$$

Integrating f proves (9.54). Positivity follows from $\tanh a < a$ for $a > 0$.

All terms in the even Gaussian moment expansion are nonnegative, so $\mathbb{E} Y^{4n} \ge v_Y^{2n} b_n$. This proves the lower bound with $c_- = 3 D v_Y^2 / 24$.

Dividing (9.53) by $n!$ proves the radius and exact factorial order. The integral Taylor remainder for $e^{-g\mathcal{V}}$ has absolute value at most $|g\mathcal{V}|^N / N!$ on the closed right half-plane, which proves the asymptotic estimate.

The upper moment bound also implies exponential square-root integrability: use

$$e^{a\sqrt{\mathcal{V}}} \le 2\cosh(a\sqrt{\mathcal{V}}), \qquad \mathbb{E}\cosh(a\sqrt{\mathcal{V}}) = \sum_{n \ge 0} \frac{a^{2n} \mathbb{E}\mathcal{V}^n}{(2n)!} < \infty$$

for $a^2 c_+ < 1$, since $(2n)! \ge (n!)^2$. Lemma 9.16 applies.

Along the positive ray the reconstructed kernel is bounded in absolute value by p_ℓ. Dominated convergence makes it continuous in L^2. Its Bochner integral therefore exists in L^2, and the endpointwise identity identifies it with (9.44). The resulting equality of kernels transfers their actual convolution law. □

The analytic germ in this theorem is endpointwise. Its radius may depend on the endpoints. The positive-ray Bochner integral uses the uniform domination by p_ℓ. No endpoint-independent analytic radius in L^2 is needed for that assertion.

The second connected response has an explicit form as well. Let X_1, X_2 be smoothed field values with means a, b, variances v, w and covariance C. Gaussian expansion gives

$$\begin{aligned}\operatorname{Cov}(X_1^4, X_2^4) = {} & 16ab(a^2 + 3v)(b^2 + 3w)C + 72(a^2 + v)(b^2 + w)C^2 \\ & + 96abC^3 + 24C^4.\end{aligned} \tag{9.55}$$

For a proof, expand X_1^4 in the Wick polynomials of its centered Gaussian part. The coefficients of Wick degrees 1, 2, 3, 4 are $4a(a^2 + 3v)$, $6(a^2 + v)$, $4a$, 1. Cross expectation of equal Wick degrees r is $r! C^r$, while unequal degrees have zero cross expectation. This proves (9.55). Its double integral divided by 24^2 is $\partial_g^2 \log F_{\ell,x,y}(0+) = \operatorname{Var}(\mathcal{V})$.

9.8 A unitary comparison of the scalar states

The scalar kernel has a Hilbert Fock realization with the same interaction and vacuum normalization. Construct its Hamiltonian on the symmetric Hilbert Fock space

$$\mathcal{F}_{\Omega_0} = \widehat{\bigoplus}_{n\geq 0} \operatorname{Sym}^n \ell^2(\{j\}),$$

where the summands are symmetric Hilbert tensor powers. Its orthonormal occupation basis $|\mathbf{n}\rangle$ is indexed by finitely supported sequences of nonnegative integers. The creation and annihilation operators $b_j^\dagger, b_j$ obey $[b_i, b_j^\dagger] = \delta_{ij}$, with all other commutators zero. The energy $d\Gamma(\Omega_0)$ has eigenvalue $\sum_j \omega_j n_j$.

For $\sigma > 0$ define the selfadjoint Segal field

$$Q_\sigma(z) = \sum_j e^{-\sigma\omega_j} e_j(z) \sqrt{c_j}\,(b_j + b_j^\dagger).$$

The real coefficient sequence is square-summable uniformly in z. On $\mathcal{F}_{\Omega_0}$ define the maximal operator

$$\mathsf{T}_\sigma : \mathcal{F}_{\Omega_0} \longrightarrow L^2(X; \mathcal{F}_{\Omega_0}), \qquad (\mathsf{T}_\sigma \psi)(z) = Q_\sigma(z)^2 \psi,$$

whose domain consists of vectors for which the displayed expression is defined almost everywhere and is square-integrable. This operator is closed. A convergent graph sequence has a subsequence converging pointwise almost everywhere in the target. Closedness of each $Q_\sigma(z)^2$ identifies the limit.

On particle sectors of number at most K, the creation and annihilation bounds give $\|Q_\sigma(z)^2\psi\| \leq C_\sigma(K+2)\|\psi\|$ uniformly in z. Thus finite occupation vectors lie in its domain.

For $g > 0$ define the quadratic form

$$\begin{aligned} q_{g,\sigma}(\psi) &= \|(d\Gamma(\Omega_0))^{1/2}\psi\|^2 + \frac{g}{24}\|\mathsf{T}_\sigma\psi\|^2, \\ D(q_{g,\sigma}) &= D((d\Gamma(\Omega_0))^{1/2}) \cap D(\mathsf{T}_\sigma). \end{aligned} \tag{9.56}$$

It is nonnegative and closed: its form norm is the joint graph norm of two closed operators. Its domain contains the algebraic span of the occupation basis and is therefore dense.

Let $K_{g,\sigma}$ be its associated nonnegative selfadjoint operator. For $g = 0$ set $K_{0,\sigma} = d\Gamma(\Omega_0)$ on $D(d\Gamma(\Omega_0))$. The form representation and increasing-form convergence apply to these densely defined nonnegative forms.[4]

Let He_n be the probabilists' Hermite polynomial, defined by $e^{ty - t^2/2} = \sum_n \mathrm{He}_n(y) t^n/n!$. For a finitely supported occupation sequence set

$$h_{\mathbf{n}}(x) = \prod_j \frac{\mathrm{He}_{n_j}(x_j/\sqrt{c_j})}{\sqrt{n_j!}}.$$

Proposition 9.18. *The map*

$$\mathcal{U} : L^2(\mu) \longrightarrow \mathcal{F}_{\Omega_0}, \qquad \mathcal{U} h_{\mathbf{n}} = |\mathbf{n}\rangle$$

[4] See T. Kato, *Perturbation Theory for Linear Operators*, second edition, corrected printing. The results are VI.2.1, VI.2.6, VI.2.23 and VIII.3.13a.

extends to a unitary isomorphism. On polynomial cylinder vectors it preserves the following operators and their common commutation relations:

$$\begin{aligned} \mathcal{U}1 &= |0\rangle, \\ \mathcal{U}\, x_j\, \mathcal{U}^{-1} &= \sqrt{c_j}(b_j + b_j^\dagger), \\ \mathcal{U}\, \sqrt{c_j}\partial_{x_j}\, \mathcal{U}^{-1} &= b_j, \\ \mathcal{U} P_\ell \mathcal{U}^{-1} &= e^{-\ell d\Gamma(\Omega_0)}. \end{aligned}$$

At fixed smoothing, the multiplication form of V_σ is carried to $\|\mathsf{T}_\sigma\psi\|^2/24$, with equality of the full form domains. For $g \geq 0$ one has

$$\mathcal{U} P_\ell^{g,\sigma} \mathcal{U}^{-1} = e^{-\ell K_{g,\sigma}}. \tag{9.57}$$

This comparison preserves operator composition, adjoints, traces and normalized positive states.

Proof. The generating function and Gaussian integration give $\mathbb{E}\mathrm{He}_n(Z)\mathrm{He}_r(Z) = n!\delta_{nr}$. Finite products form an orthonormal basis because conditional expectations onto finitely many coordinates converge in L^2 and finite-dimensional Hermite polynomials are complete. Thus $\mathcal{U}$ is unitary. The recurrence $y\mathrm{He}_n = \mathrm{He}_{n+1} + n\mathrm{He}_{n-1}$ and $\mathrm{He}_n' = n\mathrm{He}_{n-1}$ prove the field and derivative identities. In particular, the Gaussian adjoint of $\sqrt{c_j}\partial_{x_j}$ is $x_j/\sqrt{c_j} - \sqrt{c_j}\partial_{x_j}$, matching $b_j^\dagger$. The free heat identity follows on the same basis and hence on the Hilbert completion.

Exponential smoothing makes the coefficient sequence of $Q_\sigma(z)$ square-summable, with all fixed spatial derivatives. The multiplication operator $R_\sigma\phi(z)$ is its unitary image under the preceding identities, first on cylinders and then by closure.

Finite occupation vectors are analytic for every real Segal field, by the creation and annihilation bounds. Their Gaussian polynomials are analytic for the corresponding real linear multiplication operator, so equality on this core identifies the selfadjoint closures.

The spectral theorem identifies the full domain of $Q_\sigma(z)^2$ with multiplication by $(R_\sigma\phi(z))^2$. Tonelli's theorem then identifies the domain of T_σ with $D(V_\sigma^{1/2})$.

Let H_{OU} be the nonnegative generator of P_ℓ. Together with the free heat identity, this identifies the full closed form (9.56) with the Gaussian form

$$\|H_{\mathrm{OU}}^{1/2} f\|^2 + g\|V_\sigma^{1/2} f\|^2.$$

For each bounded potential $\min(V_\sigma, M)$, expansion in its bounded multiplication operator and the Markov law give the Feynman–Kac semigroup. The forms increase to the displayed closed form as $M \to \infty$. Their associated resolvents and semigroups converge strongly, while their kernels converge by bounded convergence of the path weights. This is precisely the preparation in Theorem 9.13, proving (9.57).

Trace-class positivity and unitary invariance of the trace then give, for every bounded A,

$$\frac{\mathrm{Tr}(P_\ell^{g,\sigma} A)}{\mathrm{Tr}\, P_\ell^{g,\sigma}} = \frac{\mathrm{Tr}\big(e^{-\ell K_{g,\sigma}}(\mathcal{U} A \mathcal{U}^{-1})\big)}{\mathrm{Tr}\, e^{-\ell K_{g,\sigma}}}.$$

This is an equality of prepared states under the specified unitary, with the same reference vacuum and coupling. □

9.9 The comparison with native quantum fields

The compact residual state, determinant-line Gaussian, BRST quartet and positive scalar collar each have a specified domain and sewing law. Their constructions supply concrete targets for the native comparison of Problem 8.22. That comparison must preserve more than the vacuum amplitude.

The invariant functions O_r of Chapter 3 already lift classical matrix invariants to smooth Neumann fields. When the chosen real quadratic form is $\langle J, J\rangle = c_G\,\mathrm{tr}(J^2)$, its potential is $vc_G O_2$. Its quantum

image must specify the ordering, including the lower-Casimir corrections. The half-density computation also requires a map between the holonomy coordinate X and the residual field J.

Problem 9.19. Construct a renormalized observable complex for the native Neumann fields and a boundary state map from its action. Specify a common regulator, the determinant or Pfaffian line, the residual measure and the boundary BRST polarization. Compare the resulting invariant degree-zero states with $L^2(G, dg)^G$ and the heat operator (9.2).

For a cut surface $U = U_1 \cup_\Sigma U_2$, let F_i be regular observables supported inside U_i. Construct source-dependent sewing of the form

$$\Pi_U(F_1F_2) = \mathrm{BV}_{\mathrm{res}}\big\langle \Pi_{U_1}(F_1), \Pi_{U_2}(F_2)\big\rangle_\Sigma,$$

where the pairing includes line transport and the residual BV integral satisfies Stokes. Require adjoint boundary differentials and compatibility with the supported insertion operations. Determine the disk normalization, vacuum energy, holonomy density and ordering of the lifted O_r. If the prepared states differ, compute their first invariant defect modulo the allowed local cyclic counterterms.

The scalar interaction poses a separate analytic limit: construct local counterterms and a state for which its sewing maps converge as $\sigma \downarrow 0$. At fixed positive smoothing, the divergent coupling series has a Borel reconstruction and the scalar Fock comparison is unitary. The local limit and the comparison with the physical BRST fields, their residual sector and their boundary pairing remain to be constructed.

Chapter 10

Hamiltonian fields and continuous coefficients

A Hamiltonian function can serve as a field on a symplectic surface or as an independent coefficient of another field. These uses have different differentials and support conditions. Their formal coordinates alone do not identify the resulting theories.

Finite Hamiltonian jets provide explicit coefficient algebras. Their duals, quantizations and continuous cochains require separate constructions. The first issue is already visible before quantization: discarding a Hamiltonian jet reverses direction on its dual. For cochains, the output jet and the input jet must be allowed to differ.

10.1 Pointed Hamiltonians and their finite quotients

Work over $\mathbb{C}$ and put $\mathfrak{m} = (u, v)$ in $\mathbb{C}[[u, v]]$. Give this ring its $\mathfrak{m}$-adic topology, with discrete finite quotients. The Poisson bracket is

$$\{f, g\} = \partial_u f\, \partial_v g - \partial_v f\, \partial_u g. \tag{10.1}$$

The Hamiltonian vector field is $X_f = \{f, -\}$. Constants have zero vector field. Thus $\mathfrak{h}_{\mathrm{aff}} = \mathbb{C}[[u, v]]/\mathbb{C}$ contains every formal Hamiltonian vector field on the plane.

The pointed algebra is the Lie subalgebra $\mathfrak{h} = \mathfrak{m}^2$: its vector fields vanish at the origin. The filtration $F^a\mathfrak{h} = \mathfrak{m}^{a+2}$, for $a \geq 0$, satisfies

$$\{F^a\mathfrak{h}, F^b\mathfrak{h}\} \subset F^{a+b}\mathfrak{h}.$$

For $d \geq 2$, write W_r for homogeneous polynomials of degree r and set

$$\mathfrak{h}_d = \mathfrak{m}^2/\mathfrak{m}^{d+1} = \bigoplus_{r=2}^{d} W_r.$$

These coefficient Lie algebras have cohomological degree zero. Polynomial degree is a separate grading.

Proposition 10.1. *The bracket descends to every $\mathfrak{h}_d$, and the degree projections are Lie maps. Moreover,*

$$\mathfrak{h} = \varprojlim_d \mathfrak{h}_d, \qquad \dim \mathfrak{h}_d = \frac{d(d+3)}{2} - 2.$$

Every $\mathfrak{h}_d$ is unimodular: $\mathrm{Tr}_{\mathfrak{h}_d}(\mathrm{ad}_x) = 0$ *for every x.*

Proof. The bracket sends $\mathfrak{m}^r \times \mathfrak{m}^s$ into $\mathfrak{m}^{r+s-2}$. Hence $\mathfrak{m}^{d+1}$ is an ideal in $\mathfrak{m}^2$. Compatible jets determine one formal series, and summing $\dim W_r = r + 1$ gives the dimension.

Set $E = u^2/2$, $F = -v^2/2$ and $H = -uv$. Direct differentiation gives

$$\{H, E\} = 2E, \qquad \{H, F\} = -2F, \qquad \{E, F\} = H.$$

On $u^{r-j}v^j$, the H-weights are $r-2j$. They sum to zero. The operators E, F move between distinct weights, so their traces vanish.

Higher-degree Hamiltonians strictly raise polynomial degree and have triangular adjoint matrices with zero diagonal. Linearity proves unimodularity. □

The quadratic algebra is $\mathfrak{sl}_2$. Its presence means that $\mathfrak{h}$ is not pronilpotent. The ideal $\mathfrak{m}^3 = F^1\mathfrak{h}$ is pronilpotent, since an iterated bracket of r of its elements lies in $F^r\mathfrak{h}$.

The first nonlinear layers exhibit this filtration. Put $v_i = u^{3-i}v^i$, for $0 \le i \le 3$, and $w_j = u^{4-j}v^j$, for $0 \le j \le 4$. Then

$$\{H, v_i\} = (3-2i)v_i, \qquad \{E, v_i\} = iv_{i-1}, \qquad \{F, v_i\} = (3-i)v_{i+1},$$
$$\{v_i, v_j\} = 3(j-i)w_{i+j-1}.$$

Terms outside the indicated index ranges vanish.

Consequently $\mathfrak{h}_3 = \mathfrak{sl}_2 \ltimes W_3$ with abelian W_3. In $\mathfrak{h}_4$, the cubic brackets generate W_4, while $\{W_3, W_4\} = \{W_4, W_4\} = 0$. For example, $\{u^3, v^3\} = 9u^2v^2$ and $\{u^2v, uv^2\} = 3u^2v^2$.

Translations change the permitted quotients:

$$\{u, v^{d+1}\} = (d+1)v^d.$$

The right side survives the degree-d projection although v^{d+1} does not. The same jet ideal therefore cannot be used after adjoining the linear Hamiltonians.

10.2 Residue duals and cotangent degrees

A continuous functional from $\mathfrak{h}$ to discrete $\mathbb{C}$ annihilates some $\mathfrak{m}^{d+1}$. Its continuous dual is

$$P = \mathfrak{h}^\vee_{\mathrm{cont}} = \varinjlim_d \mathfrak{h}^\vee_d = \bigoplus_{a+b\ge 2} \mathbb{C}\rho_{ab}.$$

Here ρ_{ab} extracts the coefficient of $u^a v^b$. It can be represented by the principal part $u^{-a-1}v^{-b-1}\,du\,dv$, with residue convention

$$\mathrm{Res}_0(u^i v^j \rho_{ab}) = \delta_{ia}\delta_{jb}.$$

The algebraic direct sum has the discrete linear topology. The residue pairing with the formal product is continuous.

Use the coadjoint convention $\langle \mathrm{ad}^*_f \rho, g\rangle = -\langle \rho, \{f, g\}\rangle$. The monomial bracket gives

$$\mathrm{ad}^*_{u^iv^j} \rho_{ab} = \big(j(a+1) - i(b+1)\big)\rho_{a-i+1,b-j+1}. \tag{10.2}$$

The expression is zero when an index is negative or the output degree is less than two. Indeed, pairing with $u^k v^\ell$ forces $k = a-i+1$ and $\ell = b-j+1$, and negating the bracket coefficient gives the displayed factor.

Example 10.2. The simultaneous degree projection

$$\mathfrak{h}_4 \ltimes \mathfrak{h}_4^\vee[-1] \longrightarrow \mathfrak{h}_3 \ltimes \mathfrak{h}_3^\vee[-1]$$

is not a Lie map. Formula (10.2) gives $\mathrm{ad}^*_{u^3}\rho_{22} = -9\rho_{03}$. The projection kills ρ_{22} while preserving u^3 and ρ_{03}. Thus projecting their bracket differs from bracketing their projections.

The actual dual of $\mathfrak{h}_4 \to \mathfrak{h}_3$ is the injection $\mathfrak{h}_3^\vee \to \mathfrak{h}_4^\vee$. One may retain both systems, with their opposite directions. They do not form the simultaneous inverse system in the example.

This distinction also controls completion. For a countable basis (e_i), the vectors e_i tend to zero in the coordinate product. A continuous bilinear pairing of two such products cannot satisfy $B(e_i, e_j) = \delta_{ij}$, since $B(e_i, e_i) = 1$. Reciprocal rescaling of two dual bases leaves their contractions unchanged. In particular, contracting the first n diagonal terms still gives n. Weights alone do not define that infinite contraction.

For a finite-dimensional Lie algebra $\mathfrak{l}$, two cotangent shifts have different Chevalley degrees. With $V[k]^j = V^{j+k}$ and $C^\bullet_{\mathrm{CE}}(\mathfrak{g}) = \mathrm{Sym}(\mathfrak{g}^\vee[-1])$, put

$$\mathfrak{g}_- = \mathfrak{l} \ltimes \mathfrak{l}^\vee[-1], \qquad \mathfrak{g}_+ = \mathfrak{l} \ltimes \mathfrak{l}^\vee[1].$$

The shifted dual is abelian, with its coadjoint action. The exterior ghost c^a has degree one in both cases. The coordinate dual to the cotangent summand has degree zero for $\mathfrak{g}_-$ and degree two for $\mathfrak{g}_+$.

For the former there is an isomorphism of commutative complexes

$$C^\bullet_{\mathrm{CE}}(\mathfrak{g}_-) = C^\bullet_{\mathrm{CE}}(\mathfrak{l}, \mathrm{Sym}\,\mathfrak{l}).$$

Its differential is the adjoint Chevalley differential.

Subtracting twice the cotangent weight from the degree in $C^\bullet_{\mathrm{CE}}(\mathfrak{g}_+)$ gives the same coordinate degrees. This is an explicit regrading. It is not an identification of the two original graded field theories.

There is also a finite BV model whose antifields have degree -2. Choose $[e_a, e_b] = f^c{}_{ab} e_c$, ghosts c^a of degree one and antifields b_a of degree -2. With left odd derivatives, set

$$\Delta = \sum_a \partial_{b_a} \partial_{c^a}, \qquad S = -\frac{1}{2} \sum_{a,b,c} b_a f^a{}_{bc} c^b c^c.$$

The BV bracket is fixed by the second-order identity

$$\Delta(FG) = (\Delta F)G + (-1)^{|F|} F \Delta G + (-1)^{|F|} \{F, G\}.$$

Proposition 10.3. *If $\mathfrak{l}$ is unimodular, then $\{S, S\}/2 + \hbar \Delta S = 0$. In particular this holds for every $\mathfrak{h}_d$ with its constant Berezinian.*

Proof. The Hamiltonian differential sends c^a to $-f^a{}_{bc} c^b c^c/2$ and sends the b variables by the cotangent lift. Jacobi gives its square zero. Equivalently, the coefficients of $\{S, S\}$ are the Jacobi sums, so they vanish. Direct contraction gives

$$\Delta S = \sum_k \mathrm{Tr}(\mathrm{ad}_{e_k}) c^k.$$

Unimodularity annihilates this term. □

The finite odd cotangent model has its declared coordinates, measure and differential. Its quantum master equation supplies no measure or diagonal contraction on an unrestricted infinite coefficient product.

10.3 Horizontal physical jets

Let $z = (z_1, z_2)$ be holomorphic coordinates on a flat chart U. Introduce independent formal variables w_1, w_2 and odd variables ϑ_1, ϑ_2 of degree one. Define

$$\mathcal{J} = \mathcal{O}(U)[[w_1, w_2]] \otimes \Lambda(\vartheta_1, \vartheta_2), \qquad \nabla = \sum_i \vartheta_i (\partial_{z_i} - \partial_{w_i}).$$

The formal coefficients have the product topology, using compact-set holomorphic seminorms on each coefficient. Exterior degree is auxiliary and adds no dimension to the physical integration measure.

Define $T = \exp(w \cdot \partial_z)$ coefficientwise, and let $j(f) = Tf$ be the formal Taylor series of $f(z + w)$. Let $p(F)$ evaluate the exterior-degree-zero part at $w = 0$. Put $\delta = \sum_i \vartheta_i \partial_{w_i}$ and $\kappa = \sum_i w_i \iota_{\vartheta_i}$, where ι is the left exterior derivative. On w-degree m and exterior degree r, set

$$h_0 = -\frac{\kappa}{m + r} \quad (m + r > 0), \qquad h_0 = 0 \quad (m + r = 0), \qquad h = T h_0 T^{-1}.$$

Proposition 10.4. *These continuous maps satisfy*

$$pj = 1, \qquad \nabla h + h\nabla = 1 - jp, \qquad h^2 = hj = ph = 0.$$

The Taylor inclusion preserves the constant Poisson bracket and its formal Moyal product.

Proof. The anticommutator $\delta\kappa + \kappa\delta$ is multiplication by $m + r$. Thus h_0 contracts $-\delta$ onto the constants in w, ϑ. Differentiation gives $\nabla = T(-\delta)T^{-1}$, and evaluation at $w = 0$ commutes with T^{-1}. Conjugating proves all the contraction identities.

At fixed output w-degree, T and T^{-1} use finitely many coefficient derivatives. The operator h_0 raises that degree by one and lowers exterior degree by one. These facts prove continuity. Finally $\partial_{w_i} j(f) = j(\partial_{z_i} f)$. Applying this equality to each derivative proves the Poisson and Moyal assertions coefficientwise. □

For example, $j(z_1^2) = z_1^2 + 2z_1 w_1 + w_1^2$. Evaluation on arbitrary independent coefficients is not Poisson:

$$p\{w_1, w_2\}_w = 1, \qquad \{p(w_1), p(w_2)\}_z = 0.$$

The contraction therefore compares geometric functions with the horizontal complex. Imposing that complex changes a theory of unconstrained independent coefficient fields. The same formulas work for smooth coefficients and the total mixed differential, with its tensor signs and coefficientwise support.

10.4 The abstract Rees algebra and the Moyal product

For a finite-dimensional Lie algebra $\mathfrak{l}$, let $R_N = \mathbb{C}[\hbar]/(\hbar^{N+1})$ and define

$$U_N(\mathfrak{l}) = T_{R_N}(R_N \otimes \mathfrak{l})/(xy - yx - \hbar[x, y]).$$

Here N is parameter precision, unrelated to matrix rank. The Poincaré–Birkhoff–Witt map is the module isomorphism

$$\mathrm{sym}_N : \mathrm{Sym}(\mathfrak{l}) \otimes R_N \longrightarrow U_N(\mathfrak{l}), \qquad x_1 \cdots x_r \longmapsto \frac{1}{r!} \sum_{\sigma \in S_r} x_{\sigma(1)} \cdots x_{\sigma(r)}.$$

The source here has its ordinary symmetric product. Symmetrization is not generally an algebra map.

For completeness, choose an ordered basis of $\mathfrak{l}$. Interchanging an inverted neighboring pair adds its shorter $\hbar$-bracket term. Reduction terminates by word length and then lexicographic order at fixed

length. The only overlapping ambiguity has three factors, and its difference reduces to the Jacobi sum. Disjoint interchanges commute. Thus ordered monomials form a basis, over R_N as well as over $\mathbb{C}[\hbar]$.

PBW degree differs from powers of the augmentation ideal. For example, let $\mathfrak{n}$ be the Heisenberg Lie algebra with basis x, y, z, bracket $[x, y] = z$ and central z. In its ordinary enveloping algebra, the augmentation sends these three generators to zero. Its kernel I contains $z = xy - yx \in I^2$, although z has PBW degree one. The two filtrations therefore define different truncations.

The coefficient limit is $U_\hbar(\mathfrak{l}) = \varprojlim_N U_N(\mathfrak{l})$. Its PBW module is $(\operatorname{Sym}\mathfrak{l})[[\hbar]]$: each coefficient is polynomial, but its degree can increase with parameter order. For example, $\sum_{n\ge 0}\hbar^n x^n$ belongs to this module and need not belong to $\mathbb{C}[[\hbar]][x]$.

Quantizing functions on the symplectic plane gives a different operation. On $\mathbb{C}[u,v][[\hbar]]$, define

$$f \star g = \sum_{r\ge 0}\frac{\hbar^r}{2^r r!}\sum_{j=0}^{r}(-1)^j\binom{r}{j}(\partial_u^{r-j}\partial_v^{j}f)(\partial_u^{j}\partial_v^{r-j}g). \tag{10.3}$$

For polynomial inputs the sum terminates. In formal $\hbar$ series each coefficient still has finitely many contributions. The commuting constant bidifferential operators on three factors prove associativity by multiplying their exponentials.

Proposition 10.5. *Quadratic Hamiltonians have $[f,g]_\star = \hbar\{f,g\}$. The full pointed polynomial subspace is not closed under $\hbar^{-1}[-,-]_\star$, even modulo constants:*

$$[u^3, v^3]_\star = 9\hbar u^2v^2 + \frac{3}{2}\hbar^3,$$
$$[u^3, v^4]_\star = 12\hbar u^2v^3 + 6\hbar^3 v.$$

Proof. Reversing the inputs multiplies the derivative-order-r term by $(-1)^r$. The commutator retains precisely the odd orders. Orders at least three vanish on quadratics. For u^a, v^b the surviving coefficient is

$$2\frac{\hbar^r}{2^r r!}\frac{a!}{(a-r)!}\frac{b!}{(b-r)!}u^{a-r}v^{b-r}.$$

Taking $(a,b) = (3,3)$ and $(3,4)$ gives the formulas. The second normalized commutator retains the linear term $6\hbar^2 v$ modulo constants. □

The Rees algebra of $\mathfrak{h}_d$ uses its specified truncated classical bracket. It does not make the pointed subspace of Moyal functions closed. A quantum Hamiltonian action retaining all polynomial functions belongs to the larger affine recipient.

10.5 The finite relative cochain comparison

Put $A = U_N(\mathfrak{l})$ and let M be an A-bimodule. The relative Hochschild cochains are

$$C^n_{\mathrm{HH}}(A/R_N, M) = \operatorname{Hom}_{R_N}(A^{\otimes_{R_N} n}, M).$$

Their differential is

$$\begin{aligned}(bF)(a_1,\dots,a_{n+1}) &= a_1F(a_2,\dots,a_{n+1})\\ &\quad+\sum_{i=1}^{n}(-1)^iF(a_1,\dots,a_ia_{i+1},\dots,a_{n+1})\\ &\quad+(-1)^{n+1}F(a_1,\dots,a_n)a_{n+1}.\end{aligned}$$

The coefficient ring is held fixed in every input.

For $e \geq d$, abbreviate $A_{d,N} = U_N(\mathfrak{h}_d)$. Give $A_{d,N}$ its $A_{e,N}$-bimodule structure through the quotient. On

$$P^n_{e,d,N} = \operatorname{Hom}_{\mathbb{C}}(\Lambda^n \mathfrak{h}_e, \operatorname{Sym} \mathfrak{h}_d) \otimes R_N,$$

the coefficient action factors through $\mathfrak{h}_d$. Define

$$\begin{aligned}(d_{\mathrm{CE}}\phi)(x_0, \dots, x_n) = &\sum_i (-1)^i x_i \cdot \phi(x_0, \dots, \widehat{x_i}, \dots, x_n) \\ &+ \sum_{i<j} (-1)^{i+j} \phi(\{x_i, x_j\}, x_0, \dots, \widehat{x_i}, \dots, \widehat{x_j}, \dots, x_n).\end{aligned}$$

The bracket occupies the first slot in the second sum.

Theorem 10.6. *Output PBW symmetrization and input antisymmetrization define a quasi-isomorphism of R_N-complexes*

$$\mathrm{Alt}_{e,d,N} : C^\bullet_{\mathrm{HH}}(A_{e,N}/R_N, A_{d,N}) \longrightarrow (P^\bullet_{e,d,N}, \hbar d_{\mathrm{CE}}),$$

where

$$(\mathrm{Alt}_{e,d,N}\, F)(x_1, \dots, x_n) = \operatorname{sym}_N^{-1} \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) F(x_{\sigma(1)}, \dots, x_{\sigma(n)}).$$

These maps commute with input refinement, output quotient and parameter reduction.

Proof. Write $U = A_{e,N}$ and $\mathfrak{l}_N = R_N \otimes \mathfrak{h}_e$ with bracket $[x, y]_N = \hbar\{x, y\}$. Consider the bimodule complex $K_n = U \otimes_{R_N} \Lambda^n \mathfrak{l}_N \otimes_{R_N} U$. For $X = x_1 \wedge \cdots \wedge x_n$, its boundary is

$$\begin{aligned}\partial(a \otimes X \otimes b) = &\sum_i (-1)^{i+1} \big(a x_i \otimes X_{\widehat{i}} \otimes b - a \otimes X_{\widehat{i}} \otimes x_i b\big) \\ &+ \sum_{i<j} (-1)^{i+j} a \otimes [x_i, x_j]_N \wedge X_{\widehat{ij}} \otimes b.\end{aligned}$$

Omitted exterior factors retain their order. The augmentation $K_0 \to U$ is multiplication.

Antisymmetrizing the middle factors maps K into the relative bar resolution. The endpoint bar faces give the first sum. Adjacent internal faces cancel in pairs except for their commutators, giving the second sum. This proves the chain identity. The antisymmetrization is injective because $\mathfrak{l}_N$ is a free PBW summand and factorials are invertible. Thus $\partial^2 = 0$ follows from the bar identity.

Filter K_n by the two outer PBW degrees plus n. The associated graded is the diagonal Koszul resolution of $\operatorname{Sym}_{R_N} \mathfrak{l}_N$. Its diagonal differences are independent polynomial variables after a linear change of coordinates. They form a regular sequence even over the ring R_N with nilpotents.

A cycle has finite filtration degree. Lift a primitive of its leading symbol and subtract its boundary. Induction lowers the degree and proves exactness.

Both K and the relative bar are free bimodule resolutions. Their comparison lifts the identity and is a chain homotopy equivalence, by lifting successively through the surjective maps onto cycles. Applying bimodule Hom into $A_{d,N}$ therefore preserves that equivalence.

The resulting coefficient action is the commutator with the image of x. For a symmetric polynomial f, PBW gives

$$[x, \operatorname{sym}_N(f)] = \hbar \operatorname{sym}_N(x \cdot f).$$

The internal bracket term has the same factor $\hbar$. Hence the differential is precisely $\hbar d_{\mathrm{CE}}$. All formulas use only the quotient Lie maps and the coefficient ring, so the three stated naturalities hold. □

The factor $\hbar$ is necessary. For the quadratic jet, $[H, E] = 2\hbar E$, whereas $(d_{\mathrm{CE}}E)(H) = 2E$. The Hochschild differential of E becomes zero on the commutative special fiber. An unscaled Chevalley differential would not.

The comparison is relative to R_N, and its formal limit is relative to $R = \mathbb{C}[[\hbar]]$. Already for $\mathfrak{l} = 0$, the absolute continuous derivation $\partial_\hbar$ on R has no relative counterpart.

10.6 Continuous input and output cutoffs

Form the complete algebra and its quotient tensor powers

$$\mathcal{A} = \varprojlim_{d,N} A_{d,N}, \qquad \mathcal{A}^{\widehat{\otimes}_R n} = \varprojlim_{d,N} A_{d,N}^{\otimes_{R_N} n}.$$

The quotients are discrete, and no PBW-degree completion is taken at a fixed (d, N). For $n = 0$ the tensor source is R. Distinct indices in the tensor factors give the same limit, because a common upper bound is cofinal.

Definition 10.7. The continuous cochain space is

$$C^n_{\mathrm{uc}}(\mathcal{A}/R, \mathcal{A}) = \varprojlim_{d,N} \varinjlim_{e \geq d} \mathrm{Hom}_{R_N}(A_{e,N}^{\otimes_{R_N} n}, A_{d,N}).$$

Input refinement is precomposition with a quotient. The inverse maps reduce the output and parameter precision. Each direct limit is discrete, and the cochain space has their inverse-limit topology.

Proposition 10.8. *There is a natural identification*

$$C^n_{\mathrm{uc}}(\mathcal{A}/R, \mathcal{A}) = \mathrm{Hom}_{R,\mathrm{cont}}(\mathcal{A}^{\widehat{\otimes}_R n}, \mathcal{A}).$$

All quotient projections of the displayed tensor source are onto.

Proof. Compatible PBW bases identify the underlying module with $(\varprojlim_d \mathrm{Sym}\, \mathfrak{h}_d)[[\hbar]]$. The tensor powers have the analogous coefficientwise description. Lift a finite PBW polynomial by the same monomials at all higher cutoffs. This proves surjectivity. Coefficientwise division also gives $\mathcal{A}/\hbar^{N+1}\mathcal{A} = \varprojlim_d A_{d,N}$, and the same identity for each completed tensor.

A continuous map to the discrete output (d, N) factors through an input quotient (e, M). Enlarge these indices so that $e \geq d$ and $M \geq N$. The factored map is R-linear by surjectivity of the input projection. Its target is annihilated by $\hbar^{N+1}$, so the input precision reduces to N.

Two factorizations agree after a common refinement. This gives the stated inverse and direct limits. Conversely, such a family defines a continuous map to every output quotient. □

The larger input cutoff is essential. Choose a basis Hamiltonian x of degree $e > d$. Extract its coefficient as a PBW monomial and multiply by the unit of $\mathcal{A}$. This is a continuous linear cochain with scalar output. It detects x and therefore cannot factor through $A_{d,N}$.

For an arity-n cochain use reduced degree $n - 1$. The brace $F\{G_1, \ldots, G_r\}$ sums over ordered, disjoint insertions into F. If G_j has arity q_j and its block begins after i_j input arguments, the sign is $(-1)^{\sum_j (q_j - 1) i_j}$. The cup product is

$$(F \smile G)(a_1, \ldots, a_{p+q}) = F(a_1, \ldots, a_p) G(a_{p+1}, \ldots, a_{p+q}).$$

Proposition 10.9. *The Hochschild differential, cup product and every finite brace are defined on $C^\bullet_{\mathrm{uc}}$ and satisfy their usual identities.*

Proof. Fix output (d, N) and an input cutoff e for the outer cochain. Each inner cochain is needed only at output (e, N). Choose a common input cutoff for this finite family. Every insertion then factors through one tensor quotient, and further refinement leaves it unchanged.

Multiplication descends at each quotient, so the same argument defines the differential and cup product. In each identity choose one common quotient for all intermediate outputs. The usual insertion bijection and its reduced-degree signs prove the identity there. Equality at every output proves it in the complete complex. □

A single space $\mathrm{Hom}(A_{e,N}^{\otimes n}, A_{d,N})$ with $e \neq d$ has no self-valued brace composition. The complete family supplies the intermediate output required by the proof.

There is a relative derived interpretation with the same topology. Use complete separated linearly topologized R-modules and their quotient-completed tensors. A complete $\mathcal{A}$-bimodule has a continuous action on that tensor. Declare a short exact sequence admissible when it has a continuous R-linear splitting. This splitting need not commute with the bimodule actions.

Proposition 10.10. *In this exact structure the augmented completed bar is a relative projective resolution of $\mathcal{A}$. Consequently,*

$$\mathrm{RHom}^{\mathrm{rel,cont}}_{\mathcal{A}^e}(\mathcal{A}, \mathcal{A}) \simeq C^\bullet_{\mathrm{uc}}(\mathcal{A}/R, \mathcal{A}), \qquad \mathcal{A}^e = \mathcal{A}\widehat{\otimes}_R\mathcal{A}^{\mathrm{op}}.$$

Proof. Induction from a complete coefficient module E has the adjunction

$$\mathrm{Hom}_{\mathcal{A}^e,\mathrm{cont}}(\mathcal{A}^e\widehat{\otimes}_R E, M) = \mathrm{Hom}_{R,\mathrm{cont}}(E, M).$$

The extension sends $a \otimes e \otimes b$ to $af(e)b$. Refine first the quotient for the action and then that for f. This proves continuity. Restriction to $1 \otimes E \otimes 1$ is the inverse. Composing with an admissible splitting proves relative projectivity of induced modules.

The degree-n bar term is induced from $\mathcal{A}^{\widehat{\otimes}_R n}$. Its boundary is the alternating sum of adjacent multiplications. Inserting the unit on the left gives a continuous R-linear contraction of the augmented complex. On cycles it splits the preceding boundary map, so exactness is admissible. Continuous bimodule Hom into $\mathcal{A}$ gives the cochain spaces and differential above. □

This relative exact structure is part of the assertion. An absolute topological derived category, or a different analytic tensor product, requires a separate comparison.

Theorem 10.11. *The finite antisymmetrizations induce a quasi-isomorphism*

$$C^\bullet_{\mathrm{uc}}(\mathcal{A}/R, \mathcal{A}) \longrightarrow \varprojlim_{d,N} \varinjlim_{e \geq d} (P^\bullet_{e,d,N}, \hbar d_{\mathrm{CE}}).$$

For both sides the ordinary inverse limit computes the derived inverse limit of the indicated direct-limit complexes.

Proof. Filtered direct limits of modules are exact, so Theorem 10.6 gives a quasi-isomorphism at each output (d, N). Use a countable cofinal diagonal for these two indices.

Each transition is degreewise surjective. Represent a cochain at one input cutoff and increase it if necessary. Lift its values on the PBW tensor basis to the next output and parameter precision. Extend over the new coefficient ring. For the Chevalley side, use the exterior basis instead. These are lifts of elements, not chain sections or R_N-linear splittings of coefficient reduction.

For a degreewise surjective tower, the map $1-$ shift on its product is degreewise onto. Indeed, solve its equations successively using the surjective transition maps. Its kernel is the ordinary inverse limit, so the derived-limit fiber complex is quasi-isomorphic to that kernel. Derived limits preserve the levelwise quasi-isomorphisms, which proves the comparison. □

At a fixed jet cutoff this reduces to

$$C^{\bullet}_{\mathrm{HH,cont}}(U_\hbar(\mathfrak{h}_d)/R, U_\hbar(\mathfrak{h}_d)) \simeq \big(C^{\bullet}_{\mathrm{CE}}(\mathfrak{h}_d, \operatorname{Sym}\mathfrak{h}_d)[[\hbar]], \hbar d_{\mathrm{CE}}\big).$$

The coefficientwise PBW description is used throughout. The inverse limit over jets has the additional input cutoff. A diagonal tower of self-valued finite cochains would omit it.

10.7 Integral torsion and even traces

Scaling a differential changes integral cohomology. Let $(P^\bullet, d)$ be a complex of complex vector spaces and write $Z^n = \ker d$, $B^n = \operatorname{im} d$. All formal series below mean coefficientwise products.

Proposition 10.12. *There is a natural exact sequence*

$$0 \longrightarrow B^n \longrightarrow H^n(P^\bullet[[\hbar]], \hbar d) \longrightarrow H^n(P^\bullet, d)[[\hbar]] \longrightarrow 0.$$

The first term is annihilated by $\hbar$ and is exactly the $\hbar$-power torsion. A vector-space complement of B^n in Z^n splits the sequence. The first Bockstein of the special-fiber complex, whose differential is zero, is d.

Proof. Multiplication by $\hbar$ is injective on every coefficient module. Cycles and boundaries are therefore $Z^n[[\hbar]]$ and $\hbar B^n[[\hbar]]$. Coefficientwise passage to Z^n/B^n gives the surjection, with kernel $B^n[[\hbar]]/\hbar B^n[[\hbar]] = B^n$. A complement lifts the coefficients and splits it.

The quotient has no $\hbar$-power torsion, while the kernel is killed by $\hbar$. Finally lift $x \in P^n$ by its constant series. Its differential is $\hbar dx$. Dividing by $\hbar$ and reducing gives the Bockstein dx. □

For example, $R \xrightarrow{\hbar} R$ has free cochain modules and first cohomology $R/(\hbar)$. At finite precision its degree-zero kernel is $\mathbb{C}\hbar^N$. The next reduction sends that kernel to zero. Those extra cycles disappear in the limit, while the degree-one torsion class persists.

Unimodularity controls the first adjoint trace, but the even traces have another limit. Fix $m \geq 1$. For $x \in \mathfrak{h}$, let x_2 be its quadratic part and let ρ_r denote its action on W_r. Higher Hamiltonians raise degree, so

$$T_{2m,d}(x) = \operatorname{Tr}_{\mathfrak{h}_d}(\operatorname{ad}_x^{2m}) = \sum_{r=2}^{d} \operatorname{Tr}_{W_r}(\rho_r(x_2)^{2m}).$$

Proposition 10.13. *The expression $T_{2m,d}$ is polynomial in the cutoff d. Its constant coefficient, denoted* FP_d*, is*

$$\mathrm{FP}_d\, T_{2m,d}(x) = -\operatorname{Tr}_{W_1}(\rho_1(x_2)^{2m}).$$

For $H = -uv$,

$$T_{2,d}(H) = \frac{d(d+1)(d+2)(d+3)}{12} - 2.$$

Proof. For $x_2 = aH$ the sum is $a^{2m}\sum_{r=2}^{d}\sum_{j=0}^{r}(r-2j)^{2m}$. Power-sum identities make it polynomial in d. The sum starting at $r = 1$ vanishes at $d = 0$. Removing its first term subtracts $2a^{2m}$. This proves the constant coefficient on the Cartan line. Both sides are invariant polynomials on $\mathfrak{sl}_2$, so density of semisimple elements proves the identity everywhere.

For $m = 1$ the inner sum is $r(r+1)(r+2)/3$. Summing from $r = 2$ gives the displayed polynomial. □

This defines a subtraction scheme, not a canonical limit. Let B_{2m} be the Bernoulli numbers defined by $z/(e^z - 1) = \sum_n B_n z^n/n!$. The formal half-determinant logarithm has coefficients $c_m = B_{2m}/(4m(2m)!)$. Subtracting the nonconstant cutoff polynomial from each trace gives

$$\ell_{\mathrm{ren}}(x) = -\frac{1}{2}\operatorname{Tr}_{W_1}\log\left(\frac{\sinh(\hbar\rho_1(x_2)/2)}{\hbar\rho_1(x_2)/2}\right).$$

The quotient is its power series with constant term one, without inverting an eigenvalue. On the Cartan line its exponential is $(\hbar a/2)/\sinh(\hbar a/2)$.

Replacing the cutoff by $d + b$ changes the two-wheel finite part by $b(b+1)(b+2)(b+3)/12$.

An ordinary even dual has the same even traces and doubles them when added to the original representation. A parity-reversed dual cancels them in the supertrace. These two coefficient choices therefore have different subtraction problems.

10.8 Support and the remaining field comparison

Let $\mathcal{L}_c(U)$ be compactly supported smooth sections of a local differential graded Lie algebra on an open set U. Assume its differential and bracket are differential operators. Extension by zero then gives a covariant Lie map $\mathcal{L}_c(U) \to \mathcal{L}_c(V)$ for $U \subset V$. For disjoint opens, the bracket of their extended sections is zero.

With the algebraic tensor product, the Chevalley chains $C_*^{\mathrm{Lie}}(\mathfrak{l}) = \mathrm{Sym}(\mathfrak{l}[1])$ have their internal and bracket differential. They take direct sums of Lie algebras to tensor products. Consequently the maps

$$\bigotimes_i C_*^{\mathrm{Lie}}(\mathcal{L}_c(U_i)) \longrightarrow C_*^{\mathrm{Lie}}(\mathcal{L}_c(V))$$

define a prefactorization algebra. Weiss descent is an additional property.

Extension by zero pulls Chevalley cochains from V to U. A covariant observable model therefore requires its own coefficient duality and support construction.

For the usual smooth topologies, $(C^\infty(U))'_b = \mathcal{E}'(U)$ consists of compactly supported distributions. In contrast, $(C_c^\infty(U))'_b = \mathcal{D}'(U)$ allows arbitrary support. The first assertion follows because a continuous functional uses finitely many seminorms on a common compact set. The second follows by compatibility on all compact test supports. The subscript indicates the strong dual topology.

The Hochschild side of Theorem 10.11 carries braces and a relative derived-center interpretation. An operation-preserving comparison with independently defined polyvectors requires further maps.

Problem 10.14. Construct a coherent refinement of the continuous Rees comparison for independently defined polyvector products and brackets. Require compatibility with all input and output refinements, and specify any even-wheel subtraction used by a multiplicative PBW correction.

Construct a map from supported mixed field observables to this coefficient complex. Preserve horizontal jets, normal derivatives, the vacuum and collision operations. Make the subtraction compatible with these operations and nonlinear symplectic coordinate descent.

Chapter II

Coherent Hamiltonian actions

Hamiltonian vector fields act on the commutative plane by a strict Lie action. Replacing functions by noncommutative words changes the problem. Symmetrized cyclic derivatives still preserve the commutativity relation, but their brackets acquire a defect. The variable resolving that relation supplies the first homotopy. A contraction of the same resolution supplies every higher one.

The construction also works for the Weyl relation. We keep its coefficient parameter separate from quantization of matrix entries. After constructing the actions, we ask which homotopies preserve the volume and which are Hamiltonian functions on the matrix gauge complex. These are additional structures on an action.

II.1 The first bracket defect

Work over $\mathbb{C}$, with differentials of cohomological degree one. Write q, p for the coordinates of the symplectic plane, and put

$$A = \mathbb{C}[q, p], \qquad \mathfrak{l} = A/\mathbb{C}1, \qquad \{f, g\} = f_q g_p - f_p g_q.$$

Thus $X_f = \{f, -\}$ is independent of the constant term of f. For a monomial $q^a p^b$, let $\sigma(q^a p^b)$ be the average of all distinct words with a factors q and b factors p. For a word $w = a_1 \cdots a_m$, define

$$\partial_q^{\mathrm{cyc}} w = \sum_{a_i = q} a_{i+1} \cdots a_m a_1 \cdots a_{i-1},$$

and define $\partial_p^{\mathrm{cyc}}$ in the same way.

Introduce the free graded associative algebra

$$R = \mathbb{C}\langle q, p, t\rangle, \qquad |q| = |p| = 0, \quad |t| = -1, \qquad dq = dp = 0, \quad dt = qp - pq. \tag{II.1}$$

Here t is a relation generator. For homogeneous words, brackets denote graded commutators: $[a, b] = ab - (-1)^{|a||b|} ba$. The projection $\pi : R \to A$ kills t and makes q, p commute.

The cyclic lift is the degree-zero derivation

$$F_1(f) q = -\partial_p^{\mathrm{cyc}} \sigma f, \qquad F_1(f) p = \partial_q^{\mathrm{cyc}} \sigma f, \qquad F_1(f) t = 0. \tag{II.2}$$

It kills constants. Telescoping the rotations of a word gives

$$[q, \partial_q^{\mathrm{cyc}} w] + [p, \partial_p^{\mathrm{cyc}} w] = 0.$$

Consequently $[d, F_1(f)] = 0$ and $\pi F_1(f) = X_f \pi$.

Put $\mu = qp - pq$. Cyclic differentiation of the cubic pair gives

$$\big([F_1(q^3), F_1(p^3)] - F_1(9q^2 p^2)\big)(q, p, t) = \big(3[\mu, q], 3[\mu, p], 0\big). \tag{II.3}$$

This defect is nonzero. Substitution of $X = \mathrm{diag}(1, 0)$ and $Y = E_{12}$ for q, p gives $-3E_{12}$ in its first component.

The defect is exact as a derivation. The degree-minus-one derivation

$$\eta(q, p, t) = \big(3[t, q], 3[t, p], 3t^2\big) \tag{11.4}$$

satisfies $[d, \eta]$ equal to (11.3). On q, p this follows from $dt = \mu$. On t, the terms cancel because $d(t^2) = \mu t - t\mu$ and $\eta(\mu) = 3[t, \mu]$. The value on t is necessary.

Changing the polynomial cyclic lift cannot make it a Lie section. Let $\mathcal{N}$ be the vector space of cyclic words in q, p, modulo constants. Equivalently, quotient the free algebra by the span of commutators and then by its constant class. Its necklace bracket is

$$\{[U], [V]\}_{\mathcal{N}} = \big[\partial_q^{\mathrm{cyc}} U\, \partial_p^{\mathrm{cyc}} V - \partial_p^{\mathrm{cyc}} U\, \partial_q^{\mathrm{cyc}} V\big].$$

Expanding the derivatives identifies its Hamiltonian commutator. The cyclic Euler identity $[q\partial_q^{\mathrm{cyc}} U + p\partial_p^{\mathrm{cyc}} U] = m[U]$ in degree m makes this action faithful modulo constants. Jacobi therefore follows from Jacobi for derivations. Abelianization is a Lie map $\mathcal{N} \to \mathfrak{l}$.

Proposition 11.1. *Abelianization $\mathcal{N} \to \mathfrak{l}$ has no linear Lie section, even without a degree-preservation requirement.*

Proof. Abelianization is injective in degrees at most three. A lift of a cubic therefore differs from its unique cyclic cubic by kernel terms of degree at least four. Their brackets with cubics have degree at least five.

The cyclic classes $C = [q^2 p^2]$ and $D = [qpqp]$ are distinct, and

$$\{[q^3], [p^3]\}_{\mathcal{N}} = 9C, \qquad \{[q^2 p], [q p^2]\}_{\mathcal{N}} = C + 2D.$$

The commutative brackets are $9q^2 p^2$ and $3q^2 p^2$. Thus the degree-four part of a Lie lift of $q^2 p^2$ would be both C and $(C + 2D)/3$. This would identify the two distinct cyclic words. □

The proposition concerns sections through cyclic words. To construct an action on R, we instead solve the bracket defects in the complex of derivations.

11.2 A word contraction for the plane and its deformation

Introduce a degree-zero parameter ε and set $k_\varepsilon = \mathbb{C}[\varepsilon]$. The family

$$\mathcal{R}_\varepsilon = k_\varepsilon\langle q, p, t\rangle, \qquad dt = qp - pq - \varepsilon, \qquad W_\varepsilon = k_\varepsilon\langle q, p\rangle/(qp - pq - \varepsilon) \tag{11.5}$$

has the same generator degrees as R. Write π_ε for the projection killing t. The rule $qp \mapsto pq + \varepsilon$ has no overlapping leading words. It gives the normal basis $p^b q^a$ of W_ε and a k_ε-linear section i of π_ε.

This section is a chain map, not an algebra map:

$$i(q)i(p) - i(qp) = qp - (pq + \varepsilon) = dt.$$

Here qp inside i denotes the product in W_ε. We now construct a homotopy for this linear section.

First replace dt by $d_0 t = qp$. Expand each t in a word into qp, marking the occurrence expanded. Occurrences of qp do not overlap. For each expanded word, the subsets of marked occurrences form a Boolean complex. The differential d_0 removes a mark, with sign $(-1)^a$ when a marks precede it.

Define h_0 by adding a mark at the leftmost occurrence if it is unmarked. Give zero if that occurrence is already marked, or if there is no occurrence. Let p_0 retain only the unmarked normal words. Removing and adding the distinguished mark gives

$$d_0 h_0 + h_0 d_0 = 1 - i p_0, \qquad h_0^2 = h_0 i = p_0 h_0 = 0.$$

Every removal at another occurrence cancels with the corresponding addition, with the opposite tensor sign.

Write $d = d_0 + \delta$, where $\delta t = -pq - \varepsilon$. The exchange term lowers the number of pairs with a q before a p. The scalar term lowers expanded-word length. Ordered first by length and then by this inversion number, every term of δh_0 decreases. Hence the sums

$$h = h_0 \sum_{j \geq 0} (-\delta h_0)^j, \qquad p' = p_0 \sum_{j \geq 0} (-\delta h_0)^j \tag{11.6}$$

terminate on every polynomial word.

Proposition 11.2. *The map p' equals π_ε, and*

$$\pi_\varepsilon i = 1, \qquad dh + hd = 1 - i\pi_\varepsilon, \qquad h^2 = hi = \pi_\varepsilon h = 0.$$

Thus $\mathcal{R}_\varepsilon$ resolves W_ε over k_ε. The same identities hold continuously after ε-adic completion.

Proof. Insert the finite inverse $(1 + \delta h_0)^{-1}$ into the Boolean contraction. Expanding its product with $1 + \delta h_0$ cancels the perturbation terms. Since $\delta i = 0$, the inclusion and the zero differential on the normal words stay unchanged. This gives the homotopy identity and its side conditions.

On degree-zero words, p' performs exactly the terminating rule $qp \mapsto pq + \varepsilon$. Its result is their image in the normal basis of W_ε. Negative-degree words have zero projection. Thus $p' = \pi_\varepsilon$.

Every operation has nonnegative ε-order. Modulo ε^m, only finitely many coefficients of an input contribute, each with finite polynomial support. The formulas therefore define compatible continuous maps on all these quotients. Their identities pass to the separated inverse limit. □

The completed carrier is $\widehat{\mathcal{R}}_\varepsilon = \varprojlim_m \mathcal{R}_\varepsilon/\varepsilon^m$. It permits polynomial word support to grow with parameter order. At $\varepsilon = 0$, the proposition gives a contraction of $R \to A$. Both the polynomial family and its classical fiber retain this explicit homotopy.

11.3 Contracting the derivation kernel

Give $\mathrm{Der}_{k_\varepsilon}(\mathcal{R}_\varepsilon)$ the graded commutator and differential $\partial D = [d, D]$. Let $J_\varepsilon = \ker \pi_\varepsilon$, and define

$$K_\varepsilon = \{D : \pi_\varepsilon D(q) = \pi_\varepsilon D(p) = \pi_\varepsilon D(t) = 0\}.$$

These are derivations with all values in J_ε. For a degree-j derivation, write $U = Dq, V = Dp, W = Dt$, and put $L(U, V) = [U, p] + [q, V]$. Then

$$(\partial D)(q, p, t) = \big(dU, dV, dW - (-1)^j L(U, V)\big). \tag{11.7}$$

Lemma 11.3. *The operator of degree minus one defined by*

$$(HD)(q, p, t) = \big(hU, hV, h(W - (-1)^j L(hU, hV))\big) \tag{11.8}$$

preserves K_ε and satisfies $\partial H + H\partial = 1$ there. It has finite polynomial values and extends continuously in ε.

Proof. The ideal J_ε contains every argument of h in the formula, and $\pi_\varepsilon h = 0$. On q, p, the identity is $dh + hd = 1$ on J_ε. Since q, p are closed, $dL(U, V) = L(dU, dV)$. Substitution in the third component leaves $(dh + hd)W$ and the additional term

$$(-1)^j hL\big((dh + hd - 1)U, (dh + hd - 1)V\big).$$

The first expression equals W and the second vanishes. This proves the identity on arbitrary kernel derivations, including those which are not closed. Finiteness and continuity follow from Proposition II.2. □

II.4 Solving every action equation

The Moyal product on $k_\varepsilon[q, p]$ is

$$f \star g = m \exp\Big(\frac{\varepsilon}{2}(\partial_q \otimes \partial_p - \partial_p \otimes \partial_q)\Big)(f \otimes g).$$

Its divided commutator $B_\varepsilon(f, g) = (f \star g - g \star f)/\varepsilon$ is polynomial. This quotient belongs to $k_\varepsilon[q, p]$. Quotienting constant polynomials defines the Lie algebra $\mathfrak{l}_\varepsilon = k_\varepsilon[q, p]/k_\varepsilon 1$.

Weyl symmetrization identifies this product with W_ε. One can check the identification on auxiliary exponential series: the product of $\exp(sq + rp)$ and $\exp(s'q + r'p)$ acquires the factor $\exp(\varepsilon(sr' - rs')/2)$. The relation $[q, p] = \varepsilon$ gives this identity by differentiation in the auxiliary variables. Comparing their coefficients gives the polynomial identity.

Use (II.2) for F_1 over k_ε. The cyclic identity still proves $[d, F_1] = 0$. Cyclic differentiation of a symmetrized polynomial is the symmetrization of its ordinary derivative. Thus its projection is the strict derivation $B_\varepsilon(f, -)$ of W_ε.

We specify the signs of the higher equations by a coalgebra. For a differential graded Lie algebra $\mathfrak{g}$, let $s : \mathfrak{g} \to \mathfrak{g}[1]$ have degree -1. The codifferential of the cofree conilpotent symmetric coalgebra $S^c(\mathfrak{g}[1])$ has corestrictions

$$b_1(sa) = -s\partial a, \qquad b_2(sa\, sb) = (-1)^{|a|} s[a, b].$$

An L_∞ morphism is a coaugmented coalgebra map commuting with these codifferentials. Its component on n shifted degree-zero inputs is written sF_n. The unshifted alternating map F_n has degree $1 - n$.

Suppose the components below arity n have been chosen. Extend them by the set-partition formula to a coalgebra map $\mathcal{F}$, temporarily setting $F_n = 0$. Let Ω_n be the desuspended arity-n corestriction of $b\mathcal{F} - \mathcal{F}b$. Set

$$F_n = H\Omega_n \qquad (n \geq 2). \tag{II.9}$$

The first two residuals are

$$\Omega_2(f, g) = [F_1(f), F_1(g)] - F_1(B_\varepsilon(f, g)),$$
$$\Omega_3(f, g, k) = \sum_{\mathrm{cyc}} \big([F_1(f), F_2(g, k)] - F_2(B_\varepsilon(f, g), k)\big).$$

At arity four the residual includes brackets of two binary components. The coalgebra definition fixes these terms and their signs along with all subsequent ones.

Theorem II.4. *The recursion defines an L_∞ morphism*

$$F : \mathfrak{l}_\varepsilon \longrightarrow \mathrm{Der}_{k_\varepsilon}(\mathcal{R}_\varepsilon)$$

with first component (II.2)*. Every fixed-arity value on polynomial inputs is polynomial. The maps extend continuously to the parameter completions. Specialization at $\varepsilon = 0$ gives an action of $\mathfrak{l}$ on R.*

Proof. The arity-one identity is $[d, F_1] = 0$. Suppose the equations hold below arity n. The square-zero identities for the two codifferentials imply

$$b(b\mathcal{F} - \mathcal{F}b) + (b\mathcal{F} - \mathcal{F}b)b = 0.$$

Its first nonzero corestriction says $\partial\Omega_n = 0$.

The binary residual projects to zero because the action on W_ε is strict. Moreover, F_1 preserves J_ε and normalizes K_ε. Higher components already constructed lie in K_ε. Every later residual therefore lies in K_ε^{2-n}.

Lemma 11.3 gives $\partial F_n = \Omega_n$. The term $-s\partial F_n$ from b_1 cancels the residual. This proves the equations inductively. There are finitely many partitions at each arity, and the word contraction terminates. All operations preserve nonnegative parameter order, so the same formulas define the completed action coefficient by coefficient. □

Several values make the higher structure visible. At $\varepsilon = 0$, the chosen $F_2(q^3, p^3)$ is precisely η from (11.4). For the triple (q^4, p^4, q^2p), the recursion gives

$$\begin{aligned} 5F_3q &= -128tq^2t + 16tqtq + 112t^2q^2, \\ 5F_3p &= -128ptqt - 112pt^2q, \\ 5F_3t &= -64t^2qt - 56t^3q. \end{aligned} \tag{11.10}$$

Substitution of $dt = qp - pq$ gives $\partial F_3 = \Omega_3$ on all three generators. The triple (q^3, p^3, q^2p) instead has zero residual and zero third component.

For $\mathbf{f} = (q^3, p^3, qp^3)$ and $\mathbf{g} = (q^3, p^3, qp^3, q^2p^2)$, another pair of values is

$$\begin{aligned} F_3(\mathbf{f})q &= -\tfrac{9}{2}(tpt + pt^2), \\ F_4(\mathbf{g})q &= \tfrac{33}{5}t^3p + \tfrac{39}{5}t^2pt + \tfrac{69}{5}tpt^2 + \tfrac{99}{5}pt^3. \end{aligned} \tag{11.11}$$

Both components vanish on p, t. Applying the differential to these words gives their respective coalgebra residuals, including the binary–binary terms at arity four. Thus the specified action does not truncate after its second or third component.

The deformation has its own first correction:

$$B_\varepsilon(q^3, p^4) = 12q^2p^3 + 6\varepsilon^2p.$$

For this pair the recursion gives

$$\begin{aligned} F_2q &= 6(-pqt + ptq - qtp + tpq - \varepsilon t), \\ F_2p &= 6(-p^2t + tp^2), \\ F_2t &= 6(pt^2 + tpt). \end{aligned} \tag{11.12}$$

For (q^3, p^4, q^2p) it gives

$$F_3q = -12tqt + 12t^2q, \qquad F_3p = -12pt^2, \qquad F_3t = -6t^3.$$

These identities follow by differentiating the displayed words with $dt = qp - pq - \varepsilon$. In particular, the term $-6\varepsilon t$ contributes $6\varepsilon^2$ to the first binary equation. Omitting it loses the derivative of the Moyal correction.

11.5 Choices of lifts and pointed completion

Return to $\varepsilon = 0$. Let $\mathrm{Der}_\pi(R)$ be the derivations preserving $J = \ker \pi$. They induce derivations of A, giving an exact sequence of complexes

$$0 \longrightarrow K \longrightarrow \mathrm{Der}_\pi(R) \xrightarrow{\pi_*} \mathrm{Der}(A) \longrightarrow 0.$$

Here $K = K_0$ is understood in the classical construction.

Surjectivity is degreewise: lift the values of a derivation on q, p and set its value on t to zero. This preserves J, although the chosen lift need not commute with d. The contracted kernel makes π_* a quasi-isomorphism.

To specify a space of choices, use the complete convolution Lie algebra

$$\mathcal{G}(L) = \prod_{n\geq 1} \mathrm{Hom}\big(S^n(\mathfrak{l}[1]), L\big), \qquad \mathcal{F}^r\mathcal{G}(L) = \prod_{n\geq r} \mathrm{Hom}\big(S^n(\mathfrak{l}[1]), L\big).$$

Its differential uses that of L and the source Chevalley codifferential. Its bracket uses the reduced coproduct with the tensor Koszul sign. Degree-one Maurer–Cartan elements are the L_∞ morphisms in the preceding convention.

Define its simplicial Maurer–Cartan set by tensoring each arity quotient with polynomial differential forms on a simplex, then taking the inverse limit in arity. The derived space of lifts is the homotopy fiber over $f \mapsto X_f$ of the map induced by π_*. This fixes the commutative action.

Proposition 11.5. *The derived space of lifts of X to $\mathrm{Der}_\pi(R)$ is nonempty and contractible.*

Proof. The recursion supplies a lift. The map of convolution complexes is degreewise onto, and its arity-n associated-graded kernel is

$$\mathrm{Hom}\big(S^n(\mathfrak{l}[1]), K\big).$$

Postcomposition with H contracts it.

Consider a compatible boundary or horn of polynomial simplex forms and a lift already chosen in lower arities. Polynomial forms extend from that boundary or horn. In barycentric coordinates the ideal of face i is (x_i, dx_i). Successive face extensions preserve the restrictions already fixed. Choose a tentative next component with the prescribed boundary and projection.

Its Maurer–Cartan defect is a closed kernel element tensored with relative simplex forms. Closedness is the first nonzero arity of the Bianchi identity. The tensor extension of H contracts this relative kernel. Subtracting its primitive corrects the component without changing its boundary or projection. Arity completeness gives all the compatible corrections. The resulting simplicial map has lifts for all boundaries, so its nonempty fiber is contractible. □

This fiber imposes neither a symplectic form nor Hamiltonian functions on the homotopies. Its completion in the formal coordinates has an additional continuity requirement. Give q, p weight one and t weight two, and write R_w for the homogeneous weight space. Define $\widehat{R} = \prod_{w\geq 0} R_w$, with discrete homogeneous pieces and their product topology.

Proposition 11.6. *For homogeneous classical inputs of polynomial degrees $d_1, \dots, d_n$, the derivation F_n shifts weight by $\sum_i (d_i - 2)$. At every fixed arity the action therefore extends continuously to*

$$\widehat{\mathfrak{l}}_{\mathrm{pt}} = \prod_{d\geq 2} \mathrm{Sym}^d(\mathbb{C}q \oplus \mathbb{C}p)$$

acting on $\widehat{R}$.

Proof. The differential and word contraction preserve weight. Cyclic differentiation gives the asserted shift for F_1. Brackets add shifts, and H preserves them. Induction proves the formula for every F_n.

For pointed inputs, each $d_i - 2$ is nonnegative. At fixed arity and fixed output weight, only finitely many input degrees and source weights contribute. Each homogeneous piece is finite dimensional. The formulas and identities therefore extend coefficient by coefficient and are continuous. □

This fixed-arity extension gives no analytic summation over all arities. Its pointed classical source must also be distinguished from the quantum one. The term $6\varepsilon^2 p$ above leaves the subspace of polynomials whose ordinary degree is at least two.

11.6 Matrix actions and Hamiltonian functions

For $N \geq 1$, the commutative differential graded algebra of derived representations of (11.5) is

$$\mathcal{R}_{N,\varepsilon} = \mathbb{C}[\varepsilon, X_{ij}, Y_{ij}] \otimes \Lambda(\Psi_{ij}), \qquad |\Psi_{ij}| = -1, \qquad Q\Psi = [X, Y] - \varepsilon I_N, \tag{11.13}$$

with $QX = QY = 0$. Matrix multiplication uses graded commutative coefficients. Substitution of X, Y, Ψ in a derivation's generator values defines a derivation of their entries. The graded Leibniz rule makes substitution preserve brackets and differentials.

It follows that F defines a GL_N-equivariant L_∞ action on $\mathcal{R}_{N,\varepsilon}$ and its invariant algebra. At $\varepsilon = 0$, giving X, Y weight one and Ψ weight two also gives the continuous pointed action on the weight completion.

The matrix complex is not contracted by the universal word homotopy. At $\varepsilon = 0$, the cycle $\operatorname{Tr}\Psi$ represents a nonzero class in degree -1. Evaluation at $X = Y = 0$ maps it to the same nonzero element of $\Lambda(\Psi_{ij})$ with zero differential. It cannot be a boundary. Keeping the relation complex is therefore essential.

To include gauge ghosts classically, introduce an odd matrix c of degree one and put

$$\mathcal{B}_N = \mathbb{C}[X_{ij}, Y_{ij}] \otimes \Lambda(c_{ij}, \Psi_{ij}).$$

Its even Poisson bracket is specified by

$$\{X_{ij}, Y_{kl}\} = \delta_{il}\delta_{jk}, \qquad \{c_{ij}, \Psi_{kl}\} = \{\Psi_{kl}, c_{ij}\} = \delta_{il}\delta_{jk}.$$

All other generator brackets vanish. Set $\Theta = \operatorname{Tr}(c[X, Y] - \Psi c^2)$ and $Q = \{\Theta, -\}$. Then

$$QX = -[c, X], \quad QY = -[c, Y], \quad Qc = -c^2, \qquad Q\Psi = [X, Y] - [c, \Psi]. \tag{11.14}$$

Here $[c, \Psi] = c\Psi + \Psi c$. Expanding on the four generators, with the graded Leibniz rule, gives $Q^2 = 0$. These are the Chevalley–Eilenberg gauge differential and the moment Koszul differential on one algebra.

Every matrix-word derivation extends by $Dc = 0$. For a degree-j derivation and a homogeneous matrix word M,

$$D[c, M] = (-1)^j [c, DM].$$

Thus extension commutes with the gauge part of Q in the graded sense. It preserves the differential and bracket, and extends the classical L_∞ action to $\mathcal{B}_N$.

The first action is Hamiltonian. For $f \in A$, put $J_N(f) = \operatorname{Tr} \sigma f(X, Y)$ and $H_{J_N(f)} = \{J_N(f), -\}$. Cyclic differentiation gives $H_{J_N(f)} = F_1(f)$, and conjugation invariance gives $QJ_N(f) = 0$. For the cubic pair, a Hamiltonian primitive is

$$K = -\tfrac{3}{2} \operatorname{Tr}(\Psi\mu), \qquad QK = -\tfrac{3}{2} \operatorname{Tr} \mu^2 = \{J_N(q^3), J_N(p^3)\} - J_N(9q^2p^2), \qquad \mu = [X, Y]. \tag{11.15}$$

The gauge terms cancel under the graded trace. Among the six words in $\sigma(q^2p^2)$, four have trace $\operatorname{Tr}(X^2Y^2)$ and two have trace $\operatorname{Tr}(XYXY)$, which proves the last equality.

This Hamiltonian homotopy has components

$$H_K X = \tfrac{3}{2}[\Psi, X], \quad H_K Y = \tfrac{3}{2}[\Psi, Y], \quad H_K c = -\tfrac{3}{2}\mu, \quad H_K \Psi = 0.$$

The extension of (11.4) instead has

$$DX = 3[\Psi, X], \quad DY = 3[\Psi, Y], \quad Dc = 0, \quad D\Psi = 3\Psi^2.$$

Both have the same differential, the cubic action defect. For $N \geq 2$, D is not Hamiltonian. Indeed $Dc = 0$ would make a Hamiltonian independent of Ψ. Such a polynomial has no component of degree -1, while D is a nonzero derivation of that degree.

Proposition 11.7. *There is no L_∞ morphism from $\mathfrak{l}$ to $(\mathcal{B}_N, Q, \{-,-\})$ whose first component is J_N on the representatives with zero constant term. The same obstruction survives exact changes of these closed charges.*

Proof. The classes of q, p have zero bracket in $\mathfrak{l}$, but $\{\operatorname{Tr} X, \operatorname{Tr} Y\} = N$. The binary morphism equation would make N a Q-boundary. Evaluation at $X = Y = c = \Psi = 0$ is a chain map to $\mathbb{C}$, since all values in (11.14) have positive polynomial degree. It sends that proposed boundary to the nonzero scalar N. Exact changes do not change the induced bracket on cohomology. □

Retaining the central extension $A \to \mathfrak{l}$ with $J_N(1) = N$ removes this particular source obstruction. Choosing representatives vanishing at the origin gives its scalar cocycle $c(f, g) = \{f, g\}(0)$. This cocycle vanishes on $(q, p)^2$. Neither change by itself constructs the remaining Hamiltonian homotopies.

11.7 The volume in the normalized mixed complex

Return first to the classical resolution R. Its normalized Hochschild chains are $C_n(R) = R \otimes \overline{R}^{\otimes n}$, where $\overline{R} = R/\mathbb{C}1$. Write a chain as $a_0[a_1|\cdots|a_n]$. Its internal degree is $m = \sum_i |a_i|$, and its total cohomological degree is $m - n$.

Use the raw tensor convention

$$d_{\mathrm{H}} = (-1)^n d_\otimes + b. \tag{11.16}$$

The tensor differential $d_\otimes$ has its usual Koszul signs. The other operators are

$$\begin{aligned} b(a_0[a_1|\cdots|a_n]) &= \sum_{j=0}^{n-1} (-1)^j a_0[\cdots|a_j a_{j+1}|\cdots|a_n] \\ &\quad + (-1)^{n+|a_n|\sum_{i<n}|a_i|} a_n a_0[a_1|\cdots|a_{n-1}], \\ B(a_0[a_1|\cdots|a_n]) &= \sum_{j=0}^{n} (-1)^{nj+(\sum_{i<j}|a_i|)(\sum_{i\geq j}|a_i|)} 1[a_j|\cdots|a_n|a_0|\cdots|a_{j-1}]. \end{aligned}$$

In the $j = 0$ term of b, the product is the outer coefficient. A scalar in a bar slot gives zero. Adjacent mergers cancel in pairs, as do the cyclic faces, giving $d_{\mathrm{H}}^2 = B^2 = d_{\mathrm{H}}B + Bd_{\mathrm{H}} = 0$.

Let ζ have degree two. The negative-cyclic complex is

$$M_R = \big(C_\bullet(R)[[\zeta]], d_{\mathrm{H}} + \zeta B\big).$$

Its volume cycle is

$$\omega_0 = 1[q|p] - 1[p|q] - 1[t]. \tag{II.17}$$

The Hochschild boundary of the first two terms is $-1[qp - pq]$. In this convention the internal differential of $-1[t]$ is $+1[qp - pq]$. Every Connes term has a unit in a bar slot. Hence $d_{\mathrm{H}}\omega_0 = B\omega_0 = 0$.

Put Ω^a_A in degree $-a$. The normalized Hochschild–Kostant–Rosenberg projection is

$$P(a_0[a_1|\cdots|a_n]) = \frac{1}{n!}\pi(a_0)\, d_{\mathrm{dR}}\pi(a_1) \wedge \cdots \wedge d_{\mathrm{dR}}\pi(a_n). \tag{II.18}$$

It satisfies $Pd_{\mathrm{H}} = 0$, $PB = d_{\mathrm{dR}}P$ and $P\omega_0 = dq \wedge dp$.

Give q, p, dq, dp weight one, t weight two, and ζ weight zero. Every normalized bar slot has positive weight. At fixed weight, bar length and internal degree are bounded, so each chain space is finite dimensional.

Proposition II.8. *There is a weight-preserving contraction (P, I, T) from $(C_\bullet(R), d_{\mathrm{H}})$ to $(\Omega^\bullet_A, 0)$, with $I(dq \wedge dp) = \omega_0$. It extends to a contraction from M_R to $(\Omega^\bullet_A[[\zeta]], \zeta d_{\mathrm{dR}})$, with unchanged projection and*

$$I_\zeta = (1 + T\zeta B)^{-1} I, \qquad T_\zeta = T(1 + \zeta BT)^{-1}. \tag{II.19}$$

The inclusion still sends $dq \wedge dp$ to ω_0.

Proof. The contraction of $R \to A$ induces a quasi-isomorphism of Hochschild chains. Indeed the bar-length filtration is finite in each weight, and its associated-graded comparison is the tensor product of quasi-isomorphisms over $\mathbb{C}$.

For A, the bimodule Koszul complex on $q_L - q_R$, $p_L - p_R$ resolves the diagonal. The first element is a non-zero-divisor in A^e. The second remains one in its quotient. Successive two-term resolutions prove exactness.

Tensoring with the diagonal kills both differentials and gives $\Omega^\bullet_A$. Antisymmetrization identifies the projection with (II.18). The normalized formula for B becomes d_{dR}.

Thus P is a surjective quasi-isomorphism at each weight. Lift a basis of forms to cycles, choosing ω_0 for the volume. Split its acyclic kernel in each degree as

$$(\ker P)^r = B^r \oplus L^r, \qquad d_{\mathrm{H}} : L^r \xrightarrow{\sim} B^{r+1}.$$

Let T be this inverse on B^{r+1} and zero on L and the chosen lifts. Then $d_{\mathrm{H}}T + Td_{\mathrm{H}} = 1 - IP$ and $T^2 = TI = PT = 0$. These are finite linear-algebra choices.

Perturb by ζB. Since $P\zeta BT = \zeta d_{\mathrm{dR}}PT = 0$, the projection stays P and the transferred differential has no higher terms beyond ζd_{dR}. The series in (II.19) converge in ζ. At fixed weight and total degree they are finite, because bar length and internal degree bound the possible powers of ζ. Finally $B\omega_0 = 0$ keeps its inclusion fixed. □

II.8 Coherent invariance of the volume

For a degree-r derivation D, its action on raw chains is

$$\mathcal{L}_D(a_0[a_1|\cdots|a_n]) = \sum_{j=0}^{n} (-1)^{nr + r\sum_{i<j}|a_i|} a_0[\cdots|D(a_j)|\cdots|a_n]. \tag{II.20}$$

The $j = 0$ term acts on the outer coefficient. The factor $(-1)^{nr}$ matches the internal differential in (11.16). Tensor signs give

$$[d_{\mathrm{H}}, \mathcal{L}_D] = \mathcal{L}_{[d,D]}, \qquad [B, \mathcal{L}_D] = 0, \qquad [\mathcal{L}_D, \mathcal{L}_E] = \mathcal{L}_{[D,E]}.$$

The action F therefore acts on the negative-cyclic complex.

More explicitly, regard F_n as the values of a degree-one twisting cochain $\alpha : S^c(\mathfrak{l}[1]) \to \mathrm{Der}(R)$. Its equation, with the tensor sign in the convolution bracket, is

$$\partial\alpha + \alpha b_{\mathfrak{l}} + \tfrac{1}{2}[\alpha, \alpha] = 0.$$

Its arity-two equation is exactly $[d, F_2(f, g)] = [F_1(f), F_1(g)] - F_1\{f, g\}$.

Use the arity product of maps from this coaugmented coalgebra to M_R. For a homogeneous map ϕ, its differential is

$$(d_{\mathrm{H}} + \zeta B)\phi - (-1)^{|\phi|}\phi b_{\mathfrak{l}} + \mathcal{L}_\alpha \star \phi.$$

The last product uses the unshuffle coproduct and its tensor Koszul sign. Let δ_F denote the terms beyond the internal differential. They raise arity.

The projection P intertwines F_1 with X_f on forms. It kills the action of every F_n for $n \geq 2$, since their negative-degree generator values are killed by π. Thus it intertwines δ_F with the strict Hamiltonian action on the corresponding convolution of forms.

Theorem 11.9. *For the fixed classical action F, the volume $dq \wedge dp$ has a coherent negative-cyclic lift*

$$\omega_{\mathrm{eq}} = (1 + \mathcal{T}_\zeta \delta_F)^{-1} I_\zeta(dq \wedge dp), \tag{11.21}$$

where $\mathcal{T}_\zeta$ is the contraction on convolution induced by T_ζ. Its arity-n component has values in M_R^{-n-2}. The derived fiber over this specified invariant volume is contractible. The construction extends continuously to pointed formal Hamiltonians at fixed arity.

Proof. Extend Proposition 11.8 to convolution and perturb by δ_F. This series converges in arity. Since the projection intertwines the actions and $P\mathcal{T}_\zeta = 0$, one has $P\delta_F\mathcal{T}_\zeta = 0$. The transferred differential therefore consists of ζd_{dR} and the strict Chevalley action on forms.

The volume is de Rham closed and Hamiltonian invariant: $\mathrm{div}(X_f) = -f_{pq} + f_{qp} = 0$. Its perturbed inclusion is consequently closed. The same perturbation contracts the projection kernel. The homotopy fiber over the specified closed volume is an affine translate of this acyclic complex, so its derived mapping space is contractible. Both the action and its projection are fixed in this assertion.

At fixed arity only finitely many arity perturbations contribute. At fixed weight and degree the mixed perturbation is finite. Proposition 11.6 gives nonnegative input weight shifts, leaving finitely many input degrees at any fixed output weight. This proves continuity for the pointed completion. □

The first two coherence equations have explicit representatives. Write $\langle a_1| \cdots |a_n\rangle = 1[a_1| \cdots |a_n]$. The arity-one equation is $d_{\mathrm{H}}\omega_1(f) = -\mathcal{L}_{F_1(f)}\omega_0$. For cubics choose

$$\omega_1(q^3) = -3\langle q|q|q\rangle, \qquad \omega_1(p^3) = -3\langle p|p|p\rangle.$$

A compatible quartic representative is

$$\begin{aligned}\omega_1(q^2p^2) = {} & -\tfrac{2}{3}\langle tt\rangle \\ & + \tfrac{2}{3}\big(\langle q|pt\rangle - \langle q|tp\rangle - \langle p|qt\rangle + \langle p|tq\rangle\big) \\ & + \tfrac{2}{3}\big(\langle qt|p\rangle - \langle pt|q\rangle - \langle tq|p\rangle + \langle tp|q\rangle\big) \\ & - 2\big(\langle q|p|qp\rangle + \langle p|qp|q\rangle + \langle qp|q|p\rangle\big).\end{aligned} \tag{11.22}$$

All these chains have zero B-image and zero HKR projection.

For $f = q^3$, $g = p^3$, the next equation in the same convention is

$$d_{\mathrm{H}}\omega_2(f,g) = \omega_1(\{f,g\}) - \mathcal{L}_{F_1(f)}\omega_1(g) + \mathcal{L}_{F_1(g)}\omega_1(f) - \mathcal{L}_{F_2(f,g)}\omega_0. \tag{11.23}$$

The minus sign uses $F_2 = \eta$ as in (11.4). One solution is

$$\begin{aligned}\tfrac{1}{9}\omega_2(q^3,p^3) = {}& \langle t|t\rangle + \langle q|p|t\rangle - \langle p|q|t\rangle - \langle q|t|p\rangle + \langle p|t|q\rangle \\ & + \langle t|q|p\rangle - \langle t|p|q\rangle - \langle q|q|p|p\rangle + \langle q|p|q|p\rangle \\ & + \langle q|p|p|q\rangle + \langle p|q|q|p\rangle - \langle p|q|p|q\rangle - \langle p|p|q|q\rangle.\end{aligned} \tag{11.24}$$

Its Connes image and HKR projection vanish. Substitution in the raw differential verifies both displayed coherence equations: the internal terms use $dt = qp - pq$, and the adjacent mergers cancel the cyclic faces. The terms $\langle tt\rangle$ and $\langle t|t\rangle$ belong to different equations and are both necessary.

These prescribed lower components extend to all arities because their later residuals lie in the contracted projection kernel. They need not coincide with the representatives selected by a different finite row-reduction convention.

The volume also records nondegeneracy. The diagonal Koszul resolution of A is finite free over A^e and self-dual, with its top generator paired with $dq \wedge dp$. Hence

$$R\operatorname{Hom}_{A^e}(A, A^e) \simeq A[-2].$$

Transport through $R \to A$ gives the smooth 2-Calabi–Yau orientation of the associative resolution. This duality uses the diagonal bimodule. A proper scalar trace would require a different finiteness condition.

11.9 The quantum cycle and the remaining lift

For $\mathcal{R}_\varepsilon$, take normalized chains relative to k_ε with the same raw convention. The expression $1[q|p] - 1[p|q] - 1[t]$ is still a negative-cyclic cycle. Its differential is $-\varepsilon 1[1]$, which vanishes in the normalized complex, and its Connes image is zero.

This produces the quantum volume cycle and the coherent quantum derivation action separately. Their compatible equivariant cyclic comparison requires its own construction. On the classical matrix gauge complex, a Hamiltonian-function lift must retain the central rank or specify the pointed source. Its moment-map equations still require compatible solutions. The associative derivation contraction supplies neither conclusion by forgetting the extra structure.

Chapter 12

Matrix branes, relations and central rank

The relation resolving the commutative plane becomes a matrix moment equation. Its scalar component vanishes identically, so the relation complex retains a negative-degree class at every rank. Ordinary invariant functions forget this class. Gauge cochains add further states. These differences determine the recipient of a Hamiltonian action.

We first compute the ordinary trace algebra and its stable coordinate. We then quantize matrix entries and obtain a global ordinary reduction with a strict Moyal trace map. The full BRST complex requires additional homotopies.

Framing and coefficient quantization change the equations themselves. On a stable rank-two chart we construct regular classical and quantum Hamiltonian maps. Their global extensions depend on which charges and which parameter reductions are prescribed. Two obstruction theorems isolate those dependencies. The remaining global comparison keeps its full scope.

12.1 The derived commuting pair

All polynomial algebras in this chapter are over $\mathbb{C}$. Differentials raise cohomological degree by one.

Fix $N \geq 1$, put $G = GL_N$, and let X, Y be matrices of independent commuting coordinates. Matrix substitution in (11.1) gives

$$\mathcal{K}_N = \mathbb{C}[X_{ij}, Y_{ij}] \otimes \Lambda(\Psi_{ij}), \qquad |\Psi_{ij}| = -1, \qquad QX = QY = 0, \quad Q\Psi = [X, Y]. \tag{12.1}$$

The entries of Ψ anticommute. The group acts by simultaneous conjugation on all three matrices. Give X, Y weight one and Ψ weight two, so that Q preserves weight.

Let I_N^{cl} be the ideal generated by the entries of $[X, Y]$. Then $H^0(\mathcal{K}_N) = \mathbb{C}[X, Y]/I_N^{\mathrm{cl}}$. The full complex is needed even before taking gauge cochains:

$$Q \operatorname{Tr} \Psi = 0, \qquad [\operatorname{Tr} \Psi] \neq 0 \quad \text{in } H^{-1}(\mathcal{K}_N^G).$$

Indeed, setting $X = Y = 0$ is a chain map to $\Lambda(\Psi_{ij})$ with zero differential, and the trace has nonzero image. More precisely, splitting Ψ into its traceless and scalar parts factors $\mathcal{K}_N$ as the traceless constraint complex tensor $\Lambda(\operatorname{Tr} \Psi)$.

Infinitesimal gauge transformations are imposed by $C^\bullet(\mathfrak{gl}_N, \mathcal{K}_N)$, the Chevalley–Eilenberg complex. Introduce its matrix c of odd generators of degree one. With graded matrix commutators the total differential is

$$QX = [X, c], \quad QY = [Y, c], \quad Qc = -c^2, \quad Q\Psi = [X, Y] - [c, \Psi]. \tag{12.2}$$

These are the full gauge and relation terms. In each fixed weight, the complex is bounded and its two differentials give

$$E_2^{a,b} = H^a\big(\mathfrak{gl}_N; H^b(\mathcal{K}_N)\big) \Longrightarrow H^{a+b} C^\bullet(\mathfrak{gl}_N, \mathcal{K}_N).$$

Exactness of rational G-invariants identifies $H^b(\mathcal{K}_N^G)$ with $H^b(\mathcal{K}_N)^G$. It does not erase the positive Lie-cochain degrees.

The same information appears in the tangent complex at a commuting pair:

$$\underbrace{\mathfrak{gl}_N}_{-1} \xrightarrow{d_0} \underbrace{\mathrm{Mat}_N \oplus \mathrm{Mat}_N}_{0} \xrightarrow{d_1} \underbrace{\mathfrak{gl}_N}_{1}, \qquad \begin{cases} d_0(a) = ([a,X],[a,Y]), \\ d_1(U,V) = [U,Y] + [X,V]. \end{cases} \tag{12.3}$$

Its composite is $[a,[X,Y]] = 0$. On the middle term use

$$\omega((U,V),(U',V')) = \mathrm{Tr}(UV' - VU'),$$

and pair the outer terms by $\mathrm{Tr}(ab)$, symmetrically under reversal. This is graded skew because their degrees are odd. Trace invariance gives $\omega(d_0 a,(U,V)) = \mathrm{Tr}(a\, d_1(U,V))$. The pairings are perfect on each term.

Both the stabilizer and relation directions participate in this degree-zero symplectic pairing.

12.2 Traces after the relation

Let $F = \mathbb{C}\langle q,p\rangle$ and let $F_{\mathrm{cyc}} = F/[F,F]$ denote its quotient by the linear span of commutators. A free cyclic class need not remain nonzero in the matrix relation complex. For example,

$$Q\,\mathrm{Tr}(\Psi YX) = \mathrm{Tr}(X^2Y^2 - XYXY). \tag{12.4}$$

The classes $[q^2p^2]$ and $[qpqp]$ are distinct cyclic words. Their trace difference takes value -1 at $X = E_{12}, Y = E_{21}$, also after embedding these matrices in rank four. Thus the right side is a nonzero free invariant polynomial which becomes a boundary. Stable rank does not remove this relation.

Write $A_+ = (q,p)\mathbb{C}[q,p]$, graded by total degree. Write x_i, y_i for the coordinates of a diagonal matrix pair. For a monomial $f = q^a p^b$ of positive degree, define

$$p_f^{(N)} = \sum_{i=1}^{N} x_i^a\, y_i^b.$$

Let $\mathrm{Sym}(A_+)$ be the symmetric algebra on linear symbols τ_f, with τ_f of the same weight as f. Its multiplication is the product of trace symbols. Thus τ_{fg} and $\tau_f\tau_g$ are different elements.

Give $\mathrm{Sym}(A_+)$ the coproduct $\Delta^{\oplus}(\tau_f) = \tau_f \otimes 1 + 1 \otimes \tau_f$. An element z is called *block-sum primitive* if $\Delta^{\oplus} z = z \otimes 1 + 1 \otimes z$.

Proposition 12.1. *Restriction to diagonal pairs gives*

$$H^0(\mathcal{K}_N^G) \cong \mathbb{C}[x_1,y_1,\dots,x_N,y_N]^{S_N}.$$

The homomorphism

$$\mathrm{Sym}(A_+) \longrightarrow H^0(\mathcal{K}_N^G), \qquad \tau_{q^a p^b} \longmapsto [\mathrm{Tr}\, X^a Y^b],$$

is surjective and is injective in every weight $d \le N$. Its degreewise stable value is $\mathrm{Sym}(A_+)$. *In this stable algebra the block-sum primitive subspace is A_+.*

Proof. Rational reductive exactness gives $H^{\circ}(\mathcal{K}_N^G) = (\mathbb{C}[X,Y]/I_N^{\mathrm{cl}})^G$. The invariant part of the moment ideal equals that of its radical, and restriction of the reduced commuting invariants to diagonal pairs is an isomorphism. These are Gan–Ginzburg, Theorem 1.2.1 and Section 2.8. Only the invariant ideal is asserted radical here.

We prove the remaining statements directly in the diagonal ring. Expand a product $p_{f_1}^{(N)} \cdots p_{f_r}^{(N)}$ by assigning a particle index to each factor. The terms with distinct indices give the corresponding invariant monomial sum. All other terms use fewer indices and replace some f_i by their products. Induction on r, followed by division by the integer multiplicities of repeated monomials, expresses every invariant monomial sum as a polynomial in the $p_f^{(N)}$.

For injectivity in weight d, take a relation and choose its largest number r of factors τ_f. Since every f has positive degree, $r \le d \le N$. Terms occupying exactly r distinct labeled pairs can only come from products with r factors. Their exponent multiset determines that product, with a nonzero integer multiplicity. Its coefficient must therefore vanish. Descending induction on r proves injectivity.

Block sum sends each τ_f to $\tau_f \otimes 1 + 1 \otimes \tau_f$. In characteristic zero, the bidegree $(1, r-1)$ part of the reduced coproduct on symmetric degree $r \ge 2$ is the polynomial differential. It is nonzero on every nonzero homogeneous polynomial. Constants contribute no primitives. Hence the primitive subspace is precisely the span of the τ_f. □

The primitive coordinate is thus the vector space of nonconstant commutative Hamiltonians, after imposing the matrix relation. The map from $\mathrm{Sym}((F_{\mathrm{cyc}})_{>0})$ first factors through abelianization to $\mathrm{Sym}(A_+)$, and then through the fixed-rank power-sum map. The first kernel contains the cyclic difference $[q^2p^2] - [qpqp]$ in (12.4). Only the second kernel vanishes in the stated stable range.

At rank two the first relations occur in weight three. For example, Cayley–Hamilton and its polarization give

$$\begin{aligned} \operatorname{Tr} X^3 &= \tfrac{3}{2} \operatorname{Tr} X \ \operatorname{Tr} X^2 - \tfrac{1}{2} (\operatorname{Tr} X)^3, \\ \operatorname{Tr} X^2 Y &= \operatorname{Tr} X \ \operatorname{Tr} XY + \tfrac{1}{2} \operatorname{Tr} Y \ \operatorname{Tr} X^2 - \tfrac{1}{2} (\operatorname{Tr} X)^2 \operatorname{Tr} Y. \end{aligned}$$

Interchanging X, Y gives the other two cubic identities. Thus these four single traces vanish in the indecomposable quotient of the positive-degree rank-two invariant algebra. Their stable symbols are nonzero. Neither this quotient nor ordinary invariant cohomology describes the entire complex (12.2).

12.3 Central rank and the quadratic actions

Give matrix phase space the canonical bracket

$$\{X_{ij}, Y_{kl}\} = \delta_{il}\delta_{jk}.$$

The full classical gauge algebra also pairs c with Ψ, symmetrically on these odd variables, as in (12.2). For a plane polynomial f put $J_N(f) = \operatorname{Tr}(\sigma f)(X,Y)$, where σ is the word average of Chapter 11. These are closed Hamiltonian functions. They retain

$$J_N(1) = N, \qquad \{\operatorname{Tr} X, \operatorname{Tr} Y\} = N.$$

Evaluation of all variables at zero is a cochain map to $\mathbb{C}$, so N is not exact. Consequently the first components $q \mapsto \operatorname{Tr} X$ and $p \mapsto \operatorname{Tr} Y$ cannot define a homotopy moment map from $\mathbb{C}[q,p]/\mathbb{C}1$: their source bracket has become zero. The central extension retaining 1, or the pointed source $(q,p)^2$, avoids this particular obstruction.

The scalar polynomials of degree at most two form a six-dimensional Lie algebra. Their trace assignment is strict:

$$1 \mapsto N, \quad q \mapsto \operatorname{Tr} X, \quad p \mapsto \operatorname{Tr} Y, \quad \frac{q^2}{2} \mapsto \frac{\operatorname{Tr} X^2}{2}, \quad qp \mapsto \operatorname{Tr} XY, \quad \frac{p^2}{2} \mapsto \frac{\operatorname{Tr} Y^2}{2}.$$

Differentiating these six functions verifies their brackets. For example, the two pure quadratics bracket to $\operatorname{Tr} XY$, and this function acts with weights $-1, 1$ on the two linear traces.

Allowing a matrix coefficient inside the trace gives a larger, generally noninvariant space, and changes the closure question. At $N = 2$ let $\mathcal{S}$ be spanned by 1 and

$$\operatorname{Tr}(AX), \quad \operatorname{Tr}(AY), \quad \operatorname{Tr}(AX^2), \quad \operatorname{Tr}(AY^2), \quad \tfrac{1}{2}\operatorname{Tr}(A(XY+YX)), \qquad A \in \operatorname{Mat}_2 .$$

For $A = \operatorname{diag}(1, -1)$ one obtains

$$\left\{\tfrac{1}{2}\operatorname{Tr}(AX^2), \tfrac{1}{2}\operatorname{Tr}(AY^2)\right\} = X_{11}Y_{11} + X_{22}Y_{22}.$$

This is not in $\mathcal{S}$, even modulo I_2^{cl}. Indeed, bidegree forces a putative expression to be $\operatorname{Tr}(C(XY + YX))/2$. Setting $Y = I_2$ forces $C = I_2$. Setting $X = Y = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ then gives zero on the left and two on the right, at a commuting pair in both cases.

Proposition 12.2. *The Lie algebra generated by $\mathcal{S}$ is the full space of polynomials of degree at most two in the eight matrix entries. It has dimension $36 + 8 + 1 = 45$, with constants as its center. For a constant symplectic bracket, Weyl quantization preserves the normalized Lie bracket on every such quadratic space.*

Proof. Write $\sigma_0 = I_2$ and set

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

These satisfy $\operatorname{Tr}(\sigma_i\sigma_j) = 2\delta_{ij}$. Set $X = \sum r_i\sigma_i$ and $Y = \frac{1}{2}\sum s_i\sigma_i$, so $\{r_i, s_j\} = \delta_{ij}$ for $0 \le i, j \le 3$. The Jordan operators $U \mapsto (AU + UA)/2$ for $A = I_2, \sigma_i$ have matrices I_4 and $S_i = E_{0i} + E_{i0}$.

For symmetric T and arbitrary C put $U(T) = r^t T r/2$, $V(T) = s^t T s/2$, $W(C) = r^t C s$. The initial space contains these forms for $T, C \in \operatorname{span}(I_4, S_1, S_2, S_3)$. Their brackets are

$$\begin{aligned} \{U(T), V(T')\} &= W(TT'), \\ \{W(C), U(T)\} &= -U(CT + TC^t), \\ \{W(C), V(T)\} &= V(C^tT + TC). \end{aligned}$$

For $i \neq j$, $S_iS_j + S_jS_i = E_{ij} + E_{ji}$, while $S_i^2 = E_{00} + E_{ii}$ and $\sum_i S_i^2 - I_4 = 2E_{00}$. The brackets therefore generate all symmetric T in both $U(T)$ and $V(T)$. Products of symmetric matrices span all matrices, since $E_{ii}(E_{ij} + E_{ji}) = E_{ij}$ for $i \neq j$. They generate every $W(C)$.

The quadratic dimensions are $10 + 10 + 16$. All eight linear coordinates were already present, and their brackets supply the constant. Commuting with all linear coordinates forces every derivative to vanish, so the center consists of constants.

For a constant Poisson tensor Π and central parameter h, the polynomial Moyal product over $\mathbb{C}[h]$ is $m\exp(h\Pi/2)$. Its commutator contains only odd powers of Π. On quadratic inputs, the terms of order at least three vanish, leaving $h\{F, G\}$. Weyl ordering identifies this product with operator multiplication, and therefore preserves the bracket divided by h. □

Projection to quadratics does not extend this result to nonlinear Hamiltonian actions. For $H = \operatorname{Tr} X^3/6$, $U = \operatorname{Tr} Y^2/2$ and $V = \operatorname{Tr} X$, let D be $\{H, -\}$ followed by that projection. Then

$$DU = DV = 0, \qquad D\{U, V\} = -\tfrac{1}{2}\operatorname{Tr} X^2.$$

The discarded cubic $\{H, U\} = \operatorname{Tr}(YX^2)/2$ is exactly the term needed for the derivation identity. The quadratic space is a Lie subalgebra, not a quotient of the full Hamiltonian action.

12.4 Quantizing matrix entries

Let $k_h = \mathbb{C}[h]$. The matrix Weyl algebra $W_{N,h}$ is generated over k_h by X_{ij}, Y_{ij} with

$$[X_{ij}, Y_{kl}] = h\delta_{il}\delta_{jk}, \qquad [X_{ij}, X_{kl}] = [Y_{ij}, Y_{kl}] = 0. \tag{12.5}$$

The ordered monomials, with all X before all Y, span the algebra. Their differential-operator images $X_{ij} = x_{ij}$, $Y_{ij} = -h\partial_{x_{ji}}$ are independent after inverting h. Thus they form a basis. The matrix Weyl algebra is h-torsion free before any localization.

The normal-ordered moment matrix is

$$M_h = YX - XY + hNI_N, \qquad \operatorname{Tr} M_h = 0. \tag{12.6}$$

For $a \in \mathfrak{g} = \mathfrak{gl}_N$, put $\ell_a = -\operatorname{Tr}(aM_h)$. In the differential representation, $\ell_a = h\tau(a)$, where $\tau(a)X = [X, a]$. Hence $[\ell_a, \ell_b] = h\ell_{[a,b]}$. Scalar conjugation has zero moment.

Define the left ideal and the ordinary invariant reduction by

$$I_{N,h} = \sum_{i,j} W_{N,h}(M_h)_{ij}, \qquad \mathcal{A}_{N,h} = W_{N,h}^G/(I_{N,h} \cap W_{N,h}^G).$$

The intersection is two-sided in the invariant algebra, because invariants commute with all ℓ_a. We do not impose the moment as a two-sided ideal of the full Weyl algebra.

Let $D(\mathfrak{g})$ denote polynomial differential operators on $\mathfrak{g}$. Let $\mathfrak{t} \subset \mathfrak{g}$ be the diagonal subspace. The order filtration $F_m D(\mathfrak{g})$ bounds the number of derivatives and permits arbitrary polynomial coefficients. The assignment above identifies $W_{N,h}$ with $\bigoplus_{m\geq 0} h^m F_m D(\mathfrak{g})$. Let $J_D = (D(\mathfrak{g})\tau(\mathfrak{g}))^G$.

On the regular Cartan, ordinary radial restriction acts on invariant functions. Put $\delta = \prod_{i<j}(x_i - x_j)$ and $\Phi(D) = \delta \operatorname{rad}(D)\delta^{-1}$. The filtered Harish–Chandra theorem states that

$$D(\mathfrak{g})^G/J_D \xrightarrow{\Phi} D(\mathfrak{t})^{S_N} \quad \text{is a strict filtered isomorphism,} \qquad \operatorname{gr} J_D = (\sqrt{I_N^{\mathrm{cl}}})^G. \tag{12.7}$$

Its associated graded map is diagonal restriction (Etingof–Ginzburg, Theorems 6.8–6.10). The invariant-ideal equality used in Proposition 12.1 removes the radical in this formula.

An ideal is h-saturated if hu belonging to it implies that u belongs to it.

Theorem 12.3. *Conjugated radial restriction gives an isomorphism*

$$\Phi_{N,h} : \mathcal{A}_{N,h} \xrightarrow{\sim} \big(k_h\langle x_i, \pi_i\rangle_{i=1}^N\big)^{S_N}, \qquad [x_i, \pi_j] = h\delta_{ij},$$

with all other pair commutators zero. The invariant moment ideal is h-saturated, the quotient is free over k_h, and its special fiber is $\mathbb{C}[x_i, \pi_i]^{S_N}$.

Proof. We first control coefficient orders in J_D. If $D \in J_D \cap F_m D(\mathfrak{g})$, its leading symbol belongs to $(I_N^{\mathrm{cl}})^G$. Write it as $\sum_a p_a \sigma_1(\tau(a))$, with each p_a homogeneous of momentum degree $m-1$. This is a finite expression.

Exactness of rational invariants chooses an invariant coefficient vector mapping to that symbol. Normal quantization is equivariant, so it gives invariant $\sum_a P_a\tau(a)$ with the same symbol and ord $P_a \leq m-1$. Subtract and repeat at lower order.

There is no order-zero remainder in the moment ideal. We have proved a finite expression $D = \sum_a P_a\tau(a)$ with those coefficient bounds.

Taking invariants in the Rees algebra is degreewise. Every element of $I_{N,h} \cap W_{N,h}^G$ has its differential-operator coefficient in J_D. Conversely, the expression just proved gives

$$h^m D = \sum_a (h^{m-1} P_a)(h\tau(a)) \in I_{N,h}.$$

Thus $I_{N,h} \cap W_{N,h}^G = \mathrm{Rees}_F J_D$. Strictness of (12.7) identifies the quotient with the Cartan invariant Rees algebra in the statement.

Cartan Weyl symmetrization is an S_N-equivariant k_h-linear isomorphism from the ordinary polynomial module. A complex basis of its invariant symbols gives a k_h-basis of the quotient. Freeness implies h-saturation of the kernel: if hu maps to zero, then so does u. Taking the intrinsic Rees special fiber proves the last assertion. □

In that special fiber, π_i remains a momentum symbol. Setting $h = 0$ in the external representation $\pi_i = -h\partial_{x_i}$ would instead kill it. Specializing that representation does not compute the Rees special fiber.

12.5 The radial trace and its Moyal bracket

Use the same word average to define J_N in $W_{N,h}$. For $q^a p^b$, it averages traces of distinct words with a factors X and b factors Y. Thus $J_N(1) = N$.

Let Op_W denote the analogous scalar Weyl ordering in one pair. On $k_h[q,p]$ put

$$f \star_h g = m \exp\!\left(\frac{h}{2}(\partial_q \otimes \partial_p - \partial_p \otimes \partial_q)\right)(f \otimes g), \qquad B_h(f,g) = \frac{f \star_h g - g \star_h f}{h}.$$

The exponential terminates on polynomials, and the numerator is divisible by h. This notation inverts no parameter.

Theorem 12.4. *For every N and all polynomial f, g,*

$$\Phi_{N,h}([J_N(f)]) = \sum_i \mathrm{Op}_W f(x_i, \pi_i), \qquad h^{-1}[J_N(f), J_N(g)] - J_N(B_h(f,g)) \in I_{N,h} \cap W_{N,h}^G.$$

The induced map $k_h[q,p]/k_h 1 \longrightarrow \mathcal{A}_{N,h}/k_h 1$ is injective as a map of Lie algebras with normalized brackets.

Proof. Pure coordinate traces restrict to $\sum_i x_i^d$. The mixed second variation of an eigenvalue in entries (i,j), (j,i) is $\pm(x_i - x_j)^{-1}$, obtained by differentiating $(z - x_i)(z - x_j) - uv = 0$. Therefore

$$\mathrm{rad}(\mathrm{Tr}\, Y^2) = h^2\left(\sum_i \partial_i^2 + 2\sum_{i<j} \frac{\partial_i - \partial_j}{x_i - x_j}\right).$$

The Vandermonde δ is harmonic: its Laplacian is an alternating polynomial of smaller degree, hence is divisible by δ and must be zero. Conjugating this radial operator by δ gives $\Phi_{N,h}(\mathrm{Tr}\, Y^2) = \sum_i \pi_i^2$.

The derivation $D = h^{-1}[-\mathrm{Tr}\, Y^2/2, -]$ satisfies $DX = Y$, $DY = 0$. Its exponential terminates on each polynomial. Multiplicativity of $\Phi_{N,h}$ and torsion freeness intertwine D with $x_i \mapsto \pi_i$, $\pi_i \mapsto 0$. Apply this exponential to $\mathrm{Tr}\, X^{a+b}$ and extract its coefficient of t^b. The same binomial factor on both sides gives the claimed mixed trace formula.

Distinct particle pairs commute. In one pair, Weyl ordering intertwines the Moyal product with operator multiplication. The displayed defect thus has zero radial image, and Theorem 12.3 puts it in the actual invariant moment ideal.

If the image of $J_N(f)$ is scalar, inverse Weyl symmetrization gives $\sum_i f(x_i, \pi_i) = a \in k_h$. Set all pairs except the first to zero. Then $f(q,p) + (N-1)f(0,0) = a$, so f is constant. This proves injectivity of the scalar Lie quotient. □

This proves ideal membership in every bidegree. It does not choose coefficients in a prescribed finite span of matrix words, or compatible primitives in every BRST arity.

All these ordinary maps extend to h-adic completion, with a polynomial coefficient at each order and no uniform degree bound. To see the kernel assertion, put $J = I_{N,h} \cap W^G_{N,h}$. Saturation gives $J \cap h^m W^G_{N,h} = h^m J$ for every m. The short exact sequence remains exact modulo h^m, and $J/h^{m+1}J \to J/h^m J$ is onto. Successively correcting lifts in these kernels proves exactness of the inverse limit. The completed kernel is therefore the closure of J.

12.6 Rank splitting and the collective algebra

The Cartan description separates particles. Set $\mathcal{A}_{0,h} = k_h$ and $J_0(f) = 0$. Inclusion of S_{N+M}-invariants into $S_N \times S_M$-invariants, transported through Φ, gives injective unital maps

$$\Delta^{\mathrm{red}}_{N,M} : \mathcal{A}_{N+M,h} \longrightarrow \mathcal{A}_{N,h} \otimes_{k_h} \mathcal{A}_{M,h}.$$

Finite-group averaging identifies the target invariants with this tensor product. Both ways of splitting three blocks are the same subgroup inclusion, so the maps are coassociative across ranks. The trace formula gives

$$\Delta^{\mathrm{red}}_{N,M}[J_{N+M}(f)] = [J_N(f)] \otimes 1 + 1 \otimes [J_M(f)].$$

In particular the empty trace splits additively as $N + M$. These maps do not define a coproduct on one fixed-rank algebra.

To retain products of one-particle sums, put $L = k_h[q, p]$ and introduce a linear symbol $e(f)$ for each $f \in L$. Define

$$\mathcal{U}_h = T_{k_h}(L) \,/\, \big(e(f)e(g) - e(g)e(f) - h\, e(B_h(f, g))\big). \tag{12.8}$$

The generator $\nu = e(1)$ is central and is distinct from the multiplicative unit.

Theorem 12.5. *Ordered products of the monomial generators $e(q^a p^b)$ form a k_h-basis of $\mathcal{U}_h$. It is a Hopf algebra with*

$$\Delta e(f) = e(f) \otimes 1 + 1 \otimes e(f), \qquad \epsilon e(f) = 0, \qquad S e(f) = -e(f).$$

The maps $j_N : e(f) \mapsto [J_N(f)]$ are surjective algebra maps to $\mathcal{A}_{N,h}$ and satisfy

$$\bigcap_{N \geq 1} \ker j_N = 0, \qquad (j_N \otimes j_M)\Delta = \Delta^{\mathrm{red}}_{N,M}\, j_{N+M}.$$

These statements of joint injectivity and rank compatibility also hold after h-adic completion, with completed tensor products.

Proof. The bracket hB_h satisfies Jacobi because it is the Moyal commutator. Order the monomial basis of L. Replacing an inverted adjacent pair by the ordered pair and its bracket decreases word length or, at fixed length, inversion number. Disjoint replacements commute, and the difference for an overlapping triple is Jacobi. These terminating reductions give unique normal forms. Their leading coefficients are one, so this proves the basis over k_h without parameter inversion.

Primitive coproducts preserve the defining commutator relation: the cross terms commute and the remaining terms are its copies in the two tensor factors. Counit and antipode preserve it as well. The Hopf identities hold on generators and, by induction, on their products.

Theorem 12.4 gives the maps j_N. For surjectivity, filter the Cartan Weyl algebra by total polynomial degree, putting h in the coefficient ring. Its invariant associated graded is generated by polarized power sums, by the distinct-index induction in Proposition 12.1. Lift an invariant leading symbol by a polynomial in the trace operators and subtract it. Repeating at lower total degree proves generation.

Modulo h, the maps have source $\mathrm{Sym}_{\mathbb{C}}(\mathbb{C}[q,p])$ and send $e(f)$ to $p_f^{(N)}$, including $\nu \mapsto N$. They are jointly injective. Indeed, group any proposed relation by the largest number r of nonconstant symbols, with coefficients polynomial in ν. For $N \geq r$, terms occupying exactly r distinct pairs identify each coefficient polynomial at $\nu = N$, up to a nonzero integer. Vanishing for all such N makes every coefficient polynomial zero. Descend in r.

For a nonzero element of $\mathcal{U}_h$, factor the common lowest power of h from its PBW coefficients. Torsion freeness of every target would force its nonzero class modulo h to vanish in all classical targets, a contradiction. A nonzero completed series also has a first nonzero coefficient, which is a finite polynomial in PBW generators, so the same argument applies. Finally the two rank maps in the statement agree on $e(f)$ by splitting its one-particle sum. They agree on the entire algebra. □

No finite collection of ranks is faithful: for a finite set T of ranks, the nonzero central polynomial $\prod_{n\in T}(\nu - n)$ vanishes in all those representations. Moreover ν is primitive, whereas $\Delta 1 = 1 \otimes 1$. The rank-N quotient for $N > 0$ is therefore not a Hopf quotient with the same coproduct and counit.

Products cannot be removed by a linear projection which preserves all quantum constants. Such a projection, if it killed both $J_N(q)J_N(p)$ and $J_N(p)J_N(q)$, would send their commutator $hN1$ to zero while fixing it. The collective algebra retains these products instead.

12.7 Quantum constraints and the full BRST algebra

The ordinary quotient is the degree-zero part of a constraint complex, but the complex has further cohomology. For $\mathfrak{g} = \mathfrak{gl}_N$ define $C^{-r} = W_{N,h} \otimes \Lambda^r \mathfrak{g}$, with

$$d(w \otimes a_1 \wedge \cdots \wedge a_r) = \sum_i (-1)^{i+1} w\ell_{a_i} \otimes a_{\widehat{i}} + h \sum_{i<j} (-1)^{i+j} w \otimes [a_i, a_j] \wedge a_{\widehat{ij}}.$$

The omitted entries retain their order. This is the Chevalley chain differential for the right action $w \cdot a = w\ell_a$ and bracket $h[-,-]$. In degree two its square is $w[\ell_a, \ell_b] - hw\ell_{[a,b]} = 0$. In higher degrees the remaining overlaps are the same identity and Jacobi. Hence $d^2 = 0$, and reductive exactness gives $H^0(C^G) = \mathcal{A}_{N,h}$.

Splitting $\mathfrak{g} = \mathfrak{sl}_N \oplus \mathbb{C} I_N$ factors this complex as

$$C(\mathfrak{g}) = C(\mathfrak{sl}_N) \otimes \Lambda(\theta_0), \qquad |\theta_0| = -1, \quad d\theta_0 = 0.$$

The scalar comoment is zero. Thus $H^{-1}(C^G)$ has the direct summand $\mathcal{A}_{N,h}\theta_0$. Exterior multiplication with multiplication of coefficients does not make C a differential algebra: its differential contains right multiplication by the constraints.

To obtain a differential algebra, choose a basis a_i with $[a_i, a_j] = f_{ij}^{\ k} a_k$ and write $\ell_i = \ell_{a_i}$. Adjoin odd ghosts c^i, b_i of degrees 1, −1. Their k_h-algebra, denoted $\mathrm{Cl}_h(c,b)$, has relations

$$c^i c^j = -c^j c^i, \qquad b_i b_j = -b_j b_i, \qquad b_i c^j + c^j b_i = h\delta_i^{\ j}.$$

Set $\mathcal{B}_{N,h} = W_{N,h} \otimes_{k_h} \mathrm{Cl}_h(c,b)$, with the graded tensor product. Repeated indices are summed in

$$\Theta = \sum_i c^i \ell_i - \tfrac{1}{2} f_{ij}^{\ k} c^i c^j b_k, \qquad Q = h^{-1}[\Theta, -]_{\mathrm{gr}}. \tag{12.9}$$

Normal exterior words in c, b form a free k_h-basis: represent c^i by exterior multiplication and b_i by $h\partial_{c^i}$. For one pair the four normal words give independent two-by-two matrices after inverting h. Tensoring proves independence for all pairs. The relations give spanning before inversion. Consequently $\mathcal{B}_{N,h}$ is h-torsion free. Modulo h it is supercommutative, so the divided commutator defining Q is integral.

Proposition 12.6. *The charge satisfies $\Theta^2 = 0$, and Q is a degree-one derivation with $Q^2 = 0$. At rank one the full $\mathfrak{gl}_1$ model has $Q = 0$ and its degree-zero ghost factor is free on $1, u = cb$ with $u^2 = hu$.*

Proof. Use the faithful ghost representation above and the left regular matter representation. Then $\Theta = \sum c^i \ell_i + h\, d_{\mathrm{CE}}$. Jacobi gives $d_{\mathrm{CE}}^2 = 0$. The square of the first term is $\frac{h}{2} f_{ij}{}^k c^i c^j \ell_k$, and the mixed term is its negative. Thus $\Theta^2 = 0$ in a faithful representation and hence in the algebra.

The inner graded commutator is a derivation, and its square is $[\Theta^2, -]$. Torsion freeness gives $Q^2 = 0$.

At rank one both the moment and the structure constants vanish. The ghost relation gives $cbcb = c(h - cb)b = hcb$, with $1, cb$ independent in the normal basis. □

Already at rank one, $H^0(\mathcal{B}_{N,h}, Q)$ therefore differs from ordinary reduction. Choosing the effective traceless gauge algebra removes this particular scalar ghost factor by changing the gauge input.

The cubic trace defect has a primitive in the full model. Use a trace-dual pair of bases a_i, a^i, raise moment indices with that pairing, and put

$$T_N = \sum_i b_i \ell^i, \qquad C_N = \sum_i \ell_i \ell^i.$$

From (12.9), $Qb_i = \ell_i - f_{ij}{}^k c^j b_k$ and $Q\ell^i = c^j [\ell_j, \ell^i]/h$. The graded Leibniz rule, invariance of the pairing, and $\sum_i [\ell_i, \ell^i] = 0$ give

$$QT_N = C_N. \tag{12.10}$$

Indeed the two terms containing $c^j b_k$ cancel after moving b_i past c^j. The resulting contraction is the vanishing sum of moment commutators.

Proposition 12.7. *For every rank, with the Moyal source bracket,*

$$h^{-1}[J_N(q^3), J_N(p^3)] - J_N(B_h(q^3, p^3)) = Q\big(-\tfrac{3}{2} T_N\big).$$

Proof. Let $a = \mathrm{Tr}(X^2Y^2)$ and $b = \mathrm{Tr}(XYXY)$ denote commuting entry symbols. Write $\mathrm{Op}_W^{\mathrm{ent}}$ for entrywise Weyl quantization. The entries of $X + sY$ commute with one another, so polarization proves that the matrix-word average equals the entrywise quantization of its commutative symbol. In particular $J_N(q^2p^2) = \mathrm{Op}_W^{\mathrm{ent}}(2a/3 + b/3)$.

The first entrywise contraction of the two cubic traces is $9a$. Contracting their third derivatives gives $18N^3 + 18N$. Each third derivative has two cyclic index identifications, with multiplicity three. The normalized third-order coefficient is $h^2/24$. Thus

$$h^{-1}[\mathrm{Tr}\, X^3, \mathrm{Tr}\, Y^3] = 9\, \mathrm{Op}_W^{\mathrm{ent}}(a) + \tfrac{3}{4} h^2 (N^3 + N).$$

For the paired quadratic moment symbols the first contraction cancels. The second is

$$-\frac{h^2}{4} \sum_i \mathrm{Tr}(\mathrm{ad}\, a_i\, \mathrm{ad}\, a^i).$$

This trace is on $\mathfrak{g}$. Writing each adjoint as left minus right multiplication gives the sum $2N(N^2 - 1)$, and hence

$$C_N = \mathrm{Op}_W^{\mathrm{ent}}\, \mathrm{Tr}([X, Y]_{\mathrm{cl}}^2) - \tfrac{1}{2} h^2 (N^3 - N).$$

Finally,

$$3(a-b) = -\tfrac{3}{2}\operatorname{Tr}([X,Y]_{\mathrm{cl}}^{2}).$$

Subtracting $J_N(9q^2p^2)$ therefore leaves $-\frac{3}{2}C_N + \frac{3}{2}h^2N$. But $B_h(q^3,p^3) = 9q^2p^2 + \frac{3}{2}h^2$, and $J_N(1) = N$. The scalar terms cancel. Equation (12.10) now gives the primitive. □

The choice of source bracket is essential. At rank one,

$$B_h(q^3,p^4) - \{q^3,p^4\} = 6h^2p,$$

and $Q = 0$ in the full BRST model. The unchanged Weyl-symbol map from the undeformed Poisson source therefore has a nonzero binary defect which no homotopy can remove. This does not exclude a changed first map or a different recipient.

12.8 A formal slice with stabilizer ghosts

Fix a labeled diagonal matrix with distinct entries x_i^0 and complete the matrix base at that point. Formal eigenvectors with fixed normalizations give coordinates

$$X = g(s)D(x)g(s)^{-1}, \qquad D(x) = \operatorname{diag}(x_i).$$

There are $N^2 - N$ orbit coordinates s_j. To construct them, solve the eigenvector equations order by order. The off-diagonal linear terms divide by the nonzero constants $x_i^0 - x_k^0$. This recursive solution also proves invertibility of the formal coordinate change.

On the super base (X,c), the differential is $QX = [X,c]$, $Qc = -c^2$. Put

$$\gamma_j = Qs_j, \qquad c' = g^{-1}cg + g^{-1}Qg.$$

Differentiating the diagonalization gives $QD = [D,c']$. Its diagonal part is $Qx = 0$ and its off-diagonal part forces c' to be diagonal.

The ghost change from c to (γ_j, c'_i) is invertible: its linearization separates the orbit directions from the diagonal stabilizer. The transformed Maurer–Cartan equation gives $Qc' = -(c')^2 = 0$. Also $Q\gamma = 0$. Thus the vector field on this base is $\sum_j \gamma_j \partial_{s_j}$.

Take the order Rees algebra of differential operators with formal base coefficients, and complete it in h. Its symbols have polynomial momentum degree at each parameter order. The chain rule induces an exact algebra isomorphism, with $Y = -h\partial_X$ and $b = h\partial_c$. Matter momenta include the ghost terms required by the change of c. Changing X alone would not produce this isomorphism.

With $\pi_j = -h\partial_{s_j}$, $\beta_j = h\partial_{\gamma_j}$, the charge becomes $\Theta = -\sum_j \gamma_j\pi_j$. The orbit factors have

$$Qs_j = \gamma_j, \quad Q\beta_j = -\pi_j, \quad Q\gamma_j = Q\pi_j = 0, \qquad [s_j,\pi_k] = h\delta_{jk}, \quad \beta_j\gamma_k + \gamma_k\beta_j = h\delta_{jk}.$$

The residual factor is

$$\mathcal{R}_{N,h} = W_h(T^*\mathfrak{t})^{\wedge}_{x^0} \widehat{\otimes} \operatorname{Cl}_h(c'_1,\dots,c'_N,b'_1,\dots,b'_N), \qquad Q|_{\mathcal{R}_{N,h}} = 0.$$

Here the first factor has formal eigenvalue coefficients and polynomial momenta at each parameter order. The second factor retains the diagonal stabilizer ghosts and their conjugates. The residual momenta are $p_i = -h\partial_{x_i}$ and $b'_i = h\partial_{c'_i}$.

Proposition 12.8. *The residual inclusion is a strict differential-algebra quasi-isomorphism on this labeled formal chart. On it,*

$$f \longmapsto \sum_i \mathrm{Op}_W f(x_i, p_i)$$

is a strict Lie map from the plane Moyal algebra, with the normalized operator bracket.

Proof. Pass to Weyl symbols in the orbit variables. Let N count all four kinds of orbit generators and define on positive count

$$H = \mathsf{N}^{-1} \sum_j (s_j \partial_{\gamma_j} - \beta_j \partial_{\pi_j}).$$

The anticommutator of Q with the numerator is N. Hence $QH + HQ = 1 - ip$, where p extracts orbit count zero and i includes the residual factor. This calculation is coefficientwise in formal s and h, keeps finite momentum degree at each order, and inverts only positive integers. The separate residual factor makes i a strict algebra map.

Finally, distinct particle pairs commute and each individual pair realizes the plane Moyal product. Their sum gives the stated Lie identity. □

The projection p in this proof is linear, not multiplicative: annihilating both s_j and π_j cannot preserve $[s_j, \pi_j] = h$. The action is defined on a labeled formal chart. Extending it across the discriminant, with comparisons between charts and with prescribed global matrix charges, requires compatible BRST homotopies in every arity. The global ordinary reduction of Theorem 12.3 supplies a different, already defined recipient.

12.9 Representations of the coefficient Weyl algebra

Let ε denote the parameter in the coefficient relation $qp - pq = \varepsilon$. It is independent of the matrix quantization parameter h. Its derived N-dimensional representation algebra has commuting coordinate entries and differential

$$\mathcal{K}_{N,\varepsilon} = \mathbb{C}[\varepsilon, X, Y] \otimes \Lambda(\Psi), \qquad Q\Psi = [X, Y] - \varepsilon I_N.$$

The element $v = -\operatorname{Tr}\Psi/N$ satisfies $Qv = \varepsilon$. Consequently $Q(va) + vQa = \varepsilon a$ for every homogeneous a. All its cohomology is killed by ε, and after inverting ε, multiplication by v/ε contracts the complex. This is the homological form of the finite-dimensional trace obstruction to the Weyl relation. It does not apply to the entrywise relations (12.5).

An infinite-dimensional recipient survives integrally. Complete the coefficient Weyl algebra in ε and write it as $\widehat{W}_\varepsilon$. On $\mathcal{P} = \mathbb{C}[p][[\varepsilon]]$ let p act by multiplication and q by $\varepsilon\partial_p$. Use complete modules, continuous maps, strict exact sequences and resolutions by topologically free modules. The PBW basis gives the strict resolution

$$0 \longrightarrow \widehat{W}_\varepsilon \xrightarrow{R_q} \widehat{W}_\varepsilon \longrightarrow \mathcal{P} \longrightarrow 0.$$

Right multiplication by q is injective with closed image: in the normal basis $p^a q^b$ it increases b by one. The quotient consists of the $b = 0$ terms. Continuous Hom into $\mathcal{P}$ gives $\mathcal{P} \xrightarrow{\varepsilon\partial_p} \mathcal{P}$.

Polynomial integration coefficient by coefficient is a continuous right inverse to ∂_p. Therefore

$$\operatorname{Ext}^0(\mathcal{P}, \mathcal{P}) = \mathbb{C}[[\varepsilon]], \qquad \operatorname{Ext}^1(\mathcal{P}, \mathcal{P}) = \mathbb{C}[p], \qquad \operatorname{Ext}^i = 0 \quad (i \neq 0, 1). \tag{12.11}$$

The degree-one group is annihilated by ε and disappears after its inversion. It is part of the integral module, not an additional generic-parameter state.

12.10 A framed moment equation

A single framing vector changes the finite-dimensional equation. Fix $\zeta \in \mathbb{C}$, add a column I and a row J, and impose

$$[X, Y] + IJ = \zeta I_N. \tag{12.12}$$

Here X, Y, I, J have commuting coordinate entries. Use $\{X_{ij}, Y_{kl}\} = \delta_{il}\delta_{jk}$, $\{I_i, J_j\} = \delta_{ij}$, and zero for the other generator brackets. A point is stable if the vectors $W(X, Y)I$, for words W, span $\mathbb{C}^N$. The action is

$$g \cdot (X, Y, I, J) = (gXg^{-1}, gYg^{-1}, gI, Jg^{-1}).$$

Let $\mathcal{X}_{N,\zeta}$ be the stable quotient.

This quotient is smooth and symplectic. Indeed, an element fixing X, Y, I fixes every spanning word vector, and is the identity. The same argument gives injectivity of the infinitesimal action. Its symplectic transpose is the moment differential, which is therefore surjective.

The quotient has dimension $(2N^2 + 2N) - 2N^2 = 2N$. Choosing N word vectors which form a basis gives a gauge chart. Changes between these bases give its regular transition maps.

On the ordered chart where X has distinct eigenvalues, diagonalize it. Stability forces every I_i to be nonzero. For if $I_i = 0$, the off-diagonal moment equations give $Y_{ij} = 0$ for all $j \neq i$. Every word applied to I would then have zero ith component.

Diagonal gauge sets $I_i = 1$, and the equation becomes

$$X = \operatorname{diag}(x_i), \quad I_i = 1, \quad J_i = \zeta, \qquad Y_{ii} = y_i, \quad Y_{ij} = -\frac{\zeta}{x_i - x_j} \quad (i \neq j). \tag{12.13}$$

The ambient form $\operatorname{Tr}(dY \wedge dX) + dJ \wedge dI$ restricts to $\sum_i dy_i \wedge dx_i$, with $\{x_i, y_j\} = \delta_{ij}$. Permutations give the unlabeled spectral chart. Setting $\zeta = \varepsilon$ also gives a framed recipient over the coefficient-parameter ring, on this same discriminant-localized chart.

The necklace algebra, with its constant class retained, has a global strict Hamiltonian map on $\mathcal{X}_{N,\zeta}$:

$$[W] \longmapsto H_W = \operatorname{Tr} W(X, Y), \qquad H_1 = N. \tag{12.14}$$

Its necklace bracket is $[\partial_q^{\mathrm{cyc}} W \, \partial_p^{\mathrm{cyc}} V - \partial_p^{\mathrm{cyc}} W \, \partial_q^{\mathrm{cyc}} V]$. Cutting the two cyclic words at their differentiated factors proves $\{H_W, H_V\} = H_{\{W,V\}_{\mathrm{neck}}}$. The associated field has $\delta X = -\partial_Y^{\mathrm{cyc}} W$, $\delta Y = \partial_X^{\mathrm{cyc}} W$ and $\delta I = \delta J = 0$. The cyclic derivative identity makes $\delta[X, Y] = 0$. Thus it preserves the constraint and commutes with conjugation.

The full framed BFV algebra adds odd matrices c, Ψ of degrees $1, -1$, with $\{c_{ij}, \Psi_{kl}\} = \delta_{il}\delta_{jk}$. Its differential is the bracket with

$$\Theta_{\mathrm{fr}} = \operatorname{Tr}\big(c([X, Y] + IJ - \zeta I_N) - \Psi c^2\big).$$

Equivariance of the moment gives $Q^2 = 0$ as in the unframed model. The Hamiltonians H_W are closed: they are gauge invariant, commute with the moment functions, and contain no ghosts. No completeness of their polynomial flows is asserted.

12.11 The all-rank defect of symmetrized traces

For commuting entry symbols, average the traces of words containing a copies of X and b copies of Y. Denote the result by J_{ab}, with $J_{00} = N$. The frame in (12.12) changes which trace defects vanish.

Proposition 12.9. *On $\mathcal{X}_{N,\zeta}$,*

$$\{J_{30}, J_{03}\} - 9J_{22} = -\tfrac{3}{2}N(N-1)\zeta^2, \tag{12.15}$$

$$\{J_{30}, J_{04}\} - 12J_{23} = -6(N-1)\zeta^2 J_{01}. \tag{12.16}$$

For $N \geq 2$, a plane Moyal bracket of parameter θ compatible with these unchanged trace images would have to satisfy $\theta^2 = -(N-1)\zeta^2$. With that value,

$$\{J_{40}, J_{04}\} - 16J_{33} - 24\theta^2 J_{11} = \frac{56}{5}\zeta^2(NJ_{11} - J_{10}J_{01}). \tag{12.17}$$

For $\zeta \neq 0$ this is nonzero at every rank $N \geq 2$. Hence no single Moyal parameter makes the unchanged map from commutative polynomials to symmetrized traces a Lie map.

Proof. Put $C = [X, Y]$ and $U = IJ = \zeta I_N - C$. Tracing the constraint gives $JI = N\zeta$, so $U^2 = N\zeta U$. Expansion of the six words of bidegree $(2, 2)$ and the ten of bidegree $(2, 3)$ gives

$$\{J_{30}, J_{03}\} - 9J_{22} = -\tfrac{3}{2}\operatorname{Tr} C^2, \qquad \{J_{30}, J_{04}\} - 12J_{23} = -6\operatorname{Tr}(YC^2).$$

The rank-one expression for U implies

$$C^2 = (N-1)\zeta^2 I_N - (N-2)\zeta C.$$

Since $\operatorname{Tr} C = \operatorname{Tr}(YC) = 0$, these identities prove the two cubic formulas.

For the quartic pair use $\operatorname{Tr}(UXUY) = \operatorname{Tr}(UX)\operatorname{Tr}(UY)$. Together with $\operatorname{Tr}(CX) = \operatorname{Tr}(CY) = \operatorname{Tr}(C(XY + YX)) = 0$, this gives

$$\operatorname{Tr}(C^2(XY + YX)) = 2(N-1)\zeta^2 J_{11},$$
$$\operatorname{Tr}(CXCY) = \zeta^2(J_{10}J_{01} - J_{11}).$$

The twenty words in J_{33} divide into four cyclic classes of sizes 6, 6, 6, 2, represented by X^3Y^3, X^2YXY^2, X^2Y^2XY and $(XY)^3$. Expansion in these four classes yields

$$\operatorname{Tr}(X^3Y^3) - J_{33} = -\tfrac{2}{5}\operatorname{Tr}(C^2(XY + YX)) - \tfrac{7}{10}\operatorname{Tr}(CXCY).$$

Since $\{J_{40}, J_{04}\} = 16\operatorname{Tr}(X^3Y^3)$, substitution gives

$$\{J_{40}, J_{04}\} - 16J_{33} = \tfrac{8}{5}\zeta^2\big((15 - 8N)J_{11} - 7J_{10}J_{01}\big).$$

The scalar Moyal formulas are $B_\theta(q^3, p^3) = 9q^2p^2 + \frac{3}{2}\theta^2$ and $B_\theta(q^4, p^4) = 16q^3p^3 + 24\theta^2 qp$. Because $J_{00} = N$, the cubic equation forces the stated θ^2. Subtracting the quartic Moyal term gives (12.17). On (12.13), choose $x_i = i$, $y_1 = 1$ and $y_i = 0$ for $i > 1$. Then $NJ_{11} - J_{10}J_{01} = -N(N-1)/2$, proving nonvanishing. □

The next unequal-degree pair records the same loss of a product of traces. With the forced value of θ,

$$\{J_{30}, J_{05}\} - 15J_{24} - 15\theta^2 J_{02} = 6\zeta^2(NJ_{02} - J_{01}^2). \tag{12.18}$$

Indeed, the fifteen words in J_{24} have cyclic class sizes 6, 6, 3, and give

$$\operatorname{Tr}(X^2Y^4) - J_{24} = -\tfrac{3}{5}\operatorname{Tr}(C^2Y^2) - \tfrac{2}{5}\operatorname{Tr}(CYCY).$$

The same rank-one identities evaluate these two traces as $(N-1)\zeta^2 J_{02}$ and $\zeta^2(J_{01}^2 - J_{02})$. Together with $B_\theta(q^3, p^5) = 15q^2p^4 + 15\theta^2 p^2$, they prove (12.18).

The quartic residual also excludes a homotopy moment map into the framed BFV complex with these fixed first components. Its binary identity would make the residual a boundary. Evaluation at a constraint point, with ghosts and antighosts zero, is a chain map to $\mathbb{C}$. At the point used in the proof the residual has nonzero value. The obstruction concerns these fixed first components.

12.12 A particle map and a stable collision

The distinct-eigenvalue chart has a strict map with a different first component. Define spectral projectors and measurements by

$$P_i(X) = \prod_{j\neq i} \frac{X - x_j I_N}{x_i - x_j}, \qquad y_i = \operatorname{Tr}(P_i Y), \qquad \widehat{J}(f) = \sum_i f(x_i, y_i).$$

The sum is conjugation invariant and invariant under permutations, so it is a regular function on the spectral chart.

Proposition 12.10. *The map $\widehat{J}$ preserves the plane Poisson bracket and sends 1 to N. For the product Darboux Moyal structure of parameter h on that chart it also satisfies*

$$h^{-1}[\widehat{J}(f), \widehat{J}(g)]_{\star_{\mathrm{chart}}} = \widehat{J}(B_h(f, g)).$$

At $N = 2$ and $\zeta \neq 0$, its image of p^2 has no regular extension to the entire stable quotient.

Proof. The Darboux coordinates of (12.13) give $\{x_i, y_j\} = \delta_{ij}$. Different pairs Poisson-commute, so the bracket of the two sums is $\sum_i \{f, g\}(x_i, y_i)$. For the chart Moyal product use the same exponential tensorwise on the discriminant-localized coordinate ring, completed in h. Each coefficient is a finite-order differential operator and respects localization and permutations. Different pairs star-commute, proving the quantum formula.

The change from matrix traces is already visible in degree two:

$$\widehat{J}(p^2) = \operatorname{Tr} Y^2 + 2\zeta^2 \sum_{i<j} (x_i - x_j)^{-2}.$$

To test regular extension, consider the family

$$X(a) = \begin{pmatrix} 0 & 1 \\ a & 0 \end{pmatrix}, \quad Y(a) = \begin{pmatrix} 0 & 0 \\ \zeta & 0 \end{pmatrix}, \quad I = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad J = (0, 2\zeta).$$

It satisfies (12.12) for every a. The columns I, XI have determinant -1, so the family remains in one stable gauge chart even at $a = 0$.

For $a = b^2 \neq 0$, the eigenvalues are $\pm b$, with projectors $(I_N \pm X/b)/2$. Since $\operatorname{Tr} Y = 0$ and $\operatorname{Tr}(XY) = \zeta$, the two measurements are $\pm\zeta/(2b)$. Therefore $\widehat{J}(p^2) = \zeta^2/(2a)$, whereas $\operatorname{Tr} Y^2 = 0$. A regular function on the full stable quotient would pull back regularly at $a = 0$. This pole excludes that extension. □

The particle sum is a Lie map. Products have terms on distinct particles, so it is not an associative map from the one-particle algebra. Its chart quantization has not been identified with a global quantization of $\mathcal{X}_{N,\zeta}$.

12.13 Regular maps on the cyclic collision chart

The local classification works over any field K of characteristic zero. Fix $\zeta \in K$. Functions have cohomological degree zero. The bracket on matrix entries is $\{X_{ij}, Y_{kl}\} = \delta_{il}\delta_{jk}$ and the framing bracket is $\{I_i, J_j\} = \delta_{ij}$. The quotient is the stable rank-two quotient of $[X, Y] + IJ = \zeta \mathbf{1}$. Stability means that words in X, Y applied to I span K^2.

Let U_+ be the open subset on which I, XI are independent. Put $s = \operatorname{Tr} X/2$. The basis $((X - s)I, I)$ gives the unique representative

$$X = s\mathbf{1} + \begin{pmatrix} 0 & 1 \\ a & 0 \end{pmatrix}, \qquad Y = t\mathbf{1} + \begin{pmatrix} 0 & k \\ \zeta + ak & 0 \end{pmatrix}, \qquad I = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad J = (0, 2\zeta). \tag{12.19}$$

Consequently $U_+ = \operatorname{Spec} K[s,t,a,k]$.

To check this assertion, Cayley–Hamilton first gives the displayed form of X. Write the traceless part of Y as $\left(\begin{smallmatrix} r & k \\ \ell & -r \end{smallmatrix}\right)$. The top-right moment equation gives $r = 0$, the top-left gives $\ell = \zeta + ak$, and the bottom row gives the displayed J. Conversely every point in (12.19) satisfies the moment equation and $\det(I, XI) = -1$.

The symplectic form pulls back to

$$\operatorname{Tr}(dY \wedge dX) + dJ \wedge dI = 2\, dt \wedge ds + dk \wedge da.$$

Thus $\{s,t\} = 1/2$, $\{a,k\} = 1$, and all other coordinate brackets vanish.

Define centered invariants

$$b = \tfrac{1}{2}\operatorname{Tr}((X-s)(Y-t)) = \zeta/2 + ak, \qquad c = \tfrac{1}{2}\operatorname{Tr}(Y-t)^2 = \zeta k + ak^2.$$

Then

$$ac - b^2 = -\zeta^2/4, \qquad \{a,b\} = a, \quad \{b,c\} = c, \quad \{a,c\} = 2b. \tag{12.20}$$

The collision of Proposition 12.10 is the line $s = t = k = 0$. To recover its measured entries explicitly, adjoin r with $r^2 = a$ and restrict to $r \neq 0$. Then

$$X = \begin{pmatrix} 0 & 1 \\ a & 0 \end{pmatrix}, \qquad Y = \begin{pmatrix} 0 & 0 \\ \zeta & 0 \end{pmatrix}, \qquad P_\pm = \tfrac{1}{2}(\mathbb{1} \pm X/r).$$

The spectral projectors obey

$$p_\pm = \operatorname{Tr}(P_\pm Y) = \pm\frac{\zeta}{2r}, \qquad p_+^2 + p_-^2 = \frac{\zeta^2}{2a}, \qquad \operatorname{Tr} Y^2 = 0.$$

Thus the required measurements have their pole at a stable point: I, XI remain independent when $a = 0$. The spectral square is a different charge from the regular matrix square.

The symmetrized traces already defined above have $J_{00} = 2$ in rank two. On the cyclic chart,

$$J_{10} = 2s, \quad J_{01} = 2t, \quad J_{20} = 2s^2 + 2a, \quad J_{11} = 2st + \zeta + 2ak, \quad J_{02} = 2t^2 + 2\zeta k + 2ak^2.$$

For a parameter $\tau \in K$, let $\mathfrak{g}_\tau$ be $K[q,p]$ with the bracket

$$\mathcal{B}_\tau(f,g) = \sum_{r \geq 0} \frac{\tau^r}{2^{2r}(2r+1)!}\, m\Big(\partial_q \otimes \partial_p - \partial_p \otimes \partial_q\Big)^{2r+1}(f \otimes g). \tag{12.21}$$

Here m multiplies the two tensor factors. The sum is finite. Jacobi follows by adjoining θ with $\tau = \theta^2$ and taking the divided commutator of the associative Moyal product. The polynomial identity descends over $\mathbb{Q}[\tau]$. Thus $\mathcal{B}_\tau = B_\theta$ in the earlier notation when $\tau = \theta^2$, and $\mathcal{B}_0$ is the Poisson bracket.

The target bracket in this section is always the ordinary Poisson bracket, including when $\tau \neq 0$. No target star product is being identified with it.

Consider the five-dimensional Lie subalgebra

$$\mathfrak{b} = \langle 1, q, p, q^2, qp \rangle \subset \mathfrak{g}_\tau.$$

Its bracket is independent of τ. Suppose that a K-linear Lie map $\Phi : \mathfrak{g}_\tau \to K[[s,t,a,k]]$ agrees with J on $\mathfrak{b}$. A germ of a regular algebraic map is a special case.

Set $F = \Phi(p^2)$. The source brackets with q, p, q^2, qp give

$$F_t = 4t, \qquad F_s = 0, \qquad F_k = 2\zeta + 4ak,$$
$$\{J_{11}, F\} = 2F.$$

The first three equations imply $F = J_{02} + H(a)$. The last is $aH'(a) + H(a) = 0$. In formal power series the coefficient of a^n is multiplied by $n+1$, so $H = 0$. On the punctured chart its only Laurent solution is $H = C/a$.

This proves that the first square charge has a unique regular solution, $F = J_{02}$. It also identifies the complete polar ambiguity of that equation. The spectral-projector square charge is $J_{02} + \zeta^2/(2a)$. Replacing it by J_{02} cancels its pole and solves the first compatibility equation.

The linear functional

$$K((a))/K[[a]] \longrightarrow K, \qquad [h] \longmapsto \operatorname{Res}_{a=0}(h\,da)$$

detects this ambiguity. The statement concerns the class of the function in the quotient. It is not a residue of the symplectic form.

12.13.1 *Rigidity relative to five fixed traces*

Theorem 12.11 (Local rigidity). *Every Lie map $\Phi : \mathfrak{g}_\tau \to K[[s,t,a,k]]$ agreeing with J on $\mathfrak{b}$ must satisfy $\Phi(q^m p^n) = J_{mn}$ for every $m, n \geq 0$. If $\zeta \neq 0$, there is no such map, for any τ.*

Proof. The preceding calculation forces $\Phi(p^2) = J_{02}$. Translation covariance gives

$$\Phi(q^3) = J_{30} + A(a,k), \qquad J_{30} = 2s^3 + 6sa.$$

Indeed the brackets with q and p say respectively that its t derivative is zero and its s derivative is $3J_{20}$.

The bracket with q^2 gives $A_k = 0$. The bracket with qp gives $2aA' = 3A$. For $A = \sum_{n\geq 0} A_n a^n$, the equation is $(2n-3)A_n = 0$ for every n. Every factor is nonzero in K, so $A = 0$. This proof permits arbitrary formal series and makes no degree or finite-weight assumption.

The three quadratic Hamiltonians act by the usual linear changes of the pair (X, Y). In particular

$$\{J_{02}/2, J_{mn}\} = -mJ_{m-1,n+1}.$$

The same formula holds in the source, since the source input $p^2/2$ has degree two. Hence knowing $\Phi(q^d) = J_{d0}$ determines every degree-d monomial by repeated bracketing with $p^2/2$.

The degree-three case is therefore fixed. Inductively, for $d \geq 4$,

$$\mathcal{B}_\tau(q^3, q^{d-2}p) = 3q^d, \qquad \{\operatorname{Tr} X^3, \operatorname{Tr}(X^{d-2}Y)\} = 3\operatorname{Tr} X^d.$$

The first formula has no higher Moyal term because its second input has p degree one. Its second input has total degree $d-1$, so the induction applies. This fixes q^d, and then the whole degree-d sector. Thus every component equals its symmetrized trace.

For the cubic pair one has

$$\{J_{30}, J_{03}\} - 9J_{22} = -3\zeta^2, \qquad \mathcal{B}_\tau(q^3, p^3) = 9q^2p^2 + \tfrac{3}{2}\tau.$$

Since $\Phi(1) = 2$, compatibility forces $\tau = -\zeta^2$. At that parameter the quartic residual is

$$\{J_{40}, J_{04}\} - 16J_{33} - 24\tau J_{11} = \frac{224}{5}\zeta^2 b = \frac{112}{5}\zeta^3 + \frac{224}{5}\zeta^2 ak. \tag{12.22}$$

It is nonzero when $\zeta \neq 0$.

For completeness, this identity can be derived entirely in this chart by polarizing

$$\operatorname{Tr}(uX + vY)^n = 2 \sum_{j=0}^{\lfloor n/2 \rfloor} \binom{n}{2j} (us + vt)^{n-2j} (au^2 + 2buv + cv^2)^j$$

and using (12.20). It is also the rank-two specialization of (12.17). The value at $s = t = a = k = 0$ is $112\zeta^3/5$, proving the contradiction. □

The cyclic-word derivation of Proposition 12.9 gives the same residual without choosing a chart.

The theorem excludes every correction that is regular as a formal germ and fixes the stated five charges. It consequently excludes every correction fixing the full degree-at-most-two scalar trace sector, or the sector $K[q] \oplus K[q]p$. The correction need not be a trace, have bounded polynomial degree, or be linear in framing invariants. Its order in ζ is unrestricted.

12.13.2 *Allowing the mixed quadratic to change*

The preceding obstruction is relative to a specified sector. A regular local map exists when the charge qp is allowed to change. In fact all such maps can be classified.

Theorem 12.12 (Four-charge classification). *Let $\Phi : \mathfrak{g}_\tau \to K[[s, t, a, k]]$ be a Lie map satisfying*

$$\Phi(1) = 2, \qquad \Phi(q) = 2s, \qquad \Phi(p) = 2t, \qquad \Phi(q^2) = 2s^2 + 2a.$$

Then $\tau = 0$. There is a unique $h(a) \in K[[a]]$ such that

$$\Phi(f) = f(s + \sqrt{a}, t + (k + h(a))\sqrt{a}) + f(s - \sqrt{a}, t - (k + h(a))\sqrt{a}). \tag{12.23}$$

The expression on the right belongs to $K[[s, t, a, k]]$: its odd powers of $\sqrt{a}$ cancel. Conversely, every $h \in K[[a]]$ defines such a strict Poisson Lie map. If the images are regular functions on U_+, then $h \in K[a]$.

Proof. Put $G = \Phi(qp)$. The brackets with q, p, q^2 give

$$G = 2st + 2ak + \alpha(a)$$

for some $\alpha \in K[[a]]$. Set $\alpha_0 = \alpha(0)$ and $h = (\alpha - \alpha_0)/(2a)$. This is a formal power series.

The variable $K_0 = k + h(a)$ satisfies $\{a, K_0\} = 1$. This identity is all that is needed below. When $h(0) \neq 0$, K_0 is not a parameter vanishing at the original formal base point.

The equations for $F = \Phi(p^2)$ now give

$$F = 2t^2 + 2ak^2 + 2\alpha(a)k + H(a), \qquad (aH)' = \alpha\alpha'.$$

Regularity fixes the integration constant, so

$$H = \frac{\alpha(a)^2 - \alpha_0^2}{2a}, \qquad F = 2t^2 + 2aK_0^2 + 2\alpha_0 K_0, \qquad G = 2st + 2aK_0 + \alpha_0.$$

In the variables (s, t, a, K_0) these are precisely the scalar quadratic traces for (12.19) with effective parameter α_0 in place of ζ. The rigidity proof of Theorem 12.11 uses only their displayed brackets and the power-series equation $2aA' = 3A$, so it applies here as well. It forces all images to be those effective symmetrized traces, and it forces $\tau = -\alpha_0^2$. The quartic residual is $224\alpha_0^2(\alpha_0/2 + aK_0)/5$, so $\alpha_0 = 0$ and $\tau = 0$.

When $\alpha_0 = 0$, the effective matrices commute. Their simultaneous eigenvalues on $a \neq 0$ are exactly the pairs in (12.23).

For existence, on the double cover $a = r^2 \neq 0$, put $x_\pm = s \pm r$ and $y_\pm = t \pm K_0 r$. Direct differentiation gives $\{x_i, y_j\} = \delta_{ij}$ and zero for the other pairs. The sum of evaluation maps is therefore a Poisson Lie map.

Its values and bracket defects are formal series containing only integral powers of a. The identity on the localization is an identity before localization, because multiplication by a is injective.

This proves existence and strictness. The argument already proves uniqueness. In the polynomial case G is polynomial, hence α is polynomial and so is h. □

For $h = 0$ the map has

$$\Phi(qp) = J_{11} - \zeta, \qquad \Phi(p^2) = J_{02} - 2\zeta k.$$

It is regular at the specified stable collision, retains the source constant, and specializes to the ordinary trace map at $\zeta = 0$. It has no higher components because it is strict. The mixed quadratic charge has changed. Its quantum lift will be constructed in Theorem 12.23, after the target product has been defined.

12.14 The second collision prevents global extension

Theorem 12.13 (Global four-charge obstruction)**.** *Suppose $\zeta \neq 0$. No K-linear Lie map from $\mathfrak{g}_\tau$ to the global regular functions on the stable rank-two quotient can satisfy $\Phi(1) = 2$, $\Phi(q) = \operatorname{Tr} X$, $\Phi(p) = \operatorname{Tr} Y$, and $\Phi(q^2) = \operatorname{Tr} X^2$.*

Proof. The restriction of a global map to U_+ is polynomial. By Theorem 12.12, it must have $\tau = 0$ and must be (12.23) for a polynomial $h(a)$.

Consider the independent regular family

$$X_-(a) = \begin{pmatrix} 0 & 1 \\ a & 0 \end{pmatrix}, \qquad Y_-(a) = \begin{pmatrix} 0 & 0 \\ -\zeta & 0 \end{pmatrix}, \qquad I_- = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad J_- = (2\zeta, 0).$$

It satisfies the same moment equation. It is stable for every a, since $\det(I_-, Y_- I_-) = -\zeta$. When $a \neq 0$ it also lies in U_+. There its coordinates are $s = t = 0$ and $k = -\zeta/a$, since $b = \operatorname{Tr}(X_- Y_-)/2 = -\zeta/2$.

Consequently the pullback of the square charge is

$$\Phi(p^2)|_{(X_-, Y_-, I_-, J_-)} = 2a\left(-\frac{\zeta}{a} + h(a)\right)^2 = \frac{2\zeta^2}{a} - 4\zeta h(a) + 2a h(a)^2.$$

Its residue at $a = 0$ is $2\zeta^2 \neq 0$. A global regular function must pull back to a regular function on this affine line, a contradiction. The residue is independent of every permitted h. □

The four fixed values are fewer than a full scalar quadratic sector. The proof permits arbitrary changes of qp, p^2, all higher charges, and all polynomial dependence on the framing variables after reduction. It does not exclude maps changing q^2. For example the global map

$$f \longmapsto 2f\left(\frac{\operatorname{Tr} X}{2}, \frac{\operatorname{Tr} Y}{2}\right)$$

is a strict Poisson Lie map retaining 1, q, p. It changes q^2 and does not specialize to the full trace map at $\zeta = 0$. Thus neither the center nor the first moments alone give the obstruction.

For any proposed Hamiltonian L_∞ map into a dg Poisson model, passing to degree-zero cohomology gives a strict Lie map. Suppose this cohomology is independently identified, as a Lie algebra, with the stated regular functions. The obstruction then applies if the four cohomology classes remain fixed. This is conditional on that recipient comparison. No BRST-to-reduction identification is being inferred solely from a shared name.

12.14.1 *A relative complex for the global polar obstruction*

For the remaining global constructions take $K = \mathbb{C}$. The second family supplies a quotient-valued obstruction. Let $L = \mathbb{C}[a, a^{-1}]$, $L_{\mathrm{reg}} = \mathbb{C}[a]$, and use the two-term complex

$$\mathcal{P}^\bullet = (L_{\mathrm{reg}} \hookrightarrow L), \qquad |L_{\mathrm{reg}}| = 0, \quad |L| = 1.$$

For every admitted polynomial h, the pullback of $\Phi_h(p^2)$ has the same class

$$[2\zeta^2/a] \in H^1(\mathcal{P}^\bullet) = L/L_{\mathrm{reg}}.$$

It is closed because the complex has no degree-two term.

The well-defined functional $[F] \mapsto \mathrm{Res}_{a=0}(F\,da)$ sends it to $2\zeta^2$. Changing h adds the regular polynomial $-4\zeta h + 2ah^2$ and hence does not alter the class. The formal version uses $\mathbb{C}[[a]] \hookrightarrow \mathbb{C}((a))$.

This complex controls extension of the classified square charge along the independent second family. It is a necessary relative quotient of the global gluing problem, not an identification of its full derived mapping space.

12.14.2 *The global quadratic correction problem*

Fix $\zeta \neq 0$ and put $\kappa = \zeta/2$. The following presentation is an isomorphism of algebraic Poisson varieties.

An independent presentation of the quotient

Write

$$s = \tfrac{1}{2}\operatorname{Tr} X, \quad t = \tfrac{1}{2}\operatorname{Tr} Y, \quad S = X - s\mathbf{I}_2, \quad T = Y - t\mathbf{I}_2,$$
$$a = \tfrac{1}{2}\operatorname{Tr} S^2, \quad b = \tfrac{1}{2}\operatorname{Tr}(ST), \quad c = \tfrac{1}{2}\operatorname{Tr} T^2.$$

For traceless two-by-two matrices,

$$S^2 = a\mathbf{I}_2, \quad T^2 = c\mathbf{I}_2, \quad ST + TS = 2b\mathbf{I}_2.$$

Taking the trace of the framed equation gives $JI = 2\zeta$. Consequently $(\zeta\mathbf{I}_2 - IJ)^2 = \zeta^2\mathbf{I}_2$. On the other hand $[S, T]^2 = 4(b^2 - ac)\mathbf{I}_2$, whence

$$ac - b^2 + \kappa^2 = 0.$$

Lemma 12.14 (The global rank-two carrier)**.** *The trace coordinates identify the quotient with*

$$\mathbb{A}^2_{s,t} \times Q_\kappa, \qquad Q_\kappa = \operatorname{Spec} R_\kappa, \qquad R_\kappa = \mathbb{C}[a, b, c]/(ac - b^2 + \kappa^2).$$

Its Poisson bracket is

$$\{s, t\} = \tfrac{1}{2}, \qquad \{a, b\} = a, \qquad \{a, c\} = 2b, \qquad \{b, c\} = c,$$

with all brackets between s, t and a, b, c equal to zero.

Proof. The columns I, SI or I, TI form a basis. Indeed, if both pairs were dependent, I would be a simultaneous eigenvector for S, T. Then $[S, T]I = 0$, whereas the framed equation gives $[S, T]I = -\zeta I \neq 0$.

On the first chart choose the basis so that $I = e_2$ and $SI = e_1$. The relation and tracelessness then force

$$S = \begin{pmatrix} 0 & 1 \\ a & 0 \end{pmatrix}, \qquad T = \begin{pmatrix} 0 & k \\ 2\kappa + ak & 0 \end{pmatrix}, \qquad J = (0, 4\kappa).$$

Thus

$$b = \kappa + ak, \qquad c = 2\kappa k + ak^2.$$

The chart is exactly $Q_\kappa \setminus D_-$, where $D_- = (a = 0, b = -\kappa)$. The inverse coordinate is $k = (b-\kappa)/a = c/(b+\kappa)$. These formulas are used on their respective open sets and agree on the overlap.

On the second chart take $I = e_2, TI = e_1$. Then

$$T = \begin{pmatrix} 0 & 1 \\ c & 0 \end{pmatrix}, \qquad S = \begin{pmatrix} 0 & \ell \\ c\ell - 2\kappa & 0 \end{pmatrix},$$

$$b = c\ell - \kappa, \qquad a = c\ell^2 - 2\kappa\ell.$$

This chart is the complement of $(c = 0, b = \kappa)$.

The two indicated complements cover Q_κ, since $\kappa \neq 0$. Their matrices, together with the change of basis on the overlap, establish the claimed isomorphism. The invariant canonical matrix bracket gives the displayed relations directly. □

Every regular quadratic correction is a shear

Let a putative Lie map satisfy the four fixed charges. Brackets with $2s = \Phi(q)$ and $2t = \Phi(p)$ imply that

$$\Phi(qp) = 2st + H(a,b,c), \qquad \Phi(p^2) = 2t^2 + G(a,b,c).$$

The brackets with $\Phi(q^2)/2 = s^2 + a$ give

$$\{a, H\} = 2a, \qquad \{a, G\} = 2H.$$

Lemma 12.15 (All global quadratic changes)**.** *There is a unique $u(a) \in \mathbb{C}[a]$ such that*

$$H = 2b + au(a), \qquad G = 2c + 2u(a)b + \tfrac{a}{2}u(a)^2.$$

Conversely these formulas satisfy every scalar quadratic Lie identity. They are the standard trace quadratics followed by the Poisson automorphism

$$\alpha_u(a) = a, \qquad \alpha_u(b) = b + \tfrac{a}{2}u(a), \qquad \alpha_u(c) = c + u(a)b + \tfrac{a}{4}u(a)^2,$$

which fixes s, t and has inverse α_{-u}.

Proof. Localizing at a gives $R_\kappa[a^{-1}] = \mathbb{C}[a, a^{-1}, b]$ with $c = (b^2 - \kappa^2)/a$ and $\{a, -\} = a\partial_b$. Its kernel is $\mathbb{C}[a, a^{-1}]$. Intersection with R_κ is $\mathbb{C}[a]$. To see the latter intersection, restrict a Laurent polynomial to the generic point of either divisor $D_\pm$ defined by $a = 0, b = \pm\kappa$. A negative power of a is a pole.

It follows that

$$H = 2b + h(a), \qquad G = 2c + \frac{2h(a)b}{a} + r(a),$$

where $h \in \mathbb{C}[a]$ and $r \in \mathbb{C}[a, a^{-1}]$.

The expression $2h(a)b/a$ has at worst a simple pole. Hence regularity of G forces r to have no pole of order greater than one. The residues of $2h(a)b/a + r$ at D_+ and D_- are

$$2\kappa h(0) + r_{-1}, \qquad -2\kappa h(0) + r_{-1}.$$

Both must vanish. Since $\kappa \neq 0$, this gives $h(0) = 0$ and $r_{-1} = 0$. Thus $h = au$ with $u \in \mathbb{C}[a]$ and r is polynomial.

Finally $\{H, G\} = 2G$ is exactly

$$(ar)' = \left(\frac{a^2u^2}{2}\right)'.$$

Polynomiality at $a = 0$ eliminates the integration constant, giving $r = au^2/2$. Substitution verifies preservation of the quadric equation and all three centered Poisson brackets under α_u. Derivatives of $u(a)$ cancel in the bracket of its last two images. Substitution also verifies $\alpha_{-u}\alpha_u = 1$. □

The two residues are independent conditions. Removing either divisor changes this classification: a constant $h(0)$ can then be cancelled on the remaining branch. Thus the argument allows arbitrary corrections and locates their global compatibility obstruction, rather than merely measuring a pole of one selected formula.

Quadratic normalization forces all polynomial traces

Compose a putative map with α_{-u}. It now agrees with the standard trace map on all polynomials of degree at most two. Let J_{mn} denote the symmetrized trace with m copies of X and n copies of Y. Here $J_{00} = 2$.

Lemma 12.16 (Rigidity after quadratic normalization)**.** *Suppose a Lie map to this global recipient agrees with the trace map in degrees at most two. Then $\Phi(q^m p^n) = J_{mn}$ for every $m, n \geq 0$.*

Proof. For a homogeneous cubic f, put $D(f) = \Phi(f) - J(f)$. Brackets with q, p show that $D(f)$ is annihilated by ∂_s, ∂_t, hence belongs to R_κ.

Brackets with the quadratic polynomials show that D intertwines their $\mathfrak{sl}_2$ actions. In particular, the source cubic weights under $\{qp, -\}$ are $-3, -1, 1, 3$. In R_κ, the action $\{2b, -\}$ gives weights $-2, 0, 2$ on a, b, c. Every monomial has an even weight, and the defining relation has weight zero. Thus R_κ has no odd weight eigenspace, so $D = 0$.

Proceed by induction on total degree $n \geq 4$. The Moyal bracket

$$\mathcal{B}_\tau(q^3, q^{n-2}p) = 3q^n$$

has no higher correction, since the second input has p-degree one. The inductive hypothesis and the canonical matrix bracket give

$$3\Phi(q^n) = \{\operatorname{Tr} X^3, \operatorname{Tr}(X^{n-2}Y)\} = 3 \operatorname{Tr} X^n.$$

Repeated bracketing with $p^2/2$ gives all homogeneous degree-n monomials. On matrices the same operation polarizes $\operatorname{Tr}(X - vY)^n$ and therefore gives the symmetrized traces with exactly the same binomial factors. This completes the induction. □

12.15 An obstruction invariant under every formal first-component change

The formal assertion uses a specified reduction parameter. Put

$$A = \mathbb{C}[[\zeta]], \qquad \widehat{R} = \varprojlim_m \mathbb{C}[s, t, a, b, c, \zeta]/(ac - b^2 + \zeta^2/4, \zeta^m),$$

with

$$\{s, t\} = \tfrac{1}{2}, \quad \{a, b\} = a, \quad \{a, c\} = 2b, \quad \{b, c\} = c.$$

The completion permits a polynomial coefficient at every order. The polynomial degrees need not be uniformly bounded.

The normal form $b^2 = ac + \zeta^2/4$ shows that this completion is ζ-torsion free. Its reduction is

$$R_0 = \mathbb{C}[s, t] \otimes Q_0, \qquad Q_0 = \mathbb{C}[a, b, c]/(ac - b^2).$$

The algebra $\widehat{R}$ is specified independently of any candidate Hamiltonian comparison.

Let $\tau \in \zeta A$ be the *squared* Moyal parameter and let

$$\widehat{\mathfrak{g}}_\tau = (\mathbb{C}[q, p][[\zeta]], \mathcal{B}_\tau)$$

have the coefficientwise bracket

$$\mathcal{B}_\tau(f,g)=\sum_{r\ge 0}\frac{\tau^r}{2^{2r}(2r+1)!}m\Big(\partial_q\otimes\partial_p-\partial_p\otimes\partial_q\Big)^{2r+1}(f\otimes g).$$

Thus $\tau=\theta^2$ in the usual divided Moyal commutator of parameter θ. The parameter τ deforms the source bracket. The parameter ζ is the framed moment parameter. Choosing $\tau(\zeta)$ is explicit additional data.

Write $J_\zeta(q^m p^n)=J_{mn}$ for the trace polarization formulas on the rank-two carrier, extended continuously. At $\zeta=0$ the cover

$$a=u^2,\quad b=uv,\quad c=v^2,\qquad \{u,v\}=\tfrac{1}{2}$$

gives

$$J_0(f)=f(s+u,t+v)+f(s-u,t-v).$$

The two particle pairs have bracket one internally and zero with each other, so J_0 is a Lie map for the Poisson source.

Consider a continuous A-linear Lie map

$$\Phi:\widehat{\mathfrak{g}}_\tau\longrightarrow\widehat{R},\qquad \Phi(1)=2,\qquad \Phi \bmod \zeta=J_0.$$

The affine and quadratic charges of any such map can be normalized without altering its reduction.

12.15.1 *Formal normalization of the affine charges*

Lemma 12.17. *A continuous Poisson automorphism of $\widehat{R}$, equal to the identity modulo ζ, conjugates any such candidate to one with $\Phi(q)=2s$ and $\Phi(p)=2t$.*

Proof. Suppose the affine charges have already been normalized modulo ζ^n, and write their order-n errors as $u,v\in R_0$. The central identity $\{\Phi(q),\Phi(p)\}=2$ gives

$$\partial_s u+\partial_t v=0.$$

Polynomial integration over the coefficient algebra Q_0 gives $K\in R_0$ satisfying

$$\partial_t K=u,\qquad \partial_s K=-v.$$

For example, integrate u in t and correct the resulting s-derivative by a polynomial depending only on s.

Choose any lift $\widetilde{K}\in\widehat{R}$. Conjugation by $\exp(\zeta^n\{\widetilde{K},-\})$ changes the two errors by $-K_t$, K_s, hence removes them. Its exponential converges ζ-adically, since its derivation raises the ζ-order by n. The successive automorphisms converge coefficientwise. At every fixed coefficient only finitely many steps contribute, so no degree bound on K is required. □

12.15.2 *The quadratic normalization and its explicit contraction*

With the affine charges fixed, their brackets imply that

$$\Phi(q^2/2)-s^2,\qquad \Phi(qp)-2st,\qquad \Phi(p^2/2)-t^2$$

belong to the centered algebra $\widehat{Q}=\mathbb{C}[a,b,c][[\zeta]]/(ac-b^2+\zeta^2/4)$. They give an $\mathfrak{sl}_2$ realization reducing to $a,2b,c$.

Lemma 12.18 (The quadratic contraction)**.** *For the standard Hamiltonian action,*

$$H^1(\mathfrak{sl}_2, Q_0) = 0.$$

Every order-n perturbation of the standard centered quadratic realization is removed by a centered Hamiltonian conjugation equal to the identity modulo ζ^n.

Proof. Give a, b, c degree one. The degree-m component of Q_0 is the space of homogeneous polynomials of degree $2m$ in u, v under $a = u^2, b = uv, c = v^2$. It is the irreducible module V_{2m}. Hence $Q_0 = \bigoplus_{m\geq 0} V_{2m}$. The sum is algebraic. Every cocycle has image in a finite sum because it has three polynomial values and the action preserves degree.

Use the conventional basis $E = c, F = -a, H = 2b$, satisfying $[H, E] = 2E$, $[H, F] = -2F$, $[E, F] = H$. On a module write

$$C = \tfrac{1}{2}H^2 + EF + FE.$$

It acts on V_{2m} by $2m(m+1)$. For a cocycle $\eta : \mathfrak{sl}_2 \to V_{2m}$, put

$$S_\eta = \tfrac{1}{2}H \cdot \eta(H) + E \cdot \eta(F) + F \cdot \eta(E).$$

The invariance of $\frac{1}{2}H \otimes H + E \otimes F + F \otimes E$ and the cocycle identity give

$$x \cdot S_\eta = C\eta(x) \qquad (x \in \mathfrak{sl}_2).$$

Indeed, commuting x past the first action produces two terms that cancel by invariance, leaving the Casimir acting on $\eta(x)$. Thus for $m > 0$, $\eta(x) = x \cdot (S_\eta/[2m(m+1)])$. For $m = 0$ the module is trivial, and a cocycle vanishes because $\mathfrak{sl}_2 = [\mathfrak{sl}_2, \mathfrak{sl}_2]$. This proves the vanishing with an explicit polynomial primitive.

If a quadratic realization is standard modulo ζ^n, its order-n difference is a cocycle for this module, by expansion of its three Lie identities. Write it as $x \mapsto \{j(x), K\}$ using the preceding primitive. Conjugation by $\exp(\zeta^n\{\widetilde{K}, -\})$, for a centered lift $\widetilde{K}$, removes it and preserves s, t. Successive conjugations converge as before. □

Theorem 12.19 (Formal nonexistence with fixed reduction)**.** *There is no continuous A-linear Lie map*

$$\Phi : \widehat{\mathfrak{g}}_\tau \longrightarrow \widehat{R}$$

such that $\Phi(1) = 2$ and $\Phi \bmod \zeta = J_0$. Every higher first-component correction is admitted, including corrections to q, p, q^2, qp, p^2. The obstruction already occurs modulo ζ^3.

Proof. After these two normalizations, every affine and scalar quadratic charge is exactly its standard trace. The parity and pure-power induction from Lemma 12.16 work coefficientwise in $\widehat{Q}$: $\{2b, -\}$ still has only even weights, and the relation $ac - b^2 + \zeta^2/4$ has weight zero. Therefore the entire normalized candidate must equal J_ζ. The cubic identity forces $\tau = -\zeta^2$, and the quartic defect is then

$$\frac{224}{5}\zeta^2 b \neq 0 \quad \text{in } \widehat{R}.$$

This proves Theorem 12.19.

For the finite-order obstruction, work modulo ζ^3. Normalize only at orders one and two, and apply rigidity through degree six. The cubic relation then gives $\tau \equiv -\zeta^2 \pmod{\zeta^3}$, while the quartic residual remains $(224/5)\zeta^2 b \neq 0 \pmod{\zeta^3}$. Thus the contradiction requires no convergence argument. □

12.15.3 *The controlling complex and the detector*

There is a finite-order cohomological form of the obstruction. Let

$$\mathfrak{g}_0 = (\mathbb{C}[q,p],\{-,-\}), \qquad \mathfrak{h} = \mathbb{C}[q,p]_{\le 2},$$

and let R_0 be a $\mathfrak{g}_0$-module by $f \cdot r = \{J_0(f), r\}$. The normalized correction complex is the relative Chevalley– Eilenberg complex

$$C^\bullet(\mathfrak{g}_0, \mathfrak{h}; R_0).$$

For a one-cochain,

$$(d\eta)(f,g) = \{J_0(f), \eta(g)\} - \{J_0(g), \eta(f)\} - \eta(\{f,g\}).$$

The cubic identity after normalization modulo ζ^2 first forces $\tau \equiv 0 \pmod{\zeta^2}$: its trace defect is zero to order one whereas its source scalar correction is 3τ. It is therefore legitimate, for any remaining candidate, to set

$$\Omega_\zeta(f,g) = \{J_\zeta(f), J_\zeta(g)\} - J_\zeta(\mathcal{B}_\tau(f,g)), \qquad \omega_2 = \zeta^{-2}\Omega_\zeta \bmod \zeta.$$

The quotient is well-defined. The target relation starts in order two, and the cubic test forces the same lower bound on the source correction. Since J_0 is a Lie map, the curvature is divisible by ζ^2.

The trace map is exactly $\mathfrak{h}$-equivariant, so ω_2 is relative. The Jacobi identities in source and target give $d\omega_2 = 0$ by taking the first nonzero coefficient of the curvature identity.

The same argument below with zero curvature shows that every relative one-cocycle is zero. In particular the normalized order-one correction vanishes. The order-two equation is exactly $\omega_2 + d\eta_2 = 0$, with no quadratic contribution from an unretained order-one correction.

Suppose $\omega_2 = -d\eta$ with η relative. Its values on cubics are centered, by translation equivariance, and have odd $\{2b, -\}$ weights. Hence they vanish. For every $n \ge 4$,

$$\omega_2(q^3, q^{n-2}p) = 0.$$

Induction in the displayed formula for $d\eta$, followed by quadratic polarization, gives $\eta = 0$. Thus a relative primitive would force $\omega_2 = 0$.

If $\tau = \lambda\zeta^2 + O(\zeta^3)$, the cubic pair gives

$$\omega_2(q^3, p^3) = -3(1+\lambda).$$

If this scalar vanishes, then

$$\omega_2(q^4, p^4) = \frac{224}{5}b.$$

Evaluation of this normalized representative at the point $s = t = 0$, $a = b = c = 1$ of Spec R_0 detects the latter representative with value $224/5$. Its nontrivial cohomology class is established by the preceding exclusion of every relative primitive. Evaluation by itself is not a functional on arbitrary cohomology classes. This evaluates the residual Hamiltonian. It is not asserted to be a morphism of Poisson algebras. The affine polynomial contraction and the explicit quadratic Casimir contraction show why the nonzero relative class remains an obstruction after all admitted formal first-component changes.

12.15.4 Global functions on the completed stable framed family

The formal Hamiltonian recipient can be identified with the actual global functions on the completed stable rank-two framed family. This is an identification of Poisson algebras of functions. The underlying stable family is not identified with its affinization at the zero moment level.

Let $\mathscr{X} \to \mathbb{A}^1_\zeta$ be the stable quotient of

$$[X, Y] + IJ = \zeta \mathbf{1}_2$$

in rank two. Write $s = \operatorname{Tr} X/2$ and $t = \operatorname{Tr} Y/2$, and put $S = X - s\mathbf{1}_2$, $T = Y - t\mathbf{1}_2$. Stability makes at least one of (SI, I) and (TI, I) a basis. If both pairs were dependent, I would generate a common invariant line. This argument also applies at $\zeta = 0$.

The two basis choices give affine charts

$$U_+ = \operatorname{Spec} \mathbb{C}[\zeta, s, t, a, k], \qquad U_- = \operatorname{Spec} \mathbb{C}[\zeta, s, t, c, \ell].$$

In the first chart,

$$S = \begin{pmatrix} 0 & 1 \\ a & 0 \end{pmatrix}, \quad T = \begin{pmatrix} 0 & k \\ \zeta + ak & 0 \end{pmatrix}, \quad I = e_2, \quad J = (0, 2\zeta).$$

In the second,

$$T = \begin{pmatrix} 0 & 1 \\ c & 0 \end{pmatrix}, \quad S = \begin{pmatrix} 0 & \ell \\ c\ell - \zeta & 0 \end{pmatrix}, \quad I = e_2, \quad J = (0, 2\zeta).$$

On the overlap the change of basis is diagonal with entries $k, 1$. Hence

$$\ell = k^{-1}, \qquad c = ak^2 + \zeta k, \qquad a = c\ell^2 - \zeta \ell. \tag{12.24}$$

In particular the overlap is $k \neq 0$, equivalently $\ell \neq 0$.

The invariant trace

$$b = \tfrac{1}{2} \operatorname{Tr}(ST) = \zeta/2 + ak = -\zeta/2 + c\ell$$

is globally regular and satisfies $ac - b^2 + \zeta^2/4 = 0$.

For $m \geq 1$ put

$$\mathscr{X}_m = \mathscr{X} \times_{\mathbb{C}[\zeta]} \mathbb{C}[\zeta]/(\zeta^m), \qquad A_m = \mathbb{C}[s, t, \zeta]/(\zeta^m).$$

Theorem 12.20 (The geometric formal recipient)**.** *The trace functions induce a unital Poisson algebra isomorphism*

$$A_m[a, b, c]/(ac - b^2 + \zeta^2/4) \xrightarrow{\sim} \Gamma(\mathscr{X}_m, \mathcal{O}_{\mathscr{X}_m})$$

for every $m \geq 1$. Moreover

$$H^1(\mathscr{X}_m, \mathcal{O}_{\mathscr{X}_m}) = 0.$$

For the ζ-adic completion $\widehat{\mathscr{X}}$ along $\mathscr{X}_1$, this gives the continuous unital Poisson algebra isomorphism

$$\widehat{R} \xrightarrow{\sim} \Gamma(\widehat{\mathscr{X}}, \mathcal{O}_{\widehat{\mathscr{X}}}), \qquad \widehat{R} = \varprojlim_m \mathbb{C}[s, t, a, b, c, \zeta]/(ac - b^2 + \zeta^2/4, \zeta^m).$$

All these maps are compatible with reduction in ζ.

Proof. The affine cover has affine overlap. Its quasicoherent cohomology is therefore computed by the two-term Čech complex

$$C_m^0 = A_m[k,a] \oplus A_m[\ell,c] \longrightarrow C_m^1 = A_m[k,k^{-1},a],$$

where the second summand is embedded by $\ell = k^{-1}$, $c = k^2 a + \zeta k$. The differential is the difference of the two restrictions. No field hypothesis on A_m is used.

Filter each term by degree at most n in the affine fiber variable a, or c on the second chart. The affine transition (12.24) preserves this filtration. Call the resulting finite-degree complex $F_n C_m^\bullet$. Its degree-n quotient is

$$A_m[k]a^n \oplus A_m[k^{-1}]c^n \longrightarrow A_m[k,k^{-1}]a^n,$$

with $c^n = k^{2n}a^n$ in the quotient. This is the Čech complex of $\mathcal{O}_{\mathbb{P}^1_{A_m}}(2n)$, but its cohomology follows directly from Laurent coefficients:

$$A_m[k] + k^{2n}A_m[k^{-1}] = A_m[k,k^{-1}],$$

$$A_m[k] \cap k^{2n}A_m[k^{-1}] = \bigoplus_{j=0}^{2n} A_m k^j.$$

The first equality gives $H^1(\mathrm{gr}_n C_m) = 0$. The second gives a free basis $1, k, \dots, k^{2n}$ for its H^0 after factoring out a^n.

The filtrations are split on each chart and on the overlap, so their successive short exact sequences are short exact sequences of these two-term complexes. Starting with $n = 0$, the resulting elementary kernel–cokernel sequence gives

$$H^1(F_n C_m) = 0,$$

$$0 \longrightarrow H^0(F_{n-1}C_m) \longrightarrow H^0(F_n C_m) \longrightarrow \bigoplus_{j=0}^{2n} A_m k^j a^n \longrightarrow 0.$$

Every polynomial cochain has finite fiber degree. Taking the union over n, which preserves exact sequences of modules, proves $H^1(C_m) = 0$ and hence the asserted sheaf-cohomology vanishing. It also determines the associated graded algebra of $H^0(C_m)$.

Now filter

$$R_m = A_m[a,b,c]/(ac - b^2 + \zeta^2/4)$$

by total degree in a, b, c, with A_m in degree zero. The monic relation $b^2 = ac + \zeta^2/4$ gives the unique A_m-basis of monomials $a^i b^\epsilon c^j$, $\epsilon \in \{0,1\}$. It follows directly that

$$\mathrm{gr}\, R_m = A_m[a,b,c]/(ac - b^2).$$

The trace map $R_m \to H^0(C_m)$ respects the filtrations. On the first chart its leading terms are a, ak, ak^2. In degree n it therefore sends the normal monomial basis to the preceding Laurent-coefficient basis as follows:

$$a^{n-j}c^j \longmapsto a^n k^{2j} \quad (0 \le j \le n),$$

$$a^{n-1-j}bc^j \longmapsto a^n k^{2j+1} \quad (0 \le j < n).$$

Thus it is an isomorphism on every associated graded piece.

This proves surjectivity by induction: match a section in its leading term, then match the remaining lower-degree section. A kernel element would likewise have zero leading term and descend to zero by the same induction. This proves the asserted global-sections isomorphism over each nonreduced ring A_m.

The Poisson structure is the one induced by the canonical framed reduction. On U_+ its brackets are $\{s,t\} = 1/2$, $\{a,k\} = 1$, with ζ central. Substitution of $b = \zeta/2 + ak$ and $c = ak^2 + \zeta k$ gives

$$\{a,b\} = a, \qquad \{a,c\} = 2b, \qquad \{b,c\} = c,$$

and all mixed brackets with s, t vanish. The same identities therefore hold globally and on every infinitesimal fiber. The algebra isomorphism preserves them on its generators, so it is a Poisson isomorphism.

A global section of the completed structure sheaf is a pair of completed chart sections agreeing on the completed overlap. This equalizer commutes with inverse limits. Equivalently, its components are precisely compatible global sections on all $\mathscr{X}_m$. Hence

$$\Gamma(\widehat{\mathscr{X}}, \mathcal{O}_{\widehat{\mathscr{X}}}) = \varprojlim_m \Gamma(\mathscr{X}_m, \mathcal{O}_{\mathscr{X}_m}) = \varprojlim_m R_m.$$

All brackets are continuous for this topology and the finite-stage maps preserve them, proving the completed Poisson assertion. This uses the displayed affine charts and equalizers directly. No theorem on formal functions is needed. □

The same calculation with coefficient ring $\mathbb{C}[s, t, \zeta]$, without imposing $\zeta^m = 0$, also computes the ordinary global function ring of $\mathscr{X}$. The special fiber $\mathscr{X}_1$ contains the exceptional projective line $a = c = 0$ in these charts. Its trace coordinates have $a = b = c = 0$. Thus the theorem does not identify the stable family or its probe categories with the affine scheme of its global functions.

Corollary 12.21 (The completed geometric target)**.** *For every permitted τ, no continuous Lie map*

$$\widehat{\mathfrak{g}}_\tau \longrightarrow \Gamma(\widehat{\mathscr{X}}, \mathcal{O}_{\widehat{\mathscr{X}}})$$

sends 1 to 2 and reduces to J_0 modulo ζ. For this specific geometric recipient, no additional comparison of global Poisson function algebras is missing.

Proof. Compose a putative map with the inverse of the continuous Poisson isomorphism in Theorem 12.20. It gives exactly the map excluded by Theorem 12.19. □

12.15.5 The meaning of the formal endpoint

The classical reduction in this theorem is specifically reduction modulo the framed parameter ζ. It must not be substituted silently for reduction in an independent matrix quantization parameter $\hbar$. A quantum or dg comparison is excluded by this theorem only when its relevant reduction induces the displayed ζ-adic map on zeroth cohomology. An isolated nonzero-level comparison changing the four-charge sector, without this formal extension to $\zeta = 0$, remains outside the theorem.

12.16 A regular quantum map on the collision chart

The classical polynomial map on the chart with coordinates (s, t, a, k) has a regular quantum lift. The construction uses differential operators on even polynomials. It specifies the quantum recipient independently.

All elements in this section have cohomological degree zero and the differential is zero. Put $R = \mathbb{C}[\zeta, \hbar]$. The moment parameter ζ is a spectator. Define

$$D_R = R\langle s, a, t, k\rangle / \big([s, t] - \hbar/2, [a, k] - \hbar, [s, a], [s, k], [t, a], [t, k]\big).$$

Its normal order is $s^i a^j t^u k^v$. Thus D_R is free over R by the ordered monomials and has the faithful differential representation

$$s \mapsto s, \qquad a \mapsto a, \qquad t \mapsto -\frac{\hbar}{2}\partial_s, \qquad k \mapsto -\hbar\partial_a \quad \text{on } R[s, a].$$

Faithfulness follows from the independence of differential operators of distinct orders and the absence of $\hbar$-torsion.

Consequently $D_R/\hbar D_R$ is the independently specified polynomial Poisson algebra with $\{s,t\}=1/2$ and $\{a,k\}=1$.

For a variable τ, write the polynomial Moyal Lie bracket as

$$\mathcal{B}_\tau(f,g)=\sum_{j\geq 0}\frac{\tau^j}{2^{2j}(2j+1)!}m\big(\partial_q\otimes\partial_p-\partial_p\otimes\partial_q\big)^{2j+1}(f\otimes g). \tag{12.25}$$

The sum is finite on every pair of polynomials. If the ordinary Moyal product has parameter $\hbar$, its commutator divided by $\hbar$ is $\mathcal{B}_{\hbar^2}$. The constant polynomial 1 is retained as a central Lie element.

Introduce the two-particle Weyl algebra $W_R^{(2)}$ on (x_+,y_+), (x_-,y_-), with $[x_\alpha,y_\beta]=\hbar\delta_{\alpha\beta}$ and the other commutators zero. Put

$$s=\frac{x_++x_-}{2},\quad \beta=\frac{x_+-x_-}{2},\quad t=\frac{y_++y_-}{2},\quad \varpi=y_+-y_-.$$

Then $[s,t]=\hbar/2$, $[\beta,\varpi]=\hbar$, and the two pairs commute. The transposition of particles fixes s,t and sends (β,ϖ) to $(-\beta,-\varpi)$. Its invariant algebra has ordered basis $s^it^j\beta^m\varpi^n$ with $m+n$ even.

Lemma 12.22 (Polynomial radial restriction)**.** *There is an R-linear unital algebra homomorphism*

$$\rho:(W_R^{(2)})^{\mathfrak{S}_2}\longrightarrow D_R$$

that fixes s,t and is given on the other ordered factors by

$$\rho(\beta^m\varpi^n)=\sum_{j=0}^{\lfloor n/2\rfloor}\frac{n!2^{n-2j}}{(n-2j)!j!}(-\hbar)^j a^{(m+n)/2-j}k^{n-j},\qquad m+n\ \text{even}. \tag{12.26}$$

Every exponent of a and $\hbar$ on the right is nonnegative. In particular,

$$\rho(\beta^2)=a,\qquad \rho(\beta\varpi)=2ak,\qquad \rho(\varpi^2)=4ak^2-2\hbar k.$$

Proof. Represent $W_R^{(2)}$ on $R[s,\beta]$ by $t=-(\hbar/2)\partial_s$ and $\varpi=-\hbar\partial_\beta$. Its invariant operators preserve the even subspace $R[s,\beta^2]$, identified with $R[s,a]$ by $a=\beta^2$.

Repeated differentiation gives, for every polynomial F,

$$\partial_\beta^nF(\beta^2)=\sum_{j=0}^{\lfloor n/2\rfloor}\frac{n!2^{n-2j}}{(n-2j)!j!}\beta^{n-2j}\partial_a^{n-j}F(a).$$

This identity follows, for example, by extracting the coefficient of u^n in $F((\beta+u)^2)=F(\beta^2+2\beta u+u^2)$.

Multiplication by $\beta^m(-\hbar)^n$ and substitution $k=-\hbar\partial_a$ give (12.26). Since $m+n$ is even and $j\leq n/2$, one has $(m+n)/2-j\geq 0$. Also $j\geq 0$.

Restriction to a preserved subspace respects composition. The formula proves that every restricted operator belongs to D_R, so restriction is the asserted algebra homomorphism. Faithfulness of the differential representation of D_R turns equality of its operators into equality in D_R. No inverse of a, ζ or $\hbar$ enters the construction. □

Let $\mathrm{Op}_\hbar$ denote Weyl symmetrization on one particle. The sum

$$\Delta(f)=\mathrm{Op}_\hbar f(x_+,y_+)+\mathrm{Op}_\hbar f(x_-,y_-)$$

belongs to $(W_R^{(2)})^{\mathfrak{S}_2}$. Set $\mathcal{J}_\hbar(f)=\rho\Delta(f)$.

Theorem 12.23 (Regular local quantum Hamiltonians)**.** *The map*

$$\mathcal{J}_\hbar : (R[q,p], \mathcal{B}_{\hbar^2}) \longrightarrow (D_R, \hbar^{-1}[-,-])$$

is a strict Lie algebra homomorphism. Division on the right denotes the unique quotient of an already $\hbar$-divisible commutator. The map retains the constant central generator with $\mathcal{J}_\hbar(1) = 2$ and has the following ordered formulas:

$$\begin{aligned} \mathcal{J}_\hbar(q) &= 2s, & \mathcal{J}_\hbar(p) &= 2t, \\ \mathcal{J}_\hbar(q^2) &= 2s^2 + 2a, & \mathcal{J}_\hbar(qp) &= 2st + 2ak - \hbar, \\ \mathcal{J}_\hbar(p^2) &= 2t^2 + 2ak^2 - \hbar k. \end{aligned}$$

Its reduction modulo $\hbar$ is the polynomial Lie map

$$\mathcal{J}_0(f) = f(s + \sqrt{a}, t + k\sqrt{a}) + f(s - \sqrt{a}, t - k\sqrt{a}). \tag{12.27}$$

The right side is interpreted as its even polynomial expansion in $\sqrt{a}$. In particular it is regular at $a = 0$. The construction extends continuously to the $\hbar$-adic and $(\zeta, \hbar)$-adic completions, allowing polynomial support to grow with parameter order.

Proof. Weyl symmetrization on one particle satisfies

$$[\mathrm{Op}_\hbar f, \mathrm{Op}_\hbar g] = \hbar \,\mathrm{Op}_\hbar \mathcal{B}_{\hbar^2}(f, g).$$

This follows by expanding the exponential definition of the Moyal product. The odd powers give (12.25). Operators from distinct particles commute. Adding their identities proves $[\Delta(f), \Delta(g)] = \hbar\Delta(\mathcal{B}_{\hbar^2}(f, g))$. Applying the algebra homomorphism ρ proves the asserted Lie identity.

The affine and quadratic formulas follow from

$$x_+^2 + x_-^2 = 2s^2 + 2\beta^2, \qquad y_+^2 + y_-^2 = 2t^2 + \varpi^2/2,$$

and

$$\frac{1}{2}\sum_{\alpha=\pm}(x_\alpha y_\alpha + y_\alpha x_\alpha) = 2st + \beta\varpi - \hbar.$$

Reduction of (12.26) modulo $\hbar$ keeps only $j = 0$, namely $\beta^m \varpi^n \mapsto 2^n a^{(m+n)/2} k^n$. This is precisely substitution $\beta = \sqrt{a}$, $\varpi = 2k\sqrt{a}$ on invariant polynomials and proves (12.27).

For each polynomial input every displayed sum is finite, and every term has nonnegative parameter order. At fixed parameter precision only finitely many input coefficients contribute. Thus the maps are continuous on the stated completions, where the algebraic identities hold coefficientwise. □

The theorem constructs Hamiltonian functions with their quantum Lie bracket. The map ρ preserves associative products. The map $\mathcal{J}_\hbar$ is a Lie map, since the product of two particle sums also contains terms from different particles. As a map into a differential graded Lie algebra concentrated in degree zero, it has all higher Taylor components equal to zero.

The polynomial quantum recipient D_R has been constructed independently. Section 12.16.1 constructs its operation-preserving identification with a specified normal-ordered quantum Hamiltonian reduction on the cyclic matrix chart, including its moment character. The classical coordinates alone would not identify the quantum products. The differential-operator coordinate map supplies that step.

12.16.1 *Quantum reduction on the cyclic matrix chart*

The polynomial differential-operator recipient of Section 12.16 is realized by a specified local quantum Hamiltonian reduction. The coordinate change is regular at the stable collision and fixes the moment character.

Let

$$Q = \{(X, I) \in \mathrm{Mat}_2 \times \mathbb{C}^2 : \det((X - sI_2)I, I) \neq 0\}, \qquad s = \tfrac{1}{2}\operatorname{Tr} X, \qquad a = s^2 - \det X.$$

Here I is a column vector, and I_2 denotes the identity matrix. Define $g = ((X - sI_2)I, I)$ and $C(s, a) = \left(\begin{smallmatrix} s & 1 \\ a & s \end{smallmatrix}\right)$. Cayley–Hamilton gives a GL_2-equivariant isomorphism

$$Q \simeq \mathrm{GL}_2 \times \mathbb{A}^2_{s,a}, \qquad (g, s, a) \longmapsto (gC(s, a)g^{-1}, ge_2). \tag{12.28}$$

Indeed $(X - sI_2)^2 = aI_2$ proves both the displayed formula for X and the formula for its inverse. The group acts by left multiplication on g and fixes s, a.

Let $R = \mathbb{C}[\zeta, \hbar]$. For a smooth affine $\mathbb{C}$-variety Z, write $\mathscr{D}_{\hbar,R}(Z) = R \otimes_{\mathbb{C}[\hbar]} \mathrm{Rees}_\hbar \mathscr{D}(Z)$. In particular $\mathscr{D}_{\hbar,R}(Q)$ is generated by regular functions and operators $\hbar v$ for algebraic vector fields v, with $[\hbar v, f] = \hbar v(f)$. In the original cotangent coordinates use

$$Y_{ij} = -\hbar\partial_{X_{ji}}, \qquad J_i = -\hbar\partial_{I_i}.$$

Thus $[X_{ij}, Y_{kl}] = \hbar\delta_{il}\delta_{jk}$ and $[I_i, J_j] = \hbar\delta_{ij}$.

For $\xi \in \mathfrak{gl}_2$, set

$$\rho_\xi(X) = [X, \xi], \qquad \rho_\xi(I) = -\xi I, \qquad \widehat{\mu}_\xi = \hbar\rho_\xi, \qquad \chi_\xi = \zeta \operatorname{Tr}\xi.$$

This specifies the quantum ordering. It has classical moment $\operatorname{Tr}\xi([X, Y] + IJ)$ and obeys

$$[\widehat{\mu}_\xi, \widehat{\mu}_\eta] = \hbar\widehat{\mu}_{[\xi,\eta]}, \qquad \chi_{[\xi,\eta]} = 0.$$

Equivalently, with matrix products taken in the displayed Weyl algebra and the framing product ordered I before J,

$$\widehat{\mu}_\xi = \operatorname{Tr}\xi\big(XY - YX - 2\hbar I_2 + IJ\big).$$

The Rees operator is kept as an element of the algebra. It is not replaced by an operator divided by $\hbar$.

Write $\mathcal{I}$ for the left ideal generated by $\widehat{\mu}_\xi - \chi_\xi$, and define the local quantum reduction by

$$\mathscr{A}_{\zeta,\hbar}(Q) = \big(\mathscr{D}_{\hbar,R}(Q)/\mathcal{I}\big)^{\mathrm{GL}_2}.$$

The product is the product of invariant classes represented in the right normalizer of the left ideal.

Theorem 12.24 (An integral local quantum reduction)**.** *Horizontal differential operators in* (12.28) *induce a unital R-algebra isomorphism*

$$D_R \xrightarrow{\sim} \mathscr{A}_{\zeta,\hbar}(Q), \qquad t = -\frac{\hbar}{2}\partial_s, \quad k = -\hbar\partial_a. \tag{12.29}$$

It is defined without inverting a, $\hbar$ or ζ and preserves the character $\chi_\xi = \zeta \operatorname{Tr}\xi$.

Its reduction modulo $\hbar$ is the canonical Poisson coordinate map on the stable X-cyclic chart of the framed quotient. On the classical reduction, the gauge section is

$$X = \begin{pmatrix} s & 1 \\ a & s \end{pmatrix}, \quad Y = \begin{pmatrix} t & k \\ ak + \zeta & t \end{pmatrix}, \quad I = e_2, \quad J = (0, 2\zeta). \tag{12.30}$$

Proof. The isomorphism of smooth affine bases (12.28) induces an exact isomorphism of Rees differential-operator algebras by the chain rule. Hence

$$\mathscr{D}_{\hbar,R}(Q) \simeq \mathscr{D}_{\hbar,R}(\mathrm{GL}_2) \otimes_R D_R.$$

The moment operators act entirely on the first factor. The vector fields $\rho_\xi(g) = -\xi g$ give a global frame of its tangent bundle. Ordered differential operators give the PBW decomposition

$$\mathscr{D}_{\hbar,R}(\mathrm{GL}_2) \simeq \mathcal{O}_R(\mathrm{GL}_2) \otimes_R U_{\hbar,R}(\mathfrak{gl}_2)$$

as an R-module, with coefficient functions on the left and moment operators on the right. Here $\mathcal{O}_R(\mathrm{GL}_2) = R \otimes_{\mathbb{C}} \mathcal{O}(\mathrm{GL}_2)$ and $U_{\hbar,R}(\mathfrak{gl}_2)$ is the R-algebra with relations $uv - vu = \hbar[u,v]_{\mathfrak{gl}_2}$.

Sending u_ξ to χ_ξ defines an algebra character of $U_{\hbar,R}(\mathfrak{gl}_2)$, since the trace of a commutator is zero. Its kernel is generated by the $u_\xi - \chi_\xi$. The PBW decomposition therefore identifies the quotient by the left moment ideal, as a module, with

$$\mathscr{D}_{\hbar,R}(Q)/\mathcal{I} \simeq \mathcal{O}_R(\mathrm{GL}_2) \otimes_R D_R.$$

This decomposition is equivariant for left translation: the moment character is invariant. A left-invariant regular function on GL_2 is constant, as follows by evaluating the equality $f(hg) = f(g)$ at $g = 1$. Thus every invariant class has a unique horizontal representative in D_R.

Horizontal operators commute with every moment operator. They consequently normalize the left ideal and their ordinary composition gives the reduction product. The preceding uniqueness proves that this algebra map is bijective. All these arguments are over R.

For the classical statement, the canonical one-form is $\lambda = \mathrm{Tr}(Y\,dX) + J\,dI$. Put $\widetilde{Y} = g^{-1}Yg$, $\widetilde{J} = Jg$ and $\vartheta = g^{-1}dg$. Direct substitution into (12.28) gives

$$\lambda = \mathrm{Tr}(\widetilde{Y}\,dC) + \mathrm{Tr}\big([C,\widetilde{Y}] + e_2\widetilde{J}\big)\vartheta.$$

On the moment level the second term is $\zeta\,\mathrm{Tr}\,\vartheta$. Its exterior derivative vanishes. In the gauge $g = 1$, the off-diagonal moment equations force equal diagonal entries of Y. The diagonal equations give $Y_{21} = ak + \zeta$ and $J = (0, 2\zeta)$. Thus (12.30) holds and $\lambda = 2t\,ds + k\,da$ on the section. The resulting brackets are $\{s,t\} = 1/2$, $\{a,k\} = 1$, exactly the semiclassical relations of D_R. □

The theorem fixes normal ordering through $\widehat{\mu}_\xi = \hbar\rho_\xi$. For Weyl ordering of the entire cotangent quadratic moment, the divergence of ρ_ξ on $\mathrm{Mat}_2 \times \mathbb{C}^2$ is $-\mathrm{Tr}\,\xi$. Consequently

$$\widehat{\mu}_\xi^{\mathrm{Weyl}} = \widehat{\mu}_\xi - \frac{\hbar}{2}\,\mathrm{Tr}\,\xi.$$

Weyl-ordered reduction at a value ζ_{W} therefore corresponds to the displayed normal convention at $\zeta = \zeta_{\mathrm{W}} + \hbar/2$. This is an explicit change of the quantum moment character, not an equality of the two conventions with their numerical parameters unchanged.

Corollary 12.25 (A local quantum BFV arrow)**.** *The horizontal inclusion of $\mathcal{J}_\hbar$ is a strict Hamiltonian Lie map from $(R[q,p], \mathcal{B}_{\hbar^2})$ to the quantum BFV algebra of the cotangent problem on Q. It has central value 2 and all higher Taylor components zero. Its image in degree-zero reduction is the map of Theorem 12.23 under (12.29).*

Proof. Choose a basis of $\mathfrak{gl}_2$ with structure constants $f_{ab}^{\ c}$, and add ghosts c^a of degree 1 and antighosts β_a of degree -1, with

$$c^a c^b + c^b c^a = 0, \qquad \beta_a\beta_b + \beta_b\beta_a = 0, \qquad \beta_a c^b + c^b\beta_a = \hbar\delta_a^b.$$

All ghosts and antighosts commute with the even differential operator algebra. The BFV carrier is the resulting graded tensor product with $\mathscr{D}_{\hbar,R}(Q)$.

The charge and differential are

$$\Theta = c^a(\widehat{\mu}_a - \chi_a) - \frac{1}{2} f_{ab}^{\ c} c^a c^b \beta_c, \qquad Q_\hbar = \hbar^{-1}[\Theta, -]_s.$$

The moment representation identity and Jacobi imply $\Theta^2 = 0$. Concretely, in the faithful differential representation with $\beta_a = \hbar \partial_{c^a}$, the charge is $\hbar$ times the Chevalley differential for ρ, minus multiplication by the degree-one character $c^a \chi_a$. That character is closed and its square is zero. Hence the charge squares to zero as an operator and, by PBW faithfulness, in the algebra.

The normalized commutator is integral because the algebra is supercommutative modulo $\hbar$.

Every horizontal element of D_R commutes with the moment operators and with the ghosts. It is therefore $Q_\hbar$-closed. Horizontal inclusion is a strict dg algebra map. Composing with the strict Lie map $\mathcal{J}_\hbar$ gives the assertion. □

No quasi-isomorphism from D_R to the full absolute BFV complex is asserted. The corollary constructs the actual closed Hamiltonians and their quantum bracket. It is local on the open cyclic chart Q. Extension to the entire stable framed quotient still requires a global comparison.

12.17 Descent of a first-component obstruction

Let $\mathfrak{g}$ be a Lie algebra in degree zero and let $(B, Q, \{-,-\})$ be a differential graded Poisson algebra whose bracket has degree zero. Suppose $J_1 : \mathfrak{g} \to Z^0(B)$ is linear. Its binary defect is

$$D_{J_1}(f, g) = \{J_1(f), J_1(g)\} - J_1([f, g]).$$

The differential is a derivation, so $QD_{J_1} = 0$.

Proposition 12.26 (The degree-zero detector)**.** *If J_1 is the first component of a Hamiltonian L_∞ morphism, then $[D_{J_1}(f, g)] = 0$ in $H^0(B)$ for every pair f, g. This necessary condition is unchanged by exact modifications $J_1(f) \mapsto J_1(f) + QK(f)$ with $|K(f)| = -1$. A cochain map $e : B \to \mathbb{C}$ with $e(D_{J_1}(f, g)) \neq 0$ therefore excludes such an extension.*

Proof. Choose the Taylor convention $QJ_2(f, g) = J_1([f, g]) - \{J_1(f), J_1(g)\}$. It makes the defect exact. For $J_1' = J_1 + QK$, the derivation rule and $QJ_1 = 0$ give

$$\begin{aligned} D_{J_1'}(f, g) - D_{J_1}(f, g) = Q\big(&\{K(f), J_1(g)\} + \{J_1(f), K(g)\} \\ &+ \{K(f), QK(g)\} - K([f, g])\big). \end{aligned}$$

Hence the cohomology class is unchanged. A cochain map to a complex concentrated in degree zero annihilates boundaries, which proves the final assertion. □

For the framed matrix BFV algebra take even coordinates X, Y, I, J and ghost–antighost coordinates (c, Ψ) of degrees $(1, -1)$. Put $\mu = [X, Y] + IJ - \zeta I_N$ and use the canonical brackets $\{X_{ij}, Y_{kl}\} = \delta_{il}\delta_{jk}$, $\{I_i, J_j\} = \delta_{ij}$ and $\{c_{ij}, \Psi_{kl}\} = \delta_{il}\delta_{jk}$. The BFV differential has

$$QX = -[c, X], \quad QY = -[c, Y], \quad QI = -cI, \quad QJ = Jc, \quad Qc = -c^2, \quad Q\Psi = \mu - [c, \Psi].$$

At every constraint point, evaluation of the even coordinates there and of all ghosts and antighosts at zero is a cochain map. Indeed every displayed image evaluates to zero. Evaluation is used only as a map of complexes. It is not a Poisson map from a symplectic manifold to a point.

Nonexact modifications of J_1 are a different question. The proposition alone gives no invariance under them. Such invariance requires a relative deformation calculation or a normalization theorem covering the admitted modifications.

Proposition 12.27 (Formal quantum specialization). *Let $\widehat{B}_\hbar$ be a complete separated $\mathbb{C}[[\hbar]]$-linear differential graded associative algebra, with no $\hbar$-torsion, such that graded commutators belong to $\hbar\widehat{B}_\hbar$. Assume its normalized commutator and differential reduce to the specified classical dg Poisson algebra B_0. Let $\widehat{\mathfrak{g}}_\hbar$ be a specified continuous $\mathbb{C}[[\hbar]]$-linear Lie source whose bracket reduces to that of $\mathfrak{g}_0$. Every continuous $\mathbb{C}[[\hbar]]$-linear Hamiltonian L_∞ map from $\widehat{\mathfrak{g}}_\hbar$ into $(\widehat{B}_\hbar, Q_\hbar, \hbar^{-1}[-,-]_s)$ induces a Hamiltonian L_∞ map into B_0 after reduction modulo $\hbar$. In particular, an obstruction to every permitted classical first component excludes every quantum lift with that classical reduction.*

Proof. Each Taylor identity is an equality of continuous multilinear maps over $\mathbb{C}[[\hbar]]$. Reducing this equality modulo $\hbar$ gives the corresponding classical Taylor identity, with the same degrees and signs. Applied to the binary identity, this is the previous proposition. No point evaluation on a nonzero-parameter Weyl algebra is used. Such an evaluation would not respect its defining commutator relation. □

Corollary 12.28 (Descent to a specified recipient). *Let B be a specified dg Poisson recipient. Fix a Lie map $\rho : H^\circ(B) \longrightarrow O(X_{2,\zeta})$ taking the scalar unit to the scalar unit. At $\zeta \neq 0$, if the composite of a proposed first component with ρ has the four values in Theorem 12.13, no Hamiltonian L_∞ extension exists. Likewise a parameter-complete version whose composite lands in $\widehat{R}$ and reduces to J_0 is excluded by Theorem 12.19.*

Proof. The binary Taylor equation makes the induced map to $H^\circ(B)$ a strict Lie map. Composing with the specified Lie realization gives the forbidden map. No higher Taylor component changes this necessary equation. The same argument applies after reducing a quantum dg algebra modulo $\hbar$ as in the preceding proposition. This takes cohomology after reduction. It assumes no unproved interchange of cohomology with specialization. □

In the polynomial framed BFV model there is a concrete degree-zero realization to reduced functions. Restrict the ghost-free part of a closed degree-zero function to the constraint locus. The coefficient of a single ghost in its differential says that this restriction is gauge invariant: the possible antighost contributions are multiples of constraints. Boundaries restrict to zero for the same reason.

Moreover a degree-zero monomial involving ghosts contains at least one ghost and one antighost. Bracketing two such terms leaves a ghost factor, so its ghost-free restriction is zero. The induced bracket is therefore the ordinary reduced bracket of the ghost-free terms.

This constructs the stated Lie realization without claiming a quasi-isomorphism of the full BFV algebra, a formality theorem, or an identification of its full center. Its coefficientwise completion is admissible only with the stated regular reductions and continuous restriction maps.

12.18 The remaining global comparisons

For the unframed matrix BRST complex, extend the formal action across the discriminant. Retain stabilizer states and coherent comparisons with the prescribed trace charges. The framed problem must also account for the new local maps and their global obstructions.

The first regular correction is not the final comparison. On the X-cyclic chart, replacing the spectral square charge by J_{02} solves its first equation. Keeping the mixed quadratic then forces all traces and gives the quartic obstruction.

Changing the mixed quadratic instead produces the strict maps Φ_h, and radial restriction gives an integral quantum lift of Φ_0. Theorem 12.24 and Corollary 12.25 identify its local quantum recipient and construct its strict BFV inclusion. The second collision detects every Φ_h with the same polar coefficient $2\zeta^2$. Thus the missing global step cannot be achieved by a different choice within that classified family.

The framing-adic theorem has a wider correction class. Suppose the full trace reduction is fixed at $\zeta = 0$. The Hamiltonian contractions normalize every affine and quadratic correction. The surviving

relative class is nonzero, so changing those charges cannot repair the comparison. Its failure at order two excludes an all-order formal lift.

Theorem 12.20 identifies the formal function algebra with the global functions of the completed stable framed family at every coefficient stage and after completion. Thus this particular geometric recipient comparison is constructed. The theorem makes no assertion of numerical convergence and requires none.

Problem 12.29 (An isolated nonzero-level comparison)**.** Fix $\zeta \neq 0$ and an independently specified global quantum reduction $\mathcal{B}_{\zeta,\hbar}$, with its normalized commutator and its classical reduction. Allow a global Hamiltonian comparison to change q^2 and every other nonlinear charge, retaining $1 \mapsto 2$. Does one exist with the prescribed classical reduction in $\hbar$, without a formal extension to $\zeta = 0$? Specify the admitted first-component changes and the operations of the full source center that the comparison must preserve.

Theorems 12.13 and 12.19 settle respectively the four-charge subproblem and the framing-adic subproblem. They do not identify Problem 12.29 with either of them. The global center-of-mass map already shows why the central constant and the two linear moments alone are insufficient. A positive solution of that problem must supply a new regular nonlinear moment sector. A negative solution must give an obstruction invariant under those additional changes. A rank-two obstruction excludes a construction claimed uniformly for all ranks, but it does not alone prove an isolated fixed-rank obstruction for every $N > 2$.

Problem 12.30 (Global quantum transitions)**.** The local comparison with the specified normal-ordered Rees reduction has been constructed in Theorem 12.24. For a different independently prepared quantum or derived recipient, construct its operation-preserving comparison with this local model, retaining the explicitly stated ordering and moment-level conversion. Determine the transition maps between cyclic charts and their effect on an admitted nonlinear Hamiltonian sector. The radial map cannot extend globally with its four marked classical charges, by Theorem 12.13. A global comparison allowing different charges must supply new compatible transition data. The horizontal inclusion into the absolute BFV algebra is not asserted to be a quasi-isomorphism.

Problem 12.31 (The full center)**.** Fix an independently defined chiral or derived center containing the polynomial Hamiltonian sector. Construct its restriction to the stated Hamiltonian recipient, preserving all specified operations. Any such restriction satisfying the hypotheses of the obstruction theorems is excluded. If its polynomial restriction has different operations or a different classical reduction, state and solve that new comparison. Neither a scalar central generator nor a Hamiltonian Lie map identifies an E_2 cochain object with a geometric center.

Identifying any of these algebraic recipients with the mixed field theory requires comparison maps for its fields, cyclic pairing, support and boundary conditions. A Lie map of Hamiltonians alone does not supply that physical comparison.

Chapter 13

Integral cyclic comparison and Hamiltonian homotopies

The coefficient Weyl algebra has cyclic operations that ordinary traces do not determine. We construct a comparison from its Hochschild chains to differential forms which preserves the circle operator. Two independent parameters remain in the comparison. Their relation connects the vacuum to the volume form before either parameter is inverted.

Hamiltonian equivariance requires two explicit homotopies. After passing to continuous duals, the same comparison gives a criterion for a current to be strictly cyclic. Point currents then give nonzero scalar classes with separate primitives for the two parameters.

13.1 Two operators on relative chains

Let $A = \mathbb{C}[q, p]$ with the direct-limit topology of its finite-degree subspaces. The second coefficient domain will be $\mathcal{A} = \mathbb{C}[[q, p]]$ with its formal topology. In either case set $R_\epsilon = \mathbb{C}[[\epsilon]]$ and

$$A_\epsilon = (A[[\epsilon]], \star), \qquad a \star b = m \exp(\epsilon P/2)(a \otimes b), \qquad P = \partial_q \otimes \partial_p - \partial_p \otimes \partial_q.$$

Here m is ordinary multiplication and $|\epsilon| = 0$. This is a deformation of the coefficient algebra, with $q \star p - p \star q = \epsilon$. It is distinct from the matrix-entry quantization in Chapter 12.

Parameter series are completed coefficientwise. For the polynomial domain, every parameter coefficient is a polynomial, with no uniform degree bound. For the formal domain, tensor products are also completed in the separate coordinates of all slots.

At each parameter order the Moyal product and the circle comparison have finite differential order in their coefficient slots. They are continuous on both domains. Continuity of the chosen cone contractions, coefficientwise antiderivatives and flow operators is justified separately below. No finite-differential-order assertion is made for them.

Use normalized Hochschild chains relative to R_ϵ:

$$C_n = A_\epsilon \widehat{\otimes}_{R_\epsilon} (A_\epsilon / R_\epsilon 1)^{\widehat{\otimes} n}.$$

The chain $a_0[a_1|\cdots|a_n]$ has cohomological degree $-n$. A scalar in a bar slot is zero. The Hochschild boundary is

$$\begin{aligned} b(a_0[a_1|\cdots|a_n]) = {} & (a_0 \star a_1)[a_2|\cdots|a_n] \\ & + \sum_{i=1}^{n-1} (-1)^i a_0[a_1|\cdots|a_i \star a_{i+1}|\cdots|a_n] \\ & + (-1)^n (a_n \star a_0)[a_1|\cdots|a_{n-1}]. \end{aligned}$$

It vanishes on C_0.

The Connes operator is

$$B(a_0[a_1|\cdots|a_n]) = \sum_{i=0}^{n}(-1)^{ni}1[a_i|\cdots|a_n|a_0|\cdots|a_{i-1}].$$

Associativity cancels the pairs of terms in b^2. Cyclic rotation and normalization give $B^2 = bB + Bb = 0$. Thus $|b| = 1$ and $|B| = -1$ define a mixed complex. Adjoin u of degree two and write

$$D_C = b + uB \quad \text{on} \quad C_\bullet[[u]].$$

This is the complete negative-cyclic chain complex.

Place a differential n-form in degree $-n$. Put $\pi = \partial_q \wedge \partial_p$, use $\iota_\pi(dq \wedge dp) = 1$, and define $\delta = \iota_\pi d - d\iota_\pi$. In coordinates,

$$\delta(a\,dq + b\,dp) = b_q - a_p, \qquad \delta(c\,dq \wedge dp) = -dc, \qquad \delta(f) = 0.$$

Direct differentiation gives $d^2 = \delta^2 = d\delta + \delta d = 0$. The form complex is

$$\big(\Omega^\bullet_A[[\epsilon, u]], D_\Omega\big), \qquad D_\Omega = \epsilon\delta + ud.$$

Write $\Omega = dq \wedge dp$.

The smallest mixed relation already contains both parameters. With $\chi = 1[q|p] - 1[p|q]$, one has

$$\begin{aligned} D_C(q[p]) &= \epsilon 1 + u\chi, & b\chi &= B\chi = 0, \\ D_\Omega(q\,dp) &= \epsilon + u\Omega, & \delta\Omega &= d\Omega = 0. \end{aligned} \tag{13.1}$$

The Hochschild boundary of χ is a scalar in a bar slot and therefore vanishes. A comparison preserving the actual circle operator must carry the first relation to the second.

13.2 Integration over a circle

For $n \geq 1$, let

$$\Delta_n = \{0 < t_1 < \cdots < t_n < 1\}, \qquad t_0 = 0,$$

with orientation $dt_1 \wedge \cdots \wedge dt_n$. Let P_{ij} apply P to tensor slots i, j. Its derivatives act on form coefficients and leave their exterior factors fixed. Define

$$\mathcal{U}_n(a_0[a_1|\cdots|a_n]) = m\int_{\Delta_n} \exp\left(\epsilon \sum_{i<j}(\tfrac{1}{2} + t_i - t_j)P_{ij}\right) a_0\,da_1\cdots da_n\,dt_1\cdots dt_n. \tag{13.2}$$

In this formula m identifies the coordinate slots and multiplies their exterior factors. Set $\mathcal{U}_0$ equal to the identity. Exterior degree makes $\mathcal{U}_n = 0$ for $n > 2$. A scalar bar input has zero differential, so the formula descends to normalized chains.

For polynomial inputs the exponential terminates. The required moments are rational:

$$\int_{\Delta_n} t_1^{a_1}\cdots t_n^{a_n}\,dt = \prod_{j=1}^{n}(a_1 + \cdots + a_j + j)^{-1}.$$

Successive integration proves this formula. In particular,

$$\mathcal{U}_n \bmod \epsilon = \frac{1}{n!}a_0\,da_1 \wedge \cdots \wedge da_n.$$

The volume of the simplex supplies the normalized Hochschild–Kostant–Rosenberg map.

At one bar slot the operator has a simpler expression:

$$\mathcal{U}_1(a_0[a_1]) = mS(P)(a_0\,da_1), \qquad S(P) = \frac{2\sinh(\epsilon P/2)}{\epsilon P} = \sum_{j\geq 0} \frac{\epsilon^{2j} P^{2j}}{2^{2j}(2j+1)!}. \tag{13.3}$$

The quotient denotes its power series, whose constant coefficient is one. For example,

$$\mathcal{U}_1(q^3[p^3]) = (3q^3p^2 + \tfrac{3}{2}\epsilon^2 q)\,dp.$$

Applying $\epsilon\partial$ gives $9\epsilon q^2p^2 + 3\epsilon^3/2$, the Moyal commutator of q^3 and p^3. The classical term alone would miss the cubic correction.

The ordered-circle kernel is the one appearing in Feigin–Felder–Shoikhet, Section 2.3. Its compatibility with B is the identity of Willwacher, Theorem 3. We prove the flat formulas with the normalization above.

13.3 The mixed comparison

Theorem 13.1. *On both specified coefficient domains,*

$$\mathcal{U}b = \epsilon\partial\mathcal{U}, \qquad \mathcal{U}B = d\mathcal{U}.$$

Consequently $\mathcal{U}$ is a quasi-isomorphism

$$(C_\bullet[[u]], b + uB) \longrightarrow (\Omega^\bullet_A[[\epsilon, u]], \epsilon\partial + ud).$$

It sends χ to Ω and preserves (13.1) *before either parameter is inverted.*

Proof. At fixed parameter order all operators have constant coefficients and finite differential order. It therefore suffices to compare their symbols on $a_i = \exp(k_i \cdot (q,p))$. The coefficients in the momenta k_i recover the polynomial identities and their continuous formal extensions.

Set $K = \sum_i k_i$, $c_{ij} = \pi(k_i, k_j)$, $\Lambda_i = k_i \cdot (dq, dp)$, and

$$S(t) = \sum_{i<j} (\tfrac{1}{2} + t_i - t_j) c_{ij}.$$

The integrand of (13.2) is $e^{K\cdot(q,p)} e^{\epsilon S(t)} \Lambda_1 \wedge \cdots \wedge \Lambda_n$. The two elementary identities are

$$\partial_{t_i} S = \pi(k_i, K),$$

$$\delta(e^{K\cdot(q,p)}\Lambda_1\cdots\Lambda_n) = e^{K\cdot(q,p)} \sum_{i=1}^{n} (-1)^i \pi(k_i, K)\Lambda_1\cdots\widehat{\Lambda}_i\cdots\Lambda_n.$$

Integrating the first identity by parts gives the boundary faces of the ordered simplex.

At $t_1 = 0$, the factor $e^{\epsilon c_{01}/2}$ is the first Moyal multiplication. At $t_i = t_{i+1}$, the two omitted-form terms combine their differentials to $\Lambda_i + \Lambda_{i+1}$. At $t_n = 1$, the cyclic multiplication has factor $e^{-\epsilon c_{0n}/2}$ and sign $(-1)^n$. These are exactly the three types of terms in b.

The face sum also proves the equation for $n = 3$, where the interior three-form vanishes. For larger n both sides vanish by form degree. This proves $\mathcal{U}b = \epsilon\partial\mathcal{U}$.

For the second identity, move the marked origin around the circle. Its possible positions are partitioned by the $n+1$ intervals between labeled inputs. Taking each interval as the first gives the corresponding cyclic rotation in B, with sign $(-1)^{ni}$ from exterior reordering. The marked origin has total interval length one.

The resulting form factor is

$$(\Lambda_0 + \cdots + \Lambda_n) \wedge \Lambda_1 \wedge \cdots \wedge \Lambda_n.$$

This is the exterior derivative of the exponential coefficient. The jump of $\frac{1}{2} + t_i - t_j$ at the origin is precisely the cyclic reordering of the kernel. Thus $\mathcal{U}B = d\mathcal{U}$, with no additional factorial.

At $\epsilon = 0$, the diagonal Koszul resolution on $q_L - q_R$, $p_L - p_R$ computes Hochschild homology as differential forms. Its comparison with the normalized bar resolution gives the displayed Hochschild–Kostant–Rosenberg projection. The completed diagonal resolution gives the same assertion for formal functions and completed chains.

Give each coordinate and its differential weight one. At fixed weight the normalized undeformed comparison cone is a finite-dimensional acyclic complex: a positive bar slot has positive weight, which bounds its length. Choose a contraction at each weight. Their direct sum preserves polynomial inputs. Their product is continuous on formal inputs.

The deformed differential differs by terms in (ϵ, u). Perturbing this contraction gives a convergent geometric series in the parameter-adic topology, with continuous coefficients. It contracts the completed comparison cone and proves the quasi-isomorphism on both stated domains. Finally, the two linear bar inputs have no nonzero contraction, so direct antisymmetrization gives $\mathcal{U}\chi = \Omega$. □

13.4 Cohomology before inversion

The integral cohomology records information hidden by either parameter specialization. Write $R = \mathbb{C}[[\epsilon, u]]$. For this calculation group the completed even and odd form degrees. The original cohomological grading is retained by assigning $|u| = 2$ and $|\Omega| = -2$.

Proposition 13.2. *The form complex has*

$$H^{\text{even}} = \frac{R1 \oplus R\Omega}{R(\epsilon 1 + u\Omega)}, \qquad H^{\text{odd}} = dA, \qquad (\epsilon, u)H^{\text{odd}} = 0.$$

Both the vacuum and volume classes are nonzero.

Proof. An even element $f + g\Omega$ is closed exactly when $d(uf - \epsilon g) = 0$. Subtract the constant spatial coefficients of f and g, leaving f', g'. Then $uf' = \epsilon g'$. Reduction modulo ϵ shows that f' is divisible by ϵ. Since neither parameter is a zero divisor,

$$f' = \epsilon h, \qquad g' = uh.$$

Every coefficient of h is a curl $b_q - a_p$. For example, integrate it with respect to q and take $a = 0$. This integration is continuous on the formal domain. Thus $f' + g'\Omega = D_\Omega(a\,dq + b\,dp)$.

The remaining constant pair has the relation $\epsilon 1 + u\Omega$, obtained from $q\,dp$. Its quotient embeds in R by

$$[a + b\Omega] \longmapsto ua - \epsilon b.$$

The kernel before taking the quotient is exactly the submodule $R \cdot (\epsilon, u) \subset R^2$, by the same regular-sequence argument. The two classes map to u and $-\epsilon$, so neither vanishes.

An odd form α is closed exactly when its curl vanishes. Indeed the scalar and volume components of $D_\Omega \alpha$ are its curl multiplied by ϵ and u. Closed one-forms on the polynomial or formal plane are exact. The odd boundaries are $u\,df - \epsilon\,dg$, precisely $(\epsilon, u)dA[[\epsilon, u]]$. □

Specializing this answer need not compute the cohomology of the specialized complex. At $u = 0$, the differential is $\epsilon\partial$, and its degree-zero cohomology is $A[[\epsilon]]/\epsilon A[[\epsilon]]$. In particular q survives there. It does not extend to an even class with nonzero u, because $D_\Omega q = u\,dq$. The vacuum–volume relation alone does not account for this failure of ordinary base change.

13.5 Hamiltonian operations

Let $\mathfrak{h}_\epsilon = \mathcal{A}[[\epsilon]]/R_\epsilon 1$ with bracket

$$\{f, g\}_\epsilon = \epsilon^{-1}(f \star g - g \star f).$$

The numerator is divisible by ϵ, so this bracket is integral. Write $\{f, g\}_0 = f_q g_p - f_p g_q$ and $X_f = \{f, -\}_0$. The quantum derivation $\rho_f = \epsilon^{-1} \operatorname{ad}_\star f$ acts in every chain slot, giving an operator L_f^C.

There is also a degree-minus-one insertion

$$h_f(a_0[a_1|\cdots|a_n]) = -\sum_{i=0}^{n}(-1)^i a_0[a_1|\cdots|a_i|f|a_{i+1}|\cdots|a_n].$$

On forms put $m_f = -df \wedge (-)$, also of degree minus one. Constants vanish in both operations. The brackets below are graded commutators of operators.

Lemma 13.3. *The source and target operations satisfy*

$$[D_C, h_f] = \epsilon L_f^C, \qquad [B, h_f] = 0, \qquad [L_f^C, h_g] = h_{\{f,g\}_\epsilon},$$
$$[D_\Omega, m_f] = \epsilon L_{X_f}, \qquad [d, m_f] = 0, \qquad [L_{X_f}, m_g] = m_{\{f,g\}_0}.$$

Moreover $[h_f, h_g] = [m_f, m_g] = 0$.

Proof. In $bh_f + h_f b$, every merger disjoint from the inserted f occurs twice with opposite signs. The two neighboring mergers leave the commutator with f in each original slot. Their sum is ϵL_f^C. In $Bh_f + h_f B$, the inserted slot may occur on either side of the cyclic origin. These terms cancel in pairs. Inserting f and g in either order gives the same chain with opposite signs, proving $[h_f, h_g] = 0$.

A derivation applied to the insertion cancels against its application to the original slots. The remaining term inserts $\rho_f(g)$, giving the third source identity. For forms, substitute the coordinate formulas for δ. They give $[\delta, -df\wedge] = L_{X_f}$. Exterior differentiation and two odd exterior multiplications anticommute, while Lie differentiation sends dg to $d\{f, g\}_0$. These prove the target identities. □

The classical vector fields do not represent the quantum bracket strictly. Define

$$R(f, g) = \frac{\{f, g\}_\epsilon - \{f, g\}_0}{\epsilon}, \qquad F_1(f) = L_{X_f}, \qquad F_2(f, g) = dR(f, g) \wedge (-).$$

The difference of brackets starts at order ϵ^2, so R and F_2 are integral. Lemma 13.3 gives

$$[D_\Omega, F_2(f, g)] = [F_1(f), F_1(g)] - F_1(\{f, g\}_\epsilon). \tag{13.4}$$

The ternary expression is the exterior differential of

$$\sum_{\mathrm{cyc}(f,g,k)} \big(\{f, R(g, k)\}_0 + R(f, \{g, k\}_\epsilon)\big).$$

Multiplying the sum by ϵ gives the difference of the quantum and classical Jacobi identities. It is zero. There are no further terms, since the odd wedge operators F_2 anticommute.

A nonconstant correction makes the binary operation visible. For $f = q^3/3$ and $g = p^4/4$,

$$\{f, g\}_\epsilon = q^2 p^3 + \tfrac{1}{2}\epsilon^2 p, \qquad F_2(f, g) = \tfrac{1}{2}\epsilon\, dp \wedge (-).$$

For the two cubics $q^3/3$, $p^3/3$, the bracket correction is constant and F_2 vanishes. Their comparison homotopy below still has a nonzero secondary value.

13.6 Two integral comparison homotopies

The first comparison homotopy has degree minus one:

$$\mathcal{K}_f = \epsilon^{-1}(\mathcal{U}h_f - m_f\mathcal{U}). \tag{13.5}$$

At $\epsilon = 0$, insertion under the normalized Hochschild–Kostant–Rosenberg map is multiplication by $-df$. Indeed its $n+1$ insertion positions cancel the extra factor in $(n+1)!$. Hence the numerator is divisible by ϵ. The mixed and Cartan identities give

$$D_\Omega\mathcal{K}_f + \mathcal{K}_f D_C = \mathcal{U}L_f^C - L_{X_f}\mathcal{U}, \qquad d\mathcal{K}_f + \mathcal{K}_f B = 0.$$

The second comparison homotopy has degree minus two:

$$\mathcal{K}^{(2)}_{f,g} = \epsilon^{-2}\big(m_f\mathcal{U}h_g - m_g\mathcal{U}h_f - m_f m_g\mathcal{U} - \mathcal{U}h_f h_g\big). \tag{13.6}$$

Its numerator vanishes on positive-length chains by exterior degree. On a zero chain a the numerator is

$$df \wedge \mathcal{U}_1(a[g]) - dg \wedge \mathcal{U}_1(a[f]) - a\,df \wedge dg \\ - \mathcal{U}_2(a[f|g] - a[g|f]).$$

Its constant coefficient is zero.

The one-bar map is even in ϵ. The linear coefficient of the two-bar integral operator is

$$(P_{01} - P_{02} + P_{12})/12.$$

Interchanging the last two slots reverses both this operator and their exterior product. Antisymmetrization therefore kills the linear coefficient. Thus $\mathcal{K}^{(2)}$ is integral as well.

We specify the higher-action signs by the completed Chevalley complex. Choose an enumeration (f_a) of the nonconstant monomials, viewed in $\mathfrak{h}_\epsilon$, and let c^a be the corresponding odd dual coordinates of degree one. Define $f^a_{bc} \in R_\epsilon$ by

$$\{f_b, f_c\}_\epsilon = \sum_a f^a_{bc} f_a \quad \text{in } \mathfrak{h}_\epsilon.$$

The scalar Chevalley differential is

$$d_{\mathrm{CE}} c^a = -\frac{1}{2}\sum_{b,c} f^a_{bc} c^b c^c.$$

An odd operator anticommutes with an odd variable from the coefficient factor. The coordinate sums denote continuous multilinear maps, coefficientwise in the parameters and in fixed finite arity.

No finite basis is assumed. We use product completion over arities.

Theorem 13.4. *The operators F_1, F_2 and zero higher components define an L_∞ action of $\mathfrak{h}_\epsilon$ on the mixed form complex. In the stated Chevalley convention,*

$$\mathbb{D}_\Omega = D_\Omega + d_{\mathrm{CE}} + \sum_a c^a F_1(f_a) - \sum_{a<b} c^a c^b F_2(f_a, f_b).$$

The comparison

$$\mathbb{U} = \mathcal{U} - \sum_a c^a \mathcal{K}_{f_a} + \sum_{a<b} c^a c^b \mathcal{K}^{(2)}_{f_a,f_b}$$

is an integral L_∞ module quasi-isomorphism from the chain action. It preserves the circle operators.

Proof. Temporarily invert ϵ and set

$$T = \epsilon^{-1} \sum_a c^a h_{f_a}, \qquad M = \epsilon^{-1} \sum_a c^a m_{f_a}.$$

These have degree zero and raise Chevalley arity. Their exponentials therefore exist in the arity completion. The Cartan identities give

$$e^T (D_C + d_{\mathrm{CE}}) e^{-T} = D_C + d_{\mathrm{CE}} + \sum_a c^a L^C_{f_a},$$
$$e^M (D_\Omega + d_{\mathrm{CE}}) e^{-M} = \mathbb{D}_\Omega.$$

On chains, the commutator with d_{CE} cancels the quadratic term because the action and insertion use the same bracket. On forms their discrepancy is F_2, with the sign displayed in the theorem. Further commutators vanish by anticommutation of the insertions.

Conjugate the mixed comparison:

$$e^M \mathcal{U} e^{-T} = \mathbb{U}.$$

Expansion through arity two gives exactly (13.5) and (13.6), with the tensor signs just specified. A term of arity j sends a chain of length n to a form of degree $n + j$. Hence every term of arity at least three vanishes on a surface.

Conjugation gives $\mathbb{D}_\Omega^2 = 0$ and $\mathbb{D}_\Omega \mathbb{U} = \mathbb{U} \mathbb{D}_C$, where $\mathbb{D}_C$ is the first conjugated differential. All displayed components are integral, and their coefficient modules have no ϵ-torsion. The identities therefore hold before localization. Their parts involving u express compatibility with the circle differentials.

The arity-zero comparison is a quasi-isomorphism by Theorem 13.1. Its continuous cone contraction also contracts each fixed-arity space of continuous multilinear maps by postcomposition. The remaining differential raises arity. Perturbing in arity therefore contracts the full cone in the product completion. □

The cubic first homotopy on a zero chain is

$$\mathcal{K}_{q^3/3}(a) = -\frac{\epsilon}{12} a_{pp}\, dq.$$

The second homotopy has an all-order vacuum formula:

$$\mathcal{K}^{(2)}_{f,g}(1) = -\frac{\{f,g\}_\epsilon - \{f,g\}_0}{\epsilon^2} \Omega. \tag{13.7}$$

To see this, the unit in the outer slot kills P_{01} and P_{02}. Put $s = t_2 - t_1$. The even moments of the remaining operator are

$$2 \int_0^1 (1 - s)\left(\tfrac{1}{2} - s\right)^{2j} ds = \frac{1}{2^{2j}(2j + 1)}.$$

Together with $(2j)!$ and the two exterior derivatives, these are the coefficients of the divided Moyal commutator. The remaining terms of (13.6) subtract its classical coefficient.

In particular,

$$\mathcal{K}^{(2)}_{q^3/3, p^3/3}(1) = -\Omega/6, \qquad \mathcal{K}^{(2)}_{q^3/3, p^3/3}(qp) = -qp\Omega/6.$$

The second identity follows by the same terminating simplex integral with outer coefficient qp. These values persist at $\epsilon = 0$ after the integral division has been made. A strict classical projection by itself does not retain this secondary comparison.

13.7 The integrated Hamiltonian kernel

The differential formulas also apply coefficientwise to smooth inputs. On an oriented real plane use the volume Ω. For complex coordinates, include a fixed complementary antiholomorphic volume form in each integral. Assume compact support or periodicity, so integration by parts has no boundary term. These are formal identities in ϵ.

For scalar f and matrix-valued inputs a_0, a_1, one has

$$\int df \wedge \operatorname{Tr} \mathcal{U}_1(a_0[a_1]) = \int \Omega \operatorname{Tr}(a_0 \rho_f(a_1)). \tag{13.8}$$

The remaining form factors and orientations agree on the two sides.

To prove the identity, give f, a_0, a_1 momenta k, l, m. Integration by parts imposes $k + l + m = 0$. If $c = \pi(k, m)$, then $\pi(l, m) = -c$, and the left multiplier is

$$c \frac{2 \sinh(-\epsilon c/2)}{-\epsilon c} = \frac{2 \sinh(\epsilon c/2)}{\epsilon}.$$

This is the multiplier of ρ_f on the right. The Taylor coefficients prove the equality, with matrix factors kept in their trace order.

The comparison for matrix algebras first uses the Morita trace

$$a_0[a_1|\cdots|a_n] \longmapsto \sum_{i_0,\dots,i_n} (a_0)_{i_0 i_1}[(a_1)_{i_1 i_2}|\cdots|(a_n)_{i_n i_0}].$$

Each adjacent matrix product sums the same repeated index on both sides. Cyclic rotation preserves the closed index cycle. The map therefore commutes with b and B, and it descends to normalized chains. In degree zero it sends I_N to N, retaining the rank normalization.

The integrated Hamiltonian insertion is a strictly cyclic cochain. Put

$$\tau(a) = \int \Omega \operatorname{Tr} a, \qquad \phi_f(a_0, a_1) = \tau(a_0 \star \rho_f(a_1)).$$

Here the unit is allowed as a multiplier of compactly supported inputs. Integration by parts gives $\tau(a \star b) = \tau(ab) = \tau(b \star a)$ and $\tau \rho_f = 0$. Since ρ_f is a derivation,

$$\phi_f(a_0, a_1) + \phi_f(a_1, a_0) = \tau \rho_f(a_0 \star a_1) = 0.$$

The same derivation identity cancels its Hochschild boundary. The equality $\rho_f(1) = 0$ gives normalization.

Equation (13.8) expresses this cochain as a form pairing. For additional graded coefficients, the trace must be closed and ρ_f must commute with their differential. The graded identities then have the corresponding Koszul signs.

13.8 Continuous duals and cyclicity

A general form current has no automatic cyclicity property. We make the dual convention explicit. Let $C^n(A_\epsilon)$ be the coefficientwise continuous R_ϵ-multilinear functionals on A_ϵ^{n+1} which vanish when a bar input is 1. Define the scalar transposes by

$$(b\phi)(c) = \phi(bc), \qquad (B\phi)(c) = \phi(Bc).$$

They have degrees 1, −1 and satisfy the mixed identities.

In particular,

$$(b\phi)(a_0,\dots,a_{n+1}) = \sum_{i=0}^{n}(-1)^i\phi(a_0,\dots,a_i\star a_{i+1},\dots,a_{n+1}) + (-1)^{n+1}\phi(a_{n+1}\star a_0,a_1,\dots,a_n).$$

For $n\geq 1$ the other operator is

$$(B\phi)(a_0,\dots,a_{n-1}) = \sum_{j=0}^{n-1}(-1)^{(n-1)j}\phi(1,a_j,\dots,a_{n-1},a_0,\dots,a_{j-1}).$$

Set $B=0$ on C^0. The completed scalar mixed complex is $(C^\bullet(A_\epsilon)[[u]], b+uB)$, with $|u|=2$. These transpose conventions fix the signs in its primitives.

A continuous functional β on forms gives

$$\Phi_\beta(a_0,\dots,a_n) = \langle\beta,\mathcal{U}_n(a_0[a_1|\cdots|a_n])\rangle.$$

Transpose the two form operators by the same convention. The mixed identities for $\mathcal{U}$ give a map from this dual form complex to the scalar cochain complex. A closed dual element gives a total mixed cocycle. This assertion does not require products of distributions: each evaluation has smooth or formal inputs and one continuous functional.

A scalar one-cochain is *strictly cyclic* when $\Phi(a,b)=-\Phi(b,a)$. At $\epsilon=0$, the point current

$$\beta(a\,dq+b\,dp)=b(0)$$

gives $\Phi_\beta(1,p)=1$ and $\Phi_\beta(p,1)=0$. Thus an arbitrary continuous current fails this condition. Smooth inputs equal to 1, p near the point give the same example with compact support.

Proposition 13.5. *For a coefficientwise continuous functional $\beta:\Omega^1_{\mathcal{A}}[[\epsilon]]\to R_\epsilon$, the cochain $\Phi_\beta(a,b)=\beta(\mathcal{U}_1(a[b]))$ is strictly cyclic if and only if $\beta d=0$.*

Proof. The series $S(P)$ in (13.3) is even. Interchanging the two coefficient slots therefore gives

$$\mathcal{U}_1(a[b])+\mathcal{U}_1(b[a]) = d\big(mS(P)(a\otimes b)\big).$$

Annihilation of exact forms proves sufficiency at every parameter order. Conversely, $\mathcal{U}_1(1[b])=db$ and $\mathcal{U}_1(b[1])=0$. Cyclicity on this pair forces $\beta(db)=0$ for every b. □

13.9 Point currents and two primitives

Let $\mathscr{J}_0'$ be the vector space of finite linear combinations of derivatives at $(q,p)=(0,0)$. A series in $\mathscr{J}_0'[[\epsilon]]$ may have unbounded derivative order as the parameter order increases. For γ in this space define

$$\beta_\gamma=\gamma\delta,\qquad \Phi_\gamma(a,b)=\gamma\delta\mathcal{U}_1(a[b]).$$

This sends point functionals in degree zero to scalar one-cochains in degree one. It specifies no output-valued operation or multiplication of currents.

Lemma 13.6. *Every coefficientwise finite point functional on one-forms which annihilates exact forms is uniquely β_γ.*

Proof. At one parameter coefficient write

$$\beta(a\,dq + b\,dp) = A(\partial_q, \partial_p)a(0) + B(\partial_q, \partial_p)b(0)$$

for polynomials $A(X, Y), B(X, Y)$. The condition $\beta d = 0$ is the polynomial identity $XA + YB = 0$. Since X, Y are relatively prime, there is a unique polynomial C with

$$A = -YC, \qquad B = XC.$$

Take $\gamma(f) = C(\partial_q, \partial_p)f(0)$ and apply the argument at every parameter coefficient. For smooth complex-coordinate currents, antiholomorphic derivative symbols may be included as polynomial coefficients. The same regular-sequence argument still applies. □

The two-bar comparison supplies a second scalar cochain. Let $\lambda_\gamma(c\Omega) = \gamma(c)$ and put

$$\Psi_\gamma(a_0, a_1, a_2) = \lambda_\gamma \mathcal{U}_2(a_0[a_1|a_2]).$$

It is a normalized cochain of degree two. The integral mixed map relates its Connes boundary to Φ_γ.

Theorem 13.7. *The point cochains satisfy*

$$b\Phi_\gamma = B\Phi_\gamma = 0, \qquad \epsilon\Phi_\gamma = b\gamma, \qquad b\Psi_\gamma = 0, \qquad B\Psi_\gamma = \Phi_\gamma.$$

Consequently

$$(b + uB)\gamma = \epsilon\Phi_\gamma, \qquad (b + uB)\Psi_\gamma = u\Phi_\gamma. \tag{13.9}$$

The class $[\Phi_\gamma]$ *is nonzero exactly when the constant* ϵ *coefficient of* γ *is nonzero. The resulting copy of* $\mathscr{J}_0'$ *in degree-one mixed cohomology is annihilated by both* ϵ *and* u.

Proof. Direct differentiation of the one-bar formula gives

$$\delta\mathcal{U}_1(a[b]) = mPS(P)(a \otimes b) = \frac{a \star b - b \star a}{\epsilon}.$$

This proves $\epsilon\Phi_\gamma = b\gamma$. Associativity gives $b^2\gamma = 0$, and the scalar coefficient module has no ϵ-torsion. Hence $b\Phi_\gamma = 0$. Both unit evaluations vanish, so $B\Phi_\gamma(a) = \Phi_\gamma(1, a) = 0$.

On a chain of length three, $\mathcal{U}_2 b = \epsilon\delta\mathcal{U}_3 = 0$. Transposing proves $b\Psi_\gamma = 0$. On a chain $a[b]$, the second mixed identity gives

$$(B\Psi_\gamma)(a, b) = \lambda_\gamma d\mathcal{U}_1(a[b]) = \gamma\delta\mathcal{U}_1(a[b]) = \Phi_\gamma(a, b).$$

The middle equality uses $d(a\,dq + b\,dp) = (b_q - a_p)\Omega$. This proves the two independent primitives in (13.9).

In total degree one a boundary can only come from a scalar zero-cochain, since cochain degrees and powers of u are nonnegative. Modulo ϵ, its Hochschild boundary vanishes because A is commutative. If the constant coefficient γ_0 is nonzero, choose a polynomial f with $\gamma_0(f) \neq 0$ and a polynomial g with $\partial_p g = f$. Then $\Phi_{\gamma_0}(q, g) = \gamma_0(f) \neq 0$. Thus Φ_γ cannot be a boundary.

Conversely, if $\gamma = \epsilon\gamma_1$, then $\Phi_\gamma = b\gamma_1$ is a mixed boundary. The assignment of a class therefore depends exactly on γ_0 and embeds $\mathscr{J}_0'$ as stated. The two displayed primitives prove its annihilation by each parameter. □

For $\gamma = \mathrm{ev}_0$, the normalizations are

$$\Phi_\gamma(q, p) = 1, \qquad \Phi_\gamma(q^3, p^3) = \tfrac{3}{2}\epsilon^2, \qquad \Psi_\gamma(1, q, p) = \tfrac{1}{2}.$$

The first primitive uses a zero-cochain and multiplication by ϵ. The second uses a two-cochain and the actual Connes operator. They have different degrees and remain distinct constructions. Neither identifies this scalar current with a geometric normal residue or a closed field.

13.10 Transport by a quantum shear

A nonlinear coordinate motion tests the comparison homotopy. For $f = q^3/3$,

$$\rho_f = q^2\partial_p + \frac{\epsilon^2}{12}\partial_p^3.$$

Its two terms commute. The corresponding quantum flow is

$$\Phi_t(a) = e^{t\epsilon^2\partial_p^3/12}a(q, p + tq^2).$$

The exponential terminates on each polynomial input, and formal inputs are treated coefficientwise in the specified completion. Since ρ_f is a derivation, Φ_t is a Moyal automorphism.

Its value on a cubic is

$$\Phi_t(p^3) = (p + tq^2)^3 + \tfrac{1}{2}t\epsilon^2.$$

Ordinary pullback alone would omit the last term.

Let ϕ_t^* be the classical pullback on forms, and let Φ_t act in each chain slot. Define

$$\mathcal{H}_t = \int_0^t \phi_{t-s}^*\mathcal{K}_f\Phi_s\, ds. \tag{13.10}$$

For polynomial inputs, each parameter coefficient of the integrand is polynomial in the flow parameter. For formal inputs this holds after fixing the coordinate coefficient too. Integration of these coefficients and the first homotopy identity give

$$\mathcal{U}\Phi_t - \phi_t^*\mathcal{U} = D_\Omega\mathcal{H}_t + \mathcal{H}_t D_C.$$

Indeed differentiating $\phi_{t-s}^*\mathcal{U}\Phi_s$ with respect to s gives $\phi_{t-s}^*(\mathcal{U}L_f^C - L_{X_f}\mathcal{U})\Phi_s$. The mixed differentials commute with both flows. Integration supplies the two endpoints and proves the formula. For several Hamiltonian motions the second comparison homotopy remains part of their compatibility data.

13.11 A logarithmic torus

The same kernel has a global example without choosing logarithms. Let $A = \mathbb{C}[x^{\pm 1}, y^{\pm 1}]$, and put

$$D_x = x\partial_x, \qquad D_y = y\partial_y, \qquad \alpha = d\log x, \qquad \beta = d\log y.$$

Define its formal product on Laurent monomials by

$$x^a y^b \star x^c y^d = e^{\epsilon(ad-bc)/2}x^{a+c}y^{b+d}.$$

Bilinearity of the exponent proves associativity.

The constant Laurent coefficient is a cyclic trace: only opposite modes contribute, and their alternating pairing is zero. Replace ∂_q, ∂_p by D_x, D_y in (13.2). The commuting derivations give the same mixed identities globally, without logarithmic functions in the algebra.

The undeformed diagonal Koszul resolution of this Laurent algebra is the localization of the polynomial diagonal resolution. It proves the Hochschild comparison, and the same completed filtration argument proves the mixed quasi-isomorphism. Each parameter coefficient has finite Laurent support. The exponential is now a formal parameter series, rather than a terminating polynomial expression.

After inverting ϵ and then completing in u, consider the Laurent mode $(m, n) \in \mathbb{Z}^2$. The form differential on this mode is

$$D_{m,n} = \epsilon \iota_{-nD_x+mD_y} + u(m\alpha + n\beta) \wedge .$$

For a nonzero mode put

$$H_{m,n} = \frac{-n\alpha + m\beta}{\epsilon(m^2 + n^2)} \wedge .$$

Contraction with $-nD_x + mD_y$ evaluates the numerator to $m^2 + n^2$. The two exterior multiplication operators anticommute. Therefore

$$D_{m,n}H_{m,n} + H_{m,n}D_{m,n} = 1.$$

The zero mode has zero differential. The mixed cohomology is consequently

$$\mathbb{C}((\epsilon))[[u]] \otimes \Lambda(\alpha, \beta). \tag{13.11}$$

The homotopy preserves Laurent support and lowers the ϵ bound by one, so it is defined on the stated coefficientwise completion.

The derivations D_x, D_y are outer on the Laurent algebra. An inner derivation preserving the mode of each input could only use a zero-mode Hamiltonian. Distinct nonzero Hamiltonian modes shift each input to distinct output modes. A zero-mode Hamiltonian is constant and has zero commutator. Thus local Hamiltonian coordinates do not remove these global vector fields or the four zero-mode form classes.

13.12 The remaining source comparison

The circle map and its Hamiltonian homotopies are constructions for the associative coefficient algebra. The integrated Hamiltonian insertion gives a closed cyclic cochain under its trace and support hypotheses. A general distribution supplies a dual mixed cochain, with strict cyclicity governed by Proposition 13.5. The point-current primitives are scalar operations on this specified recipient.

To obtain the mixed field-theory comparison, one must supply a map from the geometric closed-source complex. It must preserve its differential, interaction, boundary restriction and polar collision operations. The scalar parameter primitives above neither choose this map nor fix its physical normalization. Tensoring with a normal complex preserves the mixed identities, but cyclic insertions still require a compatible normal pairing and the stated tangential cyclicity.

For a general holomorphic symplectic surface, coordinate changes must carry the comparison and its higher homotopies through overlaps. A connection, deformation period and trace normalization enter this descent. The explicit shear supplies one transport homotopy, while the logarithmic torus exhibits global classes which a local calculation must retain. Neither example constructs the full descent or its supported chiral comparison.

Chapter 14

Normal sources and supported operations

A closed field coupled to a boundary algebra must do more than produce a scalar cyclic cochain. Its insertion has a normal support, a differential and collision operations. The action determines a source in the closed equation as well as an action on the open fields. We first construct a classical model in which these two effects are transposes of one another.

Normal jets then give explicit supported cyclic operations. Real normal currents, holomorphic principal parts and smooth source tests require different complexes. We compute their primitives and the obstruction to extending an operation across a collision. These constructions retain the data needed for a comparison with independently defined closed fields. The final section studies coefficient completion. It proves reconstruction in one normal direction and identifies the limits and duals that fail to preserve support.

14.1 A coupled Hamiltonian bulk and defect

Let

$$M = \mathbb{R}_r \times \mathbb{R}_s \times \mathbb{C}^2_{q,p}, \qquad Y = \{r = 0\}, \qquad i : Y \hookrightarrow M, \qquad \Omega = dq \wedge dp.$$

Orient the real normal line by increasing r. It is the whole line, so Y is an internal defect.

Let $\mathcal{A}_M$ denote smooth mixed forms, with the completed projective smooth topology and differential

$$D_M = d_r + d_s + \bar{\partial}.$$

Define $\mathcal{A}_Y$ similarly, with $D_Y = d_s + \bar{\partial}$. An integral always has compact support in its integration variables. An unintegrated smooth factor may act as a multiplier.

Complete coefficientwise in ϵ, of degree zero. The holomorphic Moyal product is

$$a \star b = m \exp\Big(\frac{\epsilon}{2}\mathsf{P}\Big)(a \otimes b), \qquad \mathsf{P} = \partial_q \otimes \partial_p - \partial_p \otimes \partial_q. \tag{14.1}$$

Exterior and matrix multiplication are included in m. Every parameter coefficient is a finite-order differential operator.

For scalar mixed forms the graded commutator is divisible by ϵ. Hence

$$[f, g]_\epsilon = \epsilon^{-1}[f, g]_\star, \qquad \rho_f(a) = \epsilon^{-1}[i^* f I_N, a]_\star$$

are defined before parameter inversion. The second acts on matrix-valued defect forms. Associativity gives

$$[\rho_f, \rho_g] = \rho_{[f,g]_\epsilon}, \qquad \rho_f(a \star b) = (\rho_f a) \star b + (-1)^{|f||a|} a \star \rho_f b.$$

The trace $\tau_Y(a) = \int_Y \Omega \wedge \operatorname{Tr} a$ is graded cyclic. Integration by parts kills every positive two-input Moyal correction, so $\tau_Y(a \star b) = \tau_Y(ab)$. The action ρ preserves the resulting pairing. Here q, p

are physical coordinates. Proposition 10.4 compares their jets with horizontal independent coefficients. Unconstrained coefficient fields have a different carrier.

Let $\mathcal{A}_M^{\mathrm{nor}}$ admit distributions of locally finite order in r, with smooth distribution-valued tangential dependence. It is a module over smooth mixed forms and is stable under D_M. In particular it contains every finite derivative of $\delta(r)$.

We use the fields

$$\alpha \in \mathcal{A}_M[1][[\epsilon]], \qquad \beta \in \mathcal{A}_M^{\mathrm{nor}}[2][[\epsilon]], \qquad A \in (\mathcal{A}_Y \otimes \mathrm{Mat}_N)[1][[\epsilon]].$$

The bulk cotangent pairing and the defect trace pairing have degree -1. On the distributional domain the pairing is weak: we use its Hamiltonian bracket on functionals whose variational derivatives are represented in the paired field spaces. The functionals below have this property.

At ghost number zero, α, A, β have mixed form degrees 1, 1, 2. Define

$$S_c = \int_M \Omega \wedge \beta \star \left(D_M \alpha + \tfrac{1}{2}[\alpha, \alpha]_\epsilon\right), \tag{14.2}$$

$$S_o = \tau_Y \left(\tfrac{1}{2} A \star D_Y A + \tfrac{1}{3} A \star A \star A\right), \tag{14.3}$$

$$S_{co} = \tfrac{1}{2} \tau_Y (A \star \rho_\alpha A). \tag{14.4}$$

For all BV components, use suspension of degree -1 with $b_1(sx) = -sDx$ and $b_2(sx, sy) = (-1)^{|x|} s[x, y]$. The cyclic pairing fixes the signs of the corresponding quadratic and cubic Hamiltonians.

Theorem 14.1. *The functional $S_c + S_o + S_{co}$ satisfies the classical master equation. Its Hamiltonian vector field preserves the smooth and normal-distribution domains coefficientwise in ϵ. It uses no product of two singular normal coefficients.*

Proof. Let $\mathfrak{g} = (\mathcal{A}_M[[\epsilon]], D_M, [-,-]_\epsilon)$, and let $\mathfrak{a}$ be the matrix defect dg Lie algebra with its graded star commutator. Its cyclic pairing has degree -3. Adjoin to $\mathfrak{g} \oplus i_* \mathfrak{a}$ the admissible paired dual of $\mathfrak{g}$. Integration identifies this summand with $\mathcal{A}_M^{\mathrm{nor}}[1][[\epsilon]]$ before the common suspension.

The dual summand is abelian. Let $\mathfrak{g}$ act on it coadjointly and on $\mathfrak{a}$ by ρ. Its bracket with $\mathfrak{a}$ is zero. The bracket of two open elements has its native component and a dual component μ, determined by cyclicity:

$$\langle \mu(a, b), x \rangle = \langle a, [b, x] \rangle.$$

For degree-zero inputs this reads $\mu(a, b)(x) = \langle \rho_x a, b \rangle$. The transpose of restriction is normal pushforward.

Jacobi on two bulk inputs and one open input is the representation identity for ρ. On one bulk and two open inputs it is the derivation identity and equivariance of μ. For three open inputs the native component is Jacobi in $\mathfrak{a}$. Pairing the dual component with x gives the cyclic sum of $\langle \rho_x [a, b], c \rangle$. The derivation identity and cyclicity cancel its terms with their Koszul signs. The remaining triples give coadjoint identities or zero.

The mixed differentials preserve these brackets by integration by parts. We have therefore constructed a cyclic dg Lie algebra. Its suspended Hamiltonian is exactly the sum of (14.2)–(14.4), and the dg Lie identities give the classical master equation. All singular coefficients lie in the abelian dual summand. Differentiation and multiplication by smooth fields preserve that summand, proving the domain assertion. □

The source follows directly by varying the coupling. Since A is odd, $\mathrm{Tr}(A^2) = 0$ at $\epsilon = 0$. Graded exchange removes the even powers of P, giving

$$\mathcal{M}_\epsilon(A) = \epsilon^{-1} \mathrm{Tr}(A \star A) = \sum_{k \geq 0} \frac{\epsilon^{2k}}{2^{2k+1}(2k+1)!} \mathrm{Tr}\, m \mathsf{P}^{2k+1}(A \otimes A). \tag{14.5}$$

Its leading term is $\mathrm{Tr}(\partial_q A \wedge \partial_p A)$.

Cyclic rotation rewrites the coupling as $\int_Y \Omega \wedge i^*\alpha \star \mathcal{M}_\epsilon(A)$. The ghost-zero equations are

$$\begin{aligned} D_M \alpha + \tfrac{1}{2}[\alpha, \alpha]_\epsilon &= 0, \\ D_Y A + A \star A + \rho_\alpha A &= 0, \\ D_M \beta + [\alpha, \beta]_\epsilon &= \delta(r) dr\, \mathcal{M}_\epsilon(A). \end{aligned} \tag{14.6}$$

The last sign uses dr before the orientation of Y. Equivalently, variation of S_{co} in α has the positive pairing with $\mathcal{M}_\epsilon(A)$.

The source is covariantly closed on an open solution. Indeed $D_\alpha = D_Y + \rho_\alpha$ is an odd derivation and $D_\alpha A = -A \star A$, so

$$D_\alpha(A \star A) = (-A \star A) \star A - A \star (-A \star A) = 0.$$

Trace intertwines the scalar Hamiltonian action. This is the source condition in the bulk Bianchi identity.

For two coincident blocks, the source also contains

$$\epsilon^{-1} \mathrm{Tr}(A_{12} \star A_{21} + A_{21} \star A_{12}).$$

The cubic action retains the corresponding mixed cyclic words. A block-diagonal field is a subtheory of this rank-added defect, not its full field space.

14.2 Inner actions and normal transport

An inner action can be removed from the open differential while leaving a closed cubic interaction. We first explain the algebraic change of variables. Let $\mathfrak{a}$ be an ordinary quadratic Lie algebra, and suppose $\rho_x = \mathrm{ad}_{j(x)}$ for a Lie map $j : \mathfrak{g} \to \mathfrak{a}$. Write $\kappa(x, y) = \langle jx, jy \rangle$ and let $j^\dagger$ be the adjoint into the paired dual.

Proposition 14.2. *In the double extension, the change*

$$a' = a + jx, \qquad \xi' = \xi - j^\dagger a - \tfrac{1}{2} j^\dagger jx$$

preserves the pairing and decouples the open Lie algebra. The remaining bulk-dual bracket has the cocycle $c(x, y) = \kappa([x, y], -)/2$. *Its cyclic cubic Hamiltonian is* $\kappa(x, [x, x])/12$.

Proof. Substitution in $\langle a, a \rangle + 2\langle x, \xi \rangle$ cancels both cross terms and the term $\kappa(x, x)$. The new open vector has old coordinates $(0, a, j^\dagger a)$. The new bulk vector has coordinates $(x, -jx, -j^\dagger jx/2)$.

Insert these vectors in the double-extension bracket. The mixed bracket vanishes. The open bracket is the original bracket of $\mathfrak{a}$, and the bracket of two new bulk vectors is $([x, y], 0, c(x, y))$. Invariance of κ proves the cocycle identity. Pairing with the third bulk vector and multiplying by $1/6$ gives the cubic Hamiltonian. The same calculation with the cyclic Koszul signs applies after suspension. □

In the geometric theory $j(\alpha) = \epsilon^{-1} i^*\alpha I_N$ requires inversion of ϵ. The canonical change is

$$A' = A + \epsilon^{-1} i^*\alpha I_N, \qquad \beta' = \beta - \delta(r) dr \left(\epsilon^{-1} \mathrm{Tr}\, A + \frac{N}{2\epsilon^2} i^*\alpha \right).$$

It gives

$$S = S_c(\alpha, \beta') + S_o(A') + \frac{N}{6\epsilon^3} \int_Y \Omega \wedge (i^*\alpha)^{\star_3}. \tag{14.7}$$

The coefficient can be checked without suppressing the kinetic term. Put $B = i^* \alpha I_N / \epsilon$ and $F_B = D_Y B + B \star B$. Modulo an integrated total derivative,

$$\mathrm{CS}(A + B) = \mathrm{CS}(A) + \mathrm{CS}(B) + \tau_Y(B \star A \star A) + \tau_Y(A \star F_B).$$

The cotangent shifts contribute $\tau_Y(A \star F_B) + \tau_Y(B \star F_B)/2$ when the old β is expressed in terms of β'. Consequently

$$S = S_c(\alpha, \beta') + S_o(A') - S_o(B) + \tfrac{1}{2}\tau_Y(B \star F_B).$$

The last two terms equal $\tau_Y(B^{\star 3})/6$. Taking the matrix trace gives (14.7).

The residual cubic is closed and has zero self-bracket in the cotangent theory because it does not depend on β'. The transformation is not defined integrally at $\epsilon = 0$. Thus innerness after localization does not remove the integral problem or its closed cubic vertex.

To move the defect, write a smooth background on a collar as $\alpha = dr\, \alpha_r + \alpha_Y$. Complete in a field-amplitude parameter as well as ϵ. The ordered series

$$U(r_1, r_0) = 1 + \sum_{n \geq 1} (-1)^n \int_{r_0 < t_1 < \cdots < t_n < r_1} \rho_{\alpha_r(t_n)} \cdots \rho_{\alpha_r(t_1)}\, dt_1 \cdots dt_n \tag{14.8}$$

is finite at each amplitude degree. Set

$$D_{Y,\alpha}(r) = D_Y + \rho_{\alpha_Y(r)}, \qquad F_{rY} = \partial_r \alpha_Y - D_Y \alpha_r + [\alpha_r, \alpha_Y]_\epsilon.$$

Proposition 14.3. *The operator U preserves the star product and integrated trace and composes under subdivision of the collar. Its differential defect is*

$$\begin{aligned} &D_{Y,\alpha}(r_1) U(r_1, r_0) - U(r_1, r_0) D_{Y,\alpha}(r_0) \\ &\qquad = \int_{r_0}^{r_1} U(r_1, t)\, \rho_{F_{rY}(t)}\, U(t, r_0)\, dt. \end{aligned} \tag{14.9}$$

In particular a flat background gives a cyclic dg algebra map.

Proof. Differentiating the ordered simplex gives $\partial_{r_1} U = -\rho_{\alpha_r(r_1)} U$, with initial value the identity. The multiplicative defect satisfies the same homogeneous equation with zero initial value, since ρ is a derivation. Formal recursion proves it zero. Trace invariance follows from $\tau_Y \rho = 0$. Subdivision of each simplex proves composition.

Call the left side of (14.9) $C(r_1)$. Differentiating gives $\partial_{r_1} C + \rho_{\alpha_r} C = \rho_{F_{rY}} U$ and $C(r_0) = 0$. Variation of constants proves the formula. Every step is coefficientwise finite in amplitude and ϵ. □

14.3 Real normal jets and cyclic caps

Normal differentiation of an insertion must act on every input. Its algebraic model is dual to multiplication of normal test jets. For $M \geq 1$ put

$$J_M = \mathbb{C}[r, dr]/(r^{M+1}, r^M dr), \qquad |r| = 0, \quad |dr| = 1, \quad dr = dr.$$

The second relation is the derivative of the first. Thus J_M is a finite-dimensional commutative dg algebra.

Let $C_M = J_M^\vee$ have basis $t_0, \ldots, t_M$ in degree zero and $e_0, \ldots, e_{M-1}$ in degree -1. They extract the coefficients of r^n and $r^n dr$. For a homogeneous functional use $dc = -(-1)^{|c|} c d$.

Proposition 14.4. *The dual is a counital dg coalgebra with*

$$dt_n = 0, \qquad de_n = (n+1)t_{n+1},$$

$$\Delta t_n = \sum_{a+b=n} t_a \otimes t_b, \qquad \Delta e_n = \sum_{a+b=n} (e_a \otimes t_b + t_a \otimes e_b).$$

The counit extracts t_0. If $i(1) = t_0$ and p is the counit, then

$$Ht_n = e_{n-1}/n \quad (n > 0), \qquad Ht_0 = He_n = 0$$

satisfies $dH + Hd = 1 - ip$. The inclusions $C_M \to C_{M+1}$ preserve the differential and coproduct.

Proof. Evaluate on $d(r^k) = kr^{k-1}dr$ to obtain the differential. Dualizing monomial multiplication gives the coproduct. Associativity and the unit give coassociativity and the counit identities. For $i + j = n + 1$, the two contributions to the coefficient of $t_i \otimes t_j$ in the coderivation identity have coefficients i and j. Their sum is $n + 1$.

The formula for H verifies the contraction on each basis vector. The quotient $J_{M+1} \to J_M$ is a dg algebra map, which proves compatibility between stages. □

The union $C = \varinjlim C_M$ retains every finite normal derivative. Its iterated coproduct distributes a divided derivative among all arguments:

$$\Delta^{(k-1)} t_n = \sum_{n_1+\cdots+n_k=n} t_{n_1} \otimes \cdots \otimes t_{n_k}.$$

For e_n the same sum has one e in every possible position. Normal differentiation raises the stage. It does not preserve each fixed truncation.

These functionals have supported representatives. Define

$$E_n = \frac{(-1)^n}{n!}\delta^{(n)}(r), \qquad T_n = E_n\, dr.$$

Then $\int T_n f = f^{(n)}(0)/n!$ and $\int E_n g\, dr = g^{(n)}(0)/n!$. Before shifting, $dE_n = -(n+1)T_{n+1}$. The assignments $t_n \mapsto T_n[1]$ and $e_n \mapsto E_n[1]$ therefore identify C with the finite-order supported currents in the shifted de Rham complex. Every evaluation is continuous in a finite smooth seminorm.

A smooth Thom form has the same normal cohomology. Choose $\chi \in C_c^\infty(\mathbb{R})$ with integral one and set $F_\chi(r) = \int_{-\infty}^{r} \chi(s)\, ds$. Before shifting, the homotopy on a compactly supported one-form is

$$h(g\, dr)(r) = \int_{-\infty}^{r} g(s)\, ds - F_\chi(r) \int_{\mathbb{R}} g(s)\, ds.$$

It vanishes outside a compact set. Direct differentiation gives $dh + hd = 1 - ip$, with $i(1) = \chi\, dr$ and $p(g\, dr) = \int g$. The primitive of the difference between two unit-integral choices is compactly supported and gives a homotopy between their inclusions.

Let A now be a unital dg algebra. A test ideal can instead use its unitalization and relative normalized chains. In $\mathrm{CH}(A)$, an n-chain of internal degree m has total degree $m - n$. Set

$$D = d_{\mathrm{int}} + (-1)^m b, \qquad B_{\mathrm{tot}} = (-1)^m B.$$

The faces defining b include the graded cyclic last face. The operator B is cyclic rotation with unit insertion.

For the graded dual cochains use

$$(D^\vee\phi)(c) = -(-1)^{|\phi|}\phi(Dc), \qquad (B^\vee\phi)(c) = -(-1)^{|\phi|}\phi(B_{\mathrm{tot}}c).$$

Their degrees are 1, -1 and they form a mixed complex. Completion over arities is a product.

For $A_M = J_M \otimes A$, define

$$\begin{aligned}\mathrm{Cap}(c,\phi)(j_0a_0,\dots,j_na_n) &= (-1)^\sigma\langle c, j_0\cdots j_n\rangle\phi(a_0,\dots,a_n),\\ \sigma &= \sum_{i<j}|a_i||j_j| + |\phi|\sum_i |j_i|.\end{aligned} \tag{14.10}$$

Here $|\phi|$ is the total cochain degree, not its arity.

Theorem 14.5. *The cap is a map of mixed complexes*

$$C_M \otimes \mathrm{CH}(A)^\vee \longrightarrow \mathrm{CH}(A_M)^\vee.$$

It preserves the cyclic and unit identities and normal-jet refinement. It extends coefficientwise to $D^\vee + uB^\vee$, with $|u| = 2$. Continuous tangential cochains give continuous finite-order normal operations supported at $r = 0$.

Proof. Move all normal factors to the left and multiply them. This gives a chain map

$$T : \mathrm{CH}(J_M \otimes A) \longrightarrow J_M \otimes \mathrm{CH}(A),$$

with sign $\sum_{i<j}|a_i||j_j|$. The internal differential commutes with this operation by the Leibniz rule. Each Hochschild face has the same normal product. The last face uses graded commutativity to move its normal factor around that product.

The source totalization has sign $(-1)^{m_J+m_A}$ on b. The tensor differential on the target has the same sign, $(-1)^{m_J}(-1)^{m_A}$. Graded commutativity also preserves cyclic rotation, and inserting a unit multiplies the normal product by one. Hence T commutes with B_{tot}. A degenerate source tensor has a unit tangential bar entry, so the map descends to normalized chains.

Graded transposition contributes the second term of σ. It proves the mixed identities and their descendant extension. The formulas are unchanged under refinement. Each normal coefficient uses finitely many smooth derivatives, proving the support and continuity assertions. □

For zero-form normal inputs, the cap of t_n is the sum of tangential evaluations on all Taylor coefficients whose orders add to n. If ϕ is closed, the cap of e_n is a primitive of $(n+1)\,\mathrm{Cap}(t_{n+1},\phi)$. The sign is visible for $A = \Lambda(\theta)$, $|\theta| = 1$, and the zero-arity functional $\phi(\theta) = 1$. Then $|\phi| = -1$ and $\mathrm{Cap}(e_0,\phi)(dr\,\theta) = -1$. Its dual differential evaluates to $+1$ on $r\theta$.

The bare scalar transposes in Chapter 13 use a different sign convention. For an ungraded algebra, the total-complex comparison is

$$u^j\phi_n \longmapsto (-1)^{n(n+1)/2+j}u^j\phi_n. \tag{14.11}$$

Increasing n contributes $(-1)^{n+1}$. Decreasing n and increasing j contributes the same sign. Thus (14.11) intertwines both total differentials.

It is semilinear for $u \mapsto -u$. It does not identify the two conventions over the identity map of the descendant coefficient ring.

If A has a closed graded trace τ and a degree-zero derivation X with $[d_A, X] = 0$ and $\tau X = 0$, then $\phi_X(a_0,a_1) = \tau(a_0Xa_1)$ is a closed cyclic cochain. The derivation identity cancels its Hochschild boundary. Integration by parts proves cyclicity, and $X1 = 0$ gives the normalized Connes identity. The cap constructs all its finite normal descendants. A map from a physical closed field to ϕ_X remains separate data.

14.4 Holomorphic normal residues

A complex normal coordinate has a Dolbeault differential. Orient $\mathbb{C}_z$ by $dx \wedge dy$, so that $d\bar z \wedge dz = 2i\, dx \wedge dy$. Define the higher polar distributions by $z^{-n-1} = (-1)^n \partial_z^n (z^{-1})/n!$ and put

$$\mathcal{T}_n = \frac{1}{2\pi i} \bar\partial (z^{-n-1}).$$

The Cauchy formula and differentiation give

$$z\mathcal{T}_0 = 0, \qquad z\mathcal{T}_n = \mathcal{T}_{n-1}\ (n > 0), \qquad \partial_z \mathcal{T}_n = -(n+1)\mathcal{T}_{n+1}. \tag{14.12}$$

For a test holomorphic near zero, with a compact cutoff, $\int \mathcal{T}_n \wedge dz\, f = [z^n] f$. These are complex normal residues, not the real Thom currents of the preceding section.

For $a > 0$ a smooth representative is

$$c_a(z) = \frac{1 - e^{-|z|^2/a}}{z}, \qquad \mathcal{T}_{0,a} = \frac{e^{-|z|^2/a}}{2\pi i a}\, d\bar z.$$

The numerator vanishes quadratically at zero, so c_a extends smoothly. Direct differentiation gives $(2\pi i)^{-1} \bar\partial c_a = \mathcal{T}_{0,a}$. Polar integration shows that $\mathcal{T}_{0,a} \wedge dz$ has mass one. This fixes a one-source normalization.

The algebraic counterpart is the principal-parts module

$$\mathcal{P}_z = \mathbb{C}[z, z^{-1}]/\mathbb{C}[z] = \bigoplus_{n \geq 0} \mathbb{C}[z^{-n-1}].$$

Its residue pairing with $\mathbb{C}[[z]]\, dz$ is continuous against the formal topology and separates coefficients. Several independent normal coordinates use tensor products of these modules. A product of distributions with the same normal support is not supplied by this tensor construction.

14.5 Supported operations and their normal contraction

We now keep the entire observable target while resolving the normal source differential. Let (F, Q) be a Hausdorff locally convex cochain complex with continuous differential. No lower degree bound is assumed. Formal coefficient rings may be completed coefficientwise.

For n occupied source positions in a complex affine chart V, define

$$\mathcal{K}(V) = C_c^\infty(V) \otimes \Lambda(e_1, \dots, e_n), \qquad |e_i| = -1, \qquad d_\partial = -\sum_i \iota_{e_i} \partial_{\bar z_i}.$$

The exterior contraction uses the displayed order of the markers. An operation is supported on a closed set Z when it kills every test supported away from Z. Let $\boldsymbol{O}_Z(F)$ be the continuous supported Hom complex, with uniform convergence on bounded source sets in every target seminorm. Its differential is

$$\delta A = QA - (-1)^q A d_\partial, \qquad |A| = q. \tag{14.13}$$

Choose adapted affine coordinates $V = D \times W \subset \mathbb{C}_u^r \times \mathbb{C}_w^s$, where $r + s = n$ and $Z = \{u = 0\}$. For a partial diagonal, w_b is one representative of each block and the other coordinates are $u_i = z_i - w_b$. These are source collision normals. A physical normal coordinate such as r on M remains inside the target F.

For compact $K \Subset W$, use the usual Fréchet space $C^\infty_K(W)$. The normal jet spaces are

$$\mathcal{J}_r(W) = \operatorname*{indlim}_{K \Subset W} \prod_{\mathbf{a},\mathbf{b} \in \mathbb{N}^r} C^\infty_K(W),$$

$$\mathcal{J}^{\mathrm{hol}}_r(W) = \operatorname*{indlim}_{K \Subset W} \prod_{\mathbf{a} \in \mathbb{N}^r} C^\infty_K(W).$$

The products have their product topologies and the compact exhaustion gives a strict LF topology. All coefficients at one stage have a common compact support. Taylor expansion is

$$j\kappa = \sum_{\mathbf{a},\mathbf{b}} \frac{\partial_u^{\mathbf{a}} \partial_{\bar u}^{\mathbf{b}} \kappa(0, w)}{\mathbf{a}!\mathbf{b}!} u^{\mathbf{a}} \bar u^{\mathbf{b}}.$$

A normal-flat test has every displayed coefficient zero.

Lemma 14.6. *The Taylor map $j : C^\infty_c(V) \to \mathcal{J}_r(W)$ is continuous and onto. Every bounded jet set has a bounded set of smooth lifts. For $r > 0$ there is no continuous linear right inverse at a fixed nonzero support stage.*

Proof. Derivative restriction proves continuity. A bounded jet family lies at one support stage. Write $P_l(f; u, w)$ for the sum of its monomials of total normal degree l. Choose a normal cutoff χ_0 equal to one near zero. For $m < l$, the smooth seminorm through order m of $\chi_0(u/\varepsilon)P_l$ is bounded by $C_{l,m}\varepsilon^{l-m}$ times finitely many coefficient seminorms. Tangential derivatives obey the same estimate.

Choose $\varepsilon_l \downarrow 0$ uniformly on the bounded family so that this bound is at most 2^{-l} for $m \le l/2$, apart from finitely many initial terms. The series $\sum_l \chi_0(u/\varepsilon_l)P_l$ converges uniformly in every smooth seminorm. It has common compact support and the prescribed Taylor coefficients.

A continuous linear right inverse would bound its output C^0 norm by finitely many seminorms of finitely many input coefficients. A jet supported in a coefficient outside this finite list would then lift to a function with zero C^0 norm. This contradicts its nonzero jet. □

Lemma 14.7. *Every continuous operation supported on Z factors uniquely through normal Taylor jets, componentwise in the markers. The resulting identification of Hom spaces preserves the bounded-set topologies.*

Proof. Let A be supported on Z and κ normal-flat. Then $A\kappa = A(\chi_0(u/\varepsilon)\kappa)$. For each target seminorm, continuity bounds this by finitely many test derivatives on a common compact set. Normal flatness beats every inverse power of ε from the cutoff, so $A\kappa = 0$.

The Taylor map at a fixed compact stage is a surjection of Fréchet spaces. The open mapping theorem makes the factor continuous. Passing through the compact exhaustion gives the asserted factorization. Conversely, a continuous map on jets composed with j is supported on Z.

Precomposition with j is continuous on bounded-set Hom spaces. For the inverse, lift each bounded jet set by Lemma 14.6. Taking the supremum over these bounded lifts gives exactly the required bound in each target seminorm. □

The quotient is on smooth tests. It does not restrict a physical distribution to its own singular support. In particular no uniform normal-order bound across all target seminorms is required.

Change the odd basis with the affine coordinates and put the normal markers first. The jet differential becomes

$$d = -\sum_{j=1}^{r} \iota_j \partial_{\bar u_j} - \sum_{j=1}^{s} \iota_{r+j} \partial_{\bar w_j}.$$

For a partial diagonal, the tangent marker of a block is the sum of its original markers. Let π_j set $\bar{u}_j = 0$, and define formal integration by

$$S_j(u^{\mathbf{a}}\bar{u}^{\mathbf{b}}) = \frac{u^{\mathbf{a}}\bar{u}^{\mathbf{b}+\mathbf{e}_j}}{b_j + 1}.$$

Thus $\partial_{\bar{u}_j} S_j = 1$ and $S_j \partial_{\bar{u}_j} = 1 - \pi_j$.

For an ordered marker set B, set

$$P_j(f\,\epsilon_B) = \begin{cases} (\pi_j f)\epsilon_B, & j \in B, \\ 0, & j \notin B, \end{cases}$$

$$h_j(f\,\epsilon_B) = \begin{cases} -(-1)^{\#\{i\in B : i<j\}}(S_j f)\epsilon_{B\cup\{j\}}, & j \notin B, \\ 0, & j \in B. \end{cases}$$

These maps are continuous coefficientwise.

Put

$$P = P_1 \cdots P_r, \qquad h = \sum_{j=1}^{r} P_1 \cdots P_{j-1} h_j.$$

The image $\mathcal{E}_{r,s}$ of P contains all normal markers and only holomorphic normal coefficients. If $\nu = \epsilon_1 \cdots \epsilon_r$, then

$$\mathcal{E}_{r,s}^{-r-k} = \nu\, \mathcal{J}_r^{\mathrm{hol}}(W) \otimes \Lambda^k(\epsilon_{w_1}, \ldots, \epsilon_{w_s}), \qquad d_E = (-1)^r d_w.$$

Let i be inclusion and p projection, so $ip = P$.

Proposition 14.8. *The maps satisfy*

$$pi = 1, \qquad dh + hd = 1 - ip, \qquad h^2 = hi = ph = 0.$$

They commute with holomorphic source differential operators and extension by zero in the tangent variables.

Proof. In one normal direction, integration gives $dh_j + h_j d = 1 - P_j$ and $h_j^2 = h_j P_j = P_j h_j = 0$. Its exterior sign makes h_j anticommute with the differential in every other direction. Each P_j commutes with all these operators. The telescoping sum gives

$$dh + hd = \sum_j P_1 \cdots P_{j-1}(1 - P_j) = 1 - P.$$

In a product of two distinct homotopy summands, the later one has a projection in the earlier normal direction. This kills the earlier homotopy, in either order. The side conditions follow.

Formal antiholomorphic integration and evaluation commute with holomorphic multiplication and differentiation. All maps preserve common tangent support and act on one coefficient at a time. These facts prove the remaining continuity and naturality statements. □

Write $A = \bar{A} j$ for the jet factor of an operation. Give $\mathrm{Hom}_{\mathrm{cont}}(\mathcal{E}_{r,s}, F)$ the Hom differential induced by Q, d_E.

Theorem 14.9. *There is a continuous cochain deformation retraction*

$$O_Z(F) \rightleftarrows \mathrm{Hom}_{\mathrm{cont}}(\mathcal{E}_{r,s}, F)$$

with maps

$$\mathbf{P}(A) = \bar{A}i, \qquad \mathbf{I}(B) = Bpj, \qquad \mathbf{H}(A) = (-1)^q \bar{A}hj \quad (|A| = q).$$

It is natural in continuous cochain maps on F and in the individual holomorphic right differential actions. If an operation has normal order at most N, projection and inclusion preserve that bound and the homotopy lowers it to $N - 1$. The homotopy vanishes at order zero.

Proof. Taylor factorization identifies the supported space with continuous Hom from the jet complex. Precomposition is continuous on its bounded-set topology. For degree q,

$$\delta((-1)^q Ah) = (-1)^q QAh + Ahd, \qquad (-1)^{q+1}(\delta A)h = (-1)^{q+1}QAh + Adh.$$

Their sum is $A(1 - ip)$. The contraction side conditions follow by precomposition. The formulas commute with target cochain maps and with holomorphic differentiation of tests.

Every term of h raises normal monomial degree by one, whereas i, p never lower it. This proves the order bounds, also when a bound is fixed only for one target seminorm. □

For one normal coordinate and scalar target, let c°_{ab} and c^{I}_{ab} extract $[u^a \bar{u}^b]$ on the unmarked and marked component. Then

$$\delta c^{\circ}_{ab} = (b+1)c^{\mathrm{I}}_{a,b+1}, \qquad \mathbf{H}c^{\mathrm{I}}_{ab} = b^{-1}c^{\circ}_{a,b-1} \quad (b > 0).$$

The surviving classes c^{I}_{a0} have degree one. Their densities $\eta_a = (-1)^a \partial_u^a \delta_0 / a!$ obey the right-module rules

$$\eta_a \partial_u = (a+1)\eta_{a+1}, \qquad \eta_a u = \eta_{a-1}\,(a > 0), \qquad \eta_0 u = 0.$$

The positive sign in the right derivative comes from precomposition on tests. It is distinct from the left distributional derivative in (14.12).

Formal completion permits increasing normal order. For example, $\sum_{j\geq 0} \chi^j c^{\mathrm{I}}_{j0}$ is continuous coefficientwise in χ. No fixed finite sum of principal parts represents it. Such a sum would annihilate every sole coefficient u^{N+1} beyond its order N, whereas this operation returns χ^{N+1}. Section 14.12 distinguishes this coefficientwise support from ordinary algebraic support.

Let $C_{\mathbf{a}} : \mathcal{E}_{r,s} \to \mathcal{K}(W)$ remove ν and extract its $u^{\mathbf{a}}$ coefficient. It has degree r and satisfies $C_{\mathbf{a}} d_E = (-1)^r d_w C_{\mathbf{a}}$. For any continuous degree-zero cochain map $M : \mathcal{K}(W) \to F$, the operation

$$A_{\mathbf{a}}(M) = MC_{\mathbf{a}}pj \tag{14.14}$$

is a closed element of $O^r_Z(F)$. Indeed $QM = Md_w$ gives $QA_{\mathbf{a}} = (-1)^r A_{\mathbf{a}} d_\partial$. Coefficient differentiation and multiplication give its normal right actions exactly as above.

An augmented target gives a simple nonzero class. Suppose $1 \in Z^0 F$ and a continuous cochain map $\varepsilon : F \to \mathbb{C}$ satisfies $\varepsilon(1) = 1$. Then

$$V_Z(\kappa\nu) = \pi^{-s} \int_W \kappa_{\mathbf{0},\mathbf{0}}(w)\, d^{2s}w \; 1$$

and zero on tangent-marker components defines a nonzero class of degree r. Integration kills compact tangent derivatives. After applying ε, the reduced Hom complex has no degree below r, and V_Z evaluates nontrivially on a test of nonzero integral. It cannot be a boundary.

When Z has positive codimension, every separated annular test vanishes near Z, so every supported operation kills it. Such tests do not detect this class. The class belongs to the occupied source set. The scalar of an empty configuration is separate data.

Two adapted charts give two retractions of the same supported Hom complex. Their transition is $T_{ba} = \mathbf{P}_b\mathbf{I}_a$. For three charts,

$$K_{cba} = -\mathbf{P}_c\mathbf{H}_b\mathbf{I}_a, \qquad d_cK_{cba} + K_{cba}d_a = T_{cb}T_{ba} - T_{ca}.$$

This follows by inserting $\partial\mathbf{H}_b + \mathbf{H}_b\partial = 1 - \mathbf{I}_b\mathbf{P}_b$. Taking $c = a$ gives a homotopy inverse. These transitions compare normal presentations on the same support, and preserve the intrinsic holomorphic right actions.

14.6 The tangent complex of the closed field

The preceding operation model accepts any specified target complex. It does not identify different closed theories. This distinction already appears in tangent cohomology.

Put $O = \mathbb{C}[[q, p]]$ and $V = O \otimes \Lambda(\theta_q, \theta_p)$, with $|\theta_q| = |\theta_p| = 1$ and $\pi = \theta_q\theta_p$. Fix the signs

$$\Delta(a\theta_q + b\theta_p) = a_q + b_p, \qquad \Delta(g\pi) = g_q\theta_p - g_p\theta_q,$$
$$d_\pi f = f_p\theta_q - f_q\theta_p, \qquad d_\pi(a\theta_q + b\theta_p) = (a_q + b_p)\pi.$$

The unspecified extreme components are zero. Direct differentiation gives $d_\pi^2 = \Delta^2 = d_\pi\Delta + \Delta d_\pi = 0$. Define the fixed symplectic-background tangent complex

$$C_{\mathrm{symp}} = (V[[u]][2], d_\pi + u\Delta), \qquad |u| = 2.$$

The background is fixed before completion. This is not evaluation at one of a background parameter completed at zero.

For homogeneous coefficient degree m define

$$Hf = 0, \qquad H(a\theta_q + b\theta_p) = \frac{pa - qb}{m+1}, \qquad H(g\pi) = \frac{qg\theta_q + pg\theta_p}{m+2}.$$

Apply the formulas termwise to formal series and put $p_u(f + a\theta_q + b\theta_p + g\pi) = f(0) - ug(0)$. If i includes constants, then

$$(d_\pi + u\Delta)H + H(d_\pi + u\Delta) = 1 - ip_u. \tag{14.15}$$

Indeed $\theta_q \mapsto dp$, $\theta_p \mapsto -dq$ identifies the d_π part with radial de Rham contraction. On functions and vectors, $\Delta H + H\Delta = 0$. On a positive-degree bivector its two terms cancel, while on a constant bivector it is evaluation at zero. This gives exactly the term $-ug(0)$ in the projection.

For comparison, use formal mixed jets of the unmodified geometric BF fields:

$$\mathcal{A} = O[[r, s, \bar{q}, \bar{p}]] \otimes \Lambda(dr, ds, d\bar{q}, d\bar{p}),$$

with the de Rham–Dolbeault differential in the four displayed jet variables. Before a normal-support or relative-source condition is imposed, its zero-field tangent is

$$C_{\mathrm{BF}} = \mathcal{A}[[u]][1] \oplus \mathcal{A}[[u]][2],$$

where u is a spectator. Formal radial integration gives a continuous contraction onto $O[[u]][1] \oplus O[[u]][2]$ with zero differential.

Proposition 14.10. *The complexes C_{BF} and C_{symp} admit no quasi-isomorphism over $\mathbb{C}[[u]]$.*

Proof. Equation (14.15) gives $H^\bullet(C_{\mathrm{symp}}) = \mathbb{C}[[u]][2]$. Thus its degree -1 cohomology is zero. The BF contraction gives $H^{-1}(C_{\mathrm{BF}}) = O$. A quasi-isomorphism would identify these groups. □

A normally supported cone, another background, a quotient or extra fields changes this comparison. Such a change must be specified with its map. The proposition concerns the two unchanged tangent complexes, and gives no obstruction to every possible relative closed-field model.

14.7 Closed operations with one normal current

A brane source may fail the closed Maurer–Cartan equation. Retaining that failure as curvature gives a precise local model.

Work on a complex polydisk with coordinates (p, q, z), volume $dp \wedge dq \wedge dz$ and source divisor $D = \{p = 0\}$. Let ξ_p, ξ_q, ξ_z be odd polyvector generators. Antiholomorphic differentials precede them, and every exterior generator has degree one. On smooth Dolbeault polyvectors define

$$\bar\partial = \sum_x d\bar x\, \partial_{\bar x}, \qquad \Delta = \sum_x \partial_x \partial^L_{\xi_x}.$$

The left odd derivatives include signs from all preceding exterior generators. The operators anticommute and square to zero.

Let $\mathscr{A}$ be this smooth graded algebra and $\mathscr{M}$ its module of normal distributions in p with smooth distribution-valued dependence on (q, z). Smooth multiplication and differentiation act on $\mathscr{M}$. Use the square-zero extension

$$\mathscr{B} = \mathscr{A} \ltimes \nu(\mathscr{M}), \qquad (a + \nu T)(b + \nu S) = ab + \nu(aS + Tb),$$

with graded module signs. The symbol ν names a second summand of degree zero. Its products with itself vanish.

In particular $\nu(1)$ is different from the algebra unit. The identity $p\nu(1/p) = \nu(1)$ does not invert p in the unital algebra.

Complete in a source parameter λ of degree zero and a descendant u of degree two. Set

$$E = \xi_p \xi_q, \qquad C = \nu(p^{-1}\xi_q), \qquad G = \nu(p^{-1}\xi_q\xi_z), \qquad K = \nu(\bar\partial(p^{-1})\xi_q),$$
$$P = E + aG, \qquad a = \sigma\lambda, \qquad \sigma \in \{1, -1\}.$$

The locally integrable function $1/p$ fixes the current. With the positive complex orientation,

$$\bar\partial(1/p) = \pi\delta_0\, d\bar p, \qquad d(dp/p) = 2\pi i\, \delta_0\, dx \wedge dy. \tag{14.16}$$

The common ξ_q gives $P^2 = 0$, and $\Delta P = 0$.

On $\mathscr{B}[[\lambda, u]][2]$ use symmetric Taylor operations. Before inserting P, let $b_0 = 0$, $b_1 = \bar\partial + u\Delta$ and, for $n \geq 2$, set

$$b_n(u^{k_1}x_1, \dots, u^{k_n}x_n) = \begin{cases} -\dfrac{(n-2)!}{k_1! \cdots k_n!}\Delta(x_1 \cdots x_n), & \sum_i k_i = n-2, \\ 0, & \text{otherwise.} \end{cases} \tag{14.17}$$

Each operation has degree one after the shift. They are $\mathbb{C}[[\lambda]]$-multilinear, but are not multilinear over the descendant ring.

Theorem 14.11. *Translation by P defines a curved L_∞ algebra. Its curvature and unary operation are*

$$b_0^P = aK\xi_z, \qquad b_1^P x = \bar\partial x + u\Delta x - \Delta(Px_0), \qquad x = \sum_{k\ge 0} u^k x_k.$$

For $n \ge 2$, the translated nth operation consists of (14.17) *and the term*

$$-\frac{(n-1)!}{k_1!\cdots k_n!}\Delta(Px_1\cdots x_n) \quad \textit{when} \ \sum_i k_i = n-1.$$

Restriction of the chart commutes with every operation. An operation with one normal-distribution input preserves its support, and an operation with two such inputs is zero.

Proof. For smooth compactly supported polyvectors, the classical BCOV interaction has Taylor coefficients

$$I(\mu) = \sum_{m\ge 3}\frac{1}{m!}\sum_{k_1+\cdots+k_m=m-3}\frac{(m-3)!}{k_1!\cdots k_m!}\operatorname{Tr}_\Omega(\mu_{k_1}\cdots\mu_{k_m}).$$

Here $\operatorname{Tr}_\Omega(\gamma)$ is the integral of $(\gamma \lrcorner \Omega)\wedge\Omega$ and vanishes outside top total degree. The genus-zero descendant recursion, the second-order identity for Δ and integration by parts give its classical master equation.[1]

Differentiating once and applying the primary Poisson kernel gives (14.17). The state identities are the signed unshuffle sums

$$\sum_{I\sqcup J=[n]}\chi(I,J)\,b_{|J|+1}(b_{|I|}(x_I),x_J) = 0.$$

For one second-summand input, regularize its coefficient in the normal variable. The desired identity is the term linear in an auxiliary square-zero parameter in the smooth identity. Each term contains derivatives of only one current multiplied by smooth functions, so it passes to the distributional limit. With two second-summand inputs every nested term vanishes, either in the inner operation or in the outer one.

Translation of a square-zero formal vector field preserves its square. Here the Taylor translation is finite: all products with two copies of P vanish. Its constant term is $\bar\partial P = aK\xi_z$, and its other terms give the stated formulas. Smooth multiplication and differentiation preserve distributional support and commute with restriction. □

The first curved identity is already nontrivial on a holomorphic scalar $h(z)$:

$$b_1^P h = ah'C, \qquad (b_1^P)^2 h = ah'K, \qquad b_2^P(b_0^P,h) = -ah'K.$$

The last equality uses the left odd derivative through $d\bar p$. Thus the unary operator on the whole carrier is not a differential for $a \neq 0$. The interaction itself extends to monomials with one current by pairing it against compact smooth factors. A nondegenerate BV pairing on the entire square-zero extension, or a coupled disk action, is additional data.

There is nevertheless a supported subcomplex. Let $\mathscr{N} \subset \mathscr{M}$ consist of currents supported on D, locally of bounded finite normal order. On $\nu(\mathscr{N})[[\lambda,u]]$, the singular part of P has zero product with every input. The unary operation is

$$d_E = \bar\partial + u\Delta - \Delta M_E\Pi_0, \qquad \Pi_0 x = x_0,$$

where M_E is multiplication by E. Define $S_u x = (x-x_0)/u$, $A_E = 1 - M_E S_u$ and $Q_E = \bar\partial + [E,-] + u\Delta$, with $[E,-] = M_E\Delta - \Delta M_E$.

[1] K. Costello and S. Li, *Quantum BCOV theory on Calabi–Yau manifolds and the higher genus B-model*, arXiv:1201.4501. See Section 2.10, especially Lemma 2.10.2 and its proof. The displayed state operators use the contragredient sign convention for the nonlinear differential.

Proposition 14.12. *The continuous support-preserving conjugations are*

$$d_E = A_E Q_E A_E^{-1}, \qquad A_E^{-1} = 1 + M_E S_u,$$

$$U_E = A_E u A_E^{-1} = u - M_E \Pi_0.$$

The residue K defines a nonzero class for d_E. Its first transported descendant is exact:

$$U_E K = uK = d_E(zK\xi_z). \tag{14.18}$$

Proof. The exterior factors give $M_E^2 = M_E \Delta M_E = 0$. Also $\bar\partial E = 0$, $S_u u = 1$ and $u S_u = 1 - \Pi_0$. Expanding the two conjugations proves the formulas. Hence U_E is a chain map. Ordinary multiplication by u need not be one, since

$$d_E(ux) - u d_E x = u\Delta(Ex_0).$$

To detect K, first quotient by positive descendant degree. The differential becomes $\bar\partial - \Delta M_E$. Next quotient by $\xi_p, \xi_z, d\bar q, d\bar z$. This is a quotient complex: a term removing ξ_z from Ex still has the factor E, and $Ex = 0$ if x already contains ξ_p.

Write $e_0 = d\bar p$, $e_1 = \xi_q$ and represent the distribution $\partial_p^i \partial_{\bar p}^j \delta_0$ by $R^i T^j$. The quotient is

$$\big(C^\infty(D)[R,T] \otimes \Lambda(e_0, e_1)\big)[[\lambda]], \qquad d = Te_0 - Re_1.$$

The sequence R, T is regular, so its degree-two cohomology is $C^\infty(D)[[\lambda]] e_0 e_1$. The image of K is $\pi e_0 e_1$, detected by setting $R = T = 0$. Thus K was not exact in the original complex.

Finally $\bar\partial K = \Delta K = EK = 0$ and $\Delta(zK\xi_z) = K$. The other terms of d_E on $zK\xi_z$ vanish, proving (14.18). □

The normal orientation retains a boundary term. Let χ be a radial cutoff equal to one near $p = 0$ and zero near the outer circle. For a compact tangential top form ψ_D of integral one,

$$\int d\chi \wedge \frac{dp}{p} \wedge \psi_D = -2\pi i.$$

The inner boundary has the opposite of the positive-circle orientation. Equation (14.16) supplies the corresponding positive supported term. Removing the normal divisor would discard this information.

14.8 Two sources and collision extensions

A unit one-source residue does not determine a two-source contact. Write $\delta_a(z) = e^{-|z|^2/a}/(\pi a)$ and let $\Delta_{\mathbb{R}^2} = \partial_x^2 + \partial_y^2$. For $\alpha, \beta > 0$, multiplication of the smooth densities gives

$$\delta_{\alpha s}\delta_{\beta s} = \frac{1}{\pi(\alpha+\beta)s}\delta_{\alpha\beta s/(\alpha+\beta)}.$$

The centered Gaussian has zero odd moments and second moments $a/2$ in each real coordinate. Taylor expansion through degree three therefore yields

$$\delta_{\alpha s}\delta_{\beta s} = \frac{\delta_0}{\pi(\alpha+\beta)s} + \frac{\alpha\beta}{4\pi(\alpha+\beta)^2}\Delta_{\mathbb{R}^2}\delta_0 + O(s). \tag{14.19}$$

The remainder on a compact smooth test is bounded by a constant times s and its fourth-derivative seminorm.

The finite coefficient is $1/(16\pi)$ for equal widths and $1/(18\pi)$ for width ratio two. These are scalar coefficients after contraction. The calculation does not multiply two identical Dolbeault one-forms.

More generally, let t be a distribution on $\mathbb{R}^d \setminus \{0\}$ of finite scaling degree s. This means that $\lambda^{s'} t(\lambda\,\cdot)$ tends to zero distributionally for every $s' > s$. For $s \geq d$ put $m = \lfloor s - d \rfloor$ and choose $\chi = 1$ near zero with compact support. Taylor subtraction is

$$R_\chi t(\varphi) = t\left(\varphi - \chi \sum_{|\nu| \leq m} \frac{x^\nu}{\nu!} \partial^\nu \varphi(0)\right). \tag{14.20}$$

The pairing is the limit after removing a shrinking neighborhood of zero. On dyadic annuli, the Taylor remainder gives the convergent bound $2^{-j(d+m+1-s')}$ for $s < s' < d + m + 1$. This proves existence and continuity. The rescaled estimate gives scaling degree at most s. For $s < d$ the same annular estimate requires no subtraction.

Two extensions with this scaling bound differ by derivatives of delta of order at most m. Indeed their difference is supported at zero, and a distribution supported at one point has finite derivative order. Its highest derivative has scaling degree $d + |\nu|$, which gives the bound. In particular

$$(R_{\chi'} - R_\chi) t(\varphi) = - \sum_{|\nu| \leq m} \frac{\partial^\nu \varphi(0)}{\nu!} t((\chi' - \chi) x^\nu).$$

For $s < d$ the extension with the stated bound is unique. Nested and overlapping collisions still require compatible choices on the full collision diagram.

Closedness adds an independent condition. Let $Z_0 \subset Z_1$ be closed source sets and put $V^\circ = V \setminus Z_0$. Suppose A° is a closed degree-q operation supported on $Z_1 \cap V^\circ$.

Proposition 14.13. *If A° has a continuous degree-q extension $E \in \mathcal{O}_{Z_1}(F)$, then*

$$[\delta E] \in H^{q+1} \mathcal{O}_{Z_0}(F)$$

is independent of the extension. It vanishes exactly when there is a closed extension supported on Z_1. For a linear partial diagonal, its reduced representative is $\mathbf{P}(\delta E)$.

Proof. Restriction is precomposition by extension of compact tests by zero and commutes with δ. Thus δE is supported on Z_0 and is closed. Two extensions differ by an operation supported on Z_0, so their differentials differ by a boundary there.

If $\delta E = \delta C$ for a supported degree-q operation C, then $E - C$ is a closed extension. Conversely, a closed extension subtracted from E gives such a C. The normal-operation contraction preserves the class and supplies the reduced representative. □

This cohomology uses the actual image of δ, not its topological closure. If the permitted local corrections form a subcomplex $\mathcal{L} \subset \mathcal{O}_{Z_0}(F)$, the chosen E can be corrected only when $\delta E \in \delta(\mathcal{L}^q)$. Vanishing in the larger complex need not imply this. Existence of a continuous extension is also a hypothesis.

For n occupied positions, shifting the operation complex by $[n - 1]$ puts the current of an s-block partition in degree $r - (n - 1) = 1 - s$. A three-position pair-diagonal current thus has degree -1, while a small-diagonal current has degree zero. This permits a homotopy equation between them. The degree calculation does not construct the homotopy or identify its boundary with a collision defect.

14.9 A four-position attachment

Even a scalar collision coefficient can have several cohomology modules joined by a nonzero extension. Take $X = \mathbb{A}^1_{\mathbb{C}}$ and four labeled positions. Let

$$D = \Delta_{14}, \qquad P = \Delta_{13} \cap \Delta_{24}, \qquad \Delta = D \cap P, \qquad Z = D \cup P.$$

Here D is a divisor, P a codimension-two surface, and Δ the small diagonal. Work with right algebraic differential modules and define the specific coefficient

$$\mathcal{K}_4 = R\Gamma_Z(\omega_{X^4})[3].$$

This is a local-cohomology construction. Identifying it with another coefficient system requires a comparison map.

Proposition 14.14. *The only cohomology modules of $\mathcal{K}_4$ are*

$$H^{-2}(\mathcal{K}_4) = i_{D,*}\omega_D, \qquad H^{-1}(\mathcal{K}_4) = i_{P,*}\omega_P(*\Delta).$$

The resulting extension is nonzero. Its class spans

$$\operatorname{Ext}^2_{\mathcal{D}_{X^4}}(i_{P,*}\omega_P(*\Delta), i_{D,*}\omega_D) \cong \mathbb{C}.$$

It is the principal part along Δ in P, followed by the codimension-two local-cohomology boundary in D.

Proof. Use coordinates $a = x_1 - x_4$, $b = x_1 - x_3$, $c = x_2 - x_4$, $w = x_4$. Then $D = (a)$, $P = (b, c)$ and $\Delta = (a, b, c)$. The localization triangle for the pair $D \subset Z$ is

$$R\Gamma_D\omega \longrightarrow R\Gamma_Z\omega \longrightarrow R\Gamma_P(\omega[a^{-1}]) \longrightarrow R\Gamma_D\omega[1].$$

The first term has cohomology in degree one and the third in degree two. Their regular normal sequences give the two displayed modules after the shift by three.

To compute the connecting class, tensor with the inverse density and use left differential modules in these coordinates. In one variable, write $O = \mathbb{C}[x]$ and $M = \mathbb{C}[x, x^{-1}]/\mathbb{C}[x]$. The two-term differential-module resolution of M by right multiplication by x gives $\operatorname{Hom}_{\mathcal{D}}(M, O) = 0$ and $\operatorname{Ext}^1_{\mathcal{D}}(M, O) = \mathbb{C}$. The latter is detected by evaluation modulo x.

Also $\operatorname{Hom}_{\mathcal{D}}(O[x^{-1}], M)$ is a line generated by the principal part. A map from O to M must send 1 to a vector killed by ∂_x. Differentiation is injective on M, so that map is zero. Every map therefore factors through the principal-part quotient, whose differential-module endomorphisms are scalars.

In the a direction the connecting map is this principal part. In the b, c directions it is the tensor product of the two nonzero degree-one extension classes. The w direction has the identity of O_w. The tensor-product resolutions therefore give the stated one-dimensional Ext group. The localization boundary has coefficient one in each factor, so its class is nonzero. A direct sum of the shifted cohomology modules would have zero connecting class. □

The crossing module has poles along the small diagonal. Keeping only regular coefficients on P loses its principal-part boundary. Keeping the two cohomology modules but discarding their attachment also changes $\mathcal{K}_4$. A supported operation comparison must preserve both pieces and the connecting map.

14.10 A relative polar source algebra

The square-zero closed carrier and principal parts have different source products. There is a simple reason not to multiply all normal currents in an ordinary unital algebra. If a central element z has an inverse c with $zc = 1$ and annihilates ∂, then

$$\partial = (cz)\partial = c(z\partial) = 0.$$

Thus a nonzero source satisfying $z\partial = 0$ cannot coexist with that central inverse. A relative operator algebra retains the source by making the inverse one-sided.

On $\mathcal{P}_z$ write $e_n = [z^{-n-1}]$ and define

$$Le_0 = 0, \qquad Le_n = e_{n-1}\ (n > 0), \qquad Se_n = e_{n+1}.$$

Let P_0 project onto e_0. Then $LS = 1$ and $SL = 1 - P_0$. The algebra $\mathcal{T}$ generated by L, S has finite-matrix ideal

$$\mathcal{F} = \bigoplus_{i,j\ge 0} \mathbb{C}E_{ij}, \qquad E_{ij} = S^i P_0 L^j, \qquad E_{ij}E_{kl} = \delta_{jk}E_{il}.$$

Here finite matrices are matrices with finitely many nonzero entries in this basis. They are not all finite-rank endomorphisms of an algebraic direct sum.

The symbol sequence is exact:

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{T} \xrightarrow{\sigma} \mathbb{C}[\zeta, \zeta^{-1}] \longrightarrow 0, \qquad \sigma(S) = \zeta, \quad \sigma(L) = \zeta^{-1}. \tag{14.21}$$

Indeed $LS = 1$ reduces every word to $S^i L^j$. Repeated use of $SL = 1 - P_0$ expresses it as a Laurent lift plus a finite matrix. A nonzero Laurent symbol acts on all sufficiently high basis vectors and cannot be a finite matrix. This proves exactness and faithfulness of the shift presentation.

For $N \ge 1$, the projector

$$\Pi_N = 1 - S^N L^N = [L^N, S^N] = \sum_{i=0}^{N-1} E_{ii}$$

has corner $\Pi_N \mathcal{T} \Pi_N \cong \mathrm{Mat}_N$, with corner unit Π_N and trace N. The trace pairing is perfect since E_{ij} pairs with E_{ji}. At rank two, $\mathrm{Tr}([E_{01}, E_{10}]^2) = 2$. Diagonal projection kills both inserted matrix units and gives zero, despite preserving the trace of the unit.

Choose $T_{\zeta^a} = S^a$ for $a \ge 0$ and $T_{\zeta^{-b}} = L^b$ for $b > 0$. The contact is $c(f, g) = T_f T_g - T_{fg} \in \mathcal{F}$. For $a, b > 0$,

$$c(\zeta^a, \zeta^{-b}) = -\sum_{j=\max(0,b-a)}^{b-1} E_{j+a-b,j}. \tag{14.22}$$

The product $S^a L^b$ kills e_j for $j < b$. Its Laurent lift differs exactly at the indices in the sum, proving the formula. Reverse-order and same-sign monomial contacts vanish.

Associativity gives the full factor identity

$$T_f c(g, h) - c(fg, h) + c(f, gh) - c(f, g)T_h = 0.$$

For example, $c(\zeta^3, \zeta^{-2}) = -E_{10} - E_{21}$.

The finite-matrix trace gives

$$\mathrm{Tr}_{\mathcal{F}}[T_f, T_g] = \mathrm{Res}_{\zeta=0} f\, dg =: \omega(f, g). \tag{14.23}$$

For monomials $f = \zeta^a$, $g = \zeta^b$, a nonzero trace requires $a + b = 0$ and has value b. These are exactly the residue coefficients. Changing a lift by a finite matrix changes its commutator by terms of trace zero. Antisymmetry and the product rule show that ω is a cyclic one-cocycle. No cyclic trace on all of $\mathcal{T}$ can extend this trace, since $[L, S] = P_0$ has trace one.

Every element is uniquely $T_f + F$. Define $\widetilde{\tau}(T_f + F) = \mathrm{Tr}_{\mathcal{F}} F$. With the bare scalar Hochschild differential,

$$b\widetilde{\tau} = \sigma^* \omega.$$

The finite-matrix cross commutators have zero trace, and (14.23) gives the remaining term. Consequently $(\widetilde{\tau}, \omega)$ is closed in the relative cyclic complex with differential

$$d(\tau, \gamma) = (b\tau - \sigma^*\gamma, -b\gamma).$$

Its degree-n term is $C^n_\lambda(\mathcal{T}) \oplus C^{n+1}_\lambda(\mathbb{C}[\zeta, \zeta^{-1}])$, where cyclic cochains obey $\phi(a_n, a_0, \dots, a_{n-1}) = (-1)^n \phi(a_0, \dots, a_n)$.

Normal derivatives also act on $\mathcal{P}_z$: $\partial_z e_n = -(n+1)e_{n+1}$ and $[\partial_z, L] = 1$. Adjoining this operator enlarges $\mathcal{T}$. Independent normal directions use tensor products, and their contacts compose as operators. Since $P_0^2 = P_0$, no product-preserving map can send a square-zero source to P_0. The native action must determine which source product its collisions realize.

The one-sided inverse also gives a homotopy obstruction. Let D be a unital dg algebra with closed S, L in degree zero, $LS = 1$, $SL = 1 - e$. Suppose eDe has a closed degree-zero functional τ with $\tau(e) = 1$. Require the cohomology images $\zeta \mapsto S$, $\zeta^{-1} \mapsto L$. There is no such cochain map from the Laurent algebra to D that is unital and multiplicative up to homotopy.

To prove this, write $F_1(1) = 1 + dh_0$, $F_1(\zeta) = S + dx$ and $F_1(\zeta^{-1}) = L + dy$. If $dH(a, b) = F_1(a)F_1(b) - F_1(ab)$, then

$$h = -H(\zeta, \zeta^{-1}) - h_0 + xL + Sy + x\, dy \quad \text{satisfies} \quad dh = e.$$

The closed idempotent gives $d(ehe) = e$ inside the corner. Applying τ would give $1 = 0$. This obstruction retains the prescribed generators and corner detector. It does not exclude a relative comparison such as the cyclic pair above.

14.11 The joint source and quantum comparison

The coupled classical theory admits more loops than its pure cotangent sector. In a connected graph, let v_B be the number of bulk vertices, v_C coupling vertices and v_O open cubic vertices. If b cotangent and a open legs are external, then

$$e_B = v_B - b, \qquad e_A = \frac{2v_C + 3v_O - a}{2}, \qquad b_1 = 1 - b + \frac{v_O - a}{2}.$$

Two open cubic vertices joined by three propagators already have two loops. Any vanishing of their weight requires a geometric calculation.

In a finite BV algebra, or on a specified renormalized domain where every term is defined, write $I = I_0 + \hbar I_1 + \hbar^2 I_2 + \cdots$. Suppose I_0 solves the classical master equation and put $Q_{I_0} = Q + \{I_0, -\}$. The quantum equation gives

$$\begin{aligned} Q_{I_0} I_1 &= -\Delta I_0, \\ Q_{I_0} I_n &= -\Delta I_{n-1} - \tfrac{1}{2} \sum_{a=1}^{n-1} \{I_a, I_{n-a}\}, \qquad n \geq 2. \end{aligned} \tag{14.24}$$

The BV identities make the next right side closed once the preceding equations hold. A solution requires its primitive in the permitted local cyclic complex, including its normal support and boundary conditions.

If a finite regularization realizes the change of variables in (14.7) canonically, its triangular matrix has identity diagonal and Berezinian one. It preserves the finite BV measure and Laplacian. Quantum master solutions for the separated open sector and the bulk sector with its residual cubic therefore pull back to a coupled solution. This implication requires both separated solutions. It does not construct a continuum regulator or its allowed counterterms.

A source comparison must retain its homotopy as well. For cochain maps $f : L_{\mathrm{nat}} \to L$ and $g : L_{\mathrm{cyc}} \to L$, the joint tangent complex is

$$\mathcal{P}^n = L^n_{\mathrm{nat}} \oplus L^n_{\mathrm{cyc}} \oplus L^{n-1}, \qquad D(a, b, h) = (da, db, f(a) - g(b) - dh).$$

The two cochain-map identities give $D^2 = 0$. The component h is the comparison homotopy.

For nonlinear deformation theory, use the path-object homotopy pullback of the dg Lie algebras. The linear cone alone supplies no Lie bracket. In the deformation complex, the next Maurer–Cartan obstruction has degree two.

Problem 14.15. Fix the polar action, its Neumann boundary domain and its normal lattice. Construct a map from independently defined sourced closed fields to continuous cyclic operations of the open fields. Determine the permitted normal markings and structured homotopies from these physical data. Match the quadratic pair, first ghost descendants, unit, support and cyclic pairing.

Extend the map to simultaneous source insertions and all supported collision and sewing operations. Preserve the four-position attachment and identify the image of the allowed local counterterms in the supported operation complex. For a quantum comparison, construct a common regulator and solve (14.24).

The real internal defect, the holomorphic normal source and a spectral brane have independently specified supports. The cyclic cap and the scalar relative models give constructions on their stated domains. Their identification with the full sourced closed multiplet is the additional map required by the problem.

14.12 Completion in one normal direction

Increasing normal order is compatible with formal coefficients, but changes what it means for a coefficient to be supported. We isolate this issue in one algebraic normal direction. The resulting reconstruction and convolution have explicit maps. Their comparison with the sourced fields of Problem 14.15 remains separate.

Put $R = \mathbb{C}[[\hbar]]$ and $R_r = R/(\hbar^r)$. Here $\hbar$ is a coefficient parameter independent of the normal coordinate u. Let

$$D_R = R\langle u, \partial \rangle / (\partial u - u\partial - 1), \qquad P_R = R[u, u^{-1}]/R[u].$$

We use right differential modules. In the density basis $\beta_m = [u^{-m-1}]$, their normal actions are

$$\beta_0 u = 0, \qquad \beta_m u = \beta_{m-1} \quad (m > 0), \qquad \beta_m \partial = (m+1)\beta_{m+1}.$$

Thus $P_R = D_R/uD_R$, with a free right-module resolution

$$[D_R \xrightarrow{L_u} D_R]_{[-1,0]} \longrightarrow P_R. \tag{14.25}$$

The map L_u is left multiplication by u. The shift convention is $X[s]^i = X^{i+s}$.

Work in the derived category of right D_R-modules. For a complex X, set

$$X_r = X \otimes^{\mathbb{L}}_R R_r, \qquad \Lambda X = \mathbf{R}\lim_{\longleftarrow r} X_r.$$

Ordinary open restriction $X[u^{-1}]$ localizes its underlying $R[u]$-complex and retains the induced differential action. It is exact.

We call X *derived adically supported* if $X_r[u^{-1}] \simeq 0$ for every r. Let $\mathcal{C}_\hbar$ be the full category of such complexes for which $X \to \Lambda X$ is an equivalence, and let $\mathcal{G}_\hbar$ be the category of derived-complete R-complexes. Neither category has a boundedness condition. There is no completion in u.

Lemma 14.16. *The completion map induces $(\Lambda X)_r \simeq X_r$ for every complex X and every r. Moreover,*

$$X_r[u^{-1}] \simeq 0 \text{ for all } r \quad \Longleftrightarrow \quad X_1[u^{-1}] \simeq 0 \quad \Longleftrightarrow \quad \Lambda(X[u^{-1}]) \simeq 0.$$

Completion preserves derived adic support.

Proof. The perfect scalar complex $[R \xrightarrow{\hbar^r} R]$ computes R_r, so its tensor commutes with homotopy limits. Put $Y = X_r$. For $n \geq r$, the resolution computing $Y \otimes^{\mathbb{L}}_R R_n$ has zero differential in its two-term resolution direction. Its transition is the identity on the degree-zero term and multiplication by $\hbar$ on the degree-minus-one term.

The second tower has zero composite after r transitions. In the product model for its homotopy limit, $1 -$ shift has the finite geometric-series inverse. That limit vanishes, while the first tower has limit Y. This proves the assertion about the natural completion map, also for unbounded X.

Put $N = X[u^{-1}]$. Exact localization commutes with tensoring by R_r. The condition $N_1 \simeq 0$ says that multiplication by $\hbar$ is an equivalence on N. It is equivalent to $N_r \simeq 0$ for all r, and hence to $\Lambda N \simeq 0$ by the first assertion. Applying that assertion again proves preservation of support. □

For an R-flat module, the support condition is equivalent to

$$\text{for every } v \text{ and } r \text{ there is } N \text{ such that } vu^N \in \hbar^r X.$$

For a general complex the condition concerns all cohomology of the derived reductions, including their Tor terms. Ordinary quotient modules alone do not supply that condition.

Example 14.17. The completion of P_R is the flat module

$$\widehat{P}_R = \left\{ \sum_{m \geq 0} a_m \beta_m : a_m \in R,\ a_m \longrightarrow 0 \ \hbar\text{-adically} \right\}.$$

Its reduction is P_{R_r}, so it lies in $\mathcal{C}_\hbar$. Nevertheless it has nonzero ordinary open restriction. Indeed

$$p = \sum_{m \geq 0} \hbar^m \beta_m \qquad \text{satisfies} \qquad pu = \hbar p,$$

and no power of u kills p.

More precisely, its u-torsion is exactly P_R. Lowering is surjective, so $\widehat{P}_R[u^{-1}] = \widehat{P}_R/P_R$. Multiplication by $\hbar$ is invertible on this quotient. Subtract a finite representative modulo $\hbar$ and divide to prove surjectivity. Finite support proves injectivity. The open quotient is therefore nonzero but has zero derived completion.

Any complete complex with the specified compatible reductions P_{R_r} is their derived inverse limit $\widehat{P}_R$. Consequently this tower has no complete realization with ordinary u-support. Its derived adic realization exists.

Completion is left adjoint to the inclusion of complete complexes. Taking the inverse limit of the multiplication triangles gives

$$T(X) \longrightarrow X \longrightarrow \Lambda X, \qquad T(X) = \mathbf{R}\varprojlim(\cdots \xrightarrow{\hbar} X \xrightarrow{\hbar} X).$$

Multiplication by $\hbar$ is invertible on $T(X)$. Maps from such a complex to a complete complex vanish: after reduction the same scalar action is both invertible and nilpotent on the mapping complex. This proves the stated adjunction.

14.12.1 Reconstruction from the normal ground

Define induction and restriction by

$$I(E) = \Lambda(P_R \otimes^{\mathbb{L}}_R E), \qquad K(M) = \mathrm{RHom}_{D_R}(P_R, M).$$

Resolution (14.25) computes $K(M)$ as $[M \xrightarrow{\cdot u} M]$ in normal degrees 0, 1. This arrow is R-linear, without right D_R-linearity.

Theorem 14.18. *The adjunction maps give inverse equivalences*

$$I : \mathcal{G}_\hbar \rightleftarrows C_\hbar : K.$$

No flatness or finite-rank hypothesis on the scalar complex is required.

Proof. Let B be a commutative complex algebra. Write D_B, P_B for the normal algebra and module over B, and put $K_B(V) = \mathrm{RHom}_{D_B}(P_B, V)$. First suppose V is a module with locally nilpotent u-action. The finite Taylor projector is

$$\pi(v) = \sum_{j\ge 0} \frac{(-1)^j}{j!}(vu^j)\partial^j.$$

The identity $\partial^j u = u\partial^j + j\partial^{j-1}$ gives $\pi(v)u = 0$. With $E = \ker u$, define

$$P_B \otimes_B E \longrightarrow V, \qquad \beta_m \otimes e \longmapsto e\partial^m/m!.$$

It respects both right actions. Applying the highest power of u in a finite relation proves injectivity. Surjectivity follows from

$$v = \sum_{m\ge 0} \frac{\pi(vu^m)\partial^m}{m!}.$$

On expansion, the coefficient in total degree l is $\sum_{j=0}^{l}(-1)^j/(j!(l-j)!)$, equal to zero for $l > 0$ and one for $l = 0$. In particular, $u : V \to V$ is surjective.

Now suppose the cohomology of a complex V is locally u-nilpotent. The two-term restriction complex gives $H^qK_B(V) = \ker(u : H^qV \to H^qV)$, because the preceding cokernel vanishes. Since P_B is free over B, the

preceding module calculation proves that evaluation $P_B \otimes^{\mathbb{L}}_B K_B(V) \to V$ is an equivalence. The unit is an equivalence because $u : P_B \to P_B$ is a split scalar surjection with kernel $B\beta_0$.

Apply this argument over $B = R_r$. The finite free resolution gives $K(M)_r \simeq K_r(M_r)$ and $K(\Lambda M) \simeq \Lambda K(M)$. Thus K preserves completeness, and the reductions of $I(E)$ are $P_{R_r} \otimes^{\mathbb{L}}_{R_r} E_r$. The unit and counit are equivalences at every precision. Both sides are complete, so they are equivalences before reduction as well. □

For a complete differential-module complex X, evaluation $IK(X) \to X$ is consequently right adjoint to the inclusion of $C_\hbar$. Limits taken in this supported category are

$$\varprojlim_{C_\hbar} M_i \simeq I\big(\mathbf{R}\varprojlim_i K(M_i)\big).$$

They need not agree with ambient differential-module limits.

14.12.2 *Additive convolution and relative tensors*

The normal carrier has a geometric operation defined before using the equivalence. Let $a(x, y) = x + y = t$ and let a_+ be right differential-module direct image, computed by the relative de Rham complex. Set

$$M *_+ N = \Lambda a_+(M \boxtimes^{\mathbb{L}}_R N), \qquad E\widehat{\otimes}_R F = \Lambda(E \otimes^{\mathbb{L}}_R F).$$

The second operation makes $\mathcal{G}_\hbar$ symmetric monoidal: the kernel of completion consists of the $\hbar$-invertible complexes and is a tensor ideal.

Proposition 14.19. *There is a natural equivalence*

$$I(E) *_+ I(F) \simeq I(E\widehat{\otimes}_R F).$$

It makes I a symmetric monoidal equivalence, with convolution unit $\widehat{P}_R$.

Proof. At precision r, change coordinates to $t = x + y$, $s = y$. This identifies the normal distribution module $P_{x,R_r} \boxtimes P_{y,R_r}$ with $P_{t,R_r} \boxtimes P_{s,R_r}$. The relative de Rham complex is

$$[P_{s,R_r} \xrightarrow{\partial_s} P_{s,R_r}]_{[-1,0]}.$$

Its differential sends β_m to $(m+1)\beta_{m+1}$. It is injective, with free cokernel $R_r\beta_0$ and a scalar splitting. Hence the direct image is P_{t,R_r}, with no shift, also after tensoring with arbitrary scalar complexes.

The coordinate change sends

$$\beta_i(x)\beta_j(y) \longmapsto \sum_{k=0}^{j} \binom{i+k}{k} \beta_{i+k}(t)\beta_{j-k}(s).$$

Relative de Rham projection therefore gives $\binom{i+j}{i}\beta_{i+j}(t)$. The multinomial identity gives the same result for three inputs in either order. Addition and iterated relative de Rham complexes supply the coherent associativity and symmetry maps. The origin inclusion supplies the unit.

Scalar reduction commutes with these complexes. Lemma 14.16 shows that completing the inputs changes none of their reductions. The comparison is an equivalence at each precision. Completion proves the assertion. □

Let B be an associative algebra in $\mathcal{G}_\hbar$, with a right module E and a left module F. Transporting their actual actions through the equivalence and realizing the two-sided bar gives

$$I(E) *^{\mathbb{L}}_{I(B)} I(F) \simeq I\big(\Lambda(E \otimes^{\mathbb{L}}_B F)\big). \tag{14.26}$$

Indeed the bar terms correspond under the monoidal equivalence, including every face and degeneracy. Colimits in the complete scalar category are completed ambient colimits. This proves the formula and support of the result without a flatness assumption.

Flatness itself is not preserved. Take $B = \widehat{R[t]}^{(\hbar)}$, the series with coefficients tending to zero, and the two complete flat character modules $E_0 = R$, $E_\hbar = R$ with $t = 0, \hbar$. The free resolution of E_0 has differential t, so

$$E_0 \otimes^{\mathbb{L}}_B E_\hbar = [R \xrightarrow{\hbar} R]_{[-1,0]} \simeq R/(\hbar).$$

At precision r, this complex has both $H^{-1} = \mathbb{C}\hbar^{r-1}$ and $H^0 = \mathbb{C}$. Both remain supported after normal induction. A flat-only coefficient class would lose this relative tensor.

Two other operations have different output types. Write $(-)^t$ for transpose under $u \mapsto u$, $\partial \mapsto -\partial$. Then

$$\Lambda\big(I(E) \otimes^{\mathbb{L}}_{D_R} I(F)^t\big) \simeq (E \widehat{\otimes}_R F)[1], \qquad \mathrm{RHom}_{D_R}(I(E), I(F)) \simeq \mathrm{RHom}_R(E, F).$$

For the first formula, the normal resolution gives the surjective map $u : P^t_{R_r} \to P^t_{R_r}$ in degrees $-1, 0$, with kernel R_r. The second is full faithfulness.

Both outputs are scalar complexes. Neither retains a normal coordinate on which to impose support.

14.12.3 Products, duality and the class of tests

Proposition 14.20. *The ambient product $W = \prod_{i \geq 0} \widehat{P}_R$ is flat and complete, but is not derived adically supported.*

Proof. It is torsion-free over the discrete valuation ring R, hence flat. Componentwise division and inverse limits give $W/\hbar^r W = \prod_i P_{R_r}$ and completeness. In its special fiber, let $e = (\beta_i)_{i \geq 0}$. For every N, coordinate $i = N$ of eu^N is β_0. Thus no power of u kills e, and $W_1[u^{-1}] \neq 0$. □

The product inside $C_\hbar$ is instead $U = I(R^{\mathbb{N}})$. In coefficients $a_{i,m}$ its inclusion into W imposes

$$\text{for every } r \text{ there is } M_r \text{ such that } a_{i,m} \in \hbar^r R \quad \text{for all } i \text{ and } m \geq M_r.$$

The ambient product permits the mode bound to depend on i. The diagonal family above detects the nonzero quotient $L = W/U$.

This inclusion is $\hbar$-saturated, so L is flat. As a cofiber of complete complexes it is derived complete. Lowering is surjective on U and W, with the same kernel $R^{\mathbb{N}}\beta_0$. Hence u is invertible on L and

$$K(L) = 0, \qquad \mathrm{RHom}_{D_R}(D_R, L) = L \neq 0.$$

The normal ground misses a quotient that the free differential-module test detects.

Define the ordinary normalized coefficient dual with complete target by

$$\mathbb{D}_\hbar(M) = \mathrm{RHom}_{D_R}(M, \widehat{D}_R)^t[1], \qquad \widehat{D}_R = \Lambda D_R.$$

The shift is the dimension-one normalization. This dual takes values in ordinary differential modules.

Proposition 14.21. *The flat supported complete module $M = \Lambda(\bigoplus_{i \geq 0} P_R)$ has $\mathbb{D}_\hbar(M) \simeq W$, which is not supported. If E is perfect over R, then*

$$\mathbb{D}_\hbar(I(E)) \simeq I(E^\vee), \qquad E^\vee = \mathrm{RHom}_R(E, R).$$

Proof. Dualizing (14.25) gives $[\widehat{D}_R \xrightarrow{R_u} \widehat{D}_R]$ in degrees 0, 1. Here R_u is right multiplication by u, a left-module map. In the PBW order $\partial^j u^i$, it is an injective shift of i, with cokernel $\widehat{P}^t_R$. Thus the normalized dual of P_R is $\widehat{P}_R$.

The complete target allows removal of source completion inside this derived Hom. Mapping out of the direct sum gives W. The source M is flat, and its reductions $\bigoplus_i P_{R_r}$ are supported. The preceding proposition proves the failure for its dual. For perfect E, finite projective evaluation commutes with this calculation and introduces no infinite product. It gives the second formula. □

Keeping the finite-sum duals as a pro diagram preserves their support. It cannot be replaced by an ordinary supported realization preserving every perfect test. Indeed, already over $k = \mathbb{C}$, such a realization Z would have natural equivalences

$$\mathrm{RHom}_{D_k}(F, Z) \simeq \mathbf{R}\varprojlim_n \mathrm{RHom}_{D_k}(F, P_k^{\oplus n}) \qquad (F \text{ perfect}).$$

The free test $F = D_k$ identifies its underlying complex with $\prod_i P_k$, concentrated in degree zero. Naturality under the endomorphisms L_d of D_k recovers the right D_k-action. The diagonal family then contradicts support. Restricting to supported tests would change this requirement: over R they do not detect L, and over k they do not detect its reduction.

Perfect scalar grounds also need a relative finiteness condition if relative tensors are allowed. For $B = R[\epsilon]/(\epsilon^2)$ and the character module $R = B/(\epsilon)$, the free resolution has differential ϵ at every step. Tensoring with R makes all differentials zero. Consequently $H^{-n}(\Lambda(R \otimes^{\mathbb{L}}_B R)) = R$ for every $n \geq 0$, although each input is scalar-perfect. The output remains supported after induction but is no longer scalar-perfect.

These calculations specify an algebraic normal model with reconstruction and additive convolution, together with its limits of closure. Identifying this convolution with boundary fusion requires the geometric correspondence, tangential factors, unit and supported collision maps. Its unit $\widehat{P}_R$, the scalar unit R, the operator unit $1 \in D_R$ and a prepared physical state have distinct roles. The physical comparison must also decide which probes are allowed: retaining a pro diagram and taking its supported limit preserve different test functors.

Chapter 15

Centers and boundary actions

A closed boundary observable gives a multiplication operator. Bulk reconstruction also requires its commutation with boundary operations and its action on other boundary conditions. Supported operations must extend when insertions meet. Each requirement adds data to the multiplication operator.

The diagonal bimodule gives an algebraic recipient for these data. Its endomorphisms can be computed explicitly for polynomial algebras and for the continuous Hamiltonian envelopes of Chapter 10. A geometric action has further requirements. On a curve, the choice of admissible actors can even decide whether a final action exists. The comparison with the independently defined mixed bulk must therefore specify its recipient before asserting that it is an equivalence.

15.1 The diagonal bimodule

Let k be a field of characteristic zero and let A be an ordinary unital associative k-algebra. Write $A^e = A \otimes_k A^{\mathrm{op}}$. The diagonal bimodule is A, with its left and right multiplication. Its derived endomorphism complex is

$$Z_{\mathrm{bimod}}(A) = R\,\mathrm{Hom}_{A^e}(A, A).$$

This definition specifies both the object and the category of modules in which its resolutions are formed.

Put $\bar{A} = A/k1$. The normalized bar resolution has terms

$$B_n(A) = A \otimes \bar{A}^{\otimes n} \otimes A, \qquad n \geq 0,$$

in homological degree n. Its boundary is the alternating sum of adjacent multiplications, including the two end actions. The augmentation $B_0(A) \to A$ is multiplication.

Each term is a free A^e-module. Inserting 1 on the left contracts the augmented complex as a complex of k-vector spaces. The normalized formula uses the class of the former leftmost factor in $\bar{A}$. Consequently this bar complex resolves the diagonal.

Applying $\mathrm{Hom}_{A^e}(-, A)$ gives the normalized Hochschild cochains

$$C^n(A, A) = \mathrm{Hom}_k(\bar{A}^{\otimes n}, A).$$

For a cochain F of arity n, its differential is

$$\begin{aligned}(bF)(a_1, \dots, a_{n+1}) = {} & a_1 F(a_2, \dots, a_{n+1}) \\ & + \sum_{i=1}^{n} (-1)^i F(a_1, \dots, a_i a_{i+1}, \dots, a_{n+1}) \\ & + (-1)^{n+1} F(a_1, \dots, a_n) a_{n+1}.\end{aligned}$$

The formula is evaluated on representatives, with normalized cochains vanishing when an argument is 1. Thus $C^\bullet(A, A)$ computes $Z_{\mathrm{bimod}}(A)$.

A Hochschild 0-cochain is an element $a \in A$. It is closed precisely when a is central. Multiplication by a is then a bimodule endomorphism of the diagonal. For a commutative dg algebra, a degree-zero element closed under the internal differential gives the same construction. This accounts for a boundary multiplication operator without identifying all derived endomorphisms of the diagonal.

The cochains carry the cup product and insertion operations. An arity-q insertion has reduced degree $q-1$. Inserting it after i arguments contributes $(-1)^{i(q-1)}$ in the ungraded case. The braces sum over ordered nonoverlapping insertions, with these signs. For dg inputs their internal Koszul signs are included as well.

Write $F \circ G = F\{G\}$. Our unshifted bracket convention is

$$[F, G]_G = (-1)^{(p-1)(q-1)} F \circ G - G \circ F \qquad (F \in C^p,\ G \in C^q).$$

This is the suspension normalization for which $b = [m, -]_G$ and $[X, f]_G = X(f)$ for a derivation X. The bracket has degree -1. These operations retain more information than the graded algebra $HH^\bullet(A, A)$.

15.2 Polynomial coordinates and the Poisson differential

Take $A = k[x_1, \dots, x_m]$. The diagonal ideal in $A^e = k[x_1, \dots, x_m, y_1, \dots, y_m]$ is generated by the regular sequence $x_i - y_i$. Its Koszul resolution is

$$K = A^e \otimes \Lambda(\eta_1, \dots, \eta_m), \qquad d\eta_i = x_i - y_i,$$

where the η_i have homological degree one. After applying $\mathrm{Hom}_{A^e}(-, A)$ the differential vanishes. The result is the polyvector algebra

$$\mathrm{PV}(A) = A \otimes \Lambda(\partial_1, \dots, \partial_m), \qquad |\partial_i| = 1.$$

Here ∂_i denotes the constant derivation $\partial/\partial x_i$.

Proposition 15.1. *Antisymmetrization defines a quasi-isomorphism*

$$\mathrm{HKR} : (\mathrm{PV}(A), 0) \longrightarrow (C^\bullet(A, A), b),$$

with

$$\mathrm{HKR}(f X_1 \wedge \cdots \wedge X_n)(a_1, \dots, a_n) = \frac{f}{n!} \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) \prod_{i=1}^{n} X_{\sigma(i)}(a_i).$$

On cohomology it identifies the exterior product with cup product and the Schouten bracket with the Gerstenhaber bracket.

Proof. Send $\eta_{i_1} \wedge \cdots \wedge \eta_{i_n}$ to the alternating sum of bar words $[x_{i_{\sigma(1)}} | \cdots | x_{i_{\sigma(n)}}]$. Internal bar faces cancel in pairs because the coordinates commute. The end faces give the Koszul boundary. This is a comparison of free resolutions lifting the identity of A.

Restriction of the displayed cochain to that alternating sum is the corresponding Koszul dual basis vector: the $n!$ terms cancel the denominator. It therefore induces the computed isomorphism on cohomology. Products and brackets can be checked on functions and derivations. The cup classes of derivations anticommute, the bracket of two derivations is their commutator, and their action on a function is differentiation. The derivation identities extend these formulas to all polyvectors. □

The differential in this proposition is zero on polyvectors. A Poisson tensor introduces a different complex. Fix the Schouten convention $[\partial_i, x_j]_S = \delta_{ij}$, and put $d_\pi = [\pi, -]_S$. For the linear Poisson tensor

$$\pi = y\,\partial_x \wedge \partial_y \quad \text{on } k[x, y],$$

one has $d_\pi x = -y\partial_y$. The Hochschild differential of the 0-cochain x is still zero in the commutative algebra. Hence HKR from (PV, d_π) to the undeformed Hochschild complex is not a chain map.

There is nevertheless an exact Chevalley description of this Poisson differential. Let $\mathfrak{l}$ be finite dimensional, with $[e_i, e_j] = C_{ij}^k e_k$. Give $\mathfrak{l}^*$ the coadjoint action $(x\lambda)(y) = -\lambda([x, y])$ and form $\mathfrak{l} \ltimes \mathfrak{l}^*[1]$. We use cohomological shifts, so $[1]$ lowers degree by one.

Its polynomial CE cochains have ghosts c^i of degree one and cotangent coordinates u_i of degree two. The differential is

$$dc^k = -\frac{1}{2} C_{ij}^k c^i c^j, \qquad du_k = C_{jk}^\ell u_\ell c^j. \tag{15.1}$$

The order jk is the adjoint action on the coefficient e_k. The differential acts from the left by the graded Leibniz rule.

Proposition 15.2. *Subtract twice the polynomial degree in the u_i from the CE degree. The substitutions*

$$u_i \longmapsto O_i, \qquad c^i \longmapsto \partial^i$$

identify the resulting dg algebra with

$$\big(\mathrm{PV}(\mathrm{Sym}\,\mathfrak{l}), [\pi, -]_S\big), \qquad \pi = \frac{1}{2} C_{ij}^k O_k \partial^i \partial^j.$$

They preserve the degree-minus-one brackets normalized by $\{c^i, u_j\} = \delta_j^i$. If $Y_x = \{O_x, -\}$ is the Hamiltonian derivation for the Kirillov bracket, then $d_\pi O_x = -Y_x$.

Proof. The regraded coordinates have degrees $|u_i| = 0$ and $|c^i| = 1$. Direct Schouten differentiation gives

$$d_\pi \partial^k = -\frac{1}{2} C_{ij}^k \partial^i \partial^j, \qquad d_\pi O_k = C_{jk}^\ell O_\ell \partial^j.$$

These agree with (15.1). Both differentials are derivations, so agreement on generators proves the chain identity. The brackets also agree on generators, and their biderivation rule proves the bracket identity. Finally $\{O_k, O_j\} = C_{kj}^\ell O_\ell$, which is the negative of the coefficient in $d_\pi O_k$. □

For $[e_1, e_2] = e_2$, these equations read

$$dc^1 = 0, \quad dc^2 = -c^1 c^2, \quad du_1 = -u_2 c^2, \quad du_2 = u_2 c^1.$$

They square to zero. Reversing only the lower indices in the last two equations instead gives $d^2 u_1 = -2u_2 c^1 c^2$. Thus the sign cannot be changed while retaining the displayed ghost differential. The regrading is also essential: it does not identify the original CE degrees with the polyvector degrees of the field theory.

For a formal quantization, fix a filtered continuous L_∞ quasi-isomorphism $\mathcal{U}$ from shifted polyvectors to shifted Hochschild cochains, with first term HKR. This is a formality datum. If its Taylor series converge in the $\hbar$-adic filtration, the Poisson tensor defines

$$\mu_\hbar = \sum_{r \geq 1} \frac{\hbar^r}{r!} \mathcal{U}_r(\pi, \dots, \pi), \qquad m_\hbar = m + \mu_\hbar.$$

The Maurer–Cartan equation for $\mu_\hbar$ is associativity of $m_\hbar$.

Twisting gives Taylor maps

$$\mathcal{U}_n^{\hbar\pi}(\gamma_1, \dots, \gamma_n) = \sum_{r \geq 0} \frac{\hbar^r}{r!} \mathcal{U}_{n+r}(\pi^{\otimes r}, \gamma_1, \dots, \gamma_n).$$

Their source differential is $\hbar d_\pi$ and their target differential is $[m_\hbar, -]_G$. The L_∞ equations follow by collecting coefficients in the original equations after the substitution $\hbar\pi + \gamma$.

At each parameter quotient the filtration is finite. Its graded comparison is HKR. Induction on the quotient therefore gives a quasi-isomorphism. Derived $\hbar$-complete passage gives the completed comparison. A brace refinement, a PBW identification with a particular envelope, and compatibility with another coordinate tower are additional data. In particular the formula does not construct formality in an arbitrary topological category.

15.3 Continuous cochains and the relative center

For the pointed Hamiltonian algebra of Chapter 10, put $R = k[[\hbar]]$, $R_N = R/(\hbar^{N+1})$, and $A_{d,N} = U_\hbar(\mathfrak{h}_d) \otimes_R R_N$. There is no completion in PBW degree at a fixed pair (d, N). The complete envelope is $\mathcal{A} = \varprojlim_{d,N} A_{d,N}$. Its continuous cochains are

$$C^n_{\mathrm{uc}}(\mathcal{A}/R, \mathcal{A}) = \varprojlim_{d,N} \varinjlim_{e \geq d} \operatorname{Hom}_{R_N}(A_{e,N}^{\otimes n}, A_{d,N}).$$

The inner index measures input precision and the outer pair measures output precision. A scalar-valued map may detect a high Hamiltonian jet, so the input cannot be restricted to the output cutoff.

Give complete $\mathcal{A}$-bimodules the exact structure in which admissible short exact sequences split continuously over R. Proposition 10.10 constructs the relative projective bar resolution and proves

$$R\operatorname{Hom}^{\mathrm{rel,cont}}_{\mathcal{A}^e}(\mathcal{A}, \mathcal{A}) \simeq C^\bullet_{\mathrm{uc}}(\mathcal{A}/R, \mathcal{A}).$$

This is a continuous relative center in the specified exact category. It is not defined by silently changing the tensor product or the class of admissible exact sequences.

The braces exist because every finite insertion diagram has a common input refinement. At output (d, N), first choose the input cutoff of the outer cochain, then use that cutoff as the output of each inner cochain. A common refinement of their inputs defines the composite. The usual insertion identities hold at this refinement and hence at every output.

Antisymmetrization compares the relative cochains with

$$\varprojlim_{d,N} \varinjlim_{e \geq d} \big(C^\bullet_{\mathrm{CE}}(\mathfrak{h}_e, \operatorname{Sym} \mathfrak{h}_d) \otimes R_N, \hbar d_{\mathrm{CE}}\big),$$

where $\mathfrak{h}_e$ acts through $\mathfrak{h}_d$. The complete proof and the derived-limit assertion are Theorem 10.11. This comparison of complexes does not supply a brace-preserving inverse.

The factor $\hbar$ retains information. If Z^n and B^n are cycles and boundaries of a vector-space complex (P, d), then

$$H^n(P[[\hbar]], \hbar d) = Z^n[[\hbar]]/\hbar B^n[[\hbar]].$$

In particular its $\hbar$-torsion is the copy of B^n in constant order. Dividing the differential by $\hbar$ changes this cohomology before parameter inversion.

The wheel correction has an independent limiting issue. At a finite Hamiltonian cutoff, the logarithm of the Duflo half-determinant contains

$$\sum_{m\geq 1}\frac{B_{2m}}{4m(2m)!}\hbar^{2m}\operatorname{Tr}_{\mathfrak{b}_d}(\operatorname{ad}_x^{2m}).$$

The traces need not stabilize with d.

The finite-part subtraction of Proposition 10.13 gives, on the Cartan line $x_2 = aH$,

$$\exp(\ell_{\mathrm{ren}}(aH)) = \frac{\hbar a/2}{\sinh(\hbar a/2)}.$$

A shift of the cutoff changes the two-wheel finite part. This defines a subtraction choice. It neither constructs the continuous formality maps nor proves that their braces agree.

15.4 Evaluation at a boundary condition

Let C be a small dg category over k, or a specified compact subcategory of a k-linear presentable stable category. Its diagonal bimodule is $\Delta_C(x,y) = \operatorname{Hom}_C(y,x)$. Define

$$Z_{\mathrm{bimod}}(C) = R\operatorname{Hom}_{C\otimes C^{\mathrm{op}}}(\Delta_C,\Delta_C).$$

For $b\in C$, put $A_b = \operatorname{End}_C(b)$. Restriction to the full one-object subcategory gives

$$\mathrm{ev}_b : Z_{\mathrm{bimod}}(C)\longrightarrow C^\bullet(A_b,A_b).$$

At the level of bar cochains this simply restricts every object argument to b, so it is a map of cochain complexes and braces.

Proposition 15.3. *Suppose restriction induces an equivalence of the chosen derived bimodule categories and carries Δ_C to A_b. Then ev_b is a quasi-isomorphism. In ordinary derived module categories this holds when b is a compact Morita generator. Generation is not necessary for the evaluation map on centers to be a quasi-isomorphism.*

Proof. An equivalence preserves derived mapping complexes, which proves the first assertion. For a compact generator, the tensor–Hom adjunction with b gives the module equivalence. Its unit and counit are equivalences on the generator, hence on the localizing category it generates. Applying the equivalence on both module variables gives the diagonal bimodule equivalence. The final assertion follows from the next example. □

Example 15.4. Let A be the upper triangular 2 by 2 matrix algebra. Write $e_1 = E_{11}$, $a = E_{12}$ and $e_2 = E_{22}$. For right modules take $b = e_1A$. Its endomorphism algebra is $e_1Ae_1 = k$. The diagonal has the projective bimodule resolution

$$0\longrightarrow Ae_1\otimes e_2A\xrightarrow{d}(Ae_1\otimes e_1A)\oplus(Ae_2\otimes e_2A)\xrightarrow{\mu}A\longrightarrow 0,$$

where $d(e_1\otimes e_2) = e_1\otimes a - a\otimes e_2$ and μ is multiplication.

The dimensions are 1, 4, 3. Multiplication is onto and its kernel is the displayed one-dimensional image. Each tensor summand is a direct summand of a free A^e-module.

Bimodule Hom into A gives

$$k^2\xrightarrow{(\lambda_1,\lambda_2)\mapsto\lambda_1-\lambda_2}k.$$

Restriction to b is $(\lambda_1, \lambda_2) \mapsto \lambda_1$ in degree zero and zero in degree one. Its kernel is $k \xrightarrow{-1} k$, so the restriction is a quasi-isomorphism to $C^\bullet(k, k) = k$.

Nevertheless $e_2 A \neq 0$ and $R\operatorname{Hom}_A(e_1 A, e_2 A) = e_2 A e_1 = 0$. Since $e_1 A$ is projective, this vanishing holds in every degree. It prevents b from generating the derived module category. Thus equality of center endomorphisms detects less than Morita generation.

The density quotient $C/\langle b \rangle$ and the cochain cone $\operatorname{Cone}(\mathrm{ev}_b)$ answer different questions. The former measures generation. The latter is acyclic exactly when this particular center map is a quasi-isomorphism. The example has an acyclic cone and a nonzero density quotient.

For a mixed boundary, one must first construct the category of boundary conditions on its brane-link site, its complete factorization bimodules and its diagonal. A restriction functor must preserve the declared topology and descent conditions. These are the hypotheses under which the same endomorphism argument applies. A scalar trace at one Dirac brane does not provide that bimodule equivalence.

A junction may carry both boundary actions without generating either boundary category. Section 15.15 gives an explicit example. It also distinguishes strict intertwiners from local operator fields: they have different derived comparison problems.

15.5 Compact support and the direction of a map

For this section take $k = \mathbb{R}$ or $\mathbb{C}$. Let $\mathfrak{l}$ be an ordinary Lie algebra and let I be an open interval. The complex

$$\mathfrak{l}_c(I) = \Omega_c^\bullet(I) \otimes_k \mathfrak{l}, \qquad [\alpha \otimes x, \beta \otimes y] = (\alpha \wedge \beta) \otimes [x, y]$$

is a dg Lie algebra. Extension by zero gives a covariant map for inclusions of intervals. For disjoint $I_1, I_2 \subset V$, the two images commute, since their supports are disjoint. Hence their direct sum maps to $\mathfrak{l}_c(V)$ as a dg Lie algebra.

The algebraic CE chains are $C_*^{\mathrm{CE}}(\mathfrak{l}_c(I)) = \operatorname{Sym}(\mathfrak{l}_c(I)[1])$, with direct sums in symmetric length and with their internal and bracket differentials. They are covariant. The direct-sum identity for CE chains therefore gives structure maps

$$C_*^{\mathrm{CE}}(\mathfrak{l}_c(I_1)) \otimes C_*^{\mathrm{CE}}(\mathfrak{l}_c(I_2)) \longrightarrow C_*^{\mathrm{CE}}(\mathfrak{l}_c(V)).$$

Their associativity and units follow from those of extension by zero and direct sum. They define a unital prefactorization algebra, with k at the empty configuration.

Dual CE cochains run in the other direction. With the algebraic dual, their structure map is

$$C^\bullet_{\mathrm{CE}}(\mathfrak{l}_c(V)) \longrightarrow \operatorname{Hom}_k\Big(C_*^{\mathrm{CE}}(\mathfrak{l}_c(I_1)) \otimes C_*^{\mathrm{CE}}(\mathfrak{l}_c(I_2)), k\Big).$$

Writing the target as a tensor product of cochain spaces requires an appropriate dual tensor or a finite-dimensional hypothesis. With continuous duals the topology must also be specified. Reversing this arrow is not a factorization operation.

There is a concrete contraction on compactly supported forms. Choose $\rho_I \in C_c^\infty(I)$ with integral one. Let p_I integrate a one-form and kill a zero-form, and let $i_I(1) = \rho_I(t)\, dt$. The target is $k[-1]$, concentrated in cohomological degree one. Define

$$K_I(a(t)\, dt)(t) = \int_{-\infty}^{t} \left(a(s) - \rho_I(s) \int_I a \right) ds, \qquad K_I(f) = 0.$$

The integral in parentheses is zero, so K_I has compact support. Differentiation proves

$$dK_I + K_I d = 1 - i_I p_I.$$

Integration is natural for extension by zero. Thus every interval inclusion induces a quasi-isomorphism on I_c.

It also induces a quasi-isomorphism on algebraic CE chains. Filter those chains by length at most n. The bracket lowers length, and the associated quotients are symmetric powers of the underlying complex. Tensor powers and symmetric coinvariants preserve quasi-isomorphisms over k. Induction on n proves the assertion on each finite length filtration. Exact filtered colimit proves it for the direct sum.

This is a locally constant prefactorization construction. Its completion and any required Weiss descent must be proved in the corresponding completed category.

The contraction is not an identification of all current duals. In the usual test-function topology, $C_c^\infty(I)' = \mathcal{D}'(I)$ consists of all distributions on I, while $C^\infty(I)' = \mathcal{E}'(I)$ consists of compactly supported distributions. For example integration against the constant density has noncompact support on I and belongs to the first dual. A cotangent pairing that uses compactly supported test fields must retain this distinction.

Proposition 15.2 can be applied to an actual finite coefficient model. Extending it to intervals requires compatible duals, completed tensor products and transition maps. The equation $\Phi_V = m(\Phi_{I_1} \otimes \Phi_{I_2})\Delta$ would force Φ_V to vanish on $\ker \Delta$. No local coordinate identity supplies that vanishing. A covariant bulk action therefore requires its own supported construction.

15.6 Cyclic chains and stable matrix traces

Matrix traces also have a direction. Let A first be an ordinary unital algebra. Put

$$C_n^\lambda(A) = A^{\otimes(n+1)}/(1 - t), \qquad t(a_0, \dots, a_n) = (-1)^n(a_n, a_0, \dots, a_{n-1}).$$

The cyclic boundary is

$$b(a_0, \dots, a_n) = \sum_{i=0}^{n-1} (-1)^i (a_0, \dots, a_i a_{i+1}, \dots, a_n) + (-1)^n (a_n a_0, a_1, \dots, a_{n-1}).$$

Over k, this cyclic coinvariant complex computes cyclic homology.

For a dg algebra the rotation includes the internal Koszul sign, and the internal differential is totalized with b. The reduced theory of an augmented algebra is the kernel of the map to the corresponding complex of k.

Use the homological CE convention $d(X \wedge Y) = [X, Y]$. The alternating matrix trace is

$$\gamma_{N,n}(X_1 \wedge \cdots \wedge X_n) = \sum_{\sigma \in S_{n-1}} \operatorname{sgn}(\sigma) \sum_{i_1, \dots, i_n} ((X_1)_{i_1 i_2}, (X_{\sigma(2)})_{i_2 i_3}, \dots, (X_{\sigma(n)})_{i_n i_1}) \quad \text{in } C_{n-1}^\lambda(A). \tag{15.2}$$

Here S_{n-1} permutes $2, \dots, n$. For $n = 1$ the formula is the ordinary matrix trace. It is equivalent to full antisymmetrization divided by n, using the cyclic relation.

Proposition 15.5. *For every N, the maps (15.2) satisfy $b\gamma_N = \gamma_N d$. They are compatible with stabilization by a zero block. The same statement holds for dg algebras with the total Koszul signs.*

Proof. The cyclic relation makes the expression alternating in all n inputs. A face of its boundary multiplies two successive matrix entries, and summing the intermediate index is matrix multiplication. For each unordered pair of inputs, the two orders give their commutator. The permutation signs are the signs that move this pair to the front of the CE wedge. The remaining permutations are exactly those in the alternating trace of that commutator and the remaining inputs.

The closing cyclic face supplies the pair across the chosen starting point. This partitions every summand on both sides and proves the identity. Zero blocks add only zero entries.

For homogeneous dg inputs, each reordering moves the same suspended inputs on both sides. The internal differential commutes with products and trace. This proves the total version. □

The reverse sum of cyclic matrix tensors has a different normalization. Define

$$\Theta_1(a) = \sum_i E_{ii}a, \qquad \Theta_2(a,b) = \sum_{i,j} E_{ij}a \wedge E_{ji}b.$$

Then

$$d\Theta_2(a,b) = N\Theta_1(ab - ba). \tag{15.3}$$

Indeed the forward product sums to $N\sum_i E_{ii}ab$ and the reverse product sums to $N\sum_j E_{jj}ba$. Thus the unnormalized Θ is not a chain map from the cyclic complex for a noncommutative A at $N > 1$. The factor is already visible for two free generators. A finite rank bound does not remove it.

There is a chain-level primitive projection after taking the correct stable coinvariants. Let

$$\mathfrak{gl}_\infty(A) = \varinjlim_N \mathfrak{gl}_N(A), \qquad H(A) = C_*^{\mathrm{CE}}(\mathfrak{gl}_\infty(A))_{\mathfrak{gl}_\infty(k)}.$$

The subscript means coinvariants for the scalar matrix Lie algebra. Stabilization and these coinvariants make block sum an associative, graded-commutative chain product. The CE coproduct makes $H(A)$ a connected dg Hopf algebra.

Theorem 15.6 (Stable cyclic primitives)**.** *The primitive subcomplex of $H(A)$ is naturally isomorphic to $C^\lambda_{*-1}(A)$. The Hopf logarithm*

$$P_{\mathrm{prim}} = \log^*(1) = \sum_{r\geq 1} \frac{(-1)^{r-1}}{r}(1 - u\varepsilon)^{*r}$$

is a chain idempotent onto that subcomplex. Its sum is finite in each CE length. Consequently

$$\mathrm{Prim}\, H_*^{\mathrm{CE}}(\mathfrak{gl}_\infty(A)) \cong HC_{*-1}(A).$$

For an augmented algebra, put

$$H_{\mathrm{rel}}(A) = H(A)/(H(k)^+H(A)),$$

where $H(k)^+$ is the augmentation ideal of the scalar Hopf subalgebra. Its primitive complex is $C^\lambda_{*-1}(A)/C^\lambda_{*-1}(k)$, giving reduced cyclic homology.

Proof. At CE length n, scalar matrix coinvariants have the permutation description

$$\big(k[S_n] \otimes A^{\otimes n} \otimes \mathrm{sgn}\big)_{S_n}.$$

The symmetric group conjugates the permutation label and permutes the A-inputs. This follows from matrix invariant theory: contractions are indexed by permutations, and they are independent at matrix rank at least n.

The coproduct splits the labels into subsets preserved by the permutation. A one-cycle permutation is therefore primitive. Every permutation is a disjoint collection of cycles, so the Hopf algebra is the graded symmetric algebra on the one-cycle subspace. Identifying all starting points of one cycle gives exactly the cyclic relation defining $C^{\lambda}_{n-1}(A)$.

For a cycle, choose the class represented by $E_{12}a_1 \wedge E_{23}a_2 \wedge \cdots \wedge E_{n1}a_n$, using distinct matrix indices. Its alternating trace is $(a_1, \dots, a_n)$. The CE boundary joins adjacent entries, including the last and first, with the signs of b. It follows that the primitive identification is an isomorphism of complexes. This is the chain calculation in the Loday–Quillen theorem.

On a product of n positive primitives, the Hopf logarithm has coefficient

$$\sum_{r=1}^{n} (-1)^{r-1}(r-1)!S(n,r),$$

where $S(n,r)$ counts partitions into r nonempty parts. The coefficient is 1 for $n = 1$ and 0 otherwise, as follows from $\log(1 + (e^t - 1)) = t$. This proves idempotence and its image. All the maps in its convolution formula are chain maps.

Finally the scalar matrix action on CE chains is nullhomotopic by insertion of a scalar matrix. In each finite rank its rational representations are semisimple. The nontrivial isotypic summands have zero homology, so passage to coinvariants is a quasi-isomorphism. Filtered colimit gives the stable assertion.

The algebra augmentation splits the scalar cyclic subcomplex. The symmetric-algebra description then identifies the Hopf quotient by $H(k)^+$ with the symmetric algebra on the reduced cyclic complex. This proves the relative statement. □

For dg A concentrated in nonnegative homological degrees, the proof uses total Koszul signs and the same first-quadrant filtrations. At a fixed CE length n the scalar matrix coinvariant description is valid for $N \geq n$. Computing homology through length L also uses the next chain term. Thus $N \geq L + 2$ is a sufficient bound.

A separate polynomial trace window of weight K can impose $N \geq K$. The common bound $N \geq \max(K, L + 2)$ is sufficient for both purposes. The projection acts on the specified coinvariant complex. It is not deletion of multitrace tensors in the original fixed-rank CE complex.

15.7 The full cyclic-word resolution

The scalar quotient in cyclic homology must also be distinguished from removing an antifield class. Use homological degrees in this section and set

$$R = k\langle x, y, \psi\rangle, \qquad |x| = |y| = 0, \quad |\psi| = 1, \quad d\psi = xy - yx.$$

The free-resolution contraction of Proposition II.2 identifies R with $k[x, y]$ in the derived category of associative algebras. Define its reduced cyclic-word complex by

$$R_{\mathrm{cyc,red}} = R/(k1 + [R, R]_{\mathrm{gr}}).$$

The denominator is a vector subspace of graded commutators, not a two-sided ideal.

Give x, y, ψ bidegrees $(1, 0)$, $(0, 1)$ and $(1, 1)$, respectively. The differential preserves this bidegree.

Theorem 15.7. *The cyclic-word homology is*

$$H_p(R_{\mathrm{cyc,red}}) = \begin{cases} k[x, y]/k, & p = 0, \\ k[x, y]\, dx \wedge dy, & p = 1, \\ 0, & p \geq 2. \end{cases}$$

Let $\mathsf{W}_{a,b}$ consist of words with a copies of x and b copies of y, for $a, b \geq 0$. In bidegree $(a+1, b+1)$, the degree-one group is generated by

$$\Psi_{a,b} = \sum_{w \in \mathsf{W}_{a,b}} [\psi w].$$

In particular $[\psi]$ survives.

Proof. First compare cyclic words with cyclic chains of the free dg algebra R. For a graded vector space V, the tensor algebra $T(V)$ has the bimodule resolution

$$0 \longrightarrow T(V) \otimes V \otimes T(V) \longrightarrow T(V) \otimes T(V) \longrightarrow T(V) \longrightarrow 0.$$

The first map sends $a \otimes v \otimes b$ to $av \otimes b - a \otimes vb$. At word length $m > 0$, its Hochschild kernel and cokernel are the invariants and coinvariants of the graded cyclic rotation on $V^{\otimes m}$.

The Connes operator between them is the norm $1 + t + \cdots + t^{m-1}$. In characteristic zero that norm is an isomorphism from coinvariants to invariants. Thus the reduced cyclic complex of a tensor algebra has only its cyclic-word quotient in cyclic degree zero.

Apply this calculation to the underlying graded tensor algebra of R, and then restore its internal differential. The cyclic double complex is first quadrant in homological degrees. In each positive bidegree there are only finitely many words and internal degrees. The comparison therefore gives

$$H_*(R_{\mathrm{cyc,red}}) = \overline{HC}_*(R).$$

Tensor powers preserve the quasi-isomorphism $R \to k[x, y]$ over k. The same first-quadrant comparison identifies this with $\overline{HC}_*(k[x, y])$.

For $k[x, y]$, the diagonal Koszul resolution gives Hochschild chains equal in homology to polynomial forms. The normalized map $a_0[a_1|\cdots|a_n] \mapsto a_0 da_1 \wedge \cdots \wedge da_n/n!$ intertwines the Connes operator with the de Rham differential. The cyclic bicomplex consequently gives

$$\overline{HC}_0 = k[x, y]/k, \qquad \overline{HC}_1 = \Omega^1/dk[x, y],$$

and zero in higher degrees. To see the last assertion directly, contract positive-weight polynomial forms by the Euler vector field divided by total weight. The only uncontracted de Rham class is the scalar class, removed in the reduced complex. Differentiation identifies $\Omega^1/dk[x, y]$ with $\Omega^2 = k[x, y]dx \wedge dy$.

It remains to identify the cycle. Replacing the displayed ψ by $xy - yx$ shows that $d\Psi_{a,b} = 0$: every zero-antifield cyclic word has equally many cyclic xy and yx transitions. Define a functional on one-antifield words by $[\psi w] \mapsto 1$ in this fixed bidegree. A two-antifield boundary has the four terms

$$d[\psi u \psi v] = [\psi(vxyu - vyxu - uxyv + uyxv)],$$

whose functional values sum to zero.

The value on $\Psi_{a,b}$ is $\binom{a+b}{a} \neq 0$. Hence its class is nonzero and spans the already computed one-dimensional group. This also specifies its normalization: the functional divided by $\binom{a+b}{a}$ sends $\Psi_{a,b}$ to 1. □

The class $[\psi]$ has bidegree $(1, 1)$ and corresponds to the constant area form. It is not the scalar augmentation class of bidegree $(0, 0)$. Removing it is an additional quotient of the cyclic-word complex. In the full cyclic chain complex, its representative includes the bar term that cancels $d\psi = xy - yx$. With the convention $b(x|y) = xy - yx$, it is $\psi - (x|y)$. Quotienting only the two-term subcomplex $\psi \mapsto xy - yx$ does not remove this cyclic homology class.

15.8 Primitive projections and the scalar rank

Let L^ν be the vector space of cyclic words in $k\langle x, y\rangle$, including the empty cyclic word ν. Define its necklace Lie bracket by cutting an x in the first word and a y in the second. Join the remaining words and subtract the reverse cuts. The empty word is central and

$$[x, y] = \nu.$$

Associativity of concatenation and pairing opposite orders of two cuts prove Jacobi. The symmetric algebra $\mathrm{Sym}(L^\nu)$ has the induced linear Poisson bracket. Its Hopf coproduct makes every element of L^ν primitive, including ν.

Evaluation on matrices sends ν to $\mathrm{Tr}\, I_N = N$. Before this specialization, ν counts one trace. After it, the same term is a scalar. This change of trace count affects the primitive projection.

Proposition 15.8. *Projection* $\mathrm{Sym}(L^\nu) \to L^\nu$ *onto symmetric degree one is a Lie map. The quotient* $L = L^\nu/k\nu$ *also gives a Lie projection* $\mathrm{Sym}(L) \to L$*. Abelianization of the nonempty cyclic words then gives a Lie map* $L \to k[x, y]/k$*.*

At nonzero fixed rank, the projection that kills scalar and multitrace terms is not a Lie map for the unreduced matrix Poisson bracket. On $X = \mathrm{Tr}\, x$ *and* $Y = \mathrm{Tr}\, y$*, its defect is*

$$\pi_{\mathrm{prim}}\{X^2, Y\} - \{\pi_{\mathrm{prim}}X^2, \pi_{\mathrm{prim}}Y\} = 2NX. \tag{15.4}$$

Proof. The linear Poisson bracket sends symmetric degrees (r, s) to degree $r + s - 1$. With positive degrees, its degree-one part occurs only at $(1, 1)$. Scalars have zero bracket. This proves the first projection identity. The central Lie quotient preserves it. On cyclic words, abelianization turns the cut formula into the ordinary Hamiltonian bracket, modulo its constant term.

At fixed rank, $\{X, Y\} = N$ and the Leibniz rule gives $\{X^2, Y\} = 2NX$. The projection kills X^2, retains X and Y, and kills the scalar bracket of X and Y. This proves (15.4). □

For $N \geq 2$, the degree-two trace window already separates the single-trace and double-trace terms in this example. The class X is nonzero on the commuting-matrix locus. Thus the defect cannot be made into a Koszul boundary by a homotopy between the same two projected bracket maps. The zero-central-character construction precedes specialization. It does not quotient the fixed-rank scalar $N1$ to zero in a nonzero unital algebra.

The Hamiltonian abelianization extends to a chain map from the full cyclic-word complex by killing words containing ψ. A one-antifield differential becomes a commutator, which abelianizes to zero. Higher-antifield terms still contain an antifield after one differential. Extending over the symmetric algebra and projecting to its linear part therefore preserves the differential. Its bracket scope is the one just proved. The independent one-antifield classes of Theorem 15.7 must be retained for a cotangent comparison.

There is a useful product truncation as well. For any symmetric algebra $S(L)$, projection to $k \oplus L$ is the quotient by $S^{\geq 2}(L)$, with square-zero multiplication on L. If $F, G \in S(L)$, then

$$\pi_1(FG) = \varepsilon(F)\pi_1(G) + \pi_1(F)\varepsilon(G).$$

This is the linear part of the product with its scalar unit retained. A Lie bracket on L is a separate operation. For supported observables, a factorization interpretation of this quotient also requires compatible structure maps and descent on the chosen site.

15.9 Actions on the ordered collision carrier

The geometric center problem on a curve has a different recipient. We specify it over $X = \mathbb{A}^1_k$. For a finite set I, form a scheme $\mathscr{R}_I$ from one copy of X^I for each total order of I. Two copies are glued where $x_i \neq x_j$ for every pair whose orders disagree.

Distinct positions forget their relative order. Coincident positions retain it. For two labels there are two diagonal branches.

An ordered surjection $I \twoheadrightarrow J$ includes an order on each fiber. Replacing each label of an order on J by its ordered fiber gives a collision map $\mathscr{R}_J \to \mathscr{R}_I$. Let $\mathcal{D}$ be the category of compatible derived right differential modules on these schemes, with extraordinary pullback along every such collision map. Compatibility includes its composition homotopies. It retains full normal direct images, including normal derivatives.

There are two tensor operations. Disjointness convolution $\star$ sums over ordered decompositions $I = I_1 \sqcup I_2$. For each decomposition, externally tensor the coefficients on the disjoint open and apply open direct image. Its unit K is supported at the empty configuration.

The diagonal tensor is

$$(F \otimes^! G)_I = \Delta^!(F_I \boxtimes G_I).$$

Its unit J is the compatible dualizing family. On a chart over X^I, $J_I = \omega_{X^I}[|I|]$. The map $K \to J$ is not an identification of the two units.

Use connected coaugmented counital coassociative coalgebras for $\star$, with empty coefficient k. Denote their category by $\mathcal{C}$. The forward interchange keeps common ordered decompositions in

$$(F \star G) \otimes^! (H \star L) \longrightarrow (F \otimes^! H) \star (G \otimes^! L).$$

It tensors coalgebra coproducts. Consequently $(\mathcal{C}, \otimes^!, J)$ is symmetric monoidal.

A coalgebra is *factorizing* if its coproduct becomes an equivalence for each ordered decomposition on its disjoint open. Write $\mathcal{C}^{\mathrm{f}}$ for this full subcategory.

A transverse monoid is a unital associative algebra object in $(\mathcal{C}, \otimes^!, J)$. Its product and unit are coalgebra maps. An action on C consists of $\alpha : F \otimes^! C \to C$ and all the unit and associativity homotopies of a module structure. Morphisms of actions preserve these comparisons. A *factorizing monoid center* of $C \in \mathcal{C}^{\mathrm{f}}$ means a final object among these actions with $F \in \mathcal{C}^{\mathrm{f}}$.

One can first allow all coalgebras as actors. The resulting endomorphism object always exists on this carrier.

Positive objects have zero empty coefficient, so their r-fold convolution vanishes on configurations of fewer than r positions. The normalized associative bar sums suspended words in the augmentation ideal. Its differential multiplies adjacent factors and its coproduct cuts words. The normalized cobar sums desuspended words in the reduced coalgebra, with differential induced by its coproduct. Both sums are finite at each configuration.

To compare their composites with the original object, filter by the number n of original factors. Each of the $n - 1$ gaps is either inside a block or between blocks. On associated graded its two states form an identity complex, with the suspension signs. Their tensor product is contractible for $n > 1$. Lengths zero and one retain the unit and original reduced object.

The filtration ends at $n = |I|$ on a configuration I. Thus both comparisons are equivalences. Finite normalization commutes with collision pullback, so the comparisons and their coherences belong to the declared ordered carrier.

The category of augmented convolution algebras is presentable, since nonunital algebras on positive objects are algebras for an accessible tensor-algebra monad. Bar–cobar gives presentability of C. Coalgebra colimits are computed on the underlying reduced families: equivalently, they are limits of nonunital algebras in the opposite category. The diagonal tensor preserves these colimits. The adjoint functor theorem therefore supplies an internal Hom $[C, D]_!$ in $\mathcal{C}$.

Proposition 15.9. *The object $\mathcal{E}(C) = [C, C]_!$ has a coherent composition product and unit $J \to \mathcal{E}(C)$. Its evaluation is a transverse action, and*

$$\mathrm{Map}_{\mathrm{Alg}(C)}(F, \mathcal{E}(C)) \simeq \{\textit{coherent transverse actions of } F \textit{ on } C\}.$$

If $\mathcal{E}(C)$ factorizes, it is a factorizing monoid center. Factorization of this internal Hom is an additional condition.

Proof. Adjoint to two successive evaluations is a product $\mathcal{E}(C) \otimes^! \mathcal{E}(C) \to \mathcal{E}(C)$. The identity of C gives its unit. Iterated evaluation is independent of parentheses with all its coherent comparisons. Adjunction carries the multiplication and unit equations of a map into $\mathcal{E}(C)$ to the action equations, and carries their higher simplices in the same way. This proves the mapping-space equivalence. If the object factorizes, this universal action belongs to the restricted category as well and remains final there. □

A pointed actor carries only a map $u : J \to F$. Its pointed action space is

$$\Phi_C(F, u) = \mathrm{Map}_{C/}(F \otimes^! C, C),$$

where the structure map from C is $u \otimes^! 1$. Representing this functor on factorizing pointed actors is sufficient to construct a factorizing monoid center by composition of the representing action. A pointed action does not itself include the multiplication homotopies of a monoid action. Their representability questions must therefore be compared at their actual morphism scopes.

15.10 The center under a cohomological bound

Let Q be an ordinary unital chiral algebra, flat over $\mathcal{O}_X$, and write C_Q for its Chevalley–Cousin factorization coalgebra on the ordered carrier. Its main-diagonal Lie object L_Q has singleton right coefficient $\omega_X \otimes Q$, and $C_Q = \mathrm{CE}_\star(L_Q)$.

The chiral center $Z(Q)$ consists of the states whose singular products with every state vanish. It is a commutative differential algebra. The tensor product of ordinary chiral algebras represents pairs of commuting maps: this is the Beilinson–Drinfeld tensor-product property. In particular a map $B \otimes Q \to Q$ that is the identity on Q is a map $B \to Z(Q)$.

For a derived actor, the corresponding bound is the ordinary factorization heart. On every ordered chart require

$$F_I \in D^{\leq -|I|}(\mathcal{D}_{X^I}).$$

After the dualizing shift and passage to left modules, its top cohomology has degree zero. This condition permits negative derived degrees but excludes positive degrees above that heart.

Theorem 15.10. *For pointed factorizing actors satisfying this bound, restriction of an action on C_Q to its vacuum gives an equivalence of discrete spaces*

$$\Phi_{C_Q}(F, u) \simeq \mathrm{Map}_{J/}(F, C_{Z(Q)}).$$

The central action of $C_{Z(Q)}$ is therefore final among the factorizing transverse monoid actions satisfying the same bound.

Proof. Maps from an object below the heart into an object of the heart are determined on top cohomology and have discrete mapping spaces. The same holds for the compatibility limits defining coefficient maps and their unit conditions. Write M_I for the top left module of F_I and A_I for the left factorization module of Q. An action gives $a_I : M_I \otimes A_I \to A_I$.

Let e be the image of the actor vacuum and set $\zeta(f) = a_1(f \otimes 1)$. The pointing says $a_1(e \otimes q) = q$. The two-position localization triangle supplies a chiral boundary operation μ_F on M_1. Naturality of its connecting morphism under a_2 gives

$$\mu_Q((a_1 \boxtimes a_1)(x \otimes c)) = a_1(\mu_F(x) \otimes c|_\Delta).$$

Here x is a meromorphic section of $M_1 \boxtimes M_1$ on the disjoint open and $c \in A_2$.

Take $x = (f \boxtimes e)\varphi$ and use the two vacuum sections $1 \boxtimes q$ and $q \boxtimes 1$ of A_2. Their restrictions to the diagonal agree. Hence

$$\mu_Q((\zeta(f) \boxtimes q)\varphi) = \mu_Q((a_1(f \otimes q) \boxtimes 1)\varphi).$$

At $\varphi = 1$ the right side is zero in the full normal direct image. Its normal-derivative coefficients are all singular products, so $\zeta(f) \in Z(Q)$. At $\varphi = (z_1 - z_2)^{-1}$ the vacuum residue axiom gives $a_1(f \otimes q) = \zeta(f)q$.

Restriction to the vacuum thus gives a coalgebra map $g : F \to C_Q$ whose singleton image lies in $Z(Q)$. Factorization puts its image in the external powers of $Z(Q)$ on every disjoint open. The Chevalley–Cousin kernel description gives

$$(C_{Z(Q)})_I = (C_Q)_I \cap j_{I,*} j_I^*(Z(Q)^{\boxtimes I})$$

in the corresponding left-module description. This is because the first Cousin differential on the central subalgebra is its own Cousin differential. Thus g factors through $C_{Z(Q)}$.

The factorization modules of the flat chiral algebra have no sections supported only on diagonals. Agreement on the disjoint open therefore proves that its central action recovers α. Evaluation on the vacuum recovers g in the opposite direction.

For a monoid actor, associativity of its action now says that g preserves multiplication. The action unit says that it preserves the unit. All mapping spaces involved are discrete, so these equations determine their higher coherences uniquely. This proves finality in the bounded category. □

The cohomological bound precedes the use of the residue. Without it, top cohomology can forget action homotopies. For example $\pi_1 \operatorname{Map}(k[-1], k) = k$, although $H^0(k[-1]) = 0$. The following construction shows that, for ordinary boundaries with a derivation, removing the bound changes the existence question itself.

15.11 A factorizing monoid obstruction

Work now over $k = \mathbb{C}$. Let Q be an algebraic degree-zero unital vertex algebra, and fix a nonzero vertex derivation D. Thus $D1 = 0$, $DT = TD$, and $D(x_{(n)}y) = (Dx)_{(n)}y + x_{(n)}Dy$ for every integer n. Put $C = C_Q$ and $\ell = L_k$.

For a finite-dimensional vector space W, place its exterior generators in degree one and define

$$E_W = \Lambda(W[-1]), \qquad C_W = \mathrm{CE}_\star(\ell \otimes E_W).$$

The differential and translation on the exterior generators are zero. These explicit actors have an augmentation to $J = C_0$. Arbitrary receiving actors below are not required to have one.

We need natural maps of these exterior actors. A natural operation uses products, Lie brackets, permutations and exterior coefficient maps, with each ordinary Lie input used once. Its identities hold for every test Lie algebra. Complete only in the number of exterior input generators. At m positions this number is at most $m \dim W$, so evaluation remains finite on the original coefficient diagrams.

Lemma 15.11. *Natural maps $C_W \to C_V$ under and over J are classified, including their natural homotopies, by formal matrices*

$$A(t) \in \mathrm{Hom}(W, V)[[W^*]].$$

Their components are contractible. The matrix remembers which input supplies the exterior output. With $\Phi_A(t) = A(t)t$, composition and tensor product are

$$(B \circ A)(t) = B(\Phi_A(t))A(t), \qquad A \otimes B = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}.$$

Thus an associative unital multiplication of these matrices defines a coherent transverse monoid on C_W.

Proof. The scalar-relative convolution algebra of natural operations has a weightwise model obtained from the Lie cobar of $(E_W^+)^*$. In a basis its generators y_S, for nonempty exterior subsets S, have degree $1 - |S|$. The differential splits S into two subsets and takes their signed Lie bracket. An extra scalar generator z has degree one and $dz = z^2$ in the associative envelope.

The ideal generated by z contracts by $h(z^{2j}) = z^{2j-1}$ and $h(z^{2j-1}) = 0$, applied to the first such factor in a free-product word. The remaining Lie cobar projects onto the abelian Lie algebra W^*. Its envelope is a quasi-isomorphism onto $\mathrm{Sym}(W^*)$: the dual Koszul resolution is

$$E_W \otimes k[v_1, \dots, v_r], \qquad dv_i = \alpha_i,$$

and positive total weight n contracts by $n^{-1} \sum_i v_i \partial_{\alpha_i}$. PBW symmetrization identifies the Lie length-one summands as well.

Tensoring with E_V and taking the positive target exterior ideal gives a surjective dg algebra map to

$$\left[\widehat{\mathrm{Sym}}(W^* \otimes E_V)\right]_{E_V^+}$$

with acyclic kernel in every source weight. This commutative model has no nonpositive degrees. Its degree-one part is $\widehat{\mathrm{Sym}}(W^*) \otimes W^* \otimes V$. The Maurer–Cartan equation is automatic there. Weightwise lifting through the acyclic kernel, also with polynomial forms relative to simplex boundaries, gives the contractible components.

The remaining degree-one symbol has one marked input. Substitution into a second natural map first replaces the ordinary coordinate by $A(t)t$, then carries its marked input by $A(t)$. This proves the composition formula. Independent exterior factors give the block-diagonal formula. These formulas identify the natural category with the nerve of the stated matrix category. Associativity and unit equations lift with all their coherences through the contractible fibers. □

Use the auxiliary dual-number algebra with basis a, b and products $a^2 = a$, $ab = ba = b$, $b^2 = 0$. Its unit a is not the vacuum of an actor. Let L_x be multiplication by x in this two-dimensional algebra and set

$$Q(x) = e^{L_x}, \qquad M(x) = \sum_{r \geq 0} \frac{L_x^r}{(r+1)!}.$$

The constant term of M is the identity.

The two blocks

$$\big(M(x+y)^{-1}M(x) \mid M(x+y)^{-1}Q(x)M(y)\big) \tag{15.5}$$

define a multiplication on $F = C_{ka \oplus kb}$. Its coordinate contraction is addition.

To verify associativity, use fiber coordinates $U = M(x)u$. The product becomes

$$(x, U)(y, V) = (x + y, U + Q(x)V).$$

The identity $Q(x+y) = Q(x)Q(y)$ proves associativity. The identity $M(x)x = e^x - a$ proves that the diagonal $u = x$ is closed under this product, giving the claimed coordinate contraction.

The lines $G = C_{ka}$ and $H = C_{kb}$ are submonoids. On H the product is additive, since $L_{sb}b = 0$. Write $i : G \to F$ and $j : H \to F$ for their inclusions.

Lemma 15.12. *There is a coherent unital action of F on C_Q. Its restriction to G is trivial, and its restriction to H has first derivation term D. Both inclusions are morphisms of monoid actions.*

Proof. Tensoring with C_Q is strong symmetric monoidal for $\star$. On an ordered disjoint decomposition its comparison is the factorization equivalence of C_Q, followed by open projection. Iteration retains the same ordered partition, proving all unit and collision comparisons. On main diagonals it carries ℓ to L_Q, including its chiral bracket. Thus it carries C_W to $\mathrm{CE}_\star(L_Q \otimes E_W)$ and commutes with the positive bar construction.

The vertex automorphism $1 + vD$ over $k[v]/(v^2)$ shows that these comparisons carry D to $D \otimes 1$ after exterior extension. Natural differential polynomials in D therefore evaluate on the full collision diagrams.

For r actors let t_ν, s_ν be dual to the a, b directions. Give t_ν, s_ν, D auxiliary weights $0, -1, 1$, respectively. Prescribe the entire action to be trivial when every b input is absent. The natural relative cochain complex then has the commutative model

$$[(s_1, \dots, s_r)k[[t_1, s_1, \dots, t_r, s_r]] \otimes \Lambda(kD[-1])]^{\mathbb{G}_m}.$$

Here $D[-1]$ has cohomological degree one. To obtain this model, express universal differential polynomials as convolution products of unary coefficient extractions and divided derivatives.

After the contractible scalar generator is removed, their differential is the cobar differential of $\Lambda(P[1]) \otimes \Gamma(kD)$, where P records the actor coordinates. The elementary exterior and polynomial Koszul resolutions give $\widehat{\mathrm{Sym}}(P) \otimes \Lambda(kD[-1])$. The projection is surjective with acyclic kernel weight by weight.

Restriction to the prescribed ideal and auxiliary weight is exact. All deletions of actor or exterior coordinates split and commute. The same projection therefore has acyclic kernel relative to their compatible prescribed faces. Its target has no nonpositive degrees, and its degree-one terms are exactly $\sum_\nu s_\nu f_\nu(t_1, \dots, t_r)D$. Each corresponding relative action component is contractible by weightwise Maurer–Cartan lifting. The bar and Chevalley versions give the same lifting complex: their PBW contraction is finite on each input and preserves every exterior face.

Choose sD. Pullback by (15.5) gives $(s_1 + s_2)D$. Composition gives the same class, because its operator symbol is $e^{s_1 D} e^{s_2 D} = e^{(s_1+s_2)D}$. Relative contractibility fills the action associahedra and their unit faces. It retains the trivial G action and the prescribed derivation action of H.

Every coefficient uses finitely many operations at a configuration. Evaluation through the strong tensor comparison proves the claimed geometric action and maps. □

Let $S = C_{k^2}$. Insert its two exterior generators into four copies of F as $\varepsilon_1 \mapsto a^{(1)} - a^{(3)}$ and $\varepsilon_2 \mapsto b^{(2)} - b^{(4)}$, then multiply in that order. The resulting pointed map $w : S \to F$ factors, up to pointed homotopy, through j as $w = jf$. The marked matrix of $f : S \to H$ is

$$A_f(t,s) = \big(-s\chi(t) \mid t\chi(t)\big), \qquad \chi(t) = \frac{e^t - 1}{t} = \sum_{r \geq 0} \frac{t^r}{(r+1)!}. \tag{15.6}$$

Indeed the affine commutator has zero coordinate and translation $\chi(t)(tq - sr)b$ on fiber inputs ra, qb. Negating coordinate and fiber gives the inverse since $M(-x) = Q(-x)M(x)$. Lemma 15.11 then identifies the pointed maps. The contraction $A_f(t,s)(t,s)^T$ vanishes, but its first marked form $t\,ds - s\,dt$ is nonzero.

Lemma 15.13. *Let $h : H \to Z$ be a pointed map receiving the derivation action, with Z any factorizing coassociative actor on C_Q. No cohomological bound, cocommutativity, or augmentation to J is required of Z. The pointed map hf is not homotopic to the map $S \to J \to Z$.*

Proof. Put $B = \Omega Z$ and let h_0 be the twisting component on one pure exterior actor input. Its plain singleton class lies in $H^1 R\operatorname{Hom}_X(\ell, B_{\{1\}})$. We first show that it is nonzero on every nonempty affine open Y.

A degree-zero class $\ell|_Y \to B_{\{1\}}|_Y$ acts on $L_Q|_Y$ by an endomorphism M. It obeys

$$M\mu_Q = \mu_Q(1 \star M) = \mu_Q(M \star 1). \tag{15.7}$$

To see this without a condition on B, restrict to two distinct roots. Both ordered cobar products are equivalences there by factorization. Subtract the open exchange commutator and its action nullhomotopy. The remainder factors through

$$\operatorname{cofib}(j_! j^*(\ell \boxtimes \ell) \longrightarrow j_* j^*(\ell \boxtimes \ell)).$$

Only this source must be regular holonomic. On each diagonal branch its successive terms are $\ell[1]$ and ℓ. Tensoring by C_Q replaces ℓ by L_Q. The PBW envelope $U_\star L_Q$ and its extraordinary diagonal restriction lie in the right-module heart. Negative Ext kills the first term.

The remaining principal-part map gives $N\mu_Q$ for an endomorphism N. Inserting the vacuum shows $N = M$. The principal-part vacuum map is injective on these Hom groups by the same negative-Ext argument. Interchanging the slots proves (15.7).

If $[h_0]|_Y = 0$, a relative gauge removes h_0 on Y. The first actor-weight term then has one ordinary input and one actor input. Place them on $Y \setminus \{q\}$ and at q, respectively. The finite Spencer resolution of ℓ commutes with tensoring by arbitrary coefficient complexes. Algebraic Künneth therefore decomposes the two-output class into a finite sum of bidegrees $(r, -r)$.

Since Q lies in degree zero, its endomorphism complex at the point has only degree-zero cohomology. After the action, only $r = 0$ survives. The class is thus $\sum_i M_i \otimes N_i$, with each M_i satisfying (15.7).

The action comparison identifies this class with $D \otimes 1_Q$. There is no additional zero-column boundary: maps from the main-diagonal object L_Q to the open-supported two-root PBW term vanish by open adjunction. A linear functional on the finite span of $1_Q, N_1, \dots, N_r$ expresses D as a linear combination of the M_i.

Hence D also satisfies (15.7). But it is a derivation and kills the vacuum. Subtracting the derivation identity and inserting that vacuum gives $D = 0$. Faithfulness of localization on $\mathcal{O}_X \otimes Q$ contradicts the chosen nonzero derivation.

The map f has zero pure singleton components. Restrict to the two-root slice $(X \setminus \{q\}) \times \{q\}$. The marked form (15.6) gives the binary class of hf as

$$[\iota]|_{X\setminus\{q\}} \otimes [h_0]_q - [h_0]|_{X\setminus\{q\}} \otimes [\iota]_q,$$

where ι is the vacuum pointing of B. The second term is nonzero, since $[h_0]$ is nonzero there and $[\iota]_q$ acts as 1_Q. The two terms have different Künneth bidegrees.

This binary class is invariant under arbitrary pointed homotopies with these endpoints. Write such a homotopy as $\tau(v) + c(v)\,dv$. Its singleton terms are $\partial E, \partial F$, where $E(0) = F(0) = 0$ and $\partial E(1) = \partial F(1) = 0$. Off the collision the source bracket vanishes, and the binary homotopy equation is

$$\tau'_{1,1} = \partial c_{1,1} + [\partial E, F'] + [\partial F, E'].$$

The last terms equal $\partial[E, F'] - ([E, \partial F])'$.

Integration changes the binary term by a boundary. The endpoint term vanishes. The map through J has zero binary class, so it cannot be homotopic to hf. □

Theorem 15.14. *Let Q be an ordinary algebraic degree-zero vertex algebra on $\mathbb{A}^1_{\mathbb{C}}$ with a nonzero derivation. Its full category of factorizing transverse monoid actions has no final object. There is a parallel pair of monoid-action maps that no receiving factorizing monoid action can equalize up to homotopy, even with unbounded derived receiving actors.*

Proof. In addition to $i : G \to F$, use $i_0 : G \to J \to F$. The action on G is trivial, so both are morphisms of monoid actions by Lemma 15.12. Suppose a receiving monoid-action map $r : F \to Z$ equalizes them. Multiplicativity then replaces the first and third factors of the commutator word w by units. Its second and fourth factors cancel in the additive submonoid H. Thus rw is homotopic under J to the map through J.

On the other hand $w = jf$, and rj receives the nonzero derivation action. Lemma 15.13 excludes precisely this homotopy. A final action would receive F and make the two maps from G homotopic, contradicting the result. □

For the Heisenberg vertex algebra, the derivation $D(J) = 1$ is nonzero, so the theorem applies. For $Q = k$, the action category is the slice of transverse monoids over J, and J is final. These examples concern the stated full actor category. They do not remove the bounded center of Theorem 15.10, the ambient endomorphism monoid, or a separately defined bimodule center.

15.12 Self-extensions of the Heisenberg vacuum

The failure of that universal action leaves a useful derived endomorphism calculation intact. Let V_κ be the Heisenberg vertex algebra at $\kappa \neq 0$. Its modes satisfy

$$[J_m, J_n] = \kappa m \delta_{m+n,0},$$

where the central element acts as the identity. A module is smooth if every vector is killed by J_n for all sufficiently large n. These are the modules of the vertex algebra.

They form a Grothendieck abelian category. Submodules, quotients and direct sums retain smoothness. Filtered colimits are exact, and the cyclic smooth quotients form a set of generators.

The vacuum module is $V_\kappa = k[x_1, x_2, \dots]$, with

$$J_{-n} = x_n, \qquad J_n = \kappa n \partial_{x_n} \quad (n \geq 1), \qquad J_0 = 0.$$

For a smooth module M, define

$$C^\bullet(M) = M \otimes \Lambda(\xi_0, \xi_1, \dots), \qquad dm = \sum_{n \geq 0} J_n m \, \xi_n.$$

Each element uses finitely many exterior generators and smoothness makes the displayed differential finite. The nonnegative modes commute, so $d^2 = 0$.

Proposition 15.15. *In the smooth module category,*

$$\operatorname{Ext}^*_{V_\kappa}(V_\kappa, M) = H^*(C^\bullet(M)).$$

In particular $\operatorname{Ext}^*(V_\kappa, V_\kappa) = \Lambda(\xi_0)$. *The degree-one class is the nonsplit extension on which J_0 acts by ε on $V_\kappa \otimes k[\varepsilon]/(\varepsilon^2)$.*

Proof. Let $\mathfrak{b}$ be the abelian span of J_n for $n \geq 0$, and let $\mathfrak{a}_p$ be the span for $n \geq p$. Vacuum maps into M are $M^{\mathfrak{b}}$. The complex $C^\bullet(M)$ is the union of the finite-mode CE complexes of $\mathfrak{b}/\mathfrak{a}_p$ with coefficients in $M^{\mathfrak{a}_p}$.

For a $k[J_0, \dots, J_{p-1}]$-module N, first extend the action to $\mathfrak{b}$ by making $\mathfrak{a}_p$ act by zero. Induction from this full nonnegative-mode subalgebra to the Heisenberg algebra at the fixed central character is exact by PBW. It produces smooth modules: each vector contains finitely many negative modes, and sufficiently high positive modes commute past them and annihilate N. It is left adjoint to $M \mapsto M^{\mathfrak{a}_p}$. The latter therefore carries injectives to injectives.

For an injective smooth Y, each finite Koszul cochain complex on $Y^{\mathfrak{a}_p}$ has no positive cohomology. Exact filtered colimit gives the same vanishing for $C^\bullet(Y)$. Since $M \mapsto M \otimes \Lambda$ is exact and its degree-zero cohomology is $\operatorname{Hom}(V_\kappa, M)$, dimension shifting proves the Ext comparison.

On the vacuum the differential is $\sum_{n \geq 1} \kappa n \xi_n \partial_{x_n}$. The degree-minus-one derivation $\sigma(\xi_n) = x_n/(\kappa n)$ for $n \geq 1$, with $\sigma(\xi_0) = \sigma(x_n) = 0$, satisfies $d\sigma + \sigma d =$ total degree in these positive-mode pairs. Dividing by this degree contracts every nonconstant pair monomial. Only $\Lambda(\xi_0)$ remains.

The stated extension cannot split, because J_0 vanishes on a split sum of vacuum modules and does not vanish on this extension. Its class spans the one-dimensional Ext group. Its square is zero because Ext in degree two vanishes. This identifies the Yoneda algebra. □

For a Fock module of nonzero charge μ, the same complex has an additional term $\mu\xi_0\wedge$. Contraction in ξ_0 divided by μ makes it acyclic. In contrast, imposing $J_0 = 0$ as a condition on all modules removes ξ_0 and leaves only the scalars in the vacuum self-extensions. Thus the answer depends on the module category.

There is also an explicit coordinate complex

$$B_{\mathrm{pt}} = k[u_0, u_1, \dots] \otimes \Lambda(c_0, c_1, \dots), \qquad du_n = \kappa c_{n+1}, \quad dc_m = 0.$$

The assignments $u_n \mapsto n!x_{n+1}$ and $c_m \mapsto m!\xi_m$ identify it with $C^\bullet(V_\kappa)$ as a dg algebra. On a generator both differentials give $\kappa(n+1)!\xi_{n+1}$. This point-complex identification retains the vacuum charge class. Identifying a physical bulk with this complex on all supports and collision diagrams requires a further map. The algebra $\Lambda(\xi_0)$ is not a final action in the category excluded by Theorem 15.14.

15.13 Complete comparisons and model structures

A filtered comparison can be tested without postulating a model category. Let $f_r : C_r \to D_r$ be a compatible tower of cochain maps and let $E_r = \mathrm{Cone}(f_r)$. Assume the complexes C and D are their derived inverse limits. Then

$$\mathrm{Cone}(f) \simeq R\varprojlim_r E_r.$$

If every E_r is acyclic, so is this derived limit, and f is a quasi-isomorphism.

Degreewise surjective towers compute their derived limits by ordinary limits, since $1 -$ shift is onto on the product complex. These are the explicit hypotheses used in the continuous relative comparison.

A stronger constructive criterion gives a homotopy. If $h_r : E_r \to E_r[-1]$ commute with the transitions and satisfy $dh_r + h_r d = 1$, then $\varprojlim h_r$ contracts the ordinary limit. Contractions chosen independently at each finite stage do not define that limit homotopy.

Completeness matters even for zero differentials. For nonzero finite-dimensional spaces V_r, the inclusion

$$\bigoplus_{r\ge 1} V_r \longrightarrow \prod_{r\ge 1} V_r$$

is an isomorphism on every finite tail quotient, but its ordinary cokernel is nonzero. The direct sum is not complete for the tail filtration. Thus this example does not contradict the derived-complete comparison criterion.

A Quillen presentation adds hypotheses to this calculation. In ordinary nonnegative homological complexes, let S^n be k in degree n and let D^n have k in degrees $n, n-1$ with identity differential, for $n \ge 1$. The generating sets are

$$I = \{0 \to S^0\} \cup \{S^{n-1} \to D^n : n \ge 1\}, \qquad J = \{0 \to D^n : n \ge 1\}.$$

The degree-zero sphere belongs to I, and $0 \to S^0$ is not acyclic. A filtered version must specify its filtration-cell functors as well as smallness, recognition and tensor compatibility. Exact inverse limits alone do not establish these model-category axioms.

Once actual algebra and coalgebra model categories and a Quillen bar–cobar adjunction have been constructed, a restricted equivalence has a precise criterion. Choose full derived subcategories preserved

by the two derived functors. If their derived unit and counit are equivalences on those subcategories, the functors are inverse there: their unit and counit are the required natural equivalences. An equivalence of the whole categories requires these hypotheses on every relevant object. A length filtration or a finite quotient does not supply the missing global unit or counit.

15.14 The open–closed comparison

Let $\mathcal{B}$ be the independently defined bulk observables and let C_∂ be the chosen category of boundary conditions. Suppose its complete factorization-bimodule category and diagonal have been constructed. A coherent bulk action is then a map to

$$R\operatorname{Hom}_{\operatorname{Bimod}^{\mathrm{fact}}(C_\partial)}(\Delta_\partial, \Delta_\partial),$$

with its support, coefficient base, completion and operation structure specified. Its evaluation at one brane is a further map to that brane's Hochschild cochains.

At the Lie level the first equations are explicit. Let $\mathfrak{g}$ be a dg Lie algebra of closed coefficients and A a boundary dg algebra. Put $\mathfrak{h} = C^\bullet(A, A)[1]$, with its compatible shifted differential and Gerstenhaber bracket. A degree-one twisting cochain

$$\tau : \overline{C}_*^{\mathrm{CE}}(\mathfrak{g}) \longrightarrow \mathfrak{h}$$

satisfies

$$d\tau + \frac{1}{2}[\tau, \tau] = 0. \tag{15.8}$$

The bracket uses the CE coproduct and the bracket of $\mathfrak{h}$. The first Taylor coefficient is a chain map. The second supplies its bracket homotopy. Later coefficients fill the higher action equations.

This is an L_∞ action in Hochschild cochains. A mixed geometric action also needs compatibility with the braces, units, disjoint insertions and all collision maps.

These additional identities can be treated by an operadic convolution complex once a resolution of the actual colored operation operad is supplied. If that resolution is a cobar $\Omega\mathcal{D}$, its convolution with the endomorphism operations of $(\mathcal{B}, A)$ has Maurer–Cartan elements precisely the corresponding homotopy-coherent structures. Twisting by the fixed bulk and boundary operations leaves the relative action problem. Merely listing desired operation-preserving maps does not define this deformation complex.

For any such complete filtered convolution complex, a partial solution has a closed curvature in the first unsolved filtration degree. A primitive kills that curvature and extends the solution. Completeness and compatible choices produce a full action. The complex and its filtration must be constructed on the chosen supports before their obstruction classes are defined.

Even a completed action need not be an equivalence. After the map exists, its comparison cone is the remaining test. On derived complete coefficient towers, the preceding cone criterion provides a sufficient proof. At the category level, evaluation at a brane must also be related to the diagonal by the actual bimodule functor.

Problem 15.16. Construct a bulk action for the mixed Hamiltonian theory on the declared boundary category. Retain its coefficient topology, both units, supported collision operations and their higher homotopies. Determine whether its comparison cone is acyclic. If a restricted actor or brane category is required, construct its relation to the physical boundary conditions and identify which operations it preserves.

The finite coordinate identities and continuous relative center give explicit algebraic domains for this problem. The geometric obstruction rules out a particular unrestricted final-action formulation on a curve. Neither determines the missing mixed bulk action. Quantization and reconstruction must therefore retain the chosen recipient and its comparison maps.

15.15 A normalized junction on a curve

An action of two boundary algebras on one complex does not by itself reconstruct either boundary category. The obstruction can already occur in a free theory on a holomorphic curve. We shall exhibit nonzero modules invisible to the junction, then an extension on which its four generating fields cannot act compatibly. Finally, a localization will distinguish polynomial local operators from derived mapping objects.

15.15.1 States, fields and mapping complexes

Put $R = \mathbb{C}[[q]]$ and $S_\ell = R/(q^\ell)$, where $\ell \geq 1$. The parameter q has degree, weight and energy zero. On a marked spin coordinate define the Ramond complex

$$P_\ell = S_\ell[x_i, y_n] \otimes \Lambda(\theta_i, \eta_n), \qquad i \geq 0, \quad n \geq 1,$$
$$D = \sum_{i\geq 0} x_i \partial_{\theta_i} - \sum_{n\geq 1} \eta_n \partial_{y_n}.$$

The cohomological degrees of x, y, θ, η are respectively $0, 0, -1, 1$. Give x_i, θ_i weight and energy $(1, 2i+1)$, and y_n, η_n weight and energy $(-1, 2n-1)$. Every state is a finite polynomial. The complete complex $P = \varprojlim_\ell P_\ell$ allows a different finite polynomial at each coefficient of q. It does not take products over energy at a fixed coefficient.

The even boundary algebra A_0 is the commutative differential polynomial algebra generated by γ. The odd boundary algebra A_1 is its exterior counterpart generated by $\widehat{c}$. Both have zero differential and zero self-contractions. Their generating fields on P_ℓ are

$$\Gamma(z) = \sum_{i\geq 0} x_i z^{i-1/2} - \sum_{n\geq 1} \partial_{y_n} z^{-n-1/2},$$
$$C(z) = \sum_{n\geq 1} \eta_n z^{n-1/2} + \sum_{i\geq 0} \partial_{\theta_i} z^{-i-1/2}.$$

The two additional fields

$$\widehat{\beta}(z) = \sum_{n\geq 1} y_n z^{n-1/2} + \sum_{i\geq 0} \partial_{x_i} z^{-i-1/2},$$
$$B(z) = \sum_{i\geq 0} \theta_i z^{i-1/2} + \sum_{n\geq 1} \partial_{\eta_n} z^{-n-1/2}$$

satisfy, with graded commutators,

$$[D, B] = \Gamma, \qquad [D, \widehat{\beta}] = -C, \qquad [D, \Gamma] = [D, C] = 0.$$

Writing $\Delta(u, z) = \sum_{r\in\mathbb{Z}} u^{r-1/2} z^{-r-1/2}$, the only nonzero brackets between the four generating fields are

$$[\widehat{\beta}(u), \Gamma(z)] = \Delta(u, z), \qquad [C(u), B(z)] = \Delta(u, z).$$

These identities follow coefficientwise from the Weyl and exterior contraction relations. We call the four fields with these relations the *normalized quartet*. The physical odd field $c = qC$ is a separate normalization. No construction here inverts q.

Translation is the operator

$$T = \sum_{i\ge 0}(i+\tfrac{1}{2})(x_{i+1}\partial_{x_i} + \theta_{i+1}\partial_{\theta_i}) + \sum_{n\ge 1}(n+\tfrac{1}{2})(y_{n+1}\partial_{y_n} + \eta_{n+1}\partial_{\eta_n}) - \tfrac{1}{2}y_1x_0 + \tfrac{1}{2}\eta_1\theta_0.$$

It has weight zero and energy two, commutes with D, and obeys $[T, F(z)] = \partial_z F(z)$ for each quartet field. The two multiplication terms supply the change from creation to annihilation modes. In particular,

$$T1 = -\tfrac{1}{2}y_1x_0 + \tfrac{1}{2}\eta_1\theta_0 = D(-\tfrac{1}{2}y_1\theta_0).$$

The constant state is closed but is not translation invariant. The boundary vacuum field acts as the identity. This unit condition does not turn the constant junction state into the boundary vacuum.

To specify the recipients, write $X_i = x_i$, $G_n = -\partial_{y_n}$ for the even boundary modes and $H_n = \eta_n$, $K_i = \partial_{\theta_i}$ for the odd ones. Let $\mathcal{R}_{0,\ell}$ be the category of bigraded dg modules for the commuting X_i, G_n. Require the following support condition: for every vector v, some N, h satisfy

$$J_{N,h}v = 0, \qquad J_{N,h} = (G_n : n > N) + (G_1, \dots, G_N)^h.$$

Let $\mathcal{R}_{1,\ell}$ be the corresponding category for the anticommuting H_n, K_i, where every vector is killed by all sufficiently high K_i. All modules are unital.

Differential polynomials act by products of derivatives of the generating field. Move annihilators to the right: their indices and powers are bounded on an input, and a fixed Laurent coefficient then bounds the remaining creation indices. Thus these products are fields with finite coefficients on each input.

Morphisms in these categories strictly intertwine the boundary modes, with the cohomological signs. The restricted Hom complex is the direct sum of the Hom complexes of fixed weight and energy shift. Its derived version is formed in the stated category at each S_ℓ, before taking the derived inverse limit in ℓ. One may also retain a translation operator with the displayed mode commutators and require morphisms to commute with it. These mapping complexes do not impose locality with the quartet. Local operator fields will be considered separately below.

15.15.2 Nonzero modules invisible to the junction

Define two single-boundary modules, each with zero differential, by

$$N_{0,\ell} = S_\ell[x_i, y_n], \qquad N_{1,\ell} = \Lambda_{S_\ell}(\theta_i, \eta_n).$$

Use the boundary actions just given. Their translations T_0 and T_1 are respectively the even and odd parts of T, including the terms $-y_1x_0/2$ and $\eta_1\theta_0/2$. Both modules obey the support conditions: polynomial degree bounds the even annihilators, and a finite exterior monomial has only finitely many θ indices. Their energies are nonnegative and each energy piece is finite free over S_ℓ. The constant vector gives a nonzero cohomology class.

Theorem 15.17. *For $a = 0, 1$, the restricted derived mapping complex from P_ℓ to $N_{a,\ell}$ vanishes in $\mathcal{R}_{a,\ell}$, both with and without translation. The vanishing persists after the prescribed derived coefficient completion. Consequently the junction does not generate either category, even in the full derived subcategory whose objects have nonnegative energy and finite free energy pieces.*

Proof. Put $B_0 = S_\ell[x_i]$ and $B_1 = \Lambda_{S_\ell}(\eta_n)$. The even support condition identifies its recipient with comodules in bigraded B_0-modules for the coalgebra

$$C_+ = \bigoplus_\alpha S_\ell e_\alpha, \qquad \Delta e_\alpha = \sum_{\beta+\gamma=\alpha} e_\beta \otimes e_\gamma.$$

Here α ranges over finite multiindices. The coaction of v is $\sum_\alpha e_\alpha \otimes G^\alpha v$. The support condition makes it finite.

The odd recipient is similarly the category of comodules for the exterior coalgebra $\Lambda(\theta_i)$ in bigraded B_1-modules. The field construction above identifies these mode presentations with the respective boundary actions.

In either presentation the forgetful functor U is exact. Its right adjoint $F(V) = C \otimes_{S_\ell} V$ is exact because the coalgebra is a direct sum of free coefficient modules. On the even side identify e_α with $(-y)^\alpha/\alpha!$. On the odd side use the exterior cofree module. Thus $N_a = F(B_a)$.

If I is a K-injective base-module resolution, FI is K-injective: Hom from an acyclic comodule is Hom from its acyclic underlying module to I. This proves the unbounded derived adjunction

$$R\operatorname{Hom}_{\mathcal{R}_a}(M, FV) \simeq R\operatorname{Hom}_{B_a}(UM, V)$$

in every internal shift. Direct sums of those shifts are exact.

The polynomial Euler contractions of the pairs (θ, x) and (y, η) give $UP_\ell \simeq S_\ell$ as a B_a-module. It remains to show $R\operatorname{Hom}_{B_a}(S_\ell, B_a) = 0$ in each shift.

For B_0, use the semifree Koszul resolution $B_0 \otimes \Lambda(\theta_i)$ with $d\theta_i = x_i$. A degree-p Hom cochain is an alternating family $(f_I)_{|I|=p}$ of finite polynomials. Its index support may be infinite. If ϵ_i denotes a formal dual exterior symbol, the raw differential is $(-1)^{p+1}\sum_i x_i \epsilon_i \wedge -$. Multiplication of degree p by $(-1)^{p(p+1)/2}$ changes this to $\delta = \sum_i x_i \epsilon_i \wedge -$. Every output coefficient of this formula is a finite sum.

Define

$$Q_j f = \frac{f - f|_{x_j=0}}{x_j}, \qquad h_j = \iota_{\epsilon_j} Q_j.$$

Let π_j retain the terms containing ϵ_j and set $x_j = 0$ in their coefficients. Division in Q_j is polynomial division of an expression divisible by x_j. On terms with and without ϵ_j, the two contraction identities give

$$\delta h_j + h_j \delta = 1 - \pi_j.$$

The other indices anticommute with h_j, and each π_j commutes with δ. For any m distinct consecutive indices form

$$H_m = h_0 + \pi_0 h_1 + \cdots + \pi_0 \cdots \pi_{m-2} h_{m-1}.$$

Then

$$\delta H_m + H_m \delta = 1 - \pi_0 \cdots \pi_{m-1}.$$

The last projection requires all m exterior symbols, so it kills a cochain of degree $p < m$. A degree-p cocycle therefore has the primitive $H_{p+1} f$. Negative degrees are zero. Each primitive uses finitely many queries of the family and produces a finite polynomial on every input. Removing ϵ_j cancels the internal degree of division by x_j, so the primitive preserves the prescribed shift.

For B_1, take the semifree resolution

$$B_1 \otimes S_\ell[s_1, s_2, \ldots], \qquad ds_n = \eta_n, \qquad |s_n| = 0,$$

using divided monomials and the same internal degrees for s_n and η_n. A degree-p Hom cochain is a family $F_\alpha \in B_1^p$. Its raw differential is $-\sum_{n:\alpha_n>0} \eta_n F_{\alpha-e_n}$: the sign from graded left-linearity cancels the degree-dependent Hom sign. Multiplication of degree p by $(-1)^p$ gives

$$(\delta F)_\alpha = \sum_{n:\alpha_n>0} \eta_n F_{\alpha-e_n}.$$

Set

$$(h_j F)_\alpha = \iota_{\eta_j} F_{\alpha+e_j}, \qquad (\pi_j F)_\alpha = \begin{cases} \eta_j \wedge \iota_{\eta_j} F_\alpha, & \alpha_j = 0, \\ 0, & \alpha_j > 0. \end{cases}$$

Exterior contraction again gives $\delta h_j + h_j \delta = 1 - \pi_j$. The same telescoping sum, now over $j = 1, \dots, m$, has a final projection whose output contains m distinct η's. It vanishes for $m = p + 1$. The resulting primitive has finite outputs and unchanged internal shift.

The Euler contractions prove that the two resolutions resolve S_ℓ. Their filtrations by the number of free generators prove semifreeness, hence K-projectivity. No bound on the cohomological degree of the target was used.

It remains to retain translation. Let δ_T be the derivation of the mode algebra given by commutation with T. Induction to its Ore extension has underlying module $S_\ell[T] \otimes M$, with

$$a(T^r \otimes v) = \sum_{k=0}^{r} (-1)^k \binom{r}{k} T^{r-k} \otimes \delta_T^k(a) v.$$

It is exact and preserves the support condition. Indeed, $J_{N,h} v = 0$ implies $J_{N+r,h+r}(T^r \otimes v) = 0$: at most r positive modes are differentiated in any summand, leaving at least h undifferentiated factors. High odd annihilators similarly increase their cutoff by at most r.

Every translated module has the two-term relative induced resolution whose map sends $T^r \otimes v$ to $T^{r+1} \otimes v - T^r \otimes T_M v$. Its injectivity follows from the leading power of T. Its cokernel is M. The two induced terms give acyclic derived Hom complexes by the translation-forgetting result. Their cone is therefore acyclic. This argument does not require the earlier primitives to commute with translation.

Finally the derived inverse limit of the zero mapping objects is zero. The targets remain nonzero, which proves failure of generation. □

For finitely many Koszul pairs, the last exterior class survives in top degree. Infinite vanishing uses a longer finite set of indices in each fixed degree. It is not a limit of assertions that the finite complexes are acyclic. If a tensor–Hom adjunction for the boundary junction has been constructed, its counit on N_a has source zero and nonzero target. This prevents an equivalence on the stated recipient. It says nothing by itself about dualizability in a different category with a different fusion tensor.

A smaller recipient can have a positive generation theorem. For $A = S_\ell[x]$, the complex $[A\theta \xrightarrow{x} A]$ in degrees -1, 0 is a compact nongenerator of $D(A)$: Hom into $A[x^{-1}]$ is contracted by $-x^{-1}$. It generates the independently defined subcategory of complexes whose localization at x vanishes. Indeed the local-cohomology complex $[A \to A[x^{-1}]]$ is the filtered colimit of $(A/x^n)[-1]$. Each A/x^n is built by extensions from A/x, and tensoring this local-cohomology complex with an x-torsion object recovers that object. A restriction of the recipient can therefore repair generation, but its support condition must be specified independently.

15.15.3 A stable extension without a compatible quartet

Requiring every object to admit the four fields is another possible restriction. If morphisms still intertwine only one boundary, that restriction fails to be stable. The failure is detected by two commuting even boundary modes.

Define the operator

$$\Omega = \sum_{i \geq 0} (2i + 1) \eta_{i+1} \partial_{\theta_i} = -\operatorname{Res}_z C(z) \partial_z C(z).$$

It has cohomological degree two, weight -2 and energy zero. The sum is finite on each state. It commutes with D, all even boundary modes, and T. For translation, the coefficient of each adjacent pair cancels by

$$(2i + 1)(i + \tfrac{3}{2}) - (2i + 3)(i + \tfrac{1}{2}) = 0;$$

the remaining multiplication term contains $\eta_1^2 = 0$.

Adjoin an odd variable ε of cohomological degree -1, weight two and energy zero. For $a \in S_\ell$ put

$$E_a = P_\ell \otimes \Lambda(\varepsilon), \qquad d_a = D + a\varepsilon\Omega,$$

where $D\varepsilon = T\varepsilon = 0$ and the even boundary actions are unchanged. Closedness of Ω gives $d_a^2 = 0$ and $[T, d_a] = 0$. The exact sequence $\varepsilon P_\ell \to E_a \to P_\ell$ places E_a in the stable subcategory generated by P_ℓ and its internal shifts. All its energy pieces are finite free.

The change of polynomial coordinates $x_i' = x_i + a(2i+1)\varepsilon\eta_{i+1}$ gives underlying cohomology $\Lambda_{S_\ell}(\varepsilon)$. It does not preserve the boundary action.

Proposition 15.18. *If $a \neq 0$, the even boundary module E_a is not quasi-isomorphic to one admitting a compatible normalized quartet. Let $\mathcal{S}$ be a full stable subcategory of the even boundary derived category, closed under equivalence and containing P_ℓ and its internal shifts. Some object of $\mathcal{S}$ then admits no compatible quartet, even up to quasi-isomorphism.*

Proof. Retain $x = X_0$ and $g = G_1 = -\partial_{y_1}$, and take the derived specialization $g = 0$. The two-term free resolution of S_ℓ over $S_\ell[g]$ computes this specialization. Since g is degreewise surjective, its cone is, up to the common cohomological shift, its kernel. This is a statement about complexes. Integration with respect to y_1 need not be a chain map. The kernel omits y_1.

Contract the remaining pairs after the coordinate change above. The residual complex is

$$K_a = S_\ell[x] \otimes \Lambda(\theta, \varepsilon, \eta),$$
$$d\theta = x + a\varepsilon\eta, \qquad d\varepsilon = d\eta = 0.$$

Division by the monic polynomial $x + a\varepsilon\eta$ gives its cohomology and the action of x:

$$H(K_a) = \Lambda_{S_\ell}(\varepsilon, \eta), \qquad x[1] = -a[\varepsilon\eta].$$

The cohomology is free over S_ℓ, so this action is nonzero for every nonzero a, including nilpotent a.

For example, if $u = a\varepsilon\eta$, then $u^2 = 0$ and

$$\frac{x^m - (-u)^m}{x + u} = x^{m-1} - ux^{m-2} \quad (m \geq 2).$$

Multiplying this quotient by θ gives the explicit contraction onto evaluation at $x = -u$. The cases $m = 0, 1$ are immediate. No polynomial-degree quotient is involved.

In a compatible quartet the coefficient b_0 of B commutes with every even boundary mode and satisfies $[d, b_0] = x$. It therefore gives a null-homotopy of x after derived specialization at $g = 0$. The nonzero action just calculated contradicts this. Restriction and derived specialization preserve quasi-isomorphisms, so the contradiction survives every replacement in the even boundary derived category. Stability would include E_a, which proves the final assertion. □

The same extension occurs without dividing the physical odd field by q. The expression $-\operatorname{Res} c\,\partial c = q^2\Omega$ gives $a = q^2$. It is nonzero over S_ℓ for $\ell \geq 3$ and over R. The lower two precisions do not detect it. Thus an additional support condition satisfied by P cannot remove this extension while preserving full stable closure and the original morphisms.

One can restore the differential identity alone by changing the odd field to

$$B_a = B - 2a\varepsilon\,\partial_z\widehat{\beta}.$$

It obeys $[d_a, B_a] = \Gamma$ and translation covariance. Its mixed bracket with Γ is $2a\varepsilon\partial_z\Delta$, however, so it is not the original quartet. At the two selected modes, $[g, \theta_0 - a\varepsilon y_1] = a\varepsilon$. The defect occurs in the boundary-linearity required of the homotopy, not in the underlying scalar contraction.

There is also a positive category with stricter morphisms. Let $\mathcal{W}_\ell$ be the dg Weyl–Clifford algebra of finite polynomial operators in all quartet modes. In normal order its symbols are the polynomial generators x, y, ξ, ζ and exterior generators $\theta, \eta, \kappa, \pi$, where $\xi = \partial_x$, $\zeta = \partial_y$, $\kappa = \partial_\theta$ and $\pi = \partial_\eta$. The differential pairs them by

$$d\theta = x, \qquad dy = -\eta, \qquad d\xi = -\kappa, \qquad d\pi = -\zeta.$$

On normal symbols, the reverse pairing divided by positive total polynomial degree is an Euler contraction. Every symbol uses finitely many modes, so the contraction proves $S_\ell \to \mathcal{W}_\ell$ is a quasi-isomorphism of dg algebras. Evaluation $w \mapsto w_1$ is a quasi-isomorphism $\mathcal{W}_\ell \to P_\ell$ of left dg modules. Hence

$$D(\mathcal{W}_\ell) \simeq D(S_\ell), \qquad R\operatorname{End}_{\mathcal{W}_\ell}(P_\ell) \simeq S_\ell,$$

and P_ℓ is a compact generator in this ordinary module category.

Its endomorphisms recover scalars, not the opposite boundary algebra. The category has changed its morphisms, and no boundary fusion has been constructed. The primitive diagonal action on a tensor product of two Weyl modules has $[\partial, x] = 2$. It does not define a tensor for the relation $[\partial, x] = 1$.

15.15.4 Polynomial local operators and derived descent

The preceding derived Hom complexes consist of boundary intertwiners. A different prescription uses fields local with the quartet. At each coefficient of q, use normally ordered polynomial expressions in the original creation and annihilation symbols, with bounded total degree. Mode indices may range without bound. Require every Laurent coefficient to act finitely on each input and the field to be lower truncated on that input.

Impose strict linearity for one boundary and locality with each quartet field. Translation covariance may also be imposed. Denote this complex of local operator fields from M to N by $\mathcal{A}_{\mathrm{pol}}(M, N)$, with differential the commutator with D.

Locality means that a power of $u - z$ kills each graded commutator. It does not mean that every coefficient is an intertwiner of all quartet modes. The polynomial degree restriction is part of the definition of $\mathcal{A}_{\mathrm{pol}}$. Its derived invariance must therefore be checked in that precise class. A derived bifunctor is called reduced here if it vanishes whenever either argument is zero.

Proposition 15.19. *Put $Q_\ell = P_\ell[x_0^{-1}]$, with its extended quartet and translation. Its underlying complex is acyclic, but $\mathcal{A}_{\mathrm{pol}}(Q_\ell, Q_\ell)$ has a nonzero identity class. The rectangular complex $\mathcal{A}_{\mathrm{pol}}(P_\ell, Q_\ell)$ likewise has a nonzero inclusion class. Thus this prescription does not descend, objectwise or on the displayed pair, to a reduced bifunctor on the localization at underlying quasi-isomorphisms.*

Proof. Localization leaves all polynomial-input support bounds intact. The fields and translation extend by the usual derivative of x_0^{-1}. Multiplication by $h = \theta_0/x_0$ satisfies $Dh + hD = 1$, so $Q_\ell \to 0$ is an underlying quasi-isomorphism. This contraction need not intertwine the chosen boundary or commute with translation.

Every operator in the polynomial class preserves $P_\ell \subset Q_\ell$. Conversely, such an operator on P_ℓ extends uniquely to Q_ℓ. Its ∂_{x_0}-order is bounded, and a finite input involves only finitely many other derivative indices. Normal coefficients can be recovered by evaluating on monomials of increasing degree. The diagonal entries are factorials, which are units in S_ℓ. This also proves faithfulness after localization.

Apply that argument to each Laurent coefficient. All identities for the differential, boundary-linearity, locality and translation are identities of the same finite-degree normal operators. They hold on Q_ℓ exactly when they hold on P_ℓ. The polynomial diagonal and rectangular complexes are therefore identified with the corresponding complex on P_ℓ, with identity or inclusion corresponding to the identity operator.

Let $\epsilon_P : P_\ell \to S_\ell$ be the constant-term functional. It satisfies $\epsilon_P D = 0$ and $D1 = 0$. If the identity were $[D, F]$ for a degree--1 polynomial field, evaluation on 1 would give $D(F1) = 1$. Applying ϵ_P contradicts this coefficientwise.

Thus the identity and inclusion classes are nonzero. A reduced derived bifunctor must vanish when one argument is the zero object. Thus $Q_\ell \simeq 0$ proves the claimed failure. □

The same argument holds after coefficientwise completion. A polynomial primitive over R would reduce to an impossible primitive modulo q.

This example allows negative energy, with an infinite energy-zero piece. It does not rule out a construction on a smaller positive-energy category, nor a derived replacement agreeing with the polynomial prescription only at P. Both require their own mapping objects and comparison maps.

On the even boundary, enlarging the coefficient class does produce a local contraction. Write

$$a(z) = \sum_{i \geq 0} x_i z^i, \qquad p(z) = \sum_{n \geq 1} \partial_{y_n} z^{-n}, \qquad \Gamma = z^{-1/2}(a - p).$$

Since x_0 is invertible on Q_ℓ, the formal series $a(z)^{-1}$ is defined. The operator field

$$I(z) = z^{1/2} \sum_{r \geq 0} a(z)^{-r-1} p(z)^r$$

is an inverse of $\Gamma(z)$. For a fixed input the r-sum ends at its total y-degree. Each remaining Laurent coefficient involves finitely many variables.

Consequently

$$H(z) = B(z)I(z), \qquad [D, H(z)] = 1$$

is a well-defined lower-truncated field. It commutes with Γ, is translation covariant, and is local with the quartet. Indeed

$$[\widehat{\beta}(u), H(z)] = -\Delta(u, z)B(z)I(z)^2,$$
$$[C(u), H(z)] = \Delta(u, z)I(z),$$

while its brackets with Γ and B vanish. Multiplication by $u - z$ kills the displayed distributions.

This inverse field requires unbounded orders of poles in x_0 across its Laurent coefficients. After setting $x_i = 0$ for $i \geq 2$, its θ_0 component on the constant state is

$$\frac{\theta_0}{x_0 + x_1 z} = \sum_{m \geq 0} (-1)^m \theta_0 x_1^m x_0^{-m-1} z^m.$$

It therefore lies outside the polynomial class.

Allowing a fixed finite pole order does not supply the same construction. In fact, locality with $\widehat{\beta}$ rules out a primitive in the class of finite-degree normal operators with a globally bounded Laurent pole order in x_0. Let K be a locality order at a fixed coefficient of q. Multiplication by $(u - z)^K$ gives the K-th finite-difference equation in the annihilator index. For a normal symbol Σ its solutions have the form

$$\partial_{x_i} \Sigma = \sum_{k=0}^{K-1} G_k(z)(i - \tfrac{1}{2})^{\underline{k}} z^{i-1/2-k}.$$

Here $t^{\underline{k}} = t(t-1)\cdots(t-k+1)$, and $t^{\underline{0}} = 1$. The falling factorials form a basis of polynomial sequences of degree less than K. Evaluation at $i = 1, \dots, K$ has an invertible Vandermonde matrix. It therefore expresses the row $i = 0$ as a Laurent scalar combination of those rows. Hence

$$\left(\partial_{x_0} - \sum_{i=1}^{K} c_i(z)\,\partial_{x_i}\right)\Sigma = 0$$

for suitable $c_i(z)$. For $K = 0$ use ∂_{x_0}.

If Σ has largest pole x_0^{-m} with coefficient Σ_{-m} independent of x_0, the last equation has coefficient $-m\Sigma_{-m}$ at x_0^{-m-1}. The other derivatives cannot increase that pole order. Positive integers are units in S_ℓ, so this is impossible. No pole can occur, and the polynomial augmentation argument applies.

Strict odd boundary-linearity has a stronger obstruction. For a degree--1 field F commuting with all K_i, the vector $F(z)1$ lies in every $\ker \partial_{\theta_i}$. It is therefore θ-free. Such a vector has nonnegative cohomological degree, even with Laurent coefficients in the even variables, and so $F(z)1 = 0$. It cannot satisfy $[D, F] = 1$. The nonzero bracket $[C, H] = \Delta I$ is exactly where the positive inverse-field construction leaves this strict odd boundary class.

The normalized curve model thus separates three questions. Strict derived intertwiners need not detect all boundary objects. Objects carrying a quartet need not remain so after cones of boundary-linear maps. Polynomial local fields need not define derived mapping objects.

Requiring all quartet modes to be intertwined repairs generation in an ordinary scalar module category. Allowing inverse fields repairs the even local contraction on a larger coefficient class. Neither construction identifies the physical boundary fusion, its unit, or its supported collision operations. Those comparisons remain part of Problem 15.16.

Chapter 16

Formal geometry and cyclic descent

A local formula for an observable must transform when the Darboux chart changes. A local comparison must transform with it. The two transformations can agree up to homotopy on each overlap and still require another homotopy on triple overlaps. Quantization adds the period of the algebra and the normalization of its trace to these gluing data.

The circle comparison of Chapter 13 provides a local map and two Hamiltonian homotopies. Its logarithmic torus example also gives a global comparison on a specified algebra. A general surface requires coordinate changes and allowed overlap corrections. Their successive choices need a topology of convergence. The first obstruction already appears for two one-dimensional complexes.

16.1 Coordinates that preserve the formal point

Put $R = \mathbb{C}[[q, p]]$, with maximal ideal $\mathfrak{m} = (q, p)$ and its $\mathfrak{m}$-adic topology. A pointed formal coordinate change is a continuous $\mathbb{C}$-algebra automorphism of R. It preserves $\mathfrak{m}$. It is symplectic when its pullback preserves $\omega = dq \wedge dp$. For maps ϕ followed by ψ, pullbacks satisfy

$$(\psi \circ \phi)^* = \phi^* \psi^*.$$

Thus the order of pullback operators is fixed by their action on functions, independently of the notation for a transition matrix.

The Hamiltonian convention remains $X_f = \{f, -\}$, with $\{q, p\} = 1$. It gives

$$X_f = f_q \partial_p - f_p \partial_q, \qquad -\iota_{X_f} \omega = df.$$

A symplectic formal vector field v therefore determines a closed formal one-form $-\iota_v \omega$.

Proposition 16.1. *Every symplectic formal vector field is Hamiltonian, uniquely modulo constants. It vanishes at the formal point exactly when its Hamiltonian has no linear term. A Hamiltonian in $\mathfrak{m}^3$ has a convergent formal exponential e^{X_f} acting on R. A nonzero constant translation is not a pointed formal automorphism.*

Proof. Let $E = q\partial_q + p\partial_p$. Decompose a closed formal one-form as $\alpha = \sum_{r \geq 1} \alpha_r$, where pullback by scalar dilation multiplies α_r by its rth power. Cartan's identity gives

$$d(\iota_E \alpha_r) = r \alpha_r.$$

Hence $f = \sum_{r \geq 1} r^{-1} \iota_E \alpha_r$ satisfies $df = \alpha$. The sum converges in the formal topology.

A function with zero differential is constant. The constant coefficients of v are exactly the coefficients of the linear part of f.

For $f \in \mathfrak{m}^3$, the derivation X_f sends $\mathfrak{m}^s$ into $\mathfrak{m}^{s+1}$. Each finite jet of $e^{X_f} a$ therefore uses finitely many terms. The derivation rule makes this exponential multiplicative, and e^{-X_f} is its inverse. It preserves the Poisson bracket. In contrast, $p \mapsto p + c$ with $c \neq 0$ does not preserve the maximal ideal, so it cannot define a pointed formal automorphism. □

Translations do act after adjoining a formal motion parameter. For example $e^{t\partial_p}$ acts continuously on $\mathbb{C}[[q, p, t]]$ with its (q, p, t)-adic topology. Evaluating t at a nonzero complex number is an additional operation, unavailable on an arbitrary formal series. Quadratic Hamiltonians instead generate the linear symplectic group. Their exponentiation is distinct from the pronilpotent exponential of $\mathfrak{m}^3$.

Now let (X, ω) be a holomorphic symplectic surface and let v be a global holomorphic symplectic vector field. Define $\alpha_v = -\iota_v\omega$. On sufficiently small coordinate balls choose holomorphic functions h_i satisfying $dh_i = \alpha_v$. On each connected component of an overlap, $h_j - h_i$ is constant. These differences define

$$\mathrm{per}_X(v) = [h_j - h_i] \in H^1(X, \mathbb{C}_X). \tag{16.1}$$

Equivalently this is the connecting class for the exact sequence of sheaves

$$0 \longrightarrow \mathbb{C}_X \longrightarrow \mathcal{O}_X \xrightarrow{d} \Omega^1_{X,\mathrm{cl}} \longrightarrow 0.$$

Here $\Omega^1_{X,\mathrm{cl}}$ is the sheaf of closed holomorphic one-forms. Local exactness follows from the holomorphic Poincaré lemma on a ball. An arbitrary Stein open need not have this global exactness.

Proposition 16.2. *The field v has a global holomorphic Hamiltonian if and only if* $\mathrm{per}_X(v) = 0$. *If X is compact and connected, this holds if and only if $v = 0$.*

Proof. Vanishing of the connecting class means that constants can be added to the h_i so that they agree. The resulting global function is a Hamiltonian for v. A global Hamiltonian conversely supplies such a choice. On a compact connected complex manifold every holomorphic function is constant, so its Hamiltonian vector field is zero. Nondegeneracy of ω proves the remaining implication. □

For the logarithmic torus, $\omega = d\log x \wedge d\log y$ and $v = x\partial_x$ give $\alpha_v = -d\log y$. Its integral around the second circle is $-2\pi i$, so no single-valued Hamiltonian exists. The vector field and the global cyclic comparison nevertheless exist. The period obstructs a Hamiltonian function with the stated coefficients. Global algebras and non-Hamiltonian vector fields can still exist on the surface.

16.2 An obstruction relative to fixed local maps

Take the complex $\mathbb{C}$ in degree zero on two charts and their overlap, with identity restrictions. Prescribe comparison maps given by multiplication by 1 and by 2. They are separately isomorphisms. Their overlap discrepancy is the element 1 of

$$T^0 = \mathbb{C}^2, \qquad T^1 = \mathbb{C}, \qquad D(a, b) = b - a.$$

It is a boundary in T. Removing it while keeping both prescribed chart maps requires a correction with zero chart components, and there is no such degree-zero correction. The coefficient space of allowed corrections is therefore part of the obstruction problem.

Let $\pi : (T, D) \to (Q, d_Q)$ be a degreewise split surjection of cochain complexes over a commutative ring. The quotient Q records the local data that must be preserved. Put $K = \ker\pi$, and choose a graded section $s : Q \to T$. For topological complexes all these maps, including the section, are required to be continuous.

Define the degree-one map

$$b = Ds - sd_Q : Q \longrightarrow K[1].$$

The notation indicates its degree. As operators, $Db + bd_Q = 0$. For a prescribed cycle $\theta \in Q^0$, the defect of its chosen lift is

$$\Omega = D(s\theta) = b\theta \in K^1.$$

Proposition 16.3. *The class $[\Omega] \in H^1(K)$ is independent of the section. There is a closed lift of θ to T° exactly when this class vanishes. If lifts exist, their classes modulo homotopies in K^{-1} form a torsor for $H^\circ(K)$.*

Proof. The identity $D^2 = 0$ makes Ω closed. A second section changes $s\theta$ by an element of K°, so it changes Ω by a boundary in K. Every lift is $s\theta + h$, with $h \in K^\circ$. It is closed precisely when $Dh = -\Omega$.

The difference of two closed lifts is a degree-zero cycle in K. Two such differences represent the same relative homotopy class precisely when they differ by $D(K^{-1})$. This gives the free transitive action of $H^\circ(K)$. □

In the two-chart example, $Q = \mathbb{C}^2$ lies in degree zero, $K^\circ = 0$, and $K^1 = \mathbb{C}$. Thus $[\Omega] = [1]$ is nonzero in $H^1(K)$ although $H^1(T) = 0$. The absolute complex loses the requirement that the two local maps remain fixed. Existence need not make the choices contractible. For $K = \mathbb{C}$ in degree zero, distinct elements give distinct homotopy classes of lifts.

Sometimes local changes are permitted. Specify them by a complex L and a chain map $j : L \to Q$. An allowed change replaces θ by $\theta + jw$, with $w \in Z^\circ(L)$.

Put

$$E^n = K^n \oplus L^n, \qquad d_E(k, w) = (Dk + bjw, d_L w). \tag{16.2}$$

The relation $Db + bd_Q = 0$ makes $d_E^2 = 0$. The class $[(\Omega, 0)] \in H^1(E)$ vanishes exactly when

$$d_L w = 0, \qquad Dh + bjw = -\Omega$$

has a solution. These equations say that $s(\theta + jw) + h$ is a closed lift. On the two charts, diagonal changes (w, w) leave the discrepancy unchanged, whereas allowing arbitrary changes in $\mathbb{C}^2$ removes it. The two choices define different comparison problems.

16.3 Overlap equations and their higher terms

For cochain complexes P, G, the differential on a homogeneous map $f : P \to G$ of degree r is

$$d_{\mathrm{Hom}} f = d_G f - (-1)^r f d_P.$$

Suppose first that P, G are complexes of sheaves on an ordered cover $\{U_i\}$. Their continuous Hom sheaf $\mathcal{H}$ has the product Čech total complex

$$T^n = \prod_{p \geq 0} \check{C}^p(\mathfrak{U}, \mathcal{H}^{n-p}), \qquad D = \check{\delta} + (-1)^p d_{\mathrm{Hom}}. \tag{16.3}$$

Projection to Čech degree zero is a chain map onto the complex of chart data. Its kernel is the relative complex of Proposition 16.3.

A closed element of total degree zero has components Φ_p of internal degree $-p$. Its equations begin

$$d\Phi_0 = 0, \qquad d\Phi_1 = \check{\delta}\Phi_0, \qquad d\Phi_2 = -\check{\delta}\Phi_1.$$

Each equation is a primitive problem in its actual Hom complex. To identify this totalization with derived global sections requires an acyclic resolution or the corresponding hypotheses on the cover and totalization. The displayed algebraic equations hold without that further identification.

Object transitions may themselves be coherent only up to homotopy. Write $u_{ij} : P_i \to P_j$ and $v_{ij} : G_i \to G_j$ for chain maps on overlaps. All compositions below are restricted to the full intersection. Choose degree-minus-one object homotopies satisfying

$$dt^P_{ijk} = u_{jk}u_{ij} - u_{ik}, \qquad dt^G_{ijk} = v_{jk}v_{ij} - v_{ik}.$$

For local comparison maps $\Theta_i : P_i \to G_i$, a pair homotopy $h_{ij} : P_i \to G_j$ must satisfy

$$dh_{ij} = \Theta_j u_{ij} - v_{ij}\Theta_i. \tag{16.4}$$

The right side is closed. Its exactness is an additional condition.

Lemma 16.4. *If the pair equations hold, then*

$$c_{ijk} = v_{jk}h_{ij} + h_{jk}u_{ij} - h_{ik} + t^G_{ijk}\Theta_i - \Theta_k t^P_{ijk} \tag{16.5}$$

is a degree-minus-one cycle. The next comparison homotopy has degree -2 and obeys $dh_{ijk} = -c_{ijk}$ in the totalization convention (16.3).

Proof. Applying d_{Hom} to the first three terms and using (16.4) gives

$$\Theta_k(u_{jk}u_{ij} - u_{ik}) - (v_{jk}v_{ij} - v_{ik})\Theta_i.$$

The differential of the last two terms is its negative. The sign of the next primitive is the positive internal differential in Čech degree two. □

The object data require their own next equation. For instance

$$u_{kl}t^P_{ijk} + t^P_{ikl} - t^P_{jkl}u_{ij} - t^P_{ijl}$$

is closed: substitution of the four equations for dt^P cancels each composite. A degree-minus-two object homotopy must kill this cycle before the next comparison equation is formulated.

For strict sheaves, a total comparison Φ acts on total Čech cochains by

$$(\Phi x)_{i_0\dots i_n} = \sum_{p=0}^{n} (-1)^{p(n-p)} \Phi_{i_0\dots i_p}(x_{i_p\dots i_n}).$$

The sign moves internal degree $-p$ past Čech degree $n-p$. The cup-product Leibniz identity then gives $D\Phi = \Phi D$. Each output component uses a finite sum, so this formula is continuous in the product topology when its component maps are.

An algebra comparison also has multiplication inputs. Its multiplicativity equation is a map on $P\widehat{\otimes}P$, not on P alone. Supported collision operations, normal derivatives and both vacuum insertions likewise contribute their own inputs and relations.

A deformation complex for all these operations requires a resolution of the entire operation diagram. At a linear extension stage, restriction to fixed local operations must supply the map $T \to Q$. Its kernel specifies the allowed corrections. The unary Hom calculation does not construct this larger complex or its supported maps.

16.4 Nonlinear corrections and complete filtrations

A differential graded Lie algebra $\mathfrak{g}$ has a degree-one differential d and a degree-zero graded Lie bracket. For $a \in \mathfrak{g}^1$, define

$$F(a) = da + \tfrac{1}{2}[a,a], \qquad d_a = d + [a,-].$$

The Jacobi and derivation identities give

$$d_a^2 x = [F(a), x], \qquad d_a F(a) = 0. \tag{16.6}$$

Thus d_a is a differential at a Maurer–Cartan element, meaning an element with $F(a) = 0$. For a general candidate a, a correction $b \in \mathfrak{g}^1$ must solve

$$F(a+b) = F(a) + d_a b + \tfrac{1}{2}[b,b] = 0. \tag{16.7}$$

The quadratic term can be decisive. Take degree-one elements a, b, central degree-two elements c, z, and set

$$da = c, \qquad db = -c, \qquad [b,b] = 2z,$$

with all other differentials and brackets zero. These formulas define a dg Lie algebra. Then

$$F(a) + d_a b = 0, \qquad F(a+b) = z \neq 0.$$

A primitive of the linear defect has solved only its linear equation.

For higher operations use the cohomological L_∞ convention $|\ell_n| = 2 - n$. The curvature and its unary Taylor operator are

$$F(a) = \sum_{n \geq 1} \frac{1}{n!} \ell_n(a^{\otimes n}), \qquad d_a x = \sum_{n \geq 0} \frac{1}{n!} \ell_{n+1}(a^{\otimes n}, x).$$

These series are finite in a nilpotent algebra, or converge in the complete filtrations used below.

Proposition 16.5. *Let $\widetilde{\mathfrak{g}} \to \mathfrak{g}$ be a surjective strict morphism of nilpotent L_∞ algebras over a field of characteristic zero. Let its kernel be I, and assume that every bracket containing two entries from I vanishes. For a Maurer–Cartan element a of $\mathfrak{g}$, choose a graded lift $\widetilde{a}$. Then (I, d_a) is a complex and $F(\widetilde{a}) \in I^2$ is a closed element. Its class in $H^2(I, d_a)$ is independent of the lift and vanishes exactly when a Maurer–Cartan lift exists. Relative gauge classes of such lifts, when nonempty, form a torsor for $H^1(I, d_a)$.*

Proof. The curvature lies in I because its image is $F(a) = 0$. The L_∞ identities give its Bianchi identity. The same identities express $d_a^2 x$ as a sum with both a curvature input and the input $x \in I$. Every such term vanishes.

Replacing $\widetilde{a}$ by $\widetilde{a} + b$, with $b \in I^1$, leaves d_a unchanged on I and changes the curvature by $d_a b$. This proves the obstruction criterion. Differences of lifts are cycles in I^1. Relative degree-zero gauge changes are the boundaries $d_a(I^0)$. Higher terms in the gauge action contain two kernel inputs and vanish. □

The obstruction degree is two for these objects and one for the linear comparison maps of Proposition 16.3. A shift changes the degrees of both the elements and the brackets.

Let $\mathfrak{g} = F^1\mathfrak{g}$ be complete and separated for a decreasing filtration. Assume that d preserves it and that

$$\ell_k(F^{r_1}, \dots, F^{r_k}) \subseteq F^{r_1 + \cdots + r_k}.$$

Completeness means $\mathfrak{g} \simeq \varprojlim_r \mathfrak{g}/F^r$. Separatedness means $\bigcap_r F^r = 0$.

Theorem 16.6. *Suppose $a \in F^1\mathfrak{g}^1$ satisfies $F(a) \in F^r$, with $r \geq 1$. Its leading curvature o_r is a closed degree-two element of $(\mathrm{gr}_F^r \mathfrak{g}, d)$. A correction $b_r \in F^r\mathfrak{g}^1$ makes the curvature lie in F^{r+1} exactly when $[o_r] = 0$. Successive compatible corrections at every order converge to a Maurer–Cartan element.*

Proof. Each term in the Bianchi identity involving a raises the filtration, so its leading part is $do_r = 0$. The leading correction equation is

$$o_r + d\bar{b}_r = 0.$$

Every term with two corrections lies in $F^{2r} \subseteq F^{r+1}$, and every additional background input also raises the filtration. This proves the equivalence.

The successive corrections form a Cauchy sequence. Continuity of each bracket identifies its limit curvature modulo every F^r with zero. Separatedness then makes the full curvature zero. □

The later classes depend on the earlier choices. Finding unrelated solutions at every quotient is weaker than constructing this compatible sequence. Prescribed local operations are retained by performing the correction in their relative kernel.

For a comparison of coalgebra models with square-zero coderivations Q_P, Q_G, the defect of a degree-zero map $\mathcal{X}$ is

$$\mathcal{R} = Q_G\mathcal{X} - \mathcal{X}Q_P, \qquad Q_G\mathcal{R} + \mathcal{R}Q_P = 0.$$

At a filtration stage where the new Taylor coefficients enter linearly, this is a degree-one cycle in the comparison complex. Changing those coefficients by h changes the defect by Dh. The obstruction is its relative H^1 class. Nullary terms, unit conditions and the continuity of every Taylor component remain part of the chosen model.

16.5 Compatible primitives in an inverse system

Let (C_n, d_n) be a countable inverse system of cochain complexes of vector spaces, with chain maps $r_n : C_{n+1} \to C_n$. Products in this section are algebraic products. A topological application requires an exact structure with exact products and a Roos resolution computing its derived inverse limit. All indicated maps and lifts must belong to its continuous coefficient class.

Write

$$Z_n^q = \ker(d_n : C_n^q \to C_n^{q+1}), \qquad B_n^q = d_n(C_n^{q-1}), \qquad H_n^q = Z_n^q/B_n^q.$$

For any tower A_n, its Roos operator is

$$\Delta_A : \prod_n A_n \longrightarrow \prod_n A_n, \qquad \Delta_A(a)_n = a_n - r_n a_{n+1}.$$

Its kernel is $\varprojlim A_n$ and its cokernel is the first derived inverse limit $\varprojlim{}^1 A_n$ for this countable system.

Suppose $o_n \in Z_n^1$ is compatible: $r_n o_{n+1} = o_n$. A strictly compatible primitive is a family $c_n \in C_n^0$ with

$$d_n c_n = -o_n, \qquad \Delta_C(c) = 0. \tag{16.8}$$

The first equation alone is a separate problem at each level.

Theorem 16.7. *Assume $[o_n] = 0$ for every n, and choose primitives c_n. Their discrepancies $\delta_n = c_n - r_n c_{n+1}$ lie in Z_n^0. The class*

$$\lambda_Z(o) = [(\delta_n)_n] \in \varprojlim{}^1 Z_n^0 \tag{16.9}$$

is independent of the chosen primitives. A strict primitive family exists if and only if $\lambda_Z(o) = 0$. Its image $\lambda_H(o)$ in $\varprojlim{}^1 H_n^0$ is sufficient when $\varprojlim{}^1 B_n^0 = 0$.

Proof. Compatibility of o gives $d_n\delta_n = 0$. Changing c_n by a cycle a_n changes δ by $\Delta_Z(a)$. This proves independence. If $\delta = \Delta_Z(a)$, then $c - a$ satisfies both equations (16.8). The converse follows by comparing any initial primitives with a strict family.

Apply the kernel–cokernel exact sequence to the two product rows of $0 \to B_n^0 \to Z_n^0 \to H_n^0 \to 0$ and their Roos operators. It contains

$$\varprojlim{}^1 B_n^0 \longrightarrow \varprojlim{}^1 Z_n^0 \longrightarrow \varprojlim{}^1 H_n^0 \longrightarrow 0.$$

The first vanishing hypothesis therefore makes the last map an isomorphism, proving the final assertion. □

Thus a cohomology-level discrepancy must not be substituted for a cycle-level discrepancy without a condition allowing its boundary remainders to be removed. This distinction disappears for complexes concentrated in nonnegative degrees, since then $B_n^0 = 0$.

A tower is Mittag–Leffler if, for each fixed n, the images of $A_m \to A_n$ stabilize as m increases. A countable Mittag–Leffler tower has surjective Roos operator.

To see this, take its stable-image tower. Its transition maps are surjective, so $a_n - r_n a_{n+1} = b_n$ can be solved recursively. The quotient tower has eventually zero maps to each fixed level. There the solution is the finite sum

$$a_n = b_n + r_n b_{n+1} + r_n r_{n+1} b_{n+2} + \cdots .$$

Lifting the quotient solution and correcting its discrepancy in the stable-image tower solves the original equation.

Finite-dimensional towers are Mittag–Leffler. A finite cutoff on field labels does not by itself make a completed functional space finite-dimensional.

The derived inverse limit

The two-column Roos complex of the tower C_n has terms and differential

$$\begin{aligned} \mathcal{R}(C)^k &= \prod_n C_n^k \oplus \prod_n C_n^{k-1}, \\ d_{\mathcal{R}}(x, y) &= (dx, \Delta_C x - dy). \end{aligned} \tag{16.10}$$

The chain-map identity $d\Delta_C = \Delta_C d$ proves $d_{\mathcal{R}}^2 = 0$. It is the shifted cone of the Roos operator. Its long exact cohomology sequence gives

$$0 \longrightarrow \varprojlim{}^1 H_n^{k-1} \longrightarrow H^k(\mathcal{R}(C)) \longrightarrow \varprojlim H_n^k \longrightarrow 0. \tag{16.11}$$

Products of vector spaces are exact, so cohomology commutes with the products used in this derivation. No splitting of this exact sequence is specified.

The compatible residual gives the closed element $(o, 0)$ in degree one. A primitive in $\mathcal{R}(C)$ consists of $c_n \in C_n^0$ and $h_n \in C_n^{-1}$ satisfying

$$dc = -o, \qquad \Delta_C c = dh.$$

These are homotopy-compatible primitives. After solving the first equation, alter c by cycles. The second equation is then solvable exactly when its discrepancy vanishes in H^0, which is the condition $\lambda_H(o) = 0$.

The class in the left term of (16.11) is therefore secondary to the vanishing of the image in its right term. The exact sequence does not define a canonical pair of obstruction coordinates beforehand.

If each degreewise Roos operator Δ_{C^k} is surjective, the inclusion $\varprojlim C_n \to \mathcal{R}(C)$ is a quasi-isomorphism. Indeed the short exact sequence of complexes

$$0 \longrightarrow \varprojlim C_n \longrightarrow \prod_n C_n \xrightarrow{\Delta_C} \prod_n C_n \longrightarrow 0$$

identifies its kernel with the homotopy fiber. This supplies another sufficient condition for replacing a homotopy-compatible primitive by a strict one.

Example 16.8. For $n \geq 1$, consider the acyclic complex

$$C_n^{-1} = x^n\mathbb{C}[x] \hookrightarrow C_n^0 = \mathbb{C}[x] \longrightarrow C_n^1 = \mathbb{C}[x]/(x^n),$$

with inclusion and quotient as differentials. Transitions are inclusion in degree -1, identity in degree zero, and quotient in degree one. The compatible residual

$$o_n = 1 + x + \cdots + x^{n-1} \pmod{x^n}$$

has primitives $c_n = -(1 + x + \cdots + x^{n-1})$. Their discrepancies are $c_n - c_{n+1} = x^n \in Z_n^0 = x^n\mathbb{C}[x]$.

A strict primitive would be one polynomial with every Taylor coefficient equal to -1. It would satisfy $(1 - x)c = -1$. Evaluation at $x = 1$ proves that no such polynomial exists.

Nevertheless $H_n^0 = 0$ at every level, and $h_n = x^n \in C_n^{-1}$ satisfies $dh_n = x^n$. Thus (c, h) is a primitive in the Roos complex. Replacing the degree-zero polynomial ring by $\mathbb{C}[[x]]$ admits $c = -(1 - x)^{-1}$ and changes the strict problem. The example distinguishes both kinds of compatibility and the coefficient completion that relates them.

Compatible representatives and closed exchange

Suppose a compatible cohomology class $\alpha \in \varprojlim H^1(X_n)$ has representatives $w_n \in Z^1(X_n)$. Their discrepancies

$$w_n - r_n w_{n+1} \in B^1(X_n)$$

define a class

$$\mu_B(\alpha) \in \varprojlim{}^1 B^1(X_n). \tag{16.12}$$

Changing a representative adds a boundary, and hence changes its discrepancy by a Roos coboundary in the boundary tower. It follows that μ_B is well-defined and vanishes precisely when α has strict compatible cycle representatives.

Writing $w_n - r_n w_{n+1} = du_n$ does not put u_n in $Z^0(X_n)$. In $\mathbb{C} \xrightarrow{1} \mathbb{C}$, placed in degrees zero and one, the degree-one cycles 1 and 0 differ by $d(1)$. Their degree-zero primitive is not closed. There is no class $[u_n] \in H^0(X_n)$ to use in a Roos cokernel.

This gives an explicit strict criterion for a closed-field contribution together with a counterterm. Let $\Xi_n : X_n \to Q_n$ be compatible chain maps and let $o_n \in Z^1(Q_n)$ be compatible. The equations are

$$o_n + \Xi_n W_n + dC_n = 0, \qquad dW_n = 0, \tag{16.13}$$

with W_n and C_n both strictly compatible. Define the cone

$$E_n^k = Q_n^k \oplus X_n^{k+1}, \qquad d_E(q, x) = (d_Q q + \Xi_n x, -d_X x).$$

Then (16.13) asks for a strict primitive of $(o_n, 0)$ in E_n. Theorem 16.7 applies directly.

The same criterion can be resolved into successive choices. First choose

$$\alpha \in \varprojlim H^1(X_n), \qquad H^1(\Xi_n)(\alpha_n) = -[o_n].$$

Next require $\mu_B(\alpha) = 0$ and choose compatible representatives W_n. Finally require $\lambda_Z(o + \Xi W) = 0$ in $\varprojlim{}^1 Z^0(Q_n)$.

These conditions are necessary and sufficient by their definitions and the two primitive criteria. The last condition can depend on the preceding choice of W. If the towers $B^1(X_n)$ and $Z^0(Q_n)$ are Mittag–Leffler, only the first choice remains. Neither of these cycle conditions is an automatic consequence of Mittag–Leffler cohomology groups.

16.6 Linear transport at an interacting background

Renormalization acts on interactions nonlinearly. To transport an obstruction complex one needs its derivative at the chosen interaction. The distinction is already visible in a finite Gaussian model.

Let P be a constant symmetric covariance on a finite-dimensional even field space and put $\partial_P = \frac{1}{2}P^{ij}\partial_i\partial_j$. Use polynomial interactions with a positive formal coupling, so that all expressions are interpreted coefficientwise. Define

$$Z_P(I) = e^{\hbar\partial_P}e^{I/\hbar}, \qquad W_P(I) = \hbar\log Z_P(I).$$

The exponential and logarithm here specify the connected formal graph expansion. At a fixed coupling order there are finitely many vertices. Polynomial valences make each graph sum finite. In the connected expansion every power of $\hbar$ is a loop number and is nonnegative.

Proposition 16.9. *The derivative of W_P at I is the linear operator*

$$T_{P,I}(F) = Z_P(I)^{-1}e^{\hbar\partial_P}(Fe^{I/\hbar}). \tag{16.14}$$

For another covariance Q,

$$T_{P+Q,I} = T_{Q,W_P(I)}\,T_{P,I}.$$

If the Gaussian transport intertwines the master-equation vector fields and I solves that equation, $T_{P,I}$ is a chain map between their tangent complexes.

Proof. Differentiate $\hbar\log Z_P(I+sF)$ at $s=0$ to obtain (16.14). Constant contraction operators commute, so $Z_Q(W_P(I)) = e^{\hbar\partial_Q}Z_P(I) = Z_{P+Q}(I)$. Differentiating this identity proves the composition formula. For the last assertion, differentiate the stated intertwining identity at a zero of the master-equation vector field. Its derivatives there are the tangent differentials, so the resulting identity is precisely the chain-map equation. □

For one coordinate, covariance P, and $I = gx^2/2$, the formal Gaussian identity gives

$$W_P(gx^2/2) = \frac{gx^2}{2(1-Pg)} - \frac{\hbar}{2}\log(1-Pg).$$

It follows either by completing the square formally or by solving the heat equation with the given initial exponential. The term $Pg^2x^2/2$ shows that W_P is not linear in I. Consequently $W_P(C)$ is not the transition map on a linear counterterm complex.

Example 16.10 (RG can produce nonlocal terms)**.** Consider two sites, with fields x, y and local functionals $\mathbb{C}[x]+\mathbb{C}[y]\subset\mathbb{C}[x,y]$. Choose $\partial_P = \partial_x\partial_y$ and the local interaction $I = g(x^3+y^3)/6$. Direct expansion gives

$$W_P(I) = \frac{g}{6}(x^3+y^3) + g^2\left(\frac{x^2y^2}{4} + \frac{\hbar xy}{2} + \frac{\hbar^2}{6}\right) \pmod{g^3}.$$

Indeed, the mixed part of the coefficient of g^2 in $e^{I/\hbar}$ is $x^3y^3/(36\hbar^2)$. Its successive cross derivatives have coefficients $9, 36, 36$. In the heat exponential these are multiplied by $\hbar$, $\hbar^2/2$ and $\hbar^3/6$.

The logarithm cancels the uncontracted quadratic term, giving the displayed expression after multiplication by $\hbar$.

The term $g^2x^2y^2/4$ depends on both sites and lies outside the local subspace. Thus finite-scale transport requires a space of effective functionals larger than the local counterterms. The locality condition for a field theory must specify the allowed small-scale asymptotics and singular counterterms.

Fix compatible lower-order interactions and a filtration order n. If $C_n \in F^n$, then

$$W_P(I_{<n} + C_n) = W_P(I_{<n}) + T_{P,I_{<n}} C_n \pmod{F^{n+1}}.$$

This statement requires the filtration to make terms with two new corrections lie in F^{n+1}. The order-n mismatch of the already chosen lower interactions remains an inhomogeneous term. The linear transition on the next correction complex is the induced $T_{P,I_{<n}}$, with that mismatch retained separately.

For local field theories one must construct continuous transport on the specified effective functionals. Its small-scale locality conditions and boundary identities require further analytic arguments. The finite Gaussian formula identifies the map whose properties must be proved.

16.7 A Weyl connection with prescribed central curvature

Coordinate transport also acts on the quantized coefficient algebra. A useful global construction begins with more geometric data than a symplectic form alone. Suppose that (X, ω) has a global holomorphic torsion-free symplectic connection ∇. This is a hypothesis. Local Darboux connections do not themselves give such a global connection.

Let $\mathcal{W}$ be the bundle of formal Weyl algebras on T_X. In a symplectic chart use fiber variables Q, P, with $[Q, P]_\star = \epsilon$. Complete by total weight, assigning weight one to Q, P and weight two to ϵ. Base holomorphic forms have weight zero. On $\Omega^\bullet_X(\mathcal{W})$ define

$$\delta = dq\, \partial_Q + dp\, \partial_P.$$

For a monomial of fiber degree s and form degree t, put

$$\delta^{-1} a = \begin{cases} (s+t)^{-1}(Q\iota_{\partial_q} + P\iota_{\partial_p})a, & s + t > 0, \\ 0, & s = t = 0. \end{cases}$$

These operators are coordinate independent. If σ projects to fiber degree and form degree zero, then

$$\delta\delta^{-1} + \delta^{-1}\delta = 1 - \sigma, \qquad (\delta^{-1})^2 = 0. \tag{16.15}$$

The identity follows by counting the $s + t$ occurrences of a fiber variable or a base differential.

The connection extends as a derivation of the Weyl product. Torsion-freeness and preservation of ω give

$$[\delta, \nabla] = 0, \qquad \nabla^2 = \epsilon^{-1}[R, -]_\star, \qquad \delta R = 0, \qquad \nabla R = 0,$$

where R is the quadratic fiber representative of its curvature, a two-form of total weight two. Brackets of form-valued elements use the graded product, including exterior multiplication. Division of a Weyl commutator by ϵ is integral: the commutator is divisible by ϵ.

Fix a closed central series $\Omega \in \epsilon\Omega^2_{X,\mathrm{cl}}[[\epsilon]]$. The word central means that its coefficients contain no fiber variables.

Theorem 16.11. *There is a unique one-form r of weight at least three satisfying $\delta^{-1} r = 0$ and*

$$r = \delta^{-1}\Big(R - \Omega + \nabla r + \frac{1}{2\epsilon}[r, r]_\star\Big). \tag{16.16}$$

The derivation

$$D = -\delta + \nabla + \epsilon^{-1}[r, -]_\star$$

is flat. Projection σ identifies its sheaf of flat degree-zero sections with $\mathcal{O}_X[[\epsilon]]$ as modules. The Weyl product transports to an associative deformation product on this module, with first commutator $\epsilon\{f, g\}$. The complex $(\Omega^\bullet_X(\mathcal{W}), D)$ has no positive cohomology sheaves.

Proof. The operator $\partial^{-1}\nabla$ raises weight by one. For r of weight at least three, $\partial^{-1}\epsilon^{-1}[r, -]_\star$ raises weight by at least two. Iteration of (16.16), starting with zero, therefore determines each weight after finitely many steps and gives the unique solution. It has weight at least three and satisfies the stated normalization by $(\partial^{-1})^2 = 0$.

Put

$$F = R - \partial r + \nabla r + \frac{1}{2\epsilon}[r, r]_\star.$$

Expansion gives $D^2 = \epsilon^{-1}[F, -]_\star$. The connection identities give $DF = 0$. Since Ω is central and closed, $D(F - \Omega) = 0$ as well.

The recursion and (16.15) imply $\partial^{-1}(F - \Omega) = 0$. Its form degree is two, so its σ projection vanishes. It consequently satisfies

$$F - \Omega = \partial^{-1}(\nabla(F - \Omega) + \epsilon^{-1}[r, F - \Omega]_\star).$$

The right side raises weight. The only solution in the separated completion is zero. Thus $F = \Omega$ is central and $D^2 = 0$.

For $f \in \mathcal{O}_X[[\epsilon]]$, solve

$$\tau(f) = f + \partial^{-1}\big(\nabla\tau(f) + \epsilon^{-1}[r, \tau(f)]_\star\big). \tag{16.17}$$

The same weight argument gives a unique solution. Its defect $D\tau(f)$ has zero ∂^{-1} component, is D-closed by flatness, and has positive form degree. The preceding contraction argument makes it zero. Conversely a flat section with symbol f satisfies (16.17), proving uniqueness.

Define $f \star_X g = \sigma(\tau(f) \star \tau(g))$. The product of flat sections is flat, since D is a derivation. Uniqueness of the lift gives associativity. Modulo ϵ and fiber degree at least two, $\tau(f) = f + Qf_q + Pf_p$. Their single contraction gives the first commutator $\epsilon\{f, g\}$.

Finally let a be D-closed of positive form degree. Solve

$$b = -\partial^{-1}a + \partial^{-1}(\nabla b + \epsilon^{-1}[r, b]_\star), \qquad \sigma b = 0.$$

Again the recursion converges. The error $Db - a$ is closed and has zero ∂^{-1} component and zero symbol, hence is zero by the same weight argument. This proves positive-degree acyclicity. □

This is the Weyl version of the Fedosov contraction. Its formal resolution method also underlies the covariant formality construction of Dolgushev, *Covariant and equivariant formality theorems*, §3. The theorem above uses holomorphic data supplied globally. A smooth resolution or a Dolbeault resolution has different coefficients and requires its own comparison to holomorphic sections.

Period and algebroid gluing

The central curvature retains information invisible to the flat derivation $D^2 = 0$. In a Darboux chart the linear one-form

$$\gamma_{-1} = P\,dq - Q\,dp$$

satisfies $\epsilon^{-1}[\gamma_{-1}, -]_\star = -\partial$ and $\epsilon^{-1}\gamma_{-1} \star \gamma_{-1} = \omega$. With this convention the scalar Weyl curvature is $\omega + \Omega$, and the associated period is

$$\vartheta = \epsilon^{-1}[\omega + \Omega] \in \epsilon^{-1}[\omega] + H^2_{\mathrm{dR}}(X, \mathbb{C})[[\epsilon]]. \tag{16.18}$$

The class is understood in holomorphic de Rham hypercohomology when no global form representative is available. A central one-form added to a connection representative changes its scalar curvature

by an exact form. Thus such a change preserves (16.18). A comparison allowing a different period is a different quantization problem. Flatness alone does not erase this parameter.

When no global holomorphic connection has been specified, one can instead ask for local quantized algebras A_i and algebroid descent data. In a strict presentation these consist of isomorphisms $g_{ij} : A_j \to A_i$ and invertible elements $a_{ijk} \in A_i$ with

$$g_{ij}g_{jk} = \mathrm{Ad}(a_{ijk})g_{ik},$$
$$a_{ijk}a_{ikl} = g_{ij}(a_{jkl})a_{ijl}.$$

All terms are restricted to the indicated intersection. The first equation identifies the two ways of transporting a module on a triple overlap. The second makes that identification coherent on quadruple overlaps. A strict algebra sheaf is the special case $a_{ijk} = 1$. Local existence of a flat Weyl algebra does not choose these intertwiners or prove their coherence.

Holomorphic quantization and its characteristic classes are treated by Nest–Tsygan, *Deformations of symplectic Lie algebroids, deformations of holomorphic symplectic structures, and index theorems*, §5. The local connection calculation above provides a construction on the stated class of surfaces with a global holomorphic symplectic connection. A general surface requires the additional descent or resolution data.

The density and its unit value

A cyclic density is further structure: a map from the chosen cyclic chain complex to a specified complex of forms or currents. It must also have coherent overlap homotopies. A closed integration functional in the required degree, defined on the chosen supports, then gives the corresponding cyclic functional.

The product and period do not fix its normalization. Multiplying a trace by $c(\epsilon) \in \mathbb{C}[[\epsilon]]^\times$ preserves cyclicity and multiplies its unit value by $c(\epsilon)$.

For the logarithmic torus of Chapter 13, the constant Laurent coefficient gives the explicit normalization $\tau(1) = 1$. On a general quantization, a unit-index statement must specify its density map, normalization and supports. It must also fix the cohomology class in which the unit is evaluated.

The periodic algebraic index theorem relates a localized trace density to the tangent bundle's characteristic classes and the quantization period. The relevant result is Theorem 3.3.1 of Bressler–Nest–Tsygan, *Riemann–Roch theorems via deformation quantization II*. Its characteristic expression involves the $\widehat{A}$ class and the exponential of the period, with the parameter regrading of that theorem.

It does not by itself give an integral comparison over $\mathbb{C}[[\epsilon, u]]$. The point-current classes in Chapter 13 are annihilated by both parameters. Inverting either parameter loses precisely such information. Recovering it requires an integral map and cannot be inferred from an equality after localization.

16.8 Descent of the structured comparison

The preceding constructions separate two questions: whether the objects glue, and whether the specified comparison glues with all its operations. State the latter in a category that retains the operations being compared.

Let C be an infinity-category of complete complexes with a specified algebraic or supported operation structure. Assume that the required limits exist in C. Let $P_\bullet$, $G_\bullet$ be their descent diagrams on a cover, with all overlap and higher coherence maps supplied. Effective descent means that the global objects under consideration are identified with holim $P_\bullet$ and holim $G_\bullet$. A map of diagrams includes homotopies in every simplicial degree, not merely maps on charts.

Proposition 16.12. *Under these hypotheses a homotopy-coherent map $\Phi_\bullet : P_\bullet \to G_\bullet$ induces a global map in C. If each component is an equivalence, its global map is an equivalence. Every operation or marking retained in C is preserved by this map.*

Proof. Apply the limit functor to the map of diagrams. A componentwise equivalence is an equivalence in the diagram infinity-category, so a functor sends it to an equivalence. Effective descent identifies the resulting limits with the specified global objects. Their operations and markings are preserved because the construction takes place in C. □

For unary complexes the overlap equations were written explicitly above. For a cyclic algebra comparison the category must also retain multiplication, the mixed differential, the coefficient completion, the unit and the density. For the chiral problem it must in addition retain the full supported collision diagram and normal derivatives. The proposition applies once these data and their descent theorem are supplied. An equivalence of the underlying complexes does not supply them.

A sheaf of E_2 algebras illustrates the variance issue. Fix a holomorphic open U and an E_2 algebra of sections. Its factorization realization on framed topological two-disks requires the corresponding theorem in the chosen completed tensor category.

For $V \subset U$, sheaf restriction goes from sections on U to sections on V. A factorization algebra on the product needs covariant structure maps for disjoint product opens and their descent. The sheaf restrictions therefore do not define the holomorphic part of that factorization algebra. Extension from framed disks to oriented disks also requires the appropriate rotation equivariance. A boundary category and a bulk action require additional maps even after the bulk realization exists.

The global diagonal bimodule requires a comparison with the local diagonal resolutions. This comparison is needed to identify the global Morita center with derived sections of local Hochschild cochains. The affine derived Morita theorem of Chapter 15 supplies the comparison under its own generator hypotheses. A general holomorphic algebroid has no such identification solely from its local Weyl presentation.

The global integral cyclic problem can now be stated with its data fixed. Choose a holomorphic symplectic surface, a quantized Weyl algebroid with period $\mathfrak{d}$, a normalized cyclic density, and a coefficient and support completion over $\mathbb{C}[[\epsilon, u]]$.

Construct an integral cyclic comparison extending the local circle map on this algebroid. It must preserve coherent coordinate transport, the prescribed unit-index class and the mixed theory's supported operations.

The two-chart kernel gives its first relative obstruction. Higher overlap equations and compatible filtered choices remain to be solved in their respective complexes. Finally a map from the geometric closed-source complex must identify this cyclic comparison with the physical one. The logarithmic torus and the global-connection construction establish specified portions of these data. The final physical map on a general surface remains to be constructed.

Chapter 17

Compact targets and gravitational comparisons

A formal boundary chart determines local fields and their operations. A compact theory must also glue those charts, and a gravitational theory must identify their observables with fields defined by its own action. The comparison needs a map of field complexes compatible with products and collisions. It also needs states on which amplitudes can be sewn.

The local trace construction of Chapter 12 supplies one component. The compact examples below have additional global degrees of freedom. Their Hall products and root characters can be constructed on explicit coefficient spaces. The gravitational examples likewise admit explicit connection fields, representations and prepared states. These constructions let us ask which operations a comparison must preserve.

17.1 Calabi–Yau traces and compact fields

Let C be a small stable $\mathbb{C}$-linear category. Properness means that each morphism complex is perfect over $\mathbb{C}$. Smoothness means that its diagonal bimodule is perfect. Assume C is smooth and proper. A d-Calabi–Yau trace pairs $\mathrm{RHom}(E,F)$ with $\mathrm{RHom}(F,E)$ nondegenerately in degree $-d$. We require a negative-cyclic lift when using this trace in a closed differential form on a moduli space.

At an object E, the tangent complex to the derived moduli of objects is $\mathrm{RHom}(E,E)[1]$. Composition and trace give the pairing

$$\mathrm{RHom}(E,E)[1]^{\otimes 2} \longrightarrow \mathrm{RHom}(E,E)[2] \longrightarrow \mathbb{C}[2-d].$$

On a moduli stack to which the shifted symplectic construction applies, Calabi–Yau duality makes this pairing nondegenerate, and the cyclic lift supplies its closedness. The shift $2-d$ belongs to this moduli problem. Hochschild cochains have their own composition operations. Identifying them with operations on physical fields requires a map.

A concrete compact field model starts with a K3 surface S, an elliptic curve E, and holomorphic forms σ_S and dz. Then $X = S \times E$ has volume form $\Omega_X = \sigma_S \wedge dz$. Let $\mathfrak{l}$ be a finite-dimensional Lie algebra and let $\mathfrak{d} = \mathfrak{l} \ltimes \mathfrak{l}^*$ be its cotangent Lie algebra. Its pairing is the symmetric evaluation pairing, and its bracket uses the coadjoint action on $\mathfrak{l}^*$.

The holomorphic cotangent BF fields are

$$A \in \Omega^{0,*}(X,\mathfrak{l})[1], \qquad B \in \Omega^{0,*}(X,\mathfrak{l}^*)[1],$$

with action

$$S(A,B) = \int_X \Omega_X \wedge \left\langle B, \bar\partial A + \tfrac{1}{2}[A,A] \right\rangle. \tag{17.1}$$

The integral pairs complementary Dolbeault degrees. Including the field shifts, it is a BV pairing of degree -1.

Proposition 17.1. *The action* (17.1) *satisfies the classical master equation. Choose a Kähler metric on S and a harmonic* $(0,2)$*-form* η *with* $\int_S \sigma_S \wedge \eta = 1$*. Put* $K = \mathbb{C}[\eta]/(\eta^2)$*, with* $|\eta| = 2$ *and zero differential. For every open* $U \subset E$*, inclusion of harmonic representatives gives a trace-preserving dg Lie quasi-isomorphism*

$$K \otimes \Omega^{0,*}(U) \otimes \mathfrak{d} \longrightarrow \Omega^{0,*}(S \times U) \otimes \mathfrak{d}.$$

The smooth tensor product in the S direction is completed. The maps commute with restriction in U.

Proof. The differential of the cotangent theory is the cotangent lift of $\bar\partial A + \frac{1}{2}[A,A]$. Its square vanishes by $\bar\partial^2 = 0$, the derivation rule and Jacobi. Invariance of the evaluation pairing identifies this identity with the classical master equation.

The map $i : K \to \Omega^{0,*}(S)$ is multiplicative because $(0,4)$-forms on S vanish. The Dolbeault cohomology of S is one-dimensional in degrees zero and two and vanishes in degree one. Hodge projection p and $h = \bar\partial^* G$ therefore satisfy

$$pi = 1, \qquad \bar\partial h + h\bar\partial = 1 - ip, \qquad hi = 0.$$

Tensor these continuous operators with the identity on $\Omega^{0,*}(U) \otimes \mathfrak{d}$, with the tensor differential. They give the required contraction. Multiplicativity of i proves bracket compatibility, and the normalization of η proves trace compatibility. Restriction acts on the U factor alone. □

The two K3 modes both survive this comparison. All higher transferred products in the K3 factor vanish. The first product of harmonic inputs lies in $i(K)$, and the next internal homotopy kills it. This proves a classical field comparison. Quantized observables still require compatible kernels, local extensions and observable completions.

There is a useful finite-dimensional control on loop coefficients. Suppose $\mathfrak{l}$ has a strictly positive grading and is truncated above a finite height. Each cubic vertex has one B-leg. Since an internal edge consumes one such leg, a connected graph has at most one loop.

At one loop every B-leg is internal. The Lie coefficient is a trace of adjoint operators with positive total height, hence is zero. The dual trace vanishes for the same reason. This removes those Lie coefficients. Analytic extension of the graph kernels and the full quantum master equation remain separate requirements.

17.2 Extension flags and their coefficient theory

Let $\mathcal{M}_\gamma$ be a moduli stack of objects of charge γ. The extension stack $\mathcal{E}_{\alpha,\beta}$ records exact sequences $0 \to E_\alpha \to E \to E_\beta \to 0$ and has maps

$$\mathcal{M}_\alpha \times \mathcal{M}_\beta \xleftarrow{p} \mathcal{E}_{\alpha,\beta} \xrightarrow{q} \mathcal{M}_{\alpha+\beta}.$$

Choose a coefficient theory in which $p^!$ and q_* are defined on the selected classes. This choice includes support conditions and, for critical coefficients, orientation isomorphisms. A Hall product then has the form $q_* p^!$, with its declared shifts.

The associativity comparison lives on the stack of two-step flags. If base change identifies the two iterated pull–push constructions with that same flag correspondence, their products agree. For three or more steps these identifications must be compatible under forgetting intermediate terms. Thus orientation and base change are part of the multiplication, rather than consequences of a numerical count of extensions.

An action on a field theory requires one further correspondence map. Its target is a state complex defined from the fields, independently of the Hall action. For a compact chiral state space, this map must

retain the normal oscillators at diagonals and commute with joining cycles. Local coordinate formulas must also agree through higher changes of coordinates. Defining the state space to be the image of the Hall map would omit this comparison problem.

17.3 An exchange ratio does not determine the integral algebra

The following polynomial example isolates the coefficient issue. Put

$$R = \mathbb{Q}[h_1, h_2], \qquad h_3 = -h_1 - h_2,$$
$$a = -(h_1^2 + h_1 h_2 + h_2^2), \qquad b = -h_1 h_2 (h_1 + h_2),$$
$$N(v) = v^3 + av + b.$$

The ring R is a domain and $b \neq 0$.

For $j = 0, 1$, take independent copies of $\mathcal{A}^{(j)} = \bigoplus_{n \geq 0} R[x_1, \dots, x_n]^{S_n}$. Charge is n. All cohomological degrees are zero and all parities are even. The sum has finite charge support. No parameter or root difference is inverted.

Define

$$K_0(v) = N(v)/v, \qquad K_1(v) = vN(v) = v^2 K_0(v).$$

and, for inputs of charges d, e, set

$$(F *_j G)(X) = \sum_{\substack{I \sqcup J = [d+e] \\ |I| = d}} F(X_I) G(X_J) \prod_{i \in I,\, k \in J} K_j(x_k - x_i). \tag{17.2}$$

There is no factorial divisor. The charge-zero constant is the unit. The constant polynomial in charge one is a different element.

Proposition 17.2. *Both products are polynomial, unital and associative. They have the same exchange ratio*

$$r(v) = \frac{K_j(-v)}{K_j(v)} = \frac{v^3 + av - b}{v^3 + av + b},$$

but no charge-preserving R-linear algebra isomorphism identifies their full polynomial carriers. The conclusion also holds for maps semilinear under one automorphism of R.

Proof. For K_0, pair summands across a possible pole $x_k = x_i$ by exchanging the two labels. The remaining factors agree on that diagonal and the two residues are opposite. All denominators have simple root-difference factors, so cancellation proves polynomiality. For K_1 polynomiality is immediate.

Both bracketings of three inputs sum over the same ordered three-block partitions, with one kernel factor per cross-block pair. This proves associativity in all charges. Empty blocks prove the unit equations. The even factor v^2 cancels from the exchange ratio.

In charge two write $c = (x + y)/2$, $v = y - x$, $t = v^2$ and $B = R[c, t]$. Then $R[x, y] = B \oplus vB$. The linearized multiplication maps obey

$$\mu_0(A + vC) = 2((t + a)A + bC), \qquad \mu_1(A + vC) = 2t((t + a)A + bC).$$

Their images are $J = (t + a, b)B$ and tJ. Hence

$$\operatorname{coker} \mu_0 \simeq R[c]/(b), \qquad \operatorname{coker} \mu_1 = B/tJ.$$

The first cokernel has scalar annihilator (b). The second surjects onto the faithful R-module $R[c]$ by evaluation at $t = 0$, so its annihilator is zero. An algebra isomorphism preserving charge would identify these cokernels. A semilinear isomorphism would take the nonzero annihilator b to its nonzero image under the scalar automorphism. Both possibilities contradict faithfulness. □

The obstruction concerns the full polynomial realizations. A positive comparison exists on their generated subalgebras. Put

$$D_n(X) = \prod_{i<k} (x_k - x_i)^2, \qquad D_0 = D_1 = 1.$$

The maps $\Phi_n(F) = D_n F$ are injective and multiplicative because

$$D_{d+e}(X) = D_d(X_I) D_e(X_J) \prod_{i\in I,\, k\in J} (x_k - x_i)^2.$$

They fix charge one. Their restrictions therefore identify the two subalgebras generated by all charge-one polynomials. Inverting D_n also makes them isomorphisms on the localized full carriers. Neither statement gives an invertible map on the original full lattice: the constant polynomial in charge $n \geq 2$ is not divisible by D_n.

17.4 Cooperation and the tensor of its target

The same algebras expose a separate obstruction to an ordinary bialgebra. Let $e_m = x^m$ in charge one and let ε be the projection to charge zero. An ordinary coproduct here means an R-linear, charge-preserving, unital algebra map

$$\Delta : \mathcal{A}^{(j)} \longrightarrow \mathcal{A}^{(j)} \otimes_R \mathcal{A}^{(j)}$$

with the usual tensor multiplication and both counit equations for ε. The same definition can be made on the subalgebra generated by the e_m.

Proposition 17.3. *Neither algebra admits this coproduct, even on its current-generated subalgebra. Coassociativity is not needed for the obstruction.*

Proof. The charge-zero counits force every e_m to be primitive. Direct substitution in (17.2) gives

$$[e_0, e_3] - 3[e_1, e_2] + a[e_0, e_1] = b(e_0 *_j e_0). \tag{17.3}$$

Indeed, for polynomials f, g the commutator is

$$2b\, v^{2j} \frac{f(x)g(y) - g(x)f(y)}{v}, \qquad v = y - x.$$

The numerator for the first two terms of (17.3), divided by v, is v^2. The remaining term contributes a, and $e_0 *_j e_0 = 2v^{2j}(v^2 + a)$.

In an ordinary tensor algebra a commutator of even primitive elements is primitive. The charge-$(1,1)$ part of the coproduct of the left side is therefore zero. That of the right side is $2b\, e_0 \otimes e_0$, which is nonzero over R. This contradicts multiplicativity. □

This proof uses $b \neq 0$. It makes no assertion on the fiber $b = 0$. The charge-$(1,1)$ module is free, so merely writing a derived tensor over R does not remove the displayed defect.

There is again a positive construction after changing the recipient. For two disjoint root lists put

$$Q_j(X, Y) = \prod_{x\in X,\, y\in Y} K_j(y - x), \qquad \Delta^{\mathrm{rat}}_{d,e} F(X, Y) = \frac{F(X, Y)}{Q_j(X, Y)}.$$

Work on the ordered root cover with all pairwise differences and the zeros of $N(\pm v)$ inverted, and then take symmetry within each list. Let A have bidegree (d, e) and B have bidegree (f, g). Define their product by summing

$$A(X_1, Y_1)B(X_2, Y_2)Q_j(X_1, X_2)Q_j(Y_1, Y_2) \cdot \prod_{y\in Y_1,\, x\in X_2} r(y-x) \tag{17.4}$$

over partitions of X into sizes d, f and of Y into sizes e, g. The last factor changes the interchange of the two middle blocks.

Proposition 17.4. *Formula* (17.4) *is associative and unital. With this target multiplication, the sum of* $\Delta^{\mathrm{rat}}_{d,e}$ *is multiplicative and coassociative, with counits given by empty lists.*

Proof. For three factors, either bracketing uses the same partitions of X and Y. It contributes one K_j for every ordered pair within each list. It also contributes one r whenever a root in an earlier Y block crosses a root in a later X block. These scalar factors commute, so the terms agree.

To compare a split shuffle product with the product of two splits, partition each input between X and Y. Cross factors with the order X before Y cancel against the denominator $Q_j(X, Y)$. The opposite order leaves $K_j(x-y)/K_j(y-x) = r(y-x)$, exactly the factor in (17.4). This proves multiplicativity term by term. Three-way splitting divides by $Q_j(X, Y)Q_j(X, Z)Q_j(Y, Z)$ in either order. Empty lists give the counits. □

The elementary relation now reads

$$(1 - r(v))v(v^2 + a) = b(1 + r(v)),$$

so the changed interchange cancels the earlier defect. But the split of the charge-two constant is $v/N(v)$ for $j = 0$ and $1/(vN(v))$ for $j = 1$. These are not polynomial coefficients. A comparison to supported collision operations must restore the information on the removed divisors, including any derived excess classes.

17.5 Root states and scalar automorphic products

Let $V = \bigoplus_{\alpha>0} V_\alpha$ be a graded superspace with finite-dimensional pieces. Choose a positive height so that every height truncation contains finitely many basis vectors. The completed Fock space is

$$\mathcal{H}_V = \widehat{\mathrm{Sym}}(V_{\bar 0})\widehat{\otimes}\widehat{\Lambda}(V_{\bar 1}).$$

Completion is by height. A root variable x^α records the additive degree, and the parity operator defines the supertrace. For finitely many roots, or in an absolute-convergence chamber,

$$\mathrm{str}_{\mathcal{H}_V}(x^{\mathrm{degree}}) = \prod_{\alpha>0}(1 - x^\alpha)^{-\operatorname{sdim} V_\alpha}. \tag{17.5}$$

An even oscillator contributes a geometric series, while an odd one contributes $1 - x^\alpha$. Multiplication proves the formula at each finite height. Coefficientwise stabilization, or absolute convergence, gives the stated limit.

Applied to specified Borcherds root spaces, this is the Fock product associated with the denominator identity. The Weyl monomial, the coordinate pullback and any doubling of root multiplicities must still be included in comparing automorphic normalizations. In the Igusa example, the weight-five square-root

branch and the weight-ten cusp form are sections of different modular lines. The reciprocal partition product is a further scalar convention.

The supertrace loses information even before geometry enters. One even and one odd vector of the same degree contribute 1 to (17.5). Their Fock space is nonetheless nontrivial. Nor does a graded trace determine a bracket on V. Consequently a product identity cannot construct the brane objects, their extension operations or a map to an independently defined compact state space.

Problem 17.5. For a compact Calabi–Yau target, construct reduced critical coefficients with their relative unit, cyclic pairing and orientation on all extension flags. Construct their action on the independently defined chiral state complexes, preserving charge, normal oscillators, diagonal joining and higher coordinate gluing. Compare the resulting trace line and its normalized section with the specified automorphic line. On a formal normal chart, require compatibility with the native trace coordinate and supported operations.

This problem retains the positive classical compact field reduction, the Hall products on their declared carriers and the scalar product formula. Its missing maps concern the geometric field and state recipients, rather than the existence of those algebraic constructions.

17.6 Gravitational fields and the invertible coframe

The gravitational comparison begins with its own bulk fields. Let M be an oriented three-manifold and fix $\ell > 0$. Take the Lorentz form η_{ab} and its oriented volume tensor $\epsilon_{ab}{}^{c}$. The Cartan Lie algebra has generators J_a, P_a and brackets

$$[J_a, J_b] = \epsilon_{ab}{}^{c} J_c, \qquad [J_a, P_b] = \epsilon_{ab}{}^{c} P_c, \qquad [P_a, P_b] = \ell^{-2} \epsilon_{ab}{}^{c} J_c.$$

The invariant form pairs J_a with P_b by η_{ab} and vanishes on the two diagonal summands. A connection is $\mathcal{A} = \omega^a J_a + e^a P_a$.

Write

$$R^a = d\omega^a + \tfrac{1}{2} \epsilon^a{}_{bc} \omega^b \wedge \omega^c, \qquad T^a = de^a + \epsilon^a{}_{bc} \omega^b \wedge e^c.$$

Its curvature is

$$F_{\mathcal{A}} = \Big(R^a + \frac{1}{2\ell^2} \epsilon^a{}_{bc} e^b \wedge e^c \Big) J_a + T^a P_a. \tag{17.6}$$

Proposition 17.6. *On the locus where e is an invertible coframe, flatness of $\mathcal{A}$ is equivalent to torsion-free Einstein geometry with cosmological constant $\Lambda = -\ell^{-2}$ and metric $g_{\mu\nu} = \eta_{ab} e^a_\mu e^b_\nu$.*

Proof. The P component of (17.6) gives $T = 0$. Invertibility identifies ω with the metric spin connection. The J component then says that the metric has constant sectional curvature $-\ell^{-2}$. In three dimensions the Riemann tensor is determined by the Ricci tensor, so this is the Einstein equation. Conversely a torsion-free Einstein coframe has this curvature, and both components of $F_{\mathcal{A}}$ vanish. □

Degenerate coframes remain connections but do not give this metric interpretation. The same restriction matters in the linear theory: at a flat background, a variation $a = \delta\mathcal{A}$ obeys $d_{\mathcal{A}} a = 0$, with gauge variation $a \mapsto a + d_{\mathcal{A}} \epsilon$. Flatness gives $d^2_{\mathcal{A}} = 0$. Splitting a into $\delta\omega$ and δe is the linearized Cartan complex. Passing to metric variables uses the invertible coframe and quotient by infinitesimal Lorentz rotations. Boundary conditions and the induced pairing must be carried through this passage before it compares Hilbert spaces.

The usual Chern–Simons normalization sets $k = \ell/(4G)$, and its classical Brown–Henneaux relation is $c = 6k$. These equations specify a classical normalization. A quantum coupling comparison must intertwine the renormalized current and ghost operations, including their allowed finite counterterms. A chosen scalar reparametrization is not that map.

17.7 A Weyl representation and its modular transform

Fix $b > 0$. On $L^2(\mathbb{R}, dP)$, the self-adjoint Weyl pair has Schwartz-space formulas

$$\widehat{\ell}_A = 2\pi b P, \qquad \widehat{\ell}_B = -\frac{ib}{2}\frac{d}{dP}, \qquad [\widehat{\ell}_A, \widehat{\ell}_B] = i\pi b^2.$$

Reflection acts by $P \mapsto -P$. The individual length operators exchange parity, while their squares preserve the even subspace. Identify that subspace with $\mathcal{H} = L^2(\mathbb{R}_{\geq 0}, dP)$ by $f \mapsto 2^{-1/2} f(|P|)$.

The full-line Fourier transform with kernel $\sqrt{2}\, e^{4\pi i P P'}$ is unitary and squares to reflection. Its restriction is therefore the unitary involution

$$(\mathcal{F}f)(P) = 2\sqrt{2} \int_0^\infty \cos(4\pi P P') f(P')\, dP'. \tag{17.7}$$

For $t > 0$ the Gaussian integral gives

$$\mathcal{F}(e^{-tP^2}) = \sqrt{\frac{2\pi}{t}}\, e^{-4\pi^2 P^2/t}. \tag{17.8}$$

These are representation-theoretic constructions on the specified Lebesgue space. A physical zero-mode quantization must identify its measure and observables with this representation.

Put $Q = b + b^{-1}$ and $c = 1 + 6Q^2$, so $c \geq 25$. For $q = e^{2\pi i \tau}$ with $\operatorname{Im} \tau > 0$, let $\eta(\tau) = q^{1/24} \prod_{n \geq 1}(1 - q^n)$ be the Dedekind function. The Virasoro character functions in this normalization are

$$\chi_P(\tau) = \frac{q^{P^2}}{\eta(\tau)}.$$

The vacuum character is the difference at the two imaginary momenta $iQ/2$ and $i(b^{-1} - b)/2$:

$$\chi_{\mathrm{vac}}(\tau) = q^{-c/24} \prod_{n \geq 2} (1 - q^n)^{-1}.$$

The difference removes the level-one descendant. Transforming the two Gaussians yields

$$\chi_{\mathrm{vac}}(-1/\tau) = \int_0^\infty \rho(P) \chi_P(\tau)\, dP,$$

where

$$\begin{aligned} \rho(P) &= 2\sqrt{2}\big(\cosh(2\pi Q P) - \cosh(2\pi(b^{-1} - b)P)\big) \\ &= 4\sqrt{2}\, \sinh(2\pi b P) \sinh(2\pi P/b). \end{aligned} \tag{17.9}$$

This density is positive for $P > 0$ and vanishes quadratically at zero. Its exponential growth prevents it from being a vector of $\mathcal{H}$. It does define a continuous functional on each Gaussian-weighted space $L^2(\mathbb{R}_{\geq 0}, e^{P^2/n} dP)$, by Cauchy–Schwarz. Their union, with its final locally convex topology, is a test domain for the uncollared linear cap.

17.8 Positive collars and finite sewing

Let H_0 be multiplication by P^2 on $\mathcal{H}$, and write $C_t = e^{-tH_0}$ for $t \geq 0$. This is a bounded strongly continuous semigroup. Positive collars give actual vectors

$$v_{0,t}(P) = e^{-tP^2}, \qquad v_{1,t}(P) = \rho(P)e^{-tP^2}, \qquad t > 0.$$

The Gaussian dominates the exponential growth of ρ, so both belong to $\mathcal{H}$. A preparation is the map $\mathbb{C} \to \mathcal{H}$ taking 1 to one of these vectors. Its reflected cap is the Hilbert adjoint.

Use an inner product linear in its second argument and define

$$K_{ij}(s,t) = |v_{i,s}\rangle\langle v_{j,t}|, \qquad G_{ij}(u) = \int_0^\infty \rho(P)^{i+j} e^{-uP^2}\, dP, \quad i,j \in \{0,1\}.$$

The first expression is a rank-one operator. The second records the overlap at the total collar width.

Proposition 17.7. *For positive widths,*

$$\begin{aligned} K_{ij}(s,t)K_{kl}(u,v) &= G_{jk}(t+u)K_{il}(s,v), \\ K_{ij}(s,t)^* &= K_{ji}(t,s), \\ \operatorname{Tr} K_{ij}(s,t) &= G_{ji}(s+t). \end{aligned}$$

Every finite product of these kernels is trace class. More generally, a bounded operator word with a trace-class factor has a finite cyclic trace. For a Hilbert–Schmidt operator B, $\operatorname{Tr}(B^*B) = \|B\|_{\mathrm{HS}}^2 \geq 0$.

Proof. Composition of rank-one operators contracts the two middle vectors, giving the first formula. The adjoint and trace formulas follow from the same inner product. The trace norm of $|u\rangle\langle v|$ is $\|u\|\|v\|$. Thus sums with $\sum_n \|u_n\|\|v_n\| < \infty$ converge in trace norm. The rank-one trace identity extends to this completion and gives $\operatorname{Tr}(AB) = \operatorname{Tr}(BA)$ for bounded A and trace-class B. The last assertion is the definition of the Hilbert–Schmidt norm. □

All three overlap functions have closed Gaussian expressions. Put $A = 2\pi Q$, $B = 2\pi(b^{-1} - b)$ and $J_u(k) = \sqrt{\pi}\, e^{k^2/(4u)}/(2\sqrt{u})$. Then

$$\begin{aligned} G_{00}(u) &= J_u(0), \\ G_{01}(u) &= 2\sqrt{2}\big(J_u(A) - J_u(B)\big), \\ G_{11}(u) &= 4J_u(2A) + 4J_u(2B) + 8J_u(0) \\ &\qquad - 8J_u(A+B) - 8J_u(A-B). \end{aligned}$$

The linear cap uses one factor of ρ. Reflected sewing of two caps uses ρ^2. Replacing ρ by its square root would change the character transform (17.9).

Unitary conjugation and Hilbert tensor products preserve these trace-class sewing identities. Remarking the torus conjugates the cylinder as well as its states. In particular $\mathcal{F}$ does not commute with C_t: applying (17.8) to $C_s e^{-tP^2}$ gives a different width from $C_s \mathcal{F} e^{-tP^2}$. Thus unitary transport is not invariance of a fixed cylinder operator.

17.9 The trace obstruction for the continuous cylinder

The finite amplitudes above use prepared vectors. Positive time alone does not make C_t trace class. On the infinite-dimensional subspace $L^2([1,2])$, one has $C_t \geq e^{-4t}$. The sum of its diagonal matrix coefficients over any N orthonormal vectors in this subspace is at least Ne^{-4t}. In particular C_t is not compact.

The obstruction persists for a finite trace with no continuity or positivity assumption, provided it is linear and cyclic on a full operator ideal.

Theorem 17.8. *Let $t > 0$ and let $I \subset B(\mathcal{H})$ be a two-sided ideal containing C_t. No complex-linear map $\tau : I \to \mathbb{C}$ satisfies both of the following conditions:*

$$\tau(AB) = \tau(BA) \quad (A \in I,\ B \in B(\mathcal{H})),$$
$$\tau(p) = 1 \qquad (p \in I \text{ a rank-one projection}).$$

Proof. Let q be multiplication by the indicator of $[1,2]$. The multiplier $B_t = e^{tP^2}q$ is bounded, so $q = B_tC_t \in I$. Choose an orthonormal basis of $q\mathcal{H}$ and let V be its unilateral shift, extended by zero on $(1-q)\mathcal{H}$. Then $V = qVq \in I$, and

$$V^*V = q, \qquad VV^* = q - p,$$

where p projects to the first basis vector. Thus $p \in I$. Cyclicity gives $\tau(q) = \tau(q-p)$, whereas linearity gives $\tau(q) - \tau(q-p) = \tau(p) = 1$, a contradiction. □

Linearity is essential: the constant function $A \mapsto 1$ is cyclic and takes the value one on every rank-one projection, but is not a trace. The theorem also does not exclude a weight defined only on a smaller commutative algebra of multipliers.

That smaller algebra ceases to suffice when all multipliers and the modular transform act together. Let $\mathcal{M}$ be their generated von Neumann algebra. An operator commuting with all multipliers is itself a multiplier M_f. Commutation with the cosine transform implies

$$(f(P) - f(P'))\cos(4\pi PP') = 0 \quad \text{for almost every } (P, P').$$

One obtains this identity by testing the kernels on bounded intervals. The zero set of the cosine has measure zero, so f is constant. Hence $\mathcal{M}' = \mathbb{C}$ and $\mathcal{M} = B(\mathcal{H})$.

A faithful normal semifinite trace on this algebra assigns a common finite positive value κ to rank-one projections. Finiteness follows from semifiniteness on a minimal projection, and equality from unitary conjugacy. Normality then gives $\tau(q) = \infty$ and $\tau(C_t) \geq e^{-4t}\tau(q) = \infty$. Thus a finite multiplier weight cannot extend to such a trace while retaining the full modular operator algebra.

There is a related tensor obstruction. A vector in $\overline{\mathcal{H}} \otimes \mathcal{H}$ defines a Hilbert–Schmidt operator. Contracting two such vectors gives a trace-class operator. It therefore cannot give C_t, even for $t > 0$. A two-boundary Hilbert cup for the bare continuous cylinder requires a different distributional or relative recipient. The one-boundary preparations of Proposition 17.7 remain valid.

17.10 A positivity test for the regularized vacuum sum

A different obstruction concerns a particular scalar modular sum. For $c > 1$, take the vacuum character $q^{-c/24}\prod_{n\geq 2}(1-q^n)^{-1}$ and sum its absolute square over $\Gamma_\infty \backslash \mathrm{PSL}_2(\mathbb{Z})$. Fix the Maloney–Witten regularization by analytic continuation of the Poincaré exponent to $1/2$.

Its large imaginary-modulus expansion is

$$Z_c(\tau) = |\chi_{\mathrm{vac}}(\tau)|^2 + \frac{-6 + c_1 y^{-1} + c_2 y^{-2} + O(y^{-3})}{|\eta(\tau)|^2}, \qquad \tau = x + iy,$$

with fixed x, where, on putting $u = (c-13)/12$,

$$c_1 = \frac{(\pi^3 - 6\pi)(c-13)}{9\zeta(3)},$$
$$c_2 = \frac{8\pi^2((\pi^4 - 90)u^2 + \pi^4 + 45)}{135\zeta(5)} > 0.$$

These coefficients belong to the fixed regularized sum. To see the seed signs, write $\kappa_0 = (c-1)/12$. Multiplication of the vacuum term by $|\eta|^2$ gives $e^{2\pi\kappa_0 y}(1-q)(1-\bar{q})$. The four Poincaré seeds are therefore $(\kappa_0, 0)$, $(\kappa_0 - 2, 0)$ and $(\kappa_0 - 1, \pm 1)$, with coefficients $+, +, -, -$.

For a seed $(\kappa, 0)$, its continued constant Fourier mode, after removing the identity term and dividing by $y^{1/2}$, is

$$-1 + \frac{2\pi^3\kappa}{3\zeta(3)}y^{-1} + \frac{4\pi^6\kappa^2}{135\zeta(5)}y^{-2} + O(y^{-3}).$$

For $(\kappa, \pm 1)$ it is

$$2 + \frac{4\pi\kappa}{\zeta(3)}y^{-1} + \frac{4\pi^2(2\kappa^2 - 1)}{3\zeta(5)}y^{-2} + O(y^{-3}).$$

These are the constant-mode Fourier expansions of that analytic continuation. The other Fourier modes are exponentially small at fixed x. Substitution gives -6, c_1, c_2 above. The seed signs and this regularization are part of the question about a spectral measure.

On the imaginary axis, put $\beta = 2\pi y$ and define d_N by

$$\prod_{n\geq 2}(1-q^n)^{-2} = \sum_{N\geq 0} d_N q^N.$$

Then

$$Z_c(\beta) = e^{\beta c/12}\sum_{N\geq 0} d_N e^{-\beta N} + e^{\beta/12}\bigl(-6 + O(\beta^{-1})\bigr). \tag{17.10}$$

The first coefficients are 1, 0, 2, 2, 5, 6, 13.

Proposition 17.9. *Suppose the function in* (17.10) *is the Laplace transform of a positive Borel measure. Assume its energy support is bounded below and its exponential weighting is finite. Then $c = 12N + 1$ for an integer $N \geq 5$. This necessary condition is not a sufficiency statement.*

Proof. First suppose a positive measure μ has $e^{\beta a}\int e^{-\beta E}\,d\mu(E) \to d < \infty$. Any mass below a would make this limit infinite. There is thus no such mass. Dominated convergence after one fixed exponential weighting shows that the limit is the mass of the atom at a.

Apply this observation successively to (17.10). It forces actual atoms of masses d_N at $E_N = N - c/12$ for every $N < (c-1)/12$. Subtracting these atoms preserves positivity. At the threshold $E = -1/12$, the remaining rescaled limit is -6 unless $m = (c-1)/12$ is an integer. A negative limit is impossible.

If m is an integer, the limit is $d_m - 6$, so it must be nonnegative. The values $d_0, \ldots, d_4$ are all less than six, while $d_5 = 6$. Necessarily $m \geq 5$, as claimed. □

For example, $c = 25$ gives a remaining threshold mass $2 - 6 = -4$. The cases $c = 61$ and $c = 73$ give zero and seven, respectively, and this argument does not rule them out. The test permits continuous positive measures. It assumes neither a discrete spectrum nor a previously chosen vacuum subspace. It does not apply unchanged to signed measures or to a different finite renormalization of the sum.

17.11 State normalization and physical reconstruction

Even a finite algebra can have several positive sewing theories before its cap is chosen. Let $A = \mathbb{C}e_1 \oplus \mathbb{C}e_2$, with $e_i e_j = \delta_{ij} e_i$ and $e_i^* = e_i$. For $0 < w < 1$, set

$$\varepsilon_w(a_1 e_1 + a_2 e_2) = w a_1 + (1 - w) a_2.$$

This is a positive normalized trace. Write $p_1 = w$, $p_2 = 1 - w$. The Frobenius pairing has inverse $\sum_i e_i \otimes e_i / p_i$. Its connected genus-$g$ amplitude is

$$Z_g = \sum_{i=1}^{2} p_i^{1-g}.$$

Indeed, each handle multiplies the ith idempotent by p_i^{-1}, and the final cap contributes p_i. Thus the sphere and torus amplitudes are always one and two, whereas genus two distinguishes $w = 1/2$ from $w = 1/3$: the values are 4 and $9/2$. This is freedom in choosing a cap, not nonuniqueness after the Frobenius structure has been fixed.

For a gravity comparison, the analogous data must come from the bulk construction. One needs the gauge quotient with its real structure, the physical pairing, the cylinder energy and the renormalized insertion maps. A formal vacuum module or an equality of oscillator characters does not specify these analytic objects. Conversely, the positive prepared amplitudes constructed here do not depend on identifying every bulk state with a boundary vector.

Problem 17.10. Construct the comparison from the source-dependent bulk BV/BFV complex on the solid torus to the boundary state complex. Preserve the moving boundary condition and both corner variations. Define the physical gauge quotient and its pairing, and compare its Euclidean cut with the Lorentzian state space. On this domain, construct the current, ghost and line insertions with their renormalized collision products. Determine the coupling and cylinder-energy normalization from the same theory. Require the comparison to commute with sewing and with the specified real structure.

An obstruction to this problem must survive the admitted local counterterms and changes of regulator on its physical carrier. The nonexistence of the ordinary shuffle coproduct and of the finite continuous-cylinder trace have narrower, explicit domains. They determine which coefficient or state constructions cannot serve unchanged in a comparison. The rational Hall tensor and the positive prepared sewing amplitudes exhibit alternatives whose physical realization remains a separate question.

Chapter 18

Signs, duals and coefficient completions

A coefficient calculation enters a field theory through a map of complexes. Its grading determines the signs of that map, and its topology determines which inputs the map accepts. We give the conventions needed for the Hamiltonian and boundary constructions. The examples distinguish a coordinate change from a change of representation or completion.

18.1 Degrees and deformation parameters

For a cochain complex, use

$$(V[r])^j = V^{j+r}, \qquad d_{V[r]} = (-1)^r d_V.$$

Thus an element of degree zero in V has degree -1 in $V[1]$. The tensor differential and interchange law are

$$d(v \otimes w) = dv \otimes w + (-1)^{|v|} v \otimes dw, \qquad v \otimes w \longmapsto (-1)^{|v||w|} w \otimes v.$$

Parity is cohomological degree modulo two. When a complex is written homologically, its differential lowers degree. We state this change before introducing its generators.

The matrix relation variable Υ has cohomological degree -1, and the gauge ghost c has degree 1. Both are odd. Form degree on $\mathbb{R}^2 \times \mathbb{C}^2$ is an additional grading before the field complex is totalized. In a tensor product of form and coefficient complexes, moving a coefficient past a form contributes the sign from their declared degrees.

Several parameters occur in the comparison. Write t for the Rees parameter of a filtered operator algebra and h for the coefficient in a Weyl relation. The loop parameter in the quantum master equation is $\hbar_Q$.

An equivariant parameter is specified by its cohomological degree and vector field. The polar constant κ is the value of a chosen residue normalization. These parameters can be related by a stated scalar map. The grading convention does not identify them.

18.2 Cyclic derivatives and the relation differential

Let $F = \mathbb{C}\langle x, y, c, \psi \rangle$, with degrees 0, 0, 1, -1, respectively. Its graded cyclic quotient is the vector space

$$F_{\mathrm{cyc}} = F/[F, F]_{\mathrm{gr}}, \qquad [uv] = (-1)^{|u||v|}[vu].$$

The denominator is the linear span of graded commutators. It is not the ideal imposing graded commutativity on the free algebra.

For a word $a_1 \cdots a_n$, define its left cyclic derivative by

$$\partial_q [a_1 \cdots a_n] = \sum_{a_i = q} (-1)^{|a_1 \cdots a_{i-1}|(|q| + |a_{i+1} \cdots a_n|)} a_{i+1} \cdots a_n a_1 \cdots a_{i-1}.$$

This sign moves the preceding block past the selected factor and the following block. Fix the even graded-skew pairing

$$\omega(x,y)=1,\quad \omega(y,x)=-1,\qquad \omega(c,\psi)=\omega(\psi,c)=1.$$

With $\iota_{X_G}\omega=-dG$, the induced necklace bracket is

$$\{G,H\}_{\mathrm{neck}}=\big[\partial_x G\,\partial_y H-\partial_y G\,\partial_x H \\ +(-1)^{|G|}(\partial_c G\,\partial_\psi H+\partial_\psi G\,\partial_c H)\big].$$

In particular $\{[x],[y]\}=[1]$ and $\{[c],[\psi]\}=-[1]$. The symmetry of the pairing on the two odd factors is required by graded skew symmetry.

The full matrix BRST differential of (12.2) uses graded matrix products:

$$QX=[X,c],\qquad QY=[Y,c],\qquad Qc=-c^2,\qquad Q\Psi=[X,Y]-[c,\Psi].$$

The relation term is equivariant for conjugation. For example, putting $M=[X,Y]$ gives $QM=[M,c]$, and

$$Q[c,\Psi]=[Qc,\Psi]-[c,Q\Psi].$$

Since $[c,[c,\Psi]]=[c^2,\Psi]$, substitution gives $Q^2\Psi=0$. The same computation for X, Y, and $Q(c^2)=(Qc)c-c(Qc)$, gives $Q^2=0$ on all generators. The odd Leibniz rule extends this identity to the polynomial algebra.

Without ghosts, set $dx=dy=0$ and $d\psi=xy-yx$. For even words u,v,

$$d[\psi u\psi v]=[\psi vxyu]-[\psi vyxu]-[\psi uxyv]+[\psi uyxv]. \tag{18.1}$$

The second replacement crosses one odd factor and has a minus sign. The first replacement can be rotated past the remaining ψ without a further sign because the moved block is even. Thus interchanging u,v negates the result, as required by graded cyclicity.

18.3 Suspension and the ordered bar

Let A be an augmented differential graded algebra in cohomological degrees, and let $\bar A$ be its augmentation ideal. Write $sa\in\bar A[1]$, so $|sa|=|a|-1$. The tensor coalgebra

$$BA=\bigoplus_{n\ge 0}(\bar A[1])^{\otimes n}$$

has deconcatenation coproduct, including the empty word. Its differential is the coderivation determined by

$$b_1(sa)=-s(da),\qquad b_2(sa\otimes sb)=(-1)^{|a|}s(ab).$$

Inserting a degree-one map after $sa_1,\dots,sa_{i-1}$ contributes $(-1)^{\sum_{j<i}(|a_j|-1)}$.

Proposition 18.1. *These formulas give $b^2=0$. For degree-zero closed elements,*

$$b[a|b|c]=[ab|c]-[a|bc].$$

Consequently its second boundary is $[(ab)c-a(bc)]=0$.

Proof. On one input the assertion is $d^2 = 0$. On two inputs the terms containing $d(ab)$ cancel the two terms containing da and db, using $d(ab) = (da)b + (-1)^{|a|} a\, db$. On three inputs the two multiplication terms have opposite signs and cancel by associativity. On disjoint input blocks the two degree-one insertions anticommute. These cases exhaust the square of the coderivation at every word length. □

The empty word is the coalgebra coaugmentation. An algebra augmentation is a separate map $A \to \mathbb{C}$ used to define $\bar{A}$. A local vacuum or a prepared state in a field theory requires its own map. Neither is selected by the empty bar word.

18.4 Chevalley coordinates and cotangent shifts

Let $\mathfrak{l}$ be a finite-dimensional Lie algebra with $[e_I, e_J] = C_{IJ}^K e_K$. In $\mathfrak{l} \ltimes \mathfrak{l}^\vee[1]$, the dual summand has degree -1 and is abelian. Its Chevalley algebra has coordinates

$$|c^I| = 1, \qquad |u_I| = 2,$$

and differential

$$dc^K = -\tfrac{1}{2} C_{IJ}^K c^I c^J, \qquad du_K = C_{JK}^L u_L c^J.$$

Give u_I weight one and c^I weight zero. Subtracting twice this weight from cohomological degree makes the u_I even coordinates of degree zero, without changing the degree of the differential.

For the alternative shift $\mathfrak{l} \ltimes \mathfrak{l}^\vee[-1]$, these degree-zero coordinates occur without regrading. The two original graded Lie algebras have different cotangent degrees. Their coordinate comparison is the stated regrading, not a comparison of their graded field pairings.

Write O_I for linear functions on $\mathfrak{l}^\vee$ and ∂^I for constant vector fields, with $[\partial^I, O_J]_S = \delta^I_J$. The linear Poisson tensor is

$$\pi = \tfrac{1}{2} C_{IJ}^K O_K \partial^I \partial^J.$$

Its differential $d_\pi = [\pi, -]_S$ satisfies

$$d_\pi \partial^K = -\tfrac{1}{2} C_{IJ}^K \partial^I \partial^J, \qquad d_\pi O_K = C_{JK}^L O_L \partial^J.$$

Thus $c^I \mapsto \partial^I$, $u_I \mapsto O_I$ intertwines the two differentials on generators and hence on their symmetric algebras. The vector field $d_\pi O_I$ is generally not ∂^I.

For $[e_1, e_2] = e_2$, the equations become

$$dc^1 = 0, \quad dc^2 = -c^1 c^2, \qquad du_1 = -u_2 c^2, \quad du_2 = u_2 c^1.$$

Then $d^2 u_1 = -u_2 c^1 c^2 + u_2 c^1 c^2 = 0$. This two-dimensional example detects a reversal of the coadjoint sign. For an abelian Lie algebra every $d_\pi O_I$ vanishes, while the constant fields remain nonzero.

18.5 All antifield degrees in the cyclic-word complex

Use homological degrees in this section. Let $R = \mathbb{C}\langle x, y, \psi \rangle$, with $|\psi| = 1$ and $d\psi = xy - yx$. Give the factors weights $(1, 0)$, $(0, 1)$, $(1, 1)$. In bidegree (r, s), a degree-p cyclic word has p occurrences of ψ, $r - p$ of x, and $s - p$ of y. Denote this finite complex by $K_{r,s}$.

There is an elementary model for all of its degrees. For $s > 0$, put

$$B_s = \mathbb{C}[v_0, \dots, v_{s-1}] \otimes \Lambda(e_0, \dots, e_{s-1}), \qquad de_i = v_i - v_{i+1}, \quad dv_i = 0,$$

with indices modulo s. Give each generator weight one, each v_i homological degree zero and each e_i degree one. A monomial of weight r records a labeled cyclic word. The factor $v_i^{a_i}$ records a_i copies of x immediately before the ith marker. This marker is ψ when e_i occurs and y otherwise.

Replacing the marker ψ by $xy - yx$ increases the multiplicity at vertex i or vertex $i+1$, with opposite signs. Ordering the occupied markers gives exactly the exterior signs of B_s. Rotation of the s labels gives the graded cyclic relation, because interchanging blocks with a, b occupied markers changes orientation by $(-1)^{ab}$. Hence

$$K_{r,s} \cong (B_s^{\mathrm{wt}=r})_{\mathbb{Z}/s\mathbb{Z}}.$$

Proposition 18.2. *For $r, s > 0$, the homology of $K_{r,s}$ is one-dimensional in degrees zero and one, and zero in every higher degree. A degree-one generator is*

$$\Psi_{r-1,s-1} = \sum_{w \in \mathsf{W}_{r-1,s-1}} [\psi w],$$

where $\mathsf{W}_{a,b}$ consists of the words with a factors x and b factors y.

Proof. Set $z = v_0$, $a_i = v_i - v_{i+1}$ for $0 \le i < s-1$, and $\eta = \sum_i e_i$. These are invertible linear changes of the even and odd variables. They give

$$B_s \cong \mathbb{C}[z] \otimes \Lambda(\eta) \otimes \bigotimes_{i=0}^{s-2} (\mathbb{C}[a_i] \otimes \Lambda(e_i),\ de_i = a_i).$$

The last factors are one-variable Koszul resolutions. Their positive-weight parts contract by polynomial division by a_i. Thus $H_*(B_s) = \mathbb{C}[z] \otimes \Lambda(\eta)$.

At weight $r > 0$ the surviving classes are z^r and $z^{r-1}\eta$. Rotation fixes η and sends every v_i to a representative of $[z]$, so it acts trivially on homology. Coinvariants for the finite cyclic group are exact over $\mathbb{C}$, proving the dimension assertion.

Every necklace has equally many xy and yx transitions, so $d\Psi_{a,b} = 0$. The functional taking $[\psi w]$ to 1 kills every two-antifield boundary by (18.1). Its value on $\Psi_{a,b}$ is $\binom{a+b}{a}$, which is nonzero. This identifies the generator. □

If $r = 0$ or $s = 0$, only the unique degree-zero necklace remains. Summing all weights gives

$$H_*(R/[R,R]_{\mathrm{gr}}) \cong \mathbb{C}[x,y] \oplus \mathbb{C}[x,y]\theta, \qquad |\theta| = 1, \quad \mathrm{wt}(\theta) = (1,1).$$

This is an identification of graded vector spaces. A multiplication on the polynomial label space is additional structure on those labels. The cyclic quotient itself is not an associative quotient algebra.

The classes $[1]$ and $[\psi]$ occur in different bidegrees. Removing the scalar augmentation removes $[1]$ and retains $[\psi]$. Removing the latter requires another quotient. Nor is the one-dimensional label in each bidegree the whole symmetric algebra on the Hamiltonian Lie algebra: products of Hamiltonian generators create additional summands. An identification with a PBW component must specify that component and its map.

18.6 Trace normalization and the central rank

For an ungraded associative algebra A, use $N \times N$ matrices. Write $E_{ij}a$ for the matrix with entry a in position (i,j). In ordinary homological CE chains set

$$\Theta_1(a) = \sum_i E_{ii}a, \qquad \Theta_2(a,b) = \sum_{i,j} E_{ij}a \wedge E_{ji}b.$$

The boundary is the matrix commutator, so

$$d\Theta_2(a,b) = N\Theta_1(ab - ba).$$

Both products sum over a free matrix index. Thus the raw reverse trace formula has a rank factor even in length two.

The primitive cyclic comparison in Theorem 15.6 instead uses stable scalar-matrix coinvariants and single cycles with distinct indices.

The empty cyclic word ν is central for the necklace bracket, with $[x, y] = \nu$. It is a primitive generator before rank is fixed. Matrix evaluation sends it to $N1$. Consequently, for $X = \operatorname{Tr} x$ and $Y = \operatorname{Tr} y$,

$$\{X^2, Y\} = 2NX.$$

Projection that kills multitraces and scalar values kills X^2 but retains this bracket. The Lie projection in a symmetric algebra on a zero-central-character Lie algebra must therefore be taken before the fixed-rank specialization. These are different reductions of the scalar direction.

18.7 The residue dual of formal Hamiltonians

Put $O = \mathbb{C}[[u, v]]$ and $\mathfrak{m} = (u, v)$. The formal Hamiltonian Lie algebra is $\mathfrak{h} = O/\mathbb{C}$, with Poisson bracket

$$\{f, g\} = f_u g_v - f_v g_u.$$

Give it the quotient of the $\mathfrak{m}$-adic linear topology, with discrete scalar field. The pointed subalgebra $\mathfrak{h}_0 = \mathfrak{m}^2$ has finite Lie quotients $\mathfrak{m}^2/\mathfrak{m}^{d+1}$ for $d \geq 2$. The full algebra has finite Taylor vector-space quotients, but these are not Lie quotients: $\{v, u^{d+1}\} = -(d+1)u^d$.

A continuous linear functional on $\mathfrak{h}$ kills some $\mathfrak{m}^{d+1}$. Therefore

$$P = \mathfrak{h}^\vee_{\mathrm{cont}} = \bigoplus_{a+b>0} \mathbb{C}\rho_{ab}, \qquad \rho_{ab} = u^{-a-1}v^{-b-1}\,du\,dv,$$

with $\langle \rho_{ab}, u^i v^j \rangle = \delta_{ai}\delta_{bj}$. The full principal-part module is the local-cohomology quotient

$$P^{\mathrm{full}} = \frac{O[u^{-1}, v^{-1}]}{O[u^{-1}] + O[v^{-1}]}\,du\,dv.$$

Its representatives have finite support in the displayed negative monomials. The module P^{full} also contains ρ_{00}. The subspace P is the annihilator of constants under the residue pairing. It is a coadjoint submodule, not a module over O, since $u\rho_{10} = \rho_{00}$.

The transpose of a Taylor truncation is an inclusion of finite dual spaces. Its direct limit is P. In contrast, $\prod_{a+b>0} \mathbb{C}\rho_{ab}$ is a larger vector space, formed using projections in the opposite direction. It is not the continuous dual just defined.

Fix the coadjoint convention $\langle f \cdot \rho, g \rangle = -\langle \rho, \{f, g\} \rangle$. Pairing against a monomial gives

$$u^p v^q \cdot \rho_{ab} = \big(q(a+1) - p(b+1)\big)\rho_{a-p+1,b-q+1}. \tag{18.2}$$

Negative target indices give zero. The coefficient of a scalar target also vanishes.

For $a + b \leq n$, only Hamiltonian terms of degree at most $n + 2$ can contribute, and the output degree is at most $n + 1$. This proves that a formal Hamiltonian acts on each finite principal part through a finite Taylor window. With the discrete linear topology on P, it also proves continuity of the bilinear action.

18.8 Polynomial and residual representations

Let $\mathfrak{h}_{\mathrm{pol}} = \mathbb{C}[u,v]/\mathbb{C}$. It acts on $\mathcal{M} = \mathbb{C}[u,v]/\mathbb{C}$ by the adjoint action. For polynomial monomial labels,

$$u^p v^q \cdot u^a v^b = (pb - qa)u^{a+p-1}v^{b+q-1} \pmod{\mathbb{C}}.$$

In particular u and v act locally nilpotently on $\mathcal{M}$. This polynomial space is not preserved by every formal Hamiltonian. For example, $f = \sum_{n\geq 2} u^n$ has $\{f, v\} = \sum_{n\geq 2} n u^{n-1}$, which is not a polynomial.

Theorem 18.3. *The $\mathfrak{h}_{\mathrm{pol}}$-module P is simple. If N is any module on which u acts locally nilpotently, then*

$$\mathrm{Hom}_{\mathfrak{h}_{\mathrm{pol}}}(P, N) = 0, \qquad \mathrm{Hom}_{\mathfrak{h}_{\mathrm{pol}}}(N, P) = 0.$$

Proof. Take a nonzero vector in a submodule of P, and let n be the largest total degree of its nonzero coefficients. Choose a top coefficient $c_{ab} \neq 0$, with $a + b = n$. Acting by $u^a v^{b+1}$ kills every lower-degree term. Its output has degree one, and its coefficient at ρ_{10} is $(b+1)c_{ab} \neq 0$.

The quadratic Hamiltonians act irreducibly on $\mathbb{C}\rho_{10} \oplus \mathbb{C}\rho_{01}$: uv has eigenvalues $1, -1$, while $u^2\rho_{10} = -2\rho_{01}$ and $v^2\rho_{01} = 2\rho_{10}$. Thus the submodule contains both degree-one vectors. Repeated action by u and v reaches every ρ_{ab} with $a + b > 0$, with nonzero factorial coefficients. The submodule is all of P.

The action of u on P is injective: $u\rho_{ab} = -(b+1)\rho_{a,b+1}$. A nonzero map $P \to N$ would be injective by simplicity, contradicting local nilpotence on its image. A map $N \to P$ sends each vector killed by a power of u to a vector killed by that power in P, hence to zero. □

Simplicity is essential for the first direction of the proof. An injective operator on a source can acquire nilpotent action on a quotient. Non-nilpotence alone does not rule out maps to locally nilpotent modules.

Diagonal rescaling cannot remove the opposite index directions. If $\rho'_{ab} = \lambda_{ab}\rho_{ab}$ with $\lambda_{ab} \neq 0$, then

$$u\rho'_{ab} = -(b+1)\frac{\lambda_{ab}}{\lambda_{a,b+1}}\rho'_{a,b+1}.$$

Polynomial labels instead move from b to $b - 1$. The distinction concerns the representation, before any choice of factorials or deformation coefficient.

18.9 Moyal coefficients and residue transposition

Define

$$\mathcal{P} = \partial_u \otimes \partial_v - \partial_v \otimes \partial_u, \qquad f *_h g = m\exp(\tfrac{h}{2}\mathcal{P})(f \otimes g).$$

For polynomial inputs the series is finite. Interchanging inputs negates $\mathcal{P}$, so the normalized commutator is

$$\{f, g\}_h = \sum_{\substack{r\geq 1 \\ r \text{ odd}}} \frac{h^{r-1}}{2^{r-1}r!}\mathcal{P}^r(f, g).$$

The coefficients begin $1, 1/24, 1/1920, 1/322560$.

For $f = u^p v^q$ and $g = u^m v^n$, write

$$C_r(p, q; m, n) = \sum_{j=0}^{r}(-1)^j \binom{r}{j}(p)_{r-j}(q)_j(m)_j(n)_{r-j},$$

where $(a)_j = a(a-1)\cdots(a-j+1)$ and $(a)_0 = 1$. Then $\mathcal{P}^r(f, g) = C_r u^{p+m-r}v^{q+n-r}$.

Residue transposition gives

$$f \cdot_h \rho_{ab} = -\sum_{\substack{r\geq 1\\ r \text{ odd}}} \frac{h^{r-1}}{2^{r-1}r!} C_r(p,q;a-p+r,b-q+r)\rho_{a-p+r,b-q+r}.$$

Terms with negative target indices vanish. The scalar target vanishes because every bracket with the constant test function is zero.

For linear f, all terms with $r \geq 3$ vanish, so the opposite translation actions above persist without quantum correction. For $f = u^4$ and ρ_{11}, the only nonzero term is

$$u^4 \cdot_h \rho_{11} = -24h^2\rho_{04}.$$

The polynomial adjoint action on the label uv instead gives $4u^4$. These are statements about two specified modules.

18.10 The algebraic Fourier lattice and its completion

Let $R_t = \mathbb{C}[[t]]$, and define the algebraic Rees Weyl algebra

$$D_t = R_t\langle z_1, z_2, \partial_1, \partial_2\rangle/([\partial_i, z_j] = t\delta_{ij},\ [z_i, z_j] = [\partial_i, \partial_j] = 0).$$

Algebraic here means a finite sum of Weyl monomials with coefficients in R_t. Its polynomial and delta modules are

$$O_t = R_t[z_1, z_2] = D_t/(D_t\partial_1 + D_t\partial_2), \qquad \Delta_t = D_t/(D_t z_1 + D_t z_2).$$

Write e for the cyclic vector of Δ_t. The basis of Δ_t is $\partial_1^a\partial_2^b e$, with finite support in (a,b).

The automorphism $\mathsf{F}(z_i) = \partial_i$, $\mathsf{F}(\partial_i) = -z_i$ preserves the Weyl relations. The map

$$T : O_t \longrightarrow \Delta_t, \qquad z_1^a z_2^b \longmapsto \partial_1^a\partial_2^b e$$

satisfies $T(Pf) = \mathsf{F}(P)T(f)$. This specifies the direction of the Fourier twist.

Let $L_t = \bigoplus_{a,b\geq 0} R_t\rho_{ab}$, with the usual principal-part multiplication by z_i and $\partial_i = t\partial_{z_i}$. There is an injective D_t-map

$$\iota : \Delta_t \longrightarrow L_t, \qquad \partial_1^a\partial_2^b e \longmapsto (-1)^{a+b}t^{a+b}a!b!\rho_{ab}. \tag{18.3}$$

It follows by differentiating ρ_{00}. The map becomes an isomorphism after inverting t, because each vector has only finitely many principal-part labels. Neither the scalar line nor its complementary label span is a D_t-submodule: $\partial_i e$ leaves the line, and $z_i\partial_i e = -te$ leaves the complementary span.

Parameter completion changes the assertion. Put

$$\widehat{\Delta}_t = \varprojlim_r \Delta_t/t^r\Delta_t, \qquad \widehat{L}_t = \varprojlim_r L_t/t^r L_t = P^{\mathrm{full}}[[t]].$$

A vector in either completion has coefficients $b_{ab}(t)$ such that for each r only finitely many labels have $b_{ab} \notin t^r R_t$. This is the condition that each coefficient of t have finite label support. Likewise

$$\widehat{O}_t = \mathbb{C}[z_1, z_2][[t]]$$

is larger than $R_t[z_1, z_2]$.

Proposition 18.4. *The completed map $\widehat{\iota}$ is injective. Image membership for a family $\sum c_{ab}(t)\rho_{ab}$ is equivalent to two conditions. Each c_{ab} is divisible by t^{a+b}. The coefficients*

$$\frac{(-1)^{a+b}c_{ab}(t)}{t^{a+b}a!b!}$$

tend to zero t-adically over the label set. Inverting t after completion does not make this map surjective.

Proof. Formula (18.3) acts separately on each coefficient. Multiplication by a nonzero power of t is injective in R_t. The stated divisibility and decay conditions are exactly those for its coefficientwise inverse to define a vector in $\widehat{\Delta}_t$.

The vector

$$\sigma = \sum_{n\geq 0} t^n \rho_{2n,0}$$

belongs to $\widehat{L}_t$. A preimage after localization would have coefficient $t^{-n}/(2n)!$ at $\partial_1^{2n}e$. There is no common lower bound on these powers of t. Every element of $\widehat{\Delta}_t[t^{-1}]$ has such a lower bound, so no preimage exists. □

The decay condition is independent of divisibility. The vector $\sum_{n\geq 0} n!t^n\rho_{n,0}$ satisfies divisibility, but its formal inverse has coefficients $(-1)^n$, which do not tend to zero. Thus even a coefficientwise inverse with no negative powers need not belong to the completed source.

The Fourier isomorphism holds on the algebraic modules, and T extends to their parameter completions. Its completed principal-part image is the one in the proposition. A change of action, a localization and a completion are three separate operations in this comparison.

18.11 Support, pairing and compatible limits

For an open interval I, the continuous dual of $C_c^\infty(I)$ is the distribution space $\mathcal{D}'(I)$. The continuous dual of $C^\infty(I)$ is the compactly supported distribution space $\mathcal{E}'(I)$. A smooth function acts on either by

$$(aT)(\varphi) = T(a\varphi), \qquad (\partial_t T)(\varphi) = -T(\partial_t\varphi).$$

These formulas define multiplication by a smooth test and transposition of differentiation. They do not define the product of two arbitrary distributions.

Extension by zero carries compact tests from U to V when $U \subset V$. Dualizing reverses this arrow. The same variance distinction occurs between covariant Chevalley chains and contravariant Chevalley cochains. To identify a dual cochain object with physical observables one must construct the required support maps, rather than infer their direction from the coefficient formula.

A continuous Kronecker pairing cannot pair two unrestricted coordinate products. If e_n is the nth coordinate vector, then $e_n \to 0$ in the product topology, whereas the requested pairing would have $B(e_n, e_n) = 1$. Reciprocal changes of dual bases also leave the contraction of the first n diagonal terms equal to n. A weighted Hilbert or nuclear pairing must therefore specify its spaces and prove its convergence.

For countable surjective inverse systems V_d, W_d of discrete vector spaces, put $V = \varprojlim_d V_d$ and $W = \varprojlim_d W_d$. A continuous map $V \to W_d$ factors through some input quotient V_e. The input cutoff can depend on the output cutoff. When the relevant quotient projections are onto, this gives the familiar form

$$\operatorname{Hom}_{\mathrm{cont}}(V, W) = \varprojlim_d \varinjlim_e \operatorname{Hom}(V_e, W_d).$$

This formula records continuity, not locality in a physical coordinate. For complexes, derived inverse limits also retain their degree-minus-one compatibility information. The strict cycle criterion and the derived Roos complex are distinguished in Section 16.5.

Finally, the polar trace of Chapter 4 is a relative trace. With its boundary maps r, ℓ, it obeys $\tau_{\mathcal{A}} d = \tau_B r$ and $\tau_B \ell = 0$. On the path model with $b(0) = r(a)$ and $b(1) = \ell(c)$, the corrected trace includes $\int_0^1 \tau_B(b_{dt})$. Indeed its value on a differential is

$$\tau_B r(a) + \tau_B(b(1) - b(0)) = 0.$$

The endpoint term supplies closedness. A residue value alone does not supply that term or the physical measure used in quantum sewing.

The coefficient constructions above fix the signs and domains of their maps. A native comparison must additionally preserve the full field differential, support, pairing, boundary conditions and operations. Those requirements remain at the points where the physical maps are constructed.

Chapter 19

Algebraic examples of ordered reconstruction

An augmentation selects a module whose derived endomorphisms can be computed by a bar construction. A coproduct instead specifies how an algebra acts on tensor products of modules. The examples below construct both kinds of structure and exhibit the relations that make them work.

Free and square-zero algebras give the first bar calculations. The quantum plane supplies a nontrivial Yoneda product. Hall flags, rational exchange and reflection equations then provide explicit quantum-group coefficients. Their realization by fields requires the field maps and collision compatibilities in addition to these algebraic formulas.

19.1 Augmentations and two elementary bars

Fix a field k of characteristic zero. An augmented algebra is a unital associative algebra A with a unital algebra map $\varepsilon : A \to k$. Write $I = \ker \varepsilon$. For an ordinary algebra concentrated in degree zero, use the homological normalized bar

$$B_n(A) = I^{\otimes n}, \qquad B_0(A) = k, \qquad d[a_1|\cdots|a_n] = \sum_{i=1}^{n-1} (-1)^{i-1}[a_1|\cdots|a_i a_{i+1}|\cdots|a_n].$$

Deconcatenation cuts the tensor at every position, including the two empty ends. Thus $B(A)$ is a coaugmented conilpotent differential graded coalgebra. Conilpotence here means that repeated reduced coproducts kill each element. In the cohomological convention of Chapter 18, B_n lies in degree $-n$. All tensors and sums in this section are algebraic.

Let V be finite-dimensional, give it internal weight one, and write sV for V in homological degree one. The tensor algebra $T(V) = \bigoplus_{m \geq 0} V^{\otimes m}$ is augmented by projection to k. The square-zero algebra $k \oplus V$ has the same augmentation and product $V^2 = 0$.

Proposition 19.1 (Free and square-zero bars)**.** *There is a quasi-isomorphism of coaugmented differential graded coalgebras*

$$k \oplus sV \longrightarrow B(T(V)), \qquad sv \longmapsto [v],$$

where sV is primitive and its differential is zero. For the square-zero algebra,

$$B(k \oplus V) = \bigoplus_{n \geq 0} (sV)^{\otimes n}$$

with zero differential and deconcatenation coproduct. Consequently the first augmentation has square-zero positive Yoneda algebra V^* in degree one. The second has tensor Yoneda algebra $T(V^*)$, with V^* in degree one.

Proof. Fix an underlying tensor of m factors of V. A bar expression with that tensor is specified by a subset C of its $m - 1$ internal gaps. The differential deletes the ith selected gap with sign $(-1)^{i-1}$. For $m \geq 2$, define $h(C)$ by inserting the first gap if it is absent, and by zero if it is present.

Deleting that newly inserted gap contributes C to $dh(C)$. The other deletions cancel the terms of $hd(C)$. If the first gap was already present, its deletion contributes C to $hd(C)$. Thus $dh + hd = 1$ on every weight $m \geq 2$.

Weights zero and one are k and sV. Their inclusion preserves the coproduct, proving the first assertion.

When $V^2 = 0$, every merged product in the bar differential vanishes. Dualizing deconcatenation gives concatenation of dual tensors. Dualizing the primitive coalgebra $k \oplus sV$ gives a square-zero positive ideal. These dual models compute derived endomorphisms of the augmentation module and hence the Yoneda products. □

The grading determines the completion in these formulas. The degree-n Yoneda space of $k \oplus V$ is $(V^*)^{\otimes n}$. The ordinary graded algebra is their direct sum. Its internal-weight completion is the different algebra $\prod_{n\geq 0}(V^*)^{\otimes n}$, with multiplication defined by the finite sum over two weights adding to a fixed weight. A completion must therefore be specified even in this elementary example.

19.2 Paths and the direction of extensions

For a finite quiver Q, put $S = \bigoplus_{i\in Q_0} ke_i$, where the e_i are orthogonal idempotents summing to one. An arrow $a : i \to j$ is represented in the S-bimodule M by $a = e_j a e_i$. The algebra $A = T_S(M)$ consists of finite linear combinations of composable paths, multiplied with the rightmost path traversed first. Projection to the paths of length zero is the augmentation $A \to S$. The relative bar uses tensors over S.

Proposition 19.2 (Relative path resolution)**.** *The sequence of left A-modules*

$$0 \longrightarrow A \otimes_S M \xrightarrow{\mu} A \longrightarrow S \longrightarrow 0, \qquad \mu(a \otimes m) = am,$$

is a projective resolution. The relative bar admits the coalgebra quasi-isomorphism $S \oplus sM \to B_S(A)$ given by the paths of length zero and one.

Proof. Every nonempty path has a unique first arrow, which is its rightmost factor in the chosen multiplication convention. Removing that factor gives the inverse of μ onto the augmentation ideal. Semisimplicity of S makes M projective over S, so $A \otimes_S M$ is projective over A. The gap contraction in Proposition 19.1 respects endpoints and tensors over S. It gives the relative coalgebra comparison. □

For $1 \xrightarrow{a} 2$, the projective left module Ae_1 has basis e_1, a and radical $ka \cong S_2$. The module $Ae_2 = S_2$ is simple and projective. Thus

$$0 \longrightarrow Ae_2 \xrightarrow{e_2 \mapsto a} Ae_1 \longrightarrow S_1 \longrightarrow 0$$

gives $\operatorname{Ext}^1_A(S_1, S_2) = k$, while all other degree-one extensions between the two simples vanish. Higher extensions vanish by the resolution.

There are two bimodule conventions to distinguish. The vector-space dual $M^* = \operatorname{Hom}_k(M, k)$ with $(sft)(m) = f(tms)$ reverses endpoints: $(e_j M e_i)^* = e_i M^* e_j$. By contrast, in the algebra $E = \operatorname{Ext}^*_A(S, S)$ with product $\alpha\beta$ meaning first β and then α,

$$e_j E^1 e_i = \operatorname{Ext}^1_A(S_i, S_j).$$

The extension above therefore lies in $e_2 E^1 e_1$. An identification with the endpoint-reversing dual must include the corresponding opposite-algebra convention.

19.3 The quantum plane and its Yoneda product

Let $q \in k^\times$ and set $A_q = k\langle x, y\rangle/(yx - qxy)$. Assign x, y internal weight one and use the augmentation killing them. The rule $yx \mapsto qxy$ decreases the inversion number of a tensor. Its leading pair has no overlap with itself, and reductions on disjoint pairs commute. It follows that the monomials $x^a y^b$, $a, b \geq 0$, form a basis. In particular,

$$(x^a y^b)(x^c y^d) = q^{bc} x^{a+c} y^{b+d}.$$

Proposition 19.3 (Resolution and product)**.** *The augmentation has the linear free resolution*

$$0 \longrightarrow A_q e_r \xrightarrow{d_2} A_q e_x \oplus A_q e_y \xrightarrow{d_1} A_q \longrightarrow k \longrightarrow 0,$$

where e_x, e_y have internal weight one, e_r has weight two, and

$$d_1(e_x) = x, \qquad d_1(e_y) = y, \qquad d_2(e_r) = ye_x - qxe_y.$$

With the composition convention above,

$$\operatorname{Ext}^*_{A_q}(k,k) = k\langle \xi, \eta\rangle/(\xi^2, \eta^2, \xi\eta + q\eta\xi), \qquad |\xi| = |\eta| = 1.$$

Proof. The relation gives $d_1 d_2 = 0$, and $\operatorname{im} d_1$ is the augmentation ideal. Write $u = \sum u_{ab} x^a y^b$ and $v = \sum v_{ab} x^a y^b$. The equation $ux + vy = 0$ is equivalent to

$$q^b u_{a-1,b} + v_{a,b-1} = 0,$$

with negative-index coefficients zero. It gives $u_{a0} = 0$ and $v_{ab} = -q^{b+1} u_{a-1,b+1}$. For $c = \sum u_{a,b+1} x^a y^b$, these identities say $u = cy$ and $v = -qcx$.

This proves exactness at $A_q e_x \oplus A_q e_y$. Right multiplication by y is injective on the displayed basis, so d_2 is injective.

Applying $\operatorname{Hom}_{A_q}(-, k)$ kills the differential and gives dimensions 1, 2, 1. Represent the degree-one classes by maps to k with $\xi(e_x) = 1$, $\xi(e_y) = 0$, $\eta(e_x) = 0$, $\eta(e_y) = 1$. Their degree-minus-one lifts to endomorphisms of the resolution satisfy

$$\xi(e_r) = -e_y, \qquad \eta(e_r) = qe_x.$$

Indeed $d\xi + \xi d = d\eta + \eta d = 0$ on e_r. If $\theta(e_r) = 1$ denotes the degree-two class, composition gives

$$\xi\eta = q\theta, \qquad \eta\xi = -\theta, \qquad \xi^2 = \eta^2 = 0.$$

These products give precisely the asserted four-dimensional algebra. □

The resolution is linear in internal weight, which is the Koszul property in this example. Its dual relation can also be read directly from $k(y \otimes x - qx \otimes y)$: the annihilator is spanned by $\xi^2, \eta^2, \xi\eta + q\eta\xi$. The explicit lifts fix the convention needed to identify that quadratic dual with the Yoneda algebra.

19.4 Weyl and Clifford coefficients

A scalar deformation of a quadratic relation can destroy the augmentation. To see this without a completion ambiguity, first work over $R = k[\hbar]$ and form finite sums of tensor monomials. For an even finite-dimensional space V and alternating form ω, define

$$W_\hbar(V, \omega) = T_R(V \otimes R)\,/\,(uv - vu - \hbar\omega(u, v)).$$

For a purely odd space and symmetric ω, define the Clifford algebra by $uv + vu = \hbar\omega(u, v)$. In particular, an odd generator satisfies $u^2 = \hbar\omega(u, u)/2$.

Proposition 19.4 (Ordered bases and scalar curvature)**.** *After choosing an ordered basis of V, the Weyl algebra has an R-basis of nondecreasing monomials. The Clifford algebra has the same basis with each generator occurring at most once. For the filtration by the number of generators,*

$$\operatorname{gr} W_\hbar = \operatorname{Sym}_R(V \otimes R), \qquad \operatorname{gr} \mathrm{Cl}_\hbar = \Lambda_R(V \otimes R).$$

The quadratic relation space carries a scalar functional c sending each relation to its prescribed value $\hbar\omega$. If $c \neq 0$, sending all generators to zero does not define an augmentation to R.

Proof. Replace each descending adjacent pair by its ordered pair and its scalar correction. In the odd case also replace a repeated pair by its scalar square. Order tensors first by length and then lexicographically. Every replacement decreases this order.

In a triple of distinct generators, either reduction sequence makes the same three interchanges. Each scalar correction commutes with the remaining generator. The resulting ordered terms and scalar terms therefore agree.

For repeated odd generators, the square rule agrees with the diagonal relation because $2u^2 = \hbar\omega(u,u)$. This resolves the remaining overlaps, proving the bases.

The scalar terms lower the filtration by two, so the associated graded algebras are as stated. Applying a proposed augmentation to a relation would give $0 = c(r)$, proving the last assertion. □

In a curved Koszul description, c is the degree-two curvature element in the dual of the quadratic relation space. The presentation here has no linear correction and hence no linear dual differential. The functional and its degree belong to this quadratic algebraic construction. Identifying it with a geometric anomaly requires a map of the relevant deformation complexes.

One may subsequently complete in $\hbar$. In that completion, an element can have unbounded tensor length as its $\hbar$-order increases. The finite-length filtration above is not exhaustive on such elements. The algebraic PBW basis describes this completion as formal series in $\hbar$. Each coefficient is a finite linear combination of the stated monomials.

19.5 Enveloping bars and their homology

Let $\mathfrak{g}$ be an ordinary Lie algebra over k. The universal enveloping algebra $U(\mathfrak{g})$ has augmentation killing $\mathfrak{g}$. Its normalized bar is generally much larger than its Chevalley chain complex $C_n(\mathfrak{g}) = \Lambda^n\mathfrak{g}$. Set

$$d(x_1 \wedge \cdots \wedge x_n) = \sum_{i<j} (-1)^{i+j+1} [x_i, x_j] \wedge x_1 \wedge \cdots \widehat{x_i} \cdots \widehat{x_j} \cdots \wedge x_n.$$

The exterior coalgebra coproduct splits the factors into two ordered sublists with the permutation sign.

Theorem 19.5 (Ordinary bar–Chevalley comparison)**.** *Antisymmetrization defines a coalgebra quasi-isomorphism*

$$\kappa : C_*(\mathfrak{g}) \longrightarrow B(U(\mathfrak{g})), \qquad \kappa(x_1 \wedge \cdots \wedge x_n) = \sum_{\sigma \in \Sigma_n} \operatorname{sgn}(\sigma) [x_{\sigma(1)} | \cdots | x_{\sigma(n)}].$$

If $\mathfrak{g}$ is positively graded and its bracket adds weights, this comparison is a quasi-isomorphism in each weight and on the degreewise product over weights. The product completion uses tensors completed by total weight.

Proof. Pair terms whose merged adjacent entries occur in opposite orders. Their difference is the Lie bracket, with the sign in the displayed Chevalley differential. This proves the chain-map equation. Cutting an antisymmetrized tensor is the same signed sum as choosing its left subset and antisymmetrizing both parts. Hence κ preserves coproducts.

Filter by total PBW degree, putting $\Lambda^n \mathfrak{g}$ in filtration degree n. The associated graded comparison is $\Lambda(V) \to B(\mathrm{Sym}(V))$, for $V = \mathfrak{g}$. It is the comparison of the Koszul and bar free resolutions of k. For completeness, on polynomial degree p and exterior degree q with $p + q > 0$, the Koszul contraction is

$$h = \frac{1}{p+q} \sum_i (x_i \wedge -) \frac{\partial}{\partial x_i}, \qquad \delta = \sum_i x_i \iota_i, \qquad \delta h + h \delta = 1.$$

Here ι_i contracts the exterior generator x_i. Only finitely many summands act on each element.

The free bar resolution is contracted by its extra degeneracy. The augmentation-preserving comparison between the two free resolutions has a homotopy inverse obtained by lifting basis elements degree by degree. Tensoring those homotopies with k proves the associated graded assertion. Multiplication by $(-1)^n$ in degree n identifies the two overall bar-differential sign conventions.

A cycle in the cone of κ has finite PBW degree. A primitive for its highest associated graded term lowers that degree. Iteration terminates, proving acyclicity of the cone. In positive total weight w, PBW degree is at most w, so the argument also works weight by weight. Products of complexes of vector spaces have componentwise kernels and images. The comparison consequently extends to the stated product completion, which is conilpotent in each weight. □

For every augmented algebra, $H_1 B(A) = I/I^2$. For an enveloping algebra this becomes

$$H_1 B(U(\mathfrak{g})) = \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}].$$

The inclusion of generators is surjective onto I/I^2 and kills brackets. Its inverse follows from the augmented algebra map $U(\mathfrak{g}) \to k \oplus \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$ with square-zero second summand. This degree-one calculation often decides a proposed homology formula before any higher calculation is needed.

For $\mathfrak{sl}_2 = \langle e, h, f \rangle$, with $[h, e] = 2e$, $[h, f] = -2f$ and $[e, f] = h$, the differential $\Lambda^2 \mathfrak{sl}_2 \to \mathfrak{sl}_2$ sends

$$e \wedge h \mapsto -2e, \qquad e \wedge f \mapsto h, \qquad h \wedge f \mapsto -2f.$$

It is an isomorphism, while $d(e \wedge h \wedge f) = -2e \wedge f - 2f \wedge e = 0$. The homology polynomial is therefore $1 + t^3$. The Chevalley chain dimensions are $1, 3, 3, 1$.

The bar chain space $B_n(U(\mathfrak{sl}_2))$, however, is countably infinite-dimensional for every $n \geq 1$. Its finite PBW pieces have generating function

$$\sum_{d \geq 0} \dim \mathrm{gr}_d^{\mathrm{PBW}} B_n(U(\mathfrak{sl}_2)) z^d = \left((1 - z)^{-3} - 1\right)^n.$$

Thus 3^n counts only the least PBW degree $d = n$.

Over $\mathbb{C}$, the cohomology of a finite-dimensional simple Lie algebra of rank r is exterior on generators in degrees $2m_i + 1$, where m_i are its exponents. This standard Lie-cohomology theorem gives

$$P_{\mathfrak{sl}_3}(t) = (1 + t^3)(1 + t^5), \qquad P_{\mathfrak{g}_2}(t) = (1 + t^3)(1 + t^{11}).$$

See Etingof's *Lie groups and Lie algebras, I*, §44. The invariant-form comparison, rank-two examples and exponent formula appear in Theorem 44.12, Example 44.16 and Theorem 44.17. The equality of homology and cohomology dimensions here uses finite-dimensional Chevalley chain spaces.

For the positive Witt algebra

$$L_1 = \bigoplus_{m \geq 1} \mathbb{C} e_m, \qquad [e_i, e_j] = (j - i) e_{i+j},$$

the abelianization has basis e_1, e_2. Indeed brackets have weight at least three and $e_m = (m-2)^{-1}[e_1, e_{m-1}]$ for $m \geq 3$.

The higher calculation is Goncharova's theorem: for every $n \geq 1$, the homology of $B(U(L_1))$ has one class in each weight

$$\frac{3n^2 - n}{2}, \qquad \frac{3n^2 + n}{2},$$

and vanishes in other weights.[1]

Weightwise duality and Theorem 19.5 transfer the Lie-cohomology formula to this bar. Its Euler series is consequently

$$\prod_{m \geq 1} (1 - z^m) = 1 + \sum_{n \geq 1} (-1)^n \left(z^{(3n^2-n)/2} + z^{(3n^2+n)/2} \right).$$

The homology theorem supplies the degree and multiplicity of each class. The Euler series alone does not determine them.

19.6 Planar path counts and bar gradings

A Motzkin path of length n stays at nonnegative height, starts and ends at height zero, and has steps $(1, 1)$, $(1, 0)$, $(1, -1)$. Let $M(n)$ count these paths and $R(n)$ count those with no horizontal step at height zero. Both include the empty path. Write $M(z) = \sum M(n) z^n$ and $R(z) = \sum R(n) z^n$ in $\mathbb{Q}[[z]]$.

A first-return decomposition gives

$$M = 1 + zM + z^2 M^2, \qquad R = 1 + z^2 M R.$$

In the second decomposition, the interior of the first excursion is arbitrary, while the remainder obeys the height-zero restriction. The quadratic equation implies $(1 + zM)(1 - z^2 M) = 1 + z$ and therefore

$$(1 + z) R = 1 + zM, \qquad M(n) = R(n) + R(n+1).$$

Solving with constant term one gives

$$M(z) = \frac{1 - z - \sqrt{1 - 2z - 3z^2}}{2z^2}, \qquad R(z) = \frac{1 + zM(z)}{1 + z}.$$

The square root has constant term one, and the apparent pole is removable.

For $n \geq 1$, the derived sequences

$$a_n = M(n+1) - M(n) = 1, 2, 5, 12, 30, 76, \ldots, \qquad b_n = R(n+3) = 3, 6, 15, 36, 91, \ldots$$

belong to the same quadratic generating-function field. They nevertheless disagree already in degree one with the proposed enveloping homologies:

$$\dim H_1 B(U(L_1)) = 2 \neq a_1, \qquad \dim H_1 B(U(\mathfrak{sl}_2)) = 0 \neq b_1.$$

They also cannot count the full bar chain spaces, whose positive degrees are countably infinite-dimensional.

[1] L. V. Goncharova, *The cohomologies of Lie algebras of formal vector fields on the line*. Functional Analysis and Its Applications 7 (1973), 91–97 and 194–203. The weight formula is also stated as Theorem 1.4 of D. Millionschikov, *Algebra of formal vector fields on the line and Buchstaber's conjecture*, arXiv:0807.0653.

For L_1, the finite weightwise chain count is instead

$$\sum_{w\geq 0} \dim B_n(U(L_1))_w z^w = \left(\prod_{m\geq 1}(1-z^m)^{-1} - 1\right)^n.$$

A comparison with a path complex must specify its differential, grading and map to this bar. The choice of $U(L_1)$ is itself algebraic data. A Virasoro coefficient or central charge does not determine a transverse multiplication.

19.7 Hall multiplication for one arrow

Let $Q : 1 \longrightarrow 2$ over $\mathbb{F}_q$, and choose v with $v^2 = q$ in a characteristic-zero coefficient field. A representation is a linear map between its two vertex spaces. For isomorphism classes $[A]$, $[B]$, define twisted Hall multiplication by

$$[A] * [B] = v^{\langle \dim A, \dim B\rangle} \sum_{[L]} F^L_{AB}[L], \qquad \langle (a,b),(c,d)\rangle = ac + bd - ad,$$

where F^L_{AB} counts subrepresentations isomorphic to B with quotient isomorphic to A. Associativity follows by counting the same two-step flags in either order. Bilinearity of the Euler form makes the two total twisting exponents equal.

Write S_1, S_2 for the vertex simples, $D = S_1 \oplus S_2$, and M for the nonzero map $\mathbb{F}_q \to \mathbb{F}_q$. The two representations of dimension $(1,1)$ give

$$E_1E_2 = v^{-1}([D] + [M]), \qquad E_2E_1 = [D], \qquad E_i = [S_i].$$

In particular $E_1E_2 - v^{-1}E_2E_1 = v^{-1}[M]$.

Proposition 19.6 (The two quantum Serre relations)**.** *The Hall generators satisfy*

$$E_1^2E_2 - (v + v^{-1})E_1E_2E_1 + E_2E_1^2 = 0$$

and the relation obtained by exchanging 1 *and* 2*.*

Proof. A representation of dimension $(2,1)$ is a map $f : \mathbb{F}_q^2 \to \mathbb{F}_q$, of rank zero or one. Read the successive simple quotients of a complete flag from top to bottom. The counts are

rank f	112	121	211
0	$q+1$	$q+1$	$q+1$
1	$q+1$	1	0.

In rank zero, each of the $q+1$ lines is allowed. In rank one, the middle pattern forces the kernel line, the last pattern would force $f = 0$, and the first allows every line.

The Euler exponents of these three patterns are $-1, 0, 1$. The Serre coefficient on the rank-zero representation is $(q+1)(v^{-1} - (v + v^{-1}) + v) = 0$. On the rank-one representation it is $v^{-1}(q+1) - (v + v^{-1}) = 0$.

Dualizing representations reverses flags and the arrow. Interchanging the vertices restores Q. The resulting anti-isomorphism exchanges E_1, E_2 and sends this relation to the other one. □

The relation is an exact count of extensions with its stated normalization. Identifying a Hall algebra of a geometric category with these coefficients additionally requires an extension-preserving comparison of categories and of the counting or integration measures.

19.8 Exchange in an ordered chamber

Fix a finite set of pairwise distinct spectral labels and a total order on it. Let K be a coefficient field containing their differences, and let each label carry a copy of a finite-dimensional space V. An exchange rule replaces an inverted adjacent pair by a K-linear combination with the two labels in increasing order. Write its coefficient as an invertible matrix $R(u - v)$. Tensor positions track the original labeled factors. Apply rules only to inversions.

Proposition 19.7 (Chamber confluence)**.** *Suppose no further relations are imposed on the ordered tensor spaces. The exchange rules are confluent precisely when, on each triple of distinct labels,*

$$R_{12}(u - v)R_{13}(u - w)R_{23}(v - w) = R_{23}(v - w)R_{13}(u - w)R_{12}(u - v).$$

When this equation holds, the ordered tensors form a basis of the quotient.

Proof. Every exchange decreases the inversion number of the label sequence. Disjoint exchanges commute. For three descending labels, the two sequences of three exchanges have the displayed coefficients. These are all overlapping ambiguities, so equality resolves every reduction by induction on the inversion number. Conversely, confluence and independence of the ordered tensor space force equality of the two coefficient operators. □

The rational permutation matrix supplies the basic example. Let $P(v \otimes w) = w \otimes v$ and set

$$R(u) = 1 - \frac{\hbar P}{u}.$$

The Yang–Baxter identity follows in the group algebra of Σ_3. The terms of degree zero and one in $\hbar$ agree directly. At degree two, the two three-cycles have coefficients equal by

$$\frac{1}{u(u+v)} + \frac{1}{v(u+v)} = \frac{1}{uv};$$

at degree three the equality is $P_{12}P_{13}P_{23} = P_{23}P_{13}P_{12}$. This proves the operator identity on $V^{\otimes 3}$ in every dimension.

If exchange is also imposed in the reverse direction, its compatibility requires $R_{21}(-u)R(u) = 1$ on the chosen pair space. For the rational matrix,

$$R_{21}(-u)R(u) = \left(1 - \frac{\hbar^2}{u^2}\right)1.$$

Over $k(\hbar, u)$, imposing both relations therefore kills every degree-two product between those distinct labels. Even $\dim V = 1$ exhibits this failure, although the Yang–Baxter equation is then automatic. Over a formal coefficient ring, it instead produces $\hbar^2$-torsion.

The scalar normalization

$$\widetilde{R}(u) = f(u)R(u), \qquad f(u) = \left(1 - \frac{\hbar^2}{u^2}\right)^{-1/2} \in k(u)[[\hbar]]$$

restores the reverse-crossing identity. The series f is even in u and has constant term one. Its scalar nature preserves Yang–Baxter. Its formal coefficient ring is part of the normalization.

19.9 RTT multiplication, coproduct and PBW

Work first over $R = k[\hbar]$. Define $Y_\hbar(\mathfrak{gl}_N)$ by generators $t_{ij}^{(r)}$, $r \geq 1$, and put

$$T(u) = I + \sum_{r \geq 1} T^{(r)} u^{-r}, \qquad t_{ij}^{(0)} = \delta_{ij}.$$

The defining RTT relation is

$$R(u-v)T_1(u)T_2(v) = T_2(v)T_1(u)R(u-v), \qquad R(u) = I - \frac{\hbar P}{u}.$$

It means the coefficient identities after clearing $u - v$:

$$[t_{ij}^{(r+1)}, t_{kl}^{(s)}] - [t_{ij}^{(r)}, t_{kl}^{(s+1)}] = \hbar\big(t_{kj}^{(r)} t_{il}^{(s)} - t_{kj}^{(s)} t_{il}^{(r)}\big), \qquad r, s \geq 0. \tag{19.1}$$

Indeed these are the coefficients of $u^{-r}v^{-s}$ in $(u-v)[t_{ij}(u), t_{kl}(v)]$ and its numerator. Each coefficient is a finite algebraic expression.

Proposition 19.8 (Hopf structure and evaluation)**.** *There is an algebraic Hopf structure with*

$$\Delta(t_{ij}^{(r)}) = \sum_{a=1}^{N} \sum_{p=0}^{r} t_{ia}^{(p)} \otimes t_{aj}^{(r-p)}, \qquad \varepsilon(T(u)) = I, \qquad S(T(u)) = T(u)^{-1}.$$

Let $U_\hbar(\mathfrak{gl}_N)$ be generated by E_{ij} with $[E_{ij}, E_{kl}] = \hbar(\delta_{kj}E_{il} - \delta_{il}E_{kj})$. For a central parameter a, evaluation is the homomorphism

$$T(u) \longmapsto I + \frac{E}{u-a}, \qquad t_{ij}^{(r)} \longmapsto a^{r-1}E_{ij}.$$

The inclusion $E_{ij} \mapsto t_{ij}^{(1)}$ has evaluation at $a = 0$ as a left inverse.

Proof. If matrices T, U satisfy RTT and their coefficient algebras commute, then

$$R(T_1U_1)(T_2U_2) = RT_1T_2U_1U_2 = T_2T_1RU_1U_2 = T_2U_2T_1U_1R.$$

Here only factors from distinct coefficient algebras and distinct auxiliary positions have been interchanged. Thus their matrix product satisfies RTT. This proves the coproduct identity. Associativity of matrix multiplication proves coassociativity. The counit follows by substituting I.

Since $T(u) = I + O(u^{-1})$, its inverse coefficient of order r is a finite sum of products of coefficients of total order r. Inverting the RTT identity shows that this inverse satisfies the relations for an antihomomorphism. The two matrix inverse identities are the antipode axioms. All these formulas lie in the ordinary coefficient algebra and tensor product. Only their generating series use u^{-1}-adic notation.

Substituting $I + E/(u-a)$ into RTT reduces it to the displayed commutator of the E_{ij}. Equation (19.1) with $r = 0$, $s = 1$ gives that commutator for the first modes. Evaluation fixes those modes, proving the claimed injection. □

Theorem 19.9 (PBW for the rational RTT algebra)**.** *For any total order on the generators, their ordered monomials form an R-basis of $Y_\hbar(\mathfrak{gl}_N)$. For the filtration $\deg t_{ij}^{(r)} = r - 1$,*

$$\operatorname{gr} Y_\hbar(\mathfrak{gl}_N) \cong U_\hbar(\mathfrak{gl}_N[t]).$$

For the different filtration $\deg t_{ij}^{(r)} = r$, the associated graded algebra is the polynomial algebra on the same generators.

Proof. Iterating the coefficient recurrence gives

$$[t_{ij}^{(r)}, t_{kl}^{(s)}] = \hbar \sum_{b=1}^{\min(r,s)} \Big(t_{kj}^{(b-1)} t_{il}^{(r+s-b)} - t_{kj}^{(r+s-b)} t_{il}^{(b-1)} \Big).$$

In the degree-r filtration, each term on the right has strictly smaller total degree than the commutator on the left. Sorting therefore proves spanning by ordered monomials. In the degree-$r-1$ filtration, the $b=1$ term is the current Lie bracket and all other terms have lower degree. There is consequently a surjection $U_\hbar(\mathfrak{gl}_N[t]) \to \operatorname{gr} Y_\hbar$.

To prove injectivity, iterate the coproduct and evaluation at independent parameters $a_1, \dots, a_m$. The resulting matrix is

$$\prod_{j=1}^{m} \Big(I + \frac{E^{[j]}}{u - a_j} \Big),$$

with the indicated order and commuting coefficient algebras in distinct factors. Give the parameters degree one and all $E^{[j]}$ degree zero. The highest parameter-degree part of the image of $t_{ij}^{(r)}$ is $\sum_{b=1}^{m} a_b^{r-1} E_{ij}^{[b]}$. A product using at least two nonidentity matrix factors has degree at most $r-2$. Thus the induced map on current enveloping algebras is evaluation at these m independent parameters.

These evaluation maps, as m varies, are jointly injective. Indeed, take a nonzero element of highest PBW degree d. For $m = d$, pass to the PBW associated graded in each tensor factor and take the component of degree one in every factor. A symmetric tensor of d currents maps to its polarization, with the power of a_b recording the current exponent in factor b. Distinct basis tensors have distinct coefficients of these independent parameter monomials.

Polarization is injective in characteristic zero. PBW over R applies because the current Lie algebra is free over the $\mathbb{Q}$-algebra R. Hence the original element has nonzero image in one of these evaluations.

The current enveloping surjection is injective, proving the first associated graded formula. Its PBW basis proves independence of the ordered Yangian monomials and hence also the second formula. □

For $k = \mathbb{C}$, the usual complex Yangian is obtained by setting $\hbar = 1$. In an evaluation into the ordinary $U(\mathfrak{gl}_N)[\hbar]$, the matrix is instead $I + \hbar e/(u-a)$, where $[e_{ij}, e_{kl}] = \delta_{kj} e_{il} - \delta_{il} e_{kj}$. These conventions keep the deformation parameter in its proper location. Molev's *Yangians and their applications*, §§2.1–2.5, gives the standard RTT, evaluation, PBW and Hopf formulations.[2]

19.10 An explicit reflection algebra

A reflection equation is a different quadratic relation. For a constant invertible $R \in \operatorname{End}(V^{\otimes 2})$, put $R_{21} = PRP$ and define its coefficient algebra by

$$R_{21} K_1 R K_2 = K_2 R_{21} K_1 R.$$

Even for Yang–Baxter solutions, this equation does not determine a universal ordered-monomial theorem. For $R = I$, it says that all entries of K commute. For $R = P$, it is the identity $K_2^2 = K_2^2$ and imposes no relations at all. If $\dim V > 1$, the resulting symmetric and free algebras have different Hilbert series. A PBW assertion must therefore specify the matrix and the rewriting rules.

Take $V = k^2$, $q \in k^\times$, and use the basis 11, 12, 21, 22 of $V^{\otimes 2}$. Set

$$R = \begin{pmatrix} q & 0 & 0 & 0 \\ 0 & 1 & q - q^{-1} & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & q \end{pmatrix}, \qquad K = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \gamma = 1 - q^{-2}.$$

[2] A. I. Molev, *Yangians and their applications*, Handbook of Algebra **3** (2003), 907–959. arXiv:math/0211288, Proposition 2.2 and Theorems 2.6, 2.8.

This R is invertible and satisfies the constant Yang–Baxter equation.

Proposition 19.10 (Reflection PBW basis)**.** *The reflection equation for this matrix is equivalent to*

$$\begin{aligned} ba &= q^2 ab, & ca &= q^{-2} ac, & da &= ad, \\ cb &= bc + \gamma(a^2 - ad), & db &= bd + \gamma ab, & dc &= cd - \gamma q^{-2} ac. \end{aligned}$$

The monomials $a^i b^j c^k d^\ell$ form a basis, for all $q \neq 0$.

Proof. Multiplying the four-by-four matrices gives the six displayed relations. Each remaining entry of the reflection equation is a scalar multiple of one of them or reduces by $da = ad$. Conversely, the six relations make every matrix entry vanish.

Orient them from left to right with $a < b < c < d$. Each rule decreases degree-lexicographic order. The only overlapping leading pairs occur in cba, dba, dca, dcb. Both reduction sequences give, respectively,

cba	$abc + \gamma a^3 - \gamma a^2 d$
dba	$q^2 abd + q^2 \gamma a^2 b$
dca	$q^{-2} acd - \gamma q^{-4} a^2 c$
dcb	$bcd + \gamma a^2 d - \gamma ad^2$.

Disjoint replacements commute. Induction on the decreasing order therefore gives a unique normal expression in every degree, proving the basis. The constant Yang–Baxter identity for the displayed R is verified by multiplication on $(k^2)^{\otimes 3}$. It is independent of this explicit overlap argument. □

The three constructions now have specified domains. Chamber exchange acts on labeled tensor spaces. RTT gives an algebra and its tensor-product action through a coproduct. Reflection gives the stated boundary coefficient algebra. A physical identification must supply representations, supported field operations, boundary conditions and compatible collision maps in addition to these coefficients.

Chapter 20

Compact coefficients and automorphic products

A protected trace retains signed multiplicities of states. An extension product specifies how states combine. Geometric field operations also specify the support of their insertions. For a K3 surface times an elliptic curve, the Igusa product gives an exact scalar comparison. The constructions below exhibit the additional algebraic and geometric information needed to compare operators.

We begin with the Jacobi coefficients and their normalization. They determine a root determinant and a convergent diagonal operator. We then construct correspondence actions on cohomology and examine their integral specialization. The Hall correspondence uses a common elliptic torsor, while its coefficients require a compatible orientation. Each construction has its own state space and unit.

20.1 The Jacobi seed and its scalar normalization

Write $q = e^{2\pi i\tau}$ and $r = e^{2\pi iz}$. For $\operatorname{Im}\tau > 0$, use the theta series

$$\begin{aligned}\vartheta_2(\tau,z) &= \sum_{n\in\mathbb{Z}} q^{(n+1/2)^2/2} r^{n+1/2},\\ \vartheta_3(\tau,z) &= \sum_{n\in\mathbb{Z}} q^{n^2/2} r^n,\\ \vartheta_4(\tau,z) &= \sum_{n\in\mathbb{Z}} (-1)^n q^{n^2/2} r^n.\end{aligned}$$

The normalized weak Jacobi form of weight zero and index one is

$$\phi_{0,1}(\tau,z) = 4\sum_{a=2}^{4} \frac{\vartheta_a(\tau,z)^2}{\vartheta_a(\tau,0)^2} = \sum_{n\geq 0,\,\ell\in\mathbb{Z}} f(n,\ell) q^n r^\ell. \tag{20.1}$$

The K3 elliptic genus is $2\phi_{0,1}$. The coefficients are integral, satisfy $f(n,\ell) = f(n,-\ell)$, and vanish for $4n - \ell^2 < -1$. These are the usual index-one Jacobi conventions.

Proposition 20.1. *For $0 \leq n \leq 3$, the coefficients with $\ell \geq 0$ are*

$n\backslash\ell$	0	1	2	3
0	10	1	0	0
1	108	−64	10	0
2	808	−513	108	1
3	4016	−2752	808	−64

The negative-ℓ coefficients follow by symmetry, and all other coefficients in these four rows vanish.

Proof. Put $t = q^{1/8}$ and $x = r^{1/2}$. The three theta series become

$$\sum_n t^{4n^2+4n+1} x^{2n+1}, \qquad \sum_n t^{4n^2} x^{2n}, \qquad \sum_n (-1)^n t^{4n^2} x^{2n}.$$

Remove the common factor t from the numerator and denominator of the first quotient. Each denominator now has a nonzero constant coefficient. If $A(t) = \sum A_j t^j$, $B(t) = \sum B_j t^j$ and $B_0 \neq 0$, the quotient coefficients are determined by

$$C_0 = A_0/B_0, \qquad C_j = B_0^{-1}\Big(A_j - \sum_{i=1}^{j} B_i C_{j-i}\Big).$$

Squaring the displayed series and applying this recurrence through t^{24} gives the table. Terms with exponent not divisible by eight cancel in the sum. Every surviving x-exponent is even, so replacing t^8 by q and x^2 by r gives the stated coefficients. □

For $Z = \left(\begin{smallmatrix} \tau & z \\ z & \sigma \end{smallmatrix}\right)$ in the Siegel upper half-space, put $s = e^{2\pi i \sigma}$. Call a triple (n, ℓ, m) positive when $n, m \geq 0$, either $n + m > 0$, or $n = m = 0$ and $\ell < 0$. The monic product is

$$D_5(Z) = q^{1/2} r^{1/2} s^{1/2} \prod_{(n,\ell,m)>0} (1 - q^n r^\ell s^m)^{f(nm,\ell)}. \tag{20.2}$$

The Borcherds–Gritsenko–Nikulin theorem identifies this product with the weight-five Igusa form, with a character of order two. Its square is the monic weight-ten form χ_{10}. The product converges in a cusp chamber and extends holomorphically to the Siegel upper half-space.[1]

The characteristic phase fixes the sign of the theta product. For $a, b \in \{0,1\}^2$ with $a \cdot b$ even, define

$$\Theta_{a,b}(Z) = \sum_{\nu \in \mathbb{Z}^2} \exp\big(\pi i \big[(\nu + a/2)^{\mathsf T} Z (\nu + a/2) + b^{\mathsf T}\nu\big]\big), \qquad D_\theta = \prod_{a\cdot b \text{ even}} \Theta_{a,b}.$$

There are ten factors. This convention uses $b^{\mathsf T}\nu$. Replacing it by $b^{\mathsf T}(\nu + a/2)$ changes the sign of the complete product.

Proposition 20.2. *With these phases,*

$$D_\theta = 64 D_5, \qquad \chi_{10} = 2^{-12} D_\theta^2.$$

The coefficient of $q^{1/2}s^{1/2}$ in D_θ is $64(r^{1/2} - r^{-1/2})$.

Proof. For $a = (0,0)$, all four factors have leading term one. For $a = (1,0)$, the two factors have leading term $2q^{1/8}$. For $a = (0,1)$, they have leading term $2s^{1/8}$. The two remaining leading expressions are

$$2q^{1/8}s^{1/8}(r^{1/4} + r^{-1/4}), \qquad 2q^{1/8}s^{1/8}(r^{1/4} - r^{-1/4}).$$

Their product, together with the other factors, is $64 q^{1/2}s^{1/2}(r^{1/2} - r^{-1/2})$.

The theta-product identity in the cited normalization gives $D_\theta = 64 D_5$, and squaring gives the second equality. Under the alternate phase convention only the factor $a = b = (1,1)$ changes sign. □

The factors with $m = 0$ are determined by $f(0,0) = 10$ and $f(0,\pm 1) = 1$. Thus the first Fourier–Jacobi coefficient of D_5, after removing $s^{1/2}$, is

$$b(q,r) = q^{1/2} r^{1/2} (1 - r^{-1}) \prod_{n \geq 1} (1 - q^n r^{-1})(1 - q^n)^{10}(1 - q^n r). \tag{20.3}$$

[1] V. Gritsenko and V. Nikulin, *Siegel automorphic form corrections of some Lorentzian Kac–Moody Lie algebras*. See Theorem 4.1.

The same authors' paper *The Igusa modular forms and the simplest Lorentzian Kac–Moody algebras* gives equations (1.6)–(1.10). The former paper calls the unscaled theta product Δ_5. The latter uses its monic normalization.

The triple product gives $b = \eta^{9}\vartheta_{\mathrm{II}}$ when

$$\vartheta_{\mathrm{II}}(\tau, z) = \sum_{n \in \mathbb{Z}} (-1)^{n} q^{(n+1/2)^{2}/2} r^{n+1/2}.$$

Its leading factor $r^{1/2} - r^{-1/2}$ is essential to the diagonal zero. The divisor of D_{5} on the upper half-space is the orbit of $z = 0$, with multiplicity one.

As an automorphic section it lies in $\mathcal{L}^{5} \otimes \nu_{5}$, where $\mathcal{L}$ is the Hodge line and ν_{5} is the order-two character. The character records the flat factor of this line.

20.2 Root degrees, parity and the completed bar

Let L have basis f_{2}, f_{3}, f_{-2}, with $(f_{2}, f_{-2}) = -1$, $(f_{3}, f_{3}) = 2$, and other pairings zero. The sublattice

$$L_{\mathrm{II}} = 2\mathbb{Z} f_{2} \oplus \mathbb{Z} f_{3} \oplus 2\mathbb{Z} f_{-2}$$

has signature $(2, 1)$. Put

$$\alpha(n, \ell, m) = 2n f_{2} - \ell f_{3} + 2m f_{-2}, \quad \delta_{1} = 2f_{2} - f_{3}, \quad \delta_{2} = 2f_{-2} - f_{3}, \quad \delta_{3} = f_{3}.$$

Direct substitution gives

$$(\alpha, \alpha) = -2(4nm - \ell^{2}), \qquad ((\delta_{i}, \delta_{j})) = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

The Weyl vector $\rho = f_{2} - \tfrac{1}{2} f_{3} + f_{-2}$ satisfies $(\rho, \delta_{i}) = -1$.

In the ordered basis f_{2}, f_{3}, f_{-2}, the reflections $x \mapsto x - (x, \delta_{i})\delta_{i}$ are

$$S_{1} = \begin{pmatrix} 1 & 4 & 4 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{pmatrix}, \qquad S_{2} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & -1 & 0 \\ 4 & 4 & 1 \end{pmatrix},$$

$$S_{3} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Each squares to the identity and preserves the displayed form. It also preserves L_{II}.

The Gritsenko–Nikulin construction supplies a Lie superalgebra $\mathfrak{g}$ with these real simple roots and a specified multiset of imaginary simple roots. Its generators, parity and relations supply the bracket. On its positive root spaces the denominator theorem gives

$$d_{\alpha} := \operatorname{sdim} \mathfrak{g}_{\alpha} = f(nm, \ell).$$

The scalar d_{α} need not determine the two parity dimensions. In particular, adjoining an even and an odd vector of the same degree leaves every signed multiplicity unchanged.

Write $\mathfrak{n} = \bigoplus_{\alpha > 0} \mathfrak{g}_{\alpha}$. The positive cone is generated by $\delta_{1}, \delta_{2}, \delta_{3}$. In its coordinates

$$\alpha = n\delta_{1} + m\delta_{2} + (n + m - \ell)\delta_{3}, \qquad h(\alpha) = 2n + 2m - \ell.$$

This is a positive integral height on nonzero positive degrees. Every bounded-height region has finitely many degrees. Root spaces are finite-dimensional.

For a positive graded object $V = \bigoplus_{H\geq 0} V_H$, its total-height completion means $\widehat{V}_h = \prod_{H\geq 0} V_H$. Products and coproducts use the completed tensor product graded by the sum of heights. Every coefficient then involves finitely many height decompositions.

Proposition 20.3. *Super antisymmetrization gives a quasi-isomorphism*

$$\widehat{C}_*(\mathfrak{n})_h \longrightarrow \widehat{\mathrm{Bar}(U(\mathfrak{n}))}_h$$

of complete coaugmented dg coalgebras. Its Euler supercharacter is

$$\chi_{\mathrm{root}}\widehat{C}_*(\mathfrak{n})_h = \prod_{\alpha>0}(1-e^{-\alpha})^{d_\alpha}.$$

Proof. At fixed total height, only finitely many generators and tensor lengths occur. The PBW filtration is therefore finite. The associated graded comparison is the Koszul comparison for the symmetric superalgebra. An even generator contributes the usual polynomial Koszul complex. An odd generator contributes the corresponding exterior-algebra resolution. Tensoring these one-generator comparisons proves the associated graded quasi-isomorphism.

The finite filtration proves the comparison at every height, and exactness of products of vector spaces proves it after completion. The antisymmetrization map respects deconcatenation and the Chevalley coproduct, including the super signs.

For the character use homological degree together with internal parity. An even root vector contributes $1-e^{-\alpha}$ to Chevalley chains, and an odd vector contributes its reciprocal. Multiplication over a homogeneous basis gives the formula. □

There is also a useful inverse system obtained by first forming $\mathfrak{n}^{(H)} = \mathfrak{n}/\mathfrak{n}_{>H}$ and then taking its ordinary enveloping algebra. It retains all powers of every surviving generator. Its inverse limit is a different completion. For the one-dimensional even abelian algebra of height one,

$$\varprojlim_H U(\mathfrak{n}^{(H)}) = \mathbb{C}[x], \qquad \widehat{U(\mathfrak{n})}_h = \mathbb{C}[[x]].$$

Total-height truncation of the entire algebra or bar complex gives the completion used in the proposition. The ordinary bar–Chevalley maps also give a comparison of the generator-quotient inverse systems. That comparison does not identify the two topologies.

If $P = \prod_{\alpha>0}(1-e^{-\alpha})^{d_\alpha}$, super PBW gives

$$\mathrm{sch}\,\mathrm{Sym}_{\mathrm{super}}(\mathfrak{n}) = P^{-1}, \qquad e^{-\rho}P = \mathrm{Den}_{\mathfrak{g}}\,.$$

In root-torus coordinates Q, R, S, normalize $e^{-\alpha(n,\ell,m)} = Q^n R^\ell S^m$. The chosen Gritsenko–Nikulin embedding into Siegel coordinates is doubled, so

$$\mathrm{Den}_{\mathfrak{g}}(Z) = D_5(2Z).$$

The substitution $Z \mapsto Z/2$ is part of any comparison with the monic product $D_5(Z)$.

Elliptic degree m gives the positive Lie algebra a second useful decomposition. Its degree-zero part $\mathfrak{n}_0$ is a subalgebra and its positive-degree part $\mathfrak{n}_+$ is an ideal. Root addition proves $\mathfrak{n} = \mathfrak{n}_0 \ltimes \mathfrak{n}_+$. PBW, with positive-degree generators ordered first, gives

$$\widehat{U}_h(\mathfrak{n}) \cong \widehat{U}_h(\mathfrak{n}_+)\#\widehat{U}_h(\mathfrak{n}_0).$$

Here the smash product uses the adjoint derivation action.

The Chevalley differential splits into the two internal differentials and that action, giving the total complex $\widehat{C}_*(\mathfrak{n}_0; C_*(\mathfrak{n}_+))_h$. The degree label alone does not turn a point-supported current into a sheaf whose support covers the elliptic curve.

20.3 A convergent root determinant

For each active degree choose the minimal super vector space V_α of signed dimension d_α: it is purely even if $d_\alpha \geq 0$, and purely odd otherwise. Give it an orthonormal basis and take the Hilbert direct sum $\mathcal{H}_{\mathrm{vir}} = \bigoplus_\alpha^{\ell^2} V_\alpha$. This construction uses only the signed multiplicities. It does not equip V_α with the root bracket of $\mathfrak{g}$.

Lemma 20.4. *There are constants $C_0, C_1 > 0$ such that*

$$|f(N,\ell)| \leq C_0 e^{C_1\sqrt{N+1}}.$$

Consequently $|d_\alpha| \leq C_2 e^{C_3 h(\alpha)}$.

Proof. On $|q| = e^{-t}$, $|r| = 1$, with $0 < t \leq 1$, each theta numerator in (20.1) is bounded by $Ct^{-1/2}$. For ϑ_2, first remove its common factor $q^{1/8}$. The theta product formulas bound the inverse denominators by a fixed power of $\prod_{j\geq 1}(1-e^{-tj/2})^{-1}$, up to a constant. The estimate

$$\sum_{j\geq 1} -\log(1-e^{-tj/2}) = \sum_{k\geq 1} \frac{1}{k(e^{tk/2}-1)} \leq \frac{2}{t}\sum_{k\geq 1}\frac{1}{k^2}$$

therefore bounds $|\phi_{0,1}|$ by $Ct^{-1}e^{C'/t}$.

Cauchy's coefficient formula in q and r gives

$$|f(N,\ell)| \leq Ct^{-1}e^{tN+C'/t}.$$

Set $t = (N+1)^{-1/2}$ and absorb the polynomial factor. Finally $N = nm$, with $n, m \leq h(\alpha)$, proves the height bound. □

Let λ be the additive complex energy of a root degree, so that $e^{-\alpha(Z)} = e^{-\lambda(\alpha)}$. Suppose $\operatorname{Re}\lambda(\delta_i) > 0$ for all three simple roots. If $c = \min_i \operatorname{Re}\lambda(\delta_i)$, then $\operatorname{Re}\lambda(\alpha) \geq ch(\alpha)$. At most $\binom{H+2}{2}$ degrees have height H. The lemma consequently proves

$$\sum_{\alpha>0} |d_\alpha| e^{-t\operatorname{Re}\lambda(\alpha)} < \infty \quad \text{for sufficiently large } t.$$

Thus tZ lies in a trace-class region for sufficiently large t.

For the full root superspace the same assertion requires an estimate on the sum of its two parity dimensions.

In this region define

$$K_Z|_{V_\alpha} = e^{-\alpha(Z)}\,\mathrm{id}, \qquad D_Z = 1 - K_Z.$$

Both parity restrictions of K_Z are trace class. The odd determinant is nonzero when all root eigenvalues have absolute value less than one. Thus

$$\operatorname{sdet}_F(D_Z) = \frac{\det_F(D_Z|_{\bar{0}})}{\det_F(D_Z|_{\bar{1}})} = \prod_{\alpha>0}(1-e^{-\alpha(Z)})^{d_\alpha}.$$

Local uniform summability gives holomorphic dependence in the trace-class region.

The Pfaffian normalization can be made equally explicit. On a finite parity block of dimension N, put

$$A(D) = \begin{pmatrix} 0 & D \\ -D^{\mathsf{T}} & 0 \end{pmatrix}, \qquad J_N = A(I).$$

In block order, $\operatorname{Pf} A(D) = (-1)^{N(N-1)/2} \det D$. The relative Pfaffian $\operatorname{Pf} A(D)/\operatorname{Pf} J_N$ is therefore $\det D$.

Taking the even ratio divided by the odd ratio, then the trace-norm limit of height truncations, gives

$$\operatorname{sPf}_{\mathrm{rel}}(A(D_Z)) = \operatorname{sdet}_F(D_Z), \qquad e^{-\rho} \operatorname{sPf}_{\mathrm{rel}}(A(D_Z)) = D_5(2Z).$$

This is a normalized Fredholm model on a specified Hilbert superspace. The operator $1 - K_Z$ is bounded. A first-order geometric operator requires its own symbol, domain and comparison of determinant lines. Continuing the scalar automorphic section supplies its automorphy law. An operator intertwiner is additional data.

20.4 The reduced enumerative scalar

Let $X = S \times E$, where S is a nonsingular projective K3 surface and E is an elliptic curve. Choose a nonzero primitive class β_h with $\beta_h^2 = 2h - 2$, and $\beta_h^\vee \in H^2(S, \mathbb{Q})$ with $\langle \beta_h, \beta_h^\vee \rangle = 1$. If $p \in H^2(E, \mathbb{Z})$ is the point class, define

$$N_{g,h,d} = \int_{[\overline{M}^\bullet_{g,1}(X,(\beta_h,d))]^{\mathrm{red}}} \operatorname{ev}_1^*\big(\pi_S^* \beta_h^\vee \cup \pi_E^* p\big).$$

The bullet allows disconnected domains, with the standard noncontracted-component convention. The reduced virtual class and the incidence insertion are part of this definition.

Oberdieck–Pixton's theorem gives

$$\sum_{g,h,d \geq 0} N_{g,h,d}\, u^{2g-2} q^{h-1} s^{d-1} = -\frac{1}{\chi_{10}(q, e^{iu}, s)}.$$

This is a Laurent expansion under $r = e^{iu}$, in the sector primitive in the K3 factor.[2] The minus sign and the shifts by $h - 1$, $d - 1$ belong to the stated enumerative normalization.

Let $\mathcal{F} = \operatorname{Sym}_{\mathrm{super}}(\mathfrak{n})$. The preceding root calculation gives the exact scalar identity

$$-\big(e^{2\rho} \operatorname{sch}(\mathcal{F}^{\otimes 2})\big)\big|_{Z \mapsto Z/2} = -\chi_{10}(Z)^{-1}.$$

It follows from $\operatorname{sch} \mathcal{F} = P^{-1}$ and $e^{-\rho} P = D_5(2Z)$. The two sides have independently specified sources. An identification of their states must construct a map carrying the geometric operators and their trace to the root representation.

[2] G. Oberdieck and A. Pixton, *Holomorphic anomaly equations and the Igusa cusp form conjecture*. See Section 0.2, Theorem 1, and Sections 4.4–4.7.

20.5 Surface coefficients and their integral specialization

Let Y be a projective K3 or Enriques surface. Use rational cohomology and its complex orientation. Then $H^{\mathrm{odd}}(Y)=0$, $c_1(T_Y)=0$ rationally, and

$$A=H^*(Y,\mathbb{Q})=\mathbb{Q}1\oplus H^2(Y,\mathbb{Q})\oplus\mathbb{Q}\omega,\qquad \int_Y\omega=1.$$

If $b=b_2(Y)$, then $c_2(T_Y)=e\omega$, with $e=b+2$. Thus $(b,e)=(22,24)$ for K3 and $(10,12)$ for Enriques. The parameter y below records the normalized Hodge charge, and q records elliptic energy.

The elliptic characteristic class is the Todd class times the Chern character of

$$\mathcal{E}_{q,y}(Y)=y^{-1}\Lambda_{-y}T_Y^*\bigotimes_{n\geq 1}\Big(\mathrm{Sym}_{q^n}(T_Y\oplus T_Y^*)\otimes\Lambda_{-yq^n}T_Y^*\otimes\Lambda_{-y^{-1}q^n}T_Y\Big).$$

Here Sym_t and Λ_t denote the generating series of symmetric and exterior powers. Write $\epsilon_Y=\mathrm{td}(T_Y)\,\mathrm{ch}(\mathcal{E}_{q,y}(Y))$. Its integral is the elliptic genus. This class carries more information than its integral.

Proposition 20.5. *Put $s_y=y+y^{-1}-2$ and $a_y=1+s_y/12$. Then*

$$[q^0]\epsilon_Y=s_y\,1+a_yc_2(T_Y),\qquad [q^1]\epsilon_Y=-2s_y^2\,1+\Big(\frac{5}{6}s_y^2-2s_y\Big)c_2(T_Y).$$

For Enriques, integration of the second expression is $10y^2-64y+108-64y^{-1}+10y^{-2}$.

Proof. Use the formal opposite Chern roots $x,-x$, so $c_2=-x^2$. Modulo classes above degree four, the Todd factor is $1-x^2/12$, and

$$y^{-1}(1-ye^{x})(1-ye^{-x})=s_y-x^2.$$

This proves the constant coefficient.

The first oscillator coefficient is

$$(2-y-y^{-1})(e^{x}+e^{-x})=-2s_y+s_yc_2.$$

Multiplying it by the constant coefficient gives the q-linear formula. Integrating with $\int_Y c_2=12$ gives the displayed Laurent polynomial. For K3 the integral is twice that polynomial. □

Now specialize to $q=0$, put $t=y-1$, and use $R_0=\mathbb{Q}[[t]]$. Multiplication by $\epsilon=s_y+a_yc_2$ acts on $H=A\otimes R_0$, where $s_y=t^2/(1+t)$. It is invertible over $\mathbb{Q}((t))$ and injective over R_0. Its image $\mathcal{M}=\epsilon H$ is a free lattice.

Proposition 20.6. *The cokernel of this multiplication has the decomposition*

$$\mathrm{coker}(\epsilon:H\to H)\cong R_0/(t^4)\oplus\big(R_0/(t^2)\big)^b.$$

Its derived fiber at $t=0$ has dimension $b+1$ in each of degrees $-1,0$, and vanishes in other degrees. The image of $\mathcal{M}$ in H/tH has rank one.

Proof. On $H^2(Y)$, multiplication is the scalar s_y, a unit times t^2. On the ordered basis $1,\omega$, the matrix is

$$M=\begin{pmatrix}s_y & 0\\ u & s_y\end{pmatrix},\qquad u=a_ye.$$

The coefficient u is a unit in R_0. Swap the rows, subtract s_y/u times the first row from the second, and then subtract s_y/u times the first column from the second. The resulting matrix is $\operatorname{diag}(u, -s_y^2/u)$. Its Smith form is $\operatorname{diag}(1, t^4)$.

Each module $R_0/(t^k)$ has free resolution $[R_0 \xrightarrow{t^k} R_0]$. Derived tensor with $R_0/(t)$ makes its differential zero. There are $b+1$ torsion summands. Finally $\epsilon(0) = e\omega$, whose multiplication has one-dimensional image. □

The free module $\mathcal{M}/t\mathcal{M}$ has dimension $b+2$. Its map into H/tH need not be injective. The exact sequence is

$$0 \longrightarrow \operatorname{Tor}_1^{R_0}(\operatorname{coker}\epsilon, \mathbb{Q}) \longrightarrow \mathcal{M}/t\mathcal{M} \longrightarrow H/tH \longrightarrow (\operatorname{coker}\epsilon)/t \longrightarrow 0.$$

The inverse outer block also detects a pole:

$$M^{-1} = \begin{pmatrix} s_y^{-1} & 0 \\ -a_y e\, s_y^{-2} & s_y^{-1} \end{pmatrix}, \qquad \int_Y \epsilon^{-1} = -a_y e\, s_y^{-2}.$$

This is a fourth-order pole at $t=0$, although $\int_Y \epsilon = a_y e$ is regular.

20.6 Correspondence actions and the elliptic state

Put $X_n = Y^{[n]}$, with $X_0 = \mathrm{pt}$. A cohomological kernel from X_m to X_n is $K \in H^*(X_n \times X_m, \mathbb{Q})$. It acts by

$$T_K(v) = (p_n)_*(K \cup p_m^* v).$$

All spaces are smooth and projective, so these pushforwards are proper. Künneth and Poincaré duality identify the kernel space, with its source-dimension shift, with all graded linear maps between the two cohomologies.

Pullback, cup product and integration over the middle factor therefore give associative convolution and diagonal identities. The representation-theoretic information lies in the selected geometric kernels.

For $r>0$, the incidence correspondence records

$$Z \subset Z', \quad \ell(Z) = n, \quad \ell(Z') = n+r, \quad \operatorname{Supp}(I_Z/I_{Z'}) = \{x\}.$$

Its fundamental cycle, with a class $a \in A$ inserted at x, defines creation $C_r(a)$. For the ordinary Poincaré pairing define $A_r(a) = (-1)^{r-1} C_r(a)^\dagger$.

The geometric Heisenberg theorem and Fock isomorphism give

$$[A_r(a), C_s(b)] = r\delta_{rs} \int_Y ab, \qquad \operatorname{Sym}\Big(\bigoplus_{r\ge 1} p^r A\Big) \cong \bigoplus_{n\ge 0} H^*(X_n, \mathbb{Q}).$$

Creators of the same sign commute, as do annihilators.

The Fock map sends a polynomial in $p_r(a)$ to the same polynomial in $C_r(a)$ applied to the empty vacuum.[3]

The normalization of the degree-zero unit is

$$\frac{1}{n!} C_1(1)^n |0\rangle = 1_{X_n}.$$

[3] M. Lehn and C. Sorger, *The cup product of the Hilbert scheme for K3 surfaces*. See Theorems 3.2, 3.5–3.6 and Proposition 3.9. Their surface trace is $-\int_Y$. The annihilators here use $+r\int_Y ab$.

Over n distinct points the iterated incidence orders them in $n!$ ways. Both sides have degree zero, which proves the equality.

Let $\epsilon_n = \mathrm{td}(T_{X_n})\,\mathrm{ch}(\mathcal{E}_{q,y}(X_n))$, with the factor y^{-n} appropriate to complex dimension $2n$. Use $R = \mathbb{Q}(y)[[q]]$. Its degree-zero component r_n has constant coefficient $y^{-n}(1-y)^{2n}$, a unit in R. The positive-degree part $\nu_n = \epsilon_n - r_n$ satisfies $\nu_n^{2n+1} = 0$, so

$$\epsilon_n^{-1} = r_n^{-1} \sum_{j=0}^{2n} (-r_n^{-1}\nu_n)^j.$$

This is an inverse in the cohomology ring. Multiplication by it need not preserve each ordinary cohomological degree.

Proposition 20.7. *For every cohomological kernel define*

$$\widetilde{K} = p_n^* \epsilon_n \cup K \cup p_m^* \epsilon_m^{-1}.$$

If U_j is multiplication by ϵ_j, then

$$T_{\widetilde{K}} = U_n T_K U_m^{-1}, \qquad \widetilde{L \circ K} = \widetilde{L} \circ \widetilde{K}, \qquad \widetilde{[\Delta_{X_n}]} = [\Delta_{X_n}].$$

These identities hold for every finite composable string.

Proof. In the convolution, the two characteristic classes on the middle factor multiply to one. The projection formula carries the two remaining factors through the pushforward. This proves the composite and operator identities. Pullback to the diagonal proves the identity-kernel formula. Repeating the same cancellation proves the assertion for longer strings. □

Complete the state space in length and let $\langle \mathcal{I} |$ integrate coefficientwise in the length variable p. Since $U_0 = 1$, the incidence normalization gives

$$\frac{1}{n!}\widetilde{C}_1(1)^n|0\rangle = \epsilon_n, \qquad \left\langle \mathcal{I} \middle| \exp(p\widetilde{C}_1(1)) \middle| 0 \right\rangle = \sum_{n\geq 0} p^n\, \mathrm{Ell}(X_n; q, y).$$

The scalar is a specified matrix coefficient of a prepared state. Conjugation itself leaves matrix traces unchanged.

More generally, any even invertible classes α_n with $\alpha_0 = 1$ give the same construction with $\int_{X_n} \alpha_n$ as coefficients. For $n > 0$, the choice $\alpha_n = 1 + c_n[\mathrm{pt}]$ realizes any prescribed coefficient c_n. Thus the scalar alone does not identify the geometric action.

At $q = 0$, set $\mathcal{M}_n = \epsilon_n H^*(X_n, \mathbb{Q})[[t]]$. For a kernel over $\mathbb{Q}((t))$, the transported operator preserves these image lattices exactly when the original operator preserves the original $\mathbb{Q}[[t]]$-modules. Indeed,

$$\widetilde{T}_K(\epsilon_m v) = \epsilon_n T_K(v).$$

Kernels defined over $\mathbb{Q}$ satisfy this condition. The operator $t^{-1}\mathrm{id}$ does not. Proposition 20.6 describes the nontrivial specialization even in length one.

20.7 Joining cycles and the diagonal class

Choose a basis u_i of $H^2(Y,\mathbb{Q})$, with $Q_{ij} = \int_Y u_i u_j$ and inverse matrix Q^{ij}. The product satisfies $u_i u_j = Q_{ij}\omega$. Let $\Delta : A \to A \otimes A$ be adjoint to multiplication for the trace $\int_Y$. It is the diagonal Gysin map. Testing against basis pairs gives

$$\Delta 1 = 1 \otimes \omega + \omega \otimes 1 + \sum_{i,j} Q^{ij} u_i \otimes u_j,$$

$$\Delta u_i = u_i \otimes \omega + \omega \otimes u_i, \qquad \Delta\omega = \omega \otimes \omega.$$

Consequently $m\Delta 1 = (b+2)\omega = e_A = c_2(T_Y)$ and $m\Delta(a) = e_A a$. A split followed by a join inserts this Euler class.

For permutations $\sigma, \tau \in \mathfrak{S}_N$, label each cycle by a class in A. On a joint orbit O of their action set

$$g_O = \frac{|O| + 2 - c_\sigma(O) - c_\tau(O) - c_{\sigma\tau}(O)}{2}.$$

The transitive triple $\sigma, \tau, (\sigma\tau)^{-1}$ defines a branched cover of the sphere, so Riemann–Hurwitz makes g_O a nonnegative integer. Multiplying the input labels to obtain a_O, b_O, the Frobenius permutation product on that orbit is

$$\Delta_{c_{\sigma\tau}(O)}(a_O b_O e_A^{g_O}),$$

where Δ_k is the iterated diagonal and $\Delta_1 = \mathrm{id}$. The projection formula and $m\Delta(a) = e_A a$ make this operation independent of a decomposition of the covering surface. They give the associativity of the Frobenius permutation algebra.

For $s = (12)$ and $\rho = (123)$, the first products are

$$(as)(bs) = \Delta(ab)\,\mathrm{id}, \qquad (a\rho)(b\rho) = abe_A\rho^2, \qquad (a\rho)(b\rho^{-1}) = \Delta_3(ab)\,\mathrm{id}.$$

The middle case has genus one. A joint orbit of genus at least two gives zero because $e_A^2 = 0$.

For the Hilbert-scheme product use the additional sign

$$\varepsilon(\sigma,\tau) = (-1)^{(|\sigma|+|\tau|-|\sigma\tau|)/2}, \qquad |\sigma| = N - c(\sigma).$$

Permutation parity makes the exponent integral. Its telescoping identity is the two-cocycle equation. The sign therefore preserves associativity.

The Lehn–Sorger theorem identifies the resulting invariant algebra with $H^*(Y^{[N]},\mathbb{Q})$ when the canonical class is numerically trivial, including both surfaces used here. The sector factor $i^{|\sigma|}$ removes the sign over $\mathbb{C}$. Over $\mathbb{Q}$ the sign remains explicit.

This product has a useful differential-operator expression. In a homogeneous basis a_α write

$$a_\alpha a_\beta = m^\gamma_{\alpha\beta} a_\gamma, \qquad \Delta a_\gamma = \Delta_\gamma^{\alpha\beta} a_\alpha \otimes a_\beta.$$

On the Fock polynomial algebra let $\partial_{n,\alpha}$ differentiate $p_n(a_\alpha)$, and define

$$\begin{aligned} J &= \frac{1}{2} \sum_{i,j\ge 1} ij\, m^\gamma_{\alpha\beta}\, p_{i+j}(a_\gamma)\,\partial_{i,\alpha}\partial_{j,\beta}, \\ S &= \frac{1}{2} \sum_{i,j\ge 1} (i+j)\Delta_\gamma^{\alpha\beta}\, p_i(a_\alpha)\, p_j(a_\beta)\,\partial_{i+j,\gamma}, \qquad D_H = J - S. \end{aligned} \tag{20.4}$$

Every fixed-length component has finitely many terms. There are ij choices of endpoints joining two cycles. There are $i+j$ positions for a split, with the ordered sum counting each split twice. The Hilbert sign gives D_H, which is multiplication by $-c_1(\mathcal{O}_Y^{[N]})$ under the Fock isomorphism.

Proposition 20.8. *For multiplication $M_i(a)$ by $p_i(a)$,*

$$[[D_H, M_i(a)], M_j(b)] = ijM_{i+j}(ab), \qquad [[[D_H, M_i(a)], M_j(b)], M_k(c)] = 0.$$

Moreover

$$D_H^2 p_2(a) = -p_2(e_A a), \qquad D_H^{2N+1}|_{H^*(X_N)} = 0.$$

Proof. The splitting term is first order, so its double commutator with multiplication vanishes. The two derivative slots in J give the first formula. Its right side is multiplication, giving the third commutator identity.

At length two, splitting $p_2(a)$ and then joining gives $-p_2(m\Delta a)$. Both operations raise ordinary cohomological degree by two. A join raises the cycle-age shift by two. A split inserts a degree-four diagonal and lowers that shift by two. The cohomology of X_N lies between degrees zero and $4N$, proving the nilpotence bound. □

In particular $D_H^2 p_2(1) \neq 0$. Joining is an even operator of order two. Its double commutator measures its failure to be a derivation. These are cohomological diagonal operations. Lifting them to a chiral state complex requires supported maps retaining the normal oscillators and their differential.

20.8 Extension products and their coefficients

An elementary Hall action already exists on cyclic quotients of a surface. Let $C_0(Y)$ be the category of zero-dimensional coherent sheaves on Y. For constructible functions, with Euler integration over subobjects of fixed length, set

$$(h_1 * h_2)(Q) = \int_{U \subset Q} h_1(U) h_2(Q/U)\, d\chi.$$

The order here places the subobject first. A cyclic quotient is a surjection $\mathcal{O}_Y \twoheadrightarrow Q$, and its quotient by any $U \subset Q$ remains cyclic. The formula

$$(h \cdot v)(\mathcal{O}_Y \twoheadrightarrow Q) = \int_{U \subset Q} h(U) v(\mathcal{O}_Y \twoheadrightarrow Q/U)\, d\chi$$

therefore defines a left action.

The two parenthesizations integrate the same function

$$h_1(U) h_2(W/U) v(\mathcal{O}_Y \twoheadrightarrow Q/W)$$

over $0 \subset U \subset W \subset Q$. The subobject spaces are projective Quot varieties, and Euler pushforward composes. This proves associativity, all finite-flag compatibilities, and the zero-object unit. Positive-length completion is allowed because each length has finitely many decompositions.

The construction uses constructible functions. A critical cohomological Hall action requires its own pullback, pushforward and orientation maps. There is also a change of category on passing from Y to $Y \times E$. In a local normal coordinate z, an extension of two modules killed by z can have

$$z = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Thus sheaves with zero normal operator need not be closed under the ambient extensions.

For compact sheaves modulo elliptic translation, even the base of the extension correspondence matters. Let BE be the stack of right E-torsors on the fppf site. For a left E-stack M, an object of $[M/E]$ is a torsor P together with an equivariant map $P \to M$. The forgetful map lands in BE.

Proposition 20.9. *There is a natural equivalence*

$$[(M \times N)/E_{\mathrm{diag}}] \simeq [M/E] \times_{BE} [N/E].$$

The finite-factor equivalences commute with regrouping. For coherent sheaves on $S \times E$, the fixed-polynomial extension spans over BE form a coherently associative unital correspondence object. The unit is the zero-sheaf map $BE \to [\mathcal{M}/E]$.

Proof. An object of the fiber product consists of (P, m), (Q, n) and an isomorphism $\alpha : P \to Q$. It gives the single torsor P with maps $m, n\alpha$. The inverse uses the same torsor twice and the identity. On morphisms, compatibility with α determines one torsor map from the other. This proves the equivalence, including descent and base change. Several factors give one torsor with the same list of maps, which proves the regrouping coherences.

Fix the Hilbert polynomials of the two ends of an exact sequence. Over a flat family F, its quotients of the specified polynomial form a projective Quot scheme after trivializing the torsor. The kernel is flat because both the middle and quotient families are flat. An automorphism fixing the middle object and quotient is the identity. Descent therefore gives a representable proper map from the extension stack to the middle-object stack.

Both composites of three extension spans classify $0 \subset A \subset D \subset F$ on that same torsor. Recovering D from a subobject of F/A uses its inverse image in F. This is inverse to passage to the quotient, commutes with base change, and preserves flatness. Every longer regrouping gives the same flag. The zero sheaf on a torsor gives the stated unit. □

For $M = N = E$ with translation, the diagonal quotient is E, with coordinate $y - x$, whereas the two separate quotients are points. The fiber product over BE retains their relative translation. A stability restriction can replace the Quot fiber by an open subspace, which need not remain proper. A coefficient theory must also include the nonaffine stabilizer E, or be defined relative to BE.

The determinant calculation explains the orientation data. For a three-Calabi–Yau dg category, assume that the relative Hom complexes $C(A, B)$ are perfect and that the trace gives

$$C(B, A) \simeq C(A, B)^{\vee}[-3]$$

compatibly with base change. Put $L(A, B) = \det C(A, B)$, $D(A) = L(A, A)$, with $\det C = \bigotimes_i (\det C^i)^{(-1)^i}$. The symmetry of determinant lines carries parity $\chi(C) \bmod 2$.

Proposition 20.10. *For a triangle $A \to F \to B \to A[1]$,*

$$D(F) \simeq D(A) \otimes D(B) \otimes L(A, B)^{\otimes 2}.$$

The determinant comparisons agree under refinement of flags, with the stated symmetry signs.

Proof. Apply derived Hom to the triangle in both variables. Determinant additivity expresses $D(F)$ as the tensor product of the four complexes $C(A, A)$, $C(A, B)$, $C(B, A)$, $C(B, B)$. Dualization and the odd shift each invert the determinant line, so their composite identifies $L(B, A)$ with $L(A, B)$. Reorder with determinant symmetry. For a longer flag all decompositions give the same ordered product over pairs of successive quotients. Compatibility of determinant additivity with refinement proves the assertion. □

An orientation therefore includes lines o_A with $o_A^{\otimes 2} \simeq D(A)$ and maps

$$o_A \otimes o_B \otimes L(A, B) \longrightarrow o_F$$

whose squares are the determinant comparisons. The two composites on a three-step flag can differ by an element of μ_2. Its coherent trivialization is additional to the square identities.

The cross line can itself be nontrivial. For $T = \mathbb{P}^1$, $x \in X$, and

$$A = \mathcal{O}_X \boxtimes \mathcal{O}_T(1), \qquad B = \mathcal{O}_x \boxtimes \mathcal{O}_T,$$

one has $C(A, B) = \mathcal{O}_T(-1)[0]$ and $C(B, A) = \mathcal{O}_T(1)[-3]$. Hence the cross factor in $D(A \oplus B)$ is $\mathcal{O}_T(-2)$, although the self-determinants are constant lines. Its orientation factor has degree -1, and a scalar sign cannot replace it.

20.9 Physical charge and the Gram map

The Mukai lattice of a K3 surface is

$$\Lambda = H^0(S, \mathbb{Z}) \oplus H^2(S, \mathbb{Z}) \oplus H^4(S, \mathbb{Z}), \qquad ((r, c, s), (r', c', s')) = c \cdot c' - rs' - r's.$$

It is even unimodular of rank 24 and signature $(4, 20)$.

For electric and magnetic charges $c = (Q, P) \in \Lambda \oplus \Lambda$, define

$$\Pi(c) = \big(Q^2/2, Q \cdot P, P^2/2\big) \in \mathbb{Z}^3.$$

Physical charge adds. The Gram map is quadratic:

$$\Pi(c + d) = \Pi(c) + \Pi(d) + B(c, d),$$
$$B((Q, P), (Q', P')) = (Q \cdot Q',\ Q \cdot P' + Q' \cdot P,\ P \cdot P').$$

Bilinearity proves the cocycle equation $B(c, d) + B(c + d, e) = B(d, e) + B(c, d + e)$. Thus

$$(c, \lambda)(d, \mu) = (c + d, \lambda + \mu + B(c, d))$$

defines an abelian extension of the charge lattice by $\mathbb{Z}^3$.

The graph map $j_\Pi(c) = (c, \Pi(c))$ is a homomorphism. This extension is split: the explicit isomorphism to the ordinary direct product is

$$(c, \lambda) \longmapsto (c, \lambda - \Pi(c)).$$

It records the quadratic degree without changing the additive physical charge.

For a charge-graded algebra $A = \bigoplus_c A_c$, use central symbols u^λ and the product

$$(a_c u^\lambda) \star (b_d u^\mu) = a_c b_d u^{\lambda+\mu+B(c,d)}.$$

The cocycle equation proves associativity, and

$$a_c \longmapsto a_c u^{\Pi(c)}$$

is an algebra map. The splitting sends this graph copy back to $A \otimes 1$. For a completed Hall algebra the charge map and splitting must also preserve the chosen completion. This construction regrades a supplied Hall integration map. It does not construct its coefficient theory.

The common K3 state space also carries several distinct geometric actions. Nakajima creation acts between Hilbert schemes of different lengths. The Lie algebra generated by Lefschetz triples acts on each fixed hyperkähler cohomology. Its real form is $\mathfrak{so}(4, 20)$ for K3, and $\mathfrak{so}(4, 21)$ for $K3^{[n]}$ type with $n \geq 2$.

Local ADE correspondences use a curve-supported moduli problem and its equivariant coefficient field. These statements specify different carriers. A common quantum algebra requires maps of these carriers, mixed convolution relations, and a compatible coproduct and pairing.

For clarity, fix a direct sum V of the chosen cohomology spaces and a set of operators preserving finite support. The free algebra on their symbols maps to $\mathrm{End}(V)$. Quotient by its kernel gives the algebra they generate. This faithful action does not determine a tensor structure.

If a rigid linear tensor category and a strong monoidal finite-dimensional fiber functor ω are separately specified, their matrix coefficients instead form the coend

$$H = \left(\bigoplus_{M} \omega(M)^{*} \otimes \omega(M) \right) \Big/ \left\langle [\lambda \omega(f) \otimes v]_{M} - [\lambda \otimes \omega(f) v]_{N} \right\rangle.$$

Its coproduct and counit are

$$\Delta [\lambda \otimes v]_{M} = \sum_{i} [\lambda \otimes e_{i}]_{M} \otimes [e^{i} \otimes v]_{M}, \qquad \epsilon [\lambda \otimes v]_{M} = \lambda(v).$$

The identity tensor proves basis independence and coassociativity. Tensor product supplies multiplication, and the evaluation and coevaluation identities supply the antipode. Each selected categorical morphism intertwines the resulting coactions. The action algebra records products of selected operators. The coend records natural symmetries of the specified fiber functor.

20.10 Finite Hall presentations and super relations

A finite symmetric even Borcherds–Cartan matrix $A = (a_{ij})$ determines an ordinary quiver as follows. Order the vertices. For $i < j$, place $-a_{ij}$ arrows from i to j. At an imaginary vertex place $1 - a_{ii}/2$ loops. A real vertex has none.

If

$$\langle \mathbf{d}, \mathbf{e} \rangle = \sum_{i} d_{i} e_{i} - \sum_{a : i \to j} d_{i} e_{j},$$

then its symmetrization is A. For the three real roots above the quiver has two arrows between each ordered pair of distinct vertices.

Over $\mathbb{F}_{q}$, put $v^{2} = q$ and use the twisted subobject-count product

$$[M] * [N] = v^{\langle \dim M, \dim N \rangle} \sum_{[L]} \#\{ N' \subset L : N' \simeq N,\ L/N' \simeq M \} [L].$$

This convention places the quotient first. Associativity counts the same extension flags in two orders, and bilinearity makes their twist exponents agree. The class of zero is the unit. The Hall A_{2} calculation of Chapter 19 fixes the root normalization $v^{-1}[M]$.

Select the one-dimensional simple at each vertex on which all loops act by zero. The generic composition algebra generated by those classes is the ordinary positive quantum generalized Kac–Moody algebra of the even matrix. The Kang–Schiffmann parameter is $v_{\mathrm{KS}} = v^{-1}$ for the displayed Hall convention. Their canonical basis uses the specified semisimple perverse-sheaf construction on nilpotent representation varieties.[4]

[4] S.-J. Kang and O. Schiffmann, *Canonical bases for quantum generalized Kac–Moody algebras*. See Section 2.3, Theorem 2.2 and Theorem 4.1, with the addendum.

The chosen generators are part of this theorem. A vertex with g loops has q^g distinct one-dimensional simples over $\mathbb{F}_q$. The full equivariant derived category is larger still: at a real vertex in dimension one its stack is $B\mathbb{G}_m$, with cohomology $\mathbb{Q}[c_1]$. Its total dimension is not the corresponding root multiplicity.

The Borcherds superalgebra has its own labeled simple generators and parity-dependent relation ideal. For a single even abelian generator the enveloping algebra is $\mathbb{C}[x]$. For a single odd abelian generator it is $\Lambda(\xi)$, since $2\xi^2 = [\xi, \xi] = 0$. Reassigning parity to an ordinary PBW vector space leaves its degree-two vector present. Thus the ordinary Hall presentation and a quantum supergroup require an explicit comparison respecting relations and both parity dimensions.

The same quiver has a zero-potential Ginzburg dg algebra. Use left-to-right path concatenation in this paragraph. Add reverse arrows a^* of degree -1 and vertex loops t_i of degree -2, with

$$da = da^* = 0, \qquad dt_i = \sum_{s(a)=i} aa^* - \sum_{t(a)=i} a^*a.$$

The differential squares to zero. The sum over vertices is a sum of commutators, so it vanishes in the cyclic quotient and after matrix trace. It need not vanish in the path algebra.

Projection to a full subquiver kills the omitted vertices and incident arrows and commutes with this differential. The reverse inclusion generally fails: adjoining an arrow changes dt_i by aa^* at its old source vertex. These dg projections are different maps from extension by zero in the ordinary representation category.

20.11 The eight diagonal-divisor forms

The scalar comparison has a finite arithmetic extension. For positive integers t, N, define the Hecke subgroup of the paramodular group by

$$\Gamma_t(N) = \mathrm{Sp}_4(\mathbb{Q}) \cap \begin{pmatrix} \mathbb{Z} & t\mathbb{Z} & \mathbb{Z} & \mathbb{Z} \\ \mathbb{Z} & \mathbb{Z} & \mathbb{Z} & t^{-1}\mathbb{Z} \\ N\mathbb{Z} & Nt\mathbb{Z} & \mathbb{Z} & \mathbb{Z} \\ Nt\mathbb{Z} & Nt\mathbb{Z} & t\mathbb{Z} & \mathbb{Z} \end{pmatrix}.$$

Write $\Gamma_t = \Gamma_t(1)$. For $t = 1$, this is the usual genus-two $\Gamma_0^{(2)}(N)$.

A diagonal-divisor form vanishes simply on the translates of $z = 0$ and has no other divisor on the upper half-space. The Cléry–Gritsenko classification gives, up to nonzero scalar, exactly the following eight forms in this class:

form	group	weight	character or multiplier order
Δ_5	Γ_1	5	2
Δ_2	Γ_2	2	4
Δ_1	Γ_3	1	6
$\Delta_{1/2}$	Γ_4	1/2	8
∇_3	$\Gamma_1(2)$	3	2
∇_2	$\Gamma_1(3)$	2	2
$\nabla_{3/2}$	$\Gamma_1(4)$	3/2	4
Q_1	$\Gamma_2(2)$	1	4

.

The classification concerns these Hecke subgroups and integral or half-integral weights.[5] Half-integral weights use the specified multiplier system.

At higher level the Borcherds product uses coefficients at every cusp, with cusp-width factors. The level-one formula $k = f(0,0)/2$ therefore becomes

$$k = \frac{1}{2} \sum_{\mathfrak{c}} \frac{h_{\mathfrak{c}}}{N_{\mathfrak{c}}} c_{\mathfrak{c}}(0,0),$$

in the notation of the cited product theorem. Here $h_{\mathfrak{c}}$ is the cusp width and $N_{\mathfrak{c}} = N/e$ for a cusp represented by f/e. All cusp data belong to the arithmetic input.

Integral product exponents define a minimal super vector space and its formal Fock character in each row. Under an absolute coefficient bound they also define the trace-class diagonal construction used above. A reflective denominator presentation supplies further Lie and parity data. Its ordinary quiver Hall algebra retains the even Cartan construction, while the super relations remain separately specified. Neither the eight scalar identities nor a common Cartan matrix identifies those two algebras.

20.12 The compact comparison

The compact constructions now have explicit coefficients, units and normalizations. The root determinant uses signed root multiplicities and the chosen Weyl monomial. The Hilbert-scheme state inserts the tangent characteristic class. The common-torsor extension product uses flags over BE, and its critical coefficients require orientation and base-change maps on those flags.

The remaining compact chiral comparison must act on state complexes defined independently from the Hall image. It must construct diagonal and Gysin maps preserving normal oscillators, cycle joining, charge, vacuum insertions and higher coordinate gluing. At integral specialization it must account for the t^4 and t^2 torsion of Proposition 20.6. An operator comparison must also retain the determinant cross lines and the distinction between local and elliptically wrapped support. The scalar Igusa identities provide exact tests for such a map. They do not determine the map.

[5] F. Cléry and V. Gritsenko, *The Siegel modular forms of genus 2 with the simplest divisor*. See Theorem 1.4 and Section 3.

Chapter 21

Cartan fields, boundary currents and gravitational states

A boundary stress tensor supplies a transformation law and a central charge. A gravitational comparison must also supply bulk fields, their gauge symmetry, an action and a state pairing. In three dimensions the classical field comparison is unusually explicit: an invertible coframe converts a flat Cartan connection into an Einstein metric. This conversion determines neither a quantum integration cycle nor a boundary state.

We begin with the action and its infinitesimal complexes. The boundary matrix calculation then fixes the classical central term. Finite Gaussian integrals, Virasoro states and marked parallel transport give further comparisons with specified domains. Together they identify the data needed to extend the classical correspondence to the quantum comparisons of Chapter 17.

21.1 The coframe and the action

Let M be an oriented three-manifold, and let $V \to M$ be an oriented rank-three bundle with metric η of signature $(-,+,+)$. A coframe is a bundle isomorphism $e : TM \to V$. It defines $g = e^*\eta$. Choose $\ell > 0$ and set $\Lambda = -\ell^{-2}$. Let $G > 0$ denote Newton's constant.

In an orthonormal frame, put $\epsilon_{012} = 1$ and raise indices with η. The Lorentz algebra has generators J_a satisfying $[J_a, J_b] = \epsilon_{ab}{}^c J_c$.

Write the spin connection as $\omega = \omega^a J_a$ and its curvature as

$$F_\omega = d\omega + \tfrac{1}{2}[\omega, \omega] = R^a J_a.$$

Brackets of Lie-valued forms include their exterior product. The invariant bilinear form used in each chiral sector is

$$B(J_a, J_b) = \tfrac{1}{2}\eta_{ab}.$$

It is the fundamental matrix trace in the usual real $\mathfrak{sl}_2$ realization. On a chosen trivialization define

$$\mathrm{CS}_B(A) = B(A \wedge dA) + \tfrac{1}{3}B(A \wedge [A, A]), \qquad A^{\pm} = \omega \pm \ell^{-1} e.$$

Here e is identified with a Lorentz-algebra-valued one-form by the oriented metric on V.

Proposition 21.1. *The two Chern–Simons forms satisfy*

$$\begin{aligned} \mathrm{CS}_B(A^+) - \mathrm{CS}_B(A^-) = {} & \frac{4}{\ell} B(e \wedge F_\omega) + \frac{2}{3\ell^3} B(e \wedge [e, e]) \\ & - \frac{2}{\ell} dB(\omega \wedge e). \end{aligned} \tag{21.1}$$

Consequently, for $k = \ell/(4G)$, the bulk part of $\frac{k}{4\pi}\int_M (\mathrm{CS}_B(A^+) - \mathrm{CS}_B(A^-))$ is

$$\frac{1}{8\pi G}\int_M e_a \wedge R^a + \frac{1}{48\pi G\ell^2}\int_M \epsilon_{abc} e^a \wedge e^b \wedge e^c. \tag{21.2}$$

Its displayed boundary term is $-\frac{k}{4\pi\ell}\int_{\partial M} \eta_{ab}\omega^a \wedge e^b$.

Proof. Set $a = e/\ell$ and expand in a. The quadratic terms contribute $2B(\omega \wedge da) + 2B(a \wedge d\omega)$. Since $dB(\omega \wedge a) = B(d\omega \wedge a) - B(\omega \wedge da)$, their sum is $4B(a \wedge d\omega) - 2dB(\omega \wedge a)$.

Invariance of B makes the three terms containing one a in the cubic expression equal. Their sum is $2B(a \wedge [\omega, \omega])$. The remaining odd term is $\frac{2}{3}B(a \wedge [a, a])$. These terms give (21.1).

The identities $B(e \wedge F_\omega) = \frac{1}{2}e_a \wedge R^a$ and $B(e \wedge [e, e]) = \frac{1}{2}\epsilon_{abc}e^a \wedge e^b \wedge e^c$ give (21.2) after substituting k. □

With $R^a = \frac{1}{2}\epsilon^a{}_{bc}R^{bc}$, the first term in (21.2) is $\frac{1}{16\pi G}\int \epsilon_{abc}e^a \wedge R^{bc}$. This is the first-order Einstein–Hilbert normalization. The trace convention and the relation $k = \ell/(4G)$ are the ones used in the Chern–Simons description of three-dimensional gravity.[1] A boundary variational problem must further specify its fixed fields and any additional boundary action.

The Cartan algebra $\mathfrak{g}_\Lambda$ has generators J_a, P_a with

$$[J_a, P_b] = \epsilon_{ab}{}^c P_c, \qquad [P_a, P_b] = \ell^{-2}\epsilon_{ab}{}^c J_c, \qquad \mathcal{A} = \omega^a J_a + e^a P_a.$$

For the off-diagonal pairing $b(J_a, P_b) = \eta_{ab}$, put $J_a^\pm = (J_a \pm \ell P_a)/2$. Then $b(J_a^\pm, J_b^\pm) = \pm\ell B(J_a, J_b)$ and $b(J_a^+, J_b^-) = 0$. Thus the Cartan action is $\frac{k}{4\pi\ell}\int_M \mathrm{CS}_b(\mathcal{A})$. Its level is $k/\ell = 1/(4G)$ when the two sector levels are k and $-k$.

Proposition 21.2. *On the invertible-coframe locus, $F_\mathcal{A} = 0$ is equivalent to the torsion-free Einstein equation with cosmological constant $\Lambda = -\ell^{-2}$. We use $G_{\mu\nu} = \mathrm{Ric}_{\mu\nu} - \frac{1}{2}R\,g_{\mu\nu}$ for the Einstein tensor.*

Proof. The translation component of the curvature is $T^a = de^a + \epsilon^a{}_{bc}\omega^b \wedge e^c$. Its vanishing determines the metric-compatible torsion-free connection because e is invertible. The rotation component is $R^a + (2\ell^2)^{-1}\epsilon^a{}_{bc}e^b \wedge e^c$. Its vanishing says that the metric has constant curvature $-\ell^{-2}$. In dimension three the Riemann tensor is determined by its Ricci contraction, so this is equivalent to $G_{\mu\nu} + \Lambda g_{\mu\nu} = 0$. The same identities prove the converse. □

Invertibility is essential: $e = 0$ and a flat ω solve the Cartan equations but give a degenerate metric. The classical comparison therefore applies to an open field domain. It does not identify a path integral over all connections with one over nondegenerate metrics.[2]

21.2 Eliminating the algebraic pairs

Fix a flat background $\mathcal{A}$ with invertible coframe. Its unshifted deformation complex has degrees zero through three:

$$\Omega^0(M, \mathfrak{g}_\Lambda) \xrightarrow{d_\mathcal{A}} \Omega^1(M, \mathfrak{g}_\Lambda) \xrightarrow{d_\mathcal{A}} \Omega^2(M, \mathfrak{g}_\Lambda) \xrightarrow{d_\mathcal{A}} \Omega^3(M, \mathfrak{g}_\Lambda). \tag{21.3}$$

[1] See Campoleoni, Fredenhagen, Pfenninger and Theisen, *Asymptotic symmetries of three-dimensional gravity coupled to higher-spin fields*, Section 2.2, equations (2.22)–(2.25), https://arxiv.org/abs/1008.4744.

[2] Witten, *2 + 1 dimensional gravity as an exactly soluble system*, Section 2, explicitly separates the invertible-coframe argument from the larger gauge-field domain. See https://doi.org/10.1016/0550-3213(88)90143-5.

Flatness gives $d^2_{\mathcal{A}} = [F_{\mathcal{A}}, -] = 0$.

Write a parameter as (ξ, λ), a fluctuation as $(u, v) = (\delta e, \delta\omega)$, and a curvature variation as (t, r) in the translation and rotation summands. Put $D_\omega = d + [\omega, -]$. The differentials are

$$(\xi, \lambda) \longmapsto (D_\omega \xi + [e, \lambda], D_\omega \lambda + \ell^{-2}[e, \xi]), \tag{21.4}$$

$$(u, v) \longmapsto (D_\omega u + [e, v], D_\omega v + \ell^{-2}[e, u]), \tag{21.5}$$

$$(t, r) \longmapsto (D_\omega t + [e, r], D_\omega r + \ell^{-2}[e, t]). \tag{21.6}$$

They are the components of the same flat differential.

We shall remove three pairs connected by invertible algebraic operators. The relevant cancellation also specifies the comparison maps.

Lemma 21.3. *Suppose a differential between two consecutive terms of a complex has block form* $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ *on* $U \oplus A \to V \oplus B$*, with* $a : U \to V$ *invertible. The pair* $U \to V$ *contracts, leaving differential* $d - ca^{-1}b : A \to B$*. The lift of* $x \in A$ *in that degree is* $(-a^{-1}bx, x)$*, and projection in the next degree sends* (v, y) *to* $y - ca^{-1}v$*.*

Proof. Multiply the differential on the right and left by

$$\begin{pmatrix} 1 & -a^{-1}b \\ 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 \\ -ca^{-1} & 1 \end{pmatrix},$$

respectively. It becomes $a \oplus (d - ca^{-1}b)$. The square-zero identity forces the adjacent differentials to have zero components into and out of this isolated invertible block. Its inverse is a contracting homotopy. Undoing the two basis changes gives the stated lift and projection. □

Use e to regard u as an endomorphism of TM. Its symmetric part is the metric variation

$$h_{\mu\nu} = \eta_{ab}(e^a_\mu u^b_\nu + e^a_\nu u^b_\mu).$$

The Lorentz parameter maps isomorphically to the skew part of u. Cancelling this degree-zero/degree-one pair removes the frame rotation and retains h.

Next consider the torsion map on a connection variation, $L_e(v) = v^a{}_b \wedge e^b$. In an orthonormal coframe, write $v_{ab} = v_{abc}e^c$, where $v_{abc} = -v_{bac}$, and $L_e(v)^a = \frac{1}{2}t^a{}_{bc}e^b \wedge e^c$. Then

$$t_{abc} = v_{acb} - v_{abc}, \qquad v_{abc} = \tfrac{1}{2}(t_{cab} - t_{abc} - t_{bca}). \tag{21.7}$$

This is an isomorphism between two rank-nine bundles. It cancels v against the torsion component in the next degree. The lift of u is the unique v satisfying $L_e(v) = -D_\omega u$.

After this cancellation, the translation Bianchi equation is $[e, r] = 0$. Dualizing the form indices identifies r with an endomorphism of V. The map $r \mapsto [e, r]$ identifies its skew part with a rank-three Bianchi component. Cancelling this final pair leaves a symmetric curvature tensor and the divergence identity.

Theorem 21.4. *The three cancellations give a homotopy equivalence, by local differential operators, from* (21.3) *to*

$$\Gamma(TM) \xrightarrow{K} \Gamma(S^2T^*M) \xrightarrow{E_\Lambda} \Gamma(S^2T^*M) \xrightarrow{\mathrm{div}} \Gamma(T^*M). \tag{21.8}$$

Here $K\zeta = \mathcal{L}_\zeta g$*,* E_Λ *is the variation of* $G + \Lambda g$*, and the final map is covariant divergence. On a sufficiently small geodesically convex open set, this complex resolves the sheaf of Killing fields.*

Proof. Each cancellation above has a pointwise inverse, so Lemma 21.3 supplies chain maps and homotopies. The remaining parameter is the vector field $\zeta = e^{-1}\xi$. Cartan's identity $\mathcal{L}_\zeta \mathcal{A} = d_{\mathcal{A}}(\iota_\zeta \mathcal{A})$ on a flat background shows that its metric variation is $K\zeta$. A compensating Lorentz rotation gives the symmetric-frame lift selected by the first cancellation.

Solving the torsion equation gives the variation of the Levi–Civita connection. The rotation curvature equation therefore becomes the variation of $G + \Lambda g$. Its remaining Bianchi component is covariant divergence. The explicit formulas below identify these two operators. Thus the cancelled complex is (21.8).

A flat connection has a local parallel trivialization. The ordinary Poincaré homotopy then contracts its positive-degree de Rham complex on a small convex open set. The degree-zero kernel transfers to $K\zeta = 0$ under the chain equivalence. This gives the asserted local resolution. □

The operator E_Λ can be written without commuting covariant derivatives. Put $h = g^{\mu\nu}h_{\mu\nu}$ and define

$$C^\rho{}_{\mu\nu} = \tfrac{1}{2}g^{\rho\sigma}(\nabla_\mu h_{\nu\sigma} + \nabla_\nu h_{\mu\sigma} - \nabla_\sigma h_{\mu\nu}), \tag{21.9}$$

$$r_{\mu\nu} = \nabla_\rho C^\rho{}_{\mu\nu} - \nabla_\nu C^\rho{}_{\mu\rho}, \qquad r = g^{\mu\nu}r_{\mu\nu} - 2\Lambda h. \tag{21.10}$$

Since $\mathrm{Ric} = 2\Lambda g$ and $R = 6\Lambda$,

$$(E_\Lambda h)_{\mu\nu} = r_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}r - 2\Lambda h_{\mu\nu}. \tag{21.11}$$

Diffeomorphism covariance gives $E_\Lambda K = 0$. Varying the contracted Bianchi identity at the background solution gives $\nabla^\mu(E_\Lambda h)_{\mu\nu} = 0$.

A constant rescaling checks the curvature terms: for $h_{\mu\nu} = g_{\mu\nu}$, one has $C = 0$, $r = -6\Lambda$ and $E_\Lambda h = \Lambda g$. Equivalently, $G((1+t)g) = G(g)$ as a covariant tensor, while the cosmological term varies by $t\Lambda g$. The kernel of $(u, v) \mapsto h$ consists of Lorentz variations only after imposing the linearized torsion equation. Without that condition, $(0, v)$ is in the kernel for every v.

For the Calabi presentation, use the curvature defect

$$\mathcal{R}_\Lambda(g)_{\rho\mu\sigma\nu} = R_{\rho\mu\sigma\nu} - \Lambda(g_{\rho\sigma}g_{\mu\nu} - g_{\rho\nu}g_{\mu\sigma}).$$

It vanishes at the background, so its variation kills $K\zeta$.

In dimension three, Ricci contraction followed by subtraction of half the scalar trace is an isomorphism from algebraic curvature tensors to symmetric two-tensors. Applied to $\mathcal{R}_\Lambda(g)$, it gives $G + \Lambda g$. Its linearization therefore identifies the corrected Calabi curvature operator with (21.11). This comparison concerns differential complexes on open sets. Boundary conditions and Gaussian measures require additional maps.

21.3 The projective boundary calculation

Consider the matrix connection $\partial_z + a$ on a coordinate interval or formal disc, where

$$a = \begin{pmatrix} 0 & q \\ 1 & 0 \end{pmatrix}.$$

An infinitesimal gauge parameter λ acts by $\delta a = \partial_z\lambda + [a, \lambda]$. Preserving the lower-left entry and the zero diagonal leaves one function ε:

$$\lambda_\varepsilon = \begin{pmatrix} -\varepsilon'/2 & q\varepsilon - \varepsilon''/2 \\ \varepsilon & \varepsilon'/2 \end{pmatrix}, \qquad \delta_\varepsilon q = \varepsilon q' + 2\varepsilon' q - \tfrac{1}{2}\varepsilon'''. \tag{21.12}$$

Primes denote z-derivatives. Indeed the lower-left equation gives the diagonal entries, and the diagonal equation determines the upper-right entry. Substitution gives the displayed variation of q.

Choose the chiral orientation for which the charge variation is

$$\delta Q_\varepsilon = -\frac{k}{2\pi}\int \operatorname{tr}(\lambda_\varepsilon \delta a)\, dz.$$

Since the trace is $\varepsilon\delta q$, the stress density in the convention $Q_\varepsilon = (2\pi)^{-1}\int \varepsilon T\, dz$ is $T = -kq$. Consequently

$$\delta_\varepsilon T = \varepsilon T' + 2\varepsilon' T + \frac{k}{2}\varepsilon'''. \tag{21.13}$$

The Virasoro central term $c\varepsilon'''/12$ therefore has $c = 6k = 3\ell/(2G)$. This is the classical Brown–Henneaux normalization for the specified boundary reduction.[3]

The same law follows from the scalar equation $\psi'' = q\psi$. For $z = f(w)$ and a local square root of f', set $\widetilde{\psi}(w) = (f'(w))^{-1/2}\psi(f(w))$. Two differentiations give

$$\widetilde{q}(w) = (f')^2 q(f(w)) - \tfrac{1}{2}\{f, w\}, \qquad \{f, w\} = \frac{f'''}{f'} - \frac{3}{2}\left(\frac{f''}{f'}\right)^2. \tag{21.14}$$

The Schwarzian chain rule makes this a coordinate-gluing law. It defines a projective connection, whose differences are quadratic differentials. Taking $f(w) = w + t\varepsilon(w)$ recovers (21.12). The solution half-density uses a chosen square root locally. The projective coefficient itself is independent of its sign.

For a holomorphic projective background, the infinitesimal action gives the two-term Dolbeault complex

$$\Omega^{0,\bullet}(X, T_X) \xrightarrow{\varepsilon \mapsto \varepsilon T' + 2\varepsilon' T + (c/12)\varepsilon'''} \Omega^{0,\bullet}(X, K_X^2).$$

A family of mixed boundary operations over these backgrounds also needs coordinate-equivariance for every operation and coherence map. Choose a differential graded Lie model of this action and a compatible action on the relative deformation algebra. Their semidirect product describes the coupled deformation problem. A conformal vector specifies the stress action on its vertex algebra. Compatibility with additional mixed operations is further data.

21.3.1 *The quantum affine parameter*

Let $\kappa \in \mathbb{C} \setminus \{-2\}$ be the level of the ordinary affine $\mathfrak{sl}_2$ vertex algebra with trace pairing B. Its Sugawara central charge is $3\kappa/(\kappa + 2)$. Principal quantum Drinfeld–Sokolov reduction adds the derivative of the Cartan current to the stress tensor. It also adds one fermionic ghost pair of weights zero and one.

The Cartan current has double-pole coefficient $\kappa/2$. Its derivative changes the central charge by -6κ, and the ghost pair contributes -2. Thus the algebraic reduced central charge is

$$c_{\mathrm{DS}}(\kappa) = \frac{3\kappa}{\kappa + 2} - 6\kappa - 2 = 1 - \frac{6(\kappa + 1)^2}{\kappa + 2}. \tag{21.15}$$

The free-field formula makes the normalization explicit. For $b(z)b(w) \sim (z - w)^{-2}$ put

$$T_\nu = \tfrac{1}{2} : b^2 : + \nu\, \partial b, \qquad \nu = \frac{\kappa + 1}{\sqrt{2(\kappa + 2)}}.$$

[3] The boundary constraints, charges and central term are derived in Campoleoni et al., Section 4.2, equations (4.13)–(4.26), https://arxiv.org/abs/1008.4744.

Choose either square root in this expression. Wick contraction gives the identity coefficient $\frac{1}{2} - 6\nu^2$ in the fourth-order pole of $T_\nu(z)T_\nu(w)$. The other contractions give $2T_\nu$ and ∂T_ν in the second- and first-order poles. Hence its central charge is $1 - 12\nu^2$, equal to (21.15).

This is the standard free-field realization of the reduced $\mathfrak{sl}_2$ stress tensor.[4]

The affine parameter κ and the gravitational coefficient k have distinct definitions. Their quantum relation must be derived from the physical current and ghost insertion, its finite terms and its integration convention. Equation (21.13) fixes a classical charge normalization. Equation (21.15) is an identity in the specified algebraic reduction.

21.4 Principal embeddings and holonomy

Let $V_{N-1} = \mathrm{Sym}^{N-1}\mathbb{C}^2$ and let $\mathfrak{sl}_2$ act on $\mathrm{End}(V_{N-1})$ by commutators. The defining action gives the principal embedding in $\mathfrak{sl}_N$. If E, F, H are its standard generators, E^j is a highest-weight vector of weight $2j$ for $0 \le j \le N-1$. Its iterated commutators with F span a module of dimension $2j+1$. The raising and lowering relation is

$$(\mathrm{ad}\,E)(\mathrm{ad}\,F)^r E^j = r(2j - r + 1)(\mathrm{ad}\,F)^{r-1}E^j.$$

The finite-dimensional highest-weight relations give V_{2j}.

These modules have distinct highest weights, and their dimensions sum to N^2. Removing the scalar module gives

$$\mathfrak{sl}_N = \bigoplus_{j=1}^{N-1} V_{2j}, \qquad \sum_{j=1}^{N-1}(2j+1) = N^2 - 1. \tag{21.16}$$

In a supplied $\mathfrak{sl}_N \oplus \mathfrak{sl}_N$ Chern–Simons theory, each summand gives one pair of one-form fields in the Lorentz representation V_{2j}. These are the frame variables of spin $s = j+1$. Their linearized torsion and curvature equations on an AdS background are the spin-s versions of (21.5). Eliminating the spin connection and the Lorentz-type gauge component gives the corresponding Fronsdal field with its trace constraints. This field interpretation uses the invertible background coframe and the supplied action.[5]

Principal Drinfeld–Sokolov boundary reduction has currents of weights $2, \dots, N$, forming the classical W_N algebra. The representation match identifies the possible spin labels. Realizing a specified pair of W_N boundary theories still requires their boundary action, real form, states and map from these bulk fields.

The distinction between a connection and its exponentiated holonomy is already visible for constant matrices. For example, the companion connection

$$a = \begin{pmatrix} 0 & 0 & w_3 \\ 1 & 0 & w_2 \\ 0 & 1 & 0 \end{pmatrix}$$

has characteristic polynomial $\lambda^3 - w_2\lambda - w_3$ and

$$\mathrm{tr}(a^2) = 2w_2, \qquad \mathrm{tr}(a^3) = 3w_3.$$

[4] Arakawa, *Introduction to W-algebras and their representation theory*, Section 5.4 and Example 5.22, https://arxiv.org/abs/1605.00138.

[5] The torsion solution, metric field and gauge transformation are given explicitly in Campoleoni et al., Section 2.2, equations (2.28)–(2.37), https://arxiv.org/abs/1008.4744.

These are polynomial coordinates on the constant companion slice.

Parallel transport for $\partial + a$ around a circle of length L is $\exp(-La)$. For $a = \operatorname{diag}(x, -x)$, replacing x by $x + 2\pi i n/L$ leaves this holonomy unchanged and generally changes $\operatorname{tr}(a^2)$. Thus recovery of the constant charges from holonomy needs a logarithm branch and the relevant gauge identifications. For nonconstant connections it also needs the ordered transport.

A smooth flat connection on a contractible disk has identity holonomy around its boundary in its actual gauge group. A central element is indistinguishable from the identity only after passage to a group quotient that kills it. Thermal regularity, the choice of quotient and the specified state sector therefore enter a higher-spin filling problem.

21.5 Stress tensors from an action

A stress tensor is obtained by varying a supplied matter action. Let $S_{\mathrm{m}}(g, \varphi)$ be invariant under compactly supported diffeomorphisms in the interior of M. Define $T^{\mu\nu}$ and the matter Euler–Lagrange expression $\mathcal{E}_\varphi$ by

$$\delta S_{\mathrm{m}} = \int_M \operatorname{vol}_g \Big(\tfrac{1}{2} T^{\mu\nu} \delta g_{\mu\nu} + \mathcal{E}_\varphi \delta\varphi \Big).$$

This equation fixes the sign convention for T.

For a scalar field, diffeomorphism invariance implies

$$\nabla_\mu T^{\mu\nu} = \mathcal{E}_\varphi \nabla^\nu \varphi. \tag{21.17}$$

Indeed, substitute $\delta g_{\mu\nu} = \nabla_\mu \zeta_\nu + \nabla_\nu \zeta_\mu$ and $\delta\varphi = \zeta^\mu \partial_\mu \varphi$, then integrate the metric term by parts. Since ζ is arbitrary, (21.17) follows. Stress conservation holds on the matter equations of motion. With a boundary, the same variation also determines a boundary flux.

The full metric dependence matters for a coupled action. For the Euclidean scalar kinetic density, a symmetric variation h of the identity metric gives

$$\begin{aligned} \sqrt{\det(I + th)}\,(I + th)^{-1} = I + t\big(\tfrac{1}{2}\operatorname{tr}(h) I - h\big) \\ + t^2 \left[\left(\frac{\operatorname{tr}(h)^2}{8} - \frac{\operatorname{tr}(h^2)}{4} \right) I - \frac{\operatorname{tr}(h)}{2} h + h^2 \right] + O(t^3). \end{aligned} \tag{21.18}$$

This follows by multiplying the determinant and inverse-matrix series. For $h = I$ in dimension three, the result is $(1 + t)^{1/2} I$, with nonzero quadratic coefficient $-I/8$. The linear stress coupling is therefore only the first variation of this action.

There is a precise finite Hamiltonian statement behind the equivariance condition. Let $(P, \{-,-\})$ be a symplectic manifold. Let $\mathfrak{h}$ be a finite-dimensional Lie algebra with basis (t_a) and bracket $[t_a, t_b] = f_{ab}{}^c t_c$. Choose functions μ_a on P.

Adjoin odd ghosts c^a of degree one and odd conjugates b_a of degree minus one, with even Poisson bracket $\{b_a, c^b\} = \delta_a^b$. The functions on P Poisson-commute with these new coordinates. Define the degree-one charge

$$\Omega = c^a \mu_a - \tfrac{1}{2} f_{ab}{}^c c^a c^b b_c.$$

Proposition 21.5. *The charge satisfies* $\{\Omega, \Omega\} = 0$ *precisely when the moment functions are equivariant:*

$$\tfrac{1}{2}\{\Omega, \Omega\} = \tfrac{1}{2} c^a c^b \big(\{\mu_a, \mu_b\} - f_{ab}{}^c \mu_c \big).$$

Proof. The bracket of the two terms linear in c gives $c^a c^b \{\mu_a, \mu_b\}$. The two cross terms give $-f_{ab}{}^c c^a c^b \mu_c$. The term with three ghosts and one conjugate is the Jacobi expression $f_{[ab}{}^d f_{c]d}{}^e$, which vanishes. The independent exterior monomials $c^a c^b$ give the equivalence. □

For example, on $\mathbb{R}^2$ with $\{q, p\} = 1$, the functions $\mu_E = q^2/2$, $\mu_F = -p^2/2$ and $\mu_H = -qp$ satisfy the $\mathfrak{sl}_2$ relations. They give a concrete square-zero charge. The construction starts with a symplectic structure and moment functions. Coordinate covariance by itself supplies neither.

A supplied complete L_∞ action similarly defines a semidirect deformation algebra. If its local realization and dual pairing are given, its shifted cotangent action has the form $S(\alpha, \beta) = \langle \beta, Q(\alpha) \rangle$. The identity $Q^2 = 0$ implies the classical master equation for this cotangent action. Its independent dual fields are part of its definition. Comparison with gravitational BV fields must preserve the action, the BV pairing and the boundary terms on the stated coframe domain.

21.6 Gaussian determinants and their measures

The preceding chain equivalence preserves deformation cohomology. A Gaussian integral additionally uses a quadratic form and a measure. We first compute it in finite dimensions, where these data can all be fixed.

Let $0 \to C^0 \xrightarrow{d_0} C^1 \xrightarrow{d_1} C^2 \xrightarrow{d_2} C^3 \to 0$ be an acyclic finite-dimensional real Hilbert complex. Set

$$\Delta_j = d_{j-1} d_{j-1}^* + d_j^* d_j, \qquad \tau(C) = \prod_{j=0}^{3} (\det \Delta_j)^{(-1)^{j+1} j/2}.$$

Absent differentials are zero. Suppose Hodge duality identifies the spectra of Δ_3 with Δ_0 and of Δ_2 with Δ_1. On $H = (\operatorname{im} d_0)^\perp \subset C^1$, suppose the quadratic operator D is invertible and self-adjoint, with $D^2 = \Delta_1|_H$. These are the finite data corresponding to a gauge-fixed three-dimensional Chern–Simons quadratic form.

Use the Euclidean measures of the given inner products for fields and gauge parameters. Their quotient contributes the orbit Jacobian $(\det \Delta_0)^{1/2}$. For $k \in \mathbb{R}^\times$, define the oscillatory integral by a small positive Gaussian damping, with bosonic measure $(2\pi)^{-\dim H/2} dx$.

Proposition 21.6. *With $r = \dim H$, this normalized quotient integral is*

$$Z_k(C, D) = |k|^{-r/2} \tau(C)^{1/2} \exp\left(\frac{\pi i}{4} \operatorname{sign}(kD) \right). \tag{21.19}$$

Here $\operatorname{sign}(kD)$ *is the number of positive eigenvalues minus the number of negative eigenvalues.*

Proof. Orthogonal decomposition separates the gauge orbit $\operatorname{im} d_0$ from H. The linear map from gauge parameters to the orbit has Jacobian $(\det d_0^* d_0)^{1/2} = (\det \Delta_0)^{1/2}$. Diagonalizing D reduces the remaining integral to the one-dimensional damped Fresnel formula. It gives

$$|k|^{-r/2} (\det \Delta_0)^{1/2} (\det \Delta_1|_H)^{-1/4} \exp\left(\frac{\pi i}{4} \operatorname{sign}(kD) \right).$$

The exact part of C^1 has the nonzero spectrum of Δ_0, so $\det \Delta_1 = (\det \Delta_0)(\det \Delta_1|_H)$. Hodge duality in the definition of τ gives $\tau = (\det \Delta_0)^{3/2} (\det \Delta_1)^{-1/2}$. These identities give (21.19). □

Even acyclic complexes can have different torsions. For $a, b > 0$, take dimensions $(1, 2, 2, 1)$ and differentials

$$d_0 = \begin{pmatrix} a \\ 0 \end{pmatrix}, \qquad d_1 = \begin{pmatrix} 0 & 0 \\ 0 & b \end{pmatrix}, \qquad d_2 = \begin{pmatrix} a & 0 \end{pmatrix}.$$

With the standard inner products, $\tau = a^2/b$. Changing a changes the Gaussian magnitude, although every complex remains acyclic and the determinant-line orientations can be kept fixed. An abstract quasi-isomorphism therefore does not determine the metric torsion or the Gaussian normalization.

For a closed Riemannian three-manifold and an acyclic unitary flat bundle, the analytic counterpart uses zeta determinants of the elliptic de Rham Laplacians. The modulus of the gauge-fixed one-loop expression has the same determinant exponents. Its phase uses an eta invariant and a chosen framing correction.[6]

These analytic hypotheses differ from those of a noncompact gravitational real form on a solid torus with boundary. That problem also requires closed operator domains, a specified integration cycle, residual-mode measures and normalized cut states.

The chain maps in Theorem 21.4 must transport these Gaussian data. The Jacobians of their cancelled pairs also enter the determinant comparison. The source-dependent finite terms and the determinant-line connection must then agree under the permitted local corrections. Equality of the local deformation complexes is the classical part of this comparison.

Algebraic cancellation of curvature has its own precise meaning. Let $(\mathfrak{g}, d, [-,-], \kappa)$ be a curved differential graded Lie algebra with central closed curvature κ of degree two. Adjoin a central generator u of degree one with $du = \kappa$. The extended differential squares to zero, and

$$\kappa + d(-u) + \tfrac{1}{2}[-u, -u] = 0.$$

Twisting by $-u$ therefore gives zero curvature. This algebraic construction does not supply an invertible field theory.

Geometrically, opposite line curvatures make the tensor connection flat. Trivializing it also requires its holonomy and sewing data. For example, the connection $d + ia\,d\theta$ on the trivial line over a circle has zero curvature and holonomy $e^{-2\pi i a}$. At $a = 1/2$ its holonomy is -1. Cancellation by an invertible theory thus requires the complete inverse theory and its compatible boundary maps.

21.7 Virasoro states and analytic continuation

State comparisons can be tested before a full Fock construction. Let $c \in \mathbb{R}$ and let the Virasoro generators satisfy

$$[L_m, L_n] = (m - n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0},$$

and normalize a vacuum v by $L_n v = 0$ for $n \geq -1$ and $\langle v, v \rangle = 1$. The contravariant Hermitian form obeys $L_n^* = L_{-n}$. No positivity at all weights is assumed.

Proposition 21.7. *The forms at weights two and three are $(c/2)$ and $(2c)$. At weight four, in the basis $L_{-4}v$, $L_{-2}^2 v$, the form is*

$$G_4 = \begin{pmatrix} 5c & 3c \\ 3c & c^2/2 + 4c \end{pmatrix}, \qquad \det G_4 = \frac{c^2(5c + 22)}{2}. \tag{21.20}$$

[6] Witten, *Quantum field theory and the Jones polynomial*, Section 2, pages 356–359, derives the gauge-fixed determinants and their phase. The formula here fixes the torsion convention explicitly in (21.19).

For $c > 0$, these displayed forms are positive.

Proof. The first two norms are the vacuum values of $[L_2, L_{-2}]$ and $[L_3, L_{-3}]$. At weight four,

$$L_4 L_{-2}^2 v = 3cv, \qquad L_2 L_{-2}^2 v = (8+c) L_{-2} v, \qquad \|L_{-4} v\|^2 = 5c.$$

Taking their vacuum pairings gives the matrix. Its leading principal minors are positive when $c > 0$. □

Assume $c > 0$, and let X_2, X_4 be independent centered circular complex Gaussian variables with variances $c/2$ and $5c$. In their holomorphic polynomial space, the basis X_4, X_2^2 has Gram matrix $\operatorname{diag}(5c, c^2/2)$. The finite-degree isometry fixing the first vector is

$$L_{-4} v \longmapsto X_4, \qquad L_{-2}^2 v \longmapsto \frac{3}{5} X_4 + \sqrt{1 + \frac{22}{5c}}\, X_2^2. \tag{21.21}$$

The coefficient $3/5$ gives the required cross term $3c$. The remaining squared norm is $c^2/2 + 4c - 9c/5 = (c^2/2)(1 + 22/(5c))$. The assignment sending each Virasoro monomial directly to the corresponding Gaussian monomial misses this cross term. Both displayed forms are positive, and (21.21) corrects that finite assignment. It does not yet intertwine the full Virasoro actions.

A physical unitary preserving the vacuum and conjugation must preserve the Gram forms of its marked stress descendants. Such a comparison uses the actual gauge quotient and real structure of each state space. A holomorphic bilinear covariance alone does not specify its Hermitian norm.

Direct Wick rotation also has a domain. For instance,

$$u(t, \theta) = \sum_{n \geq 1} e^{-\sqrt{n}} e^{-int + in\theta}$$

is smooth for real t, θ, since every polynomially weighted coefficient sequence is summable. At $t = i\tau$ with $\tau > 0$, its Fourier coefficients become $e^{n\tau - \sqrt{n}}$. They exceed every polynomial bound, so they are not the coefficients of a distribution on the circle. For the opposite sign of imaginary time, the opposite frequency sector gives the same obstruction. Thus direct evaluation cannot rotate arbitrary smooth solutions.

By contrast, coefficients bounded by $Ce^{-\sigma n}$ give a holomorphic continuation for $|\operatorname{Im} t| < \sigma$. Finite Fourier sums are included in this analytic domain. A comparison outside it requires another construction, such as a boundary Green operator with specified closed domains. The domain obstruction above does not decide whether such a physical comparison exists.

21.8 The Cardy expression and the BTZ area

Suppose an independently defined boundary theory has the Cardy asymptotic on a specified high-energy domain. Write its cylinder energies as $H_L = L_0 - c_L/24$ and $H_R = \bar{L}_0 - c_R/24$. The leading entropy expression is

$$S_C = 2\pi \sqrt{c_L H_L / 6} + 2\pi \sqrt{c_R H_R / 6}.$$

In a modular derivation, vacuum dominance after the modular transformation and control of the inverse Laplace asymptotic are part of this assumption.

For a nonextremal BTZ geometry, choose $G, \ell > 0$ and $r_+ > |r_-|$. In the gravitational normalization, let

$$\begin{aligned} c_L = c_R = c = \frac{3\ell}{2G}, \qquad M = \frac{r_+^2 + r_-^2}{8G\ell^2}, \qquad J = \frac{r_+ r_-}{4G\ell}, \\ H_L = \frac{\ell M + J}{2} = \frac{(r_+ + r_-)^2}{16G\ell}, \qquad H_R = \frac{\ell M - J}{2} = \frac{(r_+ - r_-)^2}{16G\ell}. \end{aligned} \tag{21.22}$$

These are cylinder charges. The vacuum shift must be retained when comparing them with plane conformal weights.

Proposition 21.8. *Substitution of* (21.22) *into* S_C *gives*

$$S_C = \frac{2\pi r_+}{4G}.$$

Proof. The two terms are $\pi(r_+ + r_-)/(4G)$ and $\pi(r_+ - r_-)/(4G)$. Their sum gives the formula. The inequalities on the radii fix the signs of the square roots. □

The right-hand side is the horizon circumference divided by $4G$. The equality identifies the leading Cardy expression with the area term wherever the assumed asymptotic applies. It does not determine the finite-energy degeneracies or establish the existence of the boundary theory.[7]

21.9 Marked classical lines

The projective matrix connection also gives an explicit classical spin-one-half line. Let q be smooth on a compact interval, set $a(t) = \left(\begin{smallmatrix} 0 & q(t) \\ 1 & 0 \end{smallmatrix}\right)$, and let $U(y,x)$ solve

$$\partial_y U(y,x) = -a(y)U(y,x), \qquad U(x,x) = I. \tag{21.23}$$

For a column solution, its second component satisfies $\psi_2'' = q\psi_2$. The endpoint bases are retained, so U is a matrix rather than its trace.

Proposition 21.9. *The transport has determinant one and satisfies*

$$U(z,x) = U(z,y)U(y,x). \tag{21.24}$$

For an arbitrary coefficient variation,

$$\delta U(y,x) = -\int_x^y U(y,t)\,\delta a(t)\,U(t,x)\,dt. \tag{21.25}$$

For $\delta a = \lambda' + [a,\lambda]$, *this becomes*

$$\delta U(y,x) = -\lambda(y)U(y,x) + U(y,x)\lambda(x). \tag{21.26}$$

Proof. Iterating the integral equation for (21.23) gives the ordered series

$$I + \sum_{n\ge 1}(-1)^n \int_{x<t_1<\cdots<t_n<y} a(t_n)\cdots a(t_1)\,dt_1\cdots dt_n.$$

Its nth norm is at most $(|y-x|\sup\|a\|)^n/n!$, so it converges uniformly. Uniqueness of the initial-value problem gives (21.24). Jacobi's determinant identity gives $\partial_y \det U = -\operatorname{tr}(a)\det U = 0$.

Differentiate the integral equation with respect to the coefficient variation to obtain (21.25). For a gauge variation the integrand is

$$\partial_t\big(U(y,t)\lambda(t)U(t,x)\big).$$

Its integral gives (21.26). □

[7] Strominger, *Black hole entropy from near-horizon microstates*, Sections 3 and 5, keeps the cylinder shift and states the high-energy regime and the existence assumption, https://arxiv.org/abs/hep-th/9712251.

The same identities give a classical ghost differential. For an odd matrix ghost c, set

$$Qa = c' + [a, c], \qquad Qc = -c^2, \qquad QU(y, x) = -c(y)U(y, x) + U(y, x)c(x).$$

The graded Jacobi identity gives $Q^2a = Q^2c = 0$. Using the Leibniz sign for $Q(cU)$ gives $Q^2U = 0$.

Applying Q to (21.24) cancels the two ghost actions at the common endpoint. Thus marked parallel transport is compatible with the classical gauge complex and ordered sewing. The identity matrix is its interval-composition unit. Comparison with a local quantum boundary vacuum requires the corresponding unit action map.

21.10 The physical comparison maps

The classical constructions above leave distinct quantum questions. Their source and target complexes must be defined with their own actions, support conditions, completions and states. The following problem specifies the maps needed to join them.

Problem 21.10. Construct the following comparisons on matched physical domains.

(i) For the sourced solid-torus Gaussian, construct its renormalized determinant or Pfaffian line, connection and phase. Compare them with an independently normalized boundary preparation, preserving source derivatives, disk-cut Green data and cap sewing. The permitted finite bulk, boundary and corner corrections must preserve the declared source identity.

(ii) Construct an insertion from the affine-current and ghost complex into the constrained native observable complex. Require the chain identity $Q_{\mathrm{phys}}\mathcal{I} = \mathcal{I}Q_{\mathrm{DS}}$, including composite current products, their simultaneous cutoff/collision limits, translation and the vacuum action. Derive the relation between the affine level and gravitational coupling from this map and the physical normalization.

(iii) Construct the physical Euclidean cut-state quotient, its Lorentzian comparison and its conjugation. On this quotient, construct both Virasoro actions and the marked stress descendants with their Hermitian forms. Compare the normalized cylinders and the sewing pairing.

(iv) Construct the coincident-junction products of physical cycle operators on a common nonzero state domain. Retain their determinant-derived framing, monodromy and local vacuum action, and determine the unreduced Weyl parameter there. Formal power-series constructions and fixed numerical levels require their own comparison of domains.

(v) Define independently an effective spin-one-half bulk-line complex $\mathscr{L}(y, x)$ and a marked boundary-operation complex $\mathrm{Op}_\partial^{\mathrm{marked}}(y, x)$. Construct a degree-zero comparison

$$\beta_{y,x} : \mathscr{L}(y, x) \longrightarrow \mathrm{Op}_\partial^{\mathrm{marked}}(y, x),$$

satisfying the chain identity and both endpoint identities:

$$\begin{aligned} Q_\partial \beta &= \beta Q_{\mathrm{eff}}, \\ \beta(a_{\mathrm{out}} u) &= \rho_{\mathrm{out}}(a_{\mathrm{out}})\beta(u), \\ \beta(u a_{\mathrm{in}}) &= \beta(u)\rho_{\mathrm{in}}(a_{\mathrm{in}}). \end{aligned}$$

Here ρ_{out}, ρ_{in} are the independently specified endpoint actions. Require these identities and ordered sewing at every formal order. The selected renormalized staple $\mathcal{W}_t^{\mathrm{ren}}$ must be a cocycle at

every positive depth, with its independently normalized quantum Hill field as boundary image. Construct a degree-minus-one depth homotopy $\mathcal{H}_t$ with $\partial_t \mathcal{W}_t^{\mathrm{ren}} = -Q_{\mathrm{eff}} \mathcal{H}_t$ and a compatible marked image.

The construction must retain normal heat profiles, ghosts, antifields and common subgraph restrictions. In particular, construct the quadratic coupling coefficient with both endpoint actions and sewing.

An obstruction to one of these maps must survive the complete admissible counterterm complex on its specified carrier. The matrix, Gaussian and finite-state examples distinguish particular constructions and their domains. They supply exact conditions that a physical comparison must preserve, while its existence remains a separate mathematical question.

Book I

Finite Hamiltonian geometry

Chapter 22

The formal symplectic plane

Formal Hamiltonian jets and matrix probes give trace coordinates and coherent actions. A full center comparison also needs the boundary category, its diagonal bimodule and the supported open–closed map. The formal coordinate theorem supplies one part of this comparison.

22.1 Hamiltonians and the Poisson bracket

Let $V = \mathbb{C}^2$ with coordinates (z_1, z_2) and symplectic form

$$\omega = \mathrm{d}z_1 \wedge \mathrm{d}z_2.$$

Let $\mathfrak{m} = (z_1, z_2)$. The pointed formal Hamiltonian Lie algebra is

$$\mathfrak{h} = \mathfrak{m}^2 \subset \mathbb{C}[\![z_1, z_2]\!], \qquad \{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g.$$

It consists of Hamiltonians whose vector fields fix the chosen Darboux point. The shifted order filtration $F^p\mathfrak{h} = \mathfrak{m}^{p+2}$ is complete and obeys

$$\{F^p\mathfrak{h}, F^q\mathfrak{h}\} \subset F^{p+q}\mathfrak{h}.$$

The affine Hamiltonian algebra $\mathfrak{h}^{\mathrm{aff}} = \mathbb{C}[\![z_1, z_2]\!]/\mathbb{C}$ also contains translations. Its linear sector is adjoined separately when a moving brane or center-of-mass coordinate is required.

Write $R_r = \mathrm{Sym}^r(V^\vee)$ for the homogeneous polynomials of degree r. The bracket obeys

$$\{R_r, R_s\} \subset R_{r+s-2}.$$

For $d \geq 2$ define the finite jet quotient

$$\mathfrak{h}_{\leq d} = \bigoplus_{r=2}^{d} R_r,$$

where every bracket component of degree greater than d is set equal to zero.

Theorem 22.1 (Finite Hamiltonian jet algebra)**.** *The truncated bracket makes $\mathfrak{h}_{\leq d}$ a finite-dimensional Lie algebra. Its dimension is*

$$\dim \mathfrak{h}_{\leq d} = \frac{d(d+3)}{2} - 2.$$

The quadratic part is

$$R_2 \cong \mathfrak{sl}_2,$$

and each R_r is the irreducible $\mathfrak{sl}_2$-module $\mathrm{Sym}^r(\mathbb{C}^2)$.

Proof. The Jacobiator of the polynomial Poisson bracket vanishes before truncation. Projection to degrees at most d preserves this identity because the discarded terms form a Lie ideal. The dimension is

$$\sum_{r=2}^{d}(r+1) = \frac{d(d+3)}{2} - 2.$$

Quadratic Hamiltonians act linearly on V and preserve ω; this identifies R_2 with $\mathfrak{sp}(V) = \mathfrak{sl}_2$. The standard action on homogeneous polynomials gives the final statement. □

22.2 Unimodularity

Theorem 22.2 (Unimodularity of every finite jet)**.** *For every $d \geq 2$ and every $x \in \mathfrak{h}_{\leq d}$,*

$$\operatorname{Tr}_{\mathfrak{h}_{\leq d}}(\operatorname{ad}_x) = 0.$$

Proof. A homogeneous Hamiltonian of degree greater than two raises polynomial degree. Its adjoint action is strictly upper triangular in the decomposition $\bigoplus_{r=2}^{d} R_r$, hence has trace zero. A quadratic Hamiltonian belongs to $\mathfrak{sl}_2$. Its action on each finite-dimensional $\mathfrak{sl}_2$-module R_r has trace zero. Linearity gives the result. □

Corollary 22.3 (Canonical finite BV measure)**.** *The linear odd symplectic space $T^*[-1]\mathfrak{h}_{\leq d}[1]$ has a translation-invariant Berezinian for which the cotangent BF Hamiltonian satisfies the quantum master equation.*

Proof. Choose dual linear coordinates on the shifted cotangent space and use their constant Berezinian. The classical term vanishes by Jacobi, while the BV divergence is the adjoint trace. Unimodularity makes that trace zero. □

Chapter 23

The shifted cotangent theory

23.1 The Lie algebra and its functions

Put

$$\mathfrak{g}_d = T^*[-1]\mathfrak{h}_{\le d} = \mathfrak{h}_{\le d} \ltimes \mathfrak{h}_{\le d}^\vee[-1],$$

where $\mathfrak{h}_{\le d}$ acts on the dual by the coadjoint action and the shifted dual is abelian.

Proposition 23.1 (Chevalley–Poisson identification)**.** *There is a canonical isomorphism of commutative dg algebras*

$$C^*_{\mathrm{CE}}(\mathfrak{g}_d) \cong C^*\big(\mathfrak{h}_{\le d}, \mathrm{Sym}(\mathfrak{h}_{\le d})\big).$$

The differential on the right is the Chevalley differential for the adjoint action on $\mathrm{Sym}(\mathfrak{h}_{\le d})$*. Under the identification* $\mathrm{Sym}(\mathfrak{h}_{\le d}) = \mathcal{O}(\mathfrak{h}^*_{\le d})$*, it is the linear Poisson differential.*

Proof. The dual of $\mathfrak{h}^\vee_{\le d}[-1]$ shifted by $[-1]$ is $\mathfrak{h}_{\le d}$ in degree zero. Chevalley cochains therefore contain polynomial functions on $\mathfrak{h}^*_{\le d}$ and exterior ghosts for $\mathfrak{h}_{\le d}$. The semidirect bracket produces the coadjoint action, which is the Hamiltonian action of the linear Poisson tensor. □

23.2 The BV action

Choose dual bases e_a and e^a. Let c^a be the ghost coordinates and b_a the cotangent coordinates. The finite action is

$$S_d = \frac{1}{2} b_c f^c{}_{ab} c^a c^b.$$

The canonical BV bracket satisfies

$$\frac{1}{2}\{S_d, S_d\} = 0$$

by Jacobi.

Theorem 23.2 (Exact finite quantum master equation)**.** *With the translation-invariant BV Laplacian,*

$$\Delta S_d = \sum_b \mathrm{Tr}(\mathrm{ad}\, e_b) c^b = 0.$$

Consequently

$$\frac{1}{2}\{S_d, S_d\} + \hbar \Delta S_d = 0$$

for every d.

Proof. Contracting one field and one antifield in the cubic Hamiltonian gives the divergence of the adjoint vector field. The preceding unimodularity theorem makes every coefficient vanish. □

Chapter 24

The universal boundary brane

24.1 Polarization

Place the cotangent theory on a half-plane. The standard Lagrangian boundary condition fixes the cotangent field and leaves the $\mathfrak{h}_{\leq d}$ field free. Boundary observables form the Rees enveloping algebra

$$U_\hbar(\mathfrak{h}_{\leq d}) = T(\mathfrak{h}_{\leq d})[\![\hbar]\!]/\langle x\,y - y\,x - \hbar[x,y]\rangle.$$

The Poincaré–Birkhoff–Witt symmetrization gives a filtered vector-space isomorphism

$$\mathrm{sym}_\hbar : \mathrm{Sym}(\mathfrak{h}_{\leq d})[\![\hbar]\!] \xrightarrow{\sim} U_\hbar(\mathfrak{h}_{\leq d}).$$

Theorem 24.1 (Boundary quantization)**.** *The factorization algebra of boundary observables of the finite cotangent theory is locally constant and its value on an interval is $U_\hbar(\mathfrak{h}_{\leq d})$. Fusion of adjacent intervals is multiplication in the enveloping algebra.*

Proof. Canonical quantization turns the linear boundary phase space $\mathfrak{h}^*_{\leq d}$ into the Gutt star algebra. For a linear Poisson bracket the Gutt product is transported by PBW to $U_\hbar(\mathfrak{h}_{\leq d})$. Excision identifies ordered interval fusion with multiplication. □

Book II

Relative Hochschild comparison

Chapter 25

Hochschild cochains of the universal brane

25.1 The coefficient ring and the Chevalley differential

The Rees relation scales both the Lie bracket and its adjoint action. Its Chevalley differential must retain the same parameter. For example, take the quadratic jet $\mathfrak{l} = \mathfrak{sl}_2$ with $[h, e] = 2e$. Modulo $\hbar$, the enveloping algebra is commutative, so the Hochschild differential of e vanishes. The unscaled Chevalley differential instead satisfies $(d_{\mathrm{CE}} e)(h) = 2e$.

Let $\mathfrak{l}$ be a finite-dimensional complex Lie algebra. For $N \geq 0$, put

$$R_N = \mathbb{C}[\hbar]/(\hbar^{N+1}), \qquad \mathfrak{l}_N = R_N \otimes_{\mathbb{C}} \mathfrak{l}, \qquad [x, y]_N = \hbar[x, y],$$

and

$$A_N = U_{R_N}(\mathfrak{l}_N) = T_{R_N}(\mathfrak{l}_N)/(xy - yx - \hbar[x, y]).$$

Here N denotes the order in $\hbar$, independently of a Chan–Paton rank. Write $\bar{A}_N$ for the augmentation ideal and $S = \mathrm{Sym}_{\mathbb{C}} \mathfrak{l}$. PBW symmetrization is the R_N-module isomorphism

$$\mathrm{sym}_N : S \otimes R_N \longrightarrow A_N, \qquad x_1 \cdots x_r \longmapsto \frac{1}{r!} \sum_{\sigma \in S_r} x_{\sigma(1)} \cdots x_{\sigma(r)}.$$

The source has its ordinary symmetric product. Use normalized relative Hochschild cochains

$$C^n_{\mathrm{HH}}(A_N/R_N, A_N) = \mathrm{Hom}_{R_N}(\bar{A}_N^{\otimes_{R_N} n}, A_N)$$

with differential

$$\begin{aligned}(bF)(a_1, \dots, a_{n+1}) = {} & a_1 F(a_2, \dots, a_{n+1}) \\ & + \sum_{i=1}^{n} (-1)^i F(a_1, \dots, a_i a_{i+1}, \dots, a_{n+1}) \\ & + (-1)^{n+1} F(a_1, \dots, a_n) a_{n+1}.\end{aligned}$$

Let d_{CE} act on $C^n(\mathfrak{l}, S) = \mathrm{Hom}_{\mathbb{C}}(\Lambda^n \mathfrak{l}, S)$ by the adjoint Chevalley differential.

Theorem 25.1 (Relative Rees comparison)**.** *Antisymmetrization defines a quasi-isomorphism of R_N-complexes*

$$\mathrm{Alt}_N : C^{\bullet}_{\mathrm{HH}}(A_N/R_N, A_N) \longrightarrow \big(C^{\bullet}(\mathfrak{l}, S) \otimes R_N, \hbar d_{\mathrm{CE}}\big),$$

given by

$$(\mathrm{Alt}_N F)(x_1, \dots, x_n) = \mathrm{sym}_N^{-1} \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) F(x_{\sigma(1)}, \dots, x_{\sigma(n)}).$$

These maps commute with coefficient reduction in N.

Proof. Write $A = A_N$, $R = R_N$, and $A^e = A \otimes_R A^{\mathrm{op}}$. In homological degree n, set

$$K_n = A \otimes_R \Lambda_R^n \mathfrak{l}_N \otimes_R A.$$

Its augmentation is multiplication. For $X = x_1 \wedge \cdots \wedge x_n$, let $X_{\widehat{i}}$ and $X_{\widehat{ij}}$ omit the indicated entries without reordering. Define

$$\begin{aligned}\partial(a \otimes X \otimes b) = {} & \sum_i (-1)^{i+1}\big(a x_i \otimes X_{\widehat{i}} \otimes b - a \otimes X_{\widehat{i}} \otimes x_i b\big) \\ & + \hbar \sum_{i<j} (-1)^{i+j} a \otimes [x_i, x_j] \wedge X_{\widehat{ij}} \otimes b.\end{aligned}$$

The normalized relative bimodule bar has terms $\mathcal{B}_n = A \otimes_R \bar{A}^{\otimes_R n} \otimes_R A$. Define

$$\iota_n(a \otimes X \otimes b) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma)\, a[x_{\sigma(1)}| \cdots |x_{\sigma(n)}]b.$$

The outer bar faces give the first sum in ∂. Pair adjacent transpositions in the internal faces and use $xy - yx = \hbar[x, y]$ to obtain the second sum. For two inputs this reads

$$\begin{aligned}b\big(1[x|y]1 - 1[y|x]1\big) = {} & x[y]1 - y[x]1 + 1[x]y - 1[y]x \\ & - \hbar\, 1[[x, y]]1.\end{aligned}$$

Thus $b\iota = \iota\partial$. PBW makes $\mathfrak{l}_N$ a free summand of $\bar{A}$. Since every factorial is invertible in R, ι is injective. The identity $b^2 = 0$ then gives $\partial^2 = 0$.

Give K_n filtration degree equal to its outer PBW degrees plus n. The associated graded is the diagonal Koszul complex of $\mathrm{Sym}_R(\mathfrak{l}_N)$. Its differential uses differences of left and right polynomial generators. A linear change of variables makes these independent polynomial variables. They form a regular sequence even when R has nilpotents. For an augmented cycle, lift a primitive of its highest-degree symbol and subtract its differential. The filtration degree decreases, so this procedure terminates. Hence $K_\bullet \to A$ is a free bimodule resolution. The relative bar is also free over A^e, with its usual unit-insertion contraction. The map ι lifts the identity of A. Projectivity constructs its chain homotopy inverse degree by degree. Applying $\mathrm{Hom}_{A^e}(-, A)$ gives a cochain homotopy equivalence.

The induced differential on $\mathrm{Hom}_R(\Lambda_R^n \mathfrak{l}_N, A)$ is

$$\begin{aligned}(\delta_N \phi)(x_1, \dots, x_{n+1}) = {} & \sum_i (-1)^{i+1} [x_i, \phi(x_1, \dots, \widehat{x_i}, \dots, x_{n+1})]_A \\ & + \hbar \sum_{i<j} (-1)^{i+j} \phi([x_i, x_j], x_1, \dots, \widehat{x_i}, \dots, \widehat{x_j}, \dots, x_{n+1}).\end{aligned}$$

Commuting x past the factors of a symmetrized monomial gives

$$[x, \mathrm{sym}_N(f)]_A = \hbar\, \mathrm{sym}_N(\mathrm{ad}_x f).$$

Both sums therefore become $\hbar$ times the ordinary Chevalley differential. Precomposition with ι gives Alt_N. All formulas commute with coefficient reduction. □

25.2 The formal coefficient limit

Put $R = \mathbb{C}[\![\hbar]\!]$ and $A = \varprojlim_N A_N$. PBW identifies its underlying module with $S[\![\hbar]\!]$. Each coefficient of $\hbar$ is a polynomial. Define

$$A^{\widehat{\otimes}_R n} = \varprojlim_N A_N^{\otimes_{R_N} n}, \qquad C^\bullet_{\mathrm{HH,cont}}(A/R, A) = \varprojlim_N C^\bullet_{\mathrm{HH}}(A_N/R_N, A_N).$$

Equivalently, these are continuous R-multilinear normalized cochains on the completed relative tensors.

Corollary 25.2 (Formal relative Rees comparison)**.** *The compatible antisymmetrizations induce a quasi-isomorphism*

$$C^{\bullet}_{\mathrm{HH,cont}}(A/R, A) \longrightarrow \big(C^{\bullet}(\mathfrak{l}, S)[\![\hbar]\!], \hbar d_{\mathrm{CE}}\big).$$

Proof. Coefficient reduction is surjective in each cochain degree. For Hochschild cochains, lift outputs on the PBW input basis. The Chevalley exterior input has finite rank, so its outputs also lift. The comparison cones are acyclic and their transition maps are degreewise surjective. The long exact sequence shows that every transition kernel is acyclic. Choose a primitive of a compatible cycle at level zero. Lift it one level and correct its differential by a primitive in the transition kernel. Iteration gives compatible primitives at every order. The inverse-limit cone is therefore acyclic. □

The coefficient ring is part of this result. Absolute $\mathbb{C}$-linear cochains also detect variations of the coefficient parameter. For $\mathfrak{l} = 0$, the continuous absolute derivation $\partial_\hbar$ on R has no relative counterpart. For nonabelian $\mathfrak{l}$, the associated graded for the $\hbar$ filtration has zero differential, whereas d_{CE} need not vanish.

Chapter 26

The Kontsevich–Duflo map

26.1 Linear Poisson formality

For $\mathfrak{l} = \mathfrak{h}_{\leq d}$, put $A_d = U_\hbar(\mathfrak{h}_{\leq d})$ and

$$\mathcal{P}_d = \big(C^*(\mathfrak{h}_{\leq d}, \operatorname{Sym} \mathfrak{h}_{\leq d})[\![\hbar]\!], \hbar d_{\mathrm{CE}}\big).$$

This is the polynomial polyvector complex on $\mathfrak{h}^*_{\leq d}$ with differential $[\hbar\pi_d, -]$ for the linear Poisson tensor π_d. The relative comparison gives $C^\bullet_{\mathrm{HH,cont}}(A_d/R, A_d) \to \mathcal{P}_d$ as a map of complexes. Higher operations require a formality map with specified operadic structure.

Fix Kontsevich's formality morphism $\mathcal{U}$ on the affine space $\mathfrak{h}^*_{\leq d}$. Its twist by $\hbar\pi_d$ defines a star product $\star_d$ and an L_∞ quasi-isomorphism on the degree-shifted complexes. Let $T_d : (\operatorname{Sym} \mathfrak{h}_{\leq d}[\![\hbar]\!], \star_d) \to A_d$ be the algebra identification fixing the linear generators. This identification includes the equivalence between the Kontsevich and PBW products. PBW symmetrization alone identifies the underlying modules. The first Taylor coefficient of the twisted comparison is

$$\Phi_d(v) = (T_d)_* \sum_{r \geq 0} \frac{1}{r!} \mathcal{U}_{r+1}(\underbrace{\hbar\pi_d, \ldots, \hbar\pi_d}_{r}, v).$$

It has source $\mathcal{P}_d$ and target $C^\bullet_{\mathrm{HH,cont}}(A_d/R, A_d)$. At each order in $\hbar$, polynomial inputs have polynomial outputs. Its reduction modulo $\hbar$ is the Hochschild–Kostant–Rosenberg map, with the normalized antisymmetrization convention. That map is a quasi-isomorphism for the polynomial algebra. The complete $\hbar$ filtration then proves that Φ_d is a quasi-isomorphism. The higher Taylor coefficients supply the L_∞ identities after shifting by $[1]$. These statements use the affine formality construction [51, Sections 6 and 8.3].

On invariant polynomials, the standard Duflo normalization gives

$$\mathrm{Duf}_\hbar = \mathrm{sym}_\hbar \circ j^{1/2}(\hbar\partial), \qquad j^{1/2}(x) = \operatorname*{det}^{1/2}_{\mathfrak{h}_{\leq d}}\left(\frac{\sinh(\mathrm{ad}_x/2)}{\mathrm{ad}_x/2}\right).$$

It is an algebra isomorphism from $(\operatorname{Sym} \mathfrak{h}_{\leq d})^{\mathfrak{h}_{\leq d}}[\![\hbar]\!]$ to $Z(A_d)$, by the Duflo theorem [51, Section 8.3.4]. The square root has constant term one. This assertion concerns invariant degree-zero cohomology. It does not identify antisymmetrization with a homotopy-E_2 map.

26.2 The boundary action

The brace operad acts on $(C^\bullet_{\mathrm{HH,cont}}(A_d/R, A_d), A_d)$. Its closed color is the relative derived centre. A homotopy-E_2 refinement means a morphism or zigzag

$$\widetilde{\Phi}_d : \mathcal{P}_d \simeq C^\bullet_{\mathrm{HH,cont}}(A_d/R, A_d)$$

in complete R-linear homotopy-E_2 algebras, with the chosen Gerstenhaber structure on $\mathcal{P}_d$. A compatible Swiss-cheese refinement also specifies the mixed operations on the pair and their coherence homotopies.

Corollary 26.1 (Open–closed structure with a refinement)**.** *Given these refinements, the pair* $(\mathcal{P}_d, A_d)$ *has the transported homotopy Swiss-cheese action. Its open product is the enveloping product. The specified closed and mixed Taylor coefficients determine its closed bracket and boundary action.*

Proof. Transport the two-colored operations through the supplied equivalence of complete operadic algebras. The coherence homotopies are part of this equivalence. □

Chapter 27

The formal Hamiltonian limit

27.1 Weight completion and comparison data

The Hamiltonian Lie algebra and its boundary algebra have quotient systems

$$\mathfrak{h} = \varprojlim_{d} \mathfrak{h}_{\le d}, \qquad \widehat{U}_\hbar(\mathfrak{h}) = \varprojlim_{d} U_\hbar(\mathfrak{h}_{\le d}).$$

A Lie quotient does not by itself define a covariant map between Chevalley cochains with adjoint coefficients. Nor does an algebra quotient define a map between all Hochschild cochains with values in the algebra itself. A cochain must descend through the input quotient and the output quotient. Consequently the coefficient limit of Corollary 25.2 is distinct from this jet limit.

A *compatible jet comparison* consists of the following data. Choose complete R-linear cochain models P_d, H_d with transition maps, levelwise comparisons $P_d \simeq \mathcal{P}_d$ and $H_d \simeq C^\bullet_{\mathrm{HH,cont}}(A_d/R, A_d)$, and compatible quasi-isomorphisms $P_d \to H_d$. Require degreewise surjective transitions, or an explicit derived inverse-limit construction with the same exactness property. Specify identifications

$$\varprojlim_{d} P_d \simeq \mathcal{P}_\infty, \qquad \varprojlim_{d} H_d \simeq C^\bullet_{\mathrm{HH,cont}}(\widehat{U}_\hbar(\mathfrak{h})/R, \widehat{U}_\hbar(\mathfrak{h})),$$

where $\mathcal{P}_\infty$ is the proposed continuous Hamiltonian polyvector complex with differential $[\hbar\pi, -]$. The topology on the right combines the stated Hamiltonian weight completion with the $\hbar$ completion. The identifications must use this same topology. For higher operations, also supply compatible homotopy-E_2 and Swiss-cheese refinements whose operations converge in that topology.

Theorem 27.1 (Continuous comparison from compatible jets)**.** *A compatible jet comparison induces a continuous quasi-isomorphism*

$$\Phi_\infty : \mathcal{P}_\infty \xrightarrow{\sim} C^\bullet_{\mathrm{HH,cont}}(\widehat{U}_\hbar(\mathfrak{h})/R, \widehat{U}_\hbar(\mathfrak{h})).$$

Compatible homotopy-E_2 and Swiss-cheese refinements give the corresponding complete open–closed operations.

Proof. For surjective transitions, the cones of the levelwise maps are acyclic inverse systems with surjective maps in each degree. The primitive-lifting argument of Corollary 25.2 proves exactness of their limit. For a derived inverse limit, use its invariance under levelwise equivalences. Apply the supplied identifications of the two limits. Compatibility and convergence of the operadic maps give the final assertion. □

The finite PBW comparison constructs the relative centre at each fixed jet. A continuous field-theory comparison additionally requires the jet datum and a map from the specified field observables to $\mathcal{P}_\infty$. Universal graph formulas alone do not provide the cochain transition maps.

Principle 27.2 (Universal brane principle). The boundary algebra $\widehat{U}_\hbar(\mathfrak{h})$ is the universal formal Hamiltonian brane. Finite matrix branes are representation-theoretic images of this algebra. Its complete relative derived centre is the algebraic target for bulk observables. A field-theory identification requires the continuous comparison data stated above.

Book III

The finite-jet laboratory

Chapter 28

The first three Hamiltonian jets

28.1 Quadratic Hamiltonians

Set

$$e = \frac{1}{2}z_1^2, \qquad f = -\frac{1}{2}z_2^2, \qquad h = -z_1 z_2.$$

A direct differentiation gives

$$\{h, e\} = 2e, \qquad \{h, f\} = -2f, \qquad \{e, f\} = h.$$

Thus $\mathfrak{h}_{\le 2} = R_2$ is $\mathfrak{sl}_2$ with its standard Chevalley basis. The finite cotangent theory for $d = 2$ is the linear BV theory of $T^*[-1]\mathfrak{sl}_2$ and its boundary algebra is $U_\hbar(\mathfrak{sl}_2)$.

28.2 Cubic Hamiltonians

Let

$$v_j = z_1^{3-j} z_2^j, \qquad 0 \le j \le 3.$$

Then

$$\{h, v_j\} = (3 - 2j)v_j, \qquad \{e, v_j\} = j v_{j-1}, \qquad \{f, v_j\} = (3 - j)v_{j+1},$$

with the boundary terms understood as zero. Hence R_3 is the irreducible highest-weight-three module.

Proposition 28.1 (The cubic jet algebra)**.** *The degree-three jet algebra is the semidirect product*

$$\mathfrak{h}_{\le 3} \cong \mathfrak{sl}_2 \ltimes \mathrm{Sym}^3(\mathbb{C}^2),$$

where the second summand is abelian.

Proof. The bracket of two cubic polynomials has degree four and vanishes in the degree-three quotient. The bracket with a quadratic polynomial is the displayed $\mathfrak{sl}_2$ action. □

28.3 The first nonlinear layer

Write $w_k = z_1^{4-k} z_2^k$, $0 \le k \le 4$. In $\mathfrak{h}_{\le 4}$ the cubic modes satisfy

$$\{v_i, v_j\} = 3(j - i)w_{i+j-1}, \tag{28.1}$$

whenever the index lies between 0 and 4; the coefficient makes every remaining case vanish.

Calculation 28.2 (Two brackets)**.**

$$\{z_1^3, z_2^3\} = 9 z_1^2 z_2^2 = 9w_2, \qquad \{z_1^2 z_2, z_1 z_2^2\} = 3 z_1^2 z_2^2 = 3w_2.$$

The two independent cubic pairs reach the same quartic channel with amplitudes in the ratio $3 : 1$.

Proof. Differentiate each monomial in the two symplectic coordinates and substitute in the Poisson formula. The first pair gives $3z_1^2 \cdot 3z_2^2$. The second gives $2z_1z_2 \cdot 2z_1z_2 - z_1^2 \cdot z_2^2 = 3z_1^2z_2^2$. □

Theorem 28.3 (The quartic jet presentation)**.** *The Lie algebra $\mathfrak{h}_{\leq 4}$ is generated by $e, f, h, v_0, v_1, v_2, v_3$. Its relations are the $\mathfrak{sl}_2$ relations, the irreducible action on the v_j, formula* (28.1)*, and the induced $\mathfrak{sl}_2$ action on R_4. The quartic subspace is abelian in the quotient.*

Proof. Quadratic and cubic generators produce every quartic monomial through (28.1). Brackets involving a quartic and a cubic have degree five, and brackets of two quartics have degree six, so both vanish after projection. Jacobi follows from the polynomial Poisson bracket before projection. □

Chapter 29

Moyal quantization in coordinates

29.1 The bidifferential formula

For polynomials on $\mathbb{C}^2$, the Moyal product is

$$f \star g = \sum_{n \geq 0} \frac{1}{n!} \left(\frac{\hbar}{2} \right)^n \sum_{r=0}^{n} (-1)^r \binom{n}{r} (\partial_1^{n-r} \partial_2^r f)(\partial_1^r \partial_2^{n-r} g). \tag{29.1}$$

Every polynomial pair gives a finite sum.

Theorem 29.1 (Quadratic exactness). *For polynomials f, g of degree at most two,*

$$[f, g]_\star = \hbar \{f, g\}.$$

The quadratic generators therefore define a Lie representation of the Rees-scaled $\mathfrak{sl}_2$. Their associative algebra has an additional Casimir relation.

Proof. The commutator retains the odd powers of the Poisson bidifferential. Derivatives of order three vanish on quadratics, so the first-order term is the entire commutator. □

Proposition 29.2 (The oscillator quotient). *Let $R_N = \mathbb{C}[\hbar]/(\hbar^{N+1})$ and let $W_N = R_N[q, p]$ have the Moyal product for $\{q, p\} = 1$. Let U_N have generators E, F, H with relations*

$$[H, E] = 2\hbar E, \qquad [H, F] = -2\hbar F, \qquad [E, F] = \hbar H.$$

Set $\Omega = EF + FE + H^2/2$. The assignments

$$E \longmapsto e = q^2/2, \qquad F \longmapsto f = -p^2/2, \qquad H \longmapsto h = -qp$$

induce an algebra isomorphism

$$U_N/(\Omega + 3\hbar^2/8) \xrightarrow{\sim} W_N^{(q,p) \mapsto (-q, -p)}.$$

These compatible isomorphisms also hold after $\hbar$-adic completion. If linear Hamiltonians are retained, the generated associative algebra is the full Weyl algebra.

Proof. Quadratic exactness proves the three commutator relations. The associative products are

$$\begin{aligned} e \star f &= -q^2 p^2/4 - \hbar qp/2 - \hbar^2/8, \\ f \star e &= -q^2 p^2/4 + \hbar qp/2 - \hbar^2/8, \\ h \star h &= q^2 p^2 - \hbar^2/4. \end{aligned}$$

Thus Ω maps to $-3\hbar^2/8$. Its commutator with each of E, F, H vanishes by the defining relations. Give these generators PBW degree one. Give an even monomial $q^a p^b$ degree $(a + b)/2$. The leading symbol of $\Omega + 3\hbar^2/8$ is

$C = 2EF + H^2/2$. It is a non-zero-divisor in $R_N[E, F, H]$, since $2C$ is monic in H. Centrality and this property give

$$\operatorname{gr}\big(U_N/(\Omega + 3\hbar^2/8)\big) = R_N[E, F, H]/(C).$$

Indeed, every element of the two-sided ideal is a multiple of the central generator. Its leading symbol is the product of the two leading symbols. The quotient has basis $E^a H^b F^c$ with $a, c \geq 0$ and $b = 0, 1$. Their leading symbols map to unit multiples of the distinct monomials $q^{2a+b} p^{2c+b}$, precisely the even monomials. Hence the associated-graded map is an isomorphism. Lifting highest-degree terms proves surjectivity and comparing them proves injectivity. Every element at fixed N has finite polynomial degree. The formulas commute with reduction in N, so their inverse limit gives the formal isomorphism. Finally q, p generate W_N and topologically generate its formal limit. □

The primitive coproduct on U_N does not descend to this quotient. Modulo $\hbar$, it sends the classical relation to

$$\Delta(C) = C \otimes 1 + 1 \otimes C + 2E \otimes F + 2F \otimes E + H \otimes H.$$

The cross-term is nonzero in the tensor square of $\mathbb{C}[E, F, H]/(C)$. For example, evaluation at $(E, F, H) = (1, 0, 0)$ in the first factor and $(0, 1, 0)$ in the second gives 2. Thus the oscillator quotient is an associative algebra, with no coproduct inherited from these primitive generators.

Calculation 29.3 (Cubic correction)**.** For $f = z_1^3$ and $g = z_2^3$,

$$f \star g - g \star f = 9\hbar z_1^2 z_2^2 + \frac{3}{2}\hbar^3.$$

The first term quantizes the quartic Hamiltonian channel. The constant is the first wheel-free ordering correction visible beyond the quadratic Lie algebra.

Proof. In (29.1), the odd orders are $n = 1$ and $n = 3$. The first contributes $9\hbar z_1^2 z_2^2$. The third contributes $2(\hbar/2)^3(3!) = 3\hbar^3/2$. □

29.2 Weyl generators

With the convention $\{q, p\} = 1$, the symbols satisfy $[q, p]_\star = \hbar$. Their Weyl-ordered differential operators are

$$\widehat{q} = q, \qquad \widehat{p} = -\hbar\partial_q, \qquad [\widehat{q}, \widehat{p}] = \hbar.$$

The abstract Weyl algebra is defined by this commutator relation. Weyl ordering transports its multiplication to the Moyal product. The differential-operator representation realizes these relations over $\mathbb{C}[\hbar]$ and its formal completion. The finite quotient argument above uses the abstract R_N-algebra, so it does not require a faithful differential-operator representation at nilpotent $\hbar$.

Chapter 30

Chevalley coordinates and the Poisson differential

Choose a basis e_a of $\mathfrak{h}_{\leq d}$ with structure constants $[e_a, e_b] = f^c{}_{ab} e_c$. Let c^a be the odd Chevalley coordinate and let x_a be the even coordinate on $\mathfrak{h}^*_{\leq d}$. Then

$$d_{\mathrm{CE}} c^c = -\frac{1}{2} f^c{}_{ab} c^a c^b, \qquad d_{\mathrm{CE}} x_a = f^c{}_{ba} c^b x_c. \tag{30.1}$$

The first equation encodes the Lie bracket. The second is the adjoint action on $\mathrm{Sym}(\mathfrak{h}_{\leq d})$, viewed as functions on the dual.

Proposition 30.1 (Hamiltonian interpretation)**.** *For a polynomial $F(x)$,*

$$d_{\mathrm{CE}} F = c^a \{x_a, F\}_{\mathrm{lin}}.$$

Consequently $d_{\mathrm{CE}}^2 = 0$ is the Jacobi identity for the linear Poisson tensor.

Proof. The linear Poisson bracket is $\{x_a, x_b\} = f^c{}_{ab} x_c$. The Leibniz rule gives (30.1) on every polynomial. The square is one half of the Schouten bracket of the Poisson tensor with itself. □

30.1 The $\mathfrak{sl}_2$ invariant

For the quadratic jet, identify the coordinate functions with e, f, h. The invariant

$$q = 2ef + \frac{1}{2} h^2$$

is d_{CE}-closed. Its formal Duflo image is $\Omega + \hbar^2/2$, where $\Omega = ef + fe + h^2/2$ in the Rees enveloping algebra. This is the degree-zero invariant comparison. Under the oscillator quotient of Proposition 29.2, its image is $\hbar^2/8$.

Chapter 31

Residue duality and the formal current algebra

31.1 The continuous dual

Let

$$\rho_{ab} = z_1^{-a-1} z_2^{-b-1}\, \mathrm{d}z_1 \mathrm{d}z_2, \qquad a, b \geq 0.$$

The Grothendieck residue pairing is

$$\mathrm{Res}_0\Big(z_1^i z_2^j \, \rho_{ab} \Big) = \delta_{ia}\delta_{jb}.$$

The continuous dual of $\mathbb{C}[\![z_1, z_2]\!]$ is the direct sum of the ρ_{ab}. The continuous dual of the pointed algebra $\mathfrak{h} = \mathfrak{m}^2$ is the span of the modes with $a + b \geq 2$. The two modes of total degree one pair with the translation sector of $\mathfrak{h}^{\mathrm{aff}}$.

Proposition 31.1 (Coadjoint current formula)**.** *For $f = z_1^i z_2^j$ and ρ_{ab},*

$$\mathrm{ad}^*_f(\rho_{ab})(g) = -\rho_{ab}(\{f, g\}).$$

In the residue basis this is a finite linear combination of modes of bidegree

$$(a - i + 1, b - j + 1).$$

Proof. The definition is the coadjoint action. The Poisson bracket lowers the total degree of $f\,g$ by two, so residue conservation fixes the displayed bidegree. The Poisson formula contains two derivative terms. □

The shifted cotangent current algebra

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[-1]$$

is therefore a complete topological Lie algebra with a continuous invariant pairing. Its finite jet quotients are precisely the $\mathfrak{g}_d$ used above.

Chapter 32

All-arity open–closed operations

32.1 Hochschild braces

For Hochschild cochains $F \in C^p(A, A)$ and $G_i \in C^{q_i}(A, A)$, the brace

$$F\{G_1, \dots, G_r\}$$

is the signed sum over ordered insertions of the G_i into the inputs of F. The brace identities state that a nested insertion is the sum over all ways of distributing the inner insertions among the outer slots.

Theorem 32.1 (Brace realization with a refinement)**.** *For $A = A_d$, supply the homotopy-E_2 and mixed refinements of Corollary 26.1. They identify the selected homotopy Gerstenhaber operations on $\mathcal{P}_d$ with the relative Hochschild centre operations. The chosen mixed refinement also identifies their action on the boundary algebra.*

Proof. The relative Hochschild complex has its brace operations. The supplied operadic equivalence preserves their coherent homotopies. An equivalence of complexes alone would not imply this conclusion. □

32.2 The first mixed identities

A relative Hochschild 1-cocycle D of an algebra concentrated in degree zero is an R-linear derivation:

$$D(ab) = D(a)b + aD(b).$$

For two such cocycles D, E, their Gerstenhaber bracket acts by

$$[D, E](a) = D(E(a)) - E(D(a)).$$

For a graded algebra and homogeneous derivations, these formulas become

$$D(ab) = D(a)b + (-1)^{|D||a|}aD(b),$$
$$[D, E](a) = D(E(a)) - (-1)^{|D||E|}E(D(a)).$$

Higher Hochschild cochains act through multilinear evaluations and braces. They need not act by derivations or carry a coproduct defining a diagonal action. A compatible mixed refinement transports these operations, including their homotopies.

Corollary 32.2 (Continuous open–closed structure with compatible jets)**.** *Under the jet comparison and mixed refinement hypotheses of Theorem 27.1, the pair $(\mathcal{P}_\infty, \widehat{U}_\hbar(\mathfrak{h}))$ has the corresponding complete homotopy Swiss-cheese action. Its closed color is equivalent to the continuous relative derived centre.*

Proof. The supplied finite operations commute with the transition maps and converge in the specified completion. Their limit acts on the two limit objects. The comparison theorem identifies the closed complex with continuous relative Hochschild cochains. □

Chapter 33

Finite Chan–Paton ranks

33.1 Rank one

For $N = 1$, the commuting equation is automatic. The boundary coordinate algebra is $\mathbb{C}[x, y]$, and

$$J_1(f) = f(x, y).$$

The necklace bracket reduces to the ordinary Poisson bracket. Restriction to $f \in \mathfrak{m}^2$ gives the defining pointed-Hamiltonian representation. The two linear Hamiltonians act by translations and recover the affine center-of-mass extension.

33.2 Rank two and Cayley–Hamilton reduction

For a 2×2 matrix X,

$$X^2 - (\operatorname{Tr} X)X + (\det X)I = 0.$$

Hence every trace containing two consecutive powers of X reduces to traces with X-degree at most one after adjoining $\operatorname{Tr} X$ and $\det X$. The same statement holds for Y. On the commuting locus, simultaneous diagonalization identifies the invariant algebra with the symmetric polynomials in two ordered pairs.

Proposition 33.1 (First finite-rank trace relation)**.** *For commuting 2×2 matrices,*

$$p_{20} = p_{10}^2 - 2 \det X, \qquad p_{02} = p_{01}^2 - 2 \det Y,$$

and

$$p_{11}^2 - p_{20} p_{02} = -\big(x_1 y_2 - x_2 y_1\big)^2.$$

The stable generators remain independent until a relation forced by the rank is reached.

Proof. Write the joint eigenvalues as (x_1, y_1) and (x_2, y_2) and expand both sides. □

33.3 Stable passage

The transition from rank N to rank $N + 1$ adds a zero eigenvalue. Each fixed weighted degree stabilizes once N exceeds that degree. The inverse limit is therefore the completed multisymmetric algebra. The stable trace is an exact quotient of the universal centre on the polynomial trace sector.

Book IV

Global holomorphic symplectic descent

Chapter 34

Fedosov descent

34.1 The Weyl algebroid

All Hochschild cochains here are continuous and relative to $R = \mathbb{C}[\![\hbar]\!]$. Fix a holomorphic symplectic surface (S, ω), its coefficient completion and a quantization period. A global holomorphic torsion-free symplectic connection, when supplied, gives the Weyl resolution of Theorem 16.11.

Its flat sections form a deformation-quantization algebra sheaf. Without such a connection, specify a Weyl algebroid by local algebras, overlap equivalences and coherent inner intertwiners. Chapter 16 gives their gluing equations. Neither presentation alone supplies a formality morphism.

Theorem 34.1 (Descent of supplied formality data)**.** *Suppose the Weyl algebroid and the symplectic polyvector model have effective descent in the chosen complete homotopy-E_2 category. Suppose further that their local formality equivalences have been extended to a homotopy-coherent map of the full descent diagrams. Then they induce a sheafwise equivalence*

$$\Phi_S : (\mathrm{PV}_S[\![\hbar]\!], [\hbar\pi, -]) \xrightarrow{\sim} \mathcal{HH}^\bullet(\mathcal{W}_{S,\hbar}), \qquad \pi = \omega^{-1},$$

where $\mathcal{HH}^\bullet$ is the local diagonal Hochschild model with the specified algebroid descent. Every asserted operation, period marking and coordinate equivariance must be retained by these supplied data.

Proof. Apply Proposition 16.12 in that category. Local equivalences give an equivalence on the descent objects. A change of connection or local presentation gives the same comparison only when a compatible equivalence of the marked descent diagrams is supplied. Varying the quantization period is not automatically such an equivalence. □

Corollary 34.2 (Global center under diagonal comparison)**.** *If in addition the global diagonal comparison is an equivalence*

$$Z_{\mathrm{Mor}}(\mathrm{Perf}(\mathcal{W}_{S,\hbar})) \xrightarrow{\sim} \mathbb{R}\Gamma(S, \mathcal{HH}^\bullet(\mathcal{W}_{S,\hbar})),$$

then the preceding formality data give

$$Z_{\mathrm{Mor}}(\mathrm{Perf}(\mathcal{W}_{S,\hbar})) \simeq \mathbb{R}\Gamma(S, \mathrm{PV}_S[\![\hbar]\!], [\hbar\pi, -]).$$

For a K3 surface these remain the required globalization and diagonal comparison hypotheses. A Hamiltonian BF identification also needs a map from its Chevalley observables to this polyvector model.

Proof. Compose the supplied diagonal equivalence with derived global sections of the sheafwise equivalence. The local Weyl presentation does not itself prove the diagonal comparison. □

Chapter 35

Factorization and descent

35.1 The algebraic mixed-HT realization

A sheaf $\mathcal{P}_S$ of complete homotopy-E_2 algebras assigns such an algebra to each fixed holomorphic open U. In a tensor category admitting the factorization realization, $\mathcal{P}_S(U)$ gives a locally constant factorization algebra on framed topological two-disks. Restriction for $V \subset U$ goes from $\mathcal{P}_S(U)$ to $\mathcal{P}_S(V)$. This contravariance does not provide the covariant holomorphic structure maps of a factorization algebra on $\mathbb{R}^2 \times S$.

Proposition 35.1 (Conditional mixed realization)**.** *Suppose continuous products on disjoint product opens and a descent equivalence on the chosen mixed site extend these topological realizations. Then they define a mixed factorization algebra* $\mathcal{A}^{\mathrm{alg}}_{S,\mathrm{bulk}}$ *on that site. A boundary category* $\mathrm{Perf}(\mathcal{W}_{S,\hbar})$ *and a bulk-to-center equivalence require a boundary action and a comparison with its derived diagonal center. They are not determined by the sheaf of* E_2 *algebras alone.*

Proof. The supplied disjoint-open maps and descent are exactly the factorization structure. Boundary actions have additional operation inputs, and their center comparison must preserve those operations. The underlying local formality map provides only its specified local component. □

35.2 BV realization

The shifted-cotangent Hamiltonian model has local action

$$S_{\mathrm{BF}}(A, B) = \int_{\mathbb{R}^2 \times S} \left\langle B, (\mathrm{d} + \bar{\partial}) A + \frac{1}{2}[A, A] \right\rangle,$$

where the pairing, supports and descent of the fields must make the integral defined. Comparing its classical observables with the algebraic model requires a map from Hamiltonian Chevalley cochains to symplectic polyvectors, compatible with the supported operations. A perturbative quantum comparison further requires compatible gauge fixing and graph weights extending to configuration-space boundaries. It must solve the full quantum master equation and provide an observable-level comparison. The finite jet measure is a finite-dimensional construction and does not supply these analytic or descent data.

Book V

Matrix branes and trace geometry

Chapter 36

The derived commuting stack

Let $X, Y \in \mathfrak{gl}_N$ and let $\psi \in \mathfrak{gl}_N[1]$. The Koszul differential

$$QX = QY = 0, \qquad Q\psi = [X, Y]$$

resolves the commuting equation. The simultaneous conjugation action gives the boundary cdga

$$A_N = C^*_{\mathrm{CE}}\big(\mathfrak{gl}_N, \mathrm{Sym}(\mathfrak{gl}_N \oplus \mathfrak{gl}_N \oplus \mathfrak{gl}_N[1])\big).$$

Proposition 36.1 (Representation of the universal brane)**.** *Substitution of matrices defines a Lie morphism*

$$\rho_N : \mathfrak{h} \longrightarrow \mathrm{Der}(A_N), \qquad f \longmapsto \delta_f,$$

which integrates to an algebra morphism

$$\widehat{U}_\hbar(\mathfrak{h}) \longrightarrow \mathrm{End}(A_N)[\![\hbar]\!], \qquad f \longmapsto \hbar\, \delta_f.$$

Thus the N-brane stack is a representation of the universal boundary algebra.

Proof. For a polynomial Hamiltonian f, cyclic differentiation gives the matrix vector field

$$\delta_f(X) = -\partial_Y^{\mathrm{cyc}} f(X, Y), \qquad \delta_f(Y) = \partial_X^{\mathrm{cyc}} f(X, Y).$$

The cyclic Euler identity implies $\delta_f([X, Y]) = 0$. Hence δ_f commutes with the Koszul differential $Q\psi = [X, Y]$. The same formulas are equivariant for simultaneous conjugation, so δ_f descends to a derivation of the Chevalley–Eilenberg quotient A_N. The trace-moment-map calculation gives

$$[\delta_f, \delta_g] = \delta_{\{f,g\}},$$

first for polynomial Hamiltonians and then continuously for the formal completion. This proves that ρ_N is a continuous Lie morphism. The assignment $f \mapsto \hbar\delta_f$ satisfies

$$[\hbar\delta_f, \hbar\delta_g] = \hbar^2 \delta_{\{f,g\}} = \hbar\,(\hbar\delta_{\{f,g\}}),$$

which is the Rees relation. The universal property of the completed Rees enveloping algebra gives the displayed morphism. It is the universal boundary action evaluated in the N-dimensional Chan–Paton representation. □

Chapter 37

Necklace Hamiltonians

Let

$$F = \mathbb{C}\langle x, y\rangle, \qquad F_{\mathrm{cyc}} = F/[F, F].$$

For a cyclic word f, define cyclic derivatives by cutting the word at every occurrence of the selected letter. Put

$$\delta_f(x) = -\partial_y^{\mathrm{cyc}} f, \qquad \delta_f(y) = \partial_x^{\mathrm{cyc}} f.$$

Theorem 37.1 (Necklace Lie algebra)**.** *The bracket*

$$\{f, g\}_{\mathrm{neck}} = \left[\partial_x^{\mathrm{cyc}} f\ \partial_y^{\mathrm{cyc}} g - \partial_y^{\mathrm{cyc}} f\ \partial_x^{\mathrm{cyc}} g\right]_{\mathrm{cyc}}$$

makes $F_{\mathrm{cyc}}/\mathbb{C}$ a Lie algebra. The assignment $f \mapsto \delta_f$ is a Lie morphism. It preserves the moment map:

$$\delta_f([x, y]) = 0.$$

Proof. The cyclic Euler identity is

$$[x, \partial_x^{\mathrm{cyc}} f] + [y, \partial_y^{\mathrm{cyc}} f] = 0.$$

It gives the moment-map statement. The canonical double Poisson bracket on the free algebra descends to the displayed necklace bracket; its Jacobi identity gives the first assertion. Evaluation on generators gives $[\delta_f, \delta_g] = \delta_{\{f,g\}_{\mathrm{neck}}}$. □

Theorem 37.2 (Trace moment map)**.** *For*

$$J_N(f) = \operatorname{Tr} f(X, Y),$$

one has

$$\{J_N(f), J_N(g)\}_{\mathcal{M}_N} = J_N\big(\{f, g\}_{\mathrm{neck}}\big),$$

where $\mathcal{M}_N = \mathfrak{gl}_N^2$ carries $\operatorname{Tr}(\mathrm{d}X \wedge \mathrm{d}Y)$.

Proof. The differential of a trace is the matrix evaluation of the cyclic derivative. Contracting the two differentials with the inverse trace symplectic form gives the necklace formula. □

Chapter 38

Stable traces

On the semisimple commuting locus write the joint eigenvalues as (x_i, y_i). Define

$$p_{ab}^{(N)} = \sum_{i=1}^{N} x_i^a \, y_i^b.$$

Theorem 38.1 (Stable multisymmetric Poisson algebra)**.** *The inverse stable trace algebra is the completed polynomial algebra*

$$\mathcal{P}_\infty = \mathbb{C}[\![\, p_{ab} \mid a, b \geq 0,\ a + b > 0]\!].$$

Its Poisson bracket is

$$\{p_{ab}, p_{cd}\} = (ad - bc)\, p_{a+c-1,b+d-1}.$$

Proof. The stable fundamental theorem for multisymmetric functions gives polynomial generators p_{ab}. The bracket follows from $\{x_i, y_j\} = \delta_{ij}$ and the Leibniz rule. □

Remark 38.2. The stable trace algebra is a quotient of the universal closed algebra after representation and scalarization. The universal centre retains Hochschild classes, stabilizer directions, and higher central operations that vanish under the trace.

Chapter 39

The two-brane cone

Separate center-of-mass and relative coordinates. Put

$$u = x_1 - x_2, \qquad v = y_1 - y_2, \qquad E = \frac{1}{2}u^2, \quad F = -\frac{1}{2}v^2, \quad H = -uv.$$

Then

$$H^2 + 4EF = 0.$$

Proposition 39.1 (The A_1 Kirillov–Kostant cone)**.** *With $\{u, v\} = 2$,*

$$\{H, E\} = 4E, \qquad \{H, F\} = -4F, \qquad \{E, F\} = 2H.$$

After the evident rescaling this is the $\mathfrak{sl}_2$ Lie–Poisson bracket on the nilpotent cone.

Proof. The three identities follow by direct differentiation. The quadratic equation is the determinant-zero equation for the traceless matrix

$$\begin{pmatrix} H & 2E \\ 2F & -H \end{pmatrix}.$$

□

Book VI

Wheel corrections and exact quantum structure

Chapter 40

The Duflo wheel

First set the coefficient parameter to 1 in the polynomial Rees algebra. For the resulting ordinary $\mathfrak{sl}_2$ enveloping algebra, take generators

$$[h, e] = 2e, \qquad [h, f] = -2f, \qquad [e, f] = h.$$

The invariant polynomial and the quadratic Casimir are

$$q = 2ef + \frac{1}{2}h^2, \qquad \Omega = ef + fe + \frac{1}{2}h^2.$$

Calculation 40.1 (The zero-point shift)**.** The Duflo map gives

$$\mathrm{Duf}(q) = \Omega + \frac{1}{2}.$$

On the irreducible highest-weight-n module,

$$\left(\Omega + \frac{1}{2}\right)\Big|_{V_n} = \frac{(n+1)^2}{2}\,\mathrm{id}\,.$$

Proof. Write $x = ue + vf + wh$. Direct adjoint matrices give $\mathrm{Tr}(\mathrm{ad}_x^2) = 8uv + 8w^2$. Thus the quadratic term of $j^{1/2}(\partial)$ is $(\partial_e\partial_f + \partial_h^2)/6$. Applied to $q = 2ef + h^2/2$, it gives $(2+1)/6 = 1/2$. Higher differential terms annihilate this quadratic polynomial. Symmetrization sends q to Ω. On a highest-weight vector, ef acts by n, fe by zero, and $h^2/2$ by $n^2/2$. Centrality gives the eigenvalue on the entire irreducible module. □

Restoring the Rees parameter gives $\mathrm{Duf}_\hbar(q) = \Omega + \hbar^2/2$. The polynomial calculation permits specialization at 1; arbitrary formal power series do not admit that specialization. Higher wheels assemble into $j^{1/2}$ on higher-degree invariants.

Chapter 41

The local centre with comparison data

Assume a compatible jet comparison and a complete homotopy-E_2 refinement as in Theorem 27.1. Put

$$\mathcal{A} = \widehat{U}_\hbar(\mathfrak{h}), \qquad \mathcal{A}^{\mathrm{alg}}_{\mathrm{bulk}} = \mathrm{Fact}_{\mathsf{E}_2}(\mathcal{P}_\infty).$$

The factorization envelope uses the two oriented topological directions. The formal holomorphic coefficient remains at the Darboux point. Let C_∂ be the category of perfect complete $\mathcal{A}$-modules in the chosen continuous R-linear Morita theory. Assume its centre is computed by $C^\bullet_{\mathrm{HH,cont}}(\mathcal{A}/R, \mathcal{A})$.

Theorem 41.1 (Local derived-centre comparison)**.** *These data give an equivalence of complete homotopy-E_2 factorization algebras*

$$\mathcal{A}^{\mathrm{alg}}_{\mathrm{bulk}} \xrightarrow{\sim} Z^{\mathrm{der}}_{\mathrm{Mor},R}(C_\partial).$$

At the formal insertion point the map is the selected refinement of Φ_∞. Every continuous boundary module has the induced derived centre action.

Proof. The compatible jet theorem gives the complete homotopy-E_2 comparison. The factorization envelope preserves this equivalence. The assumed continuous Morita model identifies its target with the centre. A centre element is a derived natural endomorphism of the identity functor, so evaluation gives the action on each boundary module. □

Corollary 41.2 (Global K3 comparison with descent data)**.** *Let S be a K3 surface with holomorphic symplectic form. Suppose the local comparisons extend to a sheafwise equivalence with the relative Hochschild centre of* $\mathrm{Perf}(\mathcal{W}_{S,\hbar})$*. Suppose further that the selected cyclic and mixed refinements extend and satisfy descent in the same completed category. Then derived global sections give the corresponding global centre, cyclic, and open–closed comparisons.*

Proof. Derived global sections preserve equivalences of sheaves. Apply them to the supplied sheafwise maps and their coherence homotopies. □

Chapter 42

Cyclic classes, stable traces, and Lie homology

A trace of the unit records the representation dimension. For $B = \mathbb{C}$, one has $\mathrm{Tr}_N(1) = N$, so ordinary traces cannot define a rank-compatible map on all of $HC_0(B)$. A scalar-reduced trace removes this dependence after choosing an augmentation.

Let B be a unital associative $\mathbb{C}$-algebra. The affine representation scheme $\mathrm{Rep}_N B$ represents unital algebra maps $B \to \mathrm{Mat}_N(D)$ for commutative $\mathbb{C}$-algebras D. Put

$$\mathcal{R}_N = \mathcal{O}(\mathrm{Rep}_N B)^{GL_N}, \qquad \mathfrak{c}_B = HC_0(B) = B/[B,B],$$

where $[B,B]$ denotes the linear span of commutators. The ordinary trace is the linear map

$$\mathrm{Tr}_N : \mathfrak{c}_B \longrightarrow \mathcal{R}_N, \qquad \mathrm{Tr}_N([b])(\rho) = \mathrm{Tr}(\rho(b)).$$

Cyclicity makes it independent of the representative, and conjugation invariance puts its image in $\mathcal{R}_N$.

For Lie and Poisson compatibility, specify a Lie bracket on $\mathfrak{c}_B$, a Poisson bracket on each $\mathcal{R}_N$, and the identities

$$\{\mathrm{Tr}_N\,\beta, \mathrm{Tr}_N\,\gamma\}_N = \mathrm{Tr}_N[\beta,\gamma]_{\mathrm{cyc}} \qquad (\beta,\gamma \in \mathfrak{c}_B). \tag{42.1}$$

A Hamiltonian comparison also specifies a Lie map $j : \mathfrak{h} \to \mathfrak{c}_B$ for this bracket. These are additional data on B. The preceding necklace bracket and trace moment map supply the bracket and trace data for $B = \mathbb{C}\langle x, y\rangle$ with its canonical symplectic bracket. The map j remains an additional comparison. An arbitrary unital algebra does not carry these structures by definition.

For rank stabilization, choose an augmentation $\epsilon : B \to \mathbb{C}$. It defines the rank-stabilization morphism

$$s_N : \mathrm{Rep}_N B \longrightarrow \mathrm{Rep}_{N+1} B, \qquad \rho \longmapsto \rho \oplus \epsilon.$$

Restriction gives $s_N^* : \mathcal{R}_{N+1} \to \mathcal{R}_N$. Define the reduced cyclic space and trace by

$$\overline{\mathfrak{c}}_B = B/(\mathbb{C}1 + [B,B]), \qquad t_N(\overline{b})(\rho) = \mathrm{Tr}(\rho(b)) - N\epsilon(b).$$

Let $\mathfrak{m}_N \subset \mathcal{R}_N$ be the ideal of functions vanishing at the representation $b \mapsto \epsilon(b) I_N$. The completed invariant rings and stable target are

$$\widehat{\mathcal{R}}_N = \varprojlim_{r \ge 1} \mathcal{R}_N/\mathfrak{m}_N^r, \qquad \mathcal{R}_\infty^\wedge = \varprojlim_N \widehat{\mathcal{R}}_N.$$

Use the inverse-limit topologies. Write J for the augmentation ideal in $\mathrm{Sym}_{\mathbb{C}} \overline{\mathfrak{c}}_B$ and put

$$\widehat{\mathrm{Sym}}\,\overline{\mathfrak{c}}_B = \varprojlim_{r \ge 1} \mathrm{Sym}_{\mathbb{C}} \overline{\mathfrak{c}}_B / J^r.$$

The stable matrix Lie algebra is $\mathfrak{gl}_\infty B = \varinjlim_N \mathfrak{gl}_N B$ under block-zero Lie embeddings. For its homology, use trivial coefficients $\mathbb{C}$. Let $V_B = \bigoplus_{n \ge 1} HC_{n-1}(B)$, with $HC_{n-1}(B)$ placed in homological degree n. The notation $\mathrm{Sym}_{\mathrm{gr}} V_B$ denotes the free graded-commutative algebra, with the Koszul sign rule.

Theorem 42.1 (Cyclic traces and stable Lie homology). *Assume the compatible jet comparison and its homotopy Gerstenhaber refinement. For the augmented algebra and completion above, there are maps*

$$\mathcal{P}_\infty \xrightarrow{\Phi_\infty} C^*_{\mathrm{HH},\mathrm{cont}}(\mathcal{A}/R,\mathcal{A}), \qquad \mathrm{Tr}_N : \mathfrak{c}_B \longrightarrow \mathcal{R}_N,$$

$$\widehat{\mathrm{Sym}}\,\bar{\mathfrak{c}}_B \longrightarrow \mathcal{R}^\wedge_\infty, \qquad \mathrm{Sym}_{\mathrm{gr}} V_B \xrightarrow{\sim} H^{\mathrm{CE}}_*(\mathfrak{gl}_\infty B;\mathbb{C}).$$

The first preserves the selected homotopy Gerstenhaber structure. The second is linear and, under the bracket data (42.1), is a Lie map. Under those data, its composite with j preserves the Hamiltonian bracket. The third is a continuous homomorphism of commutative algebras, and the fourth is an isomorphism of graded Hopf algebras. These maps are natural for morphisms preserving the stated data.

Proof. The first arrow is the chosen comparison of Theorem 27.1. Its operadic refinement gives the asserted Gerstenhaber compatibility. The trace formula above constructs the second arrow. Equation (42.1) supplies its Lie compatibility, and the Lie property of j gives the composite assertion.

The augmentation annihilates commutators. Both terms of t_N vanish on a commutator, and their values on a scalar multiple of the unit cancel. Thus t_N is well defined on $\bar{\mathfrak{c}}_B$. At the trivial representation its value is zero, so its image lies in $\mathfrak{m}_N$. Moreover,

$$\begin{aligned}(s_N^* t_{N+1})(\bar{b})(\rho) &= \mathrm{Tr}(\rho(b)) + \epsilon(b) - (N+1)\epsilon(b)\\ &= t_N(\bar{b})(\rho).\end{aligned}$$

The point $N\epsilon$ maps to $(N+1)\epsilon$, so $s_N^*(\mathfrak{m}^r_{N+1}) \subset \mathfrak{m}^r_N$. Hence restriction defines the transition maps between the completed rings. The symmetric-algebra extension of t_N sends J^r into $\mathfrak{m}^r_N$. It therefore induces a ring homomorphism on each corresponding quotient. Taking the inverse limit in r gives a continuous map to $\widehat{\mathcal{R}}_N$. The displayed stabilization identity makes these maps compatible in N. Their inverse limit is the third arrow.

The diagonal Lie map $\mathfrak{gl}_\infty B \to \mathfrak{gl}_\infty B \oplus \mathfrak{gl}_\infty B$ induces the homology coproduct. Block sum in the opposite direction induces the homology product, using the Künneth isomorphism over $\mathbb{C}$. The Loday–Quillen–Tsygan theorem identifies primitive homology in degree n with $HC_{n-1}(B)$.[1] It identifies the full stable homology with the free graded-commutative Hopf algebra on these primitives, giving the fourth arrow. The trace formulas, the augmentation-preserving rank maps, and stable Lie homology are natural. Naturality of the first arrow requires preservation of the selected jet comparison. □

Scalar reduction need not preserve the supplied Poisson bracket. For example, for $B = \mathbb{C}[x, y]$ with $\{x, y\} = 1$ and $\epsilon(x) = \epsilon(y) = 0$, traces on N points satisfy

$$\{\mathrm{Tr}_N\, x, \mathrm{Tr}_N\, y\} = N, \qquad t_N(\bar{1}) = 0.$$

A reduced Poisson comparison consequently requires additional compatibility, such as vanishing of ϵ on the selected cyclic brackets, or a restriction to a Hamiltonian subalgebra with that property. Only multiplicativity and continuity are asserted for the third arrow. For $B = \mathbb{C}$, the reduced cyclic space is zero and that arrow is the identity on $\mathbb{C}$.

[1] J.-L. Loday and D. Quillen, *Cyclic homology and the Lie algebra homology of matrices*, Commentarii Mathematici Helvetici **59** (1984), 565–591, Theorem 6.2, pp. 584–586.

Chapter 43

Synthesis

At each fixed Hamiltonian jet, the Rees enveloping algebra has relative Hochschild complex equivalent to the Chevalley complex with differential $\hbar d_{\mathrm{CE}}$. PBW antisymmetrization proves this over every R_N and in the formal coefficient limit. The Duflo theorem gives the separate multiplicative comparison on invariant functions. Quadratic Moyal Hamiltonians generate the oscillator quotient of the enveloping algebra. Their associative Casimir relation prevents the primitive coproduct from descending.

The Hamiltonian jet limit requires the compatible cochain systems of Theorem 27.1. With its operadic refinement and continuous Morita model, the centre acts on every supplied continuous boundary module M through

$$\mathcal{P}_\infty \xrightarrow{\Phi_\infty} C^\bullet_{\mathrm{HH,cont}}(\mathcal{A}/R, \mathcal{A}) \longrightarrow \mathrm{RHom}_{\mathcal{A}}(M, M).$$

The last arrow evaluates a derived natural endomorphism at M. A matrix-brane realization must construct such a module and compare its centre action with the selected trace functions. A commuting square with scalar representation invariants requires these additional comparison maps. Likewise, Fedosov descent concerns symplectic polyvectors and a Weyl algebroid. Its relation to Hamiltonian Lie cochains requires the specified map between their carriers.

Book VII

Technical appendices to the preceding book

Chapter 44

Signs in the shifted cotangent complex

Let e_a have degree zero and let $e^a[-1]$ have degree -1. The Chevalley generators dual to them have degrees 1 and 0. Thus

$$C^*_{\mathrm{CE}}(T^*[-1]\mathfrak{l}) = \mathrm{Sym}\big(\mathfrak{l}^\vee[-1] \oplus \mathfrak{l}\big).$$

The degree-zero polynomial factor is $\mathrm{Sym}\,\mathfrak{l} = \mathcal{O}(\mathfrak{l}^*)$. The differential on a polynomial p is

$$(d_{\mathrm{CE}}p)(x) = \mathrm{ad}_x\, p,$$

which is the linear Poisson differential.

Chapter 45

A direct Jacobi check for Hamiltonian jets

For monomials

$$f = z_1^a z_2^b, \qquad g = z_1^c z_2^d,$$

one has

$$\{f, g\} = (ad - bc) z_1^{a+c-1} z_2^{b+d-1}.$$

Substitution of three monomials gives a scalar multiple of

$$\det((a,b),(c,d))\ \det((a+c-1, b+d-1),(e,f)) + \text{cyclic}.$$

Writing the three coefficients in terms of the 2×2 determinants of the exponent vectors shows pairwise cancellation. The coordinate-free proof is $[\pi, \pi]_{\mathrm{Sch}} = 0$ for $\pi = \partial_{z_1} \wedge \partial_{z_2}$.

Chapter 46

Bibliographic note

The finite affine formality map and the separate symplectic globalization use Kontsevich formality and its Fedosov resolution [13, 26, 51]. The relative Hochschild model is the Cartan–Eilenberg comparison, with both Chevalley terms scaled by $\hbar$. Its finite-base resolution proof is Theorem 25.1. Jet limits and operadic refinements require the additional data stated above. The all-arity centre action is the Deligne–Tamarkin brace action [52, 59]. The global deformation-quantization category is the holomorphic Weyl algebroid. The matrix trace and necklace formulas are the noncommutative symplectic moment-map construction.

Book **VIII**

The invariant object

Chapter 47

Two directions of locality

Collision on a complex curve, order on a transverse line, exchange in a plane, and the three corresponding bar constructions.

47.1 The local experiment

Let X be a smooth complex algebraic curve and let $I \subset \mathbb{R}$ be an oriented interval. Place two protected observables at (z_1, t_1) and (z_2, t_2) in $X \times I$. The limits $z_1 \to z_2$ and $t_1 \to t_2$ carry independent information. The first limit produces a singular distribution along the diagonal of X^2. The second composes operators in the order selected by the orientation of I.

Principle 47.1 (Two-direction principle). Holomorphic collision belongs to the factorization coefficient on X. Ordered fusion belongs to an internal E_1 multiplication. Exchange belongs to an additional two-dimensional transport structure.

This separation gives five geometric carriers:

carrier	operation
diagonal in X^2	singular chiral operation
oriented interval	ordered fusion and bar construction
oriented plane	braid transport
circle	Hochschild trace
stable curve	modular sewing.

Each later structure is introduced by the datum carried by its row.

47.2 Carrier and operation

For the diagonal embedding $\Delta : X \hookrightarrow X^2$ and its complement $j : X^2 \setminus \Delta(X) \hookrightarrow X^2$, a Beilinson–Drinfeld chiral Lie operation is a morphism

$$\mu : j_* j^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_! \mathfrak{g}$$

in the chiral pseudo-tensor category of right D-modules [5]. The D-module is the carrier. The morphism μ is the operation. A factorization object packages carriers on all powers X^I together with their disjoint-locus structure maps. The complete operator product is the pair consisting of that system and its multiplication maps.

In a formal coordinate z near the diagonal, the pole filtration of $\mathcal{O}_{X^2}(*\Delta)$ has

$$\operatorname{gr}^F_m \mathcal{O}_{X^2}(*\Delta) \cong \mathcal{O}_\Delta(m\Delta) \cong N_{\Delta/X^2}^{\otimes m}, \qquad m \geq 1.$$

A principal part of order m therefore has an intrinsic normal symbol. Coordinate changes act triangularly on the filtered object. Residues recover the first symbol; normal jets recover the higher symbols.

Chapter 48

The Ran coefficient category

48.1 The fixed six-functor setting

Fix a presentable stable category $\mathcal{D}_X$ of right D-modules on $\mathrm{Ran}(X)$ with the following operations:

(i) the pointwise tensor product $\otimes^!$;

(ii) chiral convolution $\otimes^{\mathrm{ch}}$ along disjoint union;

(iii) base change and projection formulae for the disjoint-union correspondences;

(iv) geometric realizations and totalizations preserved by the tensor products used below.

The standard renormalized D-module setting of chiral Koszul duality has these operations [31]. Every construction below is formed in this fixed setting.

Let

$$\mathrm{Ran}(X)\times\mathrm{Ran}(X) \xleftarrow{\ j\ } (\mathrm{Ran}(X)\times\mathrm{Ran}(X))_{\mathrm{disj}} \xrightarrow{\ u\ } \mathrm{Ran}(X)$$

be the disjoint-union correspondence. Define

$$F\otimes^{\mathrm{ch}} G = u_! j^!(F\boxtimes G).$$

The pointwise product is induced by the diagonal of $\mathrm{Ran}(X)$.

Definition 48.1 (Aligned-decomposition projector)**.** For a finite set I, let $\mathrm{Dec}(I)$ be the finite groupoid of ordered decompositions $I = A\sqcup B$ and let

$$a_I : \mathrm{Dec}(I)\longrightarrow \mathrm{Dec}(I)\times\mathrm{Dec}(I), \qquad d\longmapsto (d,d)$$

be the diagonal. In the linear species model define

$$e_I = a_{I,!}a_I^!.$$

Thus e_I projects a pair of decompositions onto the summands in which the two decompositions agree. In the six-functor Ran model the same formula denotes extraordinary pullback to the diagonal correspondence followed by the available renormalized pushforward.

Theorem 48.2 (Aligned sector in the finite decomposition model)**.** *Let $\mathcal{S}_I$ be a stable category indexed by ordered decompositions of a finite set I, and assume the direct-sum decomposition*

$$\mathcal{M}_I \simeq \bigoplus_{d_1,d_2\in\mathrm{Dec}(I)} \mathcal{M}_{d_1,d_2}.$$

The projection

$$e_I : \mathcal{M}_I\longrightarrow\mathcal{M}_I, \qquad e_I\big|_{\mathcal{M}_{d_1,d_2}} = \begin{cases}\mathrm{id}, & d_1 = d_2,\\ 0, & d_1\neq d_2,\end{cases}$$

is idempotent. Its image is the aligned sector

$$\mathcal{M}_I^{\mathrm{al}} = \bigoplus_{d \in \mathrm{Dec}(I)} \mathcal{M}_{d,d}.$$

The family e_I is natural under bijections and compatible with iterated disjoint union. A mixed distributive law in this model is a natural transformation defined on $\mathcal{M}_I^{\mathrm{al}}$ whose two restrictions to the triple diagonal $d_1 = d_2 = d_3$ agree.

Proof. Idempotence is checked on each direct summand. Naturality follows because a bijection carries an ordered decomposition to an ordered decomposition and preserves equality of the two indices. For three products, both iterated alignments project

$$\bigoplus_{d_1,d_2,d_3} \mathcal{M}_{d_1,d_2,d_3} \quad \text{onto} \quad \bigoplus_{d} \mathcal{M}_{d,d,d}.$$

The two projectors are therefore equal. This equality is precisely the triple-alignment coherence. □

Theorem 48.3 (Global aligned projector)**.** *In the fixed six-functor Ran category, the common-decomposition inclusion is a finite clopen inclusion on every finite-set chart. The functors*

$$e_I = a_{I,!} a_I^!$$

therefore assemble to a natural idempotent e_{al} on the mixed correspondence object. Its image is the common-decomposition correspondence. The two iterated projectors on three decomposition indices coincide, and Beck–Chevalley transports this equality through every Ran transition map.

Proof. The decomposition index is a finite discrete groupoid. Equality of two decomposition indices selects a union of connected components, so the diagonal inclusion is clopen and its extension-by-zero followed by restriction is an idempotent. A surjection of finite sets pulls equal decompositions to equal decompositions, which gives naturality. For three indices, both iterated projectors select the same clopen union $d_1 = d_2 = d_3$. The six-functor base-change isomorphisms identify the corresponding pull–push composites. □

The aligned image carries the canonical interchange obtained by applying the projection formula to one common disjoint-union correspondence. Locality states that transverse multiplication is a morphism for this factorization coproduct. A strong interchange occurs on the full mixed object exactly when every off-aligned summand vanishes.

48.2 Factorization coalgebras

A chiral commutative coalgebra is a commutative coalgebra object for $\otimes^{\mathrm{ch}}$. It is factorizing when every coproduct becomes an equivalence on the relevant disjoint locus. Let

$$\mathrm{Fact}_D(X) \subset \mathrm{CoCAlg}_{\mathrm{ch}}(\mathcal{D}_X)$$

be the full subcategory of factorizing objects. Chiral Koszul duality identifies the corresponding Lie-theoretic and factorization descriptions on the completed Ran category [31, 43].

Proposition 48.4 (Pointwise closure)**.** *The componentwise tensor product of two factorization objects is a factorization object. Thus $\mathrm{Fact}_D(X)$ is symmetric monoidal under $\otimes^!$ whenever the fixed Ran enhancement is closed under that tensor product.*

Proof. For a decomposition $I = I_1 \sqcup I_2$, restrict to its disjointness locus. The factorization equivalences of F and G give

$$\begin{aligned} j^!(F_I \otimes^! G_I) &\simeq j^! F_I \otimes^! j^! G_I \\ &\simeq (F_{I_1} \boxtimes F_{I_2}) \otimes^! (G_{I_1} \boxtimes G_{I_2}) \\ &\simeq (F_{I_1} \otimes^! G_{I_1}) \boxtimes (F_{I_2} \otimes^! G_{I_2}). \end{aligned}$$

Compatibility under refinement and permutation follows componentwise from the corresponding compatibilities for F and G. □

Chapter 49

The two noncommutative chiral objects

49.1 Transverse order

A transversely ordered chiral algebra is an E_1-algebra in the pointwise coefficient category. Its multiplication composes two operators that occupy the same chiral support and are ordered in an independent real direction. This is the structure carried by a boundary or line defect in a holomorphic–topological theory on $X \times \mathbb{R}$.

Definition 49.1 (Transversely ordered chiral algebra)**.** A transversely ordered chiral algebra is an augmented object

$$A^{\perp} \in \mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}(\mathrm{Fact}_D(X), \otimes^{!}).$$

The superscript $\perp$ records the transverse oriented direction. The shorter phrase *ordered chiral algebra* will always mean this object.

49.2 Associativity inside chiral convolution

Chiral convolution combines disjoint finite subsets before allowing them to collide. Associativity for this product is intrinsic to the curve and has a different source and a different bar construction.

Definition 49.2 (Associative chiral algebra)**.** An associative chiral algebra is an augmented algebra

$$A^{\mathrm{ch}}_{\mathsf{Ass}} \in \mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}(\mathcal{D}_X, \star^{\mathrm{ch}}).$$

A chiral Lie algebra is a Lie algebra object

$$\mathfrak{g}^{\mathrm{ch}} \in \mathrm{Alg}_{\mathsf{Lie}}(\mathcal{D}_X, \star^{\mathrm{ch}}).$$

The commutator of an associative chiral algebra is a chiral Lie algebra.

The operation $m^{\perp} : A \otimes^{!} A \to A$ composes transverse intervals. The operation $m^{\mathrm{ch}} : A \star^{\mathrm{ch}} A \to A$ composes chiral configurations. They have different domains, different units, and different deformation complexes.

Theorem 49.3 (Independence of the two associative structures)**.** *Let M be an object for which the split square-zero extensions below exist. A pointwise associative product $\mu_{\perp} : M \otimes^{!} M \to M$ defines a transversely ordered algebra on $\mathbf{1}_{\perp} \oplus M$, while the reduced chiral-convolution product may independently be zero. A chiral-convolution product $\mu_{\mathrm{ch}} : M \star^{\mathrm{ch}} M \to M$ defines an associative chiral algebra on $\mathbf{1}_{\mathrm{ch}} \oplus M$, while the reduced pointwise product may independently be zero. Hence neither associative structure determines the other.*

A common carrier equipped with both products is a mixed object after a mixed distributive datum has been supplied and its unit, binary, and triple coherences have been verified.

Proof. Adjoining the unit to an associative product on the augmentation ideal gives a unital split extension. The zero product is associative in either monoidal category. These two constructions vary independently. On a common carrier, compatibility is additional structure because the mixed composites have different a priori factorizations through the two monoidal products. □

Definition 49.4 (Mixed associative chiral datum)**.** A mixed associative chiral datum is a factorization object A equipped with

(i) a pointwise multiplication $m^{\perp} : A \otimes^{!} A \to A$ and a unit map $\eta_{\perp} : \mathbf{1}_{\perp} \to A$;

(ii) a chiral-convolution multiplication $m^{\mathrm{ch}} : A \star^{\mathrm{ch}} A \to A$ and a unit map $\eta_{\mathrm{ch}} : \mathbf{1}_{\mathrm{ch}} \to A$;

(iii) an aligned mixed distributive law identifying the two composites on $\mathrm{im}(e)$.

The coherence diagrams are the two color-3 faces of the mixed forest complex. A bare symbol $\mathbf{1}$ below denotes the unit of the monoidal category in which the displayed construction is formed.

49.3 Formal-disc realization

Fix a formal coordinate on a disc. Translation-equivariance turns a chiral-convolution product into a state–field operation $Y(a, z)b$ with coefficients in Laurent series. The following theorem packages the exact relationship with the vertex-algebra language.

Theorem 49.5 (Formal-disc realizations)**.** *Let V be a translation-equivariant state–field system with vacuum and injective state–field map.*

(i) *If its fields are weakly associative in the sense of Li, then V is a nonlocal vertex algebra.*

(ii) *If, in addition, V is topologically free over $\mathbb{k}[[\hbar]]$, carries a unitary rational quantum Yang–Baxter operator $S(z)$, and satisfies S-locality and the hexagon compatibility, then V is a quantum vertex algebra.*

(iii) *A translation-equivariant associative chiral-convolution algebra produces such a state–field system whenever its formal-disc restriction satisfies the reconstruction, lower-truncation, and weak-associativity hypotheses. Conversely, a nonlocal or quantum vertex algebra produces the corresponding formal-disc factorization system by its products of fields.*

Proof. The first assertion is Li's reconstruction theorem for quasi-compatible fields [53]. The second is the quantum-vertex reconstruction theorem: unitarity gives inverse exchange, the quantum Yang–Baxter equation compares the two three-field exchange paths, and the hexagon relates exchange to the vertex operation [24, 54]. For the third assertion, the formal-disc chiral product supplies the iterated field products; the stated analytic-formal hypotheses are exactly those used by the reconstruction theorems. The converse assigns to a configuration of formal points the iterated field product and uses weak associativity, or braided associativity, for factorization under refinement. □

49.4 Three separation laws

The complete principal-part filtration, transverse order, and planar exchange form a strict hierarchy of supplied data. The mth normal symbol of a pole is recovered by the mth normal-jet map. An oriented line supplies contractible order chambers and hence E_1 fusion. A plane supplies braid paths and hence exchange. These positive constructions locate each operation in the first geometry that carries it.

Chapter 50

Transversely ordered chiral algebras

Definition 50.1 (Ordered chiral algebra)**.** An ordered chiral algebra on X is an augmented E_1-algebra object

$$A \in \mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}\big(\mathrm{Fact}_D(X), \otimes^!\big).$$

Equivalently, A is a factorization object equipped with multiplication maps

$$m_I : A_I \otimes^! A_I \longrightarrow A_I$$

and a unit, compatible with all diagonal transition maps and all disjoint-locus factorization equivalences.

For every decomposition $I = I_1 \sqcup I_2$, locality is the componentwise identity

$$\Delta_{I_1,I_2} \circ m_I = (m_{I_1} \boxtimes m_{I_2}) \circ \sigma \circ (\Delta_{I_1,I_2} \otimes^! \Delta_{I_1,I_2}),$$

where σ reorders the middle two factors. This equation says that ordered fusion commutes with separation of chiral supports.

Theorem 50.2 (Locality recognition)**.** *To give an ordered chiral algebra is equivalent to giving a factorization family A_I, an associative unital multiplication m_I on every component, and the displayed locality identities, compatibly with refinements, permutations, and diagonal transitions.*

Proof. An E_1-algebra in the symmetric monoidal category $\mathrm{Fact}_D(X)$ is, by definition, an associative multiplication and unit whose structure maps are morphisms in $\mathrm{Fact}_D(X)$. A morphism in $\mathrm{Fact}_D(X)$ is exactly a componentwise morphism compatible with the transition and factorization maps. Expanding this compatibility for a binary decomposition gives the displayed identity; the higher decompositions follow by iteration. □

50.1 Product field theories

Let $\mathrm{Fact}_{\mathrm{hol,lc}}(X^{\mathrm{an}} \times \mathbb{R})$ denote factorization algebras that are holomorphic along X and locally constant along $\mathbb{R}$. Product additivity gives

$$\mathrm{Fact}_{\mathrm{hol,lc}}(X^{\mathrm{an}} \times \mathbb{R}) \simeq \mathrm{Alg}_{\mathsf{E}_1}\big(\mathrm{HolFact}(X^{\mathrm{an}})\big)$$

for product-generated objects. Algebraization by regular holonomic Riemann–Hilbert then gives ordered chiral algebras in $\mathrm{Fact}_D(X)$ whenever the coefficient system is regular holonomic and the comparison preserves the required colimits.

Theorem 50.3 (Holomorphic–topological recognition)**.** *Let $\mathbb{F}$ be a product-generated factorization algebra on $X^{\mathrm{an}} \times \mathbb{R}$ that is holomorphic on X and locally constant on $\mathbb{R}$. Then the restriction of $\mathbb{F}$ to a transverse point is an E_1-algebra in holomorphic factorization algebras on X. A regular-holonomic algebraization produces an ordered chiral algebra on X.*

Proof. An ordered family of disjoint intervals in $\mathbb{R}$ acts on the restriction by factorization. Contractibility of each ordered configuration chamber gives the coherent E_1 structure. Factorization in the X direction is preserved because product embeddings commute. Riemann–Hilbert transports the structure in the regular-holonomic subcategory. $\square$

Chapter 51

Block-ordered operations

The preceding definition is intrinsic. A geometric model arises after one chooses an infinity-operad Ch_X representing the selected chiral factorization theory. The Boardman–Vogt tensor product

$$\mathsf{OrdCh}_X = \mathsf{E}_1 \otimes \mathsf{Ch}_X$$

then represents compatible ordered and chiral operations.

Theorem 51.1 (Operadic exponential law)**.** *Let P and Q be infinity-operads whose Boardman–Vogt tensor product exists, and let $\mathcal{V}$ be a compatible presentable symmetric monoidal infinity-category. Then*

$$\mathrm{Alg}_{P\otimes Q}(\mathcal{V}) \simeq \mathrm{Alg}_P\big(\mathrm{Alg}_Q(\mathcal{V})\big).$$

In particular, after a chiral operad Ch_X has been constructed, the tensor product $\mathsf{E}_1 \otimes \mathsf{Ch}_X$ classifies E_1-algebra objects in Ch_X-algebras.

Proof. The tensor product of infinity-operads is characterized by the displayed universal property [58]. The second statement is the specialization $P = \mathsf{E}_1$ and $Q = \mathsf{Ch}_X$. This formal equivalence applies to each geometric model of Ch_X after that model and its operadic tensor product have been constructed. □

A profile is an ordered list $(I_1, \dots, I_r)$ of finite chiral label sets. Its geometric carrier is $X^{I_1} \times \cdots \times X^{I_r}$. Chiral morphisms refine and collide labels inside one block. Transverse morphisms merge adjacent blocks. The tensor-product operad imposes their mixed squares.

51.1 Mixed forests

Fix a cofibrant dg-operad $Q_X \to \mathsf{Ch}_X$. Apply the functorial W-construction to $\mathsf{E}_1 \otimes Q_X$. A cell is represented by a levelled forest with transverse and chiral internal edges. Put

$$\mathrm{or}(F) = \det(E_\perp(F)) \otimes \det(E_{\mathrm{ch}}(F)).$$

The differential is the sum of the screen differential and one-edge contractions.

Theorem 51.2 (Two-color forest differential)**.** *Let C_* be the chain complex freely generated by oriented two-color forests, with one determinant-line generator for every transverse edge and one for every chiral edge. Suppose contraction of an edge is functorial on the coefficient system and every codimension-two contraction square commutes before orientations are inserted. With the product orientation, the transverse and chiral contraction operators satisfy*

$$d_\perp^2 = 0, \qquad d_{\mathrm{ch}}^2 = 0, \qquad d_\perp d_{\mathrm{ch}} + d_{\mathrm{ch}} d_\perp = 0.$$

Thus $d = d_{\mathrm{int}} + d_\perp + d_{\mathrm{ch}}$ squares to zero whenever the internal differential anticommutes with both contraction operators.

Proof. Every term of $d_{\perp}^{2}$ contracts the same pair of transverse edges in two orders. The underlying coefficient maps agree by functoriality, while exchanging the two transverse determinant generators reverses the sign. The terms cancel in pairs. The same argument proves $d_{\mathrm{ch}}^{2} = 0$. For one edge of each color the coefficient maps again form a commuting square and exchanging the two determinant generators contributes the Koszul sign -1, proving the mixed anticommutator identity. The internal terms vanish by the chain-map assumption. □

This construction supplies a geometric model whenever a concrete Q_X and its screen coefficients are available. The intrinsic definition of ordered chiral algebra remains independent of that choice.

Book IX

Bar, cobar, and the boundary state

Chapter 52

Three bar constructions

The phrase *the chiral bar* denotes three functors once its monoidal product and algebraic operad have been named.

Definition 52.1 (Transverse, associative-chiral, and Lie-chiral bars)**.** For augmented objects of the indicated types define

$$\mathrm{Bar}^{\perp}(A^{\perp}) := \mathbf{1} \otimes^{\mathbb{L},!}_{A^{\perp}} \mathbf{1},$$

$$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A^{\mathrm{ch}}_{\mathsf{Ass}}) := \mathbf{1} \bigstar^{\mathrm{ch},\mathbb{L}}_{A^{\mathrm{ch}}_{\mathsf{Ass}}} \mathbf{1},$$

and

$$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}(\mathfrak{g}^{\mathrm{ch}}) := C^{\mathrm{ch}}_{*}(\mathfrak{g}^{\mathrm{ch}}) = \mathrm{Sym}^{\mathrm{ch}}(\mathfrak{g}^{\mathrm{ch}}[1])$$

with its chiral Chevalley differential. The first bar is formed with $\otimes^{!}$, the second with $\bigstar^{\mathrm{ch}}$, and the third is the Lie-to-commutative-coalgebra Koszul functor.

Theorem 52.2 (Typed bar theorem)**.** *The three bars have the following chain models.*

(i) $\mathrm{Bar}^{\perp}(A^{\perp})$ *is the tensor coalgebra* $T^{c}_{\otimes^{!}}(s\bar{A})$ *with coderivation induced by* $m^{\perp}$*.*

(ii) $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A^{\mathrm{ch}}_{\mathsf{Ass}})$ *is the tensor coalgebra* $T^{c}_{\bigstar^{\mathrm{ch}}}(s\bar{A})$ *with coderivation induced by* m^{ch}*.*

(iii) $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}(\mathfrak{g}^{\mathrm{ch}})$ *is the chiral symmetric coalgebra with differential induced by the chiral Lie bracket.*

Each differential squares to zero by the defining relation of its own operad. The outputs live in different coalgebra categories and are compared by a displayed monoidal or PBW morphism.

Proof. Normalize the two simplicial two-sided bars in their respective monoidal categories. Degeneracies remove units, leaving the two tensor coalgebras, and faces contract adjacent factors using the relevant multiplication. The Lie formula is the ordinary operadic bar of the Lie operad internal to chiral convolution. Associativity and Jacobi are exactly the quadratic identities that make the corresponding coderivations square to zero. □

Definition 52.3 (PBW-admissible chiral Lie algebra)**.** A chiral Lie algebra $\mathfrak{g}^{\mathrm{ch}}$ is PBW-admissible when its chiral enveloping algebra $U^{\mathrm{ch}}(\mathfrak{g})$ has a complete separated PBW filtration, the canonical map

$$\mathrm{Sym}^{\mathrm{ch}}(\mathfrak{g}) \longrightarrow \mathrm{gr}_{\mathrm{PBW}}\, U^{\mathrm{ch}}(\mathfrak{g})$$

is an equivalence, and the bar filtration is strongly convergent in the chosen nilcomplete Ran category.

Theorem 52.4 (Filtered chiral PBW comparison)**.** *Let* $\mathfrak{g}$ *be a chiral Lie algebra in a complete filtered chiral category. Assume:*

(i) *its chiral enveloping algebra* $U^{\mathrm{ch}}(\mathfrak{g})$ *exists;*

(ii) *the PBW filtration is exhaustive, complete, and separated;*

(iii) *the chiral PBW morphism*

$$\mathrm{Sym}^{\mathrm{ch}}(\mathfrak{g}) \longrightarrow \mathrm{gr}\, U^{\mathrm{ch}}(\mathfrak{g})$$

is an isomorphism;

(iv) *the induced filtrations on the two bar complexes are strongly convergent.*

Then antisymmetrization induces a filtered quasi-isomorphism

$$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}\big(U^{\mathrm{ch}}(\mathfrak{g})\big) \simeq \mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}(\mathfrak{g}).$$

Proof. The associated graded of the associative chiral bar is the Koszul bar of $\mathrm{Sym}^{\mathrm{ch}}(\mathfrak{g})$. The usual Koszul antisymmetrization identifies this coalgebra with the chiral Chevalley coalgebra. Hypotheses (2)–(4) give a complete, separated, strongly convergent comparison spectral sequence. Since its associated-graded map is a quasi-isomorphism, the filtered map is a quasi-isomorphism. This is the PBW lane of chiral Koszul duality [5, 31]. □

Definition 52.5 (Bar-admissible mixed datum)**.** A mixed associative chiral datum is bar-admissible when its aligned distributive law acts on the full bisimplicial two-sided bar and both geometric realizations commute with the tensor products occurring in that bisimplicial object.

Theorem 52.6 (Mixed Fubini theorem)**.** *Let A be a mixed algebra in a presentable stable category. Assume the mixed distributive datum extends to the bisimplicial two-sided bar, and that both monoidal products preserve its geometric realizations separately. Then the two iterated realizations and the diagonal realization are canonically equivalent:*

$$\Big|\, p \mapsto \big|q \mapsto B_{p,q}(A)\big|_{\mathrm{ch}}\Big|_{\perp} \simeq \big|\mathrm{diag}\, B_{\bullet,\bullet}(A)\big| \simeq \big|\, q \mapsto \big|p \mapsto B_{p,q}(A)\big|_{\perp}\big|_{\mathrm{ch}}.$$

Proof. The distributive datum makes the two simplicial directions commute, so $B_{\bullet,\bullet}(A)$ is a genuine bisimplicial object. Geometric realization is a colimit. Since the two tensor products preserve the relevant colimits, Fubini for colimits identifies both iterated realizations with the realization of the diagonal simplicial object. □

The notation used in the remainder of this part is

$$\mathrm{Bar}^{\mathrm{ord}}_{X} = \mathrm{Bar}^{\perp}.$$

It always refers to transverse order.

52.1 Realization and factorization

Factorization compares the coefficient at a disjoint configuration with an external tensor product of its coefficients at the separate configurations. This condition is nonlinear. For example, suppose those coefficients are k at one point and at two distinct points, where k is a field. The componentwise sum of two copies has coefficient k^2 at both configurations. Factorization would instead require $k^2 \otimes_k k^2 \cong k^4$ at two points. A componentwise shift also fails: its two-point coefficient is $k[1]$, whereas the tensor product of its singleton coefficients is $k[2]$. Thus an ambient stable coefficient category does not make its factorization objects closed under componentwise sums or shifts.

Geometric realization has a different colimit property. A small indexing ∞-category J is *sifted* when it is nonempty and its diagonal $J \to J \times J$ is cofinal. Here cofinal means that, for every diagram D on $J \times J$, the canonical map $\mathrm{colim}_J(D \circ \Delta) \to \mathrm{colim}_{J\times J} D$ is an equivalence whenever these colimits exist. Iteration gives the same property for $J \to J^m$, for every finite $m \geq 1$. The simplex category with arrows reversed is sifted, so simplicial geometric realization is a sifted colimit.

Proposition 52.7. *Fix coefficient categories on configuration charts, with specified restriction and collision functors and their coherent identifications. Suppose the coefficient categories admit J-indexed colimits and these functors preserve them. Suppose external tensor products preserve colimits separately in each variable. Let $(F_j)_{j\in J}$ be a diagram of factorization objects and morphisms preserving their factorization equivalences. If J is sifted, the componentwise colimit is a factorization object.*

Proof. Write $F_I = \operatorname{colim}_{j\in J}(F_j)_I$ on a chart indexed by I. The collision maps and their coherences pass to the colimit by hypothesis. For a decomposition $I = I_1 \sqcup \cdots \sqcup I_m$, let u denote restriction to its disjointness locus. Restriction, factorization, cofinality, and separate preservation give

$$\begin{aligned} uF_I &\simeq \operatorname*{colim}_{j\in J} u(F_j)_I \\ &\simeq \operatorname*{colim}_{j\in J} \boxtimes_{a=1}^{m} (F_j)_{I_a} \\ &\simeq \operatorname*{colim}_{(j_1,\dots,j_m)\in J^m} \boxtimes_{a=1}^{m} (F_{j_a})_{I_a} \simeq \boxtimes_{a=1}^{m} F_{I_a}. \end{aligned}$$

Each comparison is natural in the diagram and the decomposition. Refinements and permutations therefore preserve the given coherence diagrams. If an empty-configuration unit is specified, its constant diagram also has that unit as colimit, since a sifted category has contractible groupoid completion. □

For a simplicial two-sided bar, this proposition applies when every simplicial term and every face and degeneracy are already in the specified factorization category. It also requires the actual restriction and collision functors to preserve realization. Normalization may then be computed in an ambient stable category. Its individual shifted summands need not be factorization objects. A coproduct on the normalized complex requires its own compatibility with the factorization equivalences.

Chapter 53

The internal ordered bar

Let A be an augmented ordered chiral algebra. Assume the restriction and collision functors preserve geometric realizations, and the external and pointwise tensor products preserve them separately. The two-sided simplicial bar has

$$B_r(\mathbf{1}, A, \mathbf{1}) = \mathbf{1} \otimes^! A^{\otimes^! r} \otimes^! \mathbf{1}.$$

Every face and degeneracy is a factorization morphism by the algebra and augmentation axioms. The sifted-colimit proposition therefore constructs its realization $\mathrm{Bar}_X^{\mathrm{ord}}(A)$ as a factorization object.

Theorem 53.1 (Normalized bar). *Choose a symmetric monoidal dg model for the ambient coefficients in which the displayed tensors compute derived tensors and direct-sum normalization computes realization. Write $\bar{A}$ for the augmentation ideal in that model. The underlying bar complex is*

$$\mathrm{Bar}_X^{\mathrm{ord}}(A) \simeq \big(T^c(s\bar{A}), d_{\mathrm{int}} + b_m\big).$$

Here s decreases cohomological degree by one. The direct sum and its individual shifted summands are formed in the ambient coefficient category. Deconcatenation makes this complex a conilpotent coassociative coalgebra there:

$$\Delta_{\mathrm{dec}}[a_1|\cdots|a_n] = \sum_{i=0}^{n} [a_1|\cdots|a_i] \otimes [a_{i+1}|\cdots|a_n].$$

Proof. Normalization removes words containing a unit and leaves the shifted augmentation ideal in every slot. The faces give the internal differential and adjacent multiplication. The simplicial identities give a square-zero differential. Two successive cuts enumerate the same three subwords in either order. An internal differential or multiplication lies on one side of any surviving cut, with the tensor sign of its prefix. Thus the differential is a coderivation. Every element has finite word support, so sufficiently many reduced cuts give zero. □

53.1 The bar comparison on separated configurations

The normalized word decomposition is not itself a decomposition in $\mathrm{Fact}_D(X)$. Its compatibility with factorization is expressed by a map between whole bar complexes. For augmented dg algebras P, Q in the chosen model, define

$$\mathrm{sh} : BP \otimes BQ \longrightarrow B(P \otimes Q)$$

by shuffling the letters $p_i \otimes 1$ and $1 \otimes q_j$, keeping their respective orders. The sign is the Koszul sign for the permutation of the suspended letters. For degree-zero letters,

$$\mathrm{sh}([p] \otimes [q]) = [p \otimes 1|1 \otimes q] - [1 \otimes q|p \otimes 1].$$

Proposition 53.2 (Shuffle and factorization). *The shuffle is a natural quasi-isomorphism and a coalgebra map, where its source has the tensor-product coalgebra structure. For the geometric coefficient models above,*

suppose these maps commute with the specified restriction and collision functors. Then they give factorization equivalences for the whole bar and make deconcatenation a morphism of factorization objects in the derived category.

Proof. In the bar differential of a shuffle, a merge of letters from different factors occurs twice with opposite suspended signs. The equality $(p \otimes 1)(1 \otimes q) = (-1)^{|p||q|}(1 \otimes q)(p \otimes 1)$ cancels these terms. Merges within one factor give the differential of that factor. Internal differentials commute with the same signed permutation. Thus the shuffle is a cochain map.

A cut of a shuffled word determines one cut in each input word, a shuffle of their prefixes, and a shuffle of their suffixes. Conversely these four choices determine exactly that cut shuffle. The signs on both sides come from the same permutation of suspended letters. This proves the coalgebra identity with the tensor-product coproduct.

To verify the quasi-isomorphism, form the bisimplicial object $B_p(k, P, k) \otimes B_q(k, Q, k)$, with k denoting the tensor unit. Its iterated realization is $BP \otimes BQ$. Its diagonal is the simplicial bar of $P \otimes Q$. Cofinality of the simplicial diagonal identifies these realizations. On normalized chains the map from the iterated realization to the diagonal is the shuffle just defined: the monotone paths from $(0, 0)$ to (p, q) triangulate the product of its two simplices. Each path inserts precisely the degeneracies of its shuffle. Adjacent paths have opposite orientations on a shared face, so their signed sum has exactly the product boundary. This is the realization comparison, hence an equivalence.

For $I = I_1 \sqcup I_2$ on the disjointness locus, the given algebra factorization identifies A_I with $A_{I_1} \boxtimes A_{I_2}$. The shuffle therefore gives the comparison $BA_{I_1} \boxtimes BA_{I_2} \to BA_I$. Its inverse in the derived category is the bar factorization map. The coalgebra identity just proved gives compatibility with deconcatenation. For several blocks, iterated shuffles enumerate the same ordered multishuffles, with the same Koszul signs. This proves refinement coherence. Naturality and the stated functor compatibility give the collision coherence. □

The restriction and collision compatibility in this proposition is a hypothesis on the chosen geometric model. It does not follow from naming an ambient stable category. Conilpotence here refers to finite normalized words in that model. No word-length completion is implicit.

53.2 Three coproducts

The theorem system uses three coproducts with distinct geometric meanings:

$$\Delta_{\mathrm{fact}} : \text{separation of chiral support},$$
$$\Delta_{\mathrm{dec}} : \text{cutting an ordered bar word},$$
$$\Delta_{\mathrm{Hopf}} : \text{fusion of representations}.$$

The first two are present on the transverse bar. The third is additional bialgebra data on an acting algebra.

Proposition 53.3 (Coproduct separation)**.** *Factorization separation is carried by every ordered coefficient. Deconcatenation is generated functorially by the bar construction. A Hopf coproduct is supplied by an independent bialgebra structure. A coalgebra comparison morphism relates the Hopf coproduct to either geometric coproduct whenever a theory identifies their carriers.*

Proof. Factorization separation is part of the coefficient object, and deconcatenation is the canonical cut map on tensor words. A Hopf coproduct is a map from the algebra to a completed tensor square satisfying algebraic coassociativity. Its source, target, and defining universal property differ from the first two, so a relation among them is exactly the data of a comparison morphism. □

53.3 Interval factorization homology

For a right A-module R and a left A-module L, factorization homology of an interval with endpoint labels gives

$$\int_{([0,1],\{0,1\})} (A; R, L) \simeq R \otimes^{\mathbb{L}}_{A} L.$$

This statement holds internally in every presentable monoidal category supporting factorization homology [3, 58]. Taking $R = L = \mathbf{1}$ identifies the interval amplitude with the ordered bar.

Chapter 54

Cobar and twisting cochains

Let C be a coaugmented conilpotent differential graded coalgebra in the ambient coefficient model. The ordinary, direct-sum tensor cobar algebra is

$$\mathrm{Cobar}_X^{\mathrm{ord}}(C) = \big(T(s^{-1}\bar{C}),\, d_{\mathrm{int}} + d_{\Delta}\big).$$

A twisting cochain $\tau : C \to A$ has degree 1, vanishes on the coaugmentation, and has zero composite with the augmentation of A. It satisfies

$$d\tau + \tau d + m(\tau \otimes \tau)\bar{\Delta} = 0.$$

The universal twisting cochains are

$$\tau_A : \mathrm{Bar}_X^{\mathrm{ord}}(A) \to A, \qquad \tau_C : C \to \mathrm{Cobar}_X^{\mathrm{ord}}(C).$$

Theorem 54.1 (Bar–cobar adjunction)**.** *For ordinary augmented dg algebras and coaugmented conilpotent dg coalgebras in the specified coefficient model, tensor cobar is left adjoint to tensor bar. Their mapping set is the set of twisting cochains. An internal adjunction on factorization objects additionally requires both constructions to produce factorization objects and their twisting maps to preserve the geometric data.*

Proof. An algebra map from the tensor algebra $T(s^{-1}\bar{C})$ is determined by its restriction to $s^{-1}\bar{C}$. Compatibility with the cobar differential is the Maurer–Cartan equation above. The same equation describes coalgebra maps from C to the cofree conilpotent coalgebra $T^c(s\bar{A})$. □

Chapter 55

Complete bar–cobar reconstruction

Ordinary tensor cobar uses finite words. Completion changes that carrier. Let k be a commutative ring and $A_{\mathrm{alg}} = \bigoplus_{w \geq 0} A_w$ an augmented algebra in degree zero, with $A_0 = k$, augmentation given by projection onto A_0, and multiplication adding weights. Put $D = \Omega B(A_{\mathrm{alg}})$ using ordinary direct-sum tensors, and let D_w be its weight-w subcomplex. The complete algebra and its complete resolution are

$$\widehat{A} = \prod_{w \geq 0} A_w, \qquad \widehat{D} = \prod_{w \geq 0} D_w.$$

Products of complexes are taken degreewise, with the weight topology.

Theorem 55.1 (Complete reconstruction)**.** *The componentwise counit $\widehat{D} \to \widehat{A}$ is a quasi-isomorphism of complete differential graded algebras.*

Proof. A weight-w block word has at most w positive-weight entries. Filter by their number. Multiplication lowers this number and splitting preserves it. At a fixed list of $N \geq 2$ entries, the splitting complex is the tensor product of the coefficient module with $N - 1$ copies of $[k \xrightarrow{1} k]$ in degrees $-1, 0$. Changing the first outer cut to an inner gap contracts it. The other differential terms cancel by their prefix signs. This identity survives tensoring with any coefficient module. Subtracting a primitive at the largest entry number terminates, so negative cohomology vanishes. In degree zero, the boundaries impose $c(a)c(b) = c(ab)$, leaving exactly A_w. Taking the degreewise product preserves kernels and images of these differentials, and therefore preserves the computed cohomology. □

For $k[x]$ with weight one on x, the ordinary counit recovers $k[x]$, whereas this weight completion recovers $k[[x]]$. A completed geometric bar–cobar equivalence needs a specified completed tensor product, coalgebra category, and comparison maps compatible with collision and inverse limits. An operadic nilcompletion or a boundary double centralizer is not identified with the ordinary tensor counit by this weight calculation. The coalgebra-unit and geometric duality statements require their own convergence and comparison arguments.

55.1 Quadratic recognition

Let $A = T(V)/(R)$ with $R \subset V \otimes V$. Its quadratic dual coalgebra $A^{\mathrm{q!}}$ is cogenerated by sV with corelations $s^2 R$. The canonical map $A^{\mathrm{q!}} \to \mathrm{Bar}(A)$ gives the Koszul complex.

Theorem 55.2 (Quadratic Koszul criterion)**.** *Assume positive weight and degreewise finite duals. The following conditions are equivalent:*

(i) $A^{\mathrm{q!}} \to \mathrm{Bar}(A)$ *is a quasi-isomorphism;*

(ii) $\mathrm{Cobar}(A^{\mathrm{q!}}) \to A$ *is a quasi-isomorphism;*

(iii) *the left and right Koszul complexes have homology concentrated in weight zero;*

(iv) $\mathrm{Tor}_i^A(\mathbf{1}, \mathbf{1})$ *is concentrated in internal weight i.*

Proof. The universal twisting morphism $A^{\mathrm{q}!} \to A$ is Koszul exactly when the associated twisted tensor products resolve the unit. The fundamental theorem of twisting morphisms identifies this condition with the bar and cobar quasi-isomorphisms. Weight purity is the homological form of the same statement [55, 65]. □

Chapter 56

Modules, comodules, and Morita theory

The ordinary tensor counit reconstructs an algebra. Reconstruction of its modules requires the unit and counit of the twisted module functors as well. The following weight conditions make these maps finite in every component.

Let k be a field and let $A_{\mathrm{alg}} = \bigoplus_{w \geq 0} A_w$ be an associative algebra in cohomological degree zero. Assume $A_0 = k$, multiplication adds weights, and augmentation projects onto A_0. Put $C_{\mathrm{alg}} = B(A_{\mathrm{alg}})$, with its bar weight grading, and set

$$A = \prod_{w \geq 0} A_w, \qquad C = \prod_{w \geq 0} (C_{\mathrm{alg}})_w.$$

Products are degreewise in complexes. The universal twisting map $\tau : C \to A$ has degree one, sends $[a]$ to a, and vanishes on the empty word and all longer words. The bar differential has the earlier convention $b_2[a|b] = [ab]$.

A complete weight-graded left A-module here means a product $M = \prod_{w \geq w_0} M_w$ of complexes, for some integer w_0, with differential preserving weight and actions $A_u \otimes M_v \to M_{u+v}$. A complete weight-graded left C-comodule means such a product N with a counital coassociative coaction $\rho : N \to C \mathbin{\widehat{\otimes}} N$ preserving weight and differential. The completed tensor product is defined componentwise by

$$(C \mathbin{\widehat{\otimes}} N)_w = \bigoplus_{u+v=w} (C_{\mathrm{alg}})_u \otimes N_v.$$

This sum is finite since $u \geq 0$ and $v \geq w_0$. Each reduced coaction lowers the weight of its final N-component. It therefore vanishes after more than $w - w_0$ iterations on N_w. These comodules need no chosen coaugmentation.

Write $D_{\mathrm{wt}}(A)$ and $D_{\mathrm{wt}}(C)$ for the categories obtained by inverting maps that are quasi-isomorphisms in every weight, respectively for these modules and comodules. This specifies the weak equivalences; no coderived localization is implicit. Define

$$F(M) = C \mathbin{\widehat{\otimes}}_\tau M, \qquad G(N) = A \mathbin{\widehat{\otimes}}_\tau N.$$

The first has the coaction induced by Δ_C and the second the left A-action. Writing $\Delta(c) = \sum c_{(1)} \otimes c_{(2)}$ and $\rho(n) = \sum n_{(-1)} \otimes n_{(0)}$, their differentials are

$$d_F(c \otimes m) = d_C c \otimes m + (-1)^{|c|} c \otimes d_M m + \sum (-1)^{|c_{(1)}|} c_{(1)} \otimes \tau(c_{(2)}) m,$$
$$d_G(a \otimes n) = a \otimes d_N n - \sum a\,\tau(n_{(-1)}) \otimes n_{(0)}.$$

All sums are finite in each weight. The twisting equation, associativity, and coassociativity make both differentials square to zero.

Theorem 56.1 (Koszul module equivalence in positive weights)**.** *These functors induce inverse equivalences*

$$D_{\mathrm{wt}}(A) \simeq D_{\mathrm{wt}}(C).$$

Their counit and unit are

$$\epsilon_M(a \otimes c \otimes m) = a\epsilon_C(c)m, \qquad \eta_N(n) = \sum n_{(-1)} \otimes 1 \otimes n_{(0)}.$$

Proof. The differential formulas show that these are cochain maps, respectively A-linear and C-colinear. The twisting terms cancel in the unit equation; the counit equation uses the module action. Coassociativity and the counit identities give the two triangle identities.

The complex $GF(M)$ is the two-sided normalized bar resolution $A\widehat{\otimes}B(A_{\mathrm{alg}})\widehat{\otimes}M$. For its contraction, use the simplicial boundary convention $\partial = \sum_i (-1)^i d_i$. Multiplying a bar word of length r by $(-1)^r$ translates this convention to the displayed twisting differentials. Writing $\bar{a} = a - \epsilon_A(a)1$, the extra degeneracy is

$$h\big(a_0[a_1|\cdots|a_r]m\big) = 1[\bar{a}_0|a_1|\cdots|a_r]m.$$

The first face returns the original term minus its scalar first factor. The other faces cancel against applying h after the boundary, including the module-action face. The internal differential of M cancels by the suspension sign. Thus $\partial h + h\partial = 1 - i\epsilon_M$, with $i(m) = 1[\,]m$. This is a linear contraction; i need not be A-linear. It preserves total weight, so the counit is a quasi-isomorphism in every weight.

At fixed total weight, filter $FG(N)$ by the weight of its final N-factor. This filtration is finite. The reduced-coaction term lowers that weight and disappears in the associated graded. The associated graded unit is therefore the inclusion into $B(k, A_{\mathrm{alg}}, A_{\mathrm{alg}}) \otimes N_v$. The rightmost extra degeneracy contracts this augmented bar onto k:

$$h\big([a_1|\cdots|a_r]a\big) = (-1)^{r+1}[a_1|\cdots|a_r|\bar{a}]1$$

in the same simplicial convention. The last face gives $1 - i\epsilon$; all other faces cancel in pairs. Consequently the unit is a quasi-isomorphism on each associated graded component, and the finite filtration proves the same for each weight of N.

Both functors preserve weightwise quasi-isomorphisms. Filter by the weight of the final module or comodule component. The twisting terms strictly change that weight in one direction. On associated graded components, the functors tensor complexes of k-vector spaces, which preserves quasi-isomorphisms. The filtration is finite at fixed total weight. Hence F, G descend to the stated localizations, where their unit and counit are isomorphisms. □

This theorem uses an explicit weight decomposition and a lower weight bound on every module and comodule. Derived I-adic completion, with $I = \ker(A \to k)$, instead uses $M \to \varprojlim_p (A/I^p \otimes^{\mathbb{L}}_A M)$. The two completions require a comparison of their filtrations. A coderived category also has a separately specified class of coacyclic complexes. Neither identification follows from the algebra counit alone. For right and left modules, the two-sided bar computes their relative derived tensor product in the same finite weight components. An identification with a geometric derived cotensor product additionally requires the two-sided twisting comparison and its compatibility with the chosen completion and collision maps.

Let $\mathcal{C}$ be a presentable factorization category and let b be a compact factorization generator. The internal endomorphism object

$$A_b = \mathrm{RHom}_{\mathcal{C}}(b, b)$$

has ordered multiplication by composition.

Theorem 56.2 (Factorization Morita theorem)**.** *Assume b generates each local category and the local Morita equivalences commute with factorization descent. Then*

$$\mathcal{C} \simeq \mathrm{Mod}_{A_b}(\mathrm{Fact}_D(X)).$$

Proof. The compact-generator Morita theorem applies on every factorization chart. The descent hypothesis identifies the equivalences on intersections and hence on the Ran colimit. □

Chapter 57

Derived centre and bulk action

The ordered Hochschild cochains are

$$Z_{\mathrm{ord}}(A) = \mathrm{RHom}_{A \otimes A^{\mathrm{op}}}(A, A).$$

Theorem 57.1 (Higher Deligne centre)**.** *Let the ambient coefficient category be presentable, stable, symmetric monoidal, closed, and let tensor product preserve geometric realizations separately. Then $Z_{\mathrm{ord}}(A)$ carries a canonical E_2-algebra structure. It is the universal E_2 algebra acting centrally on A.*

Proof. Hochschild cochains carry the brace operations, and the brace operad is equivalent to chains on little two-disks. This is the Deligne theorem [52, 59]. Universality is the identification of Hochschild cochains with endomorphisms of the identity A-bimodule. □

A boundary condition in a field theory gives a morphism

$$\beta : \mathrm{Obs}_{\mathrm{bulk}}|_{\partial} \longrightarrow Z_{\mathrm{ord}}(A_{\partial}).$$

Locality makes the action central. A boundary condition is open–closed complete on a protected sector when β is an equivalence on that sector.

For a Calabi–Yau category, Hochschild cochains carry the framed E_2 refinement of the cyclic Deligne theorem [10].

Chapter 58

Cyclic closure and modular sewing

A cyclic trace becomes a surface theory after dualizability and nondegeneracy are supplied.

Definition 58.1 (Perfect Calabi–Yau ordered algebra)**.** A perfect Calabi–Yau ordered algebra is a dualizable ordered algebra A together with a trace

$$\mathrm{tr} : \mathrm{HH}_0(A) \longrightarrow \mathbf{1}[-d]$$

whose induced pairing $A \otimes A \to \mathbf{1}[-d]$ is nondegenerate.

Theorem 58.2 (Ribbon-graph realization of a Calabi–Yau algebra)**.** *Let A be a homologically finite, smooth and proper cyclic A_∞ algebra with a nondegenerate pairing of degree d. Then the cyclic bar construction of A defines an open topological conformal field theory. Its operations are indexed by chains on moduli spaces of bordered surfaces, equivalently by the corresponding ribbon-graph complexes. The value on the circle is Hochschild homology* $\mathrm{HH}_*(A)$.

Proof. The nondegenerate cyclic pairing identifies outgoing and incoming boundary labels and contracts the two flags of every internal ribbon edge. The A_∞ identities are exactly the boundary identities for metric ribbon graphs. Smoothness and properness give the dualizability and finite pairing required for all contractions. Costello's construction then produces the open TCFT and identifies the closed state complex with Hochschild chains [17]. □

Remark 58.3. The trace functional supplies the annular operation. Smoothness, properness, and the nondegenerate Calabi–Yau pairing extend it to the complete ribbon-surface theory of the theorem.

Stable-curve sewing is supplied by a modular operad action. Its contractions form the modular structure carried by the chiral coefficient. The two sewing geometries coexist when their contraction maps supercommute.

Book X

Exact noncommutative models

Chapter 59

Free and square-zero pairs

Let $M \in \mathrm{Fact}_D(X)$.

$$T_X(M) = \mathbf{1} \oplus M \oplus M^{\otimes 2} \oplus \cdots$$

is the free augmented ordered algebra.

Theorem 59.1 (Free–square-zero duality)**.** *Assume exact tensor products, the indicated direct sums, and dualizable M. Interpret the bar weightwise, and assume that its dual computes the derived augmentation endomorphisms. Then*

$$\mathrm{Bar}(T_X(M)) \simeq \mathbf{1} \oplus sM, \qquad \mathrm{RHom}_{T_X(M)}(\mathbf{1}, \mathbf{1}) \simeq \mathbf{1} \oplus M^\vee[-1]$$

with square-zero multiplication on the reduced summand. Conversely, for $A = \mathbf{1} \oplus M$ with $M^2 = 0$,

$$\mathrm{Bar}(A) = T^c(sM), \qquad A^!_n \simeq (M^\vee[-1])^{\otimes n}.$$

The direct sum of these weights is the graded tensor algebra. The product over weights is its completed tensor algebra.

Proof. Move suspension symbols past homogeneous factors with the Koszul sign. At fixed tensor weight m, the bar identifies with $M^{\otimes m}$ tensored with the signed complex of subsets of its $m-1$ internal gaps. The differential deletes the ith selected gap with sign $(-1)^{i-1}$. For $m \geq 2$, insert the first gap when absent and use zero when it is present. This map h satisfies $dh + hd = 1$: deletion of the first gap gives the identity and all other deletions cancel in pairs.

The contraction on the cut complex extends by the identity on $M^{\otimes m}$ and anticommutes with the internal differential. The surviving weights zero and one form the primitive subcoalgebra $\mathbf{1} \oplus sM$. This contraction is natural in tensor powers, so it applies with the stated categorical hypotheses. Dualization gives square-zero positive multiplication. For $M^2 = 0$, every bar differential term vanishes, and dual deconcatenation is concatenation in the indicated weight convention. Proposition 19.1 gives the ordinary algebraic instance and its completion explicitly. □

Chapter 60

The quantum plane

Let

$$A_q = \Bbbk\langle x, y\rangle/(yx - qxy), \qquad q \in \Bbbk^\times.$$

Give x and y weight one.

Theorem 60.1 (PBW basis of the quantum plane)**.** *The monomials $x^a y^b$, $a, b \geq 0$, form a basis of A_q.*

Proof. Use the rewriting rule $yx \mapsto qxy$. Every reduction decreases the inversion number of a word. The pair yx cannot overlap with itself, so simultaneous reductions occur on disjoint pairs. Bergman's diamond argument therefore gives the ordered monomials as normal forms. □

The quadratic dual is

$$A_q^! = \Bbbk\langle \xi, \eta\rangle/(\xi^2, \eta^2, \xi\eta + q\eta\xi).$$

Theorem 60.2 (Koszulness)**.** *A_q is Koszul, and its Koszul complex is*

$$0 \to A_q \otimes \Bbbk(\xi\eta) \xrightarrow{d_2} A_q \otimes (\Bbbk\xi \oplus \Bbbk\eta) \xrightarrow{d_1} A_q \to \Bbbk \to 0,$$

with

$$d_1(a \otimes \xi + b \otimes \eta) = ax + by, \quad d_2(a \otimes \xi\eta) = ay \otimes \xi - qax \otimes \eta.$$

Proof. Direct substitution gives $d_1 d_2 = ayx - qaxy = 0$. The image of d_1 is the augmentation ideal (x, y), so the complex is exact at A_q and has cokernel $\Bbbk$. We now compute $\ker d_1$.

Write

$$u = \sum_{i,j\geq 0} u_{ij} x^i y^j, \qquad v = \sum_{i,j\geq 0} v_{ij} x^i y^j.$$

The relation $y^j x = q^j x y^j$ gives

$$ux + vy = \sum_{a,b\geq 0} \bigl(q^b u_{a-1,b} + v_{a,b-1}\bigr) x^a y^b,$$

where coefficients with a negative index are zero. Thus $ux + vy = 0$ implies $u_{i,0} = 0$, $v_{0,j} = 0$, and

$$v_{a,b} = -q^{b+1} u_{a-1,b+1}.$$

Define $c = \sum_{i,j\geq 0} u_{i,j+1} x^i y^j$. Then $u = cy$ and, using $x^i y^j x = q^j x^{i+1} y^j$,

$$v = -qcx.$$

Hence $(u, v) = d_2(c)$. Finally, $d_2(c) = 0$ implies $cy = 0$; the PBW basis and $q \neq 0$ show that A_q is a domain, so $c = 0$. The complex is exact, and the quadratic Koszul criterion proves that A_q is Koszul. □

Theorem 60.3 (Yoneda algebra of the quantum plane)**.** *Use the convention that a Yoneda product $\alpha\beta$ means first β and then α. The augmentation module has*

$$\operatorname{Ext}^{*}_{A_q}(\Bbbk,\Bbbk) \cong \Bbbk\langle \xi, \eta\rangle/(\xi^2, \eta^2, \xi\eta + q\eta\xi) = A_q^{!}$$

as a bigraded algebra, where $|\xi| = |\eta| = 1$ in cohomological degree and each generator retains its internal weight one.

Proof. Apply $\operatorname{Hom}_{A_q}(-,\Bbbk)$ to the minimal Koszul resolution in the preceding theorem. Every matrix entry of d_1 and d_2 lies in the augmentation ideal, so the dual differential vanishes. The cohomology has basis

$$1, \quad \xi, \eta, \quad \xi\eta$$

in cohomological degrees 0, 1, 1, 2.

Use free-resolution symbols e_x, e_y, e_r to distinguish them from the dual cochains. The cocycles $\xi(e_x) = 1$, $\xi(e_y) = 0$, $\eta(e_x) = 0$, $\eta(e_y) = 1$ lift to degree-minus-one endomorphisms by $\xi(e_r) = -e_y$ and $\eta(e_r) = qe_x$. They satisfy $d\xi + \xi d = d\eta + \eta d = 0$. For the degree-two cocycle $\theta(e_r) = 1$, composition gives $\xi\eta = q\theta$, $\eta\xi = -\theta$ and $\xi^2 = \eta^2 = 0$. These relations present the full four-dimensional cohomology. □

These formulas give the transverse algebra, its ordinary bar and its derived augmentation endomorphisms. If a commutative factorization coefficient supplies compatible exact tensor functors and geometric structure maps, the same formulas extend coefficientwise. The geometric factorization and collision compatibilities are hypotheses of that extension.

Chapter 61

Quiver path algebras

Let Q be a finite quiver and $S = \bigoplus_{i \in Q_0} \mathbf{1} e_i$. Assign $M_a \in \mathrm{Fact}_D(X)$ to each arrow $a : i \to j$ and set

$$M = \bigoplus_{a \in Q_1} e_j M_a e_i, \qquad T_S(M) = S \oplus M \oplus M \otimes_S M \oplus \cdots .$$

Words are composable paths.

For the quiver $1 \xrightarrow{a} 2$, the path algebra has basis e_1, e_2, a and relation $a = e_2 a e_1$. The simple modules S_1, S_2 satisfy

$$\mathrm{Ext}^1(S_1, S_2) \cong \Bbbk, \qquad \mathrm{Ext}^1(S_i, S_i) = 0, \qquad \mathrm{Ext}^1(S_2, S_1) = 0.$$

The bar differential records concatenation of paths. The vector-space dual of the arrow bimodule reverses endpoints, while with left modules and product given by composition the class in $\mathrm{Ext}^1(S_1, S_2)$ lies in $e_2 \,\mathrm{Ext}^1(S, S) e_1$. An identification of these bimodules must specify the opposite-algebra convention, as in Proposition 19.2.

Theorem 61.1 (Relative path-algebra bar)**.** *For an ordinary tensor path algebra $T_S(M)$ over the semisimple algebra $S = \bigoplus_i \Bbbk e_i$,*

$$S \otimes^{\mathbb{L}}_{T_S(M)} S \simeq S \oplus sM$$

as an S-bicomodule. A quadratic quotient $T_S(M)/(R)$ is Koszul precisely when its relative quadratic complex is exact away from weight zero.

Proof. The gap-insertion contraction in each path weight is S-bilinear and proves the first assertion. Equivalently, the augmentation has the relative projective resolution $0 \to T_S(M) \otimes_S M \to T_S(M) \to S \to 0$, where multiplication identifies the first term with the nonempty paths. For a quadratic quotient, its canonical quadratic coalgebra maps to the relative bar. Exactness of the associated free quadratic resolution is the relative Koszul criterion. □

Chapter 62

Weyl and Clifford deformations

Let V be finite-dimensional over a characteristic-zero field $\Bbbk$ and let ω be alternating. Put $R = \Bbbk[\hbar]$ and first form the algebraic Weyl algebra

$$W_\hbar(V,\omega) = T_R(V \otimes R)/\langle uv - vu - \hbar\omega(u,v)\rangle.$$

For a purely odd space and symmetric ω, the relation $uv + vu = \hbar\omega(u,v)$ defines a Clifford algebra. In particular $u^2 = \hbar\omega(u,u)/2$.

Theorem 62.1 (Filtered PBW theorem)**.** *Use the algebraic tensor-length filtration. The alternating Weyl form gives associated graded* $\mathrm{Sym}_R(V \otimes R)$*. The symmetric Clifford form gives associated graded* $\bigwedge_R(V \otimes R)$*.*

Proof. Filter by word length. The constant term lowers filtration by two, so the associated graded relation is the commutative or skew-commutative quadratic relation. The cubic ambiguities reduce to the cocycle identity for a central scalar form, which holds because scalars commute. The ordered monomials give the PBW basis. □

Let $\mathcal{R}_2 \subset V^{\otimes 2}$ be the quadratic relation space. Its scalar correction is the functional $c : \mathcal{R}_2 \to \hbar\Bbbk$, sending a defining relation to $\hbar\omega$. In the curved quadratic dual this gives the degree-two curvature element. The absence of a linear correction gives zero linear dual differential. A nonzero scalar correction prevents the augmentation killing all generators. A comparison with geometric anomaly curvature is separate.

After completing in $\hbar$, an element may have unbounded tensor length as its parameter order increases. The finite-length filtration is no longer exhaustive on the completed algebra. The algebraic ordered basis identifies this completion with formal parameter series whose coefficient at each order has finite monomial support. See Proposition 19.4.

Chapter 63

Associative chiral algebras on the formal disc

Let V be a complete topological vector space with vacuum $\mathbf{1}$ and translation T. A state–field map

$$Y(-,z) : V \longrightarrow \mathrm{End}(V)((z))$$

is a coordinate presentation of an associative chiral-convolution product when its iterated products are expansions of one factorization kernel in the nested regions $|z_1| > |z_2|$ and $|z_2| > |z_1|$.

Proposition 63.1 (Associative convolution from weak associativity)**.** *Suppose $Y(\mathbf{1},z) = \mathrm{id}$, $Y(a,z)\mathbf{1} \in a + zV[[z]]$, $[T, Y(a,z)] = \partial_z Y(a,z)$, and for every a, b, c there is N such that*

$$(z_1 - z_2)^N Y(a,z_1)Y(b,z_2)c = (z_1 - z_2)^N Y(Y(a,z_1 - z_2)b, z_2)c$$

in the common formal expansion ring. Then the fields define an associative chiral algebra on the formal disc. The converse holds for a vacuum-generated translation-equivariant associative chiral algebra admitting state–field reconstruction.

Proof. The displayed identity is the equality of the two nested restrictions of the three-point chiral product. Vacuum and translation give the unit and coordinate covariance. Factorization on higher configuration spaces is obtained by iterating the equality; coherence follows from associativity of the chiral-convolution product. □

Example 63.2 (Braided current fields)**.** Let $R(z)$ be an invertible solution of the spectral Yang–Baxter equation, normalized by $R^{21}(-z)R(z) = 1$. Generating fields $a_i(z)$ with exchange

$$a_i(z_1)a_j(z_2) = \sum_{k,l} R_{ij}^{kl}(z_1 - z_2)\, a_l(z_2)a_k(z_1)$$

form a quantum vertex algebra whenever the vacuum, translation, and weak associativity relations hold. The R-matrix supplies the comparison between the two formal expansion regions and records planar exchange in addition to transverse order.

Chapter 64

Zamolodchikov–Faddeev exchange

Let V be finite-dimensional and let $R(u) \in \mathrm{End}(V \otimes V) \otimes \Bbbk(u)$ be invertible. Introduce generators $Z_i(u)$ with exchange relation

$$Z_1(u)Z_2(v) = R_{12}(u-v)Z_2(v)Z_1(u).$$

Theorem 64.1 (Yang–Baxter confluence)**.** *Orient exchange only toward a fixed order on pairwise distinct spectral labels, and impose no further relations on the ordered tensor space. Then the three-input reductions agree precisely when*

$$R_{12}(u-v)R_{13}(u-w)R_{23}(v-w) = R_{23}(v-w)R_{13}(u-w)R_{12}(u-v).$$

Proof. Reverse the word $Z_1(u)Z_2(v)Z_3(w)$. Exchanging positions in the order (12), (13), (23) gives the left side. Exchanging in the order (23), (13), (12) gives the right side. These are the two reductions of the unique length-three overlap. Equality is therefore the diamond condition. □

For the rational permutation solution

$$R(u) = 1 - \frac{\hbar P}{u},$$

the Yang–Baxter equation follows in the group algebra of Σ_3 from the permutation identities and $1/(u(u+v)) + 1/(v(u+v)) = 1/(uv)$. Its reverse product is $(1 - \hbar^2/u^2)I$. Imposing the same unnormalized exchange in both directions therefore adds a scalar torsion relation. Over $\Bbbk(\hbar, u)$ it kills the corresponding pair products. Proposition 19.7 separates the chamber presentation from this two-direction requirement.

Chapter 65

The RTT Yangian of $\mathfrak{gl}_2$

Let

$$T(u) = 1 + \sum_{r \geq 1} T^{(r)} u^{-r}, \qquad R(u) = 1 - \frac{\hbar P}{u}.$$

The RTT Yangian $Y_\hbar(\mathfrak{gl}_2)$ is generated by the coefficients $t_{ij}^{(r)}$ subject to

$$R(u-v)T_1(u)T_2(v) = T_2(v)T_1(u)R(u-v).$$

Equivalently,

$$(u-v)[t_{ij}(u), t_{kl}(v)] = \hbar\big(t_{kj}(u)t_{il}(v) - t_{kj}(v)t_{il}(u)\big).$$

Proposition 65.1 (Coefficient relations). *Put $t_{ij}^{(0)} = \delta_{ij}$. The RTT identity is equivalent to*

$$[t_{ij}^{(r+1)}, t_{kl}^{(s)}] - [t_{ij}^{(r)}, t_{kl}^{(s+1)}] = \hbar\big(t_{kj}^{(r)} t_{il}^{(s)} - t_{kj}^{(s)} t_{il}^{(r)}\big), \qquad r, s \geq 0.$$

In particular,

$$[t_{ij}^{(1)}, t_{kl}^{(1)}] = \hbar(\delta_{kj} t_{il}^{(1)} - \delta_{il} t_{kj}^{(1)}).$$

The first modes generate a copy of the algebra $U_\hbar(\mathfrak{gl}_2)$ defined by this commutator. Its injectivity follows from the evaluation map below, which is a left inverse on the first modes.

Proof. Clear the denominator in the RTT relation and compare the coefficient of $u^{-r}v^{-s}$. Multiplication by $u - v$ gives the two commutators on the left. The right side consists of the corresponding coefficients of the RTT numerator. Setting $r = 0$, $s = 1$ gives the displayed first-mode relation. □

Theorem 65.2 (Hopf structure). *The formulas*

$$\Delta(T(u)) = T(u) \dot{\otimes} T(u), \qquad \varepsilon(T(u)) = 1, \qquad S(T(u)) = T(u)^{-1}$$

define a Hopf algebra structure on the ordinary coefficient algebra. The generating series use u^{-1}-adic notation, but every coproduct and inverse coefficient is a finite polynomial. Coefficientwise,

$$\Delta(t_{ij}^{(r)}) = \sum_{a=1}^{2} \sum_{p=0}^{r} t_{ia}^{(p)} \otimes t_{aj}^{(r-p)}.$$

Proof. Apply the RTT relation in each tensor factor. Matrix multiplication in the auxiliary space gives the RTT relation for $\Delta(T)$. Coassociativity is associativity of matrix multiplication. Expanding the matrix product gives the coefficient formula. The inverse series exists uniquely because $T(u) = 1 + O(u^{-1})$. Its defining identities give the antipode axioms. □

Over $\Bbbk[\hbar]$, evaluation to $U_\hbar(\mathfrak{gl}_2)$ sends $T(u)$ to $I + E/(u-a)$, where $[E_{ij}, E_{kl}] = \hbar(\delta_{kj}E_{il} - \delta_{il}E_{kj})$. At $a = 0$ this fixes the first modes. Evaluation into the ordinary enveloping algebra instead sends

$$T(u) \longmapsto I + \frac{\hbar e}{u-a}, \qquad [e_{ij}, e_{kl}] = \delta_{kj}e_{il} - \delta_{il}e_{kj}.$$

For the filtration $\deg t_{ij}^{(r)} = r - 1$, the associated graded is $U_\hbar(\mathfrak{gl}_2[t])$. After $\hbar = 1$ this is the usual current enveloping algebra. Theorem 19.9 proves the ordered basis and distinguishes this filtration from $\deg t_{ij}^{(r)} = r$ [27, 60].

This example contains three structures: ordered multiplication in the Yangian, coproduct on tensor products of modules, and exchange by the rational R-matrix.

Chapter 66

The Hall algebra of the A_2 quiver

Let $Q : 1 \to 2$ over $\mathbb{F}_q$. Write S_1, S_2 for the simples and M for the unique non-split extension

$$0 \to S_2 \to M \to S_1 \to 0.$$

In the twisted Hall algebra, multiplication counts subobjects with Euler-form normalization. Direct counting gives

$$[S_1] * [S_2] = v^{-1}[S_1 \oplus S_2] + v^{-1}[M], \qquad [S_2] * [S_1] = [S_1 \oplus S_2], \qquad v^2 = q.$$

Hence the root vector

$$E_{12} = E_1E_2 - v^{-1}E_2E_1$$

equals $v^{-1}[M]$, the normalized class of the indecomposable extension.

Theorem 66.1 (Quantum Serre relation)**.** *The Hall generators satisfy*

$$E_1^2E_2 - (v + v^{-1})E_1E_2E_1 + E_2E_1^2 = 0,$$

and the analogous relation with 1 *and* 2 *exchanged.*

Proof. Every representation of dimension vector $(2, 1)$ is a linear map $f : \mathbb{F}_q^2 \to \mathbb{F}_q$ of rank zero or one. Read a word from the top quotient to the bottom quotient. The complete-flag counts are

$\operatorname{rk} f$	112	121	211
0	$q+1$	$q+1$	$q+1$
1	$q+1$	1	0.

For rank zero every one-dimensional subspace of $\mathbb{F}_q^2$ is admissible, giving $q + 1$ flags in each word. For rank one, the word 121 forces the unique kernel line, the word 211 is incompatible with the final subrepresentation, and the word 112 allows every line at the remaining stage.

The total Euler-form exponents of the three words are respectively $-1, 0, 1$. Their twisted coefficients are therefore $v^{-1}, 1, v$. On the rank-zero stratum the Serre combination has coefficient

$$(q + 1)\big(v^{-1} - (v + v^{-1}) + v\big) = 0.$$

On the rank-one stratum it has coefficient

$$v^{-1}(q + 1) - (v + v^{-1}) = v^{-1}(v^2 + 1) - (v + v^{-1}) = 0.$$

The two isomorphism classes exhaust dimension $(2, 1)$. Dualizing representations and interchanging vertices reverses flags and exchanges the two generators, giving the other Serre relation. This verifies the displayed relations in the Hall realization. The full identification with $U_v^+(\mathfrak{sl}_3)$ also requires its basis theorem [37, 66]. □

Chapter 67

Matrix traces and the necklace bracket

Let $F = \Bbbk\langle x, y\rangle$ and $F_{\mathrm{cyc}} = F/[F, F]$. Cyclic derivatives are defined by

$$\partial_x^{\mathrm{cyc}}[a_1 \cdots a_n] = \sum_{a_i = x} a_{i+1} \cdots a_n a_1 \cdots a_{i-1}$$

and similarly for y. Define

$$\{f, g\}_{\mathrm{neck}} = \left[\partial_x^{\mathrm{cyc}} f \, \partial_y^{\mathrm{cyc}} g - \partial_y^{\mathrm{cyc}} f \, \partial_x^{\mathrm{cyc}} g\right]_{\mathrm{cyc}}.$$

Theorem 67.1 (Trace realization)**.** *For matrices $X, Y \in \mathfrak{gl}_N$ and the canonical symplectic bracket on matrix entries,*

$$\{\operatorname{Tr} f(X,Y), \operatorname{Tr} g(X,Y)\} = \operatorname{Tr}\{f, g\}_{\mathrm{neck}}(X,Y).$$

Proof. Differentiate the trace by moving the varied matrix entry cyclically to the first position. This gives the cyclic derivative. Contract the X derivative of one trace with the Y derivative of the other and subtract the reversed contraction. The matrix indices close into the trace of the cyclic product. □

In the stable commuting-matrix quotient, let

$$p_{ab} = \operatorname{Tr}(X^a Y^b).$$

Then

$$\{p_{ab}, p_{cd}\} = (ad - bc)\, p_{a+c-1, b+d-1}.$$

For two eigenvalues, the relative invariants

$$A = (x_1 - x_2)^2, \quad B = (x_1 - x_2)(y_1 - y_2), \quad C = (y_1 - y_2)^2$$

satisfy $B^2 = AC$ and carry the standard $\mathfrak{sl}_2$ Poisson bracket after linear rescaling.

Book XI

The nonlocal ladder

Chapter 68

Spectral exchange as an associative chiral operation

68.1 The coefficient field and the exchange operator

Let V be a finite-dimensional vector space and let K be a field of meromorphic functions in spectral differences. Fix an invertible operator

$$R(u) \in \operatorname{End}(V \otimes V) \otimes K$$

that satisfies the spectral Yang–Baxter equation

$$R_{12}(u-v)\,R_{13}(u-w)\,R_{23}(v-w) = R_{23}(v-w)\,R_{13}(u-w)\,R_{12}(u-v). \tag{68.1}$$

The equation states that two routes through the chamber complex of three labelled points give the same transport.

Definition 68.1 (Zamolodchikov–Faddeev algebra)**.** Fix a finite collection of distinct spectral labels and a total order on it. The chamber exchange algebra $\mathcal{Z}_R$ has generators in their copies of V and uses the following relation only for an inverted pair:

$$A_1(u)\,A_2(v) = R_{12}(u-v)\,A_2(v)\,A_1(u). \tag{68.2}$$

A normal word has spectral labels in a fixed chamber $u_1 \succ \cdots \succ u_n$.

Theorem 68.2 (Confluence is Yang–Baxter)**.** *Assume the ordered tensor spaces have no additional relations. The oriented rewriting system* (68.2) *is confluent on words with pairwise distinct labels precisely when* (68.1) *holds. In that case the ordered tensors form a basis in each such multilinear label degree. Relations imposed also on the reverse order require inverse compatibility, separately from Yang–Baxter.*

Proof. Every two-letter inversion is removed by one application of (68.2). The minimal overlap occurs in a word of length three. Moving the leftmost label through the other two gives the coefficient $R_{12}R_{13}R_{23}$, while moving the rightmost label in the opposite sequence gives $R_{23}R_{13}R_{12}$. Equality of the two coefficients is (68.1). Bergman's diamond lemma then gives the normal-word basis. □

68.2 A diagonal exchange model

Choose a basis $e_1, \dots, e_N$ of V and functions $q_{ij}(u)$ satisfying

$$q_{ij}(u)\,q_{ji}(-u) = 1, \qquad q_{ii}(u) = 1.$$

The diagonal operator

$$R(u)(e_i \otimes e_j) = q_{ij}(u)\,e_i \otimes e_j$$

satisfies the Yang–Baxter equation because all three factors are scalars. The generators $a_i(u)$ obey

$$a_i(u)a_j(v) = q_{ij}(u-v)a_j(v)a_i(u).$$

Every ordered monomial is a basis vector. The quantum plane is the constant two-generator specialization $q_{12} = q$, $q_{21} = q^{-1}$.

Calculation 68.3 (Three-particle coefficient)**.** For $u \succ v \succ w$,

$$a_i(w)a_j(v)a_k(u) = q_{ij}(w-v)q_{ik}(w-u)q_{jk}(v-u)a_k(u)a_j(v)a_i(w).$$

The coefficient is independent of the sequence of adjacent exchanges. This scalar identity is the diagonal Yang–Baxter equation.

Proof. Each inversion of two adjacent factors contributes its scalar exchange factor. Both reduced words for the longest permutation invert the same three pairs. Scalar commutativity makes the products identical. □

Chapter 69

RTT and reflection algebras

69.1 The RTT bialgebra

Let $T(u) = (t_{ij}(u))$ be a matrix of formal currents. The RTT algebra $\mathcal{T}_R$ is defined by

$$R_{12}(u-v)T_1(u)T_2(v) = T_2(v)T_1(u)R_{12}(u-v). \tag{69.1}$$

Theorem 69.1 (RTT coproduct)**.** *The formula*

$$\Delta(T(u)) = T_{12}(u)T_{13}(u), \qquad \epsilon(T(u)) = I,$$

defines a coassociative bialgebra structure on $\mathcal{T}_R$. The auxiliary matrix occupies tensor position one and the two coefficient algebras occupy positions two and three.

Proof. For two RTT matrices T, U whose coefficient algebras commute, interchanging only the cross factors gives $R(T_1U_1)(T_2U_2) = RT_1T_2U_1U_2 = T_2T_1RU_1U_2 = T_2U_2T_1U_1R$. This proves that the product matrix again satisfies RTT. The argument uses RTT and the commuting coefficient algebras. It does not require an additional Yang–Baxter identity. The two iterated coproducts are both $T_{12}T_{13}T_{14}$, proving coassociativity. Substitution of the identity matrix proves the counit equations. □

For the Yang matrix $R(u) = 1 - \hbar P/u$, expansion at $u = \infty$ gives the RTT Yangian. The coefficient recurrence with $r = 0$, $s = 1$ gives

$$[t_{ij}^{(1)}, t_{kl}^{(1)}] = \hbar\big(\delta_{kj}t_{il}^{(1)} - \delta_{il}t_{kj}^{(1)}\big),$$

so the first current layer is the Rees enveloping algebra of $\mathfrak{gl}_N$.

69.2 Boundary exchange

A reflecting boundary reverses the spectral path. For a constant invertible $R \in \mathrm{End}(V^{\otimes 2})$ satisfying Yang–Baxter, define the reflection algebra $\mathcal{K}_R$ by

$$R_{21}K_1R_{12}K_2 = K_2R_{21}K_1R_{12}. \tag{69.2}$$

Theorem 69.2 (A reflection algebra with ordered basis)**.** *For $V = \Bbbk^2$ and the explicit Hecke matrix of Proposition 19.10, the reflection algebra has basis $a^i b^j c^k d^\ell$ for every $q \neq 0$. Yang–Baxter and invertibility alone do not imply this ordered-basis assertion for an arbitrary R.*

Proof. For the specified matrix, the six rules of Proposition 19.10 decrease degree-lexicographic order and the four overlaps cba, dba, dca, dcb have equal reductions. They prove the basis in all degrees. For a general matrix, compare $R = I$ and $R = P$. Both are invertible Yang–Baxter solutions. The first gives commuting matrix entries, whereas the second turns the reflection equation into $K_2^2 = K_2^2$ and leaves the entries free. Their Hilbert series differ for $\dim V > 1$. □

The RTT algebra governs bulk line exchange. The reflection algebra governs boundary line exchange. The ZF algebra governs charged fields in a fixed representation. Their common coefficient is the same R-matrix, while their products live in three different endomorphism objects.

Chapter 70

Quantum vertex realization

70.1 Two expansion regions

Let V be a complete topological vector space with vacuum $\mathbf{1}$, translation T, state-field map Y, and a unitary rational operator

$$S(z) : V \otimes V \longrightarrow V \otimes V((z)).$$

The two formal regions $|z_1| > |z_2|$ and $|z_2| > |z_1|$ are related by $S(z_1 - z_2)$.

Definition 70.1 (Rational quantum vertex datum)**.** A rational quantum vertex datum satisfies vacuum and translation covariance together with

$$(z_1 - z_2)^N Y(a, z_1) Y(b, z_2) = (z_1 - z_2)^N Y^{21}(S(z_1 - z_2)(a \otimes b); z_2, z_1) \tag{70.1}$$

for a sufficiently large N, and weak associativity in every triple of states.

Theorem 70.2 (Exchange coherence)**.** *Suppose S is unitary,*

$$S^{21}(-z) S(z) = 1,$$

and satisfies the spectral Yang–Baxter equation. Then the pairwise continuation maps obtained from (70.1) *define a braid-group action on every n-point field product. Weak associativity makes the action compatible with iterated operator products.*

Proof. Unitarity identifies a crossing followed by its reverse with the identity. Yang–Baxter identifies the two continuations around every codimension-two triple-collision stratum. These are the defining relations of the braid group. Weak associativity identifies the collision limit in each chamber with the corresponding iterated field. □

When S is the graded flip, the datum is an ordinary local vertex algebra. A nontrivial S retains the ordered chiral product and its exchange monodromy. This is the formal-disc realization of an associative chiral algebra with braiding data in the sense of quantum vertex algebra theory [53, 54].

70.2 The Yangian field

Let $T(u)$ act on a family of evaluation modules. The RTT relation becomes S-locality for the matrix coefficients of $T(u)$. The normalized Yang operator

$$S(u) = f(u)\left(1 - \frac{\hbar P}{u}\right), \qquad f(u) f(-u) = \left(1 - \frac{\hbar^2}{u^2}\right)^{-1},$$

is unitary. Its expansion at infinity gives the quantum vertex exchange of Yangian currents. The scalar factor changes normalization and preserves the Yang–Baxter equation.

Chapter 71

Shuffle realization and the positive current algebra

71.1 Pairwise kernel

Let $K(z)$ be a rational function and put

$$\mathrm{Sh}_n = \mathbb{C}(z_1, \dots, z_n)^{\mathfrak{S}_n}.$$

For $f \in \mathrm{Sh}_m$ and $g \in \mathrm{Sh}_n$ define

$$(f * g)(z_1, \dots, z_{m+n}) = \mathrm{Sym}\left(f(z_1, \dots, z_m)\, g(z_{m+1}, \dots, z_{m+n}) \prod_{i \le m < j} K(z_i - z_j) \right). \tag{71.1}$$

Theorem 71.1 (Associativity of the shuffle product)**.** *The product* (71.1) *is associative on every subspace closed under the displayed symmetrization and pole conditions.*

Proof. In a triple product, every ordered pair of variables belonging to different blocks contributes exactly one factor $K(z_i - z_j)$. The product of all pairwise factors is independent of the parenthesization. Final symmetrization gives the same function. □

For the tripled Jordan quiver, one uses

$$K(z) = \prod_{a=1}^{3} \frac{z + \epsilon_a}{z}, \qquad \epsilon_1 + \epsilon_2 + \epsilon_3 = 0.$$

Wheel conditions remove the forbidden triple residues. The resulting shuffle algebra is the positive current algebra that appears in the critical CoHA of $\mathbb{C}^3$ and in the positive affine Yangian of $\mathfrak{gl}_1$ [67, 68].

Calculation 71.2 (Degree one against degree one)**.** For $f(z_1) = z_1^r$ and $g(z_1) = z_1^s$,

$$f * g = z_1^r z_2^s K(z_1 - z_2) + z_2^r z_1^s K(z_2 - z_1).$$

The antisymmetric part records the first current commutator. Its residues at $z_1 - z_2 = -\epsilon_a$ are the three elementary extension channels of the tripled quiver.

Proof. For two variables the symmetrization in the shuffle definition has two terms, which gives the displayed formula. Antisymmetrization exchanges the two chambers. The three numerator factors of the tripled-quiver kernel locate the three elementary shifted extension channels. □

71.2 Coproduct and double

Choose a chamber expansion of $K(z)$ at infinity. Splitting the variables into two ordered groups and expanding every cross-kernel in that chamber defines a completed coproduct. A residue pairing between positive and negative shuffle halves gives a continuous Hopf pairing. The completed Drinfeld double is then a quasitriangular current algebra. Product, coproduct, and pairing are separate operations; each is visible in the shuffle formula that defines it.

Chapter 72

The ladder in one diagram

The nonlocal structures form the chain

$$\text{ordered coefficient} \longrightarrow \mathcal{Z}_R \longrightarrow \mathcal{T}_R \longrightarrow (V, Y, S),$$

$$(V, Y, S) \longrightarrow \mathrm{Sh}^{+} \longrightarrow D\big(\mathrm{Sh}^{+}\big).$$

The arrows retain different data:

Object	Defining operation	Coherence
$\mathcal{Z}_R$	exchange of charged fields	Yang–Baxter equation
$\mathcal{T}_R$	matrix-current product and coproduct	RTT and coassociativity
$\mathcal{K}_R$	boundary exchange	reflection equation
(V, Y, S)	formal field product	S-locality and weak associativity
Sh^{+}	extension shuffle	pairwise-kernel associativity
$D(\mathrm{Sh}^{+})$	positive–negative pairing	quasitriangularity

This ladder gives a concrete generator-and-relation theory of doubly noncommutative chiral observables. The first noncommutativity is ordered composition. The second is exchange between distinct order chambers. Chiral singularity remains the diagonal coefficient throughout.

Book XII

Field theory and master equations

Chapter 73

A finite cotangent BF model

Let L be a finite-dimensional dg Lie algebra. Its cotangent BV space is

$$\mathcal{E}_L = L[1] \oplus L^*[-2],$$

with the canonical degree -1 symplectic pairing. Write $A \in L[1]$ and $B \in L^*[-2]$. The cotangent action is

$$S_L(A, B) = \langle B, d_L A + \tfrac{1}{2}[A, A] \rangle.$$

Theorem 73.1 (Cotangent master equation). *The Hamiltonian vector field of S_L is square-zero, and consequently $\{S_L, S_L\} = 0$.*

Proof. The A component is the Maurer–Cartan vector field $d_L A + \frac{1}{2}[A, A]$; the B component is its cotangent lift. The square of the first component is the sum of d_L^2, the derivation identity for d_L, and the Jacobi identity. The cotangent lift preserves commutators of vector fields, so the full Hamiltonian vector field is square-zero. □

Proposition 73.2 (Coordinate form of the modular term). *Let L be an ordinary finite-dimensional Lie algebra with basis e_a and structure constants $[e_a, e_b] = f_{ab}^{c} e_c$. Let c^a be the odd coordinates on $L[1]$ and b_a their BV-conjugate coordinates. With*

$$S = \frac{1}{2} b_c f_{ab}^{c} c^a c^b, \qquad \Delta = -\sum_a \frac{\partial}{\partial c^a} \frac{\partial}{\partial b_a},$$

one has

$$\Delta S = \sum_b \operatorname{Tr}(\operatorname{ad}_{e_b}) c^b.$$

Thus the one-loop class of the finite cotangent theory is exactly the modular character.

Proof. Differentiation first in b_a gives $\frac{1}{2} f_{ij}^{a} c^i c^j$. Odd differentiation in c^a gives $f_{aj}^{a} c^j$, and the chosen sign in Δ converts this to $f_{ja}^{a} c^j = \operatorname{Tr}(\operatorname{ad}_{e_j}) c^j$. This is the displayed formula. □

A finite-dimensional dg Lie algebra is unimodular when the supertrace of every adjoint endomorphism vanishes. Every finite-dimensional nilpotent Lie algebra in degree zero is unimodular because each ad_x is nilpotent.

Theorem 73.3 (Finite-dimensional quantum master equation). *Let L be unimodular and let Δ be the BV Laplacian determined by the translation-invariant Berezinian on $\mathcal{E}_L$. Then*

$$\Delta S_L = 0, \qquad \frac{1}{2}\{S_L, S_L\} + \hbar \Delta S_L = 0,$$

and equivalently

$$\hbar \Delta\left(e^{S_L/\hbar}\right) = 0.$$

The quantum observable differential

$$\delta_{S_L} = \{S_L, -\} + \hbar\Delta$$

satisfies $\delta_{S_L}^2 = 0$.

Proof. The classical master equation gives $\{S_L, S_L\} = 0$. The scalar ΔS_L is the divergence of the Chevalley vector field on $L[1]$. Its linear coefficient is the modular character $x \mapsto \mathrm{str}(\mathrm{ad}_x)$, which vanishes by unimodularity. The BV exponential identity

$$\hbar^2 e^{-S_L/\hbar}\Delta(e^{S_L/\hbar}) = \frac{1}{2}\{S_L, S_L\} + \hbar\Delta S_L$$

proves the exponential form. The standard BV commutator calculation then gives

$$\delta_{S_L}^2 = \left\{\frac{1}{2}\{S_L, S_L\} + \hbar\Delta S_L, -\right\} = 0.$$

□

Now take the local dg Lie algebra

$$\mathcal{L}_X = \Omega^{0,*}(X) \mathbin{\widehat{\otimes}} \Omega^*(\mathbb{R}) \otimes \mathfrak{l}$$

for a finite-dimensional nilpotent $\mathfrak{l}$, and form its shifted cotangent theory with compactly supported density dual. The preceding finite theorem governs every pairing-preserving finite-dimensional deformation retract. The local classical observables are Chevalley cochains and are holomorphic on X and locally constant on $\mathbb{R}$.

For the independent ordinary interval model, choose $U(\mathfrak{l})$ and its two augmentation modules. Their derived tensor product is

$$\mathbf{1} \otimes^{\mathbb{L}}_{U(\mathfrak{l})} \mathbf{1} \simeq \mathrm{Bar}(U(\mathfrak{l})) \simeq C_*(\mathfrak{l}),$$

the Chevalley–Eilenberg chains. A solution of the renormalized quantum master equation in the local BV deformation complex gives the infinite-dimensional quantum factorization algebra. Identifying its boundary observables with the ordinary interval model requires a supported quantum comparison preserving the boundary polarizations.

Chapter 74

BRST and ordered reduction

Let (A, Q) be a dg ordered algebra, with Q a derivation. Extend Q to the bar factorwise. Write b for the multiplication bar differential.

Proposition 74.1 (BRST–bar bicomplex)**.** *With the suspension sign convention,*

$$Q_{\mathrm{Bar}} b + b Q_{\mathrm{Bar}} = 0.$$

Hence $D = Q_{\mathrm{Bar}} + b$ satisfies $D^2 = 0$.

Proof. Apply Q to a bar face that multiplies adjacent entries. The derivation formula splits the result into the two terms obtained by applying Q before that face. The totalization sign makes their sum the anticommutator. □

The filtration by bar length gives

$$E_1 \cong \mathrm{Bar}(H^*(A, Q)) \Longrightarrow H^*(\mathrm{Bar}(A), D)$$

under boundedness and completeness. Quantum Drinfeld–Sokolov reduction is a current-algebra realization of this bicomplex.

Chapter 75

Stokes and algebraic identities

Let $C_*(K)$ be the oriented cellular chains of a regular manifold with corners. Suppose each codimension-one face F is assigned a degree-one operator ρ_F on a coefficient complex E, and the assignments are compatible with restrictions to corners.

Theorem 75.1 (Stokes-to-master)**.** *Assume that each ρ_F is a chain map of total degree one, that its restriction to a corner depends on the induced contraction, and that disjoint contractions act with the Koszul sign prescribed by the oriented normal bundle. Then the component of*

$$D^2, \qquad D = d_E + d_K + \sum_F \rho_F,$$

supported on a codimension-two corner S is the oriented sum

$$\sum_{F \supset S} \epsilon(F, S)\, \rho_{F/S} \rho_F.$$

Thus D is a differential precisely when these corner sums vanish.

Proof. Expand D^2 and group terms by support strata. The chain-map condition cancels the mixed internal terms. A term supported on S is obtained by crossing two incident faces. The two orders carry the boundary orientations of the two sides of the oriented normal square. Their signed sum is the displayed expression. □

Associahedra give Stasheff identities. Fulton–MacPherson collision strata give chiral and graph identities after a coefficient map is supplied. Stable-graph boundaries give BV and modular master equations.

Chapter 76

Two edge colors and two loop operators

Let $\mathsf{G}^{\perp,\mathrm{ch}}$ be the completed vector space spanned by stable graphs with transverse and chiral edges. Contracting an edge that joins distinct vertices gives brackets $[-,-]_{\perp}$ and $[-,-]_{\mathrm{ch}}$. Contracting a loop gives operators $\Delta_{\perp}$ and Δ_{ch}. Product orientation gives

$$\Delta_{\perp}\Delta_{\mathrm{ch}} + \Delta_{\mathrm{ch}}\Delta_{\perp} = 0.$$

Theorem 76.1 (Bicolored master equation)**.** *Let $\mathsf{P}^{\perp,\mathrm{ch}}$ be a two-colored modular operad whose edge contractions carry the product determinant orientation, and let $\mathfrak{g}_{\mathsf{P}}$ be the completed convolution Lie algebra of its Feynman transform with an endomorphism modular operad. Algebra structures on the transform are Maurer–Cartan elements $\Theta \in \mathfrak{g}_{\mathsf{P}}^{1}$ satisfying*

$$d\Theta + \frac{1}{2}[\Theta,\Theta]_{\perp} + \frac{1}{2}[\Theta,\Theta]_{\mathrm{ch}} + \hbar_{\perp}\Delta_{\perp}\Theta + \hbar_{\mathrm{ch}}\Delta_{\mathrm{ch}}\Theta = 0.$$

Proof. The Feynman-transform differential contracts one edge. Separating edges give the two brackets and loop edges give the two BV operators. The product orientation makes mixed contractions anticommute. The condition that the structure map commute with the transform differential is exactly the displayed Maurer–Cartan equation [32]. □

Chapter 77

Typed anomalies

An anomaly is a cohomology class in the deformation complex that governs the attempted construction.

construction	complex	class
central extension	local Lie algebra cochains	cocycle
BRST quantization	BRST deformation complex	nilpotence obstruction
BV quantization	local BV cochains	quantum master anomaly
conformal-block family	Atiyah algebra of its determinant line	projective curvature
curved Koszul duality	curved convolution Lie algebra	curvature m_0

A comparison theorem consists of a morphism between two complexes and an equality of the transported classes. This formulation keeps the geometric origin of every scalar visible.

Book XIII

Calabi–Yau and automorphic interfaces

Chapter 78

Calabi–Yau categories

Let C be a smooth linear category with a d-Calabi–Yau structure. Two universal consequences are available.

Theorem 78.1 (Cyclic and shifted outputs)**.** *The Hochschild cochains $CC^*(C)$ form a framed E_2 algebra. The derived moduli stack of objects carries a $(2-d)$-shifted symplectic structure whenever the representability hypotheses hold.*

Proof. The first assertion is the cyclic Deligne theorem [10]. The second is the PTVV–Brav–Dyckerhoff construction: the Calabi–Yau trace pairs the tangent complex $\mathrm{RHom}(E,E)[1]$ with itself in degree $2-d$ [9, 64]. □

A geometric BV theory or an open factorization category supplies an ordered chiral algebra through the recognition theorem of Chapter 3. This is the bridge from Calabi–Yau geometry to the present volume.

Chapter 79

Hall and BPS algebras

A Hall multiplication starts from the extension correspondence

$$\mathcal{M}_\alpha \times \mathcal{M}_\beta \xleftarrow{p} \mathcal{M}^{\mathrm{ext}}_{\alpha,\beta} \xrightarrow{q} \mathcal{M}_{\alpha+\beta}.$$

With a coefficient theory for which p^* and q_* are defined,

$$a * b = q_* p^*(a \boxtimes b).$$

Associativity follows from the stack of two-step filtrations.

For 2-Calabi–Yau categories with good moduli spaces, the BPS algebra is the positive enveloping algebra of a generalized Kac–Moody Lie algebra [22]. For quivers and surface resolutions, preprojective and surface CoHAs recover positive Yangian halves in established cases [25, 67].

The ordered chiral structure enters when Hall multiplication is promoted to a factorization coefficient over a curve or defect parameter. A coproduct, pairing, and completion then produce the corresponding double.

Chapter 80

The Igusa denominator as a bar character

Let $\mathfrak{g}_\Delta$ be the Gritsenko–Nikulin Borcherds–Kac–Moody superalgebra and let $\mathfrak{n}_+$ be its positive root part. Fix a proper positive integral height h, and put

$$F^{>H}\mathfrak{n}_+ = \bigoplus_{h(\alpha)>H} \mathfrak{g}_{\Delta,\alpha}, \qquad \mathfrak{n}_+^{(H)} = \mathfrak{n}_+/F^{>H}\mathfrak{n}_+.$$

Height additivity makes $F^{>H}\mathfrak{n}_+$ an ideal. Properness and finite-dimensional root spaces make $\mathfrak{n}_+^{(H)}$ finite-dimensional and nilpotent. For an entire algebra or complex $V = \bigoplus_d V_d$, write $\widehat{V}_{\mathrm{tot}} = \prod_d V_d$ for its total-height completion.

Theorem 80.1 (Bar–Chevalley identification)**.** *For every H, super antisymmetrization gives a natural quasi-isomorphism*

$$C_*(\mathfrak{n}_+^{(H)}) \longrightarrow \mathrm{Bar}(U(\mathfrak{n}_+^{(H)}))$$

of conilpotent dg coalgebras. It also induces a quasi-isomorphism

$$\widehat{C_*(\mathfrak{n}_+)}_{\mathrm{tot}} \longrightarrow \widehat{\mathrm{Bar}(U(\mathfrak{n}_+))}_{\mathrm{tot}}$$

of complete dg coalgebras, with tensor product completed in total height. The Euler supercharacter is

$$\chi\,\widehat{C_*(\mathfrak{n}_+)}_{\mathrm{tot}} = \prod_{\alpha>0} (1-e^{-\alpha})^{\mathrm{sdim}\,\mathfrak{g}_{\Delta,\alpha}}.$$

Proof. At each fixed total height, the PBW filtration is finite. The associated graded map is the tensor product of polynomial and exterior one-generator Koszul comparisons. Finite filtration proves the quasi-isomorphism, and exactness of products passes it to total-height completion. An even root vector contributes $1-e^{-\alpha}$ to the Euler supercharacter. An odd root vector contributes its inverse. This is the construction of Proposition 20.3.

There is also a separate comparison

$$\varprojlim_H C_*(\mathfrak{n}_+^{(H)}) \longrightarrow \varprojlim_H \mathrm{Bar}(U(\mathfrak{n}_+^{(H)})).$$

The comparison cones are acyclic and their transition maps are degreewise surjective. The inverse-limit exact sequence for products therefore makes their limit acyclic. This generator-quotient topology retains every power of a surviving generator. For one even generator of height one, the enveloping limit is $\mathbb{C}[x]$, whereas total-height completion is $\mathbb{C}[[x]]$. □

On the root torus choose $e^{-\rho} = Q^{1/2}R^{1/2}S^{1/2}$. Under the Gritsenko–Nikulin embedding into Siegel period coordinates Ω, the denominator theorem reads

$$e^{-\rho}\,\chi\,\widehat{C_*(\mathfrak{n}_+)}_{\mathrm{tot}} = D_5(2\Omega) = 2^{-6}\Delta_5(2\Omega)$$

[38, 40]. The doubled period is part of the comparison with the monic product in Chapter 20.

Chapter 81

Finite-height currents and algebraic interval models

Retain the finite-height quotients $\mathfrak{n}_+^{(H)}$ fixed in the preceding chapter.

The current Lie conformal superalgebra is

$$\operatorname{Cur}(\mathfrak{n}_+^{(H)}) = \Bbbk[\partial] \otimes \mathfrak{n}_+^{(H)}, \qquad [a_\lambda b] = [a, b].$$

Its universal enveloping vertex superalgebra is V_H. The current generators carry the cocommutative vertex-bialgebra coproduct

$$\Delta\big(a(-1)\mathbf{1}\big) = a(-1)\mathbf{1} \otimes \mathbf{1} + \mathbf{1} \otimes a(-1)\mathbf{1}, \qquad a \in \mathfrak{n}_+^{(H)},$$

extended uniquely as a vertex-algebra morphism.

Theorem 81.1 (Height-complete current realization)**.** *Let $\mathfrak{n} = \bigoplus_{\alpha>0} \mathfrak{n}_\alpha$ be a root-graded Lie superalgebra with finite-dimensional root spaces and a proper additive height function. Put*

$$\mathfrak{n}^{(H)} = \mathfrak{n} \Big/ \bigoplus_{h(\alpha)>H} \mathfrak{n}_\alpha.$$

Then every $\mathfrak{n}^{(H)}$ is finite-dimensional and nilpotent, the quotient maps form an inverse system, and

$$\widehat{U}_h(\mathfrak{n}) = \varprojlim_H U(\mathfrak{n}^{(H)}), \quad \widehat{C}_{*,h}(\mathfrak{n}) = \varprojlim_H C_*(\mathfrak{n}^{(H)}), \quad \widehat{V}_h(\mathfrak{n}) = \varprojlim_H U^{\mathrm{ch}}(\operatorname{Cur}_X \mathfrak{n}^{(H)})$$

are separated complete algebras, coalgebras, and current algebras. Their associated graded objects are the corresponding completed symmetric objects, and the bar–Chevalley comparison is compatible with the inverse system.

Proof. Properness gives finitely many roots below a fixed height. Additivity implies that the roots of height $> H$ form a Lie ideal and that the quotient is nilpotent. PBW and Chevalley antisymmetrization are natural under quotient maps. Taking inverse limits in each finite total height therefore gives the displayed complete objects and preserves the coefficientwise quasi-isomorphism. □

The inverse limits in the preceding theorem use the generator-quotient topology. They retain its distinction from total-height completion. Each finite Lie quotient also defines the cotangent BV model above. Positive height makes every even adjoint operator strictly upper triangular, and every odd adjoint operator has zero supertrace. Its modular term vanishes, so this finite model satisfies the quantum master equation.

The ordinary algebraic interval uses $U(\mathfrak{n}_+^{(H)})$ and its two augmentation modules, with derived amplitude $C_*(\mathfrak{n}_+^{(H)})$. A local holomorphic–topological theory has the additional coefficient complex $\Omega^{0,*}(X)$. Contraction in the interval direction leaves that complex. A quantum identification of its boundary and interval observables therefore requires a map preserving the field differential, support and boundary polarizations. The ordinary bar identity alone does not provide this map.

Chapter 82

Character, bracket, parity, and orientation

A character is an Euler invariant. Faithful reconstruction combines it with the bracket, parity decomposition, and orientation line carried by the intended object.

Theorem 82.1 (Three reconstruction fibres)**.** *The following fibres contain distinct structured objects with the same scalar image.*

(i) *The PBW-character fibre of a graded vector space with basis x, y, z in degrees a, b, $a + b$ contains both the abelian Lie algebra and the Heisenberg Lie algebra $[x, y] = z$.*

(ii) *The superdimension-zero fibre contains both the zero super vector space and $\Bbbk^{1|1}$.*

(iii) *The squaring fibre of a line-valued section contains s and $s \otimes \chi$ for every sign local system χ.*

Consequently a faithful reconstruction datum consists respectively of a bracket, a parity-refined dimension, and an oriented square-root line.

Proof. PBW identifies the graded vector-space character of an enveloping algebra with the symmetric-superalgebra character of its underlying graded Lie space, so the two displayed brackets have the same PBW character. The two super vector spaces in the second item both have superdimension zero and have parity dimensions $(0, 0)$ and $(1, 1)$. Finally $\chi^{\otimes 2} \cong \mathbf{1}$, hence

$$(s \otimes \chi)^{\otimes 2} \cong s^{\otimes 2}.$$

The extra structures listed in the conclusion distinguish each pair. □

Recognition Theorem 82.2 (Minimal PBW recognition)**.** *Let $\phi : \mathfrak{g} \twoheadrightarrow \mathfrak{h}$ be a surjective morphism of positively graded Lie superalgebras with finite-dimensional homogeneous pieces. If the parity-refined Hilbert series of $U(\mathfrak{g})$ and $U(\mathfrak{h})$ agree, then ϕ is an isomorphism. Equality of signed PBW characters suffices when the parity of every primitive degree is prescribed in advance.*

Proof. If the kernel is nonzero, choose its first positive degree. In that degree all decomposable PBW contributions agree by minimality, while the primitive even or odd dimension in $U(\mathfrak{h})$ is strictly smaller. This contradicts equality of parity-refined Hilbert series. With prescribed parity, the formal logarithm of the PBW product recovers the signed primitive dimension inductively in height, and prescribed parity recovers the ordinary dimension. □

Chapter 83

Exact dependency theorem

Theorem 83.1 (Ordered-chiral closure)**.** *Fix:*

(i) *a six-functor Ran category and its factorization subcategory;*

(ii) *a transversely ordered algebra, an associative chiral algebra, or a chiral Lie algebra, as appropriate;*

(iii) *every mixed distributive, completion, duality, trace, or analytic datum named by the intended construction.*

Then the three bars, their complete cobars, their module/comodule Koszul transforms, the Hochschild centre, the formal-disc realization, the ribbon theory, and the finite-height current realization are defined functorially and satisfy the identities proved in this volume. Every passage between the pointwise and chiral-convolution monoidal products is carried by the displayed mixed distributive datum.

Proof. Each output is one of the functors constructed in the preceding chapters. The source and target monoidal categories are displayed in every definition. The comparison theorems invoke the explicitly supplied distributive, PBW, completion, duality, or trace datum. Composition of these functors gives the asserted functorial system. □

Chapter 84

Synthesis

The invariant language has three algebraic axes. Transverse order is an E_1 structure for $\otimes^!$. Associative chirality is an E_1 structure for $\star^{\mathrm{ch}}$. Chiral Lie theory is a Lie structure for $\star^{\mathrm{ch}}$. Their bars are $\mathrm{Bar}^{\perp}$, $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}$, and $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}$. The aligned projector supplies the exact sector on which a mixed distributive law is defined. Every comparison in the theorem system factors through one of these typed maps.

The constructed operations form the two functorial rows

$$\text{HT boundary theory} \xrightarrow{\mathrm{Obs}_{\partial}} \mathrm{Alg}_{\mathsf{E}_1}(\mathrm{Fact}_D(X)) \xrightarrow{\mathrm{Bar}^{\mathrm{ord}}} \mathrm{Coalg}_{\mathrm{conil,cpl}}(\mathrm{Fact}_D(X)),$$

$$\mathrm{Alg}_{\mathsf{E}_1}(\mathrm{Fact}_D(X)) \xrightarrow{Z_{\mathrm{ord}}} \mathrm{Alg}_{\mathsf{E}_2}(\mathrm{Fact}_D(X)).$$

A primitive Lie algebra enters when a separate Chevalley or Hall comparison identifies the bar coalgebra with the Chevalley chains of that Lie algebra. Hall geometry supplies this comparison through extension correspondences. Quantum groups require an additional coproduct and exchange structure. Automorphic forms arise from graded Euler characters after a grading and completion have been fixed. Each arrow therefore preserves the geometric operation named at its source.

Book XIV

Technical appendices to the preceding book

Chapter 85

Sign calculation for the length-three bar

For homogeneous $a, b, c \in \bar{A}$, write $[a|b|c] = sa \otimes sb \otimes sc$. With the standard suspension convention,

$$b_m[a|b|c] = (-1)^{|a|}[ab|c] + (-1)^{|a|+|b|+1}[a|bc].$$

Applying b_m again gives

$$b_m^2[a|b|c] = (-1)^{|b|}\big[(ab)c - a(bc)\big],$$

which vanishes by associativity.

Chapter 86

Rational R-matrix verification

Let P_{ij} permute tensor factors i and j. Expanding

$$R_{12}(u)R_{13}(u+v)R_{23}(v)$$

and its reversed product shows equality coefficientwise. The cubic coefficient uses

$$P_{12}P_{13}P_{23} = P_{23}P_{13}P_{12},$$

the quadratic coefficients use the symmetric-group relations, and the linear coefficients agree termwise.

Chapter 87

Carriers and comparison maps

The five-carrier geometry, transverse bar construction, mixed-forest signs, and noncommutative examples use four kinds of categorical data:

(i) a factorization carrier is paired explicitly with its operation;

(ii) the Ran category is fixed before its tensor products are used;

(iii) the block-ordered operad is introduced after a representing chiral operad is chosen;

(iv) ribbon and stable-curve sewing are derived from nondegenerate Calabi–Yau and modular data.

A comparison between these constructions therefore specifies its source and target categories and the operations preserved by its map.

Chapter 88

Enveloping bars and planar path sequences

The dimension of a bar chain space and the dimension of its homology are different invariants. For an enveloping algebra, the Chevalley comparison gives a Lie-theoretic model for the homology. This makes it possible to compare Motzkin and Riordan path counts with the ordered bar on a specified carrier.

Bar length and Chevalley degree are homological degrees: H_n denotes degree-n homology. The cohomology of the dual Chevalley complex is written H^n. For finite-dimensional weight pieces their dimensions agree.

88.1 Chevalley comparison

Theorem 88.1 (Bar–Chevalley comparison)**.** *Let k be a field of characteristic zero and let $\mathfrak{g}$ be an ordinary Lie algebra over k. Give $U(\mathfrak{g})$ its augmentation killing $\mathfrak{g}$. There is a natural quasi-isomorphism of ordinary conilpotent differential graded coalgebras*

$$\kappa : C_*(\mathfrak{g}) \longrightarrow \mathrm{Bar}^{\perp}\big(U(\mathfrak{g})\big),$$

where both constructions use direct sums and ordinary tensors over k. It preserves the PBW filtration. If $\mathfrak{g} = \bigoplus_{w \geq 1} \mathfrak{g}_w$ and $[\mathfrak{g}_u, \mathfrak{g}_v] \subseteq \mathfrak{g}_{u+v}$, the map is a quasi-isomorphism in each total weight. It extends to the degreewise product over weights, using completed tensor products by total weight. The completed coalgebras are conilpotent in each weight; no global conilpotence is asserted for infinite sums.

Proof. Put $C_n(\mathfrak{g}) = \Lambda^n \mathfrak{g}$ in homological degree n, with differential

$$d(x_1 \wedge \cdots \wedge x_n) = \sum_{i<j} (-1)^{i+j+1} [x_i, x_j] \wedge x_1 \wedge \cdots \widehat{x_i} \cdots \widehat{x_j} \cdots \wedge x_n.$$

Its coproduct splits the exterior factors between two ordered sublists, with the sign of the corresponding permutation. The ordinary normalized bar differential joins positions $i, i+1$ with sign $(-1)^{i-1}$. Define

$$\kappa(x_1 \wedge \cdots \wedge x_n) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) [x_{\sigma(1)} | \cdots | x_{\sigma(n)}].$$

At degree zero this map preserves the unit. At degree two its differential is $[x_1 x_2 - x_2 x_1] = [[x_1, x_2]]$. In every degree, pair permutations whose merged adjacent entries occur in opposite orders. Their difference is the bracket of those entries, with the displayed exterior sign. This proves the chain-map equation. Cutting an antisymmetrized word amounts to choosing an ordered subset and antisymmetrizing its two subwords. The two procedures give the same permutations and signs, proving the coalgebra identity.

Filter $U(\mathfrak{g})$ by PBW degree, the bar by total PBW degree, and $\Lambda^n \mathfrak{g}$ by degree n. PBW identifies the associated graded algebra with $S(\mathfrak{g})$. The associated graded comparison is the antisymmetrization $\Lambda(\mathfrak{g}) \to B(S(\mathfrak{g}))$, with zero differential on its source. To verify that it is a quasi-isomorphism, consider the free Koszul resolution $S(V) \otimes \Lambda(V)$ for $V = \mathfrak{g}$. For a basis (x_i) of V, its differential is $\partial = \sum_i x_i \iota_i$, where ι_i contracts the exterior factor x_i. On polynomial degree p and exterior degree q, with $p + q > 0$, the map

$$h = \frac{1}{p+q} \sum_i (x_i \wedge -) \, \frac{\partial}{\partial x_i}$$

satisfies $\partial h + h\partial = 1$. Only finitely many summands act on each element, even when V is infinite-dimensional. The identity follows by cancellation of cross terms and the Euler identity in the two degrees. Thus this free resolution has augmentation homology k only. The free normalized bar resolution has the same homology by its extra-degeneracy contraction. Antisymmetrization defines an augmentation-preserving comparison between these free resolutions. The simplicial bar sign convention is related to the displayed bar convention by multiplication by $(-1)^n$ in degree n. Free bases permit a reverse comparison and homotopies to be lifted inductively, since every positive-degree cycle in a resolution is a boundary. Tensoring these homotopies with k proves the asserted quasi-isomorphism on associated graded complexes.

Every cycle in the cone of κ has finite PBW filtration degree. A primitive for its highest associated graded part lowers that degree; iteration terminates because the filtration is bounded below. The cone is acyclic, proving the ordinary assertion. For a positive weight grading, PBW degree is at most total weight. The same argument is therefore finite within each weight. Finally, kernels and images of a degreewise product differential are products of kernels and images, so the componentwise quasi-isomorphisms extend to the stated completion. □

The ordinary comparison does not itself construct a morphism in $\mathrm{FactD}(X)$. For geometric coefficient systems, one must give compatible enveloping and Chevalley constructions on every chart and coherent factorization equivalences for the whole Chevalley complex. On each disjointness locus, with restriction denoted by $j^!$, the required equation is

$$f_{B,I}\, j^! \kappa_I = (\kappa_{I_1} \boxtimes \kappa_{I_2})\, f_{C,I}, \qquad I = I_1 \sqcup I_2.$$

Here $f_{C,I}$ and $f_{B,I}$ are the specified Chevalley and bar factorization maps. The bar map may use the signed shuffle comparison under its geometric functor hypotheses. The Chevalley factorization maps, collision compatibility, refinement coherence, and continuity of the completed maps remain additional geometric data to construct or verify. Internal tensor formulas and PBW alone do not provide them.

Calculation 88.2 (Finite-dimensional simple case)**.** For a simple Lie algebra of rank r with exponents $m_1, \dots, m_r$, the Chevalley cohomology is the exterior algebra on primitive classes of degrees $2m_i + 1$. The corresponding homology Poincaré polynomials are

$$\mathrm{Bar}^{\perp} U(\mathfrak{sl}_2) : 1 + t^3, \qquad \mathrm{Bar}^{\perp} U(\mathfrak{sl}_3) : 1 + t^3 + t^5 + t^8, \qquad \mathrm{Bar}^{\perp} U(\mathfrak{g}_2) : 1 + t^3 + t^{11} + t^{14}.$$

The corresponding Betti sequences for $\mathfrak{sl}_2$ and $\mathfrak{sl}_3$ are $1, 0, 0, 1$ and $1, 0, 0, 1, 0, 1, 0, 0, 1$. For a finite-dimensional Lie algebra, the Chevalley chain dimensions are $\binom{\dim \mathfrak{g}}{n}$.

For an ordinary augmented algebra A over $\mathbb{C}$, let $I = \ker(A \to \mathbb{C})$. Its normalized two-sided bar with trivial coefficients has homological chain spaces

$$B_0(A) = \mathbb{C}, \qquad B_n(A) = I^{\otimes n} \quad (n \geq 1).$$

Here n is bar length, independently of any internal weight or PBW degree. If I has finite dimension d, then $\dim B_n(A) = d^n$; in particular, an augmentation ideal of dimension three gives the chain counts 3^n. In a cohomological convention that places B_n in degree $-n$, its homology is denoted H^{-n} instead of H_n.

For $A = U(\mathfrak{sl}_2)$, PBW gives the basis $e^a h^b f^c$, with $a, b, c \geq 0$, where $[h, e] = 2e$, $[h, f] = -2f$, and $[e, f] = h$. The augmentation ideal has all these basis elements except 1. Consequently $B_n(A)$ is countably infinite-dimensional for every $n \geq 1$. Its total PBW filtration has finite-dimensional associated graded pieces, with generating function

$$\sum_{d \geq 0} \dim \mathrm{gr}^{\mathrm{PBW}}_d B_n(A)\, z^d = \big((1 - z)^{-3} - 1\big)^n.$$

In particular, 3^n is the dimension in the least possible total PBW degree $d = n$. These least-degree dimensions are 3, 9, 27, 81, 243 for $n = 1, 2, 3, 4, 5$; the full bar chain spaces are infinite-dimensional. The quotient of indecomposables is zero: the perfectness of $\mathfrak{sl}_2$ implies $I/I^2 = 0$.

The Chevalley chain model is different: its degree-n space is $\Lambda^n \mathfrak{sl}_2$, of dimensions 1, 3, 3, 1 in degrees 0, 1, 2, 3. With the convention $d(x \wedge y) = [x, y]$, the map $d : \Lambda^2 \mathfrak{sl}_2 \to \mathfrak{sl}_2$ sends

$$e \wedge h \longmapsto -2e, \qquad e \wedge f \longmapsto h, \qquad h \wedge f \longmapsto -2f,$$

and is an isomorphism. The next differential vanishes, since

$$d(e \wedge h \wedge f) = -2e \wedge f - h \wedge h - 2f \wedge e = 0.$$

Together with $d : \mathfrak{sl}_2 \to \mathbb{C}$ equal to zero, this gives Chevalley homology dimensions 1, 0, 0, 1. The ordinary bar–Chevalley comparison identifies these homology groups with those of $U(\mathfrak{sl}_2)$; it does not identify the two chain spaces.

88.2 The Virasoro row

Choose as transverse algebra the enveloping algebra of the positive Witt algebra

$$L_1 = \bigoplus_{n \geq 1} \mathbb{C}\, e_n, \qquad [e_i, e_j] = (j - i) e_{i+j}.$$

Theorem 88.1 applies to this positive grading. Its weight spaces are one-dimensional, and each weight has finitely many decompositions.

Theorem 88.3 (Pentagonal rigidity)**.** *The bar homology of $U(L_1)$ satisfies*

$$\dim H_n\big(\mathrm{Bar}^{\perp} U(L_1)\big) = 2 \qquad \text{for every } n \geq 1,$$

one dimension in each of the weights $(3n^2 - n)/2$ and $(3n^2 + n)/2$. Its graded Euler character is

$$\chi_q\big(\mathrm{Bar}^{\perp} U(L_1)\big) = \prod_{n \geq 1} (1 - q^n), \tag{88.1}$$

and (88.1) *is accounted for degree by degree with no residual freedom.*

Proof. By Theorem 88.1 the bar of $U(L_1)$ is quasi-isomorphic to $C_*(L_1)$. Each weight subcomplex is finite-dimensional, so its homology has the same dimensions as the weightwise dual cohomology. Goncharova's calculation gives two dimensions in each degree $n \geq 1$, in the weights $(3n^2 \mp n)/2$. The precise reference and weightwise duality are given in Chapter 19.

Identity (88.1) is the denominator product applied to L_1, where every root space is one dimensional and even, so $\mathrm{smult}(L_1)_n = 1$. Euler's pentagonal number theorem expands the product as $\sum_{m \in \mathbb{Z}} (-1)^m q^{m(3m-1)/2}$, whose support is exactly $\{(3n^2 \pm n)/2\}$ with coefficient $(-1)^n$ at each of $(3n^2 - n)/2$ and $(3n^2 + n)/2$. Goncharova's theorem supplies the degree and multiplicity of every class. The Euler product agrees with those classes. Its coefficients alone would not exclude extra classes cancelling between homological degrees. □

For $n = 1, 2, 3, 4$, the nonzero homology groups occupy the weight pairs $(1, 2)$, $(5, 7)$, $(12, 15)$, $(22, 26)$, one dimension in each. The corresponding nonzero coefficients of $\prod_{n \geq 1}(1 - q^n)$ occur at 0, 1, 2, 5, 7, 12, 15, 22, 26 with signs $+, -, -, +, +, -, -, +, +$.

88.3 Planar path sequences

A Motzkin path is a lattice path from $(0,0)$ to $(n,0)$, staying at nonnegative height, with steps $(1,1)$, $(1,0)$, and $(1,-1)$. Let $M(n)$ count all such paths of length n, and let $R(n)$ count those with no horizontal step at height zero. Both counts include the empty path: $M(0) = R(0) = 1$, while $M(1) = 1$ and $R(1) = 0$. For $n \geq 1$ consider the two sequences

$$a_n = M(n+1) - M(n) = 1, 2, 5, 12, 30, 76, \ldots, \qquad b_n = R(n+3) = 3, 6, 15, 36, 91, \ldots.$$

Proposition 88.4 (Motzkin and Riordan identities)**.** *For every $n \geq 0$,*

$$M(n) = R(n) + R(n+1), \qquad M(n+1) - M(n) = R(n+2) - R(n).$$

The ordinary generating functions of the two path counts, and hence those of a_n and b_n, belong to $\mathbb{Q}(z, \sqrt{1-2z-3z^2})$.

Proof. Work in the formal power-series ring $\mathbb{Q}[[z]]$, and write $M(z) = \sum_{n\geq 0} M(n)z^n$ and $R(z) = \sum_{n\geq 0} R(n)z^n$. A nonempty Motzkin path either starts with a horizontal step followed by an arbitrary Motzkin path, or has a unique first-return decomposition $UPDQ$, where U and D are the up and down steps, P and Q are arbitrary Motzkin paths, and P is drawn one unit above the horizontal axis. Thus

$$M = 1 + zM + z^2M^2.$$

A nonempty path counted by R has no initial horizontal step. Its first return similarly has the form $UPDQ$, with P arbitrary and Q counted by R. Consequently

$$R = 1 + z^2MR, \qquad R = (1 - z^2M)^{-1}.$$

The inverses here exist because their denominators have constant term one. Using $z^2M^2 = M - 1 - zM$, compute

$$(1 + zM)(1 - z^2M) = 1 + zM - z^2M - z^3M^2 = 1 + z.$$

It follows that $(1+z)R = 1 + zM$. Comparing coefficients of z^{n+1} gives $R(n+1) + R(n) = M(n)$ for all $n \geq 0$. Subtracting consecutive identities gives the difference formula.

Solving the quadratic equation for the formal series with constant term one gives

$$M(z) = \frac{1 - z - \sqrt{1-2z-3z^2}}{2z^2}, \qquad R(z) = \frac{1 + zM(z)}{1+z},$$

where the square root has constant term one. The apparent pole in the formula for M is removable as a formal series. Finite index shifts and differences preserve the indicated quadratic function field, proving the last assertion. These identities are consequences solely of the path decompositions. □

Remark 88.5 (Low-degree homology). Theorem 88.3 gives homology dimensions 2, 2, 2 in degrees 1, 2, 3, whereas a_1, a_2, a_3 are 1, 2, 5. These triples agree only in degree two. The degree-one discrepancy already excludes equality of the full homology dimension sequences.

Theorem 88.6 (Enveloping bars and planar path counts)**.** *Let $B_\bullet(A)$ be the ordinary normalized bar over $\mathbb{C}$, with trivial coefficients and homological degree given by bar length. With $a_n = M(n+1) - M(n)$ and $b_n = R(n+3)$ indexed by $n \geq 1$ as above,*

$$\dim H_1B_\bullet(U(\mathfrak{sl}_2)) = 0 \neq b_1 = 3, \qquad \dim H_1B_\bullet(U(L_1)) = 2 \neq a_1 = 1,$$

where $L_1 = \bigoplus_{m\geq 1} \mathbb{C}e_m$ and $[e_i, e_j] = (j-i)e_{i+j}$. Hence neither sequence is the bar homology dimension sequence of the corresponding enveloping algebra. Neither is its full bar chain dimension sequence: in both cases B_n is countably infinite-dimensional for every $n \geq 1$.

Proof. For any augmented algebra A with augmentation ideal I, the bar differential out of $B_1 = I$ is zero and the image from $B_2 = I \otimes I$ is I^2. Therefore

$$H_1 B_\bullet(A) = I/I^2.$$

For an ordinary Lie algebra $\mathfrak{g}$, the inclusion of generators induces an isomorphism

$$\mathfrak{g}/[\mathfrak{g},\mathfrak{g}] \xrightarrow{\sim} I/I^2, \qquad I = \ker(U(\mathfrak{g}) \to \mathbb{C}).$$

Indeed the generators span I/I^2, and $[x, y] = xy - yx$ vanishes there. For injectivity, equip $\mathbb{C} \oplus \mathfrak{g}/[\mathfrak{g},\mathfrak{g}]$ with the square-zero product on its second summand. The quotient map from $\mathfrak{g}$ to that summand is a Lie map into the commutator Lie algebra of this associative algebra. The universal property of $U(\mathfrak{g})$ extends it to an augmented algebra map, which kills I^2 and provides an inverse on the displayed quotient.

The brackets $[h, e] = 2e$, $[h, f] = -2f$, and $[e, f] = h$ show that $\mathfrak{sl}_2$ equals its commutator algebra. For L_1, every nonzero bracket has weight at least three. Conversely, for $m \geq 3$,

$$e_m = \frac{1}{m-2}[e_1, e_{m-1}].$$

Thus $[L_1, L_1] = \bigoplus_{m\geq 3} \mathbb{C}e_m$, and its abelianization has basis the classes of e_1, e_2, in weights one and two. This proves both degree-one assertions without using any higher-degree calculation.

PBW gives a countably infinite basis of the augmentation ideal of either enveloping algebra. In the L_1 case the basis consists of nonempty ordered monomials in the countable set $e_1, e_2, \ldots$, each monomial having finite length. Finite tensor powers of either augmentation ideal are therefore countably infinite-dimensional. The path counts are finite in every degree, so they cannot count these full chain spaces. □

In particular, a complex whose homology dimensions are b_n cannot be quasi-isomorphic to $B_\bullet(U(\mathfrak{sl}_2))$, and one whose homology dimensions are a_n cannot be quasi-isomorphic to $B_\bullet(U(L_1))$, with the stated degree convention. A discrepancy of chain dimensions alone would not imply this conclusion; the degree-one homology discrepancy does.

The weight grading gives a different, finite comparison. An ordered PBW monomial of weight w specifies a partition of w, so for $n \geq 1$,

$$\sum_{w\geq 0} \dim B_n(U(L_1))_w q^w = \left(\prod_{m\geq 1}(1-q^m)^{-1} - 1\right)^n.$$

Each coefficient is finite. Thus a weightwise dimension, a dimension after a specified weight cutoff, and an ungraded dimension are distinct invariants. In a weight completion, dimensions must likewise be interpreted weightwise; the countable algebraic dimensions above refer to the ordinary direct-sum enveloping algebras.

Remark 88.7 (Chiral coefficients and transverse multiplication). Independently of these scalar calculations, a Virasoro chiral coefficient does not determine the transverse bar: a fixed coefficient may be combined with free, square-zero, enveloping, or quantum-plane transverse algebras whose bars differ, while the coefficient and the central charge are unchanged. A chain-level comparison with an ordered bar therefore also requires a specified transverse multiplication and a quasi-isomorphism from the proposed complex to that bar.

Chapter 89

The obstruction constitution

Each statement below names what is *not* determined, and therefore what must be supplied. Without them the preceding theorems admit false corollaries.

Proposition 89.1 (Where a mixed distributive law can live)**.** *The sector on which a mixed distributive law between $\otimes^!$ and $\star^{\mathrm{ch}}$ is geometrically defined is the common-decomposition sector, the image of the aligned idempotent e_{al}. Off that sector the two products admit no interchange, only the aligned retraction, so no distributive law can be written.*

Proof. A distributive law requires a natural transformation interchanging the two products in a fixed direction, subject to the coherence conditions of each. The canonical comparison in the additive model is the aligned inclusion ι with retraction π, and $\pi\iota = \mathrm{id}$ while $\iota\pi = e_{\mathrm{al}} \neq \mathrm{id}$ in general. A natural transformation subject to both coherence conditions therefore exists only where e_{al} acts as the identity. □

This locates a mixed law without constructing one. No example in the present corpus exhibits one; Zamolodchikov–Faddeev algebras are the natural candidate, since their spectral variable belongs to the chiral direction while their word belongs to the transverse direction.

Theorem 89.2 (No quantum group from dimension loss)**.** *The dimension of the kernel of $V^{\otimes n} \to \mathrm{Sym}^n V$ determines no R-matrix, associator, or quantum group. That kernel carries no multiplication, coproduct, monodromy, or coherence equation.*

Proof. The kernel is a representation of Σ_n and nothing more. Two algebras with the same underlying graded vector space, hence the same kernel dimensions in every arity, may have different multiplications and inequivalent representation categories; a free algebra and a quadratic quotient with the same Hilbert series is an example. A quantum-group structure is a coproduct together with coherence data, none of which is a function of a dimension. □

Theorem 89.3 (No modular object from topology alone)**.** *The identity $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$ assigns no coefficient operations to boundary strata. A modular master element exists only after cyclic contractions and clutching-compatible coefficient maps have been supplied.*

Theorem 89.4 (No bulk from a boundary centre)**.** *The derived centre is terminal among algebraic central actions on the boundary algebra. It does not detect a decoupled bulk sector, so a physical bulk is not determined by the boundary chiral algebra unless a bulk-to-centre comparison is supplied and proved to be an equivalence.*

Proof. Terminality is the universal property of the centre. It is a statement about the boundary category and is insensitive to any sector acting trivially on it, so a bulk with trivial boundary action is invisible. □

Theorem 89.5 (No classification by central charge)**.** *Central charge determines the Virasoro anomaly coefficient. It does not determine the chiral operation, the representation category, the braid monodromy, or the higher-genus conformal blocks. Distinct holomorphic theories at equal central charge exist.*

Theorem 89.6 (No identification of anomalies of different types)**.** *A current-algebra level, a Virasoro central charge, a determinant-line curvature, a BRST ghost-number anomaly, and a Borcherds-product weight are invariants in different theories, living in different deformation complexes. They may be related by a theorem in a particular family. They are not one universal scalar, and no identity among them follows from numerical agreement.*

Proof. Each is a class in a distinct complex: an Atiyah-algebra extension class, a Virasoro cocycle, the curvature of a determinant line, a class in H^1 or H^2 of a BRST convolution complex, and a weight in the character lattice of an automorphic form. A map between two of these complexes is additional data; in its absence, equality of numbers is a coincidence of values, not an identification of classes. □

Theorem 89.7 (No unnamed comparison of bars)**.** *An asserted equality* $\mathrm{Bar}^{\perp}(A) = \mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A)$ *has no meaning without a functor carrying the pointwise multiplication problem to the chiral-convolution multiplication problem. The two have different underlying gradings, counting interval subdivisions and collision trees respectively. Agreement of Euler characteristics is not a comparison map.*

Theorem 89.8 (A scalar sequence is not homology data)**.** *A sequence of integers becomes bar homology only after a chain complex, a differential, a grading convention, and a rank computation or quasi-isomorphism are specified. Matching initial values, discriminants, or growth rates constructs no such map.*

Proof. Chain-space dimensions and homology dimensions differ by the ranks of the differentials, which are not functions of the chain dimensions. For $U(\mathfrak{sl}_2)$, Calculation 88.2 gives the least-PBW-degree bar dimensions 3, 9, 27, 81, 243. Its full positive bar chain spaces are infinite-dimensional, while its homology dimensions are 1, 0, 0, 1. Proposition 88.4 exhibits two sequences with the same discriminant and growth rate that are transforms of one another and of neither homology. □

Theorem 89.6 together with the central-charge statement retracts any programme treating a tuple of scalars drawn from these sources as a single classifying invariant, and any trace identity asserted among them without a named comparison morphism.

Book XV

The automorphic determinant

Chapter 90

The protected Jacobi input

The automorphic product, its root-algebra denominator and its Fredholm model have specified carriers and normalizations. Comparing them with boundary operators and Hall correspondences requires maps preserving charge, parity and the scalar trace.

90.1 The K3 elliptic genus

Let S be a K3 surface. The right-moving supersymmetric index of a unitary $\mathcal{N}=(4,4)$ sigma model is the elliptic genus

$$Z_{K3}(\tau,z)=2\phi_{0,1}(\tau,z).$$

The normalized weak Jacobi form has weight zero, index one, and Fourier expansion

$$\phi_{0,1}(\tau,z)=\sum_{n\geq 0}\sum_{\ell\in\mathbb{Z}} f(n,\ell)q^{n}r^{\ell}.$$

In Eichler–Zagier normalization,

$$\phi_{0,1}(\tau,z)=4\sum_{a=2}^{4}\frac{\vartheta_a(\tau,z)^2}{\vartheta_a(\tau,0)^2}\in J^{\mathrm{w}}_{0,1}$$

[29]. The coefficient depends on the discriminant $4n-\ell^2$ and on ℓ modulo 2.

Theorem 90.1 (Exact low Fourier coefficients)**.** *The first four q-rows are*

$$\begin{aligned}\phi_{0,1}=\;&r^{-1}+10+r\\&+q\left(10r^{-2}-64r^{-1}+108-64r+10r^{2}\right)\\&+q^{2}\left(r^{-3}+108r^{-2}-513r^{-1}+808-513r+108r^{2}+r^{3}\right)\\&+q^{3}\left(-64r^{-3}+808r^{-2}-2752r^{-1}+4016\right.\\&\qquad\left.-2752r+808r^{2}-64r^{3}\right)+O(q^{4}).\end{aligned}$$

In particular,

$$f(0,0)=10,\qquad \frac{f(0,0)}{2}=5.$$

Proof. Insert the theta-series expansions into the preceding quotient formula. Work with $t=q^{1/8}$ and $x=r^{1/2}$; each quotient is an exact formal power series in t with Laurent-polynomial coefficients in x. The fractional powers cancel in the sum. Division through t^{24} gives the displayed rows. The arithmetic calculation is reproduced in Appendix 121. □

90.2 The virtual half-index

For every (n, ℓ) choose a finite-dimensional super vector space $U_{n,\ell}$ satisfying

$$\operatorname{sdim} U_{n,\ell} = f(n, \ell).$$

The integer $f(n, \ell)$ is its superdimension. In the ordinary Grothendieck group of finite-dimensional super vector spaces, the even and odd dimensions remain separate. The signed integer is their image under superdimension. The determinant depends only on that image.

Lemma 90.2 (Fock character and Berezinian)**.** *Let $V = \bigoplus_{\gamma} V_{\gamma}$ be a locally finite graded super vector space. Then*

$$\operatorname{sch}\big(\operatorname{Sym}(V_{\bar{0}}) \otimes \bigwedge(V_{\bar{1}})\big) = \prod_{\gamma} (1 - x^{\gamma})^{-\operatorname{sdim} V_{\gamma}},$$

and

$$\operatorname{sdet}_{V}(1 - x) = \prod_{\gamma} (1 - x^{\gamma})^{\operatorname{sdim} V_{\gamma}}.$$

Proof. An even one-dimensional summand contributes the geometric series $(1 - x^{\gamma})^{-1}$. An odd one-dimensional summand contributes $1 - x^{\gamma}$. Multiplication over a homogeneous basis proves both identities. □

Chapter 91

The Borcherds product

91.1 The effective chamber

Define

$$\Gamma_{\mathrm{eff}} = \{(n,\ell,m) \in \mathbb{Z}^3 : m > 0,\ n \geq 0\} \cup \{(n,\ell,0) : n > 0\} \cup \{(0,\ell,0) : \ell < 0\}.$$

The ordering is lexicographic in $(m, n, -\ell)$ and is precisely the ordering selected by the cusp chamber. For $\gamma = (n,\ell,m)$ put

$$V_\gamma = U_{nm,\ell}, \qquad \operatorname{sdim} V_\gamma = f(nm,\ell).$$

Definition 91.1 (Monic virtual determinant)**.** The monic determinant of the half-index is

$$D_5(Z) = q^{1/2} r^{1/2} s^{1/2} \operatorname{sdet}_V(1 - x) = q^{1/2} r^{1/2} s^{1/2} \prod_{\Gamma_{\mathrm{eff}}} (1 - q^n r^\ell s^m)^{f(nm,\ell)}.$$

Theorem 91.2 (Borcherds–Gritsenko–Nikulin product)**.** *The preceding product converges normally on the chosen cusp chamber and extends to a holomorphic genus-two modular form of weight five with a character of order two. It is the monic Igusa form D_5. The theta-normalized form is*

$$\Delta_5^{\theta} = 2^6 D_5.$$

Proof. The singular theta lift of the vector-valued form corresponding to $\phi_{0,1}$ produces the product, divisor, weight, and automorphic character. The weight is $f(0,0)/2 = 5$. The genus-two identification with the Igusa form and the product formula are the Gritsenko–Nikulin theorems [7, 8, 38, 40]. The product of the ten even theta constants has coefficient 2^6 at $q^{1/2} r^{1/2} s^{1/2}$, fixing the scalar. □

91.2 The theta constant normalization

Let $\mathfrak{E}$ be the ten pairs $a, b \in \{0,1\}^2$ with $a \cdot b$ even. Use the characteristic convention

$$\vartheta[a,b](Z,0) = \sum_{n \in \mathbb{Z}^2} \exp\!\big(\pi i \big[(n + a/2)^{\mathsf{T}} Z (n + a/2) + b^{\mathsf{T}} n\big]\big).$$

Replacing $b^{\mathsf{T}} n$ by $b^{\mathsf{T}}(n + a/2)$ changes the sign of the complete ten-factor product. Define

$$\Delta_5^{\theta}(Z) = \prod_{\mathfrak{m} \in \mathfrak{E}} \vartheta[\mathfrak{m}](Z,0).$$

In the chosen cusp order, the ten factors contribute six factors of 2 and total exponent $(1/2, 1/2, 1/2)$. At the lowest q, s orders the complete coefficient is $64(r^{1/2} - r^{-1/2})$. Thus

$$[q^{1/2} r^{1/2} s^{1/2}] \Delta_5^{\theta} = 64.$$

This coefficient selects a canonical scalar normalization of the automorphic section.

Proposition 91.3 (Ten-factor leading-term calculation)**.** *Group the ten even characteristics by the first half-characteristic $a \in (\mathbb{Z}/2)^2$. Their leading contributions in the chosen cusp chamber are*

a	#	*scalar*	*monomial*
$(0,0)$	4	1	1
$(1,0)$	2	2	$q^{1/8}$
$(0,1)$	2	2	$s^{1/8}$
$(1,1)$	2	2	$q^{1/8}r^{1/4}s^{1/8}$.

Multiplication gives

$$1^4\,2^2\,2^2\,2^2\,q^{1/2}r^{1/2}s^{1/2} = 64q^{1/2}r^{1/2}s^{1/2}.$$

Proof. Insert the declared characteristic phases into the theta series. For $a = (0,0)$ the zero lattice vector contributes one. The $a = (1,0)$ and $a = (0,1)$ factors contribute $2q^{1/8}$ and $2s^{1/8}$ respectively. For $a = (1,1)$ the two leading expressions are $2q^{1/8}s^{1/8}(r^{1/4} + r^{-1/4})$ and $2q^{1/8}s^{1/8}(r^{1/4} - r^{-1/4})$. Their product gives the stated q, s coefficient, whose leading r term in the cusp order gives the table. Proposition 20.2 includes both phases. □

Proposition 91.4 (Boundary factor of the Borcherds product)**.** *The $m = 0$ part of the monic product is*

$$\prod_{n>0,\ell\in\mathbb{Z}}(1-q^n r^\ell)^{f(0,\ell)}\prod_{\ell<0}(1-r^\ell)^{f(0,\ell)} = (1-r^{-1})\prod_{n\geq 1}(1-q^n r^{-1})(1-q^n)^{10}(1-q^n r).$$

Consequently the first Fourier–Jacobi coefficient is

$$q^{1/2}r^{1/2}(1-r^{-1})\prod_{n\geq 1}(1-q^n r^{-1})(1-q^n)^{10}(1-q^n r) = \eta(\tau)^9\vartheta_{11}(\tau,z),$$

where the odd theta function is fixed by the exact convention

$$\vartheta_{11}(\tau,z) = -q^{1/8}r^{-1/2}\prod_{n\geq 1}(1-q^{n-1}r)(1-q^n r^{-1})(1-q^n).$$

Proof. The q^0 row of $\phi_{0,1}$ has exactly three nonzero coefficients: $f(0,-1) = 1$, $f(0,0) = 10$, and $f(0,1) = 1$. Substitution into the $m = 0$ factors gives the first formula. The Jacobi triple product and $\eta(\tau) = q^{1/24}\prod_{n\geq 1}(1-q^n)$ give the second, including its leading sign and monomial. □

Corollary 91.5 (The scalar square)**.** *The monic weight-ten form is*

$$\chi_{10} = {D_5}^2 = 2^{-12}\Delta_5^{\theta^2}.$$

Its automorphic character is trivial.

Proof. The identity $\Delta_5^\theta = 2^6 D_5$ gives the scalar factor. The character of D_5 has order two, so its square is trivial. □

Chapter 92

Divisor, character, and the meaning of the square root

The diagonal Humbert divisor is

$$H_1 = \{Z \in \mathbb{H}_2 : z = 0\}/\mathrm{Sp}_4(\mathbb{Z}).$$

Theorem 92.1 (Diagonal divisor)**.** *The divisor of D_5 is the $\mathrm{Sp}_4(\mathbb{Z})$-orbit of H_1, with multiplicity one. The form transforms by the Maass character ν_5 of order two, and $\chi_{10} = D_5^2$ is scalar-valued.*

Proof. This is the defining diagonal-divisor property of the Igusa square root and the first row of the Cléry–Gritsenko classification [16, 45, 46]. □

Real roots of the denominator algebra define Weyl reflections. Imaginary roots contribute root spaces and product exponents. This division of labor is intrinsic to a Borcherds algebra: the Weyl group is generated by real simple roots, while the imaginary simple roots enter the correction term of the denominator identity.

92.1 Automorphic section, orientation line, and Pfaffian operator

An automorphic square root is a section of a line bundle with character. An orientation is a square root of a determinant line. A Pfaffian operator is a skew family whose regularized Pfaffian is a section of that orientation line. These form a sequence of successive refinements:

$$\text{automorphic section} \longrightarrow \text{determinant-line identification}$$
$$\longrightarrow \text{oriented square-root line} \longrightarrow \text{Fredholm family and Pfaffian section.}$$

Proposition 92.2 (Logical separation of the four levels)**.** *The product theorem constructs the first object in the sequence. A determinant-line comparison constructs the second. A coherent square root of that line constructs the third. A first-order operator requires a symbol, domain, ellipticity or Fredholm condition, and regularization, and thereby produces the fourth refinement.*

Proof. Each stage adds a specified geometric structure. A determinant realization adds an isomorphism from a geometric determinant line to the automorphic line. An oriented realization adds a coherent square-root line and its parity refinement. An operator realization adds a principal symbol, a Fredholm domain, a spectrum, and a regularization whose Pfaffian is the resulting section. □

The title names the monic automorphic section D_5, characterized by the exact identity $D_5^2 = \chi_{10}$. Part IV formulates an analytic Pfaffian realization as a further geometric refinement.

Book **XVI**

The denominator algebra

Chapter 93

The Lorentzian root lattice

Let

$$L = \mathbb{Z} f_2 \oplus \mathbb{Z} f_3 \oplus \mathbb{Z} f_{-2}$$

with bilinear form

$$(f_2, f_{-2}) = -1, \qquad (f_3, f_3) = 2,$$

and all remaining basis pairings zero. Its signature is $(2, 1)$. Define the type-II root lattice

$$L_{II} = 2\mathbb{Z} f_2 \oplus \mathbb{Z} f_3 \oplus 2\mathbb{Z} f_{-2} \cong U(4) \oplus \langle 2 \rangle.$$

The Gram-index map is

$$\alpha : \Gamma_{\mathrm{ind}} = \mathbb{Z}^3 \longrightarrow L_{II}, \qquad \alpha(n, \ell, m) = 2n f_2 - \ell f_3 + 2m f_{-2}.$$

Then

$$(\alpha(n, \ell, m), \alpha(n', \ell', m')) = -2\big(2nm' + 2n'm - \ell\ell'\big),$$

and

$$(\alpha(n, \ell, m), \alpha(n, \ell, m)) = -2(4nm - \ell^2).$$

93.1 Real simple roots and Weyl vector

Put

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3.$$

Proposition 93.1 (Type-II Cartan matrix)**.** *The Gram matrix of the real simple roots is*

$$\big((\delta_i, \delta_j)\big)_{i,j=1}^{3} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

The reflections

$$s_i(x) = x - (x, \delta_i)\delta_i$$

are integral involutions preserving L_{II} and its form.

Proof. Substitution into the basis pairing gives the matrix. Since $(\delta_i, \delta_i) = 2$, the reflection formula has integral coefficients. Direct expansion gives $s_i^2 = 1$ and $(s_i x, s_i y) = (x, y)$. □

The Weyl vector is

$$\rho = f_2 - \frac{1}{2} f_3 + f_{-2}, \qquad (\rho, \delta_i) = -1.$$

In the Gram-index variables, $e^{-\rho} = q^{1/2} r^{1/2} s^{1/2}$.

Chapter 94

The Gritsenko–Nikulin Lie superalgebra

The Gritsenko–Nikulin construction specifies a generalized Kac–Moody Lie superalgebra $\mathfrak{g}_{\Delta_5}$, with real simple roots δ_1, δ_2, δ_3 and the imaginary correction attached to $\phi_{0,1}$. Its generators and relations supply the bracket whose denominator is compared with the Borcherds product.

Theorem 94.1 (Root supermultiplicities and denominator)**.** *On the active positive support,*

$$\operatorname{smult} \mathfrak{g}_{\Delta_5\,\alpha(n,\ell,m)} = f(nm,\ell).$$

In the Lorentzian coordinate convention of Gritsenko–Nikulin,

$$\operatorname{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z) = 2^{-6}\Delta_5^{\theta}(2Z).$$

Proof. The real roots and Weyl chamber are the type-II case of the Gritsenko–Nikulin construction. The imaginary simple-root correction is chosen so that its Weyl–Kac–Borcherds product has exponent $f(nm,\ell)$. The resulting product is the doubled-variable monic Igusa form [38–40]. □

The theorem gives a Lie superalgebra with a bracket, Cartan, Weyl group, root parity, and denominator. The scalar product alone records the signed dimensions of the root spaces; the Gritsenko–Nikulin construction supplies the algebraic operations.

Chapter 95

The ordered bar as the denominator complex

Let $\mathfrak{n}_{\Delta_S,+}$ be the positive root subalgebra of $\mathfrak{g}_{\Delta_S}$. Fix a proper integral height h on the positive root cone and define

$$F^{>H}\mathfrak{n}_{\Delta_S,+} = \bigoplus_{h(\alpha)>H} \mathfrak{g}_{\Delta_S\,\alpha}, \qquad {\mathfrak{n}_{\Delta_S,+}}^{(H)} = \mathfrak{n}_{\Delta_S,+}/F^{>H}\mathfrak{n}_{\Delta_S,+}.$$

Height additivity makes $F^{>H}\mathfrak{n}_{\Delta_S,+}$ an ideal. Properness and local finiteness make ${\mathfrak{n}_{\Delta_S,+}}^{(H)}$ finite-dimensional and nilpotent.

Theorem 95.1 (Height-complete Bar–Chevalley comparison)**.** *For every H, super antisymmetrization gives a natural quasi-isomorphism of conilpotent dg coalgebras*

$$C_*({\mathfrak{n}_{\Delta_S,+}}^{(H)}) \longrightarrow \mathrm{Bar}\big(U({\mathfrak{n}_{\Delta_S,+}}^{(H)})\big).$$

The maps commute with the height quotients. For an entire root-graded algebra or complex V, define its total-height completion by $\widehat{V}_h = \prod_{d\geq 0} V_d$, with completed tensor product graded by total height. Then the comparison also gives

$$C^{\wedge}_{*,h}(\mathfrak{n}_{\Delta_S,+}) := \widehat{C_*(\mathfrak{n}_{\Delta_S,+})}_h \longrightarrow \mathrm{Bar}^{\wedge}_h(U(\mathfrak{n}_{\Delta_S,+})) := \mathrm{Bar}(\widehat{U(\mathfrak{n}_{\Delta_S,+})})_h$$

as a quasi-isomorphism of complete dg coalgebras. Its Euler supercharacter is

$$\chi_{\mathrm{root}} C^{\wedge}_{*,h}(\mathfrak{n}_{\Delta_S,+}) = \prod_{\alpha>0} (1-e^{-\alpha})^{\mathrm{sdim}\,\mathfrak{g}_{\Delta_S\,\alpha}}.$$

Proof. At each fixed total height, only finitely many generators and tensor lengths occur. The PBW filtration is finite, and its associated graded comparison is the Koszul comparison for the symmetric superalgebra. It is the tensor product of the polynomial and exterior one-generator comparisons. Finite filtration gives the quasi-isomorphism at that height. Exactness of products of vector spaces passes it to the completion.

An even root vector contributes $1-e^{-\alpha}$ to the Euler supercharacter of Chevalley chains, and an odd root vector contributes its reciprocal. This proves the product, as in Proposition 20.3.

The separate inverse limit $\varprojlim_H \mathrm{Bar}(U({\mathfrak{n}_{\Delta_S,+}}^{(H)}))$ also compares with $\varprojlim_H C_*({\mathfrak{n}_{\Delta_S,+}}^{(H)})$. The transition maps of the complexes and their comparison cones are degreewise surjective, so the inverse-limit exact sequence for products proves this assertion. It is a generator-quotient limit, not the total-height completion above. For a one-dimensional even abelian algebra of height one, all nonzero generator quotients give $\mathbb{C}[x]$, whereas total-height completion gives $\mathbb{C}[[x]]$. □

Let $\mathbb{T}_{L_{II}}$ be the formal root torus and write Z_{root} for its logarithmic coordinate, normalized by

$$e^{-\alpha(n,\ell,m)} = Q^n R^\ell S^m, \qquad e^{-\rho} = Q^{1/2} R^{1/2} S^{1/2}.$$

Corollary 95.2 (Bar denominator identity)**.** *On the formal root torus,*

$$e^{-\rho}\chi_{\mathrm{root}}C^{\wedge}_{*,h}(\mathfrak{n}_{\Delta_5,+}) = D_5(Z_{\mathrm{root}}).$$

The Gritsenko–Nikulin embedding of the root torus into the Siegel period domain sends Z_{root} to $2Z$ [38, 40]*. Hence*

$$e^{-\rho}\chi_{\mathrm{root}}C^{\wedge}_{*,h}(\mathfrak{n}_{\Delta_5,+}) = \mathrm{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z)$$

in Siegel period coordinates.

Proof. Multiply the Euler product in the preceding theorem by the Weyl monomial $e^{-\rho}$. The denominator theorem identifies the result on the root torus, and the stated embedding gives the Siegel-coordinate formula. □

The denominator is therefore the Weyl-shifted Euler supercharacter of the total-height-complete interval bar of the positive enveloping algebra.

Chapter 96

Finite-height currents

Retain the finite-height quotients ${\mathfrak{n}_{\Delta_{\mathrm{s}},+}}^{(H)}$ of the preceding chapter. Define the current Lie conformal superalgebra

$$\mathrm{Cur}({\mathfrak{n}_{\Delta_{\mathrm{s}},+}}^{(H)}) = \Bbbk[\partial] \otimes {\mathfrak{n}_{\Delta_{\mathrm{s}},+}}^{(H)}, \qquad [a_\lambda b] = [a, b],$$

and its universal chiral enveloping algebra

$$V_H = U^{\mathrm{ch}}\Big(\mathrm{Cur}_E({\mathfrak{n}_{\Delta_{\mathrm{s}},+}}^{(H)})\Big).$$

Theorem 96.1 (Height-complete current envelope)**.** *The quotient maps induce a coefficientwise inverse limit*

$$\widehat{V}_{\Delta_{\mathrm{s}}} = \varprojlim_{H} V_H.$$

Every V_H is PBW-admissible as an associative chiral algebra, the current generators are primitive for the standard current coproduct, and

$$\mathrm{sch}\, \widehat{V}_{\Delta_{\mathrm{s}}} = \prod_{j \geq 1} \prod_{\alpha > 0} (1 - t^j e^{-\alpha})^{-\operatorname{sdim} \mathfrak{g}_{\Delta_{\mathrm{s}}\, \alpha}}.$$

Moreover,

$$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(V_H) \simeq \mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}\Big(\mathrm{Cur}_E({\mathfrak{n}_{\Delta_{\mathrm{s}},+}}^{(H)})\Big),$$

and these equivalences pass to the separated height completion.

Proof. PBW orders the negative modes $a(-j)$, $j \geq 1$. Even modes contribute symmetric powers and odd modes contribute exterior powers, giving the product. Proper height makes every coefficient stationary. The current coproduct sends $a(z)$ to $a(z) \otimes 1 + 1 \otimes a(z)$. Chiral antisymmetrization identifies the associative-chiral bar of the chiral enveloping algebra with the chiral Chevalley coalgebra; the complete separated PBW filtration is coefficientwise finite. □

96.1 Classical local theory and a finite quantum model

For every H, the finite-dimensional nilpotent Lie superalgebra ${\mathfrak{n}_{\Delta_{\mathrm{s}},+}}^{(H)}$ defines a cotangent BF theory on $E \times \mathbb{R}$. Its local dg Lie algebra is

$$\mathcal{L}_H = \Omega^{0,*}(E) \mathbin{\widehat{\otimes}} \Omega^*(\mathbb{R}) \otimes {\mathfrak{n}_{\Delta_{\mathrm{s}},+}}^{(H)}$$

and the classical action is

$$S_H(A, B) = \int_{E \times \mathbb{R}} \Omega_E \wedge \Big\langle B, (\bar{\partial} + \mathrm{d}_{\mathbb{R}})A + \frac{1}{2}[A, A] \Big\rangle.$$

Theorem 96.2 (Finite-height BF realization)**.** *The action S_H satisfies the classical master equation and defines the classical holomorphic–topological factorization observables. Reduction along the topological interval retains the coefficient complex $\Omega^{\circ,*}(E) \otimes \mathfrak{n}_{\Delta_s,+}{}^{(H)}$ and is not finite-dimensional. The independent finite cotangent BV model of $\mathfrak{n}_{\Delta_s,+}{}^{(H)}[1]$ has vanishing modular supercharacter and satisfies the quantum master equation.*

For the ordinary algebraic interval construction, the enveloping algebra and its two augmentation modules give

$$U(\mathfrak{n}_{\Delta_s,+}{}^{(H)}), \qquad \mathbf{1} \otimes^{\mathbb{L}}_{U(\mathfrak{n}_{\Delta_s,+}{}^{(H)})} \mathbf{1} \simeq C_*(\mathfrak{n}_{\Delta_s,+}{}^{(H)}).$$

These constructions and the separately defined current envelope V_H have their natural height transition maps. A quantum comparison with the local field theory must preserve the field differential, support, polarizations and boundary maps.

Proof. Jacobi and invariance of the cotangent pairing give the classical master equation. The classical local observables have the standard factorization descent. A contraction in the interval variable acts as the identity on $\Omega^{\circ,*}(E)$, proving the retained coefficient complex.

For the finite algebraic model let q be the shifted Lie coordinate and p its cotangent coordinate. The Hamiltonian is $\langle p, \frac{1}{2}[q,q]\rangle$, with the graded cotangent signs. Its self-bracket vanishes by Jacobi, and its BV Laplacian is the modular supercharacter. Every even adjoint operator raises positive height and is strictly upper triangular. Every odd adjoint operator has zero supertrace. The modular term is therefore zero.

The ordinary augmentation bar computes the displayed derived tensor product, and super PBW antisymmetrization identifies its chains. Naturality supplies the transition maps without asserting the missing local quantum comparison. □

Book XVII

The compact scalar character

Chapter 97

Reduced curve counting on $K3 \times E$

Let S be a nonsingular projective K3 surface and E an elliptic curve. For a primitive class $\beta_h \in H_2(S, \mathbb{Z})$ with $\beta_h^2 = 2h - 2$, let $N_{g,h,d}$ be the reduced disconnected Gromov–Witten invariant of class (β_h, d) with the standard incidence insertion. Define

$$\mathcal{Z}(u, q, s) = \sum_{g,h,d} N_{g,h,d} u^{2g-2} q^{h-1} s^{d-1}.$$

Theorem 97.1 (Primitive Igusa cusp-form theorem). *Under $r = e^{iu}$, the partition function is the Laurent expansion*

$$\mathcal{Z}(u, q, s) = -\frac{1}{\chi_{10}(r, q, s)}.$$

Equivalently, in theta normalization,

$$\mathcal{Z} = -2^{12} \Delta_5^{\theta^{-2}}.$$

Proof. Oberdieck and Pixton identify the primitive reduced partition function with the Laurent expansion of $-\chi_{10}^{-1}$ [62]; their normalization is the monic Borcherds product χ_{10}. The second formula is the identity $\chi_{10} = 2^{-12} \Delta_5^{\theta^2}$. □

The theorem identifies the numerical generating series in its stated primitive sector. The Borcherds root algebra supplies a separate algebraic character to compare with that series.

Corollary 97.2 (Exact character comparison). *Let $\jmath : \mathbb{H}_2 \to \mathbb{H}_2$ be the half-coordinate map $\jmath(Z) = Z/2$, and let*

$$\mathrm{Fock}(\mathfrak{n}_{\Delta_5,+}) = \mathrm{Sym}(\mathfrak{n}_{\Delta_5,+\bar{0}}) \otimes \bigwedge(\mathfrak{n}_{\Delta_5,+\bar{1}}).$$

Then

$$\mathcal{Z}^{\mathrm{red}}_{S \times E} = -\jmath^* \Big(e^{2\rho} \operatorname{sch}\big(\mathrm{Fock}(\mathfrak{n}_{\Delta_5,+})^{\otimes 2}\big) \Big) = -\chi_{10}^{-1}.$$

Proof. Put

$$P = \prod_{\alpha > 0} (1 - e^{-\alpha})^{\operatorname{sdim} \mathfrak{g}_{\Delta_5, \alpha}}.$$

The denominator theorem gives $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = e^{-\rho} P = D_5(2Z)$, while PBW gives $\operatorname{sch} \mathrm{Fock}(\mathfrak{n}_{\Delta_5,+}) = P^{-1}$. Hence

$$e^{2\rho} \operatorname{sch}\big(\mathrm{Fock}(\mathfrak{n}_{\Delta_5,+})^{\otimes 2}\big) = \mathrm{den}(\mathfrak{g}_{\Delta_5})^{-2}.$$

Pullback by $\jmath$ sends $\mathrm{den}(\mathfrak{g}_{\Delta_5})$ to $D_5(Z)$, whose square is χ_{10}. The reduced incidence convention supplies the overall minus sign in the Oberdieck–Pixton theorem. □

The equality compares the completed root-Fock supercharacter with the numerical reduced generating series. Its proof uses the denominator identity and the half-coordinate pullback. A map from compact geometric states to this Fock space, preserving operators and their trace, is an additional comparison.

Chapter 98

Extensions over the elliptic classifying stack

Work on locally noetherian complex schemes with the fppf topology. Let BE be the stack of right E-torsors. For a left E-stack M, an object of $[M/E](T)$ is a right torsor $P \to T$ and an equivariant map $P \to M$. Forgetting the map gives $[M/E] \to BE$. All products of stacks below are two-fibre products.

Proposition 98.1 (The common translation torsor)**.** *There is a natural equivalence*

$$[(M \times N)/E_{\mathrm{diag}}] \simeq [M/E] \times_{BE} [N/E].$$

The equivalences for any finite number of factors commute with regrouping and insertion of the terminal E-stack.

Proof. An object on the right is (P, m), (Q, n) together with a torsor isomorphism $\alpha : P \to Q$. Send it to $(P, m, n\alpha)$. Conversely, use a common torsor twice and its identity isomorphism. A pair of torsor maps a, b on the right satisfies $b\alpha = \alpha' a$, so b is determined by a. The same construction identifies the maps and their descent isomorphisms. It commutes with base change. For several factors, every regrouping gives one torsor with the same list of equivariant maps, proving all stated coherences. □

For $M = N = E$ with translation, the diagonal quotient is E, with coordinate $y - x$. Both separate quotients are points. Their fibre product over BE retains the relative translation, while their ordinary product loses it.

Let $\mathfrak{M}$ be the stack of coherent sheaves on $X = S \times E$, with families flat over the parameter scheme, and put $\mathfrak{Q} = [\mathfrak{M}/E]$. An object of $\mathfrak{Q}(T)$ is a torsor $P \to T$ and a flat sheaf on $S \times P$. Fix a polarization and write $\mathfrak{Q}_\alpha$ for a Hilbert-polynomial component. Translation preserves these polynomials. Let $\mathfrak{H}_{\alpha,\beta}$ classify short exact sequences $0 \to A \to F \to B \to 0$ on a common $S \times P$, with all terms flat over T and polynomials α, $\alpha + \beta$, β. Its span over BE is

$$\mathfrak{Q}_\alpha \times_{BE} \mathfrak{Q}_\beta \xleftarrow{p} \mathfrak{H}_{\alpha,\beta} \xrightarrow{q} \mathfrak{Q}_{\alpha+\beta}.$$

The category of spans of fppf stacks over BE has composition by fibre product and tensor product $\times_{BE}$. Its unit object is the identity map of BE.

Theorem 98.2 (Geometric Hall correspondence)**.** *For fixed α, β, q is representable by algebraic spaces and proper. The spans form a coherently associative unital Hall object in the category of spans over BE. Its unit is the zero-sheaf span $BE \leftarrow BE \to \mathfrak{Q}$.*

Proof. Over a fixed (P, F), the fibre of q classifies quotients $F \twoheadrightarrow B$ of polynomial β. Its kernel is flat, because F and B are flat. An automorphism fixing the quotient and F is the identity. Trivializing P fppf-locally identifies this fibre with the fixed-polynomial Quot scheme, projective over the parameter scheme by Grothendieck's theorem [41, Theorem 3.2]. The quotient functor descends to an algebraic space, and properness descends with it.

Both composites of three Hall spans classify $0 \subset A \subset D \subset F$ on the same torsor. One bracketing concatenates inclusions. The other starts with $A \subset F$ and a subobject $B \subset F/A$, and sets $D = F \times_{F/A} B$. These constructions are inverse on objects and isomorphisms. They preserve flatness and commute with base change. For longer flags all regroupings identify the same subobjects and quotients. Their comparison maps agree, proving the pentagon and higher coherences. Finally $\mathfrak{Q}_0 = BE$. The sequences with zero subobject or zero quotient give the unit and its coherences. □

Properness here concerns the full coherent-sheaf stack on a fixed pair of Hilbert-polynomial components. A stability restriction can replace its fibre by an open part of a Quot scheme, which need not be proper. A bounded charge window also needs closure under the extensions and flags used in its operations. A reduced critical-coefficient realization must supply pullback, pushforward, and orientation comparisons on these same flags. Its unit has stabilizer E. A coefficient theory for this quotient must include that nonaffine stabilizer, or work relative to BE, on the unit and every flag.

Chapter 99

Additive physical charge and quadratic automorphic degree

Let

$$\Lambda = \widetilde{H}(S, \mathbb{Z})$$

be the Mukai lattice of signature $(4, 20)$. The electric–magnetic charge lattice is

$$\Gamma_{\mathrm{phys}} = \Lambda \oplus \Lambda, \qquad c = (Q, P).$$

Its addition is the physical superselection law. Define the Gram map

$$\Pi(Q, P) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2} \right) \in \Gamma_{\mathrm{ind}}.$$

Proposition 99.1 (Polarization identity). *For $c_i = (Q_i, P_i)$,*

$$\Pi(c_1 + c_2) = \Pi(c_1) + \Pi(c_2) + B(c_1, c_2),$$

where

$$B(c_1, c_2) = \big(Q_1 \cdot Q_2,\, Q_1 \cdot P_2 + Q_2 \cdot P_1,\, P_1 \cdot P_2 \big).$$

The cross-term is a symmetric vector-valued two-cocycle:

$$B(c_1, c_2) + B(c_1 + c_2, c_3) = B(c_2, c_3) + B(c_1, c_2 + c_3).$$

Proof. Expand the three quadratic expressions. The cocycle identity is bilinearity of B. □

99.1 The polarized central extension

Define the abelian central extension

$$0 \longrightarrow \Gamma_{\mathrm{ind}} \longrightarrow \widetilde{\Gamma}_\Pi \longrightarrow \Gamma_{\mathrm{phys}} \longrightarrow 0, \qquad \widetilde{\Gamma}_\Pi = \Gamma_{\mathrm{phys}} \times \Gamma_{\mathrm{ind}},$$

with multiplication

$$(c, \lambda) \cdot (d, \mu) = \big(c + d,\, \lambda + \mu + B(c, d) \big).$$

The cocycle identity makes this multiplication associative. The polarization identity makes the graph map

$$j_\Pi : \Gamma_{\mathrm{phys}} \longrightarrow \widetilde{\Gamma}_\Pi, \qquad c \longmapsto (c, \Pi(c))$$

a homomorphism.

Let $A = \bigoplus_{c \in \Gamma_{\mathrm{phys}}} A_c$ be a charge-graded associative algebra, and let $\Bbbk[\Gamma_{\mathrm{ind}}]$ have monomials u^λ. Define the polarized central-extension algebra

$$\mathcal{R}_\Pi(A) = A \otimes \Bbbk[\Gamma_{\mathrm{ind}}]$$

with product

$$(a_c u^\lambda) \star (b_d u^\mu) = a_c b_d \, u^{\lambda+\mu+B(c,d)}.$$

Give $a_c u^\lambda$ degree $(c, \lambda) \in \widetilde{\Gamma}_\Pi$.

Theorem 99.2 (Graph-multiplicative polarization)**.** *The product $\star$ is associative and $\widetilde{\Gamma}_\Pi$-graded. The map*

$$\iota_\Pi : A \longrightarrow \mathcal{R}_\Pi(A), \qquad a_c \longmapsto a_c u^{\Pi(c)}$$

is an algebra homomorphism whose homogeneous degree is the graph element $j_\Pi(c)$.

Proof. The two parenthesizations differ by

$$B(c,d) + B(c+d,e) - B(d,e) - B(c,d+e),$$

which vanishes by the cocycle identity. The polarization formula gives

$$\iota_\Pi(a_c) \star \iota_\Pi(b_d) = a_c b_d \, u^{\Pi(c+d)}.$$

Thus the graph of the quadratic Gram map is multiplicative inside the central extension. □

The same construction applies to Lie superalgebras and their enveloping algebras. It retains additive physical charge, Gram degree, and the interaction cross-term in one grading group.

Chapter 100

Point-local and wrapped sectors

Let $p_E : S \times E \to E$ be projection.

Lemma 100.1 (Positive elliptic degree is wrapped)**.** *Let $C \subset S \times E$ be a proper one-dimensional subscheme whose pushforward $(p_E)_*[C]$ has positive degree. Then $p_E(C) = E$.*

Proof. Properness makes $p_E(C)$ closed. Positive pushforward degree implies that at least one irreducible component of C dominates a one-dimensional closed subset of E. Every one-dimensional closed subset of the irreducible curve E is E itself, so the image of that component, and hence the image of C, is all of E. □

Projection-finite objects define point-supported sectors on $\mathrm{Ran}(E)$. Positive elliptic degree defines wrapped sectors supported over the entire elliptic curve.

Definition 100.2 (The local–wrapped factorization operad)**.** Let Ch_E denote the chosen chiral/factorization operad on E and let Ass denote the ordered associative operad. Define the colored operad Hyb_E with colors

$$\mathsf{l}, \qquad \mathsf{w}_m \quad (m \geq 1)$$

by the following operation objects.

(i) Operations with local output are

$$\mathsf{Hyb}_E(\mathsf{l}^{\otimes k}; \mathsf{l}) = \mathsf{Ch}_E(k).$$

(ii) For $r \geq 1$ and $m = m_1 + \cdots + m_r$, operations with wrapped output are

$$\mathsf{Hyb}_E(\mathsf{l}^{\otimes k}, \mathsf{w}_{m_1}, \ldots, \mathsf{w}_{m_r}; \mathsf{w}_m) = \mathsf{Ch}_E(k) \otimes \mathsf{Ass}(r).$$

All other operation objects are initial.

Operadic composition uses chiral insertion and disjoint union in the first factor and ordered substitution in the second. The equality

$$(\gamma_1 \circ \gamma_2) \otimes (a_1 \circ a_2) = (\gamma_1 \otimes a_1) \circ (\gamma_2 \otimes a_2)$$

is the mixed interchange law. The arities $(k, r) = (k, 0), (0, 2), (k, 1)$ recover, respectively, local factorization, wrapped multiplication, and the action of local insertions on a wrapped sector.

Definition 100.3 (Hybrid local–wrapped algebra)**.** A hybrid local–wrapped algebra over E is an algebra over Hyb_E. It consists of a factorization algebra $\mathbb{F}_{\mathrm{loc}}$, a height- or charge-complete family $W = \prod_{m>0} W_m$, ordered products

$$\mu_{m_1,\ldots,m_r} : W_{m_1} \otimes \cdots \otimes W_{m_r} \longrightarrow W_{m_1+\cdots+m_r},$$

and factorization-compatible actions of the local operation objects $\mathsf{Ch}_E(k)$ on every wrapped product. The operad axioms state associativity, local factorization, and compatibility of the two operations.

A compact Hall theory of $S \times E$ produces a correspondence-valued Hyb_E-algebra when disjoint-support direct sums, wrapped extensions, and the mixed extension correspondences satisfy the same base-change identities.

Book XVIII

The Borcherds algebra and its quantum group

Chapter 101

The local–wrapped root algebra

The positive cone is graded by $\gamma = (n, \ell, m)$, with elliptic degree m. Set

$$\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}} = \bigoplus_{\substack{\alpha>0\\ m(\alpha)=0}} \mathfrak{g}_{\Delta_5\,\alpha}, \qquad \mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}} = \bigoplus_{\substack{\alpha>0\\ m(\alpha)>0}} \mathfrak{g}_{\Delta_5\,\alpha}.$$

Proposition 101.1 (Semidirect local–wrapped decomposition)**.** *The space $\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}$ is a Lie subsuperalgebra, $\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}$ is a Lie ideal, and*

$$\mathfrak{n}_{\Delta_5,+} = \mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}} \ltimes \mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}.$$

The proper height filtration preserves both summands.

Proof. The bracket adds root degrees. The sum of two roots of elliptic degree zero again has degree zero, and addition of a nonnegative degree to a positive degree remains positive. Height is additive on every nonzero bracket. □

Define the complete enveloping Hopf algebra and current envelope

$$\mathbb{U}_{\Delta_5} = \widehat{U}_h(\mathfrak{n}_{\Delta_5,+}), \qquad \mathbb{V}_{\Delta_5,E} := \widehat{V}_{\Delta_5} = \varprojlim_H U^{\mathrm{ch}}\Big(\mathrm{Cur}_E(\mathfrak{n}_{\Delta_5,+}{}^{(H)})\Big).$$

The adjoint action of $\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}$ on $\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}$ extends by derivations to $\widehat{U}_h(\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}})$. Let $\#$ denote the corresponding completed smash product.

Definition 101.2 (Current-generated local–wrapped operad)**.** The colored operad $\mathsf{Hyb}^{\mathrm{cur}}_{E,\Delta}$ is generated by the chiral current operations of $\mathrm{Cur}_E(\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}})$, the ordered multiplications of $\widehat{U}_h(\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}})$, and the adjoint derivation action of the former on the latter, modulo the chiral Jacobi, associative, derivation, and semidirect-product relations.

Theorem 101.3 (Local–wrapped Borcherds decomposition)**.** *Let $\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}$ be the sum of positive root spaces of elliptic degree $m = 0$ and let $\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}$ be the sum of those with $m > 0$. Then*

$$\mathfrak{n}_{\Delta_5,+} = \mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}} \ltimes \mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}},$$

where $\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}$ is a Lie subsuperalgebra and $\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}$ is a Lie ideal. For the proper-height completion,

$$\widehat{U}_h(\mathfrak{n}_{\Delta_5,+}) \cong \widehat{U}_h(\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}) \# \widehat{U}_h(\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}),$$

and

$$\widehat{C}_{*,h}(\mathfrak{n}_{\Delta_5,+}) \simeq \mathrm{Tot}\, C_*(\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}; \widehat{C}_{*,h}(\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}})).$$

The primitive Lie superalgebra of the completed enveloping algebra is the height completion $\widehat{\mathfrak{n}_{\Delta_5,+}}_h = \varprojlim_H \mathfrak{n}_{\Delta_5,+}{}^{(H)}$, and the Weyl-shifted Euler supercharacter of the completed Chevalley complex is $D_5(2Z)$.

Proof. Root addition adds elliptic degree. Thus the $m = 0$ summand is closed under the bracket, and every bracket with an $m > 0$ root again has $m > 0$. PBW orders wrapped generators before local generators and gives the completed smash product. The Chevalley complex of a semidirect product is the Hochschild–Serre total complex. Primitive elements of a completed enveloping algebra are recovered height by height. The denominator theorem supplies the last character identity. □

Remark 101.4. The theorem constructs a Borcherds root algebra with local and wrapped colors. The word Hall is reserved for an algebra obtained from geometric extension correspondences.

Chapter 102

The formal Igusa quantum group

Let $\mathfrak{b}_+$ be the positive Borcherds Borel half. Form the restricted graded dual $\mathfrak{b}_- := \mathfrak{b}_+^{\vee,\mathrm{gr}}$ with bracket and cobracket transported by the invariant pairing, identify the two Cartan copies along their common radical quotient, and complete by a proper root height. The completed Drinfeld double of this paired Borel datum is denoted ${\mathfrak{g}_{\Delta_5}}^{\mathrm{pair}}$. Its positive and negative halves, diagonal Cartan, invariant form, and canonical tensor form a height-admissible Manin triple and hence a quasitriangular Lie superbialgebra.

Definition 102.1 (Height-admissible Lie superbialgebra)**.** A root-graded Lie superbialgebra is height-admissible when every root space is finite-dimensional, every bounded-height region contains finitely many roots, and the bracket, cobracket, invariant tensor, and all iterated operations are coefficientwise finite in the height completion.

Proposition 102.2 (Height admissibility)**.** *The paired Gritsenko–Nikulin Borcherds Lie superbialgebra* ${\mathfrak{g}_{\Delta_5}}^{\mathrm{pair}}$ *is height-admissible.*

Proof. The Borcherds root spaces are finite-dimensional and the chosen height is proper. The bracket and cobracket preserve total root degree. In a bounded total height, finitely many decompositions of a root occur. The canonical tensor is therefore a coefficientwise finite sum in every height quotient. □

Theorem 102.3 (Quantized Borcherds superalgebra)**.** *Choose the symmetrizable Borcherds–Cartan superdatum and multiplicity index set that present* $\mathfrak{g}_{\Delta_5}$*. Hong's construction gives a quantized enveloping Hopf superalgebra*

$$U_q(\mathfrak{g}_{\Delta_5})$$

over $\mathbb{Q}(q)$*, generated by its Cartan elements and the real and imaginary simple-root generators, with triangular decomposition*

$$U_q(\mathfrak{g}_{\Delta_5}) \cong U_q^-(\mathfrak{g}_{\Delta_5}) \otimes U_q^0(\mathfrak{g}_{\Delta_5}) \otimes U_q^+(\mathfrak{g}_{\Delta_5}).$$

Its PBW root spaces have the same graded ranks as the classical Borcherds superalgebra. The canonical pairing of the two Borel halves yields a universal R-matrix in the root-height completion of

$$U_q^+(\mathfrak{g}_{\Delta_5}) \mathbin{\widehat{\otimes}} U_q^-(\mathfrak{g}_{\Delta_5}).$$

Consequently the quantization preserves the root-multiplicity data that enter $D_5(2Z)$*.*

Proof. The Gritsenko–Nikulin algebra is a symmetrizable Borcherds–Kac–Moody Lie superalgebra. Apply the generators-and-relations construction of the quantized Borcherds superalgebra. The nondegenerate bilinear form, triangular decomposition, centre, and universal R-matrix are established by Hong [44]. The PBW theorem identifies the associated graded root spaces with their classical counterparts. Since the denominator exponents are the classical signed root multiplicities, they are unchanged by the flat root-graded deformation. □

Remark 102.4. The quantum group is attached to the Borcherds generalized Cartan superdatum. A geometric Hall quantum group is identified with it through a Hall coproduct, pairing, completion, and primitive comparison.

Chapter 103

A holomorphic–topological BF realization

For every height bound H, let ${\mathfrak{n}_{\Delta_5,+}}^{(H)}$ be the finite-dimensional nilpotent quotient. On $E \times \mathbb{R}$ form the shifted cotangent theory of

$$\mathcal{L}_H = \Omega^{0,*}(E) \mathbin{\widehat{\otimes}} \Omega^*(\mathbb{R}) \otimes {\mathfrak{n}_{\Delta_5,+}}^{(H)}$$

with action

$$S_H(A, B) = \int_{E \times \mathbb{R}} \Omega_E \wedge \left\langle B, (\bar{\partial} + \mathrm{d}_{\mathbb{R}})A + \frac{1}{2}[A, A] \right\rangle.$$

Theorem 103.1 (Classical local theory and finite coefficient model)**.** *For every H:*

(i) *S_H satisfies the classical master equation and defines a classical holomorphic–topological factorization algebra.*

(ii) *Interval reduction retains $\Omega^{0,*}(E) \otimes {\mathfrak{n}_{\Delta_5,+}}^{(H)}$. The independent finite cotangent BV model of ${\mathfrak{n}_{\Delta_5,+}}^{(H)}[1]$ has zero modular supercharacter and satisfies the quantum master equation.*

(iii) *The ordinary algebraic interval model uses $U({\mathfrak{n}_{\Delta_5,+}}^{(H)})$ and its two augmentation modules, with derived amplitude $\mathbf{1} \otimes^{\mathbb{L}}_{U({\mathfrak{n}_{\Delta_5,+}}^{(H)})} \mathbf{1} \simeq C_*({\mathfrak{n}_{\Delta_5,+}}^{(H)})$. The holomorphic current envelope V_H is defined separately.*

The transition maps give inverse systems of the ordinary algebraic constructions. Their total-height completions and the current inverse limit retain their separate definitions.

Proof. The classical master equation follows from Jacobi and the cotangent pairing. Interval contraction leaves the Dolbeault coefficient complex on E. For the independent finite model, even adjoint operators raise height and odd operators have zero supertrace, so the modular BV term vanishes. The ordinary derived tensor product is computed by the two-sided bar and the super PBW comparison. These are the arguments of the finite-height realization theorem above. They do not supply the quantum boundary comparison with the local field theory. □

The theorem gives a classical local theory and a separately constructed finite quantum model. A protected-sector realization requires a comparison of their fields and boundary operations.

Chapter 104

The determinant and Fredholm square-root operators

Let

$$\mathcal{V}_H = \bigoplus_{\substack{\alpha>0\\ h(\alpha)\le H}} \mathfrak{g}_{\Delta,\alpha}$$

and define

$$\mathscr{D}_H|_{\mathfrak{g}_{\Delta,\alpha}} = (1 - e^{-\alpha})\,\mathrm{id}\,.$$

Theorem 104.1 (Universal determinant section)**.** *The finite-height Berezinian is*

$$\mathrm{Ber}(\mathscr{D}_H) = \prod_{\substack{\alpha>0\\ h(\alpha)\le H}} (1 - e^{-\alpha})^{\mathrm{sdim}\,\mathfrak{g}_{\Delta,\alpha}}.$$

The coefficientwise limit satisfies

$$e^{-\rho}\,\mathrm{Ber}(\mathscr{D}_\infty) = D_5(2Z).$$

After pullback by $\jmath(Z) = Z/2$,

$$\jmath^*(e^{-\rho}\,\mathrm{Ber}(\mathscr{D}_\infty)) = D_5(Z), \qquad D_5(Z)^2 = \chi_{10}(Z).$$

Proof. The Berezinian of a diagonal endomorphism is the product of its eigenvalues with exponent equal to the superdimension of the eigenspace. Proper height makes every coefficient stationary. The denominator theorem supplies the Weyl line and the root-to-Siegel coordinate map. □

104.1 Trace-class realization on the root Hilbert superspace

Choose homogeneous orthonormal bases of the finite-dimensional root spaces and set

$$\mathcal{H}_\Delta = \widehat{\bigoplus}_{\alpha>0} \mathfrak{g}_{\Delta,\alpha}$$

with its induced $\mathbb{Z}/2$ grading. Let $\mathfrak{C}_{\mathrm{abs}}$ be the subchamber on which

$$\sum_{\alpha>0} \dim \mathfrak{g}_{\Delta,\alpha} \left| e^{-\alpha(Z)} \right| < \infty.$$

For $Z \in \mathfrak{C}_{\mathrm{abs}}$, define the trace-class diagonal operator

$$K_Z|_{\mathfrak{g}_{\Delta,\alpha}} = e^{-\alpha(Z)}\,\mathrm{id}, \qquad D_Z = 1 - K_Z.$$

On each parity summand form the skew doubling

$$\mathbb{A}_{Z,\bar\epsilon} = \begin{pmatrix} 0 & D_{Z,\bar\epsilon} \\ -D_{Z,\bar\epsilon}^{\mathsf{t}} & 0 \end{pmatrix}.$$

Orient every finite-height truncation by the fixed root order and normalize its Pfaffian to 1 at $K_Z = 0$. The oriented Fredholm super-Pfaffian is the stabilized ratio

$$\mathrm{sPf}_F(\mathbb{A}_Z) = \frac{\mathrm{Pf}_F(\mathbb{A}_{Z,\bar{0}})}{\mathrm{Pf}_F(\mathbb{A}_{Z,\bar{1}})}.$$

Theorem 104.2 (Polarized Fredholm realization)**.** *Choose the protected parity refinement and the root-space Hilbert norms above. For every Z in the absolute-convergence chamber, the diagonal operator $K_Z|_{\mathfrak{g}_{\Delta_S,\alpha}} = e^{-\alpha(Z)}\,\mathrm{id}$ is trace class on the even and odd root Hilbert spaces. For $D_Z = 1 - K_Z$,*

$$\mathrm{sdet}_F(D_Z) = \frac{\det_F(D_Z|_{\bar{0}})}{\det_F(D_Z|_{\bar{1}})} = \prod_{\alpha>0}(1 - e^{-\alpha(Z)})^{\mathrm{sdim}\,\mathfrak{g}_{\Delta_S,\alpha}}.$$

After choosing the standard polarization of $\mathcal{H}_\Delta \oplus \mathcal{H}_\Delta^*$ and its orientation, the paritywise skew block operators $\mathbb{A}_{Z,\bar{0}}$ and $\mathbb{A}_{Z,\bar{1}}$ have an oriented Fredholm super-Pfaffian

$$\mathrm{sPf}_F(\mathbb{A}_Z) = \frac{\mathrm{Pf}_F(\mathbb{A}_{Z,\bar{0}})}{\mathrm{Pf}_F(\mathbb{A}_{Z,\bar{1}})} = \mathrm{sdet}_F(D_Z).$$

Hence

$$e^{-\rho}\,\mathrm{sdet}_F(D_Z) = D_S(2Z).$$

Proof. The defining summability condition for $\mathfrak{C}_{\mathrm{abs}}$ is precisely the trace norm of the diagonal operator, summed in both parities. The Fredholm determinant of $1 - K_Z$ is the convergent product over eigenvalues; taking the even/odd ratio gives the superdeterminant. On each finite root-height truncation the Pfaffian of the oriented skew doubling equals the determinant of D_Z. Trace-norm convergence of the truncations passes this identity to the Fredholm limit. The final equality is the Gritsenko–Nikulin denominator formula. □

Remark 104.3. The polarization and orientation in this theorem belong to the root Hilbert space. A compact moduli-space orientation is a separate line bundle and is compared with this one by a specified intertwiner.

This constructs the square-root operator on the root Hilbert superspace where its stated summability condition holds. The nonemptiness estimate below concerns the minimal virtual superspace, whose dimensions are $|\mathrm{sdim}\,\mathfrak{g}_{\Delta_S,\alpha}|$. Applying that estimate to the full root superspace requires a bound on both parity dimensions. A compact geometric realization requires an operator intertwiner and a compatible map of determinant lines.

Chapter 105

What the automorphic character determines

Theorem 105.1 (Denominator data and reconstruction data)**.** *The Borcherds product determines:*

(i) *the signed root multiplicity* $\operatorname{sdim} \mathfrak{g}_{\Delta_{\mathrm{s}}\alpha}$ *in every active root degree;*

(ii) *the Weyl vector and denominator character;*

(iii) *the divisor and automorphic multiplier of the section.*

A Lie bracket is supplied by the Borcherds–Kac–Moody presentation, parity dimensions by a parity-refined root datum, Hall multiplication by extension correspondences, and an oriented square root by a square-root line with monodromy. These reconstruction data refine the scalar product independently.

Proof. The first three items are read from the exponent, prefactor, and automorphy law. Independence is witnessed by explicit pairs: two brackets on the same graded vector space have the same PBW character; an even–odd pair has signed dimension zero; distinct Hall correspondences may have the same Euler product; and twisting a square-root line by a sign local system preserves its square. Hence each refined object is reconstructed from the additional datum named in the statement. □

Recognition Theorem 105.2 (Complete primitive recognition)**.** *Let* $\mathfrak{p}$ *be a positive graded Lie superalgebra and let* $\Psi : \mathfrak{n}_{\Delta_{\mathrm{s}},+} \twoheadrightarrow \mathfrak{p}$ *be a graded Lie map. The map is an isomorphism if either:*

(i) *the parity-refined Hilbert series of* $U(\mathfrak{n}_{\Delta_{\mathrm{s}},+})$ *and* $U(\mathfrak{p})$ *agree; or*

(ii) *each primitive degree has prescribed pure parity and the signed PBW characters agree.*

The induced height completions, Chevalley complexes, and current envelopes are then isomorphic.

Proof. Choose the first height containing a homogeneous kernel vector. All decomposable PBW terms in that height are determined by lower heights and agree. The primitive parity dimension therefore drops, contradicting the first hypothesis. In the second case the logarithm of the PBW product recovers signed primitive dimensions inductively, and prescribed parity recovers ordinary dimensions. Naturality of completion, Chevalley chains, and current envelopes gives the final assertion. □

Book **XIX**

The universal Borcherds Hall and three-Calabi–Yau models

Chapter 106

Finite simple-root windows

Let I_Δ be the multiset of simple roots of the Gritsenko–Nikulin Borcherds superalgebra. A repeated imaginary simple root is written once for each copy. The parity map is

$$p : I_\Delta \longrightarrow \mathbb{Z}/2.$$

Choose a proper height function

$$h : I_\Delta \longrightarrow \mathbb{Z}_{>0}$$

whose sublevel sets are finite. Put

$$I_H = \{ i \in I_\Delta : h(i) \le H \}.$$

The restriction of the invariant form gives a finite symmetric Borcherds–Cartan matrix

$$\begin{aligned} &A_H = (a_{ij})_{i,j \in I_H}, \\ &a_{ii} = 2 \qquad\qquad \text{for real } i, \\ &a_{ii} \le 0 \qquad\qquad \text{for imaginary } i, \\ &a_{ij} \le 0 \qquad\qquad \text{for } i \ne j. \end{aligned}$$

The evenness of the root lattice makes every imaginary diagonal entry even.

Construction 106.1 (The Igusa Borcherds quiver)**.** Order I_H. For $i < j$ place $-a_{ij}$ arrows from i to j. At an imaginary vertex i place

$$g_i = 1 - \frac{a_{ii}}{2}$$

loops. A real vertex has loop number zero. Denote the resulting quiver by $Q_{\Delta,H}$.

Proposition 106.2 (Euler form)**.** *The symmetrized Euler form of $Q_{\Delta,H}$ is A_H:*

$$\langle e_i, e_j \rangle + \langle e_j, e_i \rangle = a_{ij}.$$

Proof. For $i \ne j$, the total number of arrows between the two vertices is $-a_{ij}$, so the symmetrized Euler entry is a_{ij}. At i the value is $2 - 2g_i = a_{ii}$. □

The inclusions $I_H \subset I_{H+1}$ identify $Q_{\Delta,H}$ with a full subquiver of $Q_{\Delta,H+1}$. Representations supported on the old vertices form an exact extension-closed subcategory. This gives compatible Hall embeddings.

Chapter 107

The real-root seed window

The three real simple roots are

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3,$$

with Gram matrix

$$A_{\mathrm{re}} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

Choose the order $\delta_1 < \delta_2 < \delta_3$. The real-root quiver Q_{re} has two arrows from vertex i to vertex j for each $i < j$.

Proposition 107.1 (Real-root Hall seed)**.** *The symmetrized Euler form of Q_{re} is A_{re}. The Hall composition algebra generated by its three simple representations is the positive quantum Kac–Moody algebra with Cartan matrix A_{re}.*

Proof. The diagonal Euler entries are 2. The total number of arrows between distinct vertices is two, hence the symmetrized off-diagonal entries are -2. The Ringel–Green Hall theorem identifies the composition algebra with the positive quantum algebra defined by that symmetric Cartan matrix. □

The three vertices already contain the Weyl group and the reflective divisor. Imaginary vertices add the automorphic correction. The finite-height tower therefore begins with a completely explicit hyperbolic Hall algebra and enlarges it by one finite collection of imaginary simple roots at each stage.

107.1 The first Ginzburg differential

Let $a_{ij}^{(1)}, a_{ij}^{(2)} : i \to j$ be the two arrows for $i < j$. Add reverse arrows $(a_{ij}^{(r)})^* : j \to i$ of degree -1 and loops t_i of degree -2. The canonical three-Calabi–Yau completion has

$$d a_{ij}^{(r)} = 0, \qquad d(a_{ij}^{(r)})^* = 0,$$

and

$$d t_i = \sum_{a : s(a) = i} a a^* - \sum_{a : t(a) = i} a^* a. \tag{107.1}$$

The identity $d^2 = 0$ follows immediately. Formula (107.1) is the derived moment-map equation for the doubled real-root quiver.

Calculation 107.2 (Vertex one)**.** At vertex 1,

$$d t_1 = \sum_{r=1}^{2} \Big(a_{12}^{(r)} (a_{12}^{(r)})^* + a_{13}^{(r)} (a_{13}^{(r)})^* \Big).$$

At vertex 3,

$$dt_3 = -\sum_{r=1}^{2}\Big((a_{13}^{(r)})^* a_{13}^{(r)} + (a_{23}^{(r)})^* a_{23}^{(r)}\Big).$$

The sum over vertices is $\sum_a (aa^* - a^*a)$. It vanishes in the cyclic quotient and after matrix trace, but generally not in the path algebra.

Proof. The outgoing arrows at vertex 1 have targets 2 and 3. The incoming arrows at vertex 3 have sources 1 and 2. The displayed formulas follow from these incidence lists. Each arrow contributes one commutator to the sum over vertices. The trace of a commutator vanishes, giving the asserted trace identity. □

Chapter 108

Hall realization of the positive algebra

Let $\mathcal{A}_H = \mathrm{Rep}_{\mathbb{F}_q}(Q_{\Delta,H})$. Its twisted Hall product is

$$[M] * [N] = v^{\langle \dim M, \dim N\rangle} \sum_{[L]} |\{N' \subset L : N' \cong N,\ L/N' \cong M\}|[L], \qquad v^2 = q.$$

The class $[0]$ is the unit: the required subobject is unique on either side. Associativity follows by counting flags $N_1 \subset N_2 \subset L$ in two orders. The twist factors agree because the Euler form is bilinear:

$$\langle \alpha, \beta\rangle + \langle \alpha + \beta, \gamma\rangle = \langle \beta, \gamma\rangle + \langle \alpha, \beta + \gamma\rangle.$$

Thus this normalization defines the unital twisted Hall algebra on isomorphism classes.

Theorem 108.1 (Finite-window composition algebra)**.** *At each vertex i of $Q_{\Delta,H}$, choose the one-dimensional representation S_i supported at i, with every loop acting by zero. Use the generic composition algebra generated by these classes across finite fields, with the twisted subobject-count multiplication above. After extending coefficients from $\mathbb{Z}[v, v^{-1}]$ to $\mathbb{C}(v)$, this is the positive quantum generalized Kac–Moody algebra for the even Borcherds–Cartan matrix A_H. The parameter in Kang–Schiffmann is $v_{\mathrm{KS}} = v^{-1}$. Their specified semisimple perverse-sheaf construction on nilpotent representation varieties gives the corresponding canonical basis.*

Proof. This is the selected-simple composition construction of Kang–Schiffmann, Section 2.3 and Theorem 2.2, with one selected zero-loop simple at each vertex. Their geometric realization uses its split graded Grothendieck group over $\mathbb{Z}[v_{\mathrm{KS}}, v_{\mathrm{KS}}^{-1}]$ (Theorem 4.1), with the stated inversion of parameter [48]. The theorem does not use every simple object of the full representation category. A vertex with c loops has q^c one-dimensional simple representations over $\mathbb{F}_q$, whereas only the chosen zero-loop representation is the generator here. □

A semi-derived Hall realization of a full quantum group additionally specifies two-periodic complexes, the invertible acyclic classes and Cartan reduction [2, 57]. Its coproduct, pairing and completion are separate from the positive composition algebra used here.

108.1 Parity and the quantized superdatum

The Gritsenko–Nikulin denominator algebra carries a parity map on its simple-root multiset. It is additive on the free lattice of labelled simple generators. Projection to the Lorentzian degree can identify contributions of different parity. Both parity dimensions must therefore be retained in each projected root space. Hong's construction applied to this Borcherds superdatum gives the positive Borel

$$U_q^+(\mathfrak{g}_{\Delta_5}{}^{(H)})$$

and the full quantized Borcherds superalgebra $U_q(\mathfrak{g}_{\Delta_5}{}^{(H)})$.

Theorem 108.2 (Ordinary and super presentations)**.** *The ordinary quiver Hall presentation and Hong's quantum superalgebra use the same specified Cartan matrix and labelled simple generators. Their relation ideals and parity conventions are different inputs. Equality of their Cartan gradings alone does not identify their PBW characters.*

Proof. The two presentations are those of the preceding Hall construction and Hong's superalgebra construction. The difference already occurs for a single isotropic generator. The enveloping algebra of a one-dimensional even abelian Lie algebra is $\mathbb{C}[x]$. Declaring its generator odd only as a graded vector space leaves a nonzero vector x^2 in degree two. The enveloping algebra of a one-dimensional odd abelian Lie superalgebra is $\bigwedge(\xi)$, since its defining relation is $2\xi^2 = [\xi, \xi] = 0$. Its degree-two component is zero. Thus parity assignment to an ordinary PBW basis does not produce the super relation ideal or its character. A comparison must respect these relations degree by degree. □

Chapter 109

The three-Calabi–Yau completion

Let $kQ_{\Delta,H}$ be the path algebra. Its algebraic three-Calabi–Yau completion is

$$\Gamma_{\Delta,H} = \Pi_3(kQ_{\Delta,H}).$$

Write J for the ideal generated by all arrows in its graded quiver. Its completed version is the separate topological dg algebra $\widehat{\Gamma}_{\Delta,H} = \varprojlim_n \Gamma_{\Delta,H}/J^n$. The algebraic Calabi–Yau theorem is used for $\Gamma_{\Delta,H}$; a topological Calabi–Yau assertion for $\widehat{\Gamma}_{\Delta,H}$ requires its completed bimodule category. This is the Ginzburg model with zero potential: a degree-minus-one reverse arrow for every arrow and a degree-minus-two loop at each vertex. The loop differential is the signed sum of incident arrow–reverse-arrow products.

Theorem 109.1 (Borcherds–Ginzburg projections)**.** *For each finite quiver $Q_{\Delta,H}$, its undeformed three-Calabi–Yau completion $\Gamma_{\Delta,H}$ is the Ginzburg dg algebra with zero potential. If $H \leq H'$ and $Q_{\Delta,H}$ is the full subquiver on a subset of vertices, there is a surjective dg algebra map*

$$\pi_{H'H} : \Gamma_{\Delta,H'} \longrightarrow \Gamma_{\Delta,H}.$$

It kills the added vertices, every arrow incident to them, the corresponding reverse arrows, and the added degree-minus-two loops. These maps compose strictly and extend continuously to the completed algebras.

Proof. The finite Calabi–Yau assertion is the Ginzburg model of Keller's completion theorem [33, 49]. We concatenate paths from left to right, so aa^* is based at the source of a. Original and reverse arrows have zero differential. For the degree-minus-two loop t_i, the differential is

$$dt_i = \sum_{s(a)=i} aa^* - \sum_{t(a)=i} a^*a.$$

The projection deletes exactly the terms involving an added vertex, leaving the same sum for the full subquiver. It therefore commutes with the differential. It preserves path length and extends to the completed path algebra. Successive projections kill the union of their omitted vertices and arrows, proving strict composition. □

The evident inclusion of graded quivers does not give the reverse dg map. A single old vertex has loop differential zero. Adding an arrow $a : i \to j$ changes that differential to aa^* at i. The map fixing the old loop therefore fails to commute with the differential.

Write $\Gamma_\Delta = (\Gamma_{\Delta,H}, \pi_{H'H})_H$ for this pro-system of dg algebras. It is separate from the ordinary Hall embeddings obtained by extension by zero on representations of a full subquiver. A direct system of Calabi–Yau dg categories compatible with those Hall embeddings requires additional dg functors. It does not follow from the inclusion of the displayed Ginzburg generators.

The quantum group retains its separately specified root-height completion. Its transition maps must preserve the Cartan generators, coproduct, and pairing. Quotienting the positive nilpotent Lie algebra by height and truncating a full quantum group are different operations.

Chapter 110

Hall states and protected superstates

The representation varieties of $Q_{\Delta,H}$ carry the perverse-sheaf complexes used in the composition-algebra construction. The full equivariant derived category is a larger carrier. Already for one real vertex and dimension one, its quotient is $B\mathbb{G}_m$, with $H^*(B\mathbb{G}_m;\mathbb{Q}) = \mathbb{Q}[c_1]$, where $|c_1| = 2$. Its total equivariant cohomology is infinite-dimensional. Finite root multiplicities therefore refer to the specified composition algebra, not to total equivariant cohomology of this stack.

Definition 110.1 (Igusa Hall category)**.** For every dimension vector $\mathbf{d}$, let

$$\mathfrak{R}_{H,\mathbf{d}} = [\mathrm{Rep}_{\mathbf{d}}(Q_{\Delta,H})/G_{\mathbf{d}}].$$

The finite-height Hall category is the direct sum of the equivariant constructible derived categories on the $\mathfrak{R}_{H,\mathbf{d}}$, equipped with Hall convolution along the stack of short exact sequences. Denote it by $\mathcal{B}_{\Delta,H}$ and set

$$\mathcal{B}_{\Delta} = \varinjlim_{H} \mathcal{B}_{\Delta,H}.$$

Independently, the protected one-particle superspace is the root-graded superspace

$$\mathcal{H}_{\mathrm{ip},H} = \bigoplus_{\alpha>0} \mathfrak{g}_{\Delta_5,\alpha}^{(H)},$$

with the parity of the Borcherds superalgebra. Its multiparticle completion is the PBW/Fock superspace of $U(\mathfrak{n}_{\Delta_5,+}^{(H)})$.

Theorem 110.2 (Hall and protected-state comparison)**.** *The composition subalgebra in the finite-window Hall theorem is the ordinary positive quantum generalized Kac–Moody algebra. This statement does not identify the Grothendieck group of the full category $\mathcal{B}_{\Delta,H}$ with that composition algebra. The protected one-particle superspace has signed root multiplicities*

$$\operatorname{sdim}\mathcal{H}_{\mathrm{ip},H,\alpha} = \operatorname{sdim}\mathfrak{g}_{\Delta_5,\alpha}^{(H)}.$$

Its height-complete PBW character is

$$\operatorname{sch}\mathrm{Fock}(\mathfrak{n}_{\Delta_5,+}) = \prod_{\alpha>0}(1-e^{-\alpha})^{-\operatorname{sdim}\mathfrak{g}_{\Delta_5,\alpha}},$$

and its Weyl-shifted inverse is $D_5(2Z)$. The composition canonical basis and the protected root presentation use the specified simple-root indexing and Cartan grading; the parity refinement belongs to the Borcherds superdatum.

Proof. The composition-algebra assertion is the finite-window theorem. The protected assertion is PBW for the Borcherds Lie superalgebra. Proper height passes both constructions to the limit. The Gritsenko–Nikulin denominator theorem gives the final product identity. □

Chapter III

Extension determinants and critical orientations

Let C be a three-Calabi–Yau dg category over $\mathbb{C}$, with its chosen trace. For families of objects A, B over a base T, let $C(A,B)$ denote the relative derived Hom complex on T. Suppose these complexes are perfect, and that the trace gives base-change-compatible duality

$$C(B,A) \simeq C(A,B)^{\vee}[-3].$$

Put $L(A,B) = \det C(A,B)$ and $D(A) = L(A,A)$. The determinant convention is $\det C = \bigotimes_i (\det C^i)^{(-1)^i}$, with symmetry sign specified by $\chi(C) \bmod 2$.

Theorem III.1 (Extension determinant)**.** *For an exact triangle $A \to F \to B \to A[1]$, there is a base-change-compatible isomorphism*

$$D(F) \simeq D(A) \otimes D(B) \otimes L(A,B)^{\otimes 2}.$$

The determinant comparisons obtained by refining a flag agree, with the stated determinant symmetry signs.

Proof. Apply derived Hom to the triangle in each variable. Determinant additivity gives

$$D(F) \simeq L(B,A) \otimes D(A) \otimes D(B) \otimes L(A,B).$$

Dualization and the odd shift each invert the determinant line. Thus Calabi–Yau duality identifies $L(B,A)$ with $L(A,B)$. Reorder by determinant symmetry to obtain the formula. For a full flag, all iterated decompositions identify with the same ordered tensor product of $C(A_i, A_j)$, for its successive quotients A_i. Determinant additivity for refined filtrations proves coherence. Each operation commutes with base change. □

A square-root orientation consists of lines o_A and isomorphisms $o_A^{\otimes 2} \simeq D(A)$. Extension compatibility requires maps

$$o_A \otimes o_B \otimes L(A,B) \longrightarrow o_F$$

whose squares are the determinant comparisons. Choosing coordinates on a smooth critical chart trivializes its ambient canonical line. Hall convolution requires transition maps compatible with the cross line and the zero object. On three-step flags, the two orientation maps can differ by an element of μ_2, even when their squares agree. That sign must be trivialized coherently.

Example III.2 (A nontrivial cross line)**.** Let X be a smooth projective Calabi–Yau threefold, with a chosen volume form, and let $x \in X$. On $X \times T$, where $T = \mathbb{P}^1$, take

$$A = \mathcal{O}_X \boxtimes \mathcal{O}_T(1), \qquad B = \mathcal{O}_x \boxtimes \mathcal{O}_T.$$

Then $C(A,B) = \mathcal{O}_T(-1)[0]$, and relative Serre duality gives $C(B,A) = \mathcal{O}_T(1)[-3]$. Thus $L(A,B) = L(B,A) = \mathcal{O}_T(-1)$. The self-Hom complexes of A and B have constant determinant lines, since the factors $\mathcal{O}_T(1)$ cancel in the first self-Hom. Nevertheless

$$D(A \oplus B) = D(A) \otimes D(B) \otimes \mathcal{O}_T(-2).$$

The orientation cross factor is $\mathcal{O}_T(-1)$. A sign change cannot replace this line of nonzero degree.

Corollary III.3 (Oriented critical coefficients)**.** *Suppose the chosen critical charts have square-root orientation maps on overlaps and extension flags, with unit and flag coherences. Suppose their vanishing-cycle comparison maps preserve these data. Then the local vanishing-cycle complexes descend to the specified oriented coefficient object. Their Hall convolution is associative whenever its pullback and pushforward maps are defined and satisfy base change on the flag diagrams.*

Proof. The overlap cocycle identifies the local coefficient complexes by descent. On a three-step flag, orientation coherence identifies the two Thom–Sebastiani coefficient maps. Proper or compact-support base change, on the stipulated domain, identifies the two composites of convolution. The zero flag gives the unit equation. □

For the quiver charts, the remaining construction is an equivariant system of square roots and coefficient maps on the actual representation and extension stacks, compatible with changes of height. For compact $S \times E$, the reduced complexes and the common elliptic torsor must also be retained. A coordinate volume on an affine chart specifies neither descent system by itself.

Chapter 112

The colored elliptic Ran space

The elliptic degree is a global charge carried by the whole configuration. Define

$$\mathrm{Ran}^{\mathrm{wr}}(E) = \coprod_{m \geq 0} \mathrm{Ran}(E)_m$$

with monoidal operation

$$(I, m) \sqcup (J, n) = (I \sqcup J, m + n)$$

on disjoint supports. The component $m = 0$ is point-local. The components $m > 0$ are wrapped.

Let

$$\mathfrak{n}_{\Delta_s,+} = \bigoplus_{m \geq 0} \mathfrak{n}_{\Delta_s,+}[m]$$

be the elliptic-degree grading. The root bracket obeys

$$[\mathfrak{n}_{\Delta_s,+}[m], \mathfrak{n}_{\Delta_s,+}[n]] \subset \mathfrak{n}_{\Delta_s,+}[m + n].$$

Construction 112.1 (Hybrid factorization algebra)**.** Form the chiral Chevalley factorization coalgebra

$$\mathcal{F}^{\mathrm{hyb}}_{\Delta,E} = C^{\mathrm{ch}}_*\Big(\mathrm{Cur}_E(\mathfrak{n}_{\Delta_s,+})\Big).$$

Retain the elliptic-degree grading on every Chevalley word and place its degree-m summand on $\mathrm{Ran}(E)_m$. Disjoint union adds the degree labels.

Theorem 112.2 (Local–wrapped factorization)**.** *$\mathcal{F}^{\mathrm{hyb}}_{\Delta,E}$ is a graded factorization coalgebra on $\mathrm{Ran}^{\mathrm{wr}}(E)$. Its degree-zero color is the current theory of $\mathfrak{n}_{\Delta_s,+}[0]$. Every degree-$m$ color is a module for the degree-zero color. The factorization product of degrees m and n lands in degree $m + n$. Its chiral primitives are $\mathrm{Cur}_E(\mathfrak{n}_{\Delta_s,+})$, and constant modes recover $\mathfrak{n}_{\Delta_s,+}$.*

Proof. The chiral Chevalley construction is a factorization coalgebra for every chiral Lie algebra. The bracket preserves elliptic-degree addition, so its differential and coproduct preserve the graded Ran placement. The primitive theorem for the Chevalley coalgebra recovers the current Lie algebra. □

This construction retains the nonnegative root-degree label on a point-supported current theory. It does not replace a sheaf projecting onto the whole elliptic curve by a point-supported sheaf. The compact wrapped comparison must map its common-torsor extension correspondences to these graded current operations, preserving support and charge. The bracket on the present carrier is the Borcherds bracket.

Chapter 113

Physical charge and Gram degree

Let Γ_{phys} be an additive physical charge lattice and let

$$\Pi(c) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2} \right)$$

be the Gram map. Its polarization is

$$B(c,d) = \Pi(c+d) - \Pi(c) - \Pi(d).$$

Define the central extension

$$\widetilde{\Gamma}_{\Pi} = \Gamma_{\mathrm{phys}} \times \Gamma_{\mathrm{ind}}$$

with multiplication

$$(c,\lambda)(d,\mu) = (c+d, \lambda+\mu+B(c,d)).$$

Theorem 113.1 (Graph homomorphism)**.** *The map*

$$j_{\Pi} : \Gamma_{\mathrm{phys}} \longrightarrow \widetilde{\Gamma}_{\Pi}, \qquad c \longmapsto (c, \Pi(c))$$

is a group homomorphism. For every Γ_{phys}-graded algebra A, the Rees lift

$$a_c \longmapsto a_c u^{\Pi(c)}$$

turns multiplication into an additive $\widetilde{\Gamma}_{\Pi}$-grading with cocycle $u^{-B(c,d)}$.

Proof. The defining multiplication gives

$$j_{\Pi}(c)\, j_{\Pi}(d) = (c+d, \Pi(c)+\Pi(d)+B(c,d)) = j_{\Pi}(c+d).$$

The algebra identity follows immediately. □

The universal Igusa category is graded directly by the Lorentzian root lattice. A geometric $K3 \times E$ theory enters through a charge functor to $\widetilde{\Gamma}_{\Pi}$.

Chapter 114

The local cotangent theory and finite coefficients

For each H take cotangent BF theory with gauge Lie superalgebra ${\mathfrak{n}_{\Delta_5,+}}^{(H)}$ on $\mathbb{R}^2_{\mathrm{top}} \times E$. Its fields are

$$\Omega^\bullet(\mathbb{R}^2) \mathbin{\widehat{\otimes}} \Omega^{0,\bullet}(E) \mathbin{\widehat{\otimes}} \left({\mathfrak{n}_{\Delta_5,+}}^{(H)}[1] \oplus ({\mathfrak{n}_{\Delta_5,+}}^{(H)})^\vee[2] \right).$$

The action is

$$S_H = \int_{\mathbb{R}^2 \times E} \langle B, (\mathrm{d} + \bar{\partial}) A + \frac{1}{2}[A, A] \rangle.$$

Theorem 114.1 (Classical local theory and finite quantum model)**.** *For every finite height H, the shifted-cotangent action gives the classical mixed holomorphic–topological theory on $\mathbb{R}^2_{\mathrm{top}} \times E$. Contraction on a compact topological cell retains $\Omega^{0,*}(E) \otimes {\mathfrak{n}_{\Delta_5,+}}^{(H)}$. A further global harmonic reduction on compact E retains both $H^0(E, \mathcal{O}_E)$ and $H^1(E, \mathcal{O}_E)$. The independent finite cotangent model of ${\mathfrak{n}_{\Delta_5,+}}^{(H)}[1]$ satisfies the quantum master equation. Its ordinary enveloping algebra and augmentation bar are the algebraic constructions above. Their realization by the local theory requires a supported quantum field and boundary comparison.*

Proof. The classical identity follows from Jacobi and the invariant pairing. The topological contraction acts only on the de Rham factor. Dolbeault cohomology of the compact elliptic curve is one-dimensional in degrees zero and one, so neither component disappears under a further global contraction.

In the independent finite cotangent model, the modular BV divergence is the adjoint supertrace and vanishes by positive height. The ordinary enveloping and bar claims follow from PBW. All constructions preserve their height grading. Their distinct coefficient complexes are retained. □

Chapter 115

Absolute convergence of the virtual root operator

For each positive root α, let V_α be the minimal parity realization of the signed multiplicity. It is purely even of dimension $\operatorname{sdim} \mathfrak{g}_{\Delta_5 \alpha}$ when this integer is nonnegative. Otherwise it is purely odd of dimension $-\operatorname{sdim} \mathfrak{g}_{\Delta_5 \alpha}$. Set

$$\mathcal{V}_\Delta = \widehat{\bigoplus}_{\alpha>0} V_\alpha.$$

This minimal parity realization is determined up to graded isomorphism by the denominator character. It has no preferred bases or Hilbert norms. The Lie bracket remains carried by $\mathfrak{g}_{\Delta_5}$.

Let $h(\alpha)$ be the proper height. Lemma 20.4 proves from the theta products that there are constants $C_0, C_1 > 0$ such that

$$\dim V_\alpha = \left|\operatorname{sdim} \mathfrak{g}_{\Delta_5 \alpha}\right| \le C_0 \exp(C_1 h(\alpha)) \tag{115.1}$$

for every positive root. The number of Gram triples of height H is polynomial in H.

Theorem 115.1 (Nonempty trace-class chamber)**.** *Let Z_0 lie in the interior of the positive Weyl chamber. For sufficiently large $t > 0$,*

$$\sum_{\alpha>0} \dim V_\alpha \left|e^{-\alpha(tZ_0)}\right| < \infty.$$

Hence the absolute-convergence chamber contains a conical tail in every interior direction.

Proof. Interior positivity gives a constant $c(Z_0) > 0$ with $\Re \alpha(Z_0) \ge c(Z_0) h(\alpha)$. Combine this with (115.1). The contribution of roots of height H is bounded by a polynomial in H times

$$\exp\big((C_1 - t c(Z_0))H\big).$$

The series converges when $t c(Z_0) > C_1$. □

Corollary 115.2 (Trace-class root Hamiltonian)**.** *On the conical tail, the diagonal operator*

$$K_Z\big|_{V_\alpha} = e^{-\alpha(Z)} \operatorname{id}$$

is trace class on both parity summands. The Fredholm product is holomorphic and equals the absolutely convergent Borcherds product.

Proof. The trace norm is the summable series of the preceding theorem, separated by parity. Standard trace-class determinant theory gives holomorphy, and diagonalization gives the product over root eigenvalues. □

Chapter 116

The virtual root Fredholm operator

Let

$$\mathcal{H}_{\Delta}^{\mathrm{vir}} = \widehat{\bigoplus}_{\alpha>0} V_{\alpha}$$

with an orthonormal basis in each minimal parity realization. This is the virtual root Hilbert space used in the preceding absolute-convergence theorem. In the absolute-convergence chamber define

$$K_Z\big|_{V_\alpha} = e^{-\alpha(Z)}\,\mathrm{id}, \qquad D_Z = 1 - K_Z.$$

Theorem 116.1 (Oriented Fredholm square root)**.** *The operator K_Z is trace class on the even and odd summands of the virtual root Hilbert space. Its Fredholm Berezinian is*

$$\mathrm{sdet}_F(D_Z) = \prod_{\alpha>0} (1 - e^{-\alpha(Z)})^{\mathrm{sdim}\,\mathfrak{g}_{\Delta_5\alpha}}.$$

Choose homogeneous bases in every minimal parity space, ordered by height and then by root degree. For a parity block of dimension N, normalize the Pfaffian by division by the reference value $\mathrm{Pf}\left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right) = (-1)^{N(N-1)/2}$*. The skew doubling*

$$\mathbb{A}_Z = \begin{pmatrix} 0 & D_Z \\ -D_Z^{\mathrm{t}} & 0 \end{pmatrix}$$

has an oriented Fredholm super-Pfaffian satisfying

$$\mathrm{sPf}_F(\mathbb{A}_Z) = \mathrm{sdet}_F(D_Z).$$

Consequently

$$e^{-\rho}\,\mathrm{sPf}_F(\mathbb{A}_Z) = D_5(2Z).$$

Proof. The preceding coefficient bound proves trace-norm summability on the stated tail. The determinant follows from the diagonal spectrum. In block order the finite Pfaffian is $(-1)^{N(N-1)/2}\det D$. Division by the fixed reference Pfaffian makes it $\det D$. Trace-norm convergence passes the normalized identity to the limit. The denominator theorem supplies the Weyl shift. □

116.1 Automorphic orientation

The Weyl-shifted product defines a section of the automorphic line $\mathcal{L}^5 \otimes \nu_5$. On the absolute-convergence chamber its normalized Fredholm expression represents that section in a chosen trivialization. Analytic continuation determines the scalar transition law. Gluing operator families also requires intertwiners and a compatible determinant-line comparison.

Corollary 116.2 (Orientation character)**.** *The flat character factor of the automorphic line is the order-two local system ν_5. On the root domain the Weyl denominator is anti-invariant under every real simple reflection.*

Proof. The monic form has automorphy factor $\det(CZ + D)^5 \nu_5$. Its Hodge-line factor and flat character factor are separate. The denominator identity transforms by the determinant of the Weyl element, which is -1 for each real simple reflection. This scalar transformation does not construct a geometric Dirac family. □

Chapter 117

Igusa coefficient constructions

Definition 117.1 (Igusa coefficient tuple)**.** Fix the root and current presentations associated with the K3 half-index. Define the tuple

$$\mathfrak{K}_{\Delta} = \left(\Gamma_{\Delta}, \mathcal{B}_{\Delta}, \mathcal{H}_{\mathrm{ip}}, \mathcal{F}^{\mathrm{hyb}}_{\Delta,E}, \mathbb{A}_{Z}\right).$$

Its algebraic components are the finite three-Calabi–Yau completion system, the ordinary quiver Hall category and the Borcherds root Lie superalgebra. The fourth component is the specified current factorization algebra. The fifth is the normalized bounded skew Fredholm model.

Theorem 117.2 (Properties of the Igusa coefficient tuple)**.** *The quintuple $\mathfrak{K}_{\Delta}$ satisfies:*

(i) *its one-particle Lie superalgebra is $\mathfrak{g}_{\Delta_5}$;*

(ii) *its ordinary Hall presentation and Hong's quantum supergroup retain the stated Cartan grading; comparison of their relation ideals and parity-refined PBW spaces requires an additional map;*

(iii) *its chiral primitive algebra on E is the local–wrapped current algebra of $\mathfrak{g}_{\Delta_5}$;*

(iv) *its raw multiparticle supercharacter is P^{-1}, where $e^{-\rho}P = \mathrm{Den}_{\Delta}$; equivalently $e^{\rho}\,\mathrm{sch}\,\mathrm{Fock}(\mathfrak{n}_{\Delta_5,+}) = \mathrm{Den}_{\Delta}^{-1}$;*

(v) *the half-coordinate pullback of the doubled Weyl-normalized character is $-{\chi_{10}}^{-1}$ in the Oberdieck–Pixton normalization;*

(vi) *Its normalized bounded Fredholm model has Weyl-shifted determinant Den_{Δ}. The associated automorphic line has flat character factor ν_5.*

Proof. The first three statements are the Borcherds root, quiver Hall, Hong quantum-supergroup, Ginzburg, and colored-Ran theorems. PBW gives the fourth. The identity $\chi_{10} = {D_5}^2$ and the established reduced scalar theorem give the fifth.

The normalized Fredholm formula gives the sixth. Its scalar automorphy law retains both the Hodge-line factor and the flat character. □

The tuple retains algebraic constructions associated with the Igusa presentation and their scalar characters. A geometric type-IIA or topological-string realization must construct compatible functors from its oriented extension category to these targets. Primitive states and extension amplitudes are inputs to those functors. Their action on morphisms, convolution, orientations, and chiral operations must also be specified and proved compatible.

Chapter 118

The primitive comparison and its extensions

Let $\mathfrak{p}$ be a positive-root-graded Lie superalgebra with finite-dimensional homogeneous pieces and a proper positive height. A primitive realization of the Igusa presentation is a parity-preserving homogeneous map from its labelled simple-root spaces to $\mathfrak{p}$, satisfying the defining relations.

Theorem 118.1 (Universal presentation map)**.** *Let $\mathfrak{p}$ be a positive-root-graded Lie superalgebra over $\mathbb{C}$, with finite-dimensional homogeneous pieces. Suppose a homogeneous parity-preserving map from every simple-root generating space of $\mathfrak{n}_{\Delta_s,+}$ to $\mathfrak{p}$ satisfies the defining Borcherds–Serre relations. It extends uniquely to a graded Lie-superalgebra map*

$$f : \mathfrak{n}_{\Delta_s,+} \longrightarrow \mathfrak{p}.$$

It induces continuous maps on height completions, enveloping algebras, Chevalley chains, and level-zero currents. The map f is an isomorphism if and only if it is surjective and the root-graded PBW series agree in each parity.

Proof. Extend the chosen maps to the free Lie superalgebra on the simple-root spaces. The relation hypothesis makes the defining ideal lie in its kernel, giving f. The generators prove uniqueness. On enveloping algebras the map sends $x_1 \cdots x_j$ to $f(x_1) \cdots f(x_j)$. On Chevalley chains it is the graded symmetric map on the shifted generators. It commutes with the Chevalley differential because f preserves brackets. The current map is $\partial^j x \mapsto \partial^j f(x)$, preserving $[x_\lambda y] = [x, y]$. Root degree preservation gives compatible maps at each height and hence on the stated limits.

If f is surjective, $U(f)$ is surjective. Proper positive height makes each root-graded PBW component finite-dimensional. Equality of its even and odd dimensions makes $U(f)$ bijective in every component. PBW embeds each Lie superalgebra into its enveloping algebra, so f is injective. The converse follows by applying the constructions to its inverse. □

A geometric comparison also requires functors on the actual Hall and critical categories, a compatible coproduct and Hopf pairing, and maps of determinant lines and operator domains. These categories and lines are not the domain of the primitive Lie map. The additional maps must be constructed and checked against that map. They are not consequences of its universal property.

Chapter 119

The eight diagonal-divisor forms

Cléry and Gritsenko classify eight genus-two modular forms whose divisor is the diagonal with order one [16]. Gritsenko–Nikulin and the CHL denominator constructions give the sourced reflective presentations used for the corresponding rows [34–36, 38–40]. Every row has a minimal virtual product superspace obtained from its Fourier exponents, together with a Fock determinant and a diagonal Fredholm operator. A sourced denominator presentation refines this product model by its Borcherds–Cartan bracket, Weyl group, and quantum supergroup. The half-integral rows are formed on their multiplier covers.

form	group	weight	character order
Δ_5	$\mathrm{Sp}_4(\mathbb{Z})$	5	2
Δ_2	Γ_2	2	4
Δ_1	Γ_3	1	6
$\Delta_{1/2}$	Γ_4	1/2	8
∇_3	$\Gamma_0^{(2)}(2)$	3	2
∇_2	$\Gamma_0^{(2)}(3)$	2	2
$\nabla_{3/2}$	$\Gamma_0^{(2)}(4)$	3/2	4
Q_1	$\Gamma_2(2)$	1	4

Theorem 119.1 (Eight-row algebraic realization)**.** *The integral product exponents in each diagonal-divisor row define a proper-height minimal super vector space and its formal Fock determinant. An absolute coefficient bound supplies a trace-class diagonal Fredholm model on a convergence chamber. A reflective denominator presentation supplies its root bracket and parity data. Its finite ordinary quiver Hall construction and its quantum supergroup retain their respective relation ideals. Equality of their Cartan gradings does not identify their characters.*

Proof. At each degree use an even space for a positive exponent and an odd space for a negative exponent. The elementary super-Fock calculation gives the formal product. Trace-norm summability gives the normalized Fredholm formula when the stated bound holds. The supplied root presentation carries the bracket and super relations. The ordinary quiver construction uses the even Cartan matrix, and the single isotropic odd example above prevents inferring the super character from that ordinary presentation. Chapter 20 states the eight arithmetic rows and their coefficient requirements. □

Chapter 120

Constructions and comparison maps

Theorem 120.1 (Maps from specified Igusa data)**.** *The constructions have the following inputs and outputs. An arrow labeled by a construction records its dependence on that input:*

$$\phi_{0,1} \xrightarrow{\text{Borch}} D_5, \qquad \phi_{0,1} \xrightarrow{\text{root datum}} \mathfrak{g}_{\Delta_5}, \qquad A_\Delta \xrightarrow{\text{quiver}} Q_{\Delta,H} \xrightarrow{\text{Hall}} \mathcal{B}_\Delta,$$

$$kQ_{\Delta,H} \xrightarrow{\Pi_3} \Gamma_{\Delta,H}, \qquad \mathfrak{g}_{\Delta_5} \xrightarrow{\mathrm{Cur}_E} \mathcal{F}^{\mathrm{hyb}}_{\Delta,E}, \qquad \mathfrak{g}_{\Delta_5} \xrightarrow{\text{Hong}} \widehat{U}_q(\mathfrak{g}_{\Delta_5}),$$

$$\mathfrak{g}_{\Delta_5} \xrightarrow{\text{root spectrum}} \mathbb{A}_Z, \qquad e^{-\rho}\,\mathrm{sPf}_F(\mathbb{A}_Z) = \mathrm{Den}_\Delta(Z) = D_5(2Z),$$

$$\mathcal{Z}^{\mathrm{red}}_{K3\times E} = -(J^*\,\mathrm{Den}_\Delta)^{-2} = -\chi_{10}{}^{-1}.$$

The system contains parity-refined one-particle spaces, an ordinary Hall extension product, the Borcherds root bracket, graded current operations and a quantum supergroup. Its scalar constructions give an automorphic section, a normalized bounded Fredholm determinant model and the reduced scalar square. A first-order geometric operator and a support-preserving compact collision comparison require further maps.

Proof. The Borcherds lift, Gritsenko–Nikulin presentation, and primitive reduced curve-counting theorem give the scalar identities. The enveloping and Chevalley maps are induced by the presentation. The quiver Hall algebra and finite Calabi–Yau completion have their respective quiver and path-algebra inputs. Their common grading specifies the degrees a comparison must preserve. A map between those constructions requires the additional functors, coefficient maps, and pairings described above. □

Book XX

Technical appendices to the preceding book

Chapter 121

Exact theta-series calculation

Set $t = q^{1/8}$ and $x = r^{1/2}$. Then

$$\vartheta_2(\tau, z) = \sum_{n \in \mathbb{Z}} t^{4n^2+4n+1} x^{2n+1},$$
$$\vartheta_3(\tau, z) = \sum_{n \in \mathbb{Z}} t^{4n^2} x^{2n},$$
$$\vartheta_4(\tau, z) = \sum_{n \in \mathbb{Z}} (-1)^n t^{4n^2} x^{2n}.$$

Squaring numerator and denominator, dividing in $\mathbb{Q}[x^{\pm 1}][[t]]$, and summing the three quotients gives the coefficient table in Chapter 1. The calculation uses integer convolution and formal-series inversion; fractional powers cancel before conversion from (t, x) to (q, r).

Chapter 122

The root reflection matrices

In the basis (f_2, f_3, f_{-2}), the form is

$$G = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 0 \end{pmatrix}.$$

The three reflection matrices are

$$S_1 = \begin{pmatrix} 1 & 4 & 4 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{pmatrix}, \quad S_2 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & -1 & 0 \\ 4 & 4 & 1 \end{pmatrix}, \quad S_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Direct multiplication gives

$$S_i^2 = I, \qquad S_i^{\mathsf{T}} G S_i = G.$$

Chapter 123

Automorphic, homological, and Hall comparisons

The half-index determinant, theta normalization, Borcherds denominator, reduced scalar theorem, charge Gram map, and geometric Hall comparison datum belong to the following constructions:

(i) the automorphic square root is completed entirely within the Borcherds product theorem;

(ii) the BKM bracket comes from the Gritsenko–Nikulin algebra;

(iii) the bar–Chevalley comparison gives the denominator a homological source;

(iv) the current and BF constructions give an exact first-order algebraic model;

(v) orientation data refine determinant lines, while an operator Pfaffian requires an actual Fredholm family;

(vi) Weyl reflections are attached to real roots, and imaginary roots enter through multiplicities;

(vii) the compact Hall comparison uses additive Mukai charge, the polarized central extension, and hybrid local–wrapped sectors.

Book XXI

Categorical and field-theoretic foundations

Chapter 124

Calabi–Yau categories

An extension of geometric objects defines a correspondence. Its action on cohomology depends on the coefficient theory and the allowed pushforward. Tensor products and exchange require a coproduct and compatible intertwiners. A trace on a specified representation then produces an amplitude. The reconstruction problem is to retain these operations through the comparisons between Calabi–Yau categories, boundary fields, Hall algebras, and quantum groups.

124.1 Smoothness, properness, and the Calabi–Yau class

Let C be a small stable $\Bbbk$-linear category. Smoothness means that the diagonal bimodule is perfect. Properness means that the morphism complexes are perfect over $\Bbbk$. A d-Calabi–Yau structure is a nondegenerate negative-cyclic class

$$\theta_{\mathrm{CY}} \in HC^{-}_{d}(C)$$

whose underlying Hochschild trace identifies the diagonal bimodule with its dual shifted by d.

For $E \in C$, the tangent complex of the derived moduli stack of objects is

$$T_E \mathcal{M}_C \simeq \mathrm{RHom}_C(E,E)[1].$$

The Calabi–Yau trace pairs two tangent vectors by composition and trace.

Theorem 124.1 (The two universal categorical outputs)**.** *Let C be a smooth proper d-Calabi–Yau category.*

(i) *The Hochschild cochains $CC^*(C)$ carry a framed E_2-algebra structure.*

(ii) *Whenever the moduli stack of objects is representable in the PTVV setting, $\mathcal{M}_C$ carries a $(2-d)$-shifted symplectic form.*

Proof. The cyclic Deligne theorem upgrades the brace E_2 structure on Hochschild cochains by the circle action supplied by the cyclic trace [10]. The shifted form at E is

$$\mathrm{RHom}(E,E)[1]^{\otimes 2} \xrightarrow{\circ} \mathrm{RHom}(E,E)[2] \xrightarrow{\mathrm{Tr}} \Bbbk[2-d].$$

Nondegeneracy is the Calabi–Yau duality isomorphism, and closedness is the negative-cyclic lift [9, 64]. □

The integer d controls the shift of the moduli form. The Hochschild operation remains framed E_2. These structures meet through deformation theory while retaining their separate operadic degrees.

Chapter 125

Local BV theories and factorization observables

125.1 Cyclic elliptic local Lie algebras

Let X be a complex manifold. A classical local field theory is specified by a sheaf $\mathcal{L}$ of elliptic L_∞ algebras, a local invariant pairing, and an action functional S satisfying the classical master equation. For an open set $U \subset X$, the classical observables are

$$\mathrm{Obs}^{\mathrm{cl}}(U) = C^*_{\mathrm{CE,cont}}(\mathcal{L}(U)).$$

Here $\mathcal{L}(U)$ consists of unrestricted smooth fields with its nuclear smooth-section topology. Continuous cochains use the completed symmetric powers of its continuous dual, with the completed tensor products of the elliptic-observable construction [18, 19]. The dual consists of compactly supported distributions. For $U \subset V$, restriction of fields $\mathcal{L}(V) \to \mathcal{L}(U)$ induces the covariant map on observables. For pairwise disjoint $U_i \subset V$, restrict to $\bigoplus_i \mathcal{L}(U_i)$ and multiply the pulled-back cochains. These are the prefactorization maps.

Theorem 125.1 (Factorization observables)**.** *An elliptic classical BV theory produces a factorization algebra* $\mathrm{Obs}^{\mathrm{cl}}$*. A renormalized quantization satisfying the quantum master equation produces a factorization algebra* $\mathrm{Obs}^{\mathrm{q}}$ *over* $\Bbbk[[\hbar]]$.

Proof. Restriction is a morphism of local L_∞ algebras, so its contravariant continuous Chevalley map commutes with the differential. For disjoint opens, the completed cochains of the finite direct sum are the completed tensor product of the cochains. This constructs the displayed maps and their composition identities. The elliptic descent theorem in this completed coefficient category supplies Weiss descent [18, 19]. In the quantum theory, renormalized time-ordered products define the structure maps, and the quantum master equation makes their differential square-zero and compatible with nested ultraviolet contractions [18, 19]. □

125.2 Boundary and defect restriction

Let $Y = C \times \mathbb{R}_{\geq 0}$, with C a complex curve, and let a boundary condition be given at $C \times \{0\}$. Boundary observables are holomorphic along C and ordered along the inward normal line. A compact boundary object b in the boundary factorization category gives

$$A_b = \mathrm{RHom}(b, b).$$

Theorem 125.2 (Ordered boundary algebra)**.** *When boundary observables are holomorphic on C and locally constant along the oriented normal direction, A_b is an ordered chiral algebra*

$$A_b \in \mathrm{Alg}_{\mathsf{E}_1}\left(\mathrm{Fact}_D(C), \otimes^!\right).$$

If b is a local compact generator and the Morita equivalences obey factorization descent, the boundary category is equivalent to A_b-modules in factorization coefficients.

Proof. Composition of boundary endomorphisms gives the internal E_1 multiplication. Holomorphic collision gives the factorization coefficient on C. The componentwise locality identity makes multiplication compatible with every disjoint-support factorization map. Local compact-generator Morita theory descends over the Ran diagrams. These are the recognition and Morita theorems of Volume I. □

Chapter 126

Three curve-level algebraic outputs

Let C be a smooth curve and let $\mathcal{D}_C$ be the fixed Ran category. The same geometric source can produce three algebraic outputs, and their distinction is functorial.

Definition 126.1 (Curve-level faces)**.** A transverse face is

$$A^{\perp} \in \mathrm{Alg}_{\mathsf{E}_1}(\mathrm{Fact}_D(C), \otimes^!).$$

An associative chiral face is

$$A^{\mathrm{ch}}_{\mathsf{Ass}} \in \mathrm{Alg}_{\mathsf{E}_1}(\mathcal{D}_C, \star^{\mathrm{ch}}).$$

A chiral Lie face is

$$\mathfrak{g}^{\mathrm{ch}} \in \mathrm{Alg}_{\mathsf{Lie}}(\mathcal{D}_C, \star^{\mathrm{ch}}).$$

Their bars are $\mathrm{Bar}^{\perp}$, $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}$, and $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}$.

Theorem 126.2 (Source separation)**.** *A boundary normal direction determines the transverse face. A chiral-convolution multiplication determines the associative chiral face. A Lie bracket for chiral convolution determines the Lie face. The commutator functor sends the associative chiral face to the Lie face. A chiral enveloping algebra $U^{\mathrm{ch}}(\mathfrak{g})$ satisfies*

$$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(U^{\mathrm{ch}}(\mathfrak{g})) \simeq \mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}(\mathfrak{g})$$

on the nilcomplete PBW locus. A mixed source joins the transverse and associative chiral faces through an aligned distributive law.

Proof. Each statement is the corresponding operadic restriction or adjunction in the named monoidal category. The PBW equivalence is the enveloping lane of chiral Koszul duality. The mixed statement is the distributive-law theorem of Volume I. □

This theorem replaces every dimension-based dispatch of an E_n level. The geometry of the source selects the available directions; the operadic level records the operations actually constructed.

Chapter 127

Holomorphic Chern–Simons theory

Let X be a complex threefold with holomorphic volume form Ω_X, and let $\mathfrak{a}$ be a quadratic Lie algebra. The BV fields are

$$\mathcal{E}_X = \Omega^{0,*}(X, \mathfrak{a})[1],$$

with degree -1 pairing

$$\omega(A, B) = \int_X \Omega_X \wedge \langle A, B \rangle.$$

The action is

$$S_{\mathrm{hCS}}(A) = \int_X \Omega_X \wedge \left(\frac{1}{2} \langle A, \bar{\partial} A \rangle + \frac{1}{6} \langle A, [A, A] \rangle \right).$$

Theorem 127.1 (Classical hCS factorization algebra)**.** *The action S_{hCS} satisfies the classical master equation. Its classical observables form a holomorphic factorization algebra on X.*

Proof. The Hamiltonian vector field is the Maurer–Cartan differential $\bar{\partial} A + \frac{1}{2}[A, A]$. Its square is the Dolbeault square, the derivation identity, and the Jacobi identity. The elliptic Dolbeault complex then gives factorization observables by the preceding theorem. □

A perturbative quantization is controlled by the local BV deformation complex. In complex dimension three, its one-loop gauge anomaly occupies the degree-four invariant-polynomial summand; mixed and gravitational components enter when the background fields carry the corresponding couplings. A quantization is a trivialization of the complete local anomaly class, compatible with restriction and factorization [20].

Chapter 128

Pushforward to a curve

Let $p : X \to C$ be a holomorphic map to a complex curve. For a factorization algebra $\mathbb{F}$ on X, define

$$(p_*\mathbb{F})(U) = \mathbb{F}(p^{-1}U).$$

Theorem 128.1 (Factorization pushforward)**.** *The assignment $p_*\mathbb{F}$ is a factorization algebra on C. Suppose in addition that p is proper and locally trivial in a topology compatible with the factorization theory, that $\mathbb{F}$ is locally constant in the fiber directions, and that product-chart excision and base change hold. Then on every trivializing open $U \subset C$ with fiber F_p there is a natural equivalence*

$$(p_*\mathbb{F})(U) \simeq \int_{F_p} \mathbb{F}|_{F_p \times U},$$

compatible on overlaps. If the resulting object is holomorphic and regular holonomic along C, Riemann–Hilbert algebraization gives a factorization D-module on C.

Proof. Disjoint opens in C have disjoint inverse images, so the factorization products push forward. Weiss descent is inherited from inverse images of Weiss covers. On a product chart, local constancy in the fiber and factorization excision identify the value with fiberwise factorization homology. The base-change hypothesis glues these local equivalences. Regular-holonomic Riemann–Hilbert transports the holomorphic factorization object to the algebraic D-module setting. □

Thus a curve shadow is attached to an actual map and an actual field theory. Different fibrations and boundary conditions produce different ordered chiral algebras from the same categorical source.

Chapter 129

The derived centre and the physical bulk

For an ordered boundary algebra A, define

$$Z_{\mathrm{ord}}(A) = \mathrm{RHom}_{A \otimes A^{\mathrm{op}}}(A, A).$$

It is a canonical E_2 algebra. A bulk theory acting on the boundary gives

$$\beta : \mathrm{Obs}_{\mathrm{bulk}}|_{\partial} \longrightarrow Z_{\mathrm{ord}}(A).$$

Definition 129.1 (Open–closed completeness)**.** A boundary condition is open–closed complete on a protected sector when β is an equivalence on that sector.

The definition turns the phrase ‘‘bulk equals centre’’ into a comparison theorem. The centre exists algebraically for every ordered boundary algebra; the physical bulk is identified with it by the open–closed map.

Book XXII

Hall and BPS algebras

Chapter 130

Extension correspondences

Let $\mathcal{M} = \coprod_{\gamma \in \Gamma} \mathcal{M}_\gamma$ be a moduli stack in an exact or derived category. Let $\mathcal{M}^{\mathrm{ext}}_{\alpha,\beta}$ classify extensions

$$0 \longrightarrow E_\beta \longrightarrow E \longrightarrow E_\alpha \longrightarrow 0.$$

The correspondence is

$$\mathcal{M}_\alpha \times \mathcal{M}_\beta \xleftarrow{p} \mathcal{M}^{\mathrm{ext}}_{\alpha,\beta} \xrightarrow{q} \mathcal{M}_{\alpha+\beta}.$$

Theorem 130.1 (Hall associativity)**.** *Assume a coefficient theory in which p^* and q_* are defined and satisfy base change for the two-step filtration stack. Then*

$$a * b = q_* p^* (a \boxtimes b)$$

defines an associative product.

Proof. The two parenthesizations are the two projections from the stack of filtrations $0 \subset E_1 \subset E_2 \subset E_3$. Base change identifies both composites with pullback to this stack followed by pushforward to E_3. □

For critical CoHAs, the coefficient theory includes vanishing cycles and a coherent orientation. For symplectic CoHAs, it includes the selected equivariant Borel–Moore theory. Completion is chosen so that each output degree receives a convergent sum.

Chapter 131

Orientation data and the threefold Hall product

For compact Calabi–Yau threefolds, natural orientation data on moduli of coherent sheaves and perfect complexes is constructed by Joyce and Upmeier [47]. A general cohomological Hall algebra for a 3-Calabi–Yau category uses strong orientation data, which carries the coherent multiplicativity required by the attractor and extension correspondences [50].

Theorem 131.1 (Threefold Hall algebra from strong orientation)**.** *Let C be a 3-Calabi–Yau category satisfying the geometric hypotheses of Kinjo–Park–Safronov, and assume strong orientation data. Then its cohomological Donaldson–Thomas complexes carry an associative cohomological Hall product. The product is the parabolic-induction correspondence supplied by the attractor Lagrangian correspondence.*

Proof. Kinjo–Park–Safronov prove the functoriality of the Donaldson–Thomas perverse sheaf under the attractor Lagrangian correspondence and use strong orientation to define the parabolic-induction morphism. Their base-change and associativity theorem gives the CoHA [50]. □

Definition 131.2 (Reduced $K3 \times E$ Hall datum)**.** A reduced $K3 \times E$ Hall datum is a strong-oriented threefold Hall datum together with descent along the elliptic translation action, a reduced deformation theory, and a separated completion in physical charge. A full quantum-group datum additionally includes a coproduct, Cartan extension, negative half, and nondegenerate continuous Hopf pairing.

Chapter 132

The geometric E-equivariant Hall source

Let S be a projective K3 surface, let E be an elliptic curve, and put

$$X = S \times E.$$

Write $\mathrm{Coh}_{\le 1}(X)$ for coherent sheaves whose support has dimension at most one. This is an extension-closed abelian category. Its numerical charge group is denoted Γ_X, and the degree of the projected one-cycle on E is the additive function

$$m : \Gamma_X \longrightarrow \mathbb{Z}_{\ge 0}.$$

For $\gamma \in \Gamma_X$, let $\mathcal{M}_\gamma$ be the moduli stack of objects of charge γ. Translation on the elliptic factor gives an action of E on every $\mathcal{M}_\gamma$ and on every extension stack.

Oberdieck and Shen construct the E-equivariant Grothendieck ring of stacks and its Hall convolution. In the present notation the completed algebra is

$$\mathcal{H}_X^{E,\mathrm{mot}} = \widehat{\bigoplus}_{\gamma \in \Gamma_X} K_0^E(\mathrm{St}/\mathcal{M}_\gamma),$$

where the completion is taken in effective curve class and Euler characteristic.

Theorem 132.1 (Geometric reduced Hall algebra)**.** *The extension correspondence in* $\mathrm{Coh}_{\le 1}(X)$ *defines an associative E-equivariant motivic Hall product on* $\mathcal{H}_X^{E,\mathrm{mot}}$*. The Oberdieck–Shen reduced integration morphism*

$$I_X^E : \mathcal{H}_X^{E,\mathrm{mot}} \longrightarrow \widehat{\mathbb{Q}[\Gamma_X]}[\epsilon]/(\epsilon^2)$$

is multiplicative on the regular subalgebra on which their integration theorem is defined. The coefficient of ϵ is the reduced contribution of the positive-dimensional E-orbits.

Proof. The stack of short exact sequences is E-equivariant because translation preserves exactness and numerical charge. The stack of two-step filtrations identifies the two composites of three Hall products, so equivariant base change proves associativity. The integration statement is the E-equivariant motivic Hall theorem of Oberdieck–Shen [63]. Their stabilizer stratification separates fixed contributions from moving elliptic orbits and packages the latter in the square-zero parameter ϵ. □

Chapter 133

Local and wrapped geometry inside the Hall algebra

Define the completed degree-zero and positive-degree pieces

$$\mathcal{H}_X^{\mathrm{loc}} = \widehat{\bigoplus}_{m(\gamma)=0} K_0^E(\mathrm{St}/\mathcal{M}_\gamma), \qquad \mathcal{H}_X^{\mathrm{wr}} = \widehat{\bigoplus}_{m(\gamma)>0} K_0^E(\mathrm{St}/\mathcal{M}_\gamma).$$

A pure one-dimensional support of degree zero projects to a finite subset of E. A support of positive degree projects onto E. The first sector is point-local on the elliptic curve; the second is wrapped.

Theorem 133.1 (Geometric local–wrapped decomposition)**.** *Hall multiplication respects elliptic degree:*

$$\mathcal{H}_{m_1} * \mathcal{H}_{m_2} \subseteq \mathcal{H}_{m_1+m_2}.$$

Consequently $\mathcal{H}_X^{\mathrm{loc}}$ is a subalgebra, $\mathcal{H}_X^{\mathrm{wr}}$ is a closed two-sided ideal, and

$$\mathcal{H}_X^{E,\mathrm{mot}} \cong \mathcal{H}_X^{\mathrm{loc}} \ltimes \mathcal{H}_X^{\mathrm{wr}}$$

as a complete vector space with multiplication determined by the Hall correspondences. The reduced integration morphism preserves this decomposition.

Proof. The cycle of the middle term of a short exact sequence is the sum of the cycles of its subobject and quotient. Projection to E therefore adds degrees. Degree zero is closed under addition, and positive degree absorbs addition by a nonnegative degree. The same grading is present on the target group algebra of I_X^E. □

This theorem supplies the geometric version of the local–wrapped splitting constructed from the Borcherds root algebra in the companion volume. The two constructions have the same additive elliptic degree and become comparable after a primitive root framing is chosen.

Chapter 134

Cohomological refinement and the Gram character

A cohomological refinement uses the Donaldson–Thomas sheaf on the derived moduli stack. The datum required for the convolution is the strong orientation coherence of Kinjo–Park–Safronov. A strong orientation enhancement of the natural Joyce–Upmeier orientation therefore produces

$$\mathcal{H}_X^{E,\mathrm{coh}} = \widehat{\bigoplus}_{\gamma\in\Gamma_X} H_*^{\mathrm{BM},E}(\mathcal{M}_\gamma, \Phi_\gamma),$$

with its cohomological Hall product.

Theorem 134.1 (Strong-oriented cohomological lift)**.** *For every strong orientation enhancement on the E-equivariant derived moduli problem, the attractor correspondence constructs an associative cohomological Hall algebra $\mathcal{H}_X^{E,\mathrm{coh}}$. Elliptic degree again gives a subalgebra in degree zero and a closed ideal in positive degree. Forgetting the cohomological coefficient system recovers the corresponding motivic Hall correspondences.*

Proof. The parabolic-induction theorem of Kinjo–Park–Safronov gives the cohomological product from the strong-oriented attractor Lagrangian correspondence [50]. Numerical charge and elliptic degree are constant on connected components of every correspondence, so the proof of the preceding decomposition applies verbatim. □

For the dyonic charge lattice

$$\Gamma_{\mathrm{phys}} = \widetilde{H}(S,\mathbb{Z}) \oplus \widetilde{H}(S,\mathbb{Z}),$$

write

$$\Pi(Q,P) = \left(\frac{Q^2}{2}, Q\cdot P, \frac{P^2}{2}\right).$$

To compare sheaf charges with this lattice, choose an additive map

$$c : \Gamma_X \longrightarrow \Gamma_{\mathrm{phys}}.$$

Put $B_\Pi(a,b) = \Pi(a+b) - \Pi(a) - \Pi(b)$ and give $\widetilde{\Gamma}_{\mathrm{phys}} = \Gamma_{\mathrm{phys}} \times \mathbb{Z}^3$ the group law

$$(a,u)(b,v) = (a+b, u+v+B_\Pi(a,b)).$$

Then $j_\Pi(a) = (a, \Pi(a))$ is a homomorphism. The extension is split by $(a,u)\mapsto(a,u-\Pi(a))$. It retains the quadratic Gram coordinate alongside additive physical charge.

Assume $j_\Pi c$ preserves the selected charge completions. On the regular subalgebra $\mathcal{H}_{\mathrm{reg}}$ of the reduced integration theorem, define

$$I^E_{X,\Pi,c} = (j_\Pi c)_* I^E_X : \mathcal{H}_{\mathrm{reg}} \longrightarrow \widehat{\mathbb{Q}[\widetilde{\Gamma}_{\mathrm{phys}}]}[\epsilon]/(\epsilon^2).$$

Write $I^E_{X,\Pi}$ for this map when c has been fixed.

Proposition 134.2 (Polarized reduced character)**.** *The map $I^{E}_{X,\Pi,c}$ is multiplicative. Its coefficients are the reduced invariants supplied by I^{E}_{X}, regraded by $j_{\Pi}c$. If $g \in 1 + F^{1}H$ is group-like in a complete filtered Hall bialgebra and the logarithm converges, then* $\log g$ *is primitive.*

Proof. Expanding $\Pi(a+b+d)$ gives

$$B_{\Pi}(a,b) + B_{\Pi}(a+b,d) = B_{\Pi}(b,d) + B_{\Pi}(a,b+d).$$

This proves associativity of the group law. The definition gives $j_{\Pi}(a+b) = j_{\Pi}(a)\, j_{\Pi}(b)$. Composing its group-algebra map with reduced integration proves multiplicativity on $\mathcal{H}_{\mathrm{reg}}$. For the last assertion, the two factors in $\Delta g = (g \widehat{\otimes} 1)(1 \widehat{\otimes} g)$ commute. Taking the convergent logarithm gives $\Delta \log g = \log g \widehat{\otimes} 1 + 1 \widehat{\otimes} \log g$. □

The primitive reduced Gromov–Witten theorem supplies the separate character $-\chi_{10}^{-1}$ in classes primitive in the K3 factor [62]. Comparing this character with the Hall character requires the same charge map, sector, and normalization. A primitive root framing must also preserve the extension bracket and parity of the chosen BPS states.

Chapter 135

BPS primitives

A connected graded bialgebra has primitive Lie algebra

$$\operatorname{Prim}(H) = \{x : \Delta x = x \otimes 1 + 1 \otimes x\}.$$

In cohomological Donaldson–Thomas theory, the BPS sheaf isolates the primitive part of the Hall algebra.

Theorem 135.1 (Two-Calabi–Yau BPS algebra)**.** *For the geometric two-Calabi–Yau categories covered by the BPS-algebra theorem, the BPS algebra is the positive enveloping algebra of a generalized Kac–Moody Lie algebra. This scope includes semistable sheaves on K3 and other symplectic surfaces under the stated good-moduli hypotheses.*

Proof. The theorem is the Davison–Hennecart–Schlegel Mejia construction [22]. The simple BPS sheaves give the simple-root spaces, extension correspondences give the bracket, and the integrality/PBW theorem identifies the Hall algebra with the positive enveloping algebra. □

This result supplies a global K3 generalized Kac–Moody algebra at the two-Calabi–Yau Hall level. Yangian structures arise from additional equivariant correspondences and coproducts in the surface and quiver models described below.

Chapter 136

The integral Hall–Borcherds amalgam

Let $\mathfrak{p}_X^+$ be a complete charge-graded BPS Lie superalgebra supplied by a geometric Hall source, and let $\mathfrak{n}_{\Delta_s,+}$ be the positive Igusa Borcherds Lie superalgebra. The polarized charge extension places both gradings over one abelian Cartan algebra $\mathfrak{h}_\Pi$. Form the two complete Borels

$$\mathfrak{b}_X = \mathfrak{h}_\Pi \ltimes \mathfrak{p}_X^+, \qquad \mathfrak{b}_\Delta = \mathfrak{h}_\Pi \ltimes \mathfrak{n}_{\Delta_s,+}.$$

The Cartan acts through the physical charge on the first Borel and through Gram degree on the second.

Definition 136.1 (Integral protected Lie algebra)**.** The integral Hall–Borcherds algebra is the completed amalgamated coproduct

$$\mathfrak{b}_{X\boxtimes\Delta} = \mathfrak{b}_X \widehat{\amalg}_{\mathfrak{h}_\Pi} \mathfrak{b}_\Delta$$

in complete charge-filtered dg Lie superalgebras.

Theorem 136.2 (Hall–Borcherds amalgamation)**.** *The completed amalgam exists and represents pairs of continuous Lie morphisms from $\mathfrak{b}_X$ and $\mathfrak{b}_\Delta$ that agree on $\mathfrak{h}_\Pi$. The Cartan projections of the two Borels induce canonical retractions*

$$r_X : \mathfrak{b}_{X\boxtimes\Delta} \longrightarrow \mathfrak{b}_X, \qquad r_\Delta : \mathfrak{b}_{X\boxtimes\Delta} \longrightarrow \mathfrak{b}_\Delta.$$

Its completed enveloping algebra is the amalgamated free product

$$\widehat{U}(\mathfrak{b}_{X\boxtimes\Delta}) \cong \widehat{U}(\mathfrak{b}_X) \widehat{*}_{\widehat{U}(\mathfrak{h}_\Pi)} \widehat{U}(\mathfrak{b}_\Delta).$$

The geometric Hall operations survive under r_X, while the Igusa bracket, PBW character, and automorphic determinant survive under r_Δ.

Proof. Construct the ordinary Lie coproduct from the free product of the two Borels and divide by the ideal identifying their two Cartan copies. Complete by total positive charge and proper root height. Every fixed filtered degree uses finitely many Lie words, so the completion is separated and has the stated mapping property. The projection $\mathfrak{b}_X \to \mathfrak{h}_\Pi \to \mathfrak{b}_\Delta$ together with the identity on $\mathfrak{b}_\Delta$ gives r_Δ by universality; the second retraction is symmetric. The enveloping functor preserves coproducts of Lie algebras, and PBW identifies the completed algebraic filtration with the displayed completed free product. □

Definition 136.3 (Comparison ideal)**.** A primitive comparison $\varphi : \mathfrak{n}_{\Delta_s,+} \to \mathfrak{p}_X^+$ preserving Cartan degree defines the closed ideal

$$\mathcal{J}_\varphi = \overline{\langle i_\Delta(x) - i_X(\varphi(x)) \mid x \in \mathfrak{n}_{\Delta_s,+} \rangle} \subset \mathfrak{b}_{X\boxtimes\Delta}.$$

The diagonal realization is

$$\mathfrak{b}_{X\simeq\Delta,\varphi} = \mathfrak{b}_{X\boxtimes\Delta} / \mathcal{J}_\varphi.$$

Proposition 136.4 (Graph quotient of a primitive comparison)**.** *Let $\varphi : \mathfrak{b}_\Delta \to \mathfrak{b}_X$ be a continuous Lie morphism restricting to the identity on $\mathfrak{h}_\Pi$. Use a completion in which the amalgam and closed quotient have their stated universal properties. Then*

$$\mathfrak{b}_{X\boxtimes\Delta}/\mathcal{J}_\varphi \cong \mathfrak{b}_X.$$

The two factor maps become id *and φ. If φ is an isomorphism, the quotient is also isomorphic to $\mathfrak{b}_\Delta$.*

Proof. The maps id and φ agree on $\mathfrak{h}_\Pi$. They give $q_\varphi : \mathfrak{b}_{X\boxtimes\Delta} \to \mathfrak{b}_X$, which kills the generators of $\mathcal{J}_\varphi$. The inclusion of $\mathfrak{b}_X$ gives a map s in the other direction. The composite $q_\varphi s$ is the identity. The other composite is the identity on the geometric factor. On the Borcherds factor, the defining relation identifies x with $s\varphi(x)$, so it is also the identity there. The two factor images topologically generate the quotient. Continuity proves that the maps are inverse on the completion. □

The graph quotient implements a specified primitive comparison. Constructing that comparison requires geometric simple states whose extension brackets satisfy the Borcherds relations. The quotient formula alone produces neither those states nor their orientation data.

Chapter 137

Coproducts, pairings, and doubles

A positive Hall algebra becomes a quantum group after additional structure is supplied. Let H^+ be a complete graded bialgebra, let H^0 be a Cartan algebra, and let H^- be the opposite or dual half. Let

$$\langle -, - \rangle : H^+ \mathbin{\widehat{\otimes}} H^- \longrightarrow \Bbbk$$

be a nondegenerate Hopf pairing.

Definition 137.1 (Completed Hall double)**.** The completed Drinfeld double is the topological algebra

$$\widehat{D}(H^+) = H^- \mathbin{\widehat{\otimes}} H^0 \mathbin{\widehat{\otimes}} H^+$$

with internal products in the three factors and cross-relations determined by the Hopf pairing. Its representation category is monoidal by the coproduct and braided by the universal R-matrix whenever the canonical tensor converges.

Theorem 137.2 (Double construction)**.** *Let H^+ and H^- be paired complete graded bialgebras, let H^0 be a Cartan bialgebra acting by the stated weight grading, and let*

$$\langle -, - \rangle : H^+ \mathbin{\widehat{\otimes}} H^- \longrightarrow \Bbbk$$

be a continuous nondegenerate Hopf pairing. Assume the triangular multiplication map from $H^- \mathbin{\widehat{\otimes}} H^0 \mathbin{\widehat{\otimes}} H^+$ is separated and complete. Then the pairing cross-relations determine a unique completed double with this triangular decomposition. On every continuous module category on which the canonical tensors converge, the completed canonical element gives the universal R-matrix.

Proof. The Hopf pairing defines the cross-relations between the paired halves and makes the three canonical embeddings bialgebra morphisms. The Hopf identities identify the two reductions of every triple product, proving associativity. Separated triangular decomposition gives uniqueness. Finite-degree canonical tensors form a compatible inverse system; convergence on the selected module category yields the quasitriangular element [28, 30]. □

Theorem 137.3 (Quantum Hall–Borcherds amalgam)**.** *Let H_X and H_Δ be complete Hopf algebras containing the same complete Cartan Hopf algebra H^0, with continuous Hopf retractions onto H^0. Their completed amalgamated free product*

$$H_{X \boxtimes \Delta} = H_X \mathbin{\widehat{*}_{H^0}} H_\Delta$$

has a unique complete Hopf structure extending the two coproducts, counits, and antipodes. It represents pairs of continuous Hopf morphisms agreeing on H^0 and has canonical Hopf retractions to both factors. A mixed universal R-matrix is equivalent to a continuous cross-pairing satisfying the two hexagon identities; the intrinsic R-matrices of the two retracts remain defined independently.

Proof. The algebraic amalgamated free product exists because complete filtered algebras are obtained degreewise from finite quotients. The two coproducts agree on H^{0}, so the universal property extends them to a coproduct on the free product. Coassociativity is checked on the two generating Hopf subalgebras. The same argument extends the counit and antipode. The Hopf retractions are induced by the factor retractions onto H^{0}. A quasitriangular structure extending both factors requires the additional commutation data between the two generating halves; writing this data as a cross-pairing turns the quasitriangular and Yang–Baxter equations into the two hexagon identities. □

The deconcatenation coproduct on a bar complex and the representation-theoretic coproduct on a quantum group occupy different structures. A geometric comparison between them is a theorem about a selected model.

Chapter 138

Ordered chiral bialgebras

A tensor product of representations requires compatible actions on threefold products and the unit. Retain the factorization coefficient category $(\mathrm{Fact}_D(C), \otimes^!)$ on the smooth complex curve C. Fix a category $\widehat{\mathcal{V}}_C$ of complete filtered objects with specified presentations $F \simeq \varprojlim_r F_r$, where $F_r \in \mathrm{Fact}_D(C)$. Use derived limits in the fixed Ran enhancement and define

$$F \mathbin{\widehat{\otimes}^!} G = \varprojlim_{r,s} (F_r \otimes^! G_s).$$

Assume these limits remain factorizing and give $\widehat{\mathcal{V}}_C$ a symmetric monoidal structure with unit $\mathbf{1}$. Its associativity, symmetry, and unit equivalences must preserve the presentations and the Ran structure. Morphisms are continuous maps compatible with these presentations. These are hypotheses on the chosen completion.

Definition 138.1 (Ordered chiral bialgebra)**.** An ordered chiral bialgebra is a counital E_1-coalgebra object in complete ordered chiral algebras:

$$A \in \mathrm{CoAlg}_{\mathsf{E}_1}\Big(\mathrm{Alg}_{\mathsf{E}_1}(\widehat{\mathcal{V}}_C, \mathbin{\widehat{\otimes}^!})\Big).$$

It has continuous multiplication m and unit η, and continuous morphisms of unital ordered chiral algebras

$$\Delta : A \longrightarrow A \mathbin{\widehat{\otimes}^!} A, \qquad \epsilon : A \longrightarrow \mathbf{1}.$$

The counit is the chosen augmentation. Write τ for the coefficient symmetry, including Koszul signs. Suppressing ambient associativity and unit equivalences, compatibility reads

$$\Delta m = (m \mathbin{\widehat{\otimes}^!} m)(\mathrm{id} \mathbin{\widehat{\otimes}^!} \tau \mathbin{\widehat{\otimes}^!} \mathrm{id})(\Delta \mathbin{\widehat{\otimes}^!} \Delta),$$
$$\Delta\eta = \eta \mathbin{\widehat{\otimes}^!} \eta, \qquad \epsilon m = \epsilon \mathbin{\widehat{\otimes}^!} \epsilon, \qquad \epsilon\eta = \mathrm{id}_{\mathbf{1}}.$$

Coassociativity and the counit identities read

$$(\Delta \mathbin{\widehat{\otimes}^!} \mathrm{id})\Delta = (\mathrm{id} \mathbin{\widehat{\otimes}^!} \Delta)\Delta,$$
$$(\epsilon \mathbin{\widehat{\otimes}^!} \mathrm{id})\Delta = \mathrm{id}_A = (\mathrm{id} \mathbin{\widehat{\otimes}^!} \epsilon)\Delta.$$

For a homotopy-coherent presentation, these identities include the higher coherences of the displayed coalgebra object. Every operation and coherence preserves diagonal transitions and disjoint-locus factorization maps. An antipode is a continuous convolution inverse $S : A \to A$ of id_A:

$$m(S \mathbin{\widehat{\otimes}^!} \mathrm{id})\Delta = \eta\epsilon = m(\mathrm{id} \mathbin{\widehat{\otimes}^!} S)\Delta.$$

A bialgebra with an antipode is a Hopf object in this coefficient category.

A continuous left A-module is an object $M \in \widehat{\mathcal{V}}_C$ with a unital E_1 action $\rho_M : A \mathbin{\widehat{\otimes}^!} M \to M$. Its action and coherences are morphisms in these factorization coefficients. Write $\mathrm{Mod}_A(\widehat{\mathcal{V}}_C)$ for their category.

Theorem 138.2 (Representation fusion)**.** *An ordered chiral bialgebra makes* $\mathrm{Mod}_A(\widehat{\mathcal{V}}_C)$ *monoidal. Its product is* $M \mathbin{\widehat{\otimes}^!} N$ *with action*

$$\begin{aligned} A \mathbin{\widehat{\otimes}^!} (M \mathbin{\widehat{\otimes}^!} N) &\xrightarrow{\Delta \mathbin{\widehat{\otimes}^!} \mathrm{id}} (A \mathbin{\widehat{\otimes}^!} A) \mathbin{\widehat{\otimes}^!} (M \mathbin{\widehat{\otimes}^!} N) \\ &\xrightarrow{\mathrm{id} \mathbin{\widehat{\otimes}^!} \tau_{A,M} \mathbin{\widehat{\otimes}^!} \mathrm{id}} (A \mathbin{\widehat{\otimes}^!} M) \mathbin{\widehat{\otimes}^!} (A \mathbin{\widehat{\otimes}^!} N) \\ &\xrightarrow{\rho_M \mathbin{\widehat{\otimes}^!} \rho_N} M \mathbin{\widehat{\otimes}^!} N. \end{aligned}$$

The unit is $\mathbf{1}$ *with action induced by* ϵ*. The ambient associator and unit equivalences are module equivalences.*

Suppose a continuous universal R*-matrix is also given. It is an invertible morphism* $R : \mathbf{1} \to A \mathbin{\widehat{\otimes}^!} A$ *for the tensor-product algebra, satisfying*

$$R\Delta(a) = \Delta^{\mathrm{op}}(a) R, \qquad \Delta^{\mathrm{op}} = \tau_{A,A} \Delta,$$
$$(\Delta \mathbin{\widehat{\otimes}^!} \mathrm{id})(R) = R_{13} R_{23}, \qquad (\mathrm{id} \mathbin{\widehat{\otimes}^!} \Delta)(R) = R_{13} R_{12},$$
$$(\epsilon \mathbin{\widehat{\otimes}^!} \mathrm{id})(R) = \eta = (\mathrm{id} \mathbin{\widehat{\otimes}^!} \epsilon)(R).$$

The first equation is an identity of morphisms from A *to its completed tensor square. Leg notation uses the unit and symmetry of* $\widehat{\mathcal{V}}_C$*. In a homotopy-coherent presentation, require compatible lifts of these identities. Then* $c_{M,N} = \tau_{M,N} \circ R_{M,N}$ *is a braiding, where* $R_{M,N}$ *acts through the external tensor-product module action.*

Proof. The symmetry combines the independent module actions into an $A \mathbin{\widehat{\otimes}^!} A$-action. Restriction along the unital algebra map Δ gives the displayed action. For three modules, the two actions use $(\Delta \mathbin{\widehat{\otimes}^!} \mathrm{id})\Delta$ and $(\mathrm{id} \mathbin{\widehat{\otimes}^!} \Delta)\Delta$. Coassociativity makes the ambient associator A-linear. The counit identities make the ambient unit equivalences A-linear. The coalgebra coherences give the pentagon and triangle coherences. The completion hypotheses keep these maps continuous and factorizing. Quasitriangularity makes $c_{M,N}$ an intertwiner and invertibility makes it an equivalence. The two coproduct identities for R give the hexagons. Its counit identities give compatibility with the unit. □

The tensor in this construction is the completed transverse tensor $\otimes^!$ on factorization coefficients. A chiral-convolution or Hall coproduct requires its own construction and a comparison in the specified category. A spectral version additionally requires a difference coordinate, a Laurent-series completion, and compatible domains for iterated expansions. Its identities must hold in those domains before defining module actions.

Example 138.3 (A counital algebra map without coassociativity)**.** In ordinary complex vector spaces, let $A = \mathbb{C}[x]$ and put

$$\Delta(x) = x \otimes 1 + 1 \otimes x + x \otimes x^2, \qquad \epsilon(x) = 0.$$

Polynomial substitution extends these assignments to unital algebra maps. Writing $F(u, v) = u + v + uv^2$, the equations $F(0, v) = v$ and $F(u, 0) = u$ prove both counit identities. But

$$\begin{aligned} F(F(u, v), w) - F(u, F(v, w)) &= -uv^2w^4 - uv^2w^2 - 2uvw^3 - 2uvw \\ &= -uvw(1 + w^2)(vw + 2) \neq 0. \end{aligned}$$

On three regular modules, the two actions differ already on $1 \otimes 1 \otimes 1$. Thus the canonical tensor associator is not A-linear. This ordinary example shows why multiplicativity and the counit identities must be supplemented by coassociativity. It makes no assertion about a geometric realization in factorization coefficients.

Book XXIII

Exact model cases

Chapter 139

The A_2 Hall algebra

For the quiver $1 \to 2$ over $\mathbb{F}_q$, the twisted Hall algebra has generators E_1, E_2 and $v^2 = q$. Counting subrepresentations gives

$$E_1^2 E_2 - (v + v^{-1}) E_1 E_2 E_1 + E_2 E_1^2 = 0.$$

For a representation of dimension vector $(2,1)$, the arrow has rank zero or one. The relevant flag counts are

rank	112	121	211
0	$q+1$	$q+1$	$q+1$
1	$q+1$	1	0.

After the Hall twists $v^{-1}, 1, v$, both rows give zero. This is the elementary geometric origin of the quantum Serre relation [37, 66].

Chapter 140

The tripled Jordan quiver and $\mathbb{C}^3$

Let Q have one vertex and loops x, y, z, with potential

$$W = \operatorname{Tr} x[y, z].$$

The Jacobi algebra is $\mathbb{C}[x, y, z]$. The critical CoHA uses the vanishing cycles of $\operatorname{Tr} W$ on the representation spaces. Its positive currents are generated by the one-vertex dimension sectors.

Theorem 140.1 (Affine-Yangian positive half)**.** *After the standard equivariant localization and completion, the critical CoHA of the tripled Jordan quiver is identified with the positive half of the affine Yangian of $\mathfrak{gl}_1$. The independently defined affine Yangian carries its standard Cartan and negative current halves; the positive-half comparison embeds the CoHA into that triangular presentation.*

Proof. The CoHA product becomes the shuffle product under localization. Dimensional reduction identifies the tripled-quiver critical theory with the equivariant surface CoHA, and Davison's affine-BPS theorem identifies its completed positive algebra with the strictly positive affine Yangian. The shuffle kernel satisfies the affine-Yangian current relations with equivariant parameters ϵ_1, ϵ_2, ϵ_3 and Calabi–Yau relation $\epsilon_1 + \epsilon_2 + \epsilon_3 = 0$. PBW and current comparison identify the positive algebra. The independently presented full affine Yangian is the triangular algebra generated by its negative currents, Cartan currents, and this positive half [21, 23, 67, 68]. □

The vacuum representation produces a $W_{1+\infty}$-type vertex algebra through an evaluation homomorphism. The CoHA, its double, and the vacuum algebra occupy successive levels: positive algebra, paired quantum group, and distinguished representation.

Chapter 141

Surface CoHAs and ADE Yangians

Let $S_\Gamma \to \mathbb{C}^2/\Gamma$ be the minimal resolution for a finite subgroup $\Gamma \subset \mathrm{SL}_2(\mathbb{C})$. The exceptional curves have ADE intersection matrix. Moduli of sheaves supported on these curves carry convolution correspondences.

Theorem 141.1 (Curve-supported surface CoHA)**.** *The completed positive CoHA of one-dimensional sheaves supported on the exceptional curves is identified with a nonstandard positive half of the affine Yangian of the corresponding ADE Lie algebra.*

Proof. The theorem is the curve-supported surface construction of Diaconescu–Porta–Sala–Schiffmann–Vasserot [25]. The simple curve sheaves give the simple generators, Hecke correspondences give the current operators, and the completed PBW comparison gives the affine Yangian presentation. □

This theorem is a local K3 model whenever an ADE configuration occurs on a K3 surface. The global K3 surface adds compact curve classes and monodromy beyond the local root lattice.

Chapter 142

Coulomb branches and shifted Yangians

For a three-dimensional $\mathcal{N} = 4$ gauge theory, the BFN Coulomb branch is defined by equivariant Borel–Moore homology of an affine-Grassmannian correspondence. Its convolution algebra quantizes the Coulomb branch [11]. In quiver gauge theories, shifted Yangians and their truncations provide presentations of these quantizations. Boundary conditions select the shift and truncation data.

The BFN construction supplies a second geometric source of Yangians. Its grading and coproduct arise from loop rotation and convolution, while the surface CoHA source arises from sheaf extensions. A comparison between the two is a model-specific equivalence.

Book XXIV

K3 symmetries

Chapter 143

The Mukai lattice

For a K3 surface S, the Mukai lattice is

$$\widetilde{H}(S,\mathbb{Z}) = H^0(S,\mathbb{Z}) \oplus H^2(S,\mathbb{Z}) \oplus H^4(S,\mathbb{Z})$$

with pairing

$$((r,c,s),(r',c',s')) = c\cdot c' - rs' - r's.$$

It is even unimodular of signature $(4,20)$ and rank 24.

Chapter 144

The Nakajima Heisenberg algebra

Let

$$\mathbb{H}_S = \bigoplus_{n \geq 0} H^*(S^{[n]}).$$

For $m \in \mathbb{Z} \setminus \{0\}$ and $\alpha \in H^*(S)$, the Nakajima correspondence gives an operator $p_m(\alpha)$.

Theorem 144.1 (K3 Heisenberg action)**.** *The operators satisfy*

$$[p_m(\alpha), p_n(\beta)] = m\, \delta_{m+n,0} \int_S \alpha \cup \beta \ \mathrm{id}\,.$$

They generate the rank-twenty-four Heisenberg algebra acting irreducibly on $\mathbb{H}_S$ with vacuum $1 \in H^*(S^{[0]})$.

Proof. The commutator is the excess intersection of the two incidence correspondences. Opposite creation and annihilation correspondences meet along the diagonal with multiplicity m and intersection pairing $\int_S \alpha\beta$. Correspondences of the same sign commute. The vacuum construction gives the Fock representation [61]. □

This global action is a central extension of the abelian mode algebra, with the displayed nonzero Heisenberg commutator. It gives the rank-twenty-four Fock representation. A broader K3 algebra must specify an embedding or action comparison preserving these operators.

Chapter 145

The orthogonal LLV algebra

For a compact hyperkähler manifold, every Lefschetz class gives an $\mathfrak{sl}_2$ triple on cohomology. The Lie algebra generated by all such triples is the Looijenga–Lunts–Verbitsky algebra.

Theorem 145.1 (Orthogonal LLV symmetries)**.** *For a K3 surface S, the LLV algebra acting on $H^*(S)$ is*

$$\mathfrak{g}_{\mathrm{LLV}}(S) \cong \mathfrak{so}\big(\widetilde{H}(S), (-,-)_{\mathrm{Muk}}\big) \cong \mathfrak{so}(4, 20).$$

For an irreducible hyperkähler manifold M of $K3^{[n]}$ type with $n \geq 2$, one has $b_2(M) = 23$ and

$$\mathfrak{g}_{\mathrm{LLV}}(M) \cong \mathfrak{so}(4, 21).$$

Moduli spaces of stable objects on S carry their own LLV actions; Mukai and monodromy correspondences relate these actions to the K3 lattice.

Proof. The Looijenga–Lunts–Verbitsky theorem identifies the Lie algebra generated by all Lefschetz triples with the orthogonal algebra of the Mukai extension $H^2(M) \oplus U$. For S, the extension has signature $(4, 20)$. For $K3^{[n]}$ type with $n \geq 2$, the Beauville–Bogomolov lattice has signature $(3, 20)$, so adjoining the hyperbolic plane gives signature $(4, 21)$ [56, 69]. □

The LLV algebra is a classical cohomological symmetry. A Yangian deformation of the full orthogonal action consists of two further structures: a Drinfeld current presentation and a compatible spectral coproduct.

Chapter 146

Local ADE Yangians

An ADE configuration of (-2)-curves on S gives a root sublattice of $H^2(S,\mathbb{Z})$. Sheaves supported on the configuration produce the surface CoHA of the preceding part and hence a completed positive affine ADE Yangian. Nakajima quiver varieties supply compatible Yangian actions on their equivariant cohomology.

Thus K3 geometry contains local nonabelian Yangian sectors indexed by ADE enhancements of the Picard lattice. Picard monodromy transports the ADE root sublattices over the corresponding Noether–Lefschetz loci. A family of sheaf moduli and Hecke correspondences compatible with that monodromy transports the Yangian actions themselves.

Chapter 147

The K3 symmetry trichotomy

The three exact structures are

Structure	Carrier	Operation
Mukai Heisenberg	$\bigoplus_n H^*(S^{[n]})$	creation and annihilation
LLV $\mathfrak{so}(4,20)$	$H^*(S)$; separately $\mathfrak{so}(4,21)$ on $K3^{[n]}$ type	Lefschetz Lie action
Local ADE Yangian	curve-supported equivariant moduli	Hecke and current correspondences

A global K3 quantum algebra is a system that contains these three actions with explicit comparison homomorphisms. The two-Calabi–Yau BPS theorem supplies a global generalized Kac–Moody positive algebra. A full Yangian or double is specified by a compatible coproduct, spectral parameter, negative half, Cartan part, pairing, and completion.

Chapter 148

K3 correspondences on fixed coefficients

Fix rational cohomology and a collection $\mathscr{M}_S$ of smooth projective moduli spaces on a projective K3 surface S. Include the Hilbert schemes $S^{[n]}$ and close the collection under finite products. For $M, N \in \mathscr{M}_S$, define the graded kernel space

$$K(M,N) = H^*(M \times N, \mathbb{Q})[2\dim_{\mathbb{C}} M].$$

A kernel of degree r acts from $H^j(M)$ to $H^{j+r}(N)$ by

$$T_\alpha(v) = (p_N)_*(p_M^* v \cup \alpha).$$

The shift in $K(M,N)$ accounts for integration over M. All pushforwards in this construction are proper.

Proposition 148.1 (Kernel composition)**.** *The product*

$$\beta \circ \alpha = (p_{13})_*(p_{12}^*\alpha \cup p_{23}^*\beta)$$

defines an associative graded category with diagonal identities. The kernel action identifies $K(M,N)$ *with* $\mathrm{Hom}^*(H^*(M), H^*(N))$.

Proof. The Künneth isomorphism and Poincaré duality identify $H^*(M \times N)[2\dim M]$ with $H^*(M)^\vee \otimes H^*(N)$. Under this identification, T_α is the associated linear map. Integration over the middle factor contracts its dual pair, so $T_{\beta\circ\alpha} = T_\beta T_\alpha$. Faithfulness then proves associativity and the diagonal identity. External products give tensor products with their Koszul signs. □

A geometric symmetry is specified by selected kernels in this category. Nakajima incidence kernels connect different Hilbert schemes. Lefschetz multiplication and its adjoint act on each individual cohomology space. Using every cohomology kernel would instead give all linear maps; the selected geometric kernels carry the representation-theoretic content.

Chapter 149

The algebra generated by geometric actions

Put $V_S = \bigoplus_{M \in \mathscr{M}_S} H^*(M, \mathbb{Q})$. Choose a set of homogeneous operators T_λ on V_S, each sending a vector with finite support to a vector with finite support. The Nakajima operators on the Hilbert-scheme summand satisfy this condition: each changes the point number by a fixed integer. Individual Lefschetz operators, extended by zero on other summands, also satisfy it. Let F be the free unital graded algebra on symbols t_λ.

Proposition 149.1 (Geometric action algebra)**.** *The map*

$$\rho : F \longrightarrow \mathrm{End}_{\mathbb{Q}}(V_S), \qquad t_\lambda \longmapsto T_\lambda$$

defines a faithful action of $A_S = F/\ker \rho$. For every algebra B, maps $A_S \to B$ are exactly maps $F \to B$ that kill $\ker \rho$*. Products and commutators of the selected kernels are their convolution products and commutators on the relevant moduli spaces.*

Proof. Finite words in the selected operators preserve finite support, so ρ is defined by composition. The quotient acts faithfully by definition of its kernel. The factorization property is the universal property of an algebra quotient. The convolution assertion follows from Proposition 148.1. □

An equivariant ADE construction uses a separate coefficient field, for example $\mathrm{Frac}\, H^*_T(\mathrm{pt}, \mathbb{Q})$, and its prescribed localization and proper pushforwards. Joining it to A_S requires maps between the actual state spaces that intertwine those operators and their coefficients. A completion of A_S requires a specified topology and continuity of every generator. The indefinite Mukai norm alone supplies neither condition. Propositions 20.7 and 20.6 give an explicit cohomological conjugation and its integral specialization, including the fourth-order pole and the distinct free image lattice. The desired common K3 quantum algebra must construct these local-to-global actions and their mixed convolution relations.

Chapter 150

Tensor reconstruction and geometric kernels

Let $\mathscr{C}$ be an essentially small rigid linear monoidal category and let $\omega : \mathscr{C} \to \mathrm{Vect}_{\mathrm{fd}}(\mathbb{Q})$ be a strong monoidal functor. A choice of geometric kernels specifies the morphisms of $\mathscr{C}$; it is part of the input. Define

$$H = \left(\bigoplus_{M \in \mathscr{C}} \omega(M)^* \otimes \omega(M) \right) \Big/ \left\langle [\lambda \circ \omega(f) \otimes v]_M - [\lambda \otimes \omega(f)v]_N \right\rangle,$$

where $f : M \to N$, $v \in \omega(M)$, and $\lambda \in \omega(N)^*$.

Proposition 150.1 (Matrix-coefficient coalgebra)**.** *For a basis (e_i) with dual basis (e^i), the formulas*

$$\Delta[\lambda \otimes v]_M = \sum_i [\lambda \otimes e_i]_M \otimes [e^i \otimes v]_M, \qquad \epsilon[\lambda \otimes v]_M = \lambda(v)$$

are basis-independent and make H a coalgebra. Tensor product gives its multiplication, and duals give its antipode. Each $\omega(M)$ is a comodule by

$$v \longmapsto \sum_i e_i \otimes [e^i \otimes v]_M.$$

Every selected kernel is a comodule morphism.

Proof. The identity tensor $\sum_i e_i \otimes e^i$ is basis-independent. Inserting it twice proves coassociativity; evaluation proves both counit identities. The defining relations express naturality under f and are preserved by these formulas. The strong monoidal constraints identify the coefficient of $M \otimes N$ with the product of coefficients of M and N. Their pentagon and unit identities give the bialgebra axioms. The evaluation and coevaluation identities for duals give the antipode identities. The same relations prove that $\omega(f)$ intertwines the coactions. □

The linear dual of H represents linear natural endomorphisms of ω. Tensor-compatible natural automorphisms are the corresponding multiplicative, unit-preserving families, not all elements of this dual. An equivalence with a comodule category requires the additional exactness and reconstruction hypotheses stated in the Tannaka theorem below. The action algebra A_S acts by selected kernels; the coend records symmetries commuting with those kernels.

Book XXV

The exact $K3 \times E$ protected completion

Chapter 151

The protected input

The K3 elliptic genus is $2\phi_{0,1}$. The Borcherds half determines the weight-five denominator D_5 and its Borcherds Lie superalgebra $\mathfrak{g}_\Delta$. Fix its positive root spaces with their parity and proper height. The positive algebra and its root Fock space are defined from this Lie superalgebra, independently of a compact Hall realization.

For each finite zero-potential quiver window $Q_{\Delta,H}$, form the Ginzburg algebra $\Gamma_{\Delta,H}$. Restriction to full subquivers gives the projections of Proposition 157.1. Write Γ_Δ for this specified inverse system. A non-degenerate continuous Calabi–Yau structure on a completed limit is additional data. An ordinary Hall category $\mathcal{B}_\Delta$ must carry its chosen quiver coefficients and extension correspondences. Its identification with a root presentation requires the corresponding Hall theorem on that precise datum.

Definition 151.1 (Root-datum realizations)**.** A protected root-datum realization records

$$\mathfrak{P}(K3 \times E) = (\Gamma_\Delta, \mathcal{B}_\Delta, \mathcal{H}_{\mathrm{ip}}, \mathcal{F}^{\mathrm{hyb}}_{\Delta,E}, \widehat{U}_q(\mathfrak{g}_\Delta), \mathbb{A}_Z).$$

Here $\mathcal{H}_{\mathrm{ip}} = \bigoplus_{\alpha>0}(\mathfrak{g}_\Delta)_\alpha$ with its parity. The current-factorization, paired quantum-group, and Fredholm entries retain their own coefficient categories, completions, and defining hypotheses. Maps among these entries are specified realization maps, not consequences of using the same root labels.

The enveloping and Chevalley constructions are functorial under the primitive maps of Theorem 156.1. The root Fock trace is computed from the actual graded root spaces. A Fredholm realization adds a family of operators on a specified domain and the convergence and orientation data identifying its Pfaffian with the denominator. These are the analytic inputs to the square-root comparison below.

Definition 151.2 (Integral protected realization)**.** Let $\mathfrak{p}^+_X$ be the BPS primitive Lie algebra of a selected strong-oriented cohomological lift of the geometric E-equivariant Hall source. The integral protected realization is

$$\mathfrak{P}_{\mathrm{int}}(X, \Delta) = \left(\mathcal{H}^{E,\mathrm{mot}}_X, \mathfrak{P}(K3 \times E), \mathfrak{b}_{X\boxtimes\Delta}, r_X, r_\Delta, I^E_{X,\Pi}\right).$$

At the motivic level the first, second, and character components are defined. A strong orientation enhancement supplies $\mathfrak{p}^+_X$ and hence the amalgamated primitive component.

Proposition 151.3 (Primitive graph comparison)**.** *Suppose the two primitive Borels, their common Cartan action, and their completed amalgam are supplied as above. Their Cartan retractions induce retractions from the amalgam to each factor. For each specified primitive map φ preserving Cartan action, its graph quotient is canonically the geometric Borel.*

Proof. The identity on one Borel and the Cartan projection of the other agree on the common Cartan, so the amalgam gives the retraction. The quotient statement is Proposition 136.4. □

The motivic integration map and primitive comparison are different maps. Strong orientation gives the coefficient coherence of a cohomological Hall construction under its stated geometric hypotheses. Extracting a primitive Lie algebra additionally requires the appropriate BPS or coproduct construction.

Chapter 152

Cyclic open-string operations

Let A be a finite-dimensional, uncurved cyclic A_∞ algebra over $\mathbb{C}$, with a nondegenerate pairing of degree -3. Use the cohomological suspension $s : A \to A[1]$ of degree -1. The suspended operations $b_n : (A[1])^{\otimes n} \to A[1]$ have degree one. Our sign convention is the coderivation convention $b^2 = 0$ on the tensor coalgebra, with the Koszul rule for moving homogeneous inputs. Write ω for the resulting degree--1 symplectic form on $A[1]$. Fix the Hamiltonian convention $\iota_{X_F}\omega = dF$ and $[X_F, X_G] = X_{\{F,G\}}$.

For the universal degree-zero coordinate x on $A[1]$, put

$$S(x) = \sum_{n \geq 1} \frac{1}{n+1}\omega(x, b_n(x^{\otimes n})).$$

Cyclic invariance is read with the suspended Koszul signs. All functions are formal power series at the origin. An equivalent unsuspended convention writes this as $\frac{1}{2}\langle a, m_1 a\rangle + \sum_{n\geq 2} \frac{1}{n+1}\langle a, m_n(a^{\otimes n})\rangle$.

Proposition 152.1 (Cyclic master equation)**.** *The action satisfies* $\{S, S\} = 0$.

Proof. Cyclicity makes the $n+1$ derivatives of the arity-$n+1$ term equal, with the suspended Koszul signs. Thus $X_S(x) = \sum_{n\geq 1} b_n(x^{\otimes n})$. The coderivation identity implies $X_S^2 = 0$. The Hamiltonian identity gives $X_{\{S,S\}} = 2X_S^2 = 0$. Nondegeneracy of ω makes $\{S, S\}$ constant. Uncurvedness removes b_0, so S starts in order two and its bracket has no constant term. The constant is therefore zero. □

A finite-height Ginzburg realization requires such a cyclic model on the selected proper objects before this scalar action is applied. A Hall realization additionally requires a moduli problem with critical coefficient complexes. For an extension $A_0 \to F \to B_0$, put $L(A_0, B_0) = \det R\,\mathrm{Hom}(A_0, B_0)$. Three-Calabi–Yau duality gives

$$R\,\mathrm{Hom}(B_0, A_0) \simeq R\,\mathrm{Hom}(A_0, B_0)^\vee[-3].$$

Applying the determinant functor to the two derived Hom triangles gives

$$\det R\,\mathrm{Hom}(F, F) \cong \det R\,\mathrm{Hom}(A_0, A_0) \otimes \det R\,\mathrm{Hom}(B_0, B_0) \otimes L(A_0, B_0)^{\otimes 2}.$$

Indeed, the four subquotients are the two diagonal Hom complexes and the two cross complexes. Dualization and the odd shift cancel on their determinant lines, so the two cross determinants coincide. Orientation lines require square-root isomorphisms along this formula, compatible on triple flags and with any quotient descent. The cyclic master equation does not select these isomorphisms.

Chapter 153

An algebraic open–closed comparison

A finite-dimensional ordinary Lie algebra $\mathfrak{l}$ gives a precise coefficient model for an enveloping-algebra centre. Let $U = U(\mathfrak{l})$ over $\mathbb{C}$. The adjoint module U^{ad} has action $x \cdot u = xu - ux$. Its Chevalley cochains are

$$C^n(\mathfrak{l}, U^{\mathrm{ad}}) = \mathrm{Hom}_{\mathbb{C}}(\Lambda^n \mathfrak{l}, U),$$

with differential

$$\begin{aligned}(df)(x_1, \dots, x_{n+1}) = &\sum_i (-1)^{i+1} [x_i, f(x_1, \widehat{x_i}, \dots, x_{n+1})] \\ &+ \sum_{i<j} (-1)^{i+j} f([x_i, x_j], x_1, \widehat{x_i}, \widehat{x_j}, \dots, x_{n+1}).\end{aligned}$$

The second line places the bracket first and retains the order of the remaining arguments.

Theorem 153.1 (Hochschild–Chevalley comparison)**.** *Restriction followed by antisymmetrization defines a quasi-isomorphism*

$$C^{\bullet}_{\mathrm{HH}}(U, U) \longrightarrow C^{\bullet}(\mathfrak{l}, U^{\mathrm{ad}}), \qquad \phi \longmapsto \left[(x_1, \dots, x_n) \mapsto \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) \phi(x_{\sigma(1)}, \dots, x_{\sigma(n)}) \right].$$

This is a comparison of ordinary cochain complexes.

Proof. The Chevalley bimodule resolution has terms $K_n = U \otimes \Lambda^n \mathfrak{l} \otimes U$. Its differential has the left-minus-right multiplication terms and the bracket terms with the signs dual to the displayed differential. Filter K_n by total PBW degree, counting each exterior generator as degree one. The multiplication terms preserve this degree and bracket terms lower it. The associated graded is the Koszul resolution of the diagonal in $\mathrm{Sym}(\mathfrak{l}) \otimes \mathrm{Sym}(\mathfrak{l})$, with regular sequence $x_i \otimes 1 - 1 \otimes x_i$. That Koszul complex is exact away from degree zero. The filtration is bounded below and exhaustive, and each element has finite filtration degree, so lifting a leading term proves exactness of $K_\bullet$ by induction. Its augmentation has image U. Thus $K_\bullet$ is a free bimodule resolution.

Antisymmetrization gives a map from $K_\bullet$ to the ordinary bar resolution, lifting the identity of U. Adjacent products in its bar differential pair to commutators; $xy - yx = [x, y]$ gives exactly the bracket terms of $K_\bullet$. The outer products give its left-minus-right terms. A comparison between projective resolutions lifting the identity is a homotopy equivalence. Applying $\mathrm{Hom}_{U \otimes U^{\mathrm{op}}}(-, U)$ gives the stated formula and quasi-isomorphism. □

This is the classical Cartan–Eilenberg comparison. For a finite Lie superalgebra, the same construction uses the super-exterior powers and permutation Koszul signs. Odd generators produce symmetric exterior directions, so the resolution can be unbounded; its algebraic direct sums and cochain products remain part of the convention. The PBW and super-Koszul proof applies with those conventions.

A physical mixed holomorphic–topological realization requires a map from its independently defined bulk observables to this cochain object. The theorem specifies the adjoint coefficients and the actual algebraic comparison. An E_2 refinement, a current-factorization realization, and compatibility with root-height transitions require maps preserving their additional operations. A quotient of Lie algebras alone does not provide a covariant map of Hochschild centres.

Chapter 154

The exact scalar and first-order observables

Set

$$\mathrm{Den}_\Delta(Z) = D_5(2Z), \qquad \jmath(Z) = Z/2, \qquad \jmath^*\,\mathrm{Den}_\Delta = D_5.$$

The one-particle root spaces give

$$\mathrm{sch\,Fock}(\mathfrak{g}_{\Delta_5+}) = \prod_{\alpha>0}(1-e^{-\alpha})^{-\operatorname{sdim}\mathfrak{g}_{\Delta_5\alpha}}.$$

The oriented root operator gives

$$e^{-\rho}\,\mathrm{sPf}_F(\mathbb{A}_Z) = D_5(2Z).$$

The primitive reduced curve-counting theorem gives

$$\mathcal{Z}^{\mathrm{red}}_{K3\times E} = -(\jmath^*\,\mathrm{Den}_\Delta)^{-2} = -\chi_{10}^{-1}.$$

Theorem 154.1 (Protected square-root theorem)**.** *Assume the stated root-space Fredholm comparison with its orientation, Weyl shift, and convergence domain. Within that realization, the denominator section* Den_Δ *is simultaneously:*

(i) *the Weyl denominator of* $\mathfrak{g}_{\Delta_5}$*;*

(ii) *the Euler character of the bar–Chevalley complex;*

(iii) *the oriented Fredholm Pfaffian of* $\mathbb{A}_Z$*;*

(iv) *the PBW determinant of the parity-refined root superspace* $\mathcal{H}_{\mathrm{rp}}$*;*

(v) *the automorphic denominator whose half-coordinate pullback squares to the reduced scalar denominator.*

All five identifications preserve root degree and Weyl transport.

Proof. The first two are the denominator and bar–Chevalley theorems. The third is the Fredholm theorem. PBW for the Borcherds Lie superalgebra identifies the root superspace determinant, giving the fourth. The half-coordinate identity $\jmath^*\,\mathrm{Den}_\Delta = D_5$, the theta normalization, and the reduced curve-counting theorem give the fifth. □

Chapter 155

Physical meaning

The root-datum realizations retain several distinct constructions. The finite Ginzburg algebras and their projections form an inverse system. A completed three-Calabi–Yau sector requires a continuous nondegenerate Calabi–Yau structure and compatible cyclic models on the selected proper objects. The quiver Hall category retains its chosen coefficients and extension correspondences. A Hall theorem on that datum must identify its root presentation. The Borcherds root superspace defines the root-graded Fock space and its supercharacter. Its identification with physical one-particle states requires a parity-preserving graded comparison intertwining the primitive actions.

A realization on the elliptic curve must construct the current object with its local and wrapped supports, collision operations, and comparison maps. The quantum Borcherds algebra, with its specified pairing and completion, acts as a line-operator symmetry only through a representation preserving fusion and exchange. The root operator gives the Fredholm product on its specified convergence domain. A physical determinant interpretation requires an operator comparison and an orientation-preserving identification of the determinant lines and sections. The Fock character gives the scalar square. An operator trace additionally requires its state space and trace domain.

Simple-root states satisfying the defining relations give a primitive Lie map. The explicit enveloping and Chevalley extensions preserve products and brackets. A compact string realization must additionally construct its brane functor, Hall orientation maps, paired quantum coproduct, and trace-line comparison.

Chapter 156

Primitive comparisons and their algebraic extensions

Let $\mathfrak{n}_\Delta$ be the positive Borcherds Lie superalgebra. Choose homogeneous parity-preserving images of its simple generators in a charge-graded Lie superalgebra $\mathfrak{p}$. Assume these images satisfy every defining relation.

Theorem 156.1 (Extension of a primitive comparison)**.** *The chosen images determine a unique Lie-superalgebra map $f : \mathfrak{n}_\Delta \to \mathfrak{p}$. It induces*

$$U(f)(x_1 \cdots x_r) = f(x_1) \cdots f(x_r), \qquad C_*(f) = \mathrm{Sym}^c(f[1]).$$

These are respectively an enveloping-algebra map and a Chevalley chain-coalgebra map. Filtration-preserving maps extend to the specified inverse-limit completions. If f is surjective and both algebras have finite charge components and proper positive height, equality of their parity-refined PBW characters implies that f and $U(f)$ are isomorphisms.

Proof. The defining presentation gives f. Its tensor-algebra extension carries $xy - (-1)^{|x||y|}yx - [x, y]$ to the corresponding relation. It therefore descends to $U(f)$. The shifted symmetric-coalgebra map commutes with the Chevalley differential because f preserves brackets. Compatible maps on filtration quotients give maps on their inverse limits. Surjectivity of f gives surjectivity of $U(f)$. Character equality makes each finite charge component an isomorphism, separately in both parities. The PBW embeddings then imply injectivity of f. □

At level zero the current Lie conformal algebra is $\mathbb{C}[\partial] \otimes \mathfrak{n}_\Delta$ with $[x_\lambda y] = [x, y]$ on constant generators. The map $\partial^j x \mapsto \partial^j f(x)$ preserves this bracket and sesquilinearity. A nonzero central level additionally requires preservation of the specified invariant pairing. Hall convolution, Calabi–Yau structures, quantum coproducts, and determinant lines require their own comparison data.

Chapter 157

Finite quivers and Calabi–Yau completions

For a finite quiver Q with zero potential, let $\Gamma(Q, 0)$ be its path-length-completed Ginzburg algebra. Path products are read from left to right: if $a : i \to j$ and $b : j \to k$, then $ab : i \to k$. For each arrow $a : i \to j$, add $a^* : j \to i$ of degree -1. At each vertex add a loop t_i of degree -2. Original arrows have degree zero, and

$$da = da^* = 0, \qquad dt_i = e_i\Big(\sum_a [a, a^*]\Big)e_i.$$

Proposition 157.1 (Restriction to a full subquiver)**.** *If Q_0 is a full subquiver, retaining its generators and killing all other vertices, arrows, reverse arrows, and loops defines a continuous dg-algebra projection*

$$\Gamma(Q, 0) \longrightarrow \Gamma(Q_0, 0).$$

The inclusion of retained graded generators need not commute with the differential.

Proof. The projection respects source and target idempotents and extends multiplicatively. At a retained vertex, every term involving a discarded arrow vanishes. The other terms are precisely dt_i in the smaller algebra. All arrow differentials vanish, so the projection is a chain map. It preserves the path-length filtration and is continuous. For the final assertion, start with one vertex i and no arrows, where $dt_i = 0$. Add a vertex j and an arrow $a : i \to j$. In the larger algebra $dt_i = aa^*$, a nonzero length-two path. Thus the graded inclusion is not a dg map. □

Ginzburg completion supplies a three-Calabi–Yau algebra for each finite quiver in its stated setting. The displayed projections specify an inverse system for compatible zero-potential windows. A Calabi–Yau structure on a completed limit requires its own nondegeneracy and continuity argument. Comparisons with Hall categories and quantized Borcherds algebras must also preserve orientations, root relations, and the paired Hopf data.

Book XXVI

Quantum groups, exchange, and representation theory

Chapter 158

From positive half to full quantum group

Suppose the compact or local positive algebra H^+ is paired continuously with a negative half H^-, carries a completed coproduct, and admits Cartan data and a separated triangular completion. The completed double

$$\mathcal{U} = \widehat{D}(H^+)$$

has triangular decomposition $H^- \mathbin{\widehat{\otimes}} H^0 \mathbin{\widehat{\otimes}} H^+$. Its universal R-matrix acts on continuous tensor products of modules. The classical limit is a Lie bialgebra whose cobracket is the first-order term of the coproduct asymmetry.

For $\mathbb{C}^3$, H^+ is the positive affine Yangian. For local ADE surfaces, it is the completed positive affine ADE Yangian. For the Igusa algebra, the paired Borel construction of the preceding part supplies a height-complete formal quantum group; a compact Hall realization maps to it through primitive, coproduct, and pairing comparisons.

Chapter 159

Tannaka reconstruction

Let $\mathcal{L}$ be an essentially small rigid $\Bbbk$-linear abelian monoidal category with finite-dimensional morphism spaces, finite-length objects, and exact tensor product. Let

$$\omega : \mathcal{L} \to \mathrm{Vect}_{\mathrm{fd}}$$

be a faithful exact strong monoidal functor. Assume the coend

$$H = \int^{M \in \mathcal{L}} \omega(M)^{*} \otimes \omega(M)$$

exists and that the canonical comparison functor to finite-dimensional H-comodules is comonadic. Tensor product makes H a bialgebra, and rigidity supplies its antipode. A braiding gives a coquasitriangular form; a meromorphic braiding gives its spectral version.

Theorem 159.1 (Rigid Tannaka reconstruction)**.** *Under the preceding finite and comonadic hypotheses, the comparison functor is a monoidal equivalence*

$$\mathcal{L} \simeq \mathrm{Comod}_{\mathrm{fd}}(H).$$

Braiding transports to the universal coquasitriangular form on H.

Proof. Matrix coefficients define the universal H-coaction on every $\omega(M)$. Tensor product and duality give the bialgebra and antipode structures. The Barr–Beck comonadic theorem identifies $\mathcal{L}$ with coalgebras for the induced comonad, and the coend identifies that comonad with $H \otimes -$. The hexagon identities transport to the coquasitriangular identities. □

This reconstruction explains why a line-operator category, a fiber functor, and exchange together produce a quantum group. The ordered boundary algebra provides ordered composition. A compatible bialgebra coproduct supplies tensor products of its modules, and exchange supplies additional braiding data.

Chapter 160

KZ and rational exchange

For a simple Lie algebra $\mathfrak{g}$, the Knizhnik–Zamolodchikov connection is

$$\nabla = d - \frac{\hbar}{2\pi i} \sum_{i<j} \Omega_{ij}\, d\log(z_i - z_j).$$

Flatness follows from the infinitesimal braid relations. Drinfeld–Kohno identifies its monodromy tensor category with the corresponding Drinfeld–Jimbo category after a choice of associator.

Rational difference exchange instead gives the Yangian R-matrix

$$R(u) = 1 - \frac{\hbar P}{u}$$

in the fundamental representation. The Yang–Baxter equation is the confluence identity for three exchanges. The KZ and Yangian constructions share the geometry of configuration spaces while using different flat connections.

Chapter 161

The centre and the double

The Hochschild centre $Z_{\mathrm{ord}}(A)$ controls central operations on the ordered algebra in its specified coefficient category. To form a Drinfeld centre of modules, first specify a monoidal module category. For an ordered chiral bialgebra satisfying Definition 138.1, use

$$\mathcal{M}_A = \mathrm{Mod}_A(\widehat{\mathcal{V}}_C)$$

with the monoidal product of Theorem 138.2. An object of $Z(\mathcal{M}_A)$ is a module M together with natural equivalences

$$c_{M,N} : M \widehat{\otimes}^{!} N \xrightarrow{\sim} N \widehat{\otimes}^{!} M \qquad (N \in \mathcal{M}_A)$$

satisfying the half-braiding hexagon and unit identities, with their coherences. This categorical centre is braided. A comparison with the Hochschild centre must specify a realization functor, the source and target of the comparison, and the operations it preserves. Dualizability alone does not supply such a comparison. Derived loop-space descriptions apply in the geometric categories and under the hypotheses of their respective centre theorems [6].

The centre and the double answer complementary questions. The centre classifies central actions in the chosen category. The double reconstructs a quantum group from paired positive and negative halves. A field-theory comparison requires explicit realizations of the relevant line and bulk operations and a map compatible with both constructions.

Book XXVII

A compact six-dimensional protected source

Chapter 162

Holomorphic cotangent BF theory on $K3 \times E$

Let S be a K3 surface with holomorphic symplectic form σ_S, let E be an elliptic curve with holomorphic form $\mathrm{d}z$, and put

$$X = S \times E, \qquad \Omega_X = \sigma_S \wedge \mathrm{d}z.$$

For a finite proper-height quotient $\mathfrak{n}_{\Delta_5,+}{}^{(H)}$ of the positive Igusa Borcherds algebra, take fields

$$A \in \Omega^{0,\bullet}(X, \mathfrak{n}_{\Delta_5,+}{}^{(H)})[1], \qquad B \in \Omega^{0,\bullet}(X, (\mathfrak{n}_{\Delta_5,+}{}^{(H)})^\vee)[1].$$

The invariant evaluation pairing and Ω_X define the BV action

$$S_H(A,B) = \int_X \Omega_X \wedge \langle B, \bar\partial A + \frac{1}{2}[A,A]\rangle. \tag{162.1}$$

This is the holomorphic cotangent theory of the dg Lie algebra of $\mathfrak{n}_{\Delta_5,+}{}^{(H)}$-valued Dolbeault forms.

Theorem 162.1 (Classical compact-background source)**.** *The action* (162.1) *satisfies the classical master equation and defines a holomorphic factorization algebra on the compact Calabi–Yau threefold X.*

Proof. The shifted cotangent pairing is invariant. The classical master equation is the Jacobi identity of $\mathfrak{n}_{\Delta_5,+}{}^{(H)}$ together with $\bar\partial^2 = 0$ and the derivation property of $\bar\partial$. Local observables of an elliptic cyclic L_∞ algebra form a factorization algebra by the standard BV construction. □

Chapter 163

Tree exactness and the quantum master equation

Every interaction vertex in (162.1) has one B-leg and two A-legs. The propagator pairs an A-leg with a B-leg.

Lemma 163.1 (Loop bound)**.** *A connected Feynman graph of the cotangent theory has loop number at most one.*

Proof. A graph with V interaction vertices has exactly V internal or external B-half-edges. Every internal edge consumes one of them, hence $E \leq V$. The loop number is $E - V + 1 \leq 1$. □

The positive algebra is graded by strictly positive root height. For homogeneous $x \in \mathfrak{n}_{\Delta_s,+}{}^{(H)}$, the operator ad_x raises height. Its coadjoint operator lowers dual height. A product of adjoint operators labelled by positive roots has total nonzero height shift and therefore has zero trace.

Proposition 163.2 (Vanishing of one-loop Lie weights)**.** *For every finite positive-height quotient, the Lie weight of each one-loop graph in the cotangent theory vanishes.*

Proof. A one-loop graph has $E = V$. Every B-leg is consumed by an internal edge, so its external legs are A-legs. Its Lie weight is a cyclic trace of adjoint operators, or the dual trace with the conventionally equivalent signs. Each external label has positive root height. The product therefore strictly raises height on the finite-dimensional adjoint representation and has zero trace. The same argument applies to the dual representation. □

This vanishing removes the one-loop Lie coefficients in a perturbative construction with the stated propagator and vertex types. A renormalized quantum factorization algebra still requires admissible analytic kernels, local extensions, the quantum master equation, and compatible height transitions on the chosen observable spaces. These are the remaining quantum realization data.

Chapter 164

Classical reduction along K3

Choose a Kähler metric on S and normalize a closed harmonic $(0,2)$-form η by $\int_S \sigma_S \wedge \eta = 1$. Let $K = \mathbb{C}[\eta]/(\eta^2)$ with $|\eta| = 2$ and zero differential. There is a unital dg-algebra map

$$i : K \longrightarrow \Omega^{0,*}(S), \qquad \eta \longmapsto \eta.$$

Its image is closed under multiplication because $(0,4)$-forms vanish. The Dolbeault cohomology of a K3 surface is one-dimensional in degrees zero and two and zero in degree one, so i is a quasi-isomorphism. It preserves the trace defined by integration against σ_S.

Let $\mathfrak{l}$ be a finite-dimensional Lie algebra, and write $\mathfrak{d} = \mathfrak{l} \ltimes \mathfrak{l}^*$ for its cotangent Lie algebra, with invariant evaluation pairing. For an open $U \subset E$, tensor with $\Omega^{0,*}(U)$, using the completed smooth tensor product in the S direction. The field comparison is

$$K \otimes \Omega^{0,*}(U) \otimes \mathfrak{d} \xrightarrow{i \otimes \mathrm{id}} \Omega^{0,*}(S \times U) \otimes \mathfrak{d}.$$

The bracket is multiplication of forms followed by the Lie bracket, with Koszul signs.

Proposition 164.1 (Classical field comparison)**.** *The displayed map is a trace-preserving dg Lie quasi-isomorphism, compatible with restriction in U. Its induced minimal operations in the K3 factor have no higher products.*

Proof. Multiplicativity of i proves bracket compatibility. Hodge projection p and $h = \bar\partial^* G$ on the compact surface satisfy $pi = \mathrm{id}$ and $\bar\partial h + h\bar\partial = \mathrm{id} - i\,p$. Tensoring these continuous operators with forms on U gives the same contraction, with the tensor-product signs. Thus $i \otimes \mathrm{id}$ is a quasi-isomorphism. Restriction acts only on the U factor and commutes with it. The normalized trace on K is the restriction of the Dolbeault trace. Choose the contraction with $hi = 0$. In every higher transferred tree, the first product of inputs from $i(K)$ remains in $i(K)$ and the next internal homotopy kills it. Thus every higher transferred product vanishes. □

The coefficient algebra K retains both the degree-zero and degree-two K3 modes. The proposition compares classical field complexes and their brackets. Constructing an equivalence of quantum observable factorization algebras requires compatible quantization, support conventions, and the induced maps on observable completions. The ordinary Hochschild comparison of Theorem 153.1 is a separate algebraic calculation, not that quantum field map.

Chapter 165

Protected states and the Igusa trace

Let

$$\mathcal{H}_{\mathrm{ip}} = \widehat{\bigoplus}_{\alpha>0} \mathfrak{g}_{\Delta,\alpha}$$

with its Borcherds parity. The multiparticle state space is the completed super-Fock space

$$\mathcal{H}_{\mathrm{Fock}} = \widehat{\mathrm{Sym}}(\mathcal{H}_{\mathrm{ip},\bar{0}}) \otimes \widehat{\Lambda}(\mathcal{H}_{\mathrm{ip},\bar{1}}).$$

For Z in the absolute-convergence chamber, let e^{-H_Z} act by $e^{-\alpha(Z)}$ on the root space of degree α.

Theorem 165.1 (Protected Hamiltonian trace)**.** *The supertrace is*

$$\mathrm{str}_{\mathcal{H}_{\mathrm{Fock}}}(e^{-H_Z}) = \prod_{\alpha>0} (1 - e^{-\alpha(Z)})^{-\dim \mathfrak{g}_{\Delta,\alpha}}.$$

After multiplication by the inverse Weyl line, its reciprocal is $\mathrm{Den}_\Delta(Z) = D_5(2Z)$*. After the half-coordinate pullback and doubling, the disconnected protected scalar is* $-(\jmath^* \mathrm{Den}_\Delta)^{-2} = -\chi_{10}^{-1}$ *in the Oberdieck–Pixton normalization.*

Proof. An even oscillator contributes $(1 - e^{-\alpha})^{-1}$ and an odd oscillator contributes $1 - e^{-\alpha}$. Multiplication over the root basis gives the product. The denominator identity and the reduced curve-counting theorem supply the final two equalities. □

Chapter 166

The compact-source comparison

The compact threefold provides the classical cotangent field theory and the dg Lie comparison of Proposition 164.1. The root datum separately gives the Borcherds algebra, its enveloping algebra, and the root-graded Fock space. The latter has the trace computed above. Proposition 163.2 supplies the finite one-loop Lie cancellation, while Theorem 153.1 computes the ordinary algebraic centre model with adjoint coefficients.

Problem 166.1 (Geometric compact comparison)**.** Construct objects E_i representing the simple branes and maps on their derived morphism complexes that preserve units, composition, and higher composition homotopies. Realize these maps on extension stacks with the reduced critical coefficients and orientation isomorphisms on triple flags. Construct compatible tensor and unit constraints on the resulting boundary functor. Compare the chosen quantum bulk action with the independently defined boundary centre, including support, collisions, and completion. Finally construct a trace and a map of its value line to the automorphic line in the same charge normalization.

A primitive map satisfying the root relations gives the enveloping and Chevalley maps of Theorem 156.1. A strong monoidal equivalence transports categorical centres. Neither construction determines the brane functor or its cyclic trace. These are the additional maps required for compact geometric reconstruction.

Book XXVIII

Physics of the comparison maps

Chapter 167

Holomorphic–topological boundary theories

A three-dimensional holomorphic–topological theory on $C \times \mathbb{R}$ has local observables that are holomorphic along C and locally constant along $\mathbb{R}$. A boundary produces an ordered chiral algebra. A second topological direction permits exchange and gives an E_2 enhancement or a meromorphic braided line category. The bulk acts through the derived centre.

The finite cotangent BF model gives the exact local calculation:

$$\mathrm{Bar}(U(\mathfrak{l})) \simeq C_*(\mathfrak{l}).$$

For $\mathfrak{l} = \mathfrak{n}_{\Delta_5,+}{}^{(H)}$, this calculation realizes every finite-height Igusa denominator sector.

Chapter 168

Six-dimensional hCS on $K3 \times E$

Choose a holomorphic volume form $\Omega_S \wedge dz_E$. The hCS theory has classical fields

$$\Omega^{0,*}(S \times E, \mathfrak{a})[1]$$

and action S_{hCS}. Its factorization observables are native to the threefold. Pushforward along p_E gives a factorization algebra on E whose coefficient complex contains the Dolbeault modes of S.

A protected reduction to a chiral algebra consists of a deformation retract of the K3 Dolbeault complex, compatibility of the transferred L_∞ operations with the trace, a boundary or defect selecting ordered fusion, and a quantization trivializing the local anomaly. These data produce the ordered chiral algebra to which the Hall/BPS comparison may be applied.

The Hodge equality $\chi(\mathcal{O}_{S\times E}) = 0$ controls one compact supertrace. The gauge and gravitational anomalies are classes in the local BV deformation complex and are evaluated by their own characteristic forms.

Chapter 169

Omega backgrounds

Let a torus T act holomorphically on the local geometry and preserve the selected supercharge. For a generator V of the action, the equivariant differential is

$$Q_\Omega = Q + \iota_V, \qquad Q_\Omega^2 = \mathcal{L}_V.$$

After localization, fixed loci produce deformation parameters and often a spectral R-matrix. Toric threefolds and ADE surface resolutions carry such actions. A deformation family equipped with monodromy-compatible moduli correspondences and wall-crossing equivalences transports the localized algebra from algebraic or equivariant K3 fibers to the selected generic fiber.

Chapter 170

Anomalies as typed classes

The principal anomaly classes are

construction	deformation complex
hCS quantization	local BV cochains
critical CoHA orientation	Picard/gerbe theory of the critical stack
Hall coproduct	correspondence and localization complex
projective conformal blocks	Atiyah algebra of the determinant line
automorphic multiplier	character group of the modular cover.

A comparison theorem is a morphism between the corresponding complexes together with equality of the transported classes. This formulation turns numerical coincidences into geometric identities precisely when an intertwining map exists.

Book XXIX

Automorphic characters and scalar shadows

Chapter 171

Borcherds products as characters

Let $\mathfrak{n}_+$ be a locally finite positive Lie superalgebra. The Chevalley Euler character is

$$\chi C_*(\mathfrak{n}_+) = \prod_{\alpha>0} (1 - e^{-\alpha})^{\operatorname{sdim} \mathfrak{g}_\alpha}.$$

A Borcherds product becomes the Weyl-shifted character of $C_*(\mathfrak{n}_+)$. This construction produces a character of an existing algebra. Hall geometry supplies a second algebra, and a primitive comparison theorem identifies the two.

Chapter 172

The universal Borcherds weight formula

For a weight-zero weak Jacobi form

$$\varphi(\tau, z) = \sum c(n, \ell) q^n r^\ell,$$

the Borcherds lift has weight

$$\mathrm{wt}(\mathrm{Borch}(\varphi)) = \frac{c(0,0)}{2}.$$

For the K3 half-index, $c(0,0) = 10$ and the weight is 5. This formula lives entirely on the automorphic layer. Hodge Euler characteristics, Heisenberg ranks, central charges, and BV anomalies are evaluated by their own traces.

Chapter 173

The Gritsenko–Cléry atlas

The eight diagonal-divisor forms are

$$\Delta_{5},\ \Delta_{2},\ \Delta_{1},\ \Delta_{1/2},\ \nabla_{3},\ \nabla_{2},\ \nabla_{3/2},\ Q_{1}$$

with weights

$$5,\ 2,\ 1,\ \frac{1}{2},\ 3,\ 2,\ \frac{3}{2},\ 1.$$

Each form is an automorphic product on its own paramodular or Hecke congruence group with its own character or multiplier [16]. A geometric realization is specified row by row by a charge lattice, Hall problem, trace, and primitive algebra comparison.

Chapter 174

Scalarization hierarchy

The constructions descend through four levels:

$$\text{factorization or Hall object} \longrightarrow \text{primitive Lie algebra} \longrightarrow \text{PBW character} \longrightarrow \text{scalar partition function}.$$

Each arrow retains the quotient of structure displayed at its target. The first object contains supports, extension maps, and brackets. The second retains the infinitesimal algebra. The third retains signed graded dimensions. The fourth adds a trace and normalization. Upward reconstruction is supplied by the corresponding geometric operations and comparison morphisms.

Book XXX

The unified theorem system

Chapter 175

The foundational theorem

Definition 175.1 (Complete Calabi–Yau quantum-group datum)**.** A complete comparison datum specifies independent categorical, field, boundary, Hall, Hopf, and trace objects together with:

(i) PTVV-representable moduli for the smooth proper Calabi–Yau category;

(ii) a renormalized BV quantization satisfying the quantum master equation and a local compact factorization generator b;

(iii) a boundary reduction or proper locally trivial map to a curve satisfying factorization pushforward, together with the supplied curve-level faces and aligned distributive law;

(iv) a strong-oriented Hall coefficient theory whose pull–push maps satisfy base change and whose positive algebra is separated and complete;

(v) paired negative and Cartan halves, a completed coproduct, a continuous nondegenerate Hopf pairing, and a separated triangular completion on which the canonical tensors converge;

(vi) compatible open–closed and Hall–boundary morphisms;

(vii) a parity-refined primitive comparison for every automorphic identification.

Theorem 175.2 (Calabi–Yau quantum-group assembly)**.** *Let a complete datum consist of:*

(i) *a Calabi–Yau category and a representable moduli-of-objects problem;*

(ii) *a quantized cyclic elliptic BV theory and a boundary or defect generator;*

(iii) *a Hall extension correspondence with coefficient functor, base-change theorem, strong orientation data, and separated charge completion;*

(iv) *paired positive and negative bialgebras, Cartan data, a nondegenerate continuous Hopf pairing, and a completed double;*

(v) *every curve-level distributive, PBW, and analytic comparison used below.*

Then the datum determines functorially:

(i) *framed* E_2 *Hochschild cochains and the shifted-symplectic moduli structure;*

(ii) *the three curve-level algebraic faces and their three bars;*

(iii) *the positive Hall/BPS algebra and its primitive Lie algebra;*

(iv) *the completed quantum double and braided representation category;*

(v) *every supplied comparison among categorical, field-theoretic, Hall, current, quantum, and automorphic realizations;*

(vi) *the Weyl-shifted PBW/Chevalley character of any parity-refined primitive Lie-superalgebra identification.*

Proof. Cyclic Deligne and shifted symplectic geometry give the first output [9, 10, 64]. Boundary factorization gives the curve faces, and Volume I gives their bars and centres. The Hall correspondence and its base-change theorem give the associative product; strong orientation supplies the coefficient coherence required in the 3-Calabi–Yau setting [50]. The paired bialgebra data give the double and universal R-matrix. The comparison maps commute by hypothesis. PBW and Chevalley antisymmetrization give the final character. Each construction is natural under a morphism preserving the displayed data. □

The local quiver, equivariant surface, K3 correspondence, and Igusa Borcherds constructions have the carriers stated above. A geometric $K3 \times E$ comparison requires the primitive map and determinant-line map of Problem 166.1.

Chapter 176

Landscape of established realizations

Geometry	Positive algebra	Full or centred object	Character
A_2 quiver	$U_v^+(\mathfrak{sl}_3)$	$U_v(\mathfrak{sl}_3)$	PBW root product
$\mathbb{C}^3$	$Y^+(\widehat{\mathfrak{gl}}_1)$	$Y(\widehat{\mathfrak{gl}}_1)$	plane-partition and Fock traces
$\widetilde{\mathbb{C}^2/\Gamma}$	$Y^+(\widehat{\mathfrak{g}}_\Gamma)$	paired completion when the negative half and coproduct are supplied	surface Fock traces
$K3$	$U(\mathfrak{g}^+_{\mathrm{BPS}})$	Heisenberg and LLV actions	K3 Hall characters
$K3 \times E$	Borcherds root envelope; geometric reduced character	$U_q(\mathfrak{g}_{\Delta_5})$ and the root-current model and classical BF theory	$-\chi_{10}{}^{-1}$

The A_2 and $\mathbb{C}^3$ rows provide complete standard quantum-group models. The ADE row provides an established completed positive half together with its geometric current actions; its paired full double is formed from the negative half, Cartan data, coproduct, continuous Hopf pairing, and triangular completion stated in the table. The K3 row provides global two-Calabi–Yau BPS and cohomological symmetries. The final row records independently constructed root, character, and classical field objects. Their geometric Hall realization requires the comparison in Problem 166.1.

Chapter 177

Constructed comparisons and their geometric extensions

The Hall flag argument constructs associativity once its coefficient and orientation maps are defined. The tensor-action formula constructs representation fusion from a coassociative counital coproduct. The matrix-coefficient formulas construct the coend of a supplied fibre functor. The primitive presentation gives enveloping and Chevalley maps. The classical cyclic and K3 field calculations retain their actual differentials, traces, and brackets.

The quantum field realization requires admissible quantization and observable completions. The compact Hall realization requires the brane and orientation maps of Problem 166.1. The Fredholm and automorphic comparisons require their analytic domains and value-line isomorphisms. These constructions preserve the full geometric reconstruction problem beyond its finite algebraic and classical calculations.

Chapter 178

Actions and comparison maps

For a specified boundary algebra A_b, a coherent bulk action is a map

$$\beta : \mathrm{Obs}_{\mathrm{bulk}}|_{\partial} \longrightarrow Z_{\mathrm{ord}}(A_b).$$

The target is the derived endomorphism object of the diagonal bimodule in the chosen factorization coefficient category. A Hall construction instead starts with

$$\mathcal{M}_{\alpha} \times \mathcal{M}_{\beta} \xleftarrow{p} \mathcal{E}_{\alpha,\beta} \xrightarrow{q} \mathcal{M}_{\alpha+\beta}.$$

Critical coefficients, orientations, and admissible pull–push maps give its product. Framed moduli supply a representation after their action correspondences and triple-flag identities have been constructed. A paired Hopf structure gives $D(H^{+}, H^{\circ}, H^{-})$. Its coproduct defines tensor products of continuous modules, and a convergent universal R-matrix gives exchange.

A strong monoidal equivalence between this module category and a boundary category transports categorical centres. It does not identify that categorical centre with the object $Z_{\mathrm{ord}}(A_b)$ without a realization theorem. A trace on a specified representation then gives an amplitude. Its comparison with an automorphic section requires a map of value lines with the same normalization.

Chapter 179

Comparison data for Calabi–Yau constructions

Comparisons among Calabi–Yau categories, holomorphic field theory, chiral algebras, Hall algebras, quantum groups, and automorphic products require the following structures:

(i) for a smooth proper d-Calabi–Yau category with its nondegenerate negative-cyclic class, the categorical outputs are framed E_2 Hochschild cochains and, when the moduli stack of objects is representable in the PTVV setting, $(2-d)$-shifted symplectic moduli; the shift d does not change the Hochschild operadic level to E_n;

(ii) a geometric BV theory and boundary category produce factorization observables and transverse ordered chiral algebras; associative chiral convolution is recorded as a separate supplied structure;

(iii) pushforward to a curve uses an actual map $p : X \to C$; formal cycle-specialization notation alone specifies no pushforward;

(iv) extension stacks, coefficient sheaves, orientations, and completions define the Hall algebra;

(v) coproduct, Cartan, Hopf pairing, and topology define the double;

(vi) K3 Heisenberg, LLV orthogonal symmetry, local ADE Yangians, and global BPS generalized Kac–Moody algebras occupy distinct strata joined by explicit comparison maps where constructed;

(vii) the graph of the $K3 \times E$ Gram map becomes multiplicative in the polarized central extension;

(viii) point-local and wrapped sectors form a hybrid factorization problem;

(ix) the Igusa form is the Weyl-shifted PBW/Chevalley character of $\mathfrak{g}_{\Delta_5}$, and compact Hall geometry maps to it through the primitive comparison;

(x) anomaly identities are transported through explicit maps between deformation complexes.

The categorical, geometric, Hall, and automorphic operations thus have separate carriers and comparison maps.

Book **XXXI**

Technical appendices to the preceding book

Chapter 180

Typed data table

Object	Minimal data	Primary operation
CY category	nondegenerate negative-cyclic trace	Hochschild and shifted-symplectic structures
Factorization algebra	local observables and Weiss descent	disjoint-open product
Transversely ordered chiral algebra	factorization coefficient with an internal E_1 structure for $\otimes^{!}$	ordered fusion
Associative chiral algebra	an E_1 structure for chiral convolution $\star^{\mathrm{ch}}$	nonlocal chiral composition
CoHA	extension correspondence and coefficient theory	pull–push product
Bialgebra	compatible product and coassociative counital coproduct	fusion of representations
Double	positive and negative halves, Cartan data, Hopf pairing	universal R-matrix
Borcherds product	Jacobi or vector-valued modular input	graded character

Chapter 181

The three K3 brackets

The Heisenberg bracket is the excess intersection of two Nakajima incidence kernels. The LLV bracket is the commutator of diagonal Lefschetz kernels and their adjoints. The local Yangian bracket is the convolution commutator of ADE Hecke kernels with spectral equivariance. In $\mathrm{Corr}_{K3}(S)$ all three are compositions of geometric kernels. Their mixed commutators are the corresponding triple-intersection classes. The action algebra and Tannaka coend retain these classes while preserving the three geometric constructions.

Chapter 182

Exact arithmetic checks

The companion verification calculation establishes:

(i) the low Fourier rows of $\phi_{0,1}$ through q^3 by exact theta-series division;

(ii) the vector-valued cocycle identity for the Gram polarization;

(iii) the rational Yang–Baxter equation on $(\mathbb{C}^2)^{\otimes 3}$;

(iv) the A_2 Hall flag counts at ranks zero and one;

(v) the type-II Gram matrix and reflection identities.

Each calculation uses integer or rational arithmetic and appears as a worked formula in the relevant chapter.

Coherent boundary actions and BV fields

Boundary operations determine a deformation problem once their interface and coherence maps are supplied. A local realization gives a shifted cotangent BV theory. Comparing it with a physical bulk requires a map of fields, actions, and coherent boundary operations.

Book XXXII

The full mixed boundary datum

Chapter 183

Three products and one interface

Holomorphic collisions and transverse composition act on different coefficient objects. A common boundary theory must specify their interaction on an interface. This interaction is part of the input to the following constructions.

Let X be a smooth complex curve and let $\mathcal{D}_X = D\text{-}\operatorname{Mod}(\operatorname{Ran} X)$ in a fixed six-functor enhancement. Write $\otimes^!$ for the pointwise product and $\star^{\mathrm{ch}}$ for chiral convolution.

Definition 183.1 (Complete mixed boundary datum)**.** A complete mixed boundary datum on X is a tuple

$$\mathbb{A} = (L, A^{\mathrm{ch}}, A^{\perp}, K, \iota, \lambda)$$

with the following components.

(i) L is a complete chiral Lie algebra.

(ii) A^{ch} is a complete associative algebra for $\star^{\mathrm{ch}}$.

(iii) $A^{\perp}$ is a complete E_1 algebra in $(\operatorname{Fact}_D(X), \otimes^!)$.

(iv) K is an $(A^{\mathrm{ch}}, A^{\perp})$ interface bimodule.

(v) $\iota : L \to (A^{\mathrm{ch}})_{\mathsf{Lie}}$ is a chiral Lie morphism.

(vi) λ is the aligned mixed distributive law relating the two module actions on K.

Every structure map is continuous for the chosen weight filtrations. The tuple includes the module actions, the mixed coherence homotopies, and their identities in the chosen enhancement. When a reduced bar is used, the relevant augmentations are supplied as additional data. Existence of this tuple for a physical boundary theory is a separate construction.

The three native bars are

$$\operatorname{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}(L), \qquad \operatorname{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A^{\mathrm{ch}}), \qquad \operatorname{Bar}^{\perp}(A^{\perp}).$$

The interface has a fourth bar. Its degree-r terms are alternating words in the two associative colours with one marked interface letter. Its differential contracts adjacent compatible letters.

Construction 183.2 (Interface bar)**.** Let $\overline{A}^{\mathrm{ch}}$ and $\overline{A}^{\perp}$ be the augmentation ideals. Set

$$\operatorname{Bar}^{\mathrm{mix}}(\mathbb{A}; K) = \bigoplus_{p,q \geq 0} (s\overline{A}^{\mathrm{ch}})^{\otimes p} \otimes sK \otimes (s\overline{A}^{\perp})^{\otimes q}.$$

The differential is the sum of the internal differential, the two bar differentials, the left action on K, the right action on K, and the mixed terms supplied by λ.

Theorem 183.3 (Four-colour differential)**.** *The interface differential squares to zero. Its codimension-two terms are exactly the associative relations in the two colours, the two module relations, and the mixed distributive relation.*

Proof. Filter by total word length. Every component of the square contracts two adjacent edges. Disjoint contractions cancel by the determinant-line sign. Nested contractions in one colour cancel by associativity. A contraction meeting the interface cancels by the corresponding module axiom. The remaining square has one edge of each colour and is the aligned distributive square. The datum λ identifies its two composites. □

Chapter 184

Compatible modules and the mixed centre

An operation on a boundary theory must act on boundary-changing morphisms as well as on objects. Natural transformations retain this compatibility, and derived natural transformations retain its higher homotopies.

A compatible module is a triple $(M^{\mathrm{ch}}, M^{\perp}, \kappa)$ with an A^{ch} action, an $A^{\perp}$ action, and an interface comparison over K. Let $\mathrm{Mod}^{\mathrm{mix}}_{\mathbb{A}}$ be their stable category.

Definition 184.1 (Mixed derived centre)**.** The mixed derived centre is

$$Z_{\mathrm{mix}}(\mathbb{A}) = \mathrm{End}\Big(\mathrm{id}_{\mathrm{Mod}^{\mathrm{mix}}_{\mathbb{A}}}\Big).$$

It is computed by the endomorphisms of the diagonal compatible bimodule.

Theorem 184.2 (Universal interior action)**.** *$Z_{\mathrm{mix}}(\mathbb{A})$ carries an internal E_2 algebra structure. Every action already defined as a coherent natural endomorphism of the identity determines a point of this centre. The space of its factorizations through the representing endomorphism object is contractible. Morita-equivalent complete mixed data have equivalent centres.*

Proof. Endomorphisms of the identity functor form the higher centre of a module category. Composition and insertion of natural transformations give the brace operations and hence the E_2 structure. A supplied coherent natural action defines an endomorphism of the identity. A physical bulk insertion determines such an action only after its action on modules, morphisms, and higher compositions has been constructed. The factorization is the universal property of the endomorphism object. Morita equivalence identifies the module categories and their identity functors. □

Chapter 185

Relative induction from a chiral algebra

Let V be a chiral algebra with conformal vacuum. Its algebraic collar $\mathrm{Coll}(V)$ is the free transverse E_1 extension generated by one boundary copy of V, modulo the chiral relations already present in V. Weight completion gives $\widehat{\mathrm{Coll}}(V)$.

Theorem 185.1 (Universal collar)**.** *For every complete transversely ordered chiral algebra B and every chiral morphism $V \to B$, there is a unique continuous E_1 morphism*

$$\widehat{\mathrm{Coll}}(V) \longrightarrow B$$

extending the chiral map. The construction is left adjoint to restriction from transverse ordered algebras to chiral coefficients.

Proof. The algebraic collar is the relative operadic induction from the chiral operad to the mixed operad. Relative induction is left adjoint to restriction. Completion preserves the adjunction on positive complete objects because every map is determined weight by weight. □

When relative induction supplies a regular interface and a continuous aligned distributive law, these data define a mixed envelope with $L = V_{\mathsf{Lie}}$, $A^{\mathrm{ch}} = U^{\mathrm{ch}}(L)$, and $A^{\perp} = \widehat{\mathrm{Coll}}(V)$. Constructing that interface law in the selected geometric coefficient category is part of the envelope construction.

Book XXXIII

The deformation algebra and its BV envelope

Chapter 186

The centre-extended deformation algebra

Varying a boundary product changes its compatibility with the interface and the central action. The deformation complex must therefore include all these operations. A local field realization is further data needed to turn their deformation equations into a BV action.

Fix a cofibrant coloured operad $\mathcal{P}_{\mathrm{mix}}$ presenting the structures of a complete mixed datum and its central action. Let $\mathcal{P}_{\mathrm{mix}}^{!}$ be its bar cooperad. The convolution complex

$$\mathrm{Def}_{\mathbb{A}}^{\mathrm{full}} = \mathrm{Hom}_{\mathbb{S}}\left(\mathcal{P}_{\mathrm{mix}}^{!}, \mathrm{End}_{\mathbb{A}}\right)$$

contains the chiral operation, both associative products, the interface, and the centre action.

Theorem 186.1 (Complete deformation L_∞ algebra)**.** *The convolution complex has a canonical complete L_∞ structure. Its Maurer–Cartan points are complete mixed boundary structures together with a coherent central action. Twisting by the point defined by $\mathbb{A}$ gives the tangent algebra of deformations of the full datum.*

Proof. The infinitesimal decomposition maps of the bar cooperad insert collections of operations into one another. Composition in the endomorphism operad turns these decompositions into the convolution brackets. The cooperad identities imply the L_∞ identities. A Maurer–Cartan element is precisely an operad morphism from a cofibrant resolution of $\mathcal{P}_{\mathrm{mix}}$ to $\mathrm{End}_{\mathbb{A}}$. Twisting linearizes this mapping space at the selected structure. □

186.1 Field equations and Bianchi identities

Let α be a degree-one field in the localized deformation algebra. Its curvature is

$$\mathcal{F}(\alpha) = \sum_{n \geq 1} \frac{1}{n!} \ell_n(\alpha^{\otimes n}).$$

The linearization at α is

$$d_\alpha x = \sum_{n \geq 0} \frac{1}{n!} \ell_{n+1}(\alpha^{\otimes n}, x).$$

Theorem 186.2 (Universal Bianchi identity)**.** *For every α,*

$$d_\alpha \mathcal{F}(\alpha) = 0.$$

At a Maurer–Cartan point, $d_\alpha^2 = 0$.

Proof. Both identities are the exponential form of the L_∞ Jacobi relations. Expand the coderivation square on the cofree cocommutative coalgebra and evaluate it on e^{α} and $e^{\alpha} x$. □

Chapter 187

Shifted cotangent transgression

Let $\mathfrak{l}$ be a complete elliptic L_∞ algebra on a three-manifold M. Its shifted cotangent theory has fields

$$\mathcal{E} = T^*[-1]\mathfrak{l}[1] = \mathfrak{l}[1] \oplus \mathfrak{l}^\vee[-2].$$

The canonical action is

$$S(\alpha, \beta) = \langle \beta, \mathcal{F}(\alpha) \rangle.$$

Theorem 187.1 (Universal BV envelope)**.** *The action satisfies the classical master equation. Its Euler–Lagrange equations are*

$$\mathcal{F}(\alpha) = 0, \qquad d_\alpha^\vee \beta = 0.$$

The tangent complex at a solution is the cotangent complex of the Maurer–Cartan moduli problem.

Proof. The Hamiltonian vector field of S is the cotangent lift of the homological vector field defining the L_∞ structure. The square of that vector field is zero, hence $\{S, S\} = 0$. Variation in β gives the Maurer–Cartan equation. Variation in α gives the dual linearized equation. □

Definition 187.2 (Local realization and chiral BV envelope)**.** A local realization of the complete mixed datum consists of an elliptic local L_∞ algebra $\mathfrak{l}_{\mathbb{A}}^{\mathrm{full}}$ on $M = X^{\mathrm{an}} \times \mathbb{R}$, a specified continuous filtered L_∞ quasi-isomorphism

$$\Gamma(M, \mathfrak{l}_{\mathbb{A}}^{\mathrm{full}}) \longrightarrow \mathrm{Def}_{\mathbb{A}}^{\mathrm{full}},$$

and the local pairing needed for cotangent transgression. Here Γ denotes the chosen complex of global sections with its declared topology. The comparison includes compatibility with the specified boundary operations. The dual fields use the density-valued continuous dual, with compact support or boundary conditions that make the integrated pairing and integration by parts defined.

For a supplied local realization, define

$$\mathbf{BVEnv}_3^{\mathrm{full}}(\mathbb{A}) = T^*[-1]\,\mathrm{MC}(\mathfrak{l}_{\mathbb{A}}^{\mathrm{full}}).$$

Its action is the cotangent action displayed above. Constructing a local realization from an abstract mixed datum remains an additional existence problem.

Chapter 188

Intrinsic algebraic quantization

Let $\mathfrak{X}$ be a finite perfect derived formal moduli problem with a homological orientation μ: the corresponding Berezinian is invariant under the homological vector field. Functions on its shifted cotangent are completed polyvectors:

$$O(T^*[-1]\mathfrak{X}) = \widehat{\mathrm{PV}}(\mathfrak{X}) = \prod_{p \geq 0} \Gamma(\mathfrak{X}, \wedge^p T_{\mathfrak{X}})[p].$$

The homological orientation defines the divergence operator

$$\Delta_\mu : \widehat{\mathrm{PV}}(\mathfrak{X}) \longrightarrow \widehat{\mathrm{PV}}(\mathfrak{X})[1].$$

Theorem 188.1 (Oriented BV complex)**.** *The operator Δ_μ has order two and satisfies*

$$\Delta_\mu^2 = 0, \qquad [d_{\mathfrak{X}}, \Delta_\mu] = 0,$$

$$\Delta_\mu(fg) = \Delta_\mu(f)g + (-1)^{|f|} f \Delta_\mu(g) + (-1)^{|f|}\{f, g\}.$$

Consequently

$$\mathbf{QHdg}(\mathfrak{X}, \mu) = \left(\widehat{\mathrm{PV}}(\mathfrak{X})[\![\hbar]\!], d_{\mathfrak{X}} + \hbar\Delta_\mu\right)$$

is a square-zero quantum BV complex.

Proof. In oriented Darboux coordinates, Δ_μ is the sum of paired second derivatives. Their graded commutation gives $\Delta_\mu^2 = 0$. Invariance of the Berezinian under the homological vector field gives $[d_{\mathfrak{X}}, \Delta_\mu] = 0$. Expanding the operator on a product gives the bracket as the cross term. □

The orientation identifies polyvectors with differential forms by contraction. Under this identification,

$$\iota_\mu \Delta_\mu = d_{\mathrm{dR}}\, \iota_\mu.$$

Hence the BV complex is the Hodge-completed derived de Rham complex with differential

$$d_{\mathfrak{X}} + \hbar d_{\mathrm{dR}}.$$

The Rees parameter records Hodge degree.

Definition 188.2 (Intrinsic quantum complex of a mixed datum)**.** For a complete mixed datum $\mathbb{A}$ equipped with compatible homological orientations on its finite perfect charts, define $\mathbf{QHdg}_3^{\mathrm{full}}(\mathbb{A})$ by applying the preceding BV complex to finite perfect charts of

$$\mathrm{MC}(\mathbf{l}_{\mathbb{A}}^{\mathrm{full}})$$

and taking the separated inverse limit over the chart and weight filtrations.

Theorem 188.3 (Quantum master equation)**.** *A degree-zero action $S = S_0 + \hbar S_1 + \cdots$ defines a quantum state $e^{S/\hbar}\mu$ precisely when*

$$dS + \frac{1}{2}\{S, S\} + \hbar\Delta_\mu S = 0.$$

The equation is equivalent to

$$(d + \hbar\Delta_\mu)e^{S/\hbar} = 0.$$

Proof. Apply the second-order product formula to the exponential and collect powers of $\hbar$. □

Chapter 189

Perturbative factorization realization

The algebraic quantum differential does not specify a propagator or a space of singular kernels. Fix a local realization, elliptic gauge-fixing data, support and boundary conditions, and a renormalization prescription. A heat-kernel propagator is defined away from the diagonal. Extending its graph integrals to collisions requires counterterms and compatibility with the boundary conditions.

Definition 189.1 (Factorization realization data)**.** A factorization realization consists of renormalized local observable complexes, continuous disjoint-insertion maps, and coherent homotopies for their compositions. It includes extensions of graph weights to the required collision strata, their Stokes identities, and a quantum master equation in the permitted local functional complex. Weiss descent is a further required property of these actual complexes. A boundary realization also includes a comparison with the supplied mixed datum, preserving its two products, Lie operation, interface, and mixed law.

Under these hypotheses the disjoint-insertion maps and composition homotopies define the stated factorization object. The boundary comparison identifies its boundary operations with those of $\mathbb{A}$. The construction problem is to produce these data from the physical fields and action. A graphwise identity supplies the corresponding composition equation, whereas descent and boundary identification require their own maps and proofs.

Book XXXIV

Boundary duality, exchange, and reduction

Chapter 190

The Koszul–Morita square

A change from modules to comodules must transport the interface as well as each algebra. The mixed bar supplies the comparison object. Exchange then asks for motion in an additional topological direction, while BRST reduction asks whether the differential respects every transported operation.

Let P be a dualizable boundary state for $A^{\perp}$. Its boundary dual is

$$A^{!}_{P} = \mathrm{RHom}_{A^{\perp}}(P, P).$$

The ordinary bar construction gives explicit twisted tensor functors. Their extension to the mixed datum must also transport the interface and its geometric operations.

Let k be a field and let $A_{\mathrm{alg}} = \bigoplus_{w \geq 0} A_w$ be an augmented associative algebra in cohomological degree zero. Assume $A_0 = k$, augmentation projects onto A_0, and multiplication adds weights. Set $A = \prod_{w \geq 0} A_w$ and $C = \prod_{w \geq 0} B(A_{\mathrm{alg}})_w$, with degreewise products and the weight topology. The bar suspension lowers cohomological degree by one, with $b_2[a|b] = [ab]$. The degree-one twisting map $\tau : C \to A$ sends $[a]$ to a and vanishes on the empty word and longer words.

A module or comodule has the form $\prod_{w \geq w_0} M_w$, with differential preserving weight, for some integer w_0. Actions and coactions preserve weight. Comodule coactions are counital and coassociative, and use the tensor product

$$(C \widehat{\otimes} M)_w = \bigoplus_{u+v=w} B(A_{\mathrm{alg}})_u \otimes M_v.$$

The same convention defines $A \widehat{\otimes} M$. These sums are finite in each weight. Repeated reduced coactions vanish in weight w after more than $w - w_0$ steps. Invert quasi-isomorphisms in each weight to obtain $D_{\mathrm{wt}}(A)$ and $D_{\mathrm{wt}}(C)$.

Proposition 190.1 (Ordinary twisted tensor equivalence)**.** *The functors*

$$F(M) = C \widehat{\otimes}_{\tau} M, \qquad G(N) = A \widehat{\otimes}_{\tau} N$$

induce inverse equivalences $D_{\mathrm{wt}}(A) \simeq D_{\mathrm{wt}}(C)$. Writing $\Delta c = \sum c_{(1)} \otimes c_{(2)}$ and $\rho n = \sum n_{(-1)} \otimes n_{(0)}$, their differentials are

$$d_F(c \otimes m) = d_C c \otimes m + (-1)^{|c|} c \otimes d_M m + \sum (-1)^{|c_{(1)}|} c_{(1)} \otimes \tau(c_{(2)}) m,$$
$$d_G(a \otimes n) = a \otimes d_N n - \sum a\tau(n_{(-1)}) \otimes n_{(0)}.$$

The counit and unit are

$$\epsilon_M(a \otimes c \otimes m) = a\epsilon_C(c)m, \qquad \eta_N(n) = \sum n_{(-1)} \otimes 1 \otimes n_{(0)}.$$

Proof. The twisting equation makes both differentials square to zero. The action, coaction, and counit identities show that the displayed maps are chain maps and satisfy the triangle identities. For $GF(M)$, the extra degeneracy

of the two-sided bar inserts $1[\bar{a}_0|a_1|\cdots|a_r]m$ into $a_0[a_1|\cdots|a_r]m$, where $\bar{a}_0 = a_0 - \epsilon_A(a_0)1$. With simplicial boundary $\sum_i (-1)^i d_i$, it contracts the augmented bar. Multiplying length-r words by $(-1)^r$ gives the displayed differential convention. This contraction preserves weight, so the counit is a weightwise quasi-isomorphism. For $FG(N)$, filter each weight by the weight of the final N-factor. The reduced coaction lowers that weight. The associated graded unit is the contracted right-sided bar $k \otimes N_v \to B(k, A_{\mathrm{alg}}, A_{\mathrm{alg}}) \otimes N_v$. The filtration is finite in each total weight, so the unit is also a quasi-isomorphism. The same finite filtrations show that F, G preserve weightwise quasi-isomorphisms, since tensoring complexes over k preserves them. Thus the unit and counit give the asserted inverse equivalences. □

For a right module R and a left module L with the same lower-weight bounds, the two-sided bar has underlying weight-completed complex

$$\prod_n \bigoplus_{u+v_1+\cdots+v_r+w=n} R_u \otimes s\bar{A}_{v_1} \otimes \cdots \otimes s\bar{A}_{v_r} \otimes L_w, \qquad v_i > 0,$$

where the sum also runs over $r \geq 0$ and $\bar{A} = \ker(A_{\mathrm{alg}} \to k)$. Internal differentials and the two endpoint actions and adjacent products form its bar differential. Its realization computes $R \otimes^{\mathbb{L}}_A L$ in these weight-graded categories. Comparing it with a derived cotensor product requires a two-sided twisting comparison on the chosen comodule category and completion. Weight completion here is not identified with derived augmentation-adic completion.

Proposition 190.2 (Centre transport under a mixed equivalence)**.** *Suppose an equivalence between the compatible mixed module categories is supplied, with coherent compatibility with all four bar objects and the interface operations. It identifies the derived endomorphism objects of their identity functors.*

Proof. Conjugate a natural endomorphism by the equivalence and a quasi-inverse. The unit and counit homotopies give inverse maps on derived endomorphism objects and preserve composition and its coherent operations. □

Constructing such a mixed equivalence requires the restriction, collision, interface, and distributive-law comparisons for the actual geometric coefficients. Its unit and counit must respect those maps and the chosen inverse limits. A diagonal bimodule comparison must produce an actual compatible bicomodule before tensor products can be identified with geometric cotensor products. Transport of mixed Hochschild chains and traces further requires their cyclic comparison and the trace domains used in their definition. These are additional mixed Koszul–Morita constructions. The ordinary weight-graded equivalence supplies its stated one-colour functors, without identifying the other mixed carriers or their physical traces.

Chapter 191

Planar exchange and Tannaka reconstruction

A second topological direction gives paths exchanging parallel defects. Their homotopy classes form braid groups. A rigid meromorphic line category $\mathscr{L}$ with a faithful fibre functor ω has coend

$$\mathcal{O}(\mathbb{G}) = \int^{M \in \mathscr{L}} \omega(M)^{*} \otimes \omega(M).$$

Theorem 191.1 (Quantum symmetry from exchange)**.** *The coend is a coquasitriangular bialgebra. Its continuous dual acts on every line object. The meromorphic braiding gives the universal R-matrix on completed matrix coefficients. A spectral rational braiding yields an RTT presentation.*

Proof. Tensor product in $\mathscr{L}$ gives the coproduct on matrix coefficients. Evaluation gives the counit. Rigidity gives the antipode on the Hopf envelope. The braiding defines the coquasitriangular pairing, and the hexagon identities give the Yang–Baxter equation. Writing this naturality on a generating fibre produces the RTT relation. □

Chapter 192

BRST reduction

Let Q be a square-zero derivation of a mixed datum. Extend Q factorwise to every bar word. Since Q is a derivation for each structure map, it anticommutes with every bar differential.

Theorem 192.1 (BRST–bar bicomplex)**.** *The total differential*

$$D = Q + b_{\mathrm{ch}} + b_{\perp} + b_{\mathrm{mix}}$$

satisfies $D^{2} = 0$. *A bounded-below complete filtration gives a convergent spectral sequence*

$$E_{1} = H_{Q}\big(\mathrm{Bar}^{\mathrm{mix}}(\mathbb{A}; K)\big) \Longrightarrow H_{D}\big(\mathrm{Bar}^{\mathrm{mix}}(\mathbb{A}; K)\big).$$

When Q-cohomology is concentrated and tensor-exact, the target is the mixed bar of the reduced datum.

Proof. The square vanishes by the derivation identity and the four-colour bar theorem. The filtration by BRST degree gives the spectral sequence. Concentration and tensor exactness identify the E_{1} differential with the bar differential of the reduced operations. □

Book XXXV

Conformal covariance and Einstein completion

Chapter 193

Projective connections and the Virasoro tangent complex

A conformal boundary algebra carries a stress transformation law. To compare it with gravity, one must also specify the bulk fields, their action and their boundary conditions. Let $\mathcal{P}_c(X)$ be the formal moduli problem of projective connections with central parameter c. At a holomorphic projective connection p, its tangent complex is

$$\Omega^{0,\bullet}(X, T_X) \xrightarrow{\mathcal{D}_{p,c}} \Omega^{0,\bullet}(X, K_X^{\otimes 2}), \qquad \mathcal{D}_{p,c}(\epsilon) = \epsilon\, \partial p + 2(\partial \epsilon)\, p + \frac{c}{12}\partial^3 \epsilon.$$

The scalar differential equation and its half-density transformation in (21.14) construct this law.

Theorem 193.1 (Compatible projective covariance)**.** *Suppose the conformal action on A extends to a coherent coordinate action on every mixed operation. Choose a complete differential graded Lie model $\mathfrak{p}$ of the projective deformation problem and a compatible action on the relative deformation algebra $\mathfrak{m}$. Then* $\mathrm{MC}(\mathfrak{p} \ltimes \mathfrak{m})$ *maps to* $\mathrm{MC}(\mathfrak{p})$*, with fiber* $\mathrm{MC}(\mathfrak{m})$ *at the base point.*

Proof. The semidirect differential squares to zero and its bracket satisfies Jacobi because the supplied action is a differential graded Lie action. Projection to $\mathfrak{p}$ and inclusion of its kernel give the asserted map and fiber on nilpotent test algebras. Completeness gives the corresponding formal inverse limit. A conformal vector alone specifies the action on its vertex algebra. The hypothesis supplies compatibility with the other mixed operations. □

Chapter 194

Three-dimensional gravity as a flat Cartan theory

The Chern–Simons formulation combines the coframe and spin connection in one gauge field [1, 70]. Let M be oriented, let $\Lambda = -\ell^{-2}$, and let $\mathfrak{g}_\Lambda$ have generators J_a, P_a with

$$[J_a, J_b] = \epsilon_{ab}{}^{c} J_c, \qquad [J_a, P_b] = \epsilon_{ab}{}^{c} P_c, \qquad [P_a, P_b] = \ell^{-2} \epsilon_{ab}{}^{c} J_c.$$

The off-diagonal pairing is $b(J_a, P_b) = \eta_{ab}$ and $b(J_a, J_b) = b(P_a, P_b) = 0$. For $\mathcal{A} = \omega^a J_a + e^a P_a$, its curvature is

$$F_{\mathcal{A}} = \left(R^a + \frac{1}{2\ell^2} \epsilon^a{}_{bc} e^b \wedge e^c \right) J_a + T^a P_a.$$

Theorem 194.1 (Cartan–Einstein equivalence)**.** *On the invertible-coframe locus, $F_{\mathcal{A}} = 0$ is equivalent to*

$$T^a = 0, \qquad G_{\mu\nu} + \Lambda g_{\mu\nu} = 0, \qquad g_{\mu\nu} = \eta_{ab} e^a_\mu e^b_\nu.$$

With the pairing and coefficients of Proposition 21.1, its action is the first-order Einstein–Hilbert action plus the displayed boundary term.

Proof. Torsion-freeness determines the Levi–Civita spin connection. The rotation curvature equation gives constant sectional curvature $-\ell^{-2}$. In dimension three the Riemann tensor is determined by the Ricci tensor, giving both directions of the field comparison. Equation (21.1) gives the action and its exact boundary term. □

Chapter 195

Four local gravitational complexes

Fix an invertible constant-curvature background. The flat adjoint complex, its Einstein–Cartan splitting, the metric Einstein complex and the corrected Calabi complex give four local descriptions of its deformations.

Theorem 195.1 (Chain equivalence of gravitational presentations)**.** *These four complexes are homotopy equivalent by local differential operators. On a sufficiently small geodesically convex open set they resolve the sheaf of Killing fields.*

Proof. The Einstein–Cartan splitting is a change of components in the flat adjoint complex. The metric reduction cancels three invertible algebraic pairs. They pair Lorentz parameters with skew coframes, connections with torsion, and skew curvature with its algebraic Bianchi component. Lemma 21.3 constructs the chain maps and homotopies. The surviving operators are K, E_Λ and divergence.

In dimension three, Ricci contraction identifies the corrected Calabi curvature with E_Λ. A local parallel trivialization and the Poincaré homotopy give the local resolution, as in Theorem 21.4. □

Chapter 196

Brown–Henneaux and the projective boundary

Use $A^{\pm} = \omega \pm \ell^{-1} e$ and the sector trace $B(J_a, J_b) = \eta_{ab}/2$. The Einstein normalization gives $k = \ell/(4G)$. Choose the chiral boundary reduction and orientation of (21.12).

Theorem 196.1 (Brown–Henneaux central charge)**.** *The reduced classical stress field has transformation*

$$\delta_\epsilon T = \epsilon T' + 2\epsilon' T + \frac{k}{2}\epsilon''', \qquad c = 6k = \frac{3\ell}{2G}.$$

Thus it defines the corresponding projective boundary variable.

Proof. For $a = \left(\begin{smallmatrix} 0 & q \\ 1 & 0 \end{smallmatrix}\right)$, preservation of the gauge slice gives the matrix parameter in (21.12). Its charge is $Q_\epsilon = (2\pi)^{-1} \int \epsilon T$, with $T = -kq$. The upper-right variation gives the stated law and central term [12]. □

Chapter 197

Einstein completion of a conformal pair

Suppose A_L, A_R have supplied compatible projective families with the same central parameter. Choose their maps to a model of the projective deformation problem. Their homotopy fiber product defines a relative deformation problem over that model. Equality of central parameters alone does not supply these maps.

Definition 197.1 (Candidate Einstein BV completion)**.** Suppose the relative algebra and a continuous L_∞ action of the Cartan algebra have been supplied, together with their local realization. The associated cotangent candidate is

$$\mathbf{EinBV}_3(A_L, A_R) = T^*[-1]\,\mathrm{MC}\big(\mathfrak{g}_{\mathrm{grav}} \ltimes_{\mathrm{Ham}} \mathfrak{m}_{A_L,A_R}\big).$$

Here $\mathfrak{g}_{\mathrm{grav}}$ is the Cartan deformation algebra and $\mathfrak{m}_{A_L,A_R}$ is the relative mixed deformation algebra after projective matching.

Theorem 197.2 (Cotangent action for a supplied coupled algebra)**.** *Suppose projective matching and a continuous L_∞ action have constructed the displayed complete semidirect product. Suppose its local realization and integrated dual pairing are defined. Its shifted cotangent has action*

$$S(\alpha, \beta) = \langle \beta, \mathcal{F}(\alpha) \rangle$$

and satisfies the classical master equation. Its Bianchi identity is the L_∞ identity of the coupled algebra.

Proof. The supplied semidirect product has a square-zero homological vector field. Its cotangent lift is Hamiltonian for the displayed action and remains square-zero. This proves the master equation and the Bianchi identity. □

This cotangent theory contains independent dual fields. Its comparison with Einstein BV theory must preserve the action, the BV pairing and boundary terms on the invertible-coframe domain. A coupled physical theory also needs a matter action and an equivariant Hamiltonian action on its specified phase space.

Chapter 198

Higher-spin fields for a supplied action

Choose the principal $\mathfrak{sl}_2$ embedding in $\mathfrak{sl}_N$. Its adjoint representation decomposes as

$$\mathfrak{sl}_N = \bigoplus_{s=2}^{N} V_{2s-2}, \qquad \sum_{s=2}^{N} (2s-1) = N^2 - 1.$$

Theorem 198.1 (Principal higher-spin spectrum)**.** *In a supplied $\mathfrak{sl}_N \oplus \mathfrak{sl}_N$ Chern–Simons theory with an invertible AdS background, this decomposition gives one frame-field pair of each spin $2, \dots, N$. Its principal classical boundary reduction has the corresponding W_N currents.*

Proof. The highest-weight vectors E^j in $\mathrm{End}(\mathrm{Sym}^{N-1}\mathbb{C}^2)$ give (21.16). The module V_{2s-2} is the symmetric traceless Lorentz representation of a spin-s frame field. The supplied action gives its linearized torsion and curvature equations. Eliminating the connection and Lorentz component gives the Fronsdal variables. Principal Drinfeld–Sokolov boundary reduction gives the currents of weights $2, \dots, N$ [14, 42]. □

Realizing an arbitrary pair of W_N boundary theories requires their boundary action, real form, states and comparison map. The representation decomposition fixes the spin labels.

Book XXXVI

Local gravitational complexes and Gaussian comparisons

Chapter 199

The four complexes in coordinates

The local chain maps compare classical deformation equations. The following formulas specify their operators. Quantum Gaussian comparisons also require measures and domains.

199.1 Flat adjoint complex

At a flat connection, the complex is

$$\Omega^0(M,\mathfrak{g}_\Lambda) \xrightarrow{d_{\mathcal{A}}} \Omega^1(M,\mathfrak{g}_\Lambda) \xrightarrow{d_{\mathcal{A}}} \Omega^2(M,\mathfrak{g}_\Lambda) \xrightarrow{d_{\mathcal{A}}} \Omega^3(M,\mathfrak{g}_\Lambda).$$

The degrees contain gauge parameters, fields, curvature variations and Bianchi identities.

199.2 Einstein–Cartan complex

For $(u,v) = (\delta e, \delta\omega)$, the gauge and curvature maps are

$$u = D_\omega \xi + [e,\lambda], \quad v = D_\omega \lambda + \ell^{-2}[e,\xi],$$
$$t = D_\omega u + [e,v], \quad r = D_\omega v + \ell^{-2}[e,u].$$

The third map is $(t,r) \mapsto (D_\omega t + [e,r], D_\omega r + \ell^{-2}[e,t])$.

Proposition 199.1 (Chain identity)**.** *Consecutive Einstein–Cartan differentials compose to zero on the torsion-free constant-curvature background.*

Proof. They are the components of $d_{\mathcal{A}}$, whose square is $[F_{\mathcal{A}}, -] = 0$. Equivalently, substitution of $D_\omega e = 0$ and $F_\omega = -[e,e]/(2\ell^2)$ cancels the curvature terms by Jacobi. □

199.3 Metric complex

The metric variation is $h_{\mu\nu} = 2\eta_{ab} e^a_{(\mu} u^b_{\nu)}$. Its gauge operator is $K(\zeta)_{\mu\nu} = \nabla_\mu \zeta_\nu + \nabla_\nu \zeta_\mu$. To write its curvature operator without commuting derivatives, put

$$C^\rho{}_{\mu\nu} = \tfrac{1}{2} g^{\rho\sigma}(\nabla_\mu h_{\nu\sigma} + \nabla_\nu h_{\mu\sigma} - \nabla_\sigma h_{\mu\nu}),$$
$$r_{\mu\nu} = \nabla_\rho C^\rho{}_{\mu\nu} - \nabla_\nu C^\rho{}_{\mu\rho},$$
$$r = g^{\mu\nu} r_{\mu\nu} - 2\Lambda g^{\mu\nu} h_{\mu\nu}.$$

Then

$$(E_\Lambda h)_{\mu\nu} = r_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} r - 2\Lambda h_{\mu\nu}.$$

It obeys $E_\Lambda K = 0$ and $\nabla^\mu (E_\Lambda h)_{\mu\nu} = 0$. For $h = g$ it gives $E_\Lambda h = \Lambda g$.

Theorem 199.2 (Explicit metric reduction)**.** *On the linearized torsion-free locus, solving for v and substituting into the curvature equation gives $E_\Lambda h = 0$. The kernel of $(u, v) \mapsto h$ on this locus is the infinitesimal Lorentz action.*

Proof. Equation (21.7) solves the torsion equation uniquely for v. Removing the skew part of u leaves $h/2$ and its Levi–Civita connection variation C. Varying $G + \Lambda g$ gives the displayed operator. When $h = 0$, the remaining skew coframe variation and its uniquely determined connection variation are a Lorentz rotation. □

199.4 Calabi complex

The Calabi curvature operator is the variation of $R_{\rho\mu\sigma\nu} - \Lambda(g_{\rho\sigma}g_{\mu\nu} - g_{\rho\nu}g_{\mu\sigma})$. This tensor vanishes at the background, so its variation kills the Killing operator. In dimension three, Ricci contraction and subtraction of half the scalar trace identify it with E_Λ. The corresponding Bianchi operator becomes divergence. This gives the Calabi presentation of (21.8).

Chapter 200

Boundary currents and algebraic reduction

The Chern–Simons boundary form, with the specified trace, is

$$\Omega_{\partial} = \frac{k}{4\pi} \int_{\partial M} B(\delta A \wedge \delta A).$$

After choosing a chiral polarization, its moment map is the flatness constraint. Its classical principal reduction gives (21.13).

Theorem 200.1 (Algebraic affine reduction)**.** *For the ordinary affine* $\mathfrak{sl}_2$ *vertex algebra at level* $\kappa \neq -2$*, the principal reduced stress tensor has Sugawara, Cartan-improvement and ghost terms. Its central charge is*

$$c_{\mathrm{DS}}(\kappa) = \frac{3\kappa}{\kappa + 2} - 6\kappa - 2 = 1 - \frac{6(\kappa + 1)^2}{\kappa + 2}.$$

Proof. The Sugawara term gives the first contribution. The Cartan derivative contributes -6κ, and the ghost pair of weights zero and one contributes -2. The free-field calculation following (21.15) verifies the same coefficient. □

The physical relation between κ and the gravitational coefficient k requires the regulated current and ghost insertion with its finite terms. The classical equality $c = 6k$ does not specify that quantum map.

Chapter 201

Matter stress and Hamiltonian equivariance

Supply a diffeomorphism-invariant matter action with variation $\delta S_{\mathrm{m}} = \int \mathrm{vol}_g (\frac{1}{2} T^{\mu\nu} \delta g_{\mu\nu} + \mathcal{E}_\varphi \delta\varphi)$. A Hamiltonian description additionally supplies its symplectic phase space and moment functions for the symmetry.

Theorem 201.1 (Hamiltonian and Noether identities)**.** *On an even symplectic phase space, the finite ghost charge $\Omega = c^a \mu_a - \frac{1}{2} f_{ab}{}^c c^a c^b b_c$ is square-zero exactly when the moment functions are equivariant. For the supplied diffeomorphism-invariant scalar action,*

$$\nabla_\mu T^{\mu\nu} = \mathcal{E}_\varphi \nabla^\nu \varphi.$$

Thus stress conservation holds on the matter equations.

Proof. Proposition 21.5 computes the ghost charge square. Substituting a compactly supported infinitesimal diffeomorphism in the action variation and integrating by parts gives the Noether identity. □

For a nonlinear metric coupling, the full matter action contains higher metric variations. The nonzero quadratic term in (21.18) already occurs for the scalar kinetic action. Its linear stress coupling alone is not the full gravitational BV action.

Chapter 202

One-loop determinants with specified domains

Let M be closed and Riemannian, and let the flat adjoint bundle be unitary and acyclic. Choose the elliptic de Rham gauge fixing and the corresponding zeta determinants. Write

$$\tau_{\mathrm{RS}} = \prod_{j=0}^{3} (\det_{\zeta} \Delta_j)^{(-1)^{j+1} j/2}.$$

Theorem 202.1 (One-loop determinant)**.** *After the stated Gaussian measure normalization, the gauge-fixed one-loop modulus is*

$$|Z_{1\text{-loop}}| = (\det_{\zeta} \Delta_0)^{3/4} (\det_{\zeta} \Delta_1)^{-1/4} = \tau_{\mathrm{RS}}^{1/2}.$$

Its phase requires the eta-invariant and framing convention.

Proof. The gauge-orbit Jacobian and the coexact bosonic Gaussian give the determinant exponents in Proposition 21.6. Hodge duality identifies the degree-three and degree-zero spectra and the degree-two and degree-one spectra. The same identities for zeta determinants give the displayed formula. The regularized signature of the self-adjoint quadratic operator supplies the phase. □

For a solid torus and a gravitational real form, these hypotheses must be replaced by the actual integration cycle, closed boundary domains and residual-mode measures. Transport between gravitational complexes must retain the Jacobians of the cancelled pairs, their determinant-line connections and normalized cut states. A quasi-isomorphism and a determinant-line orientation alone do not imply equality of these Gaussian data.

Chapter 203

The Cardy expression and the area law

Assume a boundary theory with the required Cardy high-energy asymptotic. Its cylinder energies are $H_L = L_0 - c_L/24$ and $H_R = \bar{L}_0 - c_R/24$, and its leading entropy expression is

$$S_C = 2\pi\sqrt{c_L H_L/6} + 2\pi\sqrt{c_R H_R/6}.$$

For the BTZ charges of (21.22), take $r_+ > |r_-|$ and $c_L = c_R = 3\ell/(2G)$.

Theorem 203.1 (Cardy–Bekenstein expression)**.** *Substitution gives*

$$S_C = \frac{2\pi r_+}{4G}.$$

Thus the leading Cardy expression equals the horizon circumference divided by $4G$ in its assumed regime.

Proof. The two Cardy terms become $\pi(r_+ + r_-)/(4G)$ and $\pi(r_+ - r_-)/(4G)$. Their sum is the displayed expression. The existence of the boundary theory and the required asymptotic are separate hypotheses [4, 15]. □

Chapter 204

Constant connections and higher-spin charges

On a constant principal companion slice, invariant polynomials of the connection matrix give its characteristic coefficients. Holonomy is the matrix exponential and retains different data.

Theorem 204.1 (Constant charge coordinates)**.** *For a traceless constant N-by-N companion matrix a, the functions* $\operatorname{tr}(a^s)$, $2 \le s \le N$, *determine its characteristic polynomial. On a chosen regular logarithm branch, the holonomy* $\exp(-La)$ *determines the same coefficients. Without that branch it need not determine them.*

Proof. Newton's identities recover the characteristic coefficients from the stated traces. A regular matrix logarithm recovers the eigenvalues of a on its selected branch. The matrices $\operatorname{diag}(x, -x)$ and $\operatorname{diag}(x + 2\pi i n/L, -x - 2\pi i n/L)$ have equal holonomy and generally different quadratic traces. The explicit cubic companion example in Chapter 21 fixes the normalization. □

For nonconstant connections one needs ordered transport rather than the exponential of a pointwise matrix. The marked transport in Proposition 21.9 gives its classical endpoint actions and sewing. Its renormalized effective-line comparison remains part of Problem 21.10.

Book XXXVII

Anomalies, Hall sources, and automorphic completion

Chapter 205

Four native anomaly classes

A local anomaly, a determinant-line curvature, and an arity-zero bar term belong to different complexes. A cancellation in one complex becomes a geometric cancellation only through a comparison preserving the relevant class and its primitive. Hall and automorphic constructions require their coefficient and orientation maps for the same reason.

The construction carries four curvature classes.

(i) A local BV anomaly lies in the degree-one local deformation complex.

(ii) A projective anomaly is the curvature of the determinant line over parameter space.

(iii) A framing anomaly is the transgression of a characteristic class to a three-manifold.

(iv) A curved Koszul term is the arity-zero component of a curved coalgebra.

Theorem 205.1 (Algebraic curvature cancellation)**.** *Let κ be a closed central degree-two cocycle in a differential graded Lie algebra. Adjoining a central degree-one generator u with $du = \kappa$ makes its cohomology class zero. If κ is the curvature of the original curved algebra, twisting the extension by $-u$ gives zero curvature.*

Proof. The extended differential squares to zero because $d\kappa = 0$, and centrality makes it a Lie derivation. The curved Maurer–Cartan expression at $-u$ is $\kappa + d(-u) + [-u, -u]/2 = \kappa - \kappa = 0$. □

Geometric cancellation requires a supplied inverse invertible theory with its complete line, boundary and sewing data. Opposite curvatures alone can leave nontrivial flat holonomy, as the circle connection in Chapter 21 demonstrates.

Chapter 206

Hall and Calabi–Yau sources

Let C be a Calabi–Yau category with a Hall moduli problem. An oriented extension correspondence gives a Hall algebra H^+. A factorization pushforward to the boundary curve gives a chiral-convolution algebra. Relative induction supplies the remaining colours.

Proposition 206.1 (Transport under a supplied mixed Morita equivalence)**.** *Suppose a Hall construction supplies a complete mixed boundary datum and a Morita equivalence of its compatible module category with that of $\mathbb{A}$. The equivalence identifies their derived centres. A comparison of local deformation algebras, BV pairings, and actions is additional data for a BV comparison. Quantum line and amplitude comparisons additionally require the monoidal and cyclic structures used to define them.*

Proof. Conjugating natural transformations by the equivalence and its inverse identifies the endomorphisms of the two identity functors. The unit and counit homotopies provide the inverse comparison and preserve composition. This proves the centre comparison. It does not construct the additional local, symplectic, or cyclic maps. □

An algebra map alone does not induce the required forward centre map. For example, the homomorphism $\mathbb{C}[x] \to M_2(\mathbb{C})$ with $x \mapsto \operatorname{diag}(0,1)$ sends the central element x to a noncentral matrix. Its commutator with the matrix unit E_{12} is $-E_{12}$. Thus even the degree-zero condition fails for a centre map extending the given algebra map. Hall multiplication maps must be supplemented by an actual coherent module comparison before centre transport can be applied.

For $X = K3 \times E$, suppose a geometric Hall source supplies the complete charge-graded BPS Lie superalgebra and Cartan action required in Volume IV. Its semidirect product with the common Cartan defines $\mathfrak{b}_X$. Volume IV then constructs the completed Lie amalgam with the Igusa Borcherds Borel $\mathfrak{b}_\Delta$. Applying the completed enveloping functor gives the cocommutative Hopf algebra

$$H_{X\boxtimes\Delta} = \widehat{U}(\mathfrak{b}_X)\widehat{*}_{\widehat{U}(\mathfrak{h}_\Pi)}\widehat{U}(\mathfrak{b}_\Delta).$$

The two Hopf retractions preserve the geometric and automorphic sectors separately. A q-deformed mixed Hopf algebra is supplied by the quantum amalgam theorem when a geometric Hall coproduct and pairing are present.

206.1 Mixed realization of the amalgam

The Hopf retractions above are maps of their stated algebraic objects. A mixed realization must construct chiral and transverse operations, an interface, and a compatible distributive law. To realize the retractions, it must then supply maps preserving those operations and the chosen completions.

A BV retraction requires further maps between the local deformation algebras, together with compatible pairings and actions. The counterexample to forward centre functoriality prevents deducing these maps from the Hopf retractions alone. The intended geometric–automorphic theory asks for this mixed realization and for a primitive comparison quotient on its actual boundary operators. Its interaction

ideal is the kernel of that constructed quotient, rather than an ideal determined by equality of scalar characters.

Chapter 207

Igusa root states and the mixed-boundary comparison

The Gritsenko–Nikulin presentation defines the root Lie superalgebra $\mathfrak{g}_{\Delta_5}$ and its positive subalgebra $\mathfrak{n}_+$. Let $d_\alpha = \operatorname{sdim}(\mathfrak{g}_{\Delta_5})_\alpha$, and fix the positive cone, its proper height, and Weyl vector ρ. Put

$$P = \prod_{\alpha>0}(1-e^{-\alpha})^{d_\alpha}, \qquad \mathrm{Den}_\Delta = e^{-\rho}P = D_5(2Z), \qquad \jmath(Z) = Z/2.$$

The products are first interpreted in the positive-height completion.

Theorem 207.1 (Root character and reduced scalar)**.** *The root Fock space* $\operatorname{Sym}(\mathfrak{n}_+)$*, formed with super symmetry, has supercharacter* P^{-1}*. Consequently*

$$-\jmath^*\big(e^{2\rho}\operatorname{sch}(\operatorname{Sym}(\mathfrak{n}_+)^{\otimes 2})\big) = -(\jmath^*\,\mathrm{Den}_\Delta)^{-2} = -\chi_{10}^{-1}.$$

For a nonsingular projective K3 surface S and an elliptic curve E, this scalar also has an enumerative interpretation. Let $N_{g,h,d}$ be the reduced disconnected Gromov–Witten invariant in class (β_h, d), where β_h is nonzero and primitive with square $2h-2$. Use the incidence insertion $\beta_h^\vee \otimes [\mathrm{point}]$, with $\langle \beta_h, \beta_h^\vee\rangle = 1$. Then the series

$$\mathcal{Z}(u,q,s) = \sum_{g,h,d\ge 0} N_{g,h,d}\,u^{2g-2}q^{h-1}s^{d-1}$$

is the Laurent expansion of $-\chi_{10}(r,q,s)^{-1}$ under $r = e^{iu}$, by [62, Theorem 1].

Proof. An even root vector contributes $(1-e^{-\alpha})^{-1}$ to the supercharacter of its symmetric algebra. An odd root vector contributes $1-e^{-\alpha}$, because its symmetric algebra in super vector spaces is an exterior algebra. Multiplying the two contributions gives P^{-1}. Proper height makes every coefficient calculation finite. The Weyl factor gives $e^{2\rho}P^{-2} = \mathrm{Den}_\Delta^{-2}$. Pullback by $\jmath$ gives $D_5(Z)^{-2} = \chi_{10}(Z)^{-1}$. Oberdieck–Pixton, Theorem 1, identifies the separately defined reduced numerical generating series with its negative [62]. This last theorem is an enumerative input; the character calculation supplies the root side. □

A primitive realization starts with parity-preserving root classes in a charge-graded Lie superalgebra $\mathfrak{p}$ satisfying the defining relations. Require the classes to preserve charge. The free Lie presentation gives an even charge-preserving map $f : \mathfrak{n}_+ \to \mathfrak{p}$.

Applying f to tensor words constructs $U(f)$. With trivial coefficients, use Chevalley chains $C_*^{\mathrm{CE}}(\mathfrak{a}) = \operatorname{Sym}(\mathfrak{a}[1])$, where the homological suspension raises degree by one and the differential contracts pairs by the bracket. Applying the suspension of f to every factor gives a chain map because f preserves brackets.

For the level-zero Lie conformal current, define $\mathrm{Cur}(\mathfrak{a}) = \mathbb{C}[\partial]\otimes\mathfrak{a}$, with ∂ even. Its generator bracket is $[x_\lambda y] = [x,y]$. Extend it by $[\partial x_\lambda y] = -\lambda[x_\lambda y]$ and $[x_\lambda \partial y] = (\partial+\lambda)[x_\lambda y]$. The formula $\partial^j x \mapsto \partial^j f(x)$ defines $\mathrm{Cur}(f)$. Charge preservation makes these maps compatible with the specified proper-height quotients and their inverse limits.

A complete mixed boundary realization requires further geometric maps. The compact source consists of sheaves and extension flags on a common elliptic torsor. Its critical realization must supply reduced coefficient complexes, orientation cross-line maps, unit and flag coherences, and the admissible pull–push operations. A chiral comparison must preserve actual support and moving-insertion operations. A quantum-double comparison must preserve coproducts, pairings and the domain of exchange operators. The finite Calabi–Yau quiver completions require their specified dg maps; ordinary full-subquiver inclusions do not in general give those maps.

The mixed datum to which a BV envelope is applied must therefore include these operations and compatible maps between them. Identifying the primitive algebra of that envelope requires an actual map of its deformation complex to the root complex. A determinant comparison requires a skew perfect family, a square-root line and section, and an isomorphism to the automorphic line preserving transport and normalization. An operator-trace interpretation further requires its state space and trace domain. Constructing this datum and these comparisons is the Igusa mixed-boundary problem. The root character identity specifies the scalar that their final evaluation must recover.

Chapter 208

The carrier lattice

The complete theory uses the carrier set

$$\mathfrak{C} = \{\mathrm{diag}, \mathrm{ch}, \bot, \mathrm{int}, \mathrm{ex}, \mathrm{cyc}, \mathrm{mod}, \mathrm{BV}, \mathrm{Ein}\}.$$

For a subset $S \subset \mathfrak{C}$, let $\mathrm{Str}_S(\mathbb{A})$ be the homotopy limit of all mapping spaces attached to incidence cells supported in S.

Theorem 208.1 (Carrier descent)**.** *When S and T form a relation-excisive pair, the square*

$$\begin{array}{ccc} \mathrm{Str}_{S\cup T}(\mathbb{A}) & \longrightarrow & \mathrm{Str}_S(\mathbb{A}) \\ \downarrow & & \downarrow \\ \mathrm{Str}_T(\mathbb{A}) & \longrightarrow & \mathrm{Str}_{S\cap T}(\mathbb{A}) \end{array}$$

is a homotopy pullback. Thus a global structure is assembled from its carrier restrictions together with their common interface.

Proof. Relation-excision identifies the incidence category of $S \cup T$ with the pushout of the incidence categories for S and T over their intersection. Homotopy limits send this pushout to the displayed pullback. □

Definition 208.2 (Relation-complete carrier cover)**.** Let $U \subseteq \mathfrak{C}$ and let $\mathcal{U} = \{S_i\}_{i\in I}$ be a finite family with $U = \bigcup_i S_i$. The family is *relation-complete* when the canonical functor from the Cech homotopy colimit

$$\operatorname*{hocolim}_{\varnothing \neq J \subseteq I} \mathcal{I}_{\bigcap_{j\in J} S_j} \longrightarrow \mathcal{I}_U$$

is an equivalence. Equivalently, every incidence cell and every relation among its faces is contained in one member of the cover, with all overlaps recorded by the intersections.

Theorem 208.3 (Power-set coherence)**.** *The assignment*

$$S \longmapsto \mathrm{Str}_S(\mathbb{A})$$

is a contravariant functor on the full power set of $\mathfrak{C}$. For every relation-complete carrier cover $\mathcal{U} = \{S_i\}_{i\in I}$ of U, the restriction map is an equivalence

$$\mathrm{Str}_U(\mathbb{A}) \xrightarrow{\sim} \operatorname*{holim}_{\varnothing \neq J \subseteq I} \mathrm{Str}_{\bigcap_{j\in J} S_j}(\mathbb{A}).$$

Thus every subtheory, every intersection of subtheories, and the complete theory are values of one descent object.

Proof. An inclusion $S \subseteq T$ induces the full inclusion $\mathcal{I}_S \subseteq \mathcal{I}_T$ and therefore the restriction map on homotopy limits. Composition of inclusions gives functoriality. A relation-complete cover presents $\mathcal{I}_U$ as the displayed homotopy colimit of its intersection diagram. Mapping that colimit into the diagram of structure spaces converts it into the stated homotopy limit. The resulting equivalence is natural under refinement of the carrier cover. □

Definition 208.4 (Exact-support deformation complex)**.** For $S \subseteq \mathfrak{C}$, let $\mathrm{Def}_{\leq S}$ be the subcomplex generated by incidence cells supported in S, and set

$$\mathrm{Def}_{=S} = \mathrm{cofib}\Big(\operatorname*{hocolim}_{T \subsetneq S} \mathrm{Def}_{\leq T} \longrightarrow \mathrm{Def}_{\leq S}\Big).$$

Thus $\mathrm{Def}_{=S}$ retains the deformation cells whose minimal carrier support is exactly S.

Theorem 208.5 (Carrier-support spectral sequence)**.** *Let* $\mathrm{Def}_{\mathbb{A}}^{\mathrm{full}}$ *be the cellular convolution complex of the complete incidence diagram. Filter it by the cardinality of minimal carrier support. The finite filtration gives a convergent spectral sequence*

$$E_1^{p,q} = \bigoplus_{\substack{S \subseteq \mathfrak{C} \\ |S| = p}} H^{p+q}(\mathrm{Def}_{=S}) \Longrightarrow H^{p+q}\Big(\mathrm{Def}_{\mathbb{A}}^{\mathrm{full}}\Big).$$

A class on the S-summand is a deformation or curvature class whose first complete geometric carrier is S. The first differential is the signed incidence map between exact-support quotients.

Proof. The support-cardinality filtration is finite because $\mathfrak{C}$ is finite. Its pth associated-graded quotient is the direct sum of the exact-support complexes $\mathrm{Def}_{=S}$ with $|S| = p$. The spectral sequence of a finite filtered complex gives the displayed first page and converges strongly to the cohomology of the total convolution complex. The cellular determinant orientation gives the incidence signs. □

Chapter 209

Sixteen structural laws

The theory is governed by the following laws.

(i) A geometric carrier determines the type of its operation.

(ii) The diagonal stores the complete principal-part filtration.

(iii) The interval stores ordered fusion.

(iv) The plane stores exchange.

(v) The circle stores trace.

(vi) The stable curve stores modular sewing.

(vii) The transverse, associative-chiral, and chiral-Lie bars are distinct.

(viii) Chiral PBW compares the last two on enveloping objects.

(ix) The interface bar records the mixed distributive law.

(x) The derived centre is the universal interior action.

(xi) The deformation algebra is the tangent algebra of the centre-extended datum.

(xii) Maurer–Cartan curvature is the field equation.

(xiii) The L_∞ identity is the Bianchi identity.

(xiv) Shifted cotangent is the universal classical BV envelope.

(xv) Hodge-completed derived de Rham theory is the intrinsic algebraic quantum complex of the oriented formal moduli problem.

(xvi) A gravitational realization requires projective matching, a coupled local field algebra, and a comparison preserving the physical BV action and pairing.

Chapter 210

Coherent actions and gravitational reconstruction

A supplied mixed boundary datum defines compatible modules and their derived natural endomorphisms. A specified coloured operad organizes deformations of that datum together with its central action. These are constructions relative to the full operations and their coherence maps. The centre alone does not recover the datum from which the deformation algebra is formed.

A supplied local realization of the deformation algebra admits shifted cotangent transgression. With compatible invariant orientations, the finite perfect charts also carry the algebraic BV differential. The physical bulk is defined independently by its fields, action, gauge symmetry, boundary conditions, and quantization. Reconstruction requires a coherent action map from its observables to an independently defined geometric centre, followed by an equivalence theorem on the stated sector.

For gravity, the invertible coframe gives the classical Cartan–Einstein correspondence. The cotangent construction supplies a formal comparison theory with extra dual fields. Its identification with gravitational BV fields requires an action-preserving and pairing-preserving map. Quantum boundary currents, a continuum measure, global holonomy sectors, allowed fillings, and compatible states remain necessary for the quantum and global comparison.

Hall and Calabi–Yau geometry can supply boundary operations when their correspondences, coefficients, orientations, and comparison maps are constructed. Exchange additionally uses a monoidal category and a fibre functor. Cyclic pairings and sewing determine amplitudes only on their stated domains. These comparisons are the mathematical steps by which local boundary operations could reconstruct physical and geometric theories.

Book XXXVIII

Technical appendices to the preceding book

Chapter 211

The Chern–Simons normalization

For the sector trace $B(J_a, J_b) = \eta_{ab}/2$, define

$$S_{\mathrm{CS},B}[A] = \frac{k}{4\pi}\int_M \Big(B(A \wedge dA) + \tfrac{1}{3} B(A \wedge [A, A]) \Big).$$

For $A^{\pm} = \omega \pm \ell^{-1} e$, Proposition 21.1 gives

$$\begin{aligned} S_{\mathrm{CS},B}[A^+] - S_{\mathrm{CS},B}[A^-] = \frac{k}{4\pi\ell}\int_M \epsilon_{abc} e^a \wedge \left(R^{bc} + \frac{1}{3\ell^2} e^b \wedge e^c \right) \\ - \frac{k}{4\pi\ell}\int_{\partial M} \eta_{ab}\omega^a \wedge e^b. \end{aligned}$$

The Einstein–Hilbert normalization is $k = \ell/(4G)$. For the off-diagonal Cartan pairing $b(J_a, P_b) = \eta_{ab}$, the coefficient in front of $\mathrm{CS}_b(\mathcal{A})$ is instead $k/(4\pi\ell)$. The boundary term displayed here is the one from the action identity. A chosen boundary variational problem may require additional terms.

Chapter 212

The principal embedding count

The exponents of $\mathfrak{sl}_N$ are $1, 2, \ldots, N-1$. Under the principal $\mathfrak{sl}_2$ embedding,

$$\mathfrak{sl}_N = \bigoplus_{m=1}^{N-1} V_{2m}.$$

The dimensions satisfy

$$\sum_{m=1}^{N-1} (2m+1) = (N-1)(N+1) = N^2 - 1.$$

Writing $s = m+1$ gives one field at every spin $s = 2, \ldots, N$.

Chapter 213

Low-arity mixed signs

Let $x|k|y$ be a suspended interface word. The two length-lowering contractions are

$$b_L[x|k|y] = (-1)^{|x|}(xk)|y, \qquad b_R[x|k|y] = (-1)^{|x|+|k|}x|(ky).$$

Applying the remaining contraction gives the two composites $(xk)y$ and $x(ky)$ with opposite determinant-line signs. Their sum vanishes by the interface associativity relation. A transverse and a chiral contraction acquire the product orientation

$$\det(E_{\mathrm{ch}} \oplus E_{\perp}) = \det(E_{\mathrm{ch}}) \otimes \det(E_{\perp}),$$

so reversing their order contributes the Koszul sign and yields anticommutation.

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